*TM 1-1520-237-23-10

TECHNICAL MANUAL

AVIATION UNIT AND INTERMEDIATE MAINTENANCE FOR

ARMY MODELS UH-60A, UH-60L, EH-60A, HH-60A AND HH-60L HELICOPTERS

CHAPTER 10
MAINTENANCE INSTRUCTIONS

*This manual together with TM 1-1520-237-23-1, TM 1-1520-237-23-2, TM 1-1520-237-23-3, TM 1-1520-237-23-4, TM 1-1520-237-23-5, TM 1-1520-237-23-6, TM 1-1520-237-23-7, TM 1-1520-237-23-8, TM 1-1520-237-23-9, TM 1-1520-237-23-11, TM 1-1520-237-23-12, TM 1-1520-237-23-13, TM 1-1520-237-23-14, TM 1-1520-237-23-15, TM 1-1520-237-23-16, TM 1-1520-237-23-17, TM 1-1520-237-23-18, TM 1-1520-237-23-19 and TM 1-1520-237-23-20, dated 17 April 2006, supersedes TM 1-1520-237-23-1, TM 1-1520-237-23-2, TM 1-1520-237-23-3, TM 1-1520-237-23-4, TM 1-1520-237-23-5, TM 1-1520-237-23-6, TM 1-1520-237-23-7, TM 1-1520-237-23-8, TM 1-1520-237-23-9, TM 1-1520-237-23-10-1, TM 1-1520-237-23-10-2, TM 1-1520-237-23-10-3, TM 1-1520-237-23-11, dated 1 May 2003 including all changes.

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HEADQUARTERS, DEPARTMENT OF THE ARMY

17 April 2006

WARNING SUMMARY

Personnel performing operations, procedures, and practices which are included or implied in this technical manual shall observe the following warnings. To disregard these warnings and precautionary information can cause serious injury, death, or an aborted mission.

WARNING

FIRST AID - refer to FM 4-25-11.

AC POWER - before applying ac power to the helicopter, make sure that the area around the stabilator is clear of personnel and equipment. Should the stabilator be in any position other than trailing edge fully down, it may reposition automatically to fully down when ac power is applied.

CARBON MONOXIDE - during cold weather operations when preheating is necessary, all personnel shall become acquainted with the various types of heaters and where they are to be used. Be careful of accumulations of carbon monoxide and damage to heat-sensitive equipment.

CONSUMABLE MATERIALS - observe all cautions and warnings on the containers when using consumables. When applicable wear necessary protective gear during handling and use. If a consumable is flammable or explosive, MAKE SURE consumable and its vapors are kept away from heat, spark, and flame. MAKE SURE helicopter is properly grounded and firefighting equipment is readily available for use.

SAFETY PINS - make sure all safety pins are installed in ejector racks prior to performing any maintenance. To disregard this warning can cause serious injury, death, or an aborted mission.

ELECTRICAL GROUNDING OF THE HELICOPTER - army regulations require the grounding of the helicopter when doing all fueling and defueling operations. Do not operate helicopter electrical switches, except those essential for servicing during fueling and defueling. Do not smoke or use flame during fueling and defueling operations.

HIGH VOLTAGE - there are dangerous voltages in the helicopter. Use extreme care when working with equipment having these voltages.

HYDRAULIC SYSTEMS - there are dangerous high hydraulic system pressures in the helicopter. Use extreme care when working on systems having this pressure.

USE OF FIRE EXTINGUISHER - when fire extinguishers are used in a confined area, ventilate immediately. Serious injury or death could result if the area is not ventilated or the user does not use a self-contained breathing device.

MAIN ROTOR PYLON SLIDING COVER - when opening and closing main pylon sliding cover, keep hands away from sharp edges near track, mating end, and especially ventilation blower inlet hole.

TECHNICAL ACETONE - is extremely flammable and toxic to eyes, skin, and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local saefty/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

ALKALINE AQUEOUS CLEANER - alkaline cleaner is harmful to skin and eyes. Use adequate ventilation. Operators should wear protective gloves, aprons, and goggles. Upon contact with cleaning compound, wash affected area for at least 20 minutes with clean water. Seek medical attention.

CLEANING COMPOUND SOLVENT - cleaning Compound Solvent is combustible and toxic to eyes, skin, and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well-ventilated areas (or use approved respirator). Keep away from open flames or other sources of ignition.

ELECTRON - use electron in well ventilated area only. Avoid prolonged breathing of vapors. Avoid bodily contact. The use of chemical gloves and chemical splash goggles are required. Do not use near heat, spark or flame. This solvent is reactive with acids and oxidizers; do not mix or cross-apply with other chemicals. Organic vapor respirator with dust and mist filter is recommended when solvent is spray applied. Keep containers closed between applications.

- Provide mechanical ventilation if used in confined spaces. Coordinate the use of this material with your supporting industrial hygiene and safety offices. Ensure you read and understand the material safety data sheet (MSDS) for electron prior to use.

ISOPROPYL ALCOHOL - isopropyl alcohol is a volatile and flammable liquid which must be kept away from flame and other ignition sources. It must be used only in well ventilated area. Personnel must wear protective gloves and eye protection. Excessive inhalation of vapors or contact with skin must be avoided. If contact occurs with eyes or skin, flush area thoroughly with water. If ingestion occurs, induce vomiting and call a doctor. Remove to fresh air if overexposed to vapor.

DS-108 - use in well ventilated area. Avoid breathing vapors. Organic vapor respirator with dust and mist filter is recommended. May be harmful by inhalation, ingestion, or skin absorption. Avoid bodily contact. Chemical safety glasses/ goggles and chemical resistant gloves are required. Combustible, keep away from heat, spark, or flame. Vapors are heavier than air and may travel over a distance to an ignition source and then flash back. Do not mix or cross apply with other chemicals.

LOCKWIRE - in critical applications use lockwire instead of safety cable which is specifically prohibited from use in critical applications. The use of safety cable is restricted in these specific applications with a warning.

PURPOSE OF A WARNING - alerts personal to operation, procedure, practice, etc., which if not strictly observed, could result in personal injury or loss of life.

CHANGE
NO. 3

HEADQUARTERS DEPARTMENT OF THE ARMY
WASHINGTON, D.C., 20 June 2008

TECHNICAL MANUAL

AVIATION UNIT AND INTERMEDIATE MAINTENANCE FOR ARMY MODELS UH-60A, UH-60L, EH-60A, HH-60A AND HH-60L HELICOPTERS

CHAPTER 10 MAINTENANCE INSTRUCTIONS

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5. Remove old pages and insert new pages as indicated below.

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a and b	a and b
A and B	A and B
i through iv	i through iv
v/vi and Blank	none

6. Replace the following work packages with their revised version.

Work Package Number

WP 1271 00
WP 1277 00
WP 1289 00
WP 1291 00
WP 1370 00

7. Add the following new work packages.

Work Package Number

WP 1342 01

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General, United States Army
Chief of Staff

Official:

Joyce E. Morrow

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TECHNICAL MANUAL

AVIATION UNIT AND INTERMEDIATE MAINTENANCE FOR ARMY MODELS UH-60A, UH-60L, EH-60A, HH-60A AND HH-60L HELICOPTERS

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TECHNICAL MANUAL

AVIATION UNIT AND INTERMEDIATE MAINTENANCE

FOR ARMY MODELS UH-60A, UH-60L, EH-60A, HH-60A, AND HH-60L HELICOPTERS

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i through iv	i through iv
--------------	v/vi Blank
Chapter 10 Title and Blank	Chapter 10 Title and Blank
Cover	Cover

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WP 1297 00
WP 1298 00
WP 1299 00
WP 1300 00
WP 1301 00

Work Package Number

WP 1302 00
WP 1303 00
WP 1304 00
WP 1305 00
WP 1306 00
WP 1307 00
WP 1308 00
WP 1309 00
WP 1324 00
WP 1325 00
WP 1326 00
WP 1327 00
WP 1342 00
WP 1343 00
WP 1344 00
WP 1345 00
WP 1346 00
WP 1347 00
WP 1348 00
WP 1349 00
WP 1350 00
WP 1351 00

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Original	0	17 April 2006
Change	1	21 March 2007
Change	2	15 December 2007
Change	3	20 June 2008

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A - B	3
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HEADQUARTERS,
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WASHINGTON, D.C., 17 April 2007

AVIATION UNIT AND INTERMEDIATE MAINTENANCE FOR

Army Model Helicopters UH-60A, UH-60L, EH-60A, HH-60A, and HH-60L

REPORTING ERRORS AND RECOMMENDING IMPROVEMENTS.

You can improve this manual. If you find any mistakes or if you know of a way to improve these procedures, please let us know. Mail your letter or DA Form 2028 (Recommended Changes to Publications and Blank Forms), located in the back of this manual, directly to: Commander, US Army Aviation and Missile Command, ATTN:AMSAM-MMC-MA-NP, Redstone Arsenal, AL 35898-5000. A reply will be furnished to you. You may also send in your comments electronically to our E-mail address: 2028@redstone.army.mil or by fax 256-842-6546/DSN 788-6546. For the World Wide Web use: https://amcom2028.redstone.army.mil. Instructions for sending an electronic 2028 may be found at the back of this manual immediately preceding the hard copy 2028.

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CHAPTER 10

MAINTENANCE INSTRUCTIONS

FOR

ARMY MODELS UH-60A, UH-60L, EH-60A, HH-60A, AND HH-60L

HELICOPTERS

AVIATION UNIT AND INTERMEDIATE LEVEL

UTILITY SYSTEM

WINDSHIELD WIPER MOTOR (AVIM)

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Electrical Repairer's Toolkit, SC 5180-99-B06
Inserter And Remover, Electrical Contact Tool, MS3447-20
Micrometer, 0 - 1 inch

Material/Parts

Cleaning Compound, Solvent, Item 71, WP 1803 00
Cloth, Cleaning, Item 86, WP 1803 00
Grease, Aircraft, Item 131, WP 1803 00
Solder, Tin Alloy, Item 299, WP 1803 00
Tags
Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

Aircraft Electrician MOS 15F (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0143 00
WP 1209 00
WP 1803 00

DISASSEMBLE

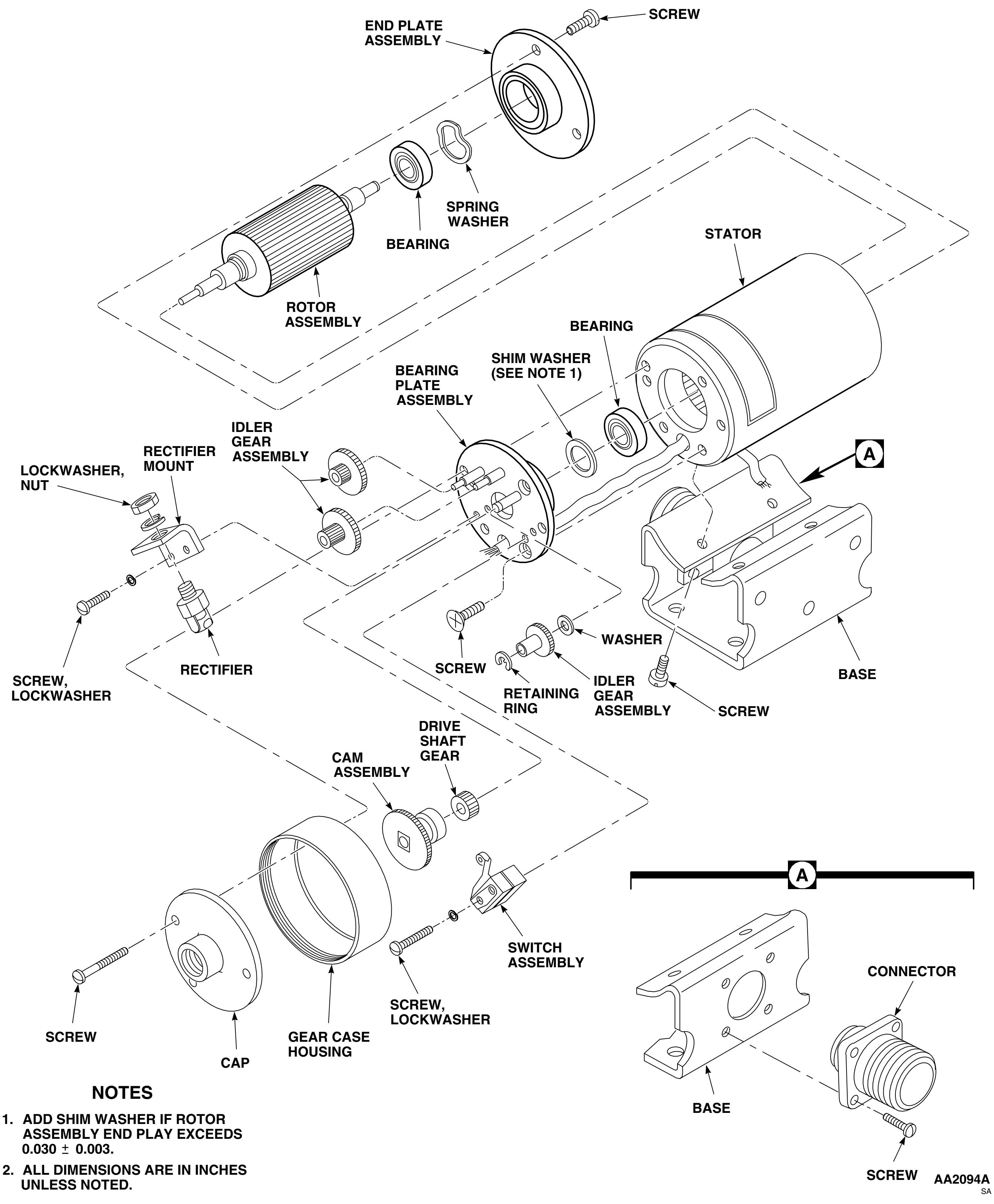
1. Remove screws, cap and gear case housing from bearing plate assembly (Figure 1).
2. Remove cam assembly and idler gear assemblies from bearing plate assembly.
3. Remove retaining ring and remove idler gear assembly and washer from bearing plate assembly.
4. Tag switch assembly and rectifier assembly wires.
5. Using soldering iron, unsolder switch assembly electrical leads and rectifier electrical leads.
6. Remove screws and lockwashers attaching switch assembly to bearing plate.
7. Remove nut and lockwasher attaching rectifier to rectifier mount.
8. Remove screws and lockwashers and remove rectifier mount from bearing plate.
9. Remove screws, end plate, and spring washer from end of stator.
10. Pull rotor assembly and shim washer, if installed, from stator assembly.

NOTE

Do not remove bearings or drive shaft gear from rotor assembly unless they are worn or damaged.

11. Remove screws attaching bearing plate to stator. Carefully thread stator wires through bearing plate and remove.
12. If stator is damaged, remove stator from base as follows:
    a. Remove screws attaching electrical connector to base (Figure 1).
    b. Tag wires to connector and remove them using inserter and remover, electrical contact tool.
    c. Remove screws securing stator to base.

DISASSEMBLE - Continued



NOTES

1. ADD SHIM WASHER IF ROTOR ASSEMBLY END PLAY EXCEEDS 0.030 ± 0.003.
2. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

Figure 1. Repair Windshield Wiper Motor (AVIM).

INSPECT

1. Clean all metal parts except stator assembly, bearings, and electrical parts using cleaning compound solvent, Item 71, WP 1803 00. Blow dry using compressed air.
2. Clean stator assembly, bearings, and electrical parts with a cleaning cloth, Item 86, WP 1803 00.
3. Inspect end plate bearing bore for wear. **REPLACE END PLATE IF BEARING BORE EXCEEDS 0.8668-INCH.**
4. Inspect bearing plate bearing bore for wear. **REPLACE END PLATE IF BEARING BORE EXCEEDS 0.8668-INCH.**
5. Check bearings for wear and fit. Bearing should fit snug, but not tight and should spin freely. **REPLACE BEARING IF WORN OR DOES NOT SPIN FREELY.**
6. If bearing(s) are to be replaced inspect rotor shaft for roundness. **REPLACE ROTOR SHAFT IF DIAMETER IS LESS THAN 0.3148-INCH.**
7. Check windings of stator for broken leads or frayed insulation. **NONE ALLOWED.** Replace stator assembly with broken leads or frayed insulation.
8. Check wiring for loose electrical connections. **NONE ALLOWED.** Repair connections using tin alloy solder, Item 299, WP 1803 00.
9. Check switch assembly by pressing and releasing the switch actuating plunger. **SWITCH SHOULD HAVE SOLID ACTION AND A DISTINCT AUDIBLE CLICK SHOULD BE HEARD WHEN SWITCH IS PRESSED IN OR RELEASED.** Replace switch if action is slow, terminals are loose or if switch plunger is worn, preventing positive action of switch assembly.
10. Check all parts for obvious defects, excessive wear, nicks, burrs, and dents which may interfere with proper operation of motor. Pay particular attention to gear and pinion teeth and gear posts. If practical, make minor repairs such as removing nicks and burrs. Replace all damaged, worn or defective parts.

ASSEMBLE

1. If removed, attach stator as follows:
   a. **QA** Install stator on base with screws (Figure 1). Using nonelectrical wire, Item 357, WP 1803 00, lockwire screws.
   b. Thread connector wires through connector mount hole.
   c. **QA** Insert wires into connector using inserter and remover, electrical contact tool. Remove tags.
   d. **QA** Install connector on base with screws (Figure 1, Detail A).
2. **QA** Carefully thread stator wires through hole in bearing plate assembly. Attach bearing plate assembly to stator with screws.

NOTE

Install drive shaft gear on rotor assembly with inside diameter chamfered end towards rotor assembly.

3. If removed, install bearings and drive shaft gear on rotor assembly.
4. Insert assembled rotor shaft assembly into stator assembly. Make sure bearing on rotor shaft assembly is fully seated against bearing plate assembly.
5. Install spring washer on rotor shaft assembly.

ASSEMBLE - Continued

6. Install end plate assembly on rotor shaft assembly. Make sure bearing on rotor shaft assembly is fully seated against end plate assembly. Attach end plate assembly to stator with screws. Using nonelectrical wire, Item 357, WP 1803 00, lockwire screws.

NOTE

Do not compress spring washer when checking rotor assembly endplay.

7. Turn motor assembly on its end and check rotor assembly for endplay. **END PLAY SHALL NOT EXCEED 0.030 ±0.003-INCH.** If endplay exceeds limits, do this:
   a. Remove bearing plate assembly from stator.
   b. Install shim washer between bearing and bearing plate assembly.
   c. Install bearing plate assembly on stator with screws.
   d. Repeat step 7. until endplay is within limits.
8. Install rectifier mount on bearing plate assembly with screws and lockwashers.
9. Attach rectifier assembly to rectifier mount with nut and lockwasher.
10. Attach switch assembly to bearing plate assembly with screws and lockwashers.
11. **QA** Using soldering iron, solder wire leads to rectifier assembly and switch assembly, using tin alloy solder, Item 299, WP 1803 00.
12. Install washer and idler gear assembly on bearing plate assembly. Secure with retaining ring.
13. Install two idler gear assemblies on bearing plate assembly.
14. Install cam assembly on rotor assembly.
15. Apply a light coat of aircraft grease, Item 131, WP 1803 00, to screws.
16. **QA** Install gear case housing and cap on stator assembly and secure with screws. Using nonelectrical wire, Item 357, WP 1803 00, lockwire screws.
17. Install windshield wiper motor in helicopter (WP 1209 00).
18. Do an operational check of windshield wiper system (WP 0143 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

UTILITY SYSTEM

BLADE DEICE TEST PANEL ROTARY SELECTOR SWITCH (AVIM)

INITIAL SETUP:

Tools and Special Tools
Electrical Repairer's Toolkit, SC 5180-99-B06

Material/Parts
Solder, Tin Alloy, Item 299, WP 1803 00
Tags

Personnel Required
Aircraft Electrician MOS 15F (1)

References
WP 0148 00
WP 1242 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Remove information plate from test panel (WP 1242 00).
3. Remove screws and bottom cover from test panel (Figure 1).

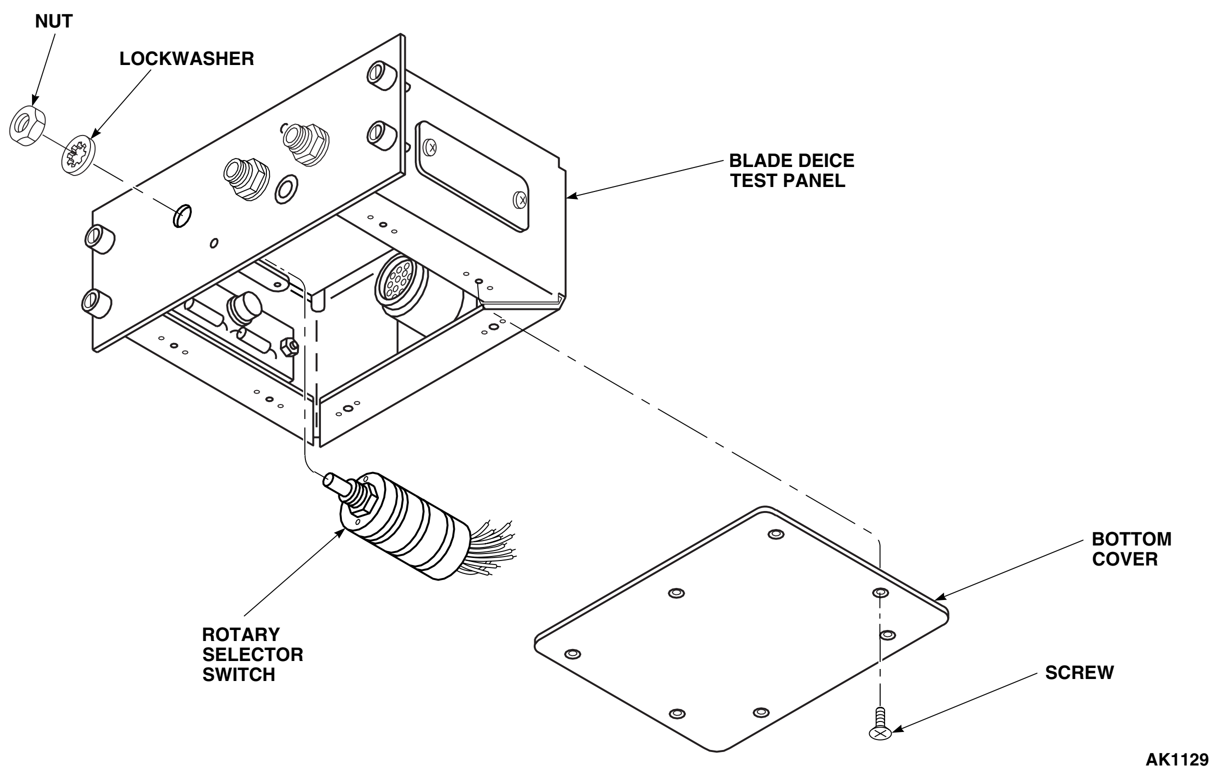


Figure 1. Replace Blade Deice Test Panel Rotary Selector Switch (AVIM).

REMOVE - Continued

4. Tag and unsolder wiring from switch.
5. Remove nut, lockwasher, and switch from test panel.

INSTALL

1. Turn off all electrical power.
2. **QA** Place switch through opening in front of test panel and fasten with nut and lockwasher (Figure 1).
3. **QA** Solder, using tin alloy solder, Item 299, WP 1803 00, wiring to switch, and remove tags.
4. **QA** Fasten bottom cover to test panel with screws.
5. Install information plate on test panel (WP 1242 00).
6. Do an operational check of blade deice test panel (WP 0148 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

UTILITY SYSTEM

BLADE DEICE TEST PANEL INDICATOR LIGHT (AVIM)

INITIAL SETUP:

Tools and Special Tools
Electrical Repairer's Toolkit, SC 5180-99-B06

Material/Parts
Solder, Tin Alloy, Item 299, WP 1803 00
Tags

Personnel Required
Aircraft Electrician MOS 15F (1)

References
WP 0148 00
WP 1242 00
WP 1803 00

NOTE

Blade deice test panel has two indicator lights. Replacement of both is the same.

REMOVE

1. Turn off all electrical power.
2. Remove information plate from test panel (WP 1242 00).
3. Remove screws and bottom cover from test panel (Figure 1).
4. Tag and unsolder wiring to indicator light.
5. Remove nut, lockwashers, and indicator light from panel.

INSTALL

1. Turn off all electrical power.
2. **QA** Place indicator light through opening in front of test panel and fasten with nut and lockwasher.
3. **QA** Solder, using tin alloy solder, Item 299, WP 1803 00, wiring to indicator light and remove tags.
4. Fasten bottom cover to test panel with screws (Figure 1).
5. Install information plate to test panel (WP 1242 00).
6. Do an operational check of blade deice test panel (WP 0148 00).

INSTALL - Continued

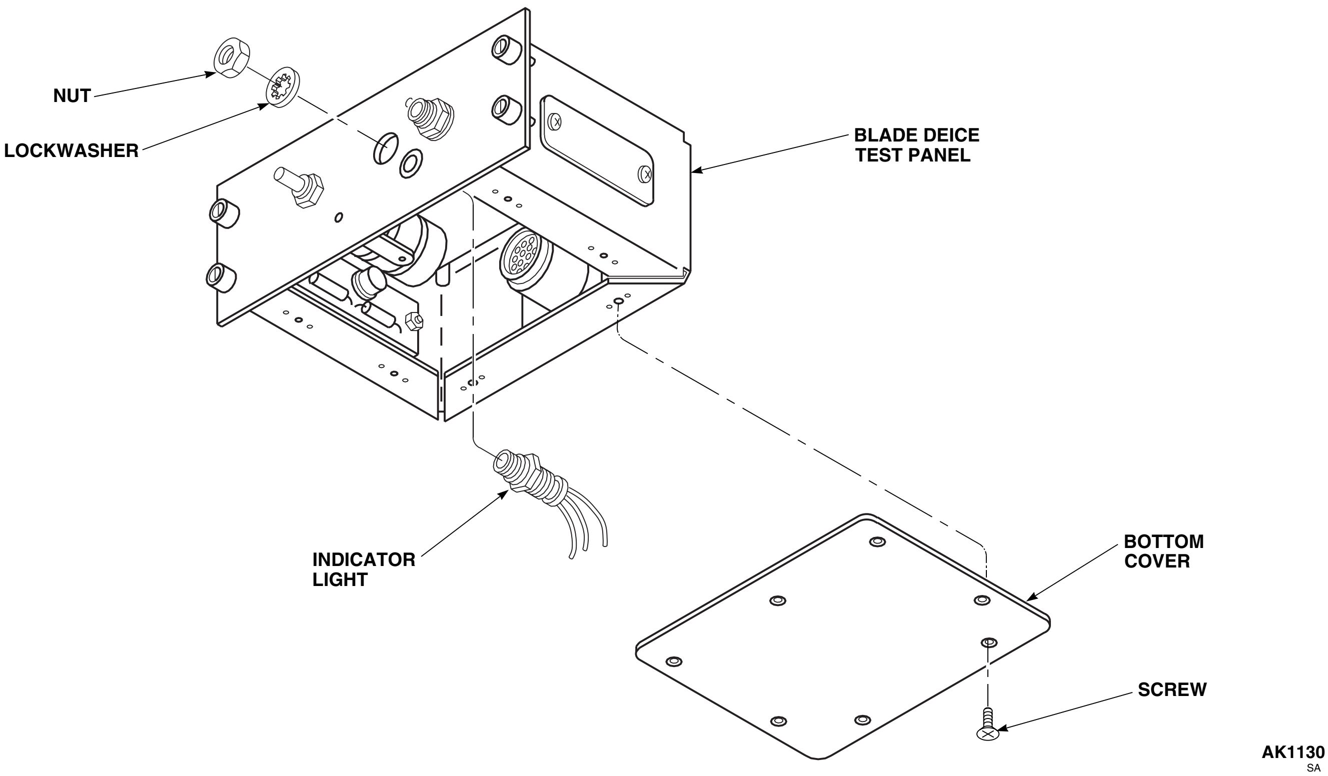


Figure 1. Replace Blade Deice Test Panel Indicator Light (AVIM).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

UTILITY SYSTEM

BLADE DEICE TEST PANEL ELECTRICAL CONNECTOR (AVIM)

INITIAL SETUP:

Tools and Special Tools

Electrical Repairer's Toolkit, SC 5180-99-B06
Inserter And Remover, Electrical Contact Tool, MS3447-18

Material/Parts

Tags

Personnel Required

Aircraft Electrician MOS 15F (1)

References

WP 0148 00
WP 1703 00

REMOVE

1. Turn off all electrical power.
2. Remove screws and remove bottom cover from test panel (Figure 1).
3. Remove screws, washers, lockwasher, and nuts securing electrical connector, J1, and ground wire to test panel.
4. Tag wires to electrical connector, J1.
5. Using inserter and remover, electrical contact tool, remove pins from electrical connector, J1.

INSTALL

1. Turn off all electrical power.
2. **QA** Using inserter and remover, electrical contact tool, insert tagged wires into electrical connector, J1. Remove tags.
3. Inspect and treat electrical connector (WP 1703 00).
4. **QA** Install electrical connector, J1, and ground wire to test panel with screws, washers, lockwasher, and nuts (Figure 1).
5. **QA** Install bottom cover to test panel and fasten with screws.
6. Do an operational check of blade deice test panel (WP 0148 00).

INSTALL - Continued

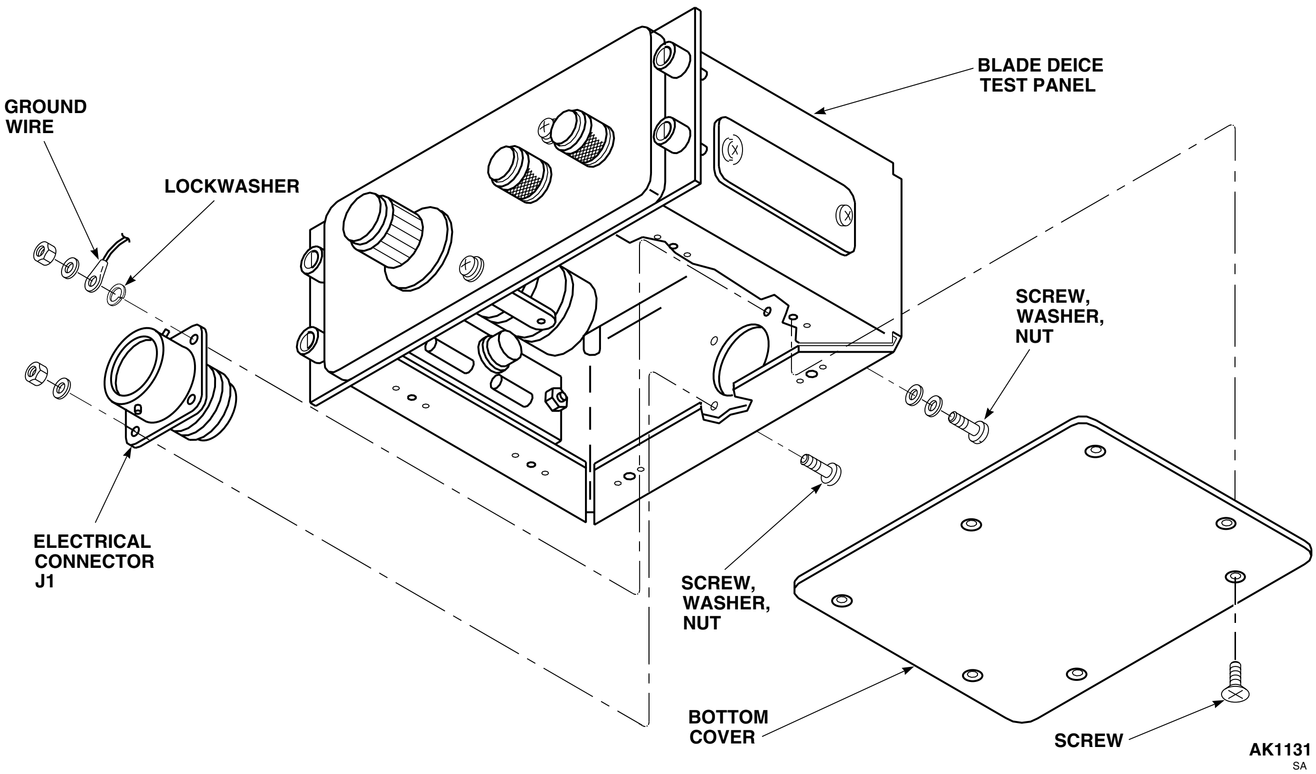


Figure 1. Replace Blade Deice Test Panel Electrical Connector (AVIM).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

UTILITY SYSTEM

BLADE DEICE TEST PANEL PRINTED WIRING BOARD (AVIM)

INITIAL SETUP:

Tools and Special Tools
Electrical Repairer's Toolkit, SC 5180-99-B06

Material/Parts
Solder, Tin Alloy, Item 299, WP 1803 00
Tags

Personnel Required
Aircraft Electrician MOS 15F (1)

References
WP 0148 00
WP 1703 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Remove screws securing top and bottom covers to test panel, and remove both covers (Figure 1).
3. Remove screws securing electrical connector, J2, to P2, on printed wiring board.
4. Disconnect electrical connector, J2, from P2.
5. Remove screws securing deck assembly to test panel, and slide deck assembly away from test panel.
6. Tag and unsolder four wires connecting printed wiring board to circuit board.
7. Tag wires to electrical connector, J2.
8. Unsolder all wires from printed wiring board connecting to electrical connector, J2.
9. Remove screws securing printed wiring board to test panel.
10. Remove printed wiring board from test panel.

INSTALL

1. Turn off all electrical power.
2. Install printed wiring board in test panel using screws (Figure 1).
3. **QA** Solder, using tin alloy solder, Item 299, WP 1803 00, four wires connecting printed wiring board to circuit board and remove tags.
4. **QA** Solder, using tin alloy solder, Item 299, WP 1803 00, all wires connecting printed wiring board to electrical connector, J2. Remove tags.
5. Inspect and treat electrical connector (WP 1703 00).
6. **QA** Join electrical connector, J2, to P2, and tighten screws.
7. Install deck assembly to test panel with screws.
8. Install top and bottom covers with screws.

INSTALL - Continued

9. Do an operational check blade deice test panel (WP 0148 00).

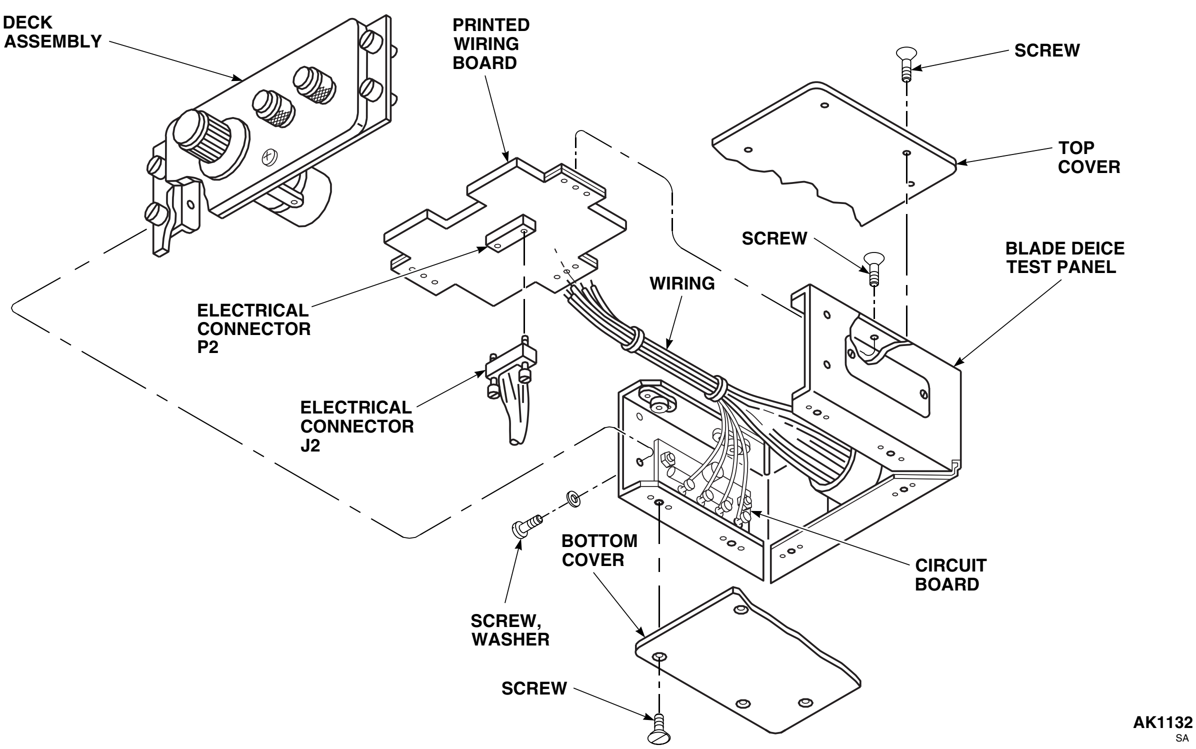


Figure 1. Replace Blade Deice Test Panel Printed Wiring Board (AVIM).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

UTILITY SYSTEM

BLADE DEICE TEST PANEL CIRCUIT BOARD (AVIM)

INITIAL SETUP:

Tools and Special Tools
Electrical Repairer's Toolkit, SC 5180-99-B06

Material/Parts
Solder, Tin Alloy, Item 299, WP 1803 00
Tags

Personnel Required
Aircraft Electrician MOS 15F (1)

References
WP 0148 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Remove screws and bottom cover from test panel (Figure 1).
3. Tag and unsolder four wires from printed wiring board connected to circuit board.
4. Remove screws, washers, spacers, and nuts securing circuit board to test panel.
5. Remove circuit board.

INSTALL

1. Turn off all electrical power.
2. **QA** Install screws, washers, spacers, and nuts securing circuit board to test panel (Figure 1).
3. **QA** Solder, using tin alloy solder, Item 299, WP 1803 00, four wires to printed wiring board leading from circuit board. Remove tags.
4. **QA** Secure bottom panel to test panel with screws.
5. Do an operational check blade deice test panel (WP 0148 00).

INSTALL - Continued

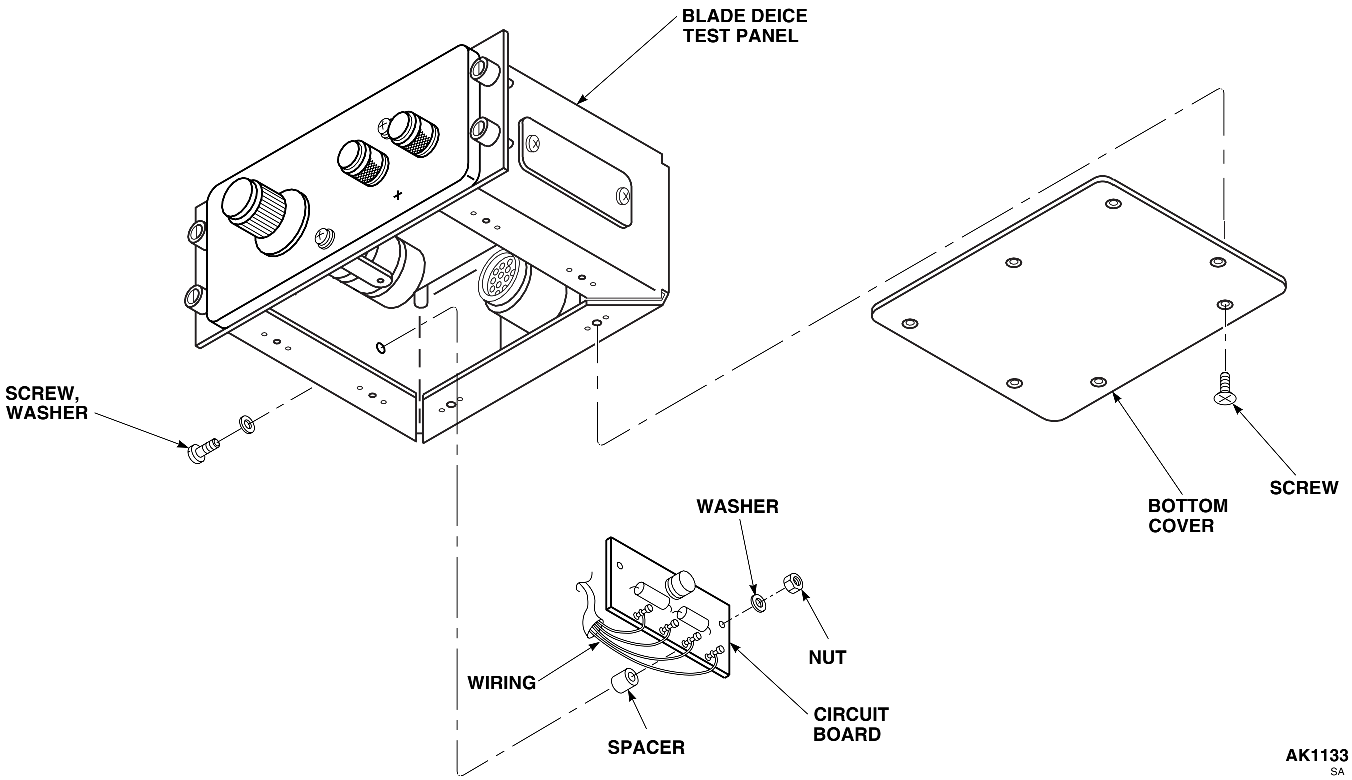


Figure 1. Replace Blade Deice Test Panel Circuit Board (AVIM).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

UTILITY SYSTEM

BLADE DEICE TEST PANEL PRINTED WIRING BOARD AND CIRCUIT BOARD COMPONENTS (AVIM)

INITIAL SETUP:

Tools and Special Tools
Electrical Repairer's Toolkit, SC 5180-99-B06

Material/Parts
Insulation Compound, Item 150, WP 1803 00
Solder, Tin Alloy, Item 299, WP 1803 00
Tags

Personnel Required
Aircraft Electrician MOS 15F (1)

References
WP 0148 00
WP 1263 00
WP 1264 00
WP 1803 00

NOTE

If replacing printed wiring board components, remove printed wiring board (WP 1263 00). If replacing circuit board components, remove circuit board (WP 1264 00).

REMOVE

1. Unsolder component leads by heating each lead and removing solder with solder-removing device (Figure 1).

NOTE

When removing components with positive and negative termination points, note the position of the component and its leads before removal.

2. Remove component from circuit board/wiring board.

INSTALL

NOTE

When installing components with positive and negative termination points, note the position of the component and its leads before installation.

1. **QA** Install replacement part on circuit board/wiring board and solder, using tin alloy solder, Item 299, WP 1803 00, leads to board (Figure 1).
2. Apply insulation compound, Item 150, WP 1803 00, to replacement part and leads.
3. **QA** Install circuit board (WP 1264 00) and/or printed wiring board (WP 1263 00) in blade deice test panel.
4. Do an operational check blade deice test panel (WP 0148 00).

INSTALL - Continued

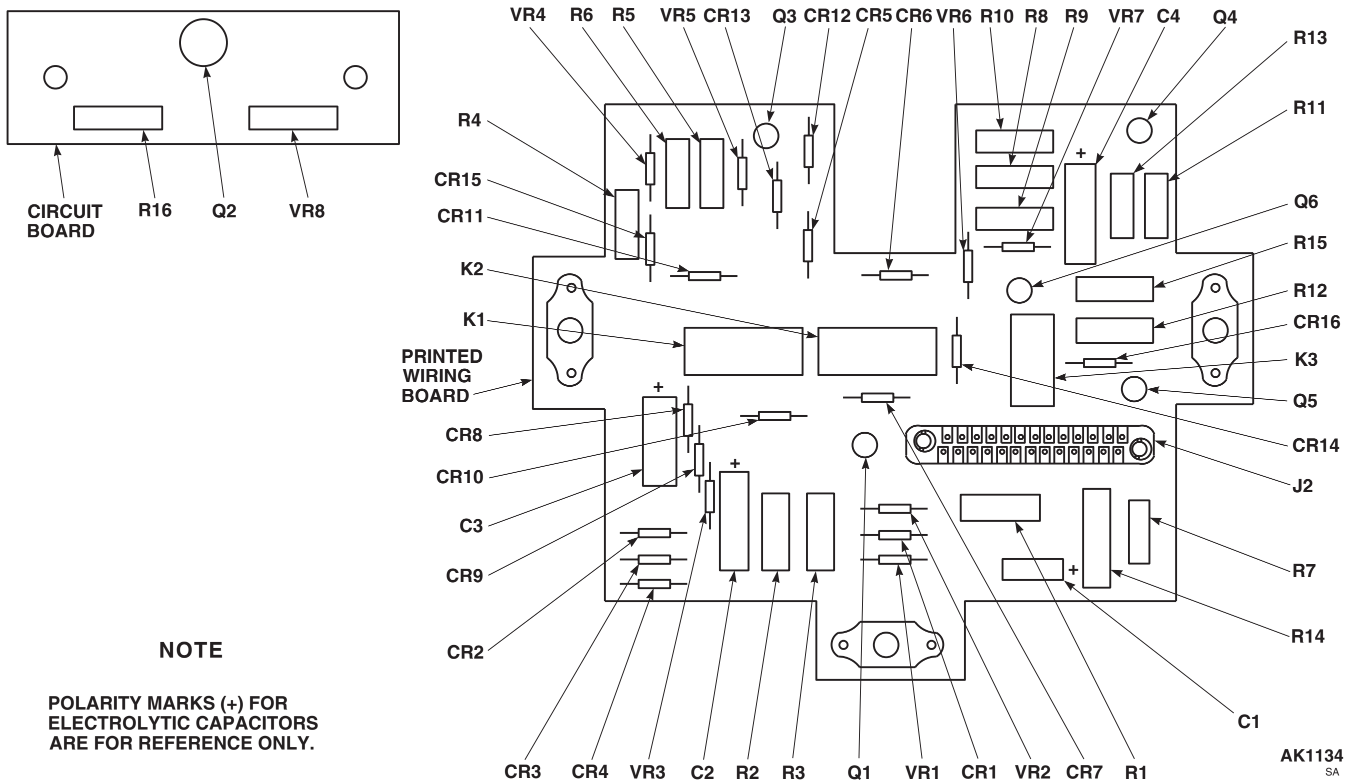


Figure 1. Replace Blade Deice Test Panel Printed Wiring Board and Circuit Board Components (AVIM).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

UTILITY SYSTEM

DIRECTIONAL CONTROL VALVE (AVIM)

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Dial Indicating, Depth Gauge, 643J
Electrical Repairer's Toolkit, SC 5180-99-B06
Torque Wrench, 30 - 200 in. lbs.
Valve Assembly Tool, T512900-ASAD-1

Material/Parts
Cleaning Compound, Solvent, Item 71, WP 1803 00
Cloth, Cleaning, Item 86, WP 1803 00
Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required
Aircraft Electrician MOS 15F (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1520-265-23
WP 1803 00

DISASSEMBLE

NOTE

Disassemble control valve only as far as necessary to remove and replace malfunctioning or damaged components.

1. Remove end cap and packing from valve body. Discard packing (Figure 1, Sheet 2).
2. Carefully remove pins from valve indicators. Discard pins. Remove valve indicators and washers. Remove and discard packings.

NOTE

Solenoid plunger and spring will remain attached to arm when solenoid is removed. Retain solenoid plunger and spring with solenoid.

3. Using valve assembly tool, T512900-ASAD-1, or equivalent, remove solenoid assembly from valve body. Remove and discard packing. If washer(s) are installed, remove and discard washers.
4. Push flapper control valve away from N.C. (Normally Closed) port until it stops; then disengage plunger from arm and release flapper.
5. Remove control nut. Remove and discard packing from nut.
6. Carefully press out and remove shaft from valve body. Remove flapper and associated components from valve body. Remove spacer and return spring and remaining packing from valve body. Discard packing.
7. Remove and discard cotter pin. Remove washer and pin. Separate arm from flapper.
8. Remove and discard return spring from arm.
9. Do not remove decal or identification plate, unless damaged or illegible.

CLEANING

1. Clean all parts as follows:

CLEANING - Continued

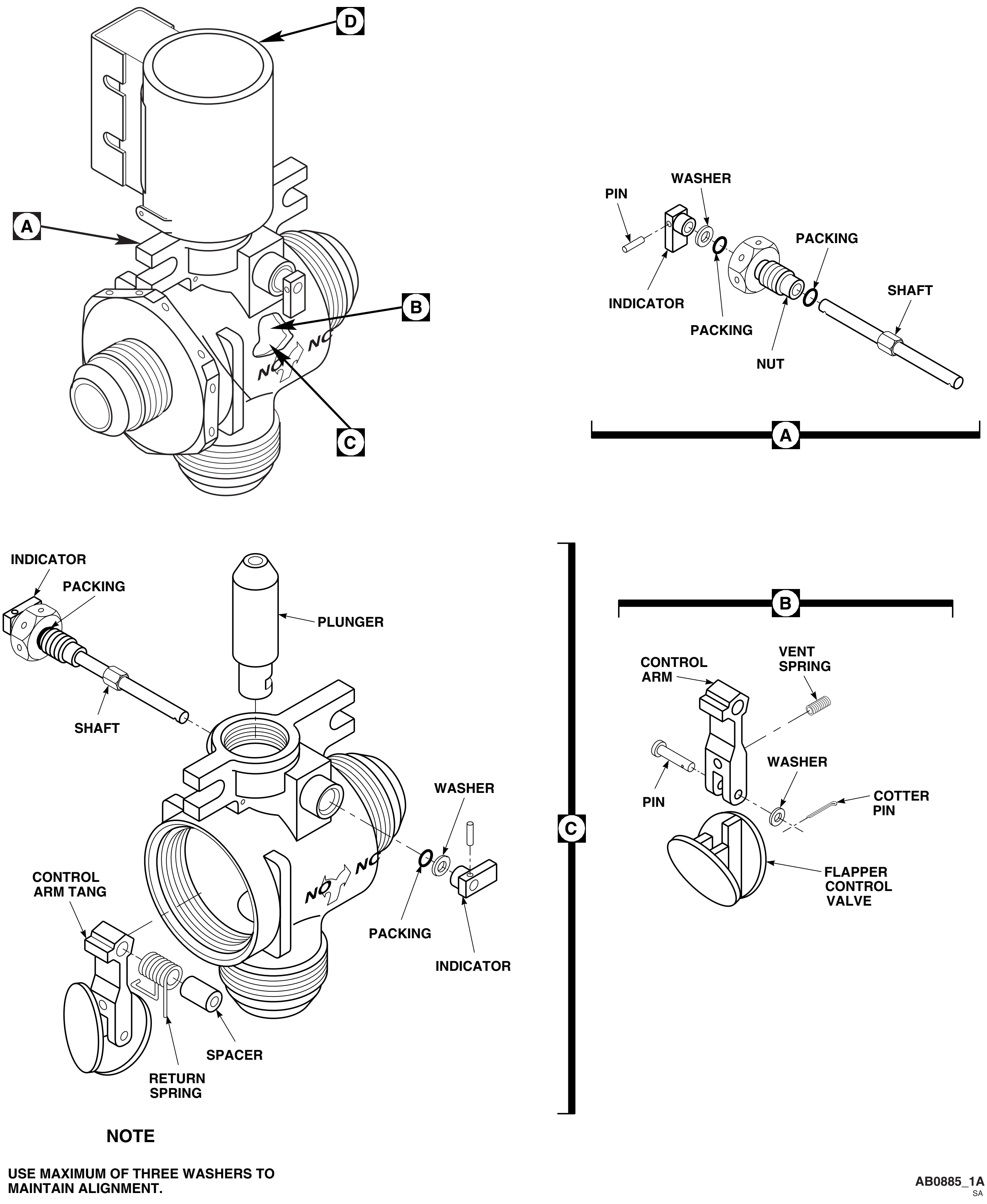


Figure 1. Repair Directional Control Valve (AVIM). (Sheet 1 of 2)

CLEANING - Continued

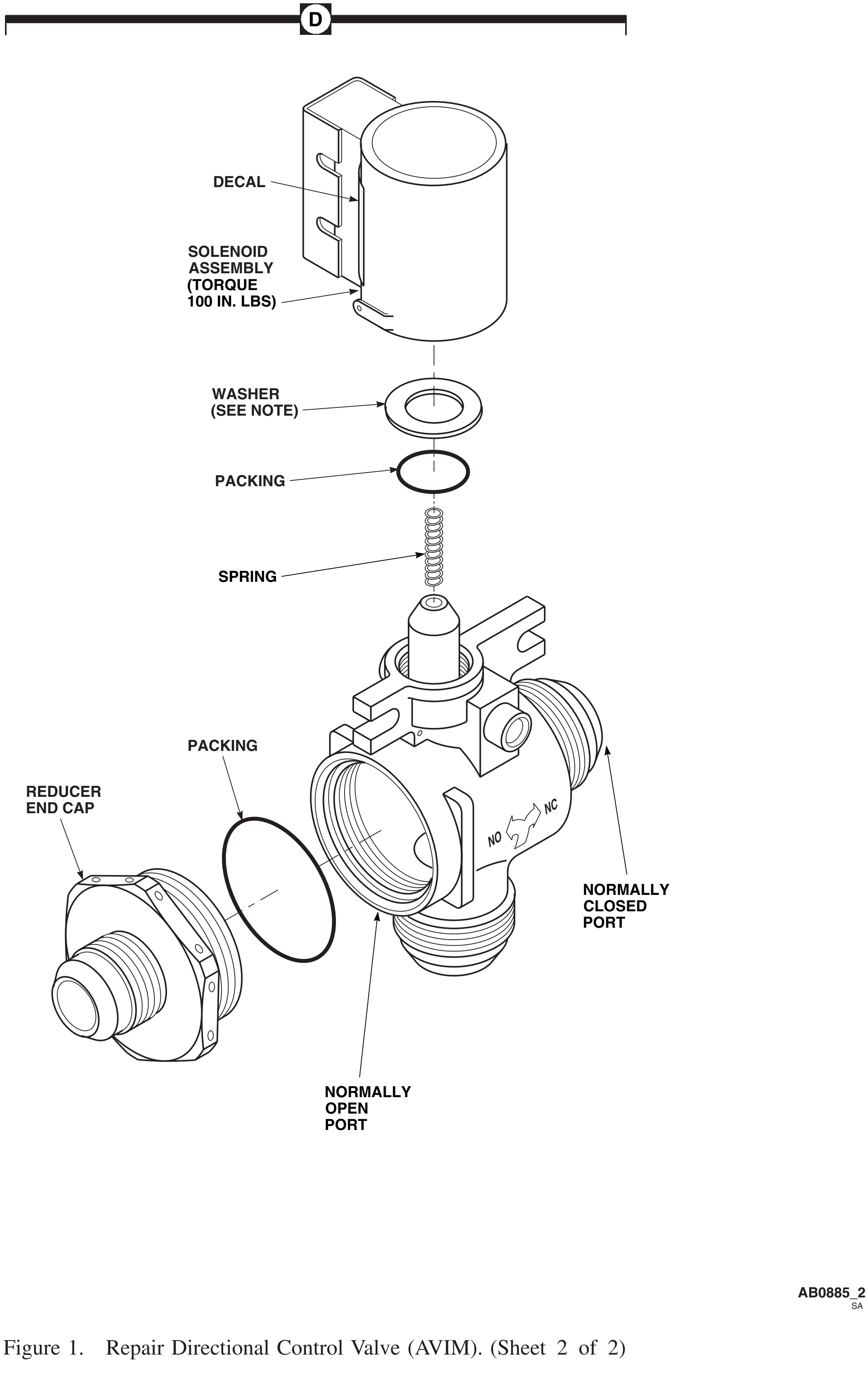


Figure 1. Repair Directional Control Valve (AVIM). (Sheet 2 of 2)

CLEANING - Continued

a. Clean metal parts using cleaning compound solvent, Item 71, WP 1803 00, or equivalent. Use nonmetallic, stiff bristle brush to remove stubborn accumulations of foreign matter.

b. Allow cleaned parts to air-dry or use low lint cleaning cloth, Item 86, WP 1803 00.

INSPECTION

1. Inspect control valve parts as follows:

a. Inspect reducer end cap as follows:

(1) Visually inspect reducer end cap for nicks, burrs and corrosion. **DAMAGE NOT TO EXCEED 0.125-INCH DEEP.** Replace end cap if out of tolerance.

(2) Visually inspect reducer end cap for crossed, stripped, or flattened threads. Repair end cap threads. If damage is over 50% of one thread, replace reducer end cap.

(3) Visually inspect reducer end cap for damaged or missing protective coating. Replace end cap if protective coating is damaged or missing.

(4) Inspect reducer end cap for cracks or fractures by fluorescent penetrant inspection method (TM 1-1500-204-23). **REPLACE REDUCER END CAP IF ANY CRACKS OR FRACTURES ARE FOUND.**

b. Inspect nut as follows:

(1) Visually inspect nut for nicks, burrs and corrosion. **DAMAGE NOT TO EXCEED 0.125-INCH DEEP.** Replace nut if out of tolerance.

(2) Visually inspect nut for crossed, stripped or flattened threads. repair nut threads. If damage is over 50% of one thread, replace nut.

(3) Visually inspect nut for damaged or missing protective coating. Replace nut if protective coating is damaged or missing.

(4) Inspect nut for cracks or fractures by fluorescent penetrant inspection method (TM 1-1500-204-23). **REPLACE NUT IF ANY CRACKS OR FRACTURES ARE FOUND.**

c. Inspect shaft as follows:

(1) Visually inspect shaft for nicks, burrs and corrosion. **DAMAGE NOT TO EXCEED 0.125-INCH DEEP.** Replace shaft if out of tolerance.

(2) Visually inspect shaft for damaged or missing protective coating. Replace shaft if protective coating is damaged or missing.

(3) Inspect shaft for cracks or fractures by fluorescent penetrant inspection method (TM 1-1500-204-23). **REPLACE SHAFT IF ANY CRACKS OR FRACTURES ARE FOUND.**

d. Inspect arm as follows:

(1) Visually inspect arm for nicks, burrs and corrosion. **DAMAGE NOT TO EXCEED 0.125-INCH DEEP.** Replace arm if out of tolerance.

(2) Visually inspect arm for damaged or missing protective coating. Replace arm if protective coating is damaged or missing.

(3) Inspect arm for cracks or fractures by fluorescent penetrant inspection method (TM 1-1500-204-23). **REPLACE ARM IF ANY CRACKS OR FRACTURES ARE FOUND.**

e. Inspect flapper control valve as follows:

INSPECTION - Continued

(1) Visually inspect flapper control valve for nicks, burrs and corrosion. **DAMAGE NOT TO EXCEED 0.125-INCH DEEP.** Replace flapper control valve if out of tolerance.

(2) Inspect flapper control valve for cracks or fractures by fluorescent penetrant inspection method (TM 1-1500-204-23). **REPLACE FLAPPER CONTROL VALVE IF ANY CRACKS OR FRACTURES ARE FOUND.**

f. Inspect solenoid assembly as follows:

(1) Visually inspect solenoid assembly for nicks, burrs and corrosion. **DAMAGE NOT TO EXCEED 0.125-INCH DEEP.** Replace solenoid assembly if out of tolerance.

(2) Visually inspect solenoid assembly for damaged or missing terminals. Replace solenoid assembly if terminals are damaged or missing.

(3) Visually inspect solenoid assembly for crossed, stripped, or flattened threads. Repair solenoid assembly threads. If damage is over 50% of one thread, replace solenoid assembly.

g. Inspect valve body as follows:

(1) Visually inspect valve body for nicks, burrs and corrosion. **DAMAGE NOT TO EXCEED 0.125-INCH DEEP.** Replace valve body if out of tolerance.

(2) Visually inspect valve body for crossed, stripped, or flattened threads. Repair valve body damaged threads. If damage is over 50% of one thread, replace valve body.

(3) Visually inspect valve body for damaged or missing protective coating. Replace valve body if protective coating is damaged or missing.

(4) Inspect valve body for cracks or fractures by fluorescent penetrant inspection method (TM 1-1500-204-23). **REPLACE VALVE BODY IF ANY CRACKS OR FRACTURES ARE FOUND.**

REPAIR

1. Use proper die set to chase threads on reducer end cap. If damage is over 50% of one thread, replace reducer end cap.
2. Use proper die set to chase threads on nut. If damage is over 50% of one thread, replace nut.
3. Use proper die set to chase threads on solenoid assembly. If damage is over 50% of one thread, replace solenoid assembly.
4. Use proper tap and die set to chase threads on valve body. If damage is over 50% of one thread, replace valve body.

ASSEMBLE

1. **QA** Install packing on nut.
2. **QA** Insert shaft through nut and install packing and washer on shaft (Figure 1, Sheet 1, Detail A).
3. **QA** Install packing, washer, and indicator onto shaft. Insert spiral pin through indicator and shaft (Figure 1, Sheet 1, Detail A).
4. Install vent spring into hole in control arm (Figure 1, Sheet 1, Detail B).
5. **QA** Line up and attach control arm to flapper control valve with pin, washer, and cotter pin (Figure 1, Sheet 1, Detail B).
6. Install spacer into return spring and place return spring into valve body and over control arm. Install shaft into valve body and line up spring and spacer (Figure 2).

ASSEMBLE - Continued

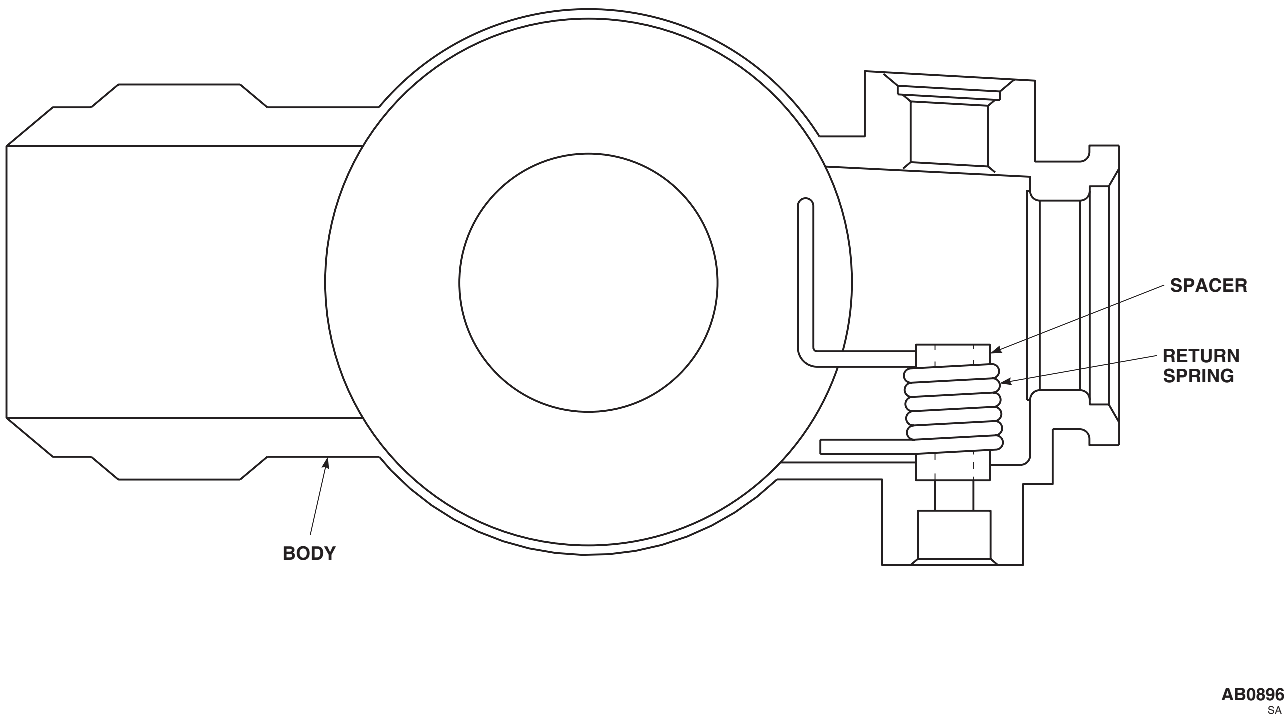


Figure 2. Install Directional Control Valve Spacer and Return Spring (AVIM).

7. Install flapper control valve into valve body, with tang on control arm facing out at N.O. (normally open) port position. Make sure shaft is installed with indicator facing back to N.C. (normally closed) position (Figure 1, Sheet 2, Detail C).
8. **QA** Engage three threads of nut into valve body.
9. Install plunger into valve body, and push flapper control valve in from N.C. (normally closed) position to engage plunger onto control arm tang and release valve (Figure 1, Sheet 2, Detail C).
10. Install solenoid spring into plunger.
11. **QA** Install packing on solenoid assembly (Figure 1, Sheet 2).

NOTE

Do not install washers between solenoid assembly and valve body at this time. Washers are only installed for alignment of solenoid terminal block. Do not use more then three washers for alignment.

12. **QA** Using valve assembly tool, T512900-ASAD-1, or equivalent, install solenoid assembly (without washers) to valve body. **TORQUE SOLENOID ASSEMBLY TO 100 INCH-POUNDS.** Check alignment of solenoid terminal block as shown (Figure 3). If proper alignment is not achieved, remove solenoid assembly and add washers (Figure 1, Sheet 2). Install solenoid assembly and retorque.

ASSEMBLE - Continued

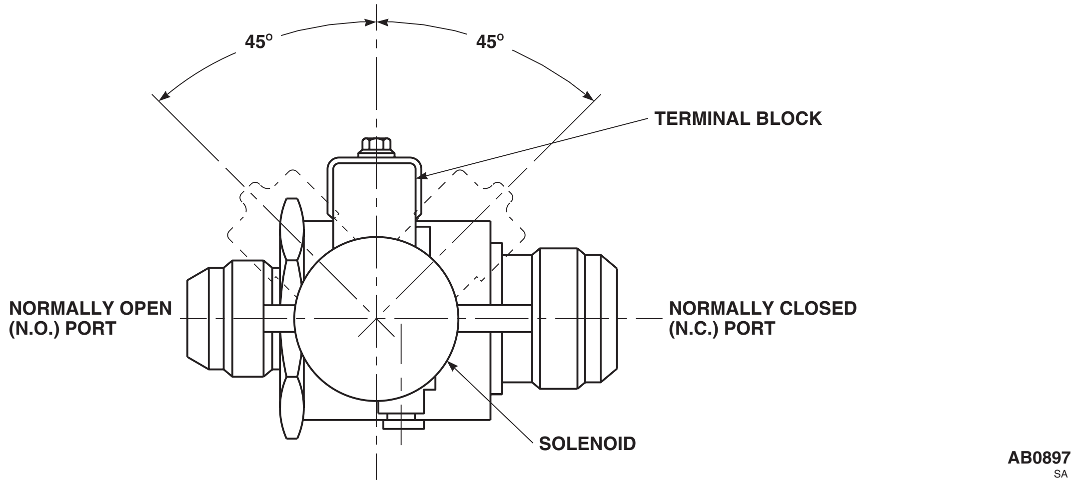


Figure 3. Directional Control Valve Terminal Block Alignment (AVIM).

NOTE

Make sure indicator is installed facing back to normally closed position (Figure 1, Sheet 2, Detail C).

13. **QA** Install packing, washer, and indicator onto shaft. Insert pin through indicator and shaft (Figure 1, Sheet 2).
14. **QA** Install packing on end cap. Install end cap into valve body. **TIGHTEN END CAP 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

TESTING

1. Do external leakage and proof pressure test of directional control valve as follows:
   a. Install plugs on N.O. (normally open) port and N.C. (normally closed) port, connect pressure source to IN port and connect solenoid to adjustable power supply.

WARNING

Injury to personnel and damage to equipment will result if electrical solenoid is submerged in water tank. Make sure electrical solenoid is not submerged in water tank.

   b. Mount directional control valve into test setup as shown (Figure 5). Make sure solenoid is not submerged in water tank.
   c. With solenoid deenergized (Figure 4) apply 880 psig pressure to IN port and submerge directional control valve into water tank. Make sure solenoid is not submerged in water tank. Hold pressure for one minute; then reduce pressure to 0 psig. **THERE SHALL BE NO PERMANENT DISTORTION OR LOOSENING OF PARTS.**
   d. With solenoid deenergized (Figure 4) apply 440 psig pressure to IN port and submerge directional control valve into water tank. Make sure solenoid is not submerged in water tank. Hold pressure for one minute; then reduce pressure to 0 psig. **THERE SHALL BE NO EVIDENCE OF LEAKAGE THROUGH BODY OR FITTINGS. LEAKAGE PAST PACKINGS ON SHAFT SHALL NOT BE OVER FIVE BUBBLES PER MINUTE.**
   e. Repeat step 1. c. and step 1. d. with solenoid energized (Figure 4) using adjustable power supply at 24 vdc. Decrease voltage to 0 vdc, reduce pressure to 0 psig, and remove directional control valve from test setup.
2. Do internal leakage test of directional control valve as follows:

TESTING - Continued

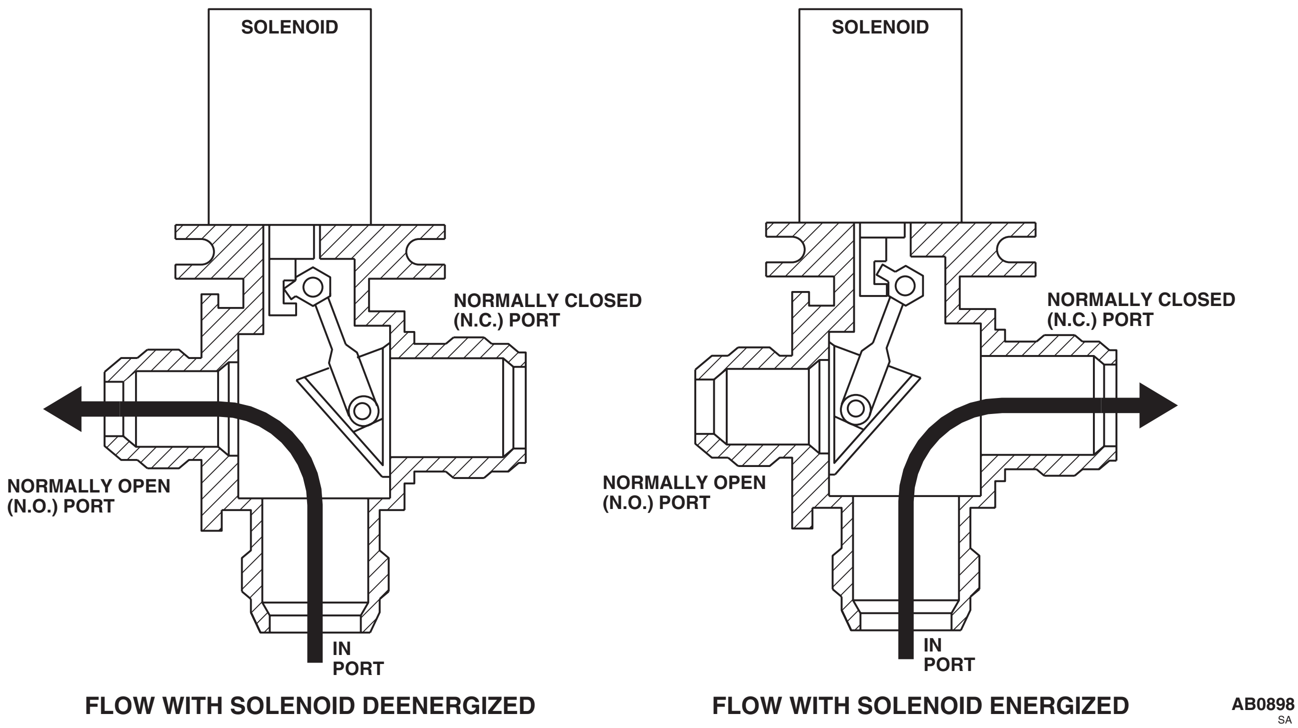


Figure 4. Directional Control Valve Flow Diagram (AVIM).

a. Install plugs on N.O. (normally open) port, connect pressure source to IN port and connect solenoid to adjustable power supply and deenergize solenoid.

b. Mount directional control valve into test setup as shown (Figure 6).

c. Close V2 control valve and open V1 control valve. Close V4 regulator valve and open V3 regulator valve (Figure 6). Apply 5.0 psig to IN port. Connect N.C. (normally closed) port to flowmeter.

NOTE

If unable to reach 5.0 psig, increase pressure until flapper valve seals; then reduce pressure to 5.0 psig.

d. Check leakage rate at 5.0 psig. Leakage rate shall not be over 0.6 Standard cubic feet per minute (SCFM).

e. Close V3 regulator valve and open V4 regulator valve. Close V1 control valve and open V2 control valve. Increase pressure at IN port to 440 psig and measure leakage flow at N.C. (normally closed) port. Leakage rate shall not be over 14.75 SCFM. Reduce pressure to 0 psig.

f. Plug N.C. (normally closed) port. Connect N.O. (normally open) port to flowmeter. Energize solenoid with 24 vdc and repeat step 2. c. through step 2. e.

g. Decrease voltage to 0 vdc, reduce pressure to 0 psig, and remove directional control valve from test setup.

3. Do functional test of directional control valve as follows:

a. Install plug on N.O. (normally open) port, Open N.C. (normally closed) port to atmosphere. Connect pressure source to IN port through flowmeter and manometer.

TESTING - Continued

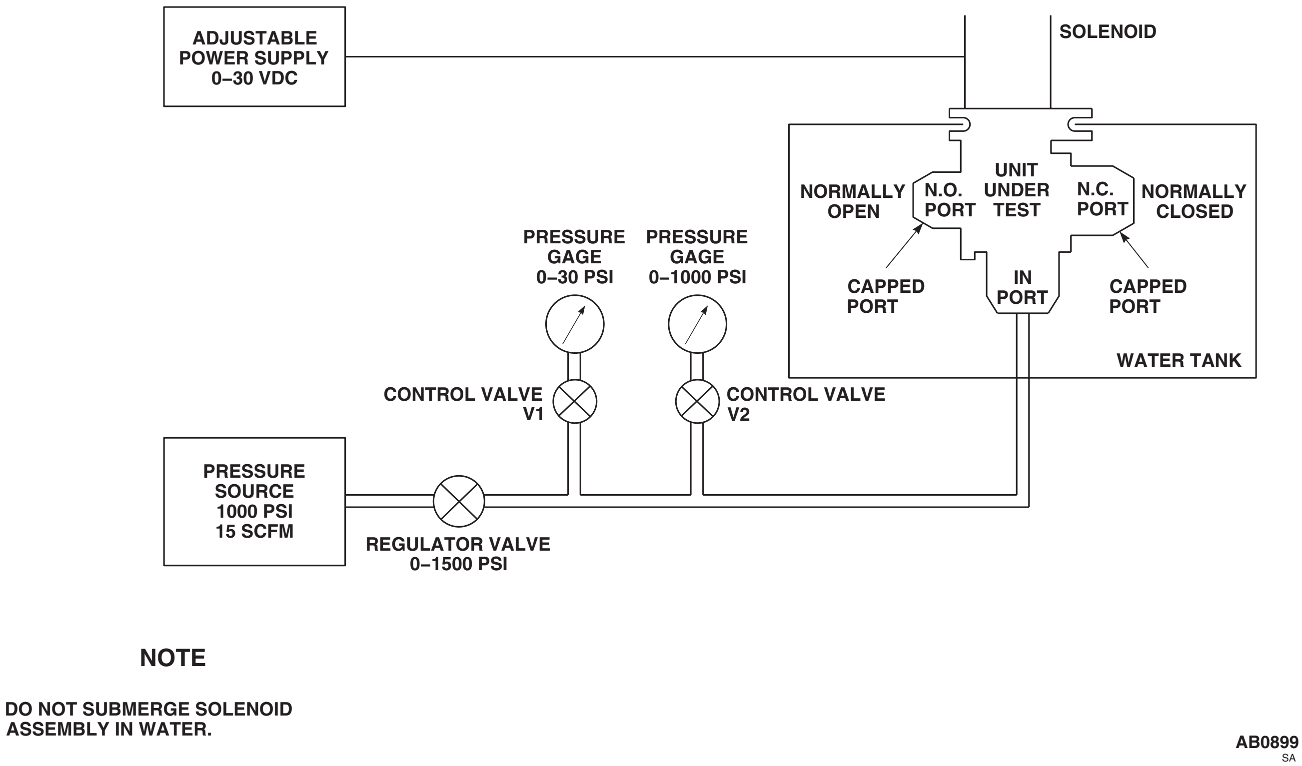


Figure 5. Directional Control Valve External Leakage and Proof Pressure Test Setup (AVIM).

b. Mount directional control valve into test setup as shown (Figure 7).

c. Apply pressure to achieve 5.0 SCFM on flowmeter. Flow reading shall be maintained with 1 psig or less.

d. Reduce pressure to 0 psig and remove directional control valve from test setup.

4. Do actuation test of directional control valve as follows:

a. Open N.C. (normally closed) port to atmosphere.

b. Mount directional control valve into test setup as shown (Figure 8).

NOTE

Make sure indicators on directional control valve indicate proper flapper valve position.

c. Energize solenoid with 24 vdc using adjustable power supply. Close V1 control valve and open V2 control valve. Apply 100 psig maximum to IN port; then deenergize solenoid.

d. Energize solenoid with 24 vdc using adjustable power supply, Make sure flapper valve moves to N.O. (normally open) port, and flow shall be from N.C. (normally closed) port. Apply 100 psig maximum to IN port; then deenergize solenoid. Flapper valve shall remain at N.O. (normally open) port.

e. Reduce pressure and note pressure at point when flapper valve moves to N.C. (normally closed) port with solenoid deenergized. Pressure shall not be over 5.0 psig. Reduce pressure to 0 psig.

TESTING - Continued

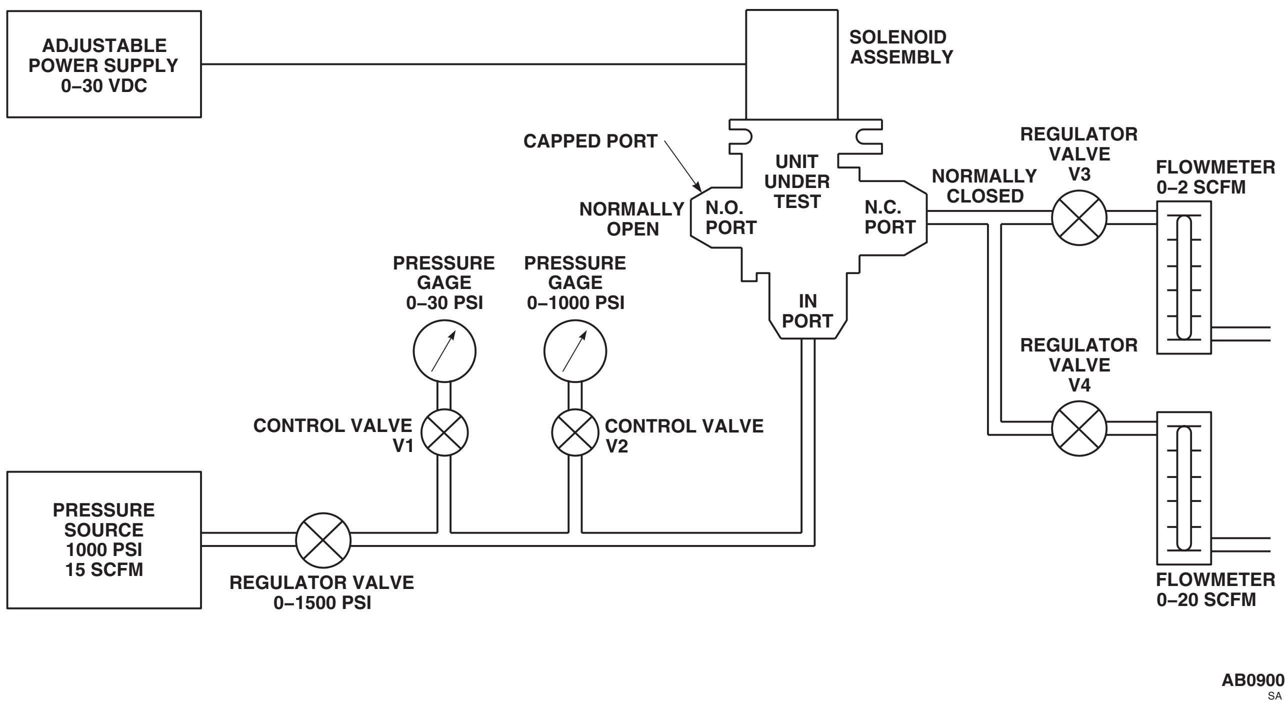


Figure 6. Directional Control Valve Internal Leakage Test Setup (AVIM).

f. Using adjustable power supply, Reduce voltage to 18.0 vdc. Then energize and deenergize solenoid several times. Flapper valve shall move instantly to N.O. (normally open) port with solenoid energized, and return to N.C. (normally closed) port when solenoid is deenergized.

g. Using adjustable power supply, increase voltage to 30.0 vdc and repeat step 4. f. with power at 30.0 vdc.

h. Using adjustable power supply, increase voltage to 24.0 vdc and repeat step 4. f. with power at 24.0 vdc.

i. With solenoid energized to 24.0 vdc, measure current draw. Current draw at 24.0 vdc shall not be over 2.1 amperes.

j. Reduce pressure to 0 psig and decrease voltage to 0 vdc. Remove directional control valve from test setup.

5. Do final assembly of directional control valve as follows:

   a. Install shipping caps on all open ports.

   b. Using nonelectrical wire, Item 357, WP 1803 00, lockwire reducer end cap to valve body.

   c. Using nonelectrical wire, Item 357, WP 1803 00, lockwire nut to valve body.

   d. Install decal and/or identification plate, if removed.

TESTING - Continued

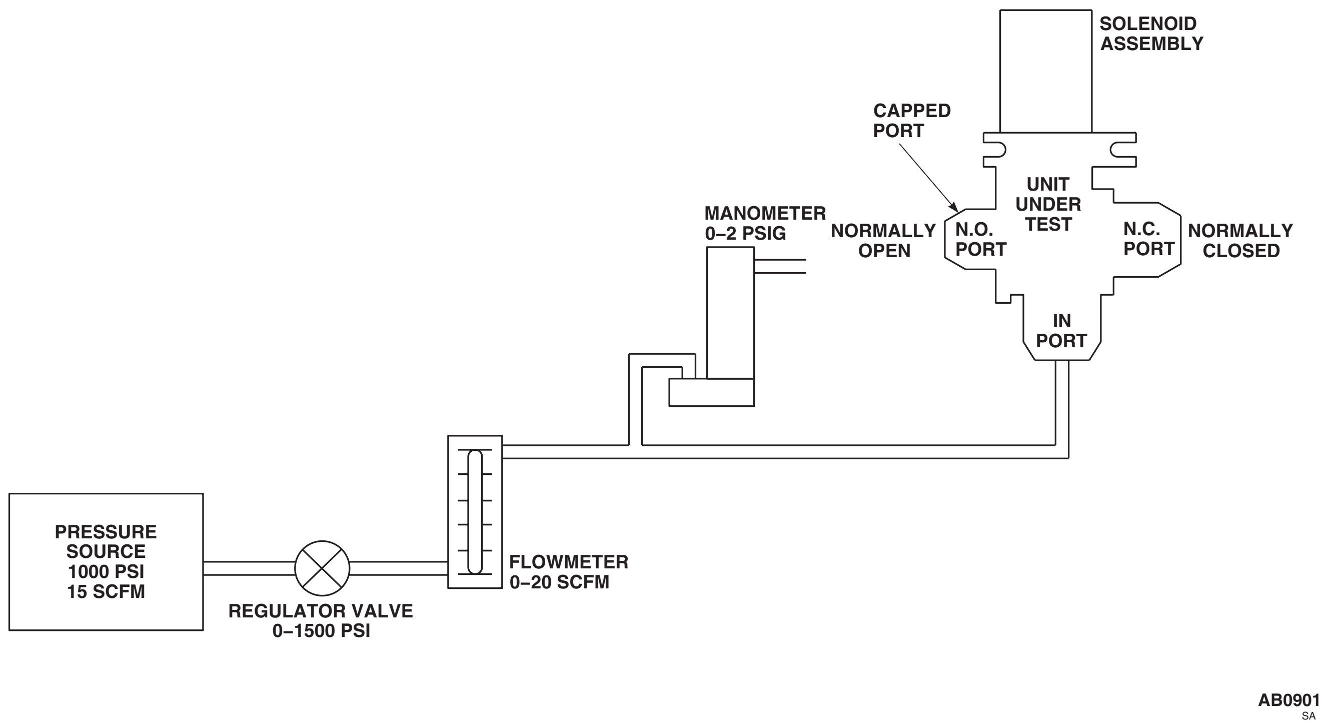


Figure 7. Directional Control Valve Functional Test Setup (AVIM).

TESTING - Continued

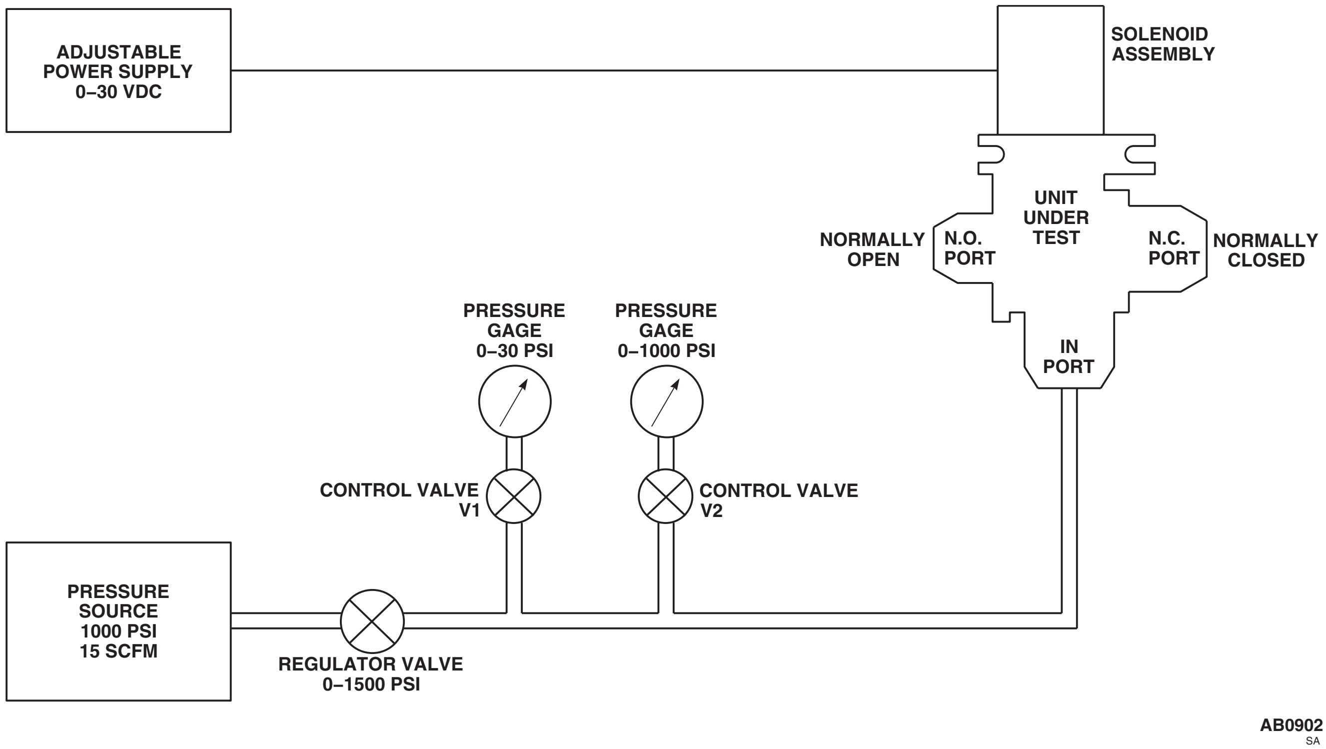


Figure 8. Directional Control Valve Actuation Test Setup (AVIM).

END OF WORK PACKAGE

UNIT LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

HEATING AND VENTILATION SYSTEM INSPECTION

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

INSPECT

1. Inspect tubes for dents. **DENTS UP TO 25% OF TUBE DIAMETER ARE ACCEPTABLE.** Replace tube if dents exceed limit.
2. Inspect all welds on tubes for cracks. **NONE ALLOWED.** Replace tubes with any cracks at weld joints.
3. Inspect tube couplings for chafing, damage, stiffness, and tears. **NONE ALLOWED.** Replace tube couplings if too stiff to slide over tube, or if chafed or torn.
4. Check brackets for cracks, bending, and security.
5. Inspect bleed-air tubing insulation for breaks and tears. Repair or replace if damaged.

END OF WORK PACKAGE

UNIT LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM INSPECTION

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

INSPECT

1. Remove aft right bulkhead soundproofing panel.
2. Inspect condenser pallet intake transition duct, exhaust transition duct, condenser exhaust duct, evaporator pallet inlet duct, and outlet transition ducts for chaffing, damage, stiffness, and tears. **NONE ALLOWED.**
3. Inspect duct couplings for chafing, damage, stiffness, and tears. **NONE ALLOWED.** Replace duct couplings if too stiff to slide over tube, or if chafed or torn.
4. Check brackets for cracks, bending, and security.
5. Install aft right bulkhead soundproofing panel.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

HEATING AND VENTILATION SYSTEM MIXING VALVE

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0149 00
WP 1703 00

REMOVE

1. Turn off all electrical power.
2. Open main rotor pylon sliding cover.
3. Disconnect electrical connector, P436, from mixing valve (Figure 1, Sheet 1, Detail B or Figure 2, Detail B).
4. Loosen clamps securing couplings to mixing valve inlet and outlet ports (Figure 1, Sheet 1, Detail B or Figure 2, Detail A).
5. Disconnect upper temperature sensing tube from elbow on mixing valve (Figure 1, Sheet 1, Detail C or Figure 2, Detail B).
6. On helicopters with mixing valve, 3224634-1, remove mixing valve as follows:
   a. Loosen clamps and remove tube coupling to disconnect front bleed-air tube from mixing valve (Figure 1, Sheet 2, Detail D).
   b. Remove bolts, washers, and mixing valve from support bracket (Figure 1, Sheet 1, Detail B).
7. On helicopters with mixing valve, 79089-2, do steps as follows:
   a. Disconnect front bleed-air tube from mixing valve (Figure 2, Detail A).
   b. Remove nut, washer, and mixing valve from support bracket.
8. If mixing valve is being replaced, remove elbow, nut, and packing (Figure 1, Sheet 1, Detail C or Figure 2, Detail B). Discard packing.

INSTALL

1. Turn off all electrical power.
2. **QA** If removed, install elbow, nut, and packing on mixing valve (Figure 2, Detail B). **TORQUE NUT TO 85 - 105 INCH-POUNDS.**
3. On helicopters with mixing valve, 3224634-1, do steps as follows:
   a. **QA** Install mixing valve on support bracket with bolts and washers (Figure 1, Sheet 1, Detail B). **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.**

INSTALL - Continued

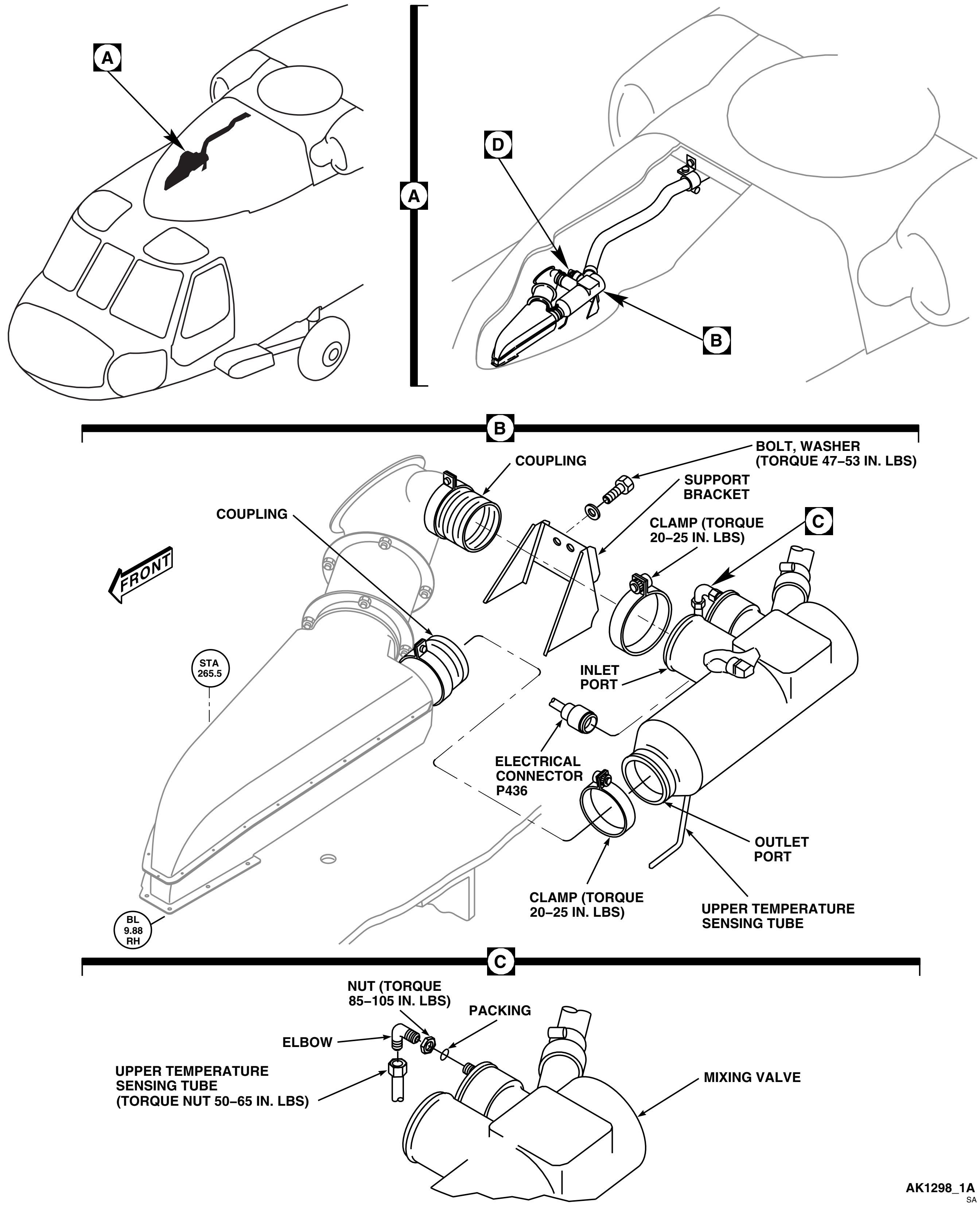


Figure 1. Replace Heating and Ventilation System Mixing Valve, 3224634-1. (Sheet 1 of 2)

INSTALL - Continued

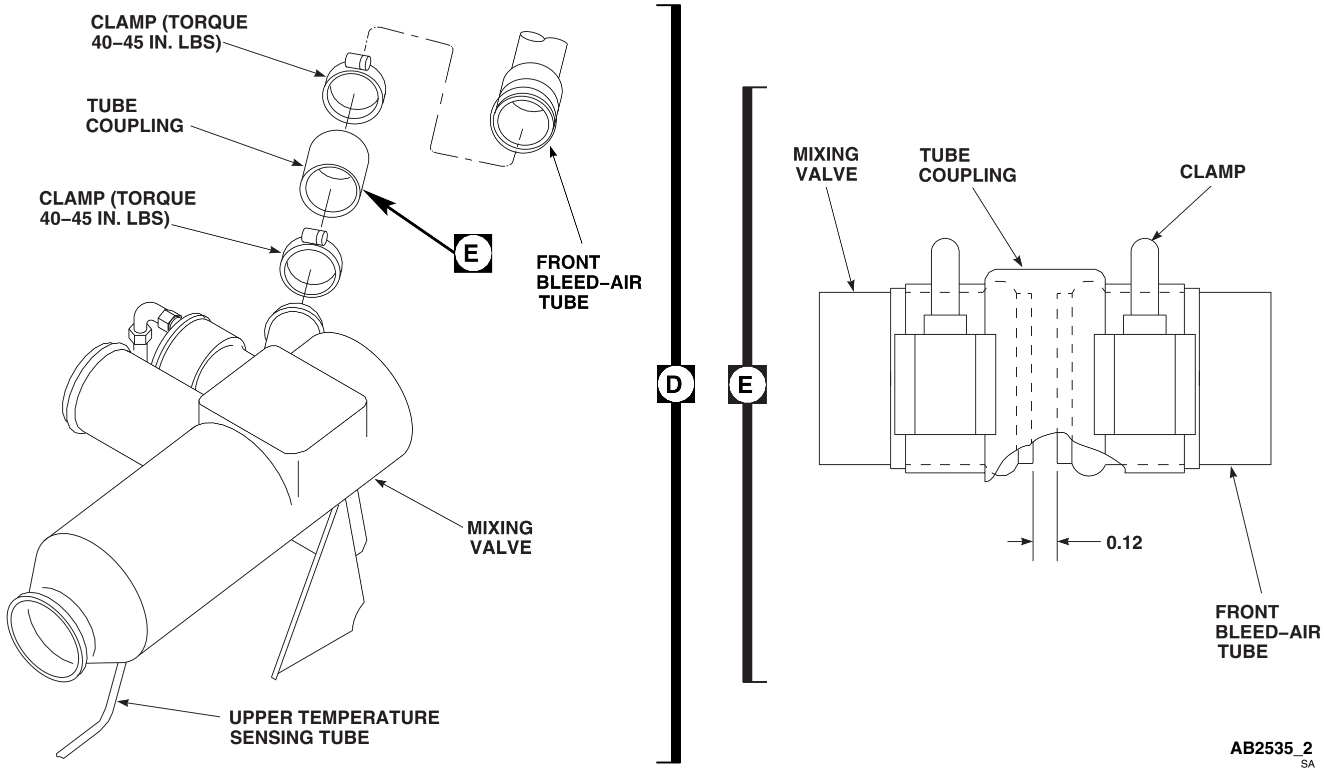


Figure 1. Replace Heating and Ventilation System Mixing Valve, 3224634-1. (Sheet 2 of 2)

NOTE

Before connecting front bleed-air tube to mixing valve, distance between tube and valve must be measured for proper gap.

b. Align front bleed-air tube with mixing valve. Measure gap between tubes. **GAP SHALL BE A MINIMUM OF 0.12-INCH (Figure 1, Sheet 2, Detail E).**

c. If gap is less than 0.12-inch, do this:

(1) Remove mixing valve.

(2) Remove metal from connecting tube on mixing valve using grinding wheel or file. Do not remove any material from bead on end of tube.

(3) Reinstall mixing valve.

d. Repeat step c.(1), step c.(2), and step c.(3) until gap is within limits.

e. **QA** Connect front bleed-air tube to mixing valve with tube coupling and clamps (Figure 1, Sheet 2, Detail D). **TORQUE CLAMPS TO 40 - 45 INCH-POUNDS.**

4. On helicopters with mixing valve, 79089-2, do this:

a. **QA** Install mixing valve in support bracket with washer and nut (Figure 2, Detail A). **TORQUE NUT 720 - 880 INCH-POUNDS.**

b. **QA** Connect front bleed-air tube to mixing valve.

INSTALL - Continued

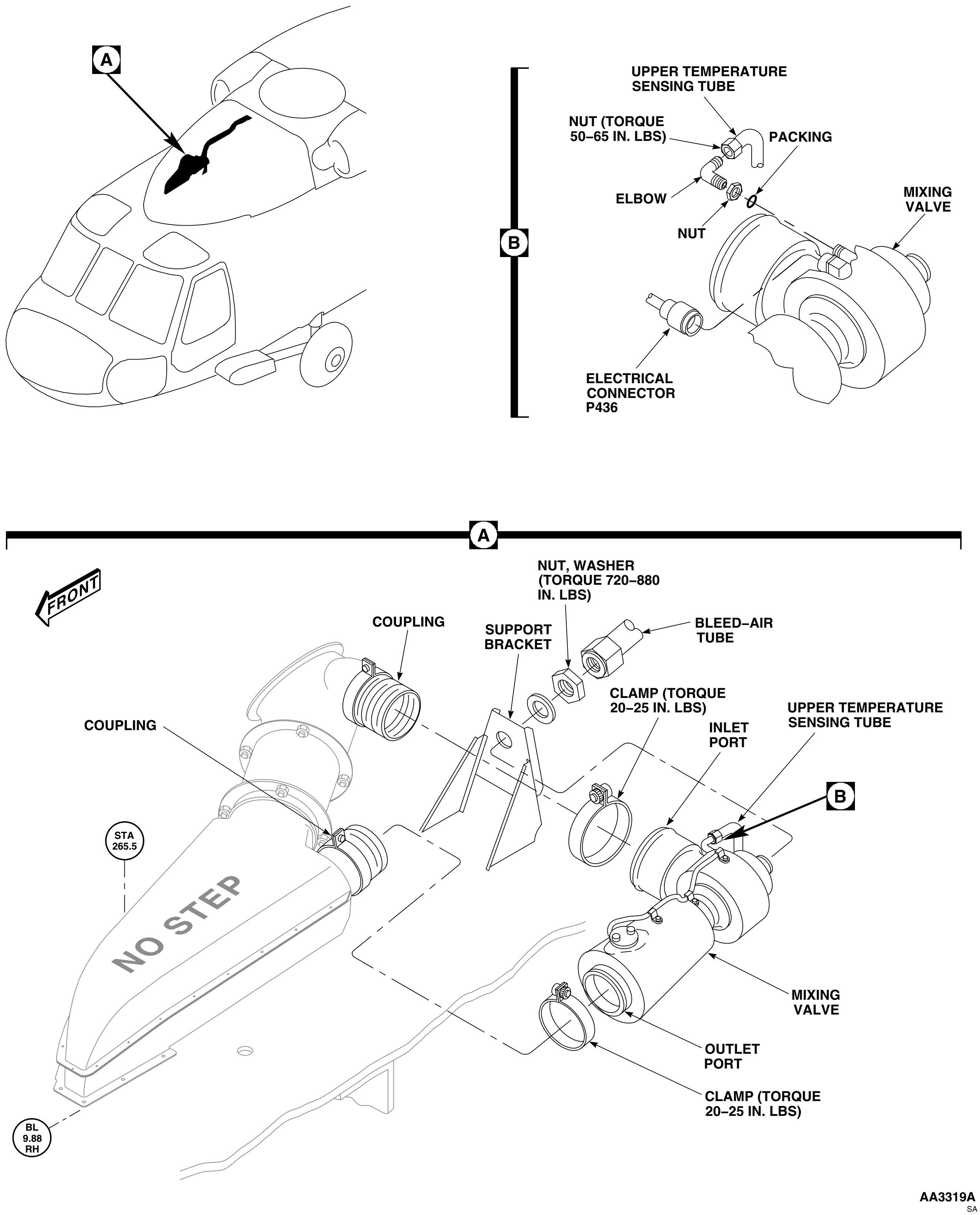


Figure 2. Replace Heating and Ventilation System Mixing Valve, 79089-2.

INSTALL - Continued

5. **QA** Connect upper temperature sensing tube to elbow on mixing valve (Figure 1, Sheet 1, Detail C or Figure 2, Detail B). **TORQUE NUT TO 50 - 65 INCH-POUNDS.**
6. **QA** Make sure tube couplings are properly positioned over inlet and outlet ports of mixing valve (Figure 1, Sheet 1, Detail B or Figure 2, Detail A). Tighten clamps. **TORQUE CLAMPS TO 20 - 25 INCH-POUNDS.**
7. Inspect and treat electrical connector (WP 1703 00).
8. **QA** Connect electrical connector, P436, to mixing valve.
9. Make sure area is clean and free of foreign material.
10. Do an operational check of heating and ventilating system (WP 0149 00).
11. Close and latch main rotor pylon sliding cover.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

HEATING AND VENTILATION SYSTEM UPPER TEMPERATURE SENSING TUBE

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Torque Wrench, 30 - 200 in. lbs.

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0149 00
WP 1271 00

REMOVE

1. Turn off all electrical power.
2. Open main rotor pylon sliding cover.
3. Disconnect upper temperature sensing tube from elbow on mixing valve (Figure 1, Detail A).
4. Disconnect upper temperature sensing tube from elbow on cabin skin.
5. Unclamp upper temperature sensing tube from cabin skin and remove from helicopter. Remove clamp, screw, nut, and spacer (Figure 1, Detail B).
6. If required, remove cabin skin elbow (WP 1271 00).

INSTALL

1. Turn off all electrical power.
2. If required, install cabin skin elbow (WP 1271 00).
3. **QA** Connect upper temperature sensing tube to elbow on cabin skin and elbow on mixing valve. **TORQUE UPPER TEMPERATURE SENSING TUBE NUTS TO 50 - 65 INCH-POUNDS (Figure 1, Detail A).**
4. **QA** Install clamp on upper temperature sensing tube (Figure 1, Detail B). Install spacer on cabin skin. Install clamp to cabin skin with screw and nut.
5. Do an operational check of heating and ventilating system (WP 0149 00).
6. Make sure area is clean and free of foreign material.
7. Close and latch main rotor pylon sliding cover.

INSTALL - Continued

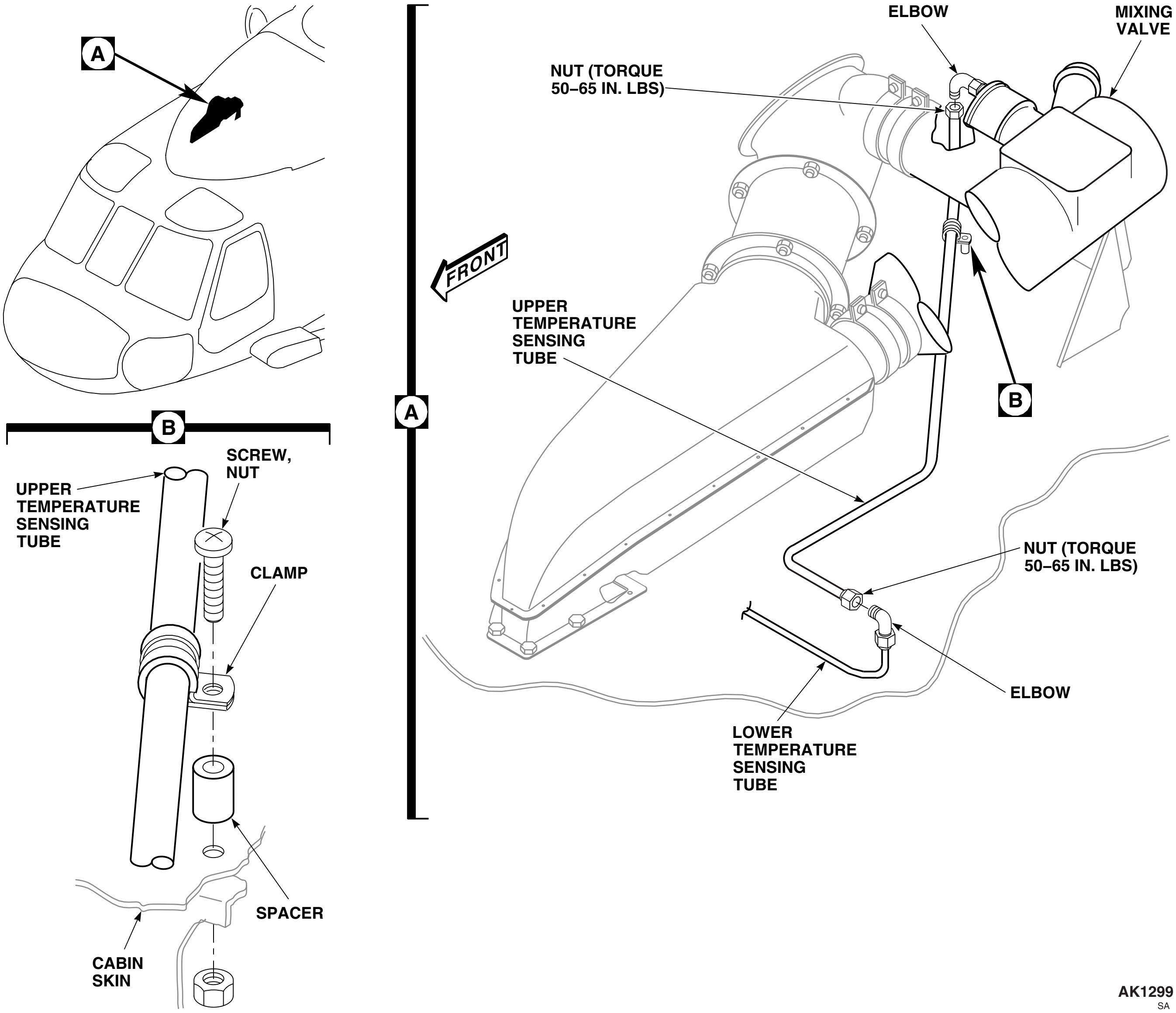


Figure 1. Replace Heating and Ventilation System Upper Temperature Sensing Tube.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

HEATING AND VENTILATION SYSTEM LOWER TEMPERATURE SENSING TUBE

This WP supersedes WP 1271 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Gloves
Goggles/Face Shield
Nonmetallic Scraper
Torque Wrench, 30 - 200 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0149 00
WP 0258 00
WP 1110 00
WP 1270 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Remove soundproofing from STA 247.0 to STA 265.0 in cabin overhead (WP 0258 00).

NOTE

Rigging of main and tail rotor controls will be affected if control rod lengths are changed. Make sure rod lengths on adjustable control rods are not changed when disconnected.

3. Disconnect and remove both cyclic control rods, and yaw control rod, to get to lower temperature sensing tube (WP 1110 00).
4. Disconnect lower temperature sensing tube from elbow in overhead cabin and from union on mixture temperature sensor (Figure 1, Detail B).
5. If required remove cabin skin elbow as follows:
   a. Remove upper temperature sensing tube at elbow (WP 1270 00).
   b. Remove sealing compound with a nonmetallic scraper.
   c. In cabin overhead, remove nut and washer from elbow.
   d. Remove sealing washers and elbow from cabin skin.

INSTALL

1. Turn off all electrical power.
2. If removed, install cabin skin elbow as follows (Figure 1, Detail B):

INSTALL - Continued

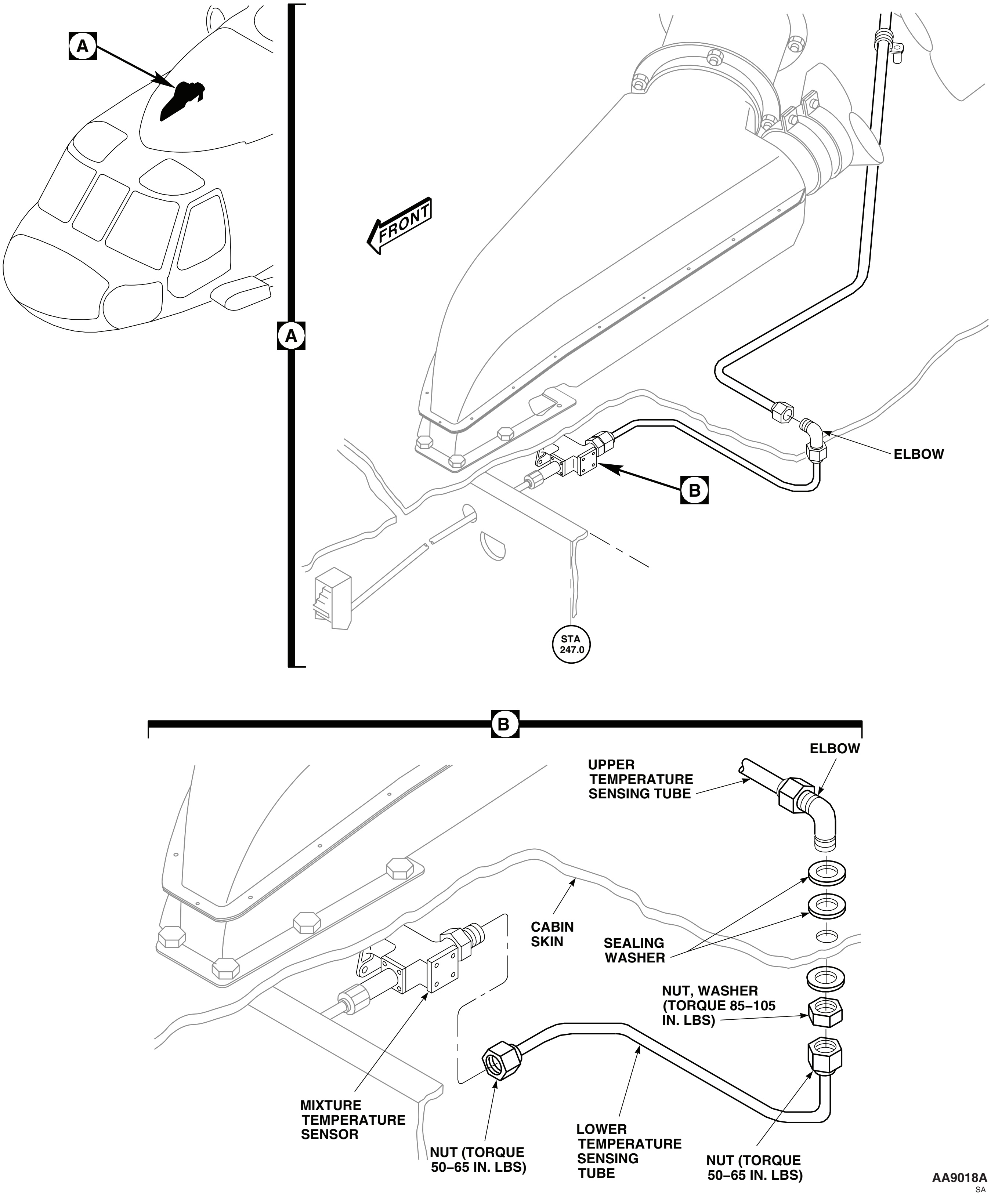


Figure 1. Replace Heating and Ventilation System Lower Temperature Sensing Tube.

INSTALL - Continued

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

a. Wipe cabin skin area with machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00. Wipe dry with a clean machinery wiping towel, Item 344, WP 1803 00.

c. **QA** Install elbow and sealing washers in cabin skin and install washer and nut from inside cabin. Position elbow for mating with upper temperature sensing tube. **TORQUE ELBOW BULKHEAD NUT TO 85 - 105 INCH-POUNDS.**

d. Connect upper temperature sensing tube (WP 1270 00).

e. Apply sealing compound, Item 273, WP 1803 00, around top of elbow where it passes through deck. Wipe off excess sealing compound.

3. **QA** Connect lower temperature sensing tube to elbow in overhead cabin and to union on mixture temperature sensor. **TORQUE NUTS TO 50 - 65 INCH-POUNDS.**
4. Reinstall two cyclic control rods and yaw control rod (WP 1110 00).
5. Do an operational check of heating and ventilating system (WP 0149 00).
6. Make sure area is clean and free of foreign material.
7. Install soundproofing (WP 0258 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

HEATING AND VENTILATION SYSTEM BLEED-AIR TUBE

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Material/Parts
Antiseize Compound, Item 49, WP 1803 00
Petrolatum, Item 216, WP 1803 00
Protective Caps and Plugs
Tape, Insulation, Electrical, Item 320, WP 1803 00
Thermal Insulation, 70309-02108-101

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0149 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Open main rotor pylon sliding cover and oil cooler compartment access doors.
3. **UH60A 77-22714 - 77-22716 UH60A 77-22719 - 77-22726** Disconnect bleed-air tube from mixing valve (Figure 1, Sheet 1, Detail B).
4. **UH60A 77-22717 - 77-22718 UH60A 77-22727 - SUBQ UH-60L EH-60A** Remove clamps holding tube coupling. Remove tube coupling to disconnect front bleed-air tube from mixing valve (Figure 1, Sheet 1, Detail B).
5. On forward main rotor pylon, remove clamp holding front bleed-air tube from support clamp (Figure 1, Sheet 2, Detail D).
6. **UH60A 83-23838 - SUBQ UH-60L EH-60A** Do this:
   a. Loosen clamps on left and right bleed-air lines (Figure 1, Sheet 2, Detail E).
   b. Disconnect bleed-air lines from tee on front bleed-air tube. Gently flex bleed-air lines over cone of tee.
   c. Remove tee from front bleed-air tube. Remove packing and jamnut from tee. Cap or plug tee.
   d. Remove clamp holding rear bleed-air tube to bracket on channel (Figure 1, Sheet 2, Detail G).
7. On right side of transmission area, remove clamps holding tube coupling connecting front and rear bleed-air tubes (Figure 1, Sheet 2, Detail F). Remove tube coupling.
8. Slide front bleed-air tube forward and remove from helicopter.
9. Remove clamps holding tube coupling connecting rear bleed-air tube to pneumatic manifold. Remove rear bleed-air tube from helicopter (Figure 1, Sheet 2, Detail H).
10. Cap or plug all tubing and manifolds.

REPAIR

1. Repair minor breaks or tears in insulation, using electrical insulation tape, Item 320, WP 1803 00. Do not compress insulation when wrapping tube.

REPAIR - Continued

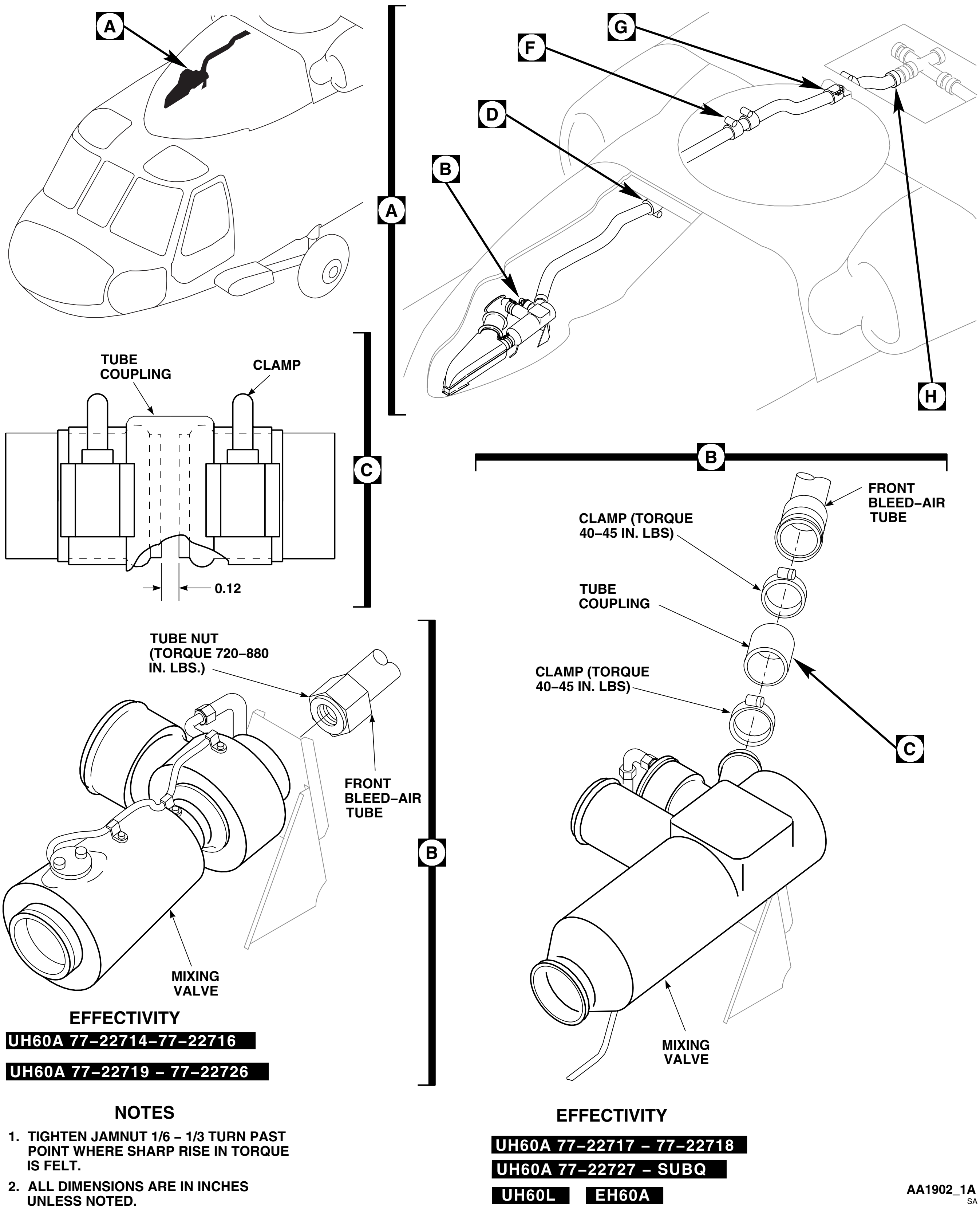


Figure 1. Replace Heating and Ventilation System Bleed-Air Tube. (Sheet 1 of 2)

REPAIR - Continued

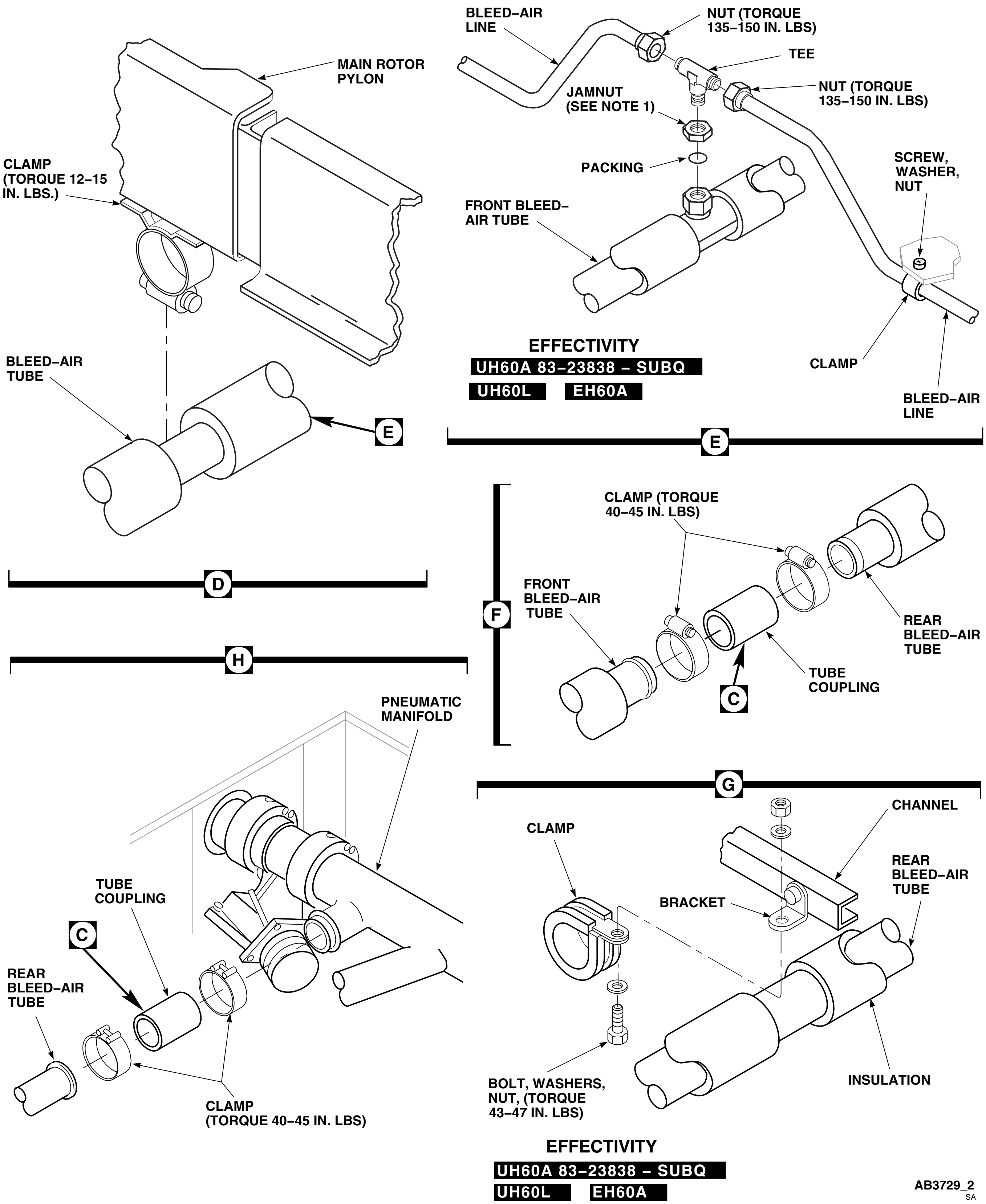


Figure 1. Replace Heating and Ventilation System Bleed-Air Tube. (Sheet 2 of 2)

REPAIR - Continued

2. To install new thermal insulation, 70309-02108-101, do this:
   a. Remove all old thermal insulation from tube.
   b. Cut wedges out of new thermal insulation, as required, to fit insulation over tube bends.
   c. Tape all seams with electrical insulation tape, Item 320, WP 1803 00, being careful not to compress insulation.
   d. Apply a full wrap (about 1 1/2 turns) of tape at each end of thermal insulation, and at one-foot intervals.

INSTALL

1. Turn off all electrical power.
2. Remove cap or plug from tubing, tee, and mixing valve ports.
3. Using filtered, dry compressed air, clean all tubes and mixing valve ports.
4. Position rear bleed-air tube.
5. **QA** Connect rear bleed-air tube to pneumatic manifold with tube coupling and clamps. Leave clamps loose (Figure 1, Sheet 2, Detail H).
6. **QA** Position front bleed-air tube between mixing valve and rear bleed-air tube (Figure 1, Sheet 2, Detail F).
7. **UH60A 83-23838 - SUBQ UH-60L EH-60A** Do this:
   a. **QA** Lubricate packing with petrolatum, Item 216, WP 1803 00. Install jamnut and packing on tee (Figure 1, Sheet 2, Detail E).
   b. Coat threads of tee with antiseize compound, Item 49, WP 1803 00, and install tee in front bleed-air tube.
   c. **QA** Connect bleed-air lines to tee in front bleed-air tube. Do not torque nuts now. **TIGHTEN JAMNUT ON TEE 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
   d. **QA** Install front bleed-air tube in support clamp. **TORQUE CLAMP TO 12 - 15 INCH-POUNDS (Figure 1, Sheet 2, Detail D).**
   e. **QA TORQUE NUTS ON BLEED-AIR LINES TO 135 - 150 INCH-POUNDS (Figure 1, Sheet 2, Detail E).** Tighten clamps on bleed air lines.

NOTE

Before connecting tube sections, distance between tubes must be measured for proper gap.

8. **QA** Align front and rear bleed-air tubes. Measure gap at rear bleed-air tube to pneumatic manifold, mixer valve connection to front bleed-air tube, and between front and rear bleed-air tubes (Figure 1, Sheet 1, Detail C). **GAP SHALL BE A MINIMUM OF 0.12-INCH AT EACH CONNECTION.**
9. If possible, obtain 0.12-inch minimum gap by readjusting tubes. If gap is still less than 0.12-inch, do this:
   a. Remove front or rear bleed-air tube.
   b. Remove metal from end of tube using grinding wheel or file. Do not remove any material from bead on end of tube.
   c. Reinstall tube.
10. Repeat step 8. and step 9. until gap is within limit.
11. **QA** Connect front and rear tubes with tube couplings and clamps. **TORQUE COUPLING CLAMPS AT FRONT AND REAR TUBE CONNECTION AND AT PNEUMATIC MANIFOLD TO 40 - 45 INCH-POUNDS (Figure 1, Sheet 2, Detail F and Detail H).**

INSTALL - Continued

12. **QA** **UH60A 83-23838 - SUBQ UH-60L EH-60A** Clamp rear bleed-air tube to channel with, clamp, bolt, washers, and nut. **TORQUE NUT TO 12 - 15 INCH-POUNDS (Figure 1, Sheet 2, Detail G).**
13. **QA** **UH60A 77-22714 - 77-22716 UH60A 77-22719 - 77-22726** Connect front tube to mixing valve. **TORQUE NUT TO 720 - 880 INCH-POUNDS (Figure 1, Sheet 1, Detail B).**
14. **QA** **UH60A 77-22717 - 77-22718 uh60a 77-22727 - SUBQ UH-60L EH-60A** Connect front bleed-air tube to mixing valve with tube coupling and clamps. **TORQUE COUPLING CLAMPS TO 40 - 45 INCH-POUNDS (Figure 1, Sheet 1, Detail B).**
15. Do an operational check of heating and ventilating system (WP 0149 00).
16. Make sure area is clean and free of foreign material.
17. Close and latch main rotor pylon sliding cover and oil cooler compartment access doors.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

HEATING AND VENTILATION SYSTEM MIXER TEMPERATURE SENSOR

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 30 - 200 in. lbs.

Personnel Required
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0149 00
WP 0258 00
WP 1110 00
WP 1277 00

REMOVE

1. Turn off all electrical power.
2. Open main rotor pylon sliding cover.
3. Remove soundproofing from cabin overhead from STA 247.0 to STA 265.0 (WP 0258 00).

NOTE

Rigging of main and tail rotor controls will be effected if control rod lengths are changed. Make sure rod length on adjustable control rods are not changed when disconnected.

4. Disconnect and remove both cyclic control rods, and yaw control rod, to get to mixture temperature sensor (WP 1110 00).
5. Remove muffler from helicopter (WP 1277 00).
6. From inside cabin, disconnect heater control shaft from mixture temperature sensor (Figure 1, Detail B).
7. Disconnect lower temperature sensing tube from mixture temperature sensor.
8. From top of helicopter, have assistant reach down inside of interior cabin duct to hold screws. Remove nuts and washers holding mixture temperature sensor to duct. Remove sensor.

INSTALL

1. Turn off all electrical power.
2. **QA** From inside cabin, position replacement mixture temperature sensor in interior duct (Figure 1, Detail B).
3. From top of helicopter, have assistant insert screws through interior duct and mixture temperature sensor.
4. **QA** Install washers and nuts on screw. Tighten nuts.
5. **QA** Connect lower temperature sensing tube to mixture temperature sensor. **TORQUE NUTS TO 50 - 65 INCH-POUNDS.**
6. **QA** Connect heater control shaft to mixture temperature sensor and adjust control knob as follows:
   a. **QA** Install slot, in end of heater control shaft, on key in mixture temperature sensor (Figure 1, Detail B). Install serrated nut and tighten.

INSTALL - Continued

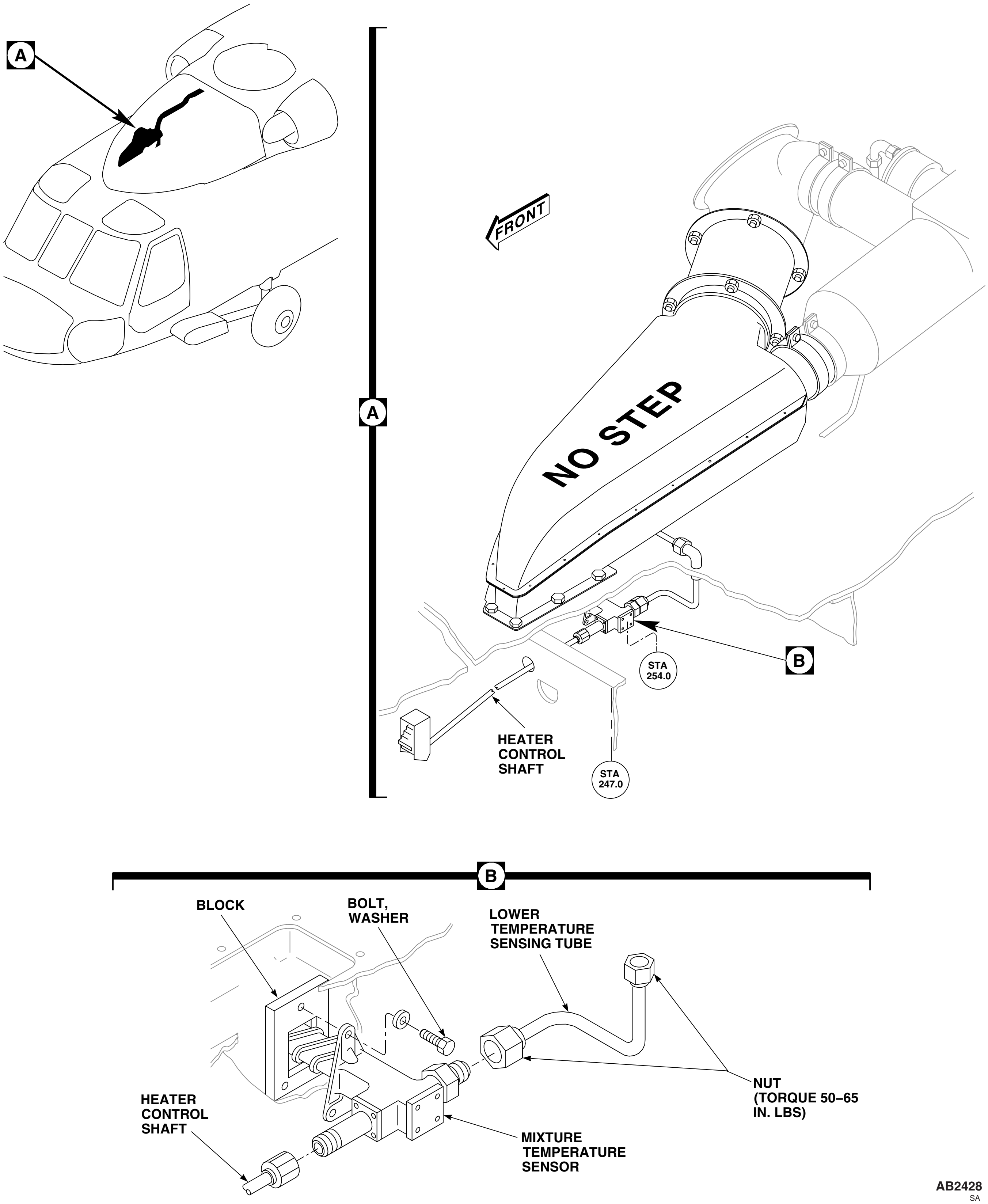


Figure 1. Replace Heating and Ventilation System Mixer Temperature Sensor.

INSTALL - Continued

b. On upper console in cockpit, turn heater control knob counterclockwise until it stops. Do not force knob.

c. Loosen setscrews of heater control knob and line up mark on knob with OFF position on panel.

d. Tighten setscrews.

7. Install two cyclic control rods, and yaw control rod (WP 1110 00).
8. Check inside of interior duct for foreign material.
9. Check condition of rubber seals before reinstalling muffler. If damaged, replace seals.
10. Install muffler on helicopter (WP 1277 00).
11. Do an operational check of heating and ventilating system (WP 0149 00).
12. Make sure area is clean and free of foreign material.
13. Install soundproofing (WP 0258 00).
14. Close and latch main rotor pylon sliding cover.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

HEATING AND VENTILATION SYSTEM AIR OUTLETS

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Blind Rivet Gun Kit, 353-51

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

NOTE

Heating and ventilating system has eight air outlets, in different locations. Removal and installation is the same at each location, except some are installed with rivets, some with screws.

REMOVE

1. Turn off all electrical power.
2. **QA** Remove screws or rivets holding air outlets to heating/ventilating duct in cabin front section ceiling (Figure 1, Detail A or Detail B).

INSTALL

1. Turn off all electrical power.
2. Install air outlet on heating/ventilating duct with screws or rivets (Figure 1, Detail A or Detail B).

INSTALL - Continued

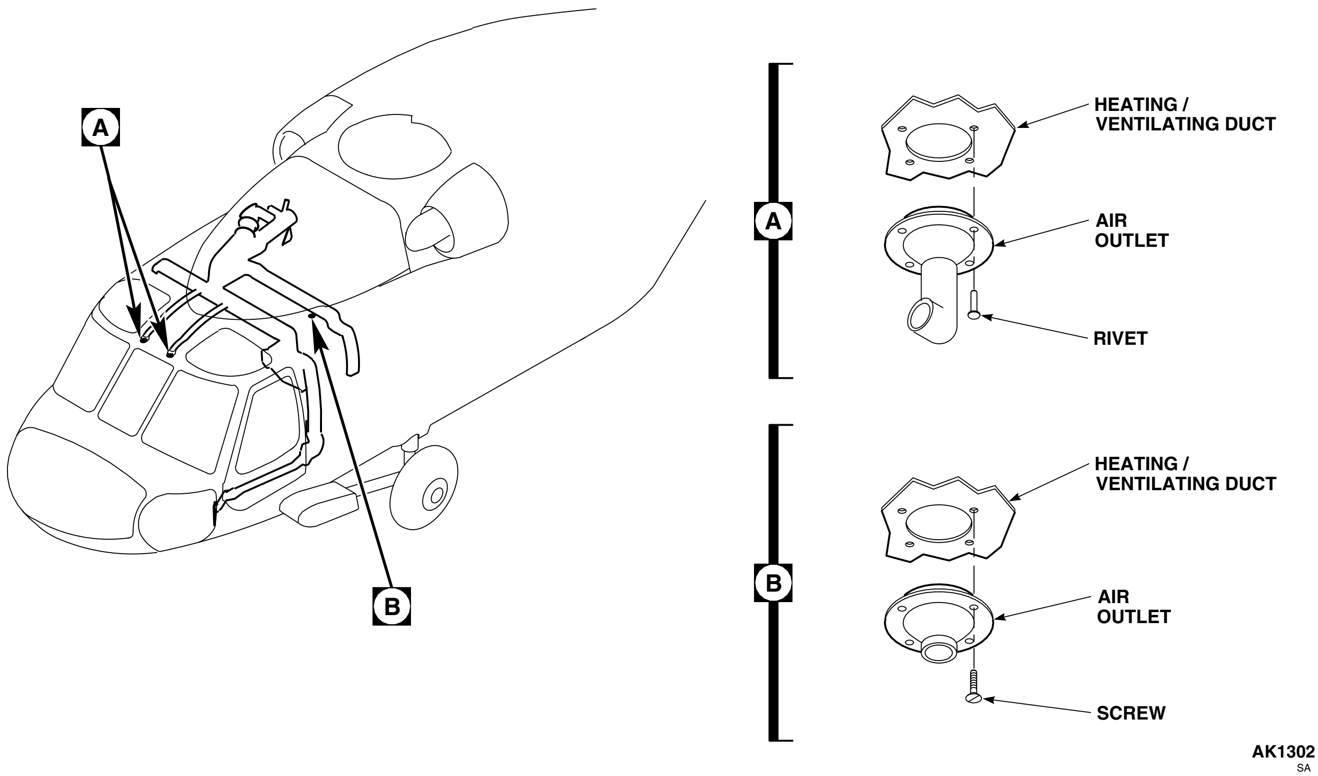


Figure 1. Replace Heating and Ventilation System Air Outlets.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

HEATING AND VENTILATION SYSTEM HEATER CONTROL SHAFT

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Material/Parts
Strap, Tiedown, Electrical, Item 301, WP 1803 00
Tubing, Plastic Spiral Wrap, Item 346, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0149 00
WP 0258 00
WP 1110 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Remove soundproofing from cabin overhead (WP 0258 00).

NOTE

Rigging of main and tail rotor controls will be effected if control rod lengths are changed. Make sure rod length on adjustable control rods are not changed when disconnected.

3. Disconnect and remove both cyclic control rods, and yaw control rod, to get to mixture temperature sensor (WP 1110 00).
4. In cockpit, unlock fasteners holding upper console to cockpit overhead. Hinge down upper console.
5. Disconnect control shaft from adapter (Figure 1, Detail B).
6. In cabin, disconnect control shaft from mixture temperature sensor.
7. Slide control shaft out of grommet in frame (Figure 1, Detail A). Remove shaft from helicopter.

INSTALL

1. Turn off all electrical power.
2. Check condition of grommet in frame. If damaged or deteriorated, replace.
3. **QA** Install plastic spiral wrap tubing, Item 346, WP 1803 00, on control shaft before installing through grommet.
4. **QA** Insert replacement control shaft through grommet in frame at rear of upper console (Figure 1, Detail A).
5. **QA** Connect shaft to mixture temperature sensor in cabin (Figure 1, Detail B).
6. Connect opposite end of shaft to adapter.
7. **QA** Install electrical tiedown straps, Item 301, WP 1803 00, on each end of control rod shaft.
8. Check inside of upper console for foreign material.
9. Close and lock fasteners holding upper console to cockpit overhead.

INSTALL - Continued

10. Line up control with mixture temperature sensor as follows:
    a. Turn adapter shaft (control knob attached) counterclockwise until it stops (Figure 1, Detail A).
    b. Loosen setscrews on heater control knob and line up mark on knob with OFF position on panel.
    c. Tighten setscrews.
11. Reinstall two cyclic control rods and yaw control rod (WP 1110 00).
12. Do an operational check of heating and ventilating system (WP 0149 00).
13. Make sure area is clean and free of foreign material.
14. Reinstall soundproofing (WP 0258 00).

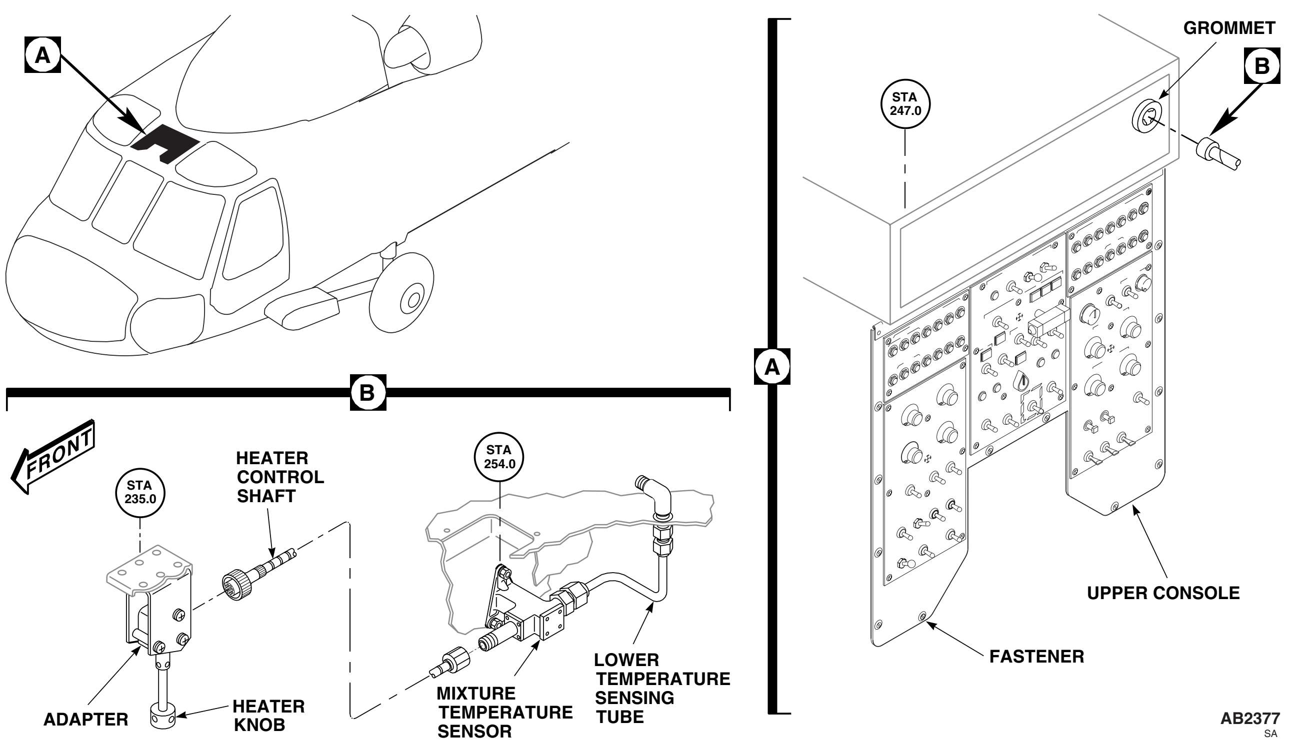


Figure 1. Replace Heating and Ventilation System Heater Control Shaft.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

HEATING AND VENTILATION SYSTEM HEATER CONTROL ADAPTER

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0149 00

REMOVE

1. Turn off all electrical power.
2. Unlock fasteners holding upper console to cockpit overhead. Hinge down upper console.
3. Loosen nut on control shaft until free of adapter (Figure 1, Detail A).
4. Loosen setscrews and remove heater control knob and shaft from adapter.
5. Remove front screw, washers, nut, and spacer from attachment bracket.
6. Loosen remaining two screws.
7. Slide adapter forward from between remaining spacers in attachment bracket and remove adapter from bracket.

INSTALL

1. Turn off all electrical power.
2. **QA** Slide adapter aft between spacers in attachment bracket (Figure 1, Detail A). Install spacer between bracket ears with front screw, washers, and nut.
3. **QA** Tighten remaining two screws.
4. **QA** Install heater control knob and shaft into adapter and tighten setscrews.
5. Install control shaft onto adapter and tighten nut.
6. Carefully hinge up upper console to make sure heater control knob and shaft line up with hole in upper console.
7. **QA** If heater control knob and shaft does not line up with hole in upper console, loosen front screw in attachment bracket and reposition adapter in bracket until they line up.
8. Make sure area is clean and free of foreign material.
9. Lock fasteners holding upper console to cockpit overhead.
10. Turn heater control knob on upper console counterclockwise until stopped. If required, loosen setscrews in knob, line up marker on knob with OFF position, and tighten setscrews.
11. Do an operational check of heating and ventilating system (WP 0149 00).

INSTALL - Continued

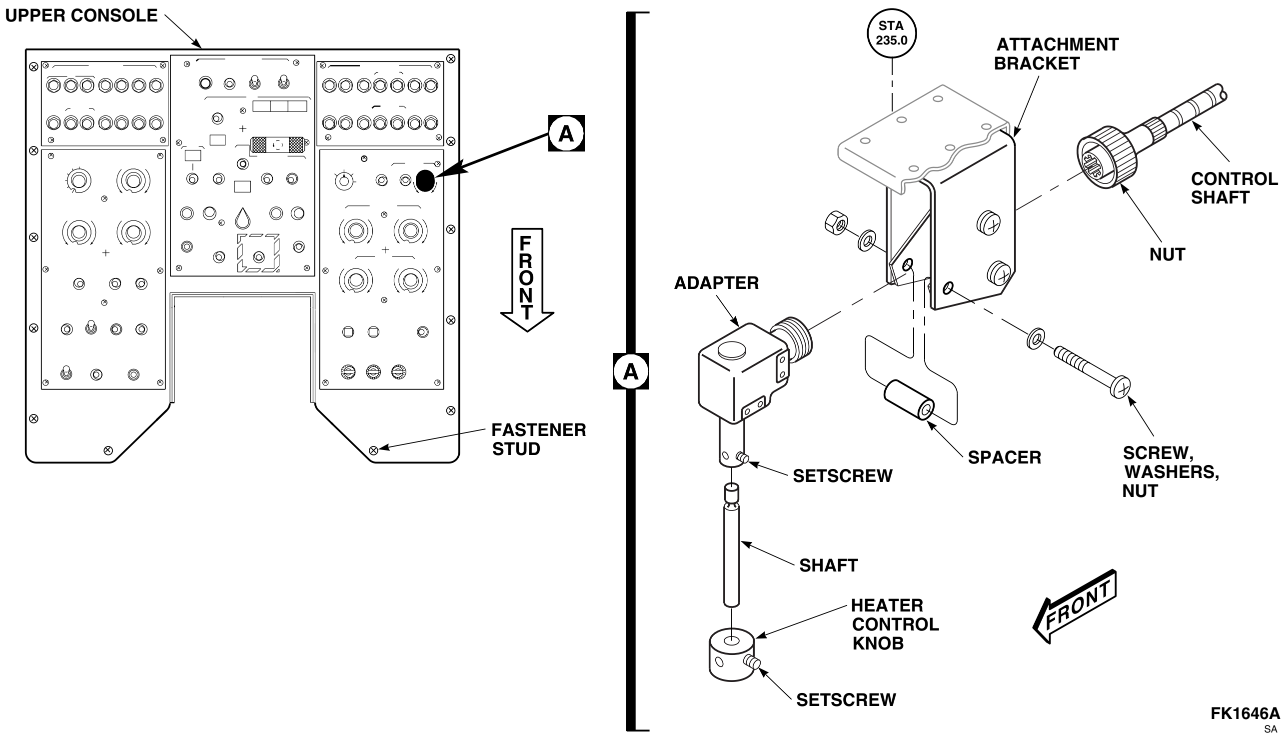


Figure 1. Replace Heating and Ventilation System Heater Control Adapter.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

HEATING AND VENTILATION SYSTEM HEATER MUFFLER

This WP supersedes WP 1277 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Gloves
Goggles/Face Shield
Nonmetallic Scraper
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 40, WP 1803 00
Cloth, Cleaning, Item 86, WP 1803 00
Sealing Compound, Item 273, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

WP 1282 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Open main rotor pylon sliding cover.
3. Remove bolts and washers from muffler and move wire and bracket to one side (Figure 1, Detail A).
4. Loosen clamp holding front mixing valve coupling to heater muffler. Slip coupling off muffler.
5. Remove bolts, washers, and nuts holding muffler to heater blower. Carefully remove muffler from helicopter.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

6. Use a cleaning cloth, Item 86, WP 1803 00, dampened in technical acetone, Item 5, WP 1803 00, and a nonmetallic scraper to remove old adhesive from hardware, muffler, and cabin skin.

INSPECT

1. Check condition of rubber seals on cabin skin before installing muffler. Replace seals if damaged (WP 1282 00).
2. Check interior of duct for foreign material.

INSTALL

1. Turn off all electrical power.

INSTALL - Continued

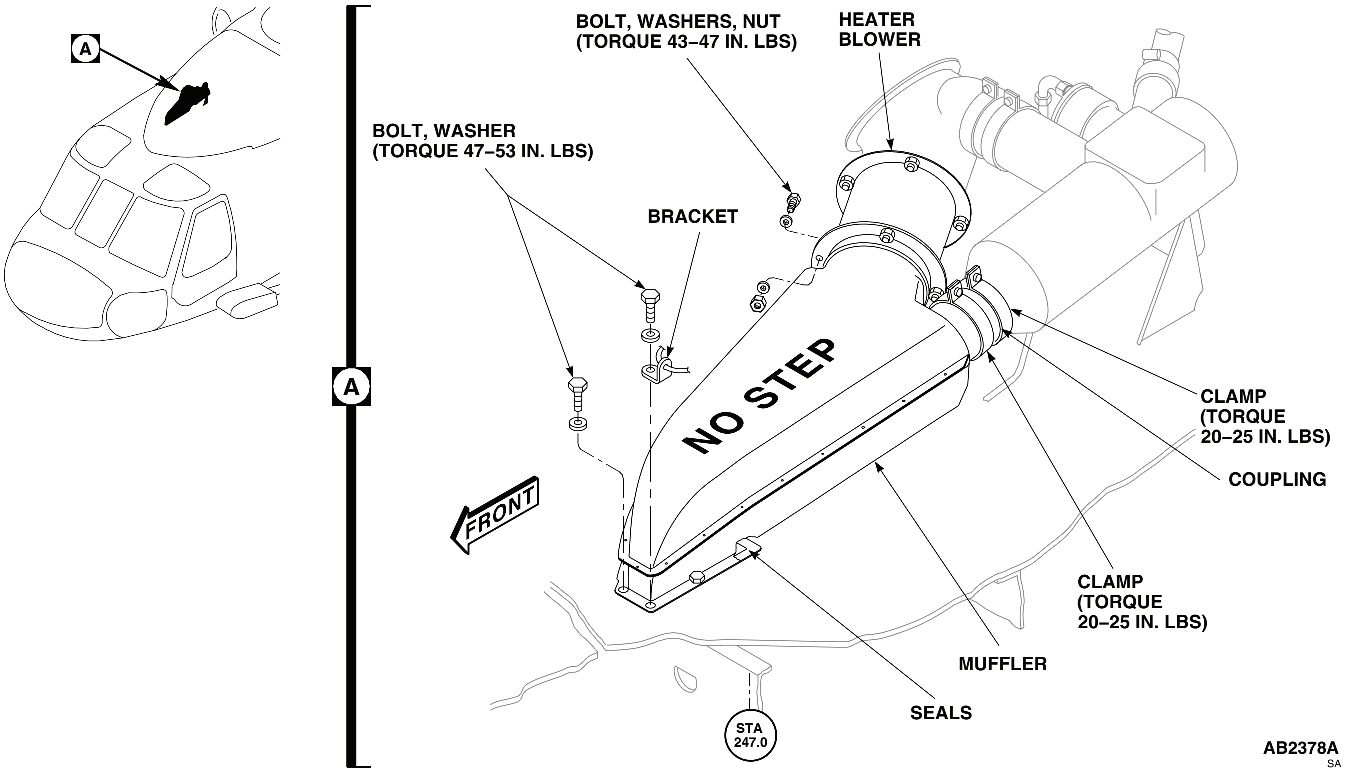


Figure 1. Replace Heating and Ventilation System Muffler.

NOTE

To prevent gaps at heater duct flanged joints, existing seal may be built up with adhesive, Item 40, WP 1803 00, to a maximum thickness of 0.060-inch.

2. Apply a light bead of sealing compound, Item 273, WP 1803 00, around boltholes in muffler.
3. **QA** Install muffler on seal on cabin skin with bracket, bolts, and washers (Figure 1, Detail A). **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.**
4. Apply a light bead of sealing compound, Item 273, WP 1803 00, around edge of muffler where it mates with cabin skin.
5. **QA** Connect muffler to heater blower with bolts, washers, and nuts. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**
6. **QA** Slide coupling over muffler and install clamps. **TORQUE NUTS TO 20 - 25 INCH-POUNDS.**
7. Make sure area is clean and free of foreign material.
8. Close and latch main rotor pylon sliding cover.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

HEATING AND VENTILATION SYSTEM COCKPIT DOOR DUCTS

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Blind Rivet Gun Kit, 353-51

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0379 00

REMOVE

1. Turn off all electrical power.
2. Remove rivets holding air outlet to heating/ventilating duct (Figure 1, Detail B or Detail C).
3. Remove screws and washers and carefully separate heating and ventilating ducts. Remove duct (Figure 1, Detail B).
4. Remove lower window duct by drilling out rivets holding duct to door frame (Figure 1, Detail C).
5. Remove heating/ventilating duct from door by removing screws, bolt, and washers. Push down on duct to disengage hidden retaining latch.

INSTALL

1. Turn off all electrical power.
2. Install lower window duct on door frame with rivets (Figure 1, Detail C).
3. **QA** Install duct on door by sliding duct forward from rear to engage hidden retaining latch. Install screws, bolt, and washers holding duct in place.
4. Trim upper edge of flange as necessary to stop interference between edge of flange and door jettison handle. Trim edge using procedures in (WP 0379 00).
5. **QA** Carefully install heating/ventilating duct with screws and washers (Figure 1, Detail B).
6. **QA** Install air outlet in heating/ventilating duct with rivets.
7. If installing new duct, do this:
   a. Trial-fit duct.
   b. Mark holes that do not line up with nutplates.
   c. Fill marked holes using procedures in (WP 0379 00).
   d. Mark and drill new holes using mounting duct as template.

INSTALL - Continued

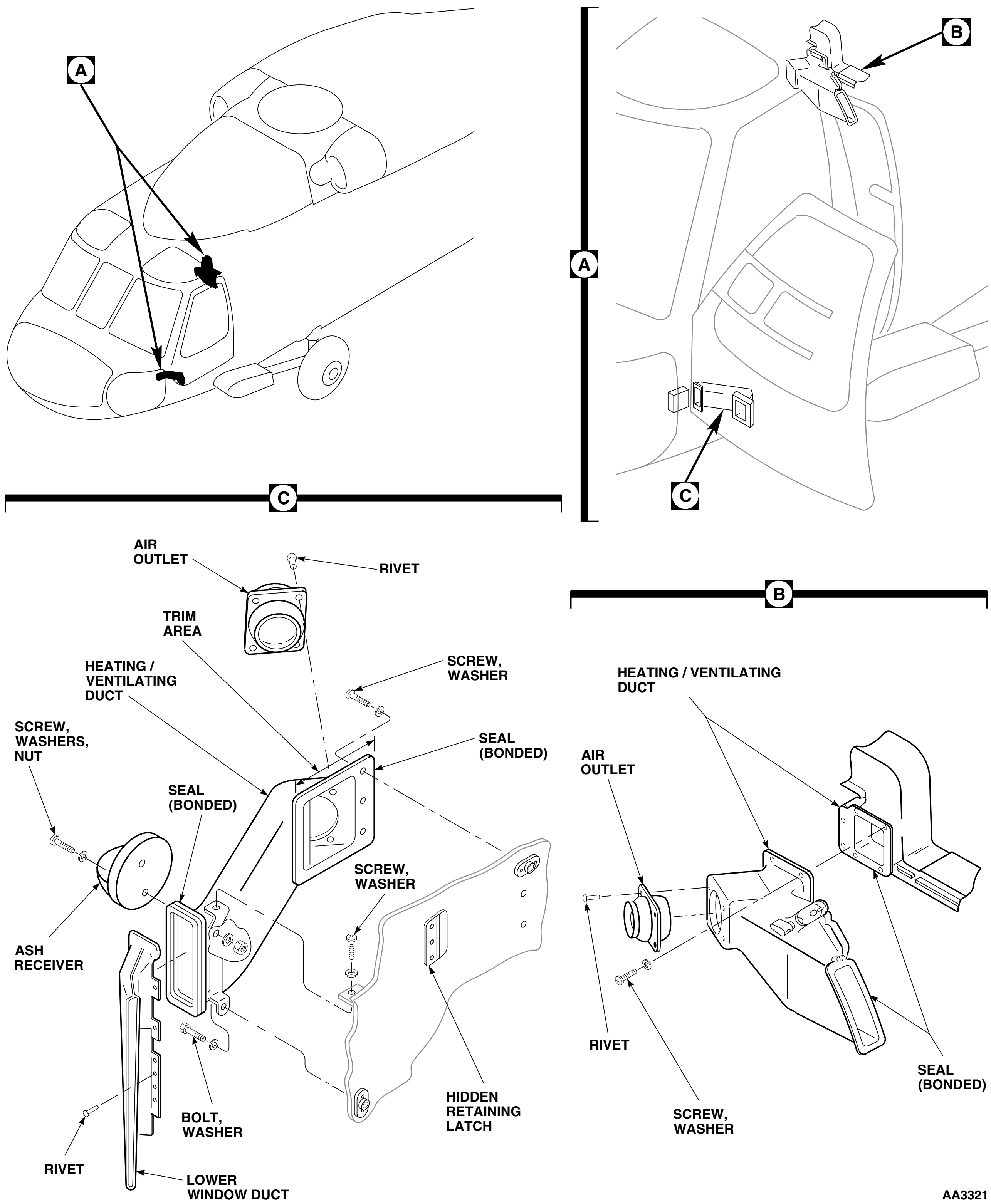


Figure 1. Replace Heating and Ventilation System Cockpit Door Ducts.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

HEATING AND VENTILATION SYSTEM HEATER DUCT

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

WP 1282 00

REMOVE

1. Turn off all electrical power.
2. Open main rotor pylon sliding cover.
3. Loosen clamps holding mixing valve coupling to heater duct. Slip coupling off duct (Figure 1, Detail A).
4. Remove bolts, washers, and nuts holding heater duct to heater blower. Carefully remove duct from helicopter.

INSPECT

1. Check condition of rubber seals on heater duct flange before installing. Replace seals if damaged (WP 1282 00).
2. Check interior of heater duct for foreign material.

INSTALL

1. Turn off all electrical power.
2. **QA** Place heater duct into position and install bolts, washers, and nuts holding duct to heater blower (Figure 1, Detail A). **TORQUE BOLTS TO 43 - 47 INCH-POUNDS.**
3. **QA** Slide coupling over heater duct and install clamps. **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**
4. Make sure area is clean and free of foreign material.
5. Close and latch main rotor pylon sliding cover.

INSTALL - Continued

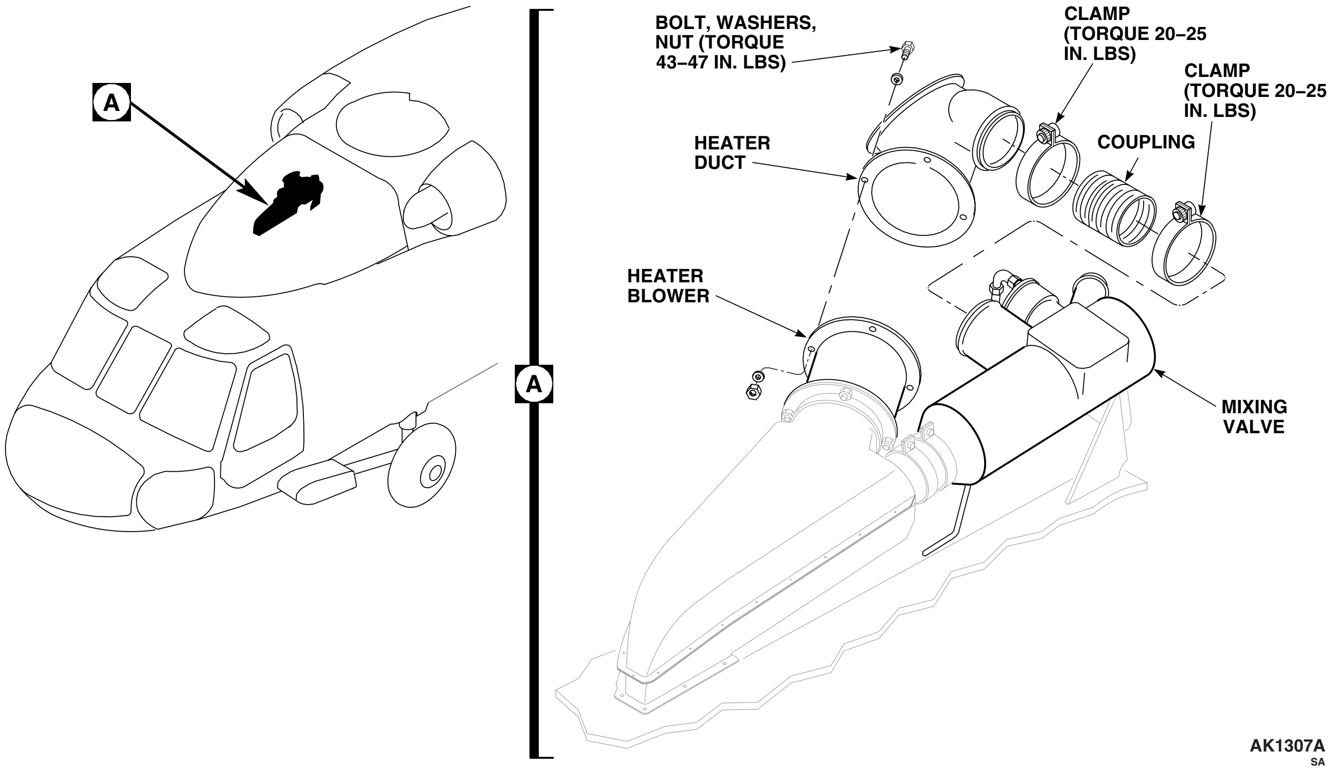


Figure 1. Replace Heating and Ventilation System Heater Duct.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

HEATING AND VENTILATION SYSTEM BLOWER

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Nonmetallic Scraper
Torque Wrench, 30 - 200 in. lbs.

Material/Parts
Adhesive, Item 40, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0149 00
WP 1703 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Open main rotor pylon sliding cover.
3. Disconnect blower electrical connector, J437 (Figure 1 Detail A).
4. Remove bolt, washer, nuts, clamp, and wiring from rear bracket.
5. Using a nonmetallic scraper, remove adhesive from flange of blower.
6. Remove bolts, washers, and nuts holding blower to duct and muffler.
7. Remove blower from helicopter.

INSTALL

1. Turn off all electrical power.
2. Inspect replacement blower and make sure impeller spins freely.
3. **QA** Place blower between duct and muffler so that air direction flow arrow points toward muffler (Figure 1, Detail A). Line up boltholes in blower and duct flanges and install bolts, washers, and nuts. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**
4. Inspect and treat electrical connector (WP 1703 00).
5. **QA** Connect blower electrical connector, J437.
6. **QA** Install blower wiring on rear bracket with bolt, washer, clamp, washer, and nut.
7. Apply a bead of adhesive, Item 40, WP 1803 00, around front flange of blower.
8. Do an operational check of heating and ventilating system (WP 0149 00).
9. Make sure area is clean and free of foreign material.
10. Close main rotor pylon sliding cover.

INSTALL - Continued

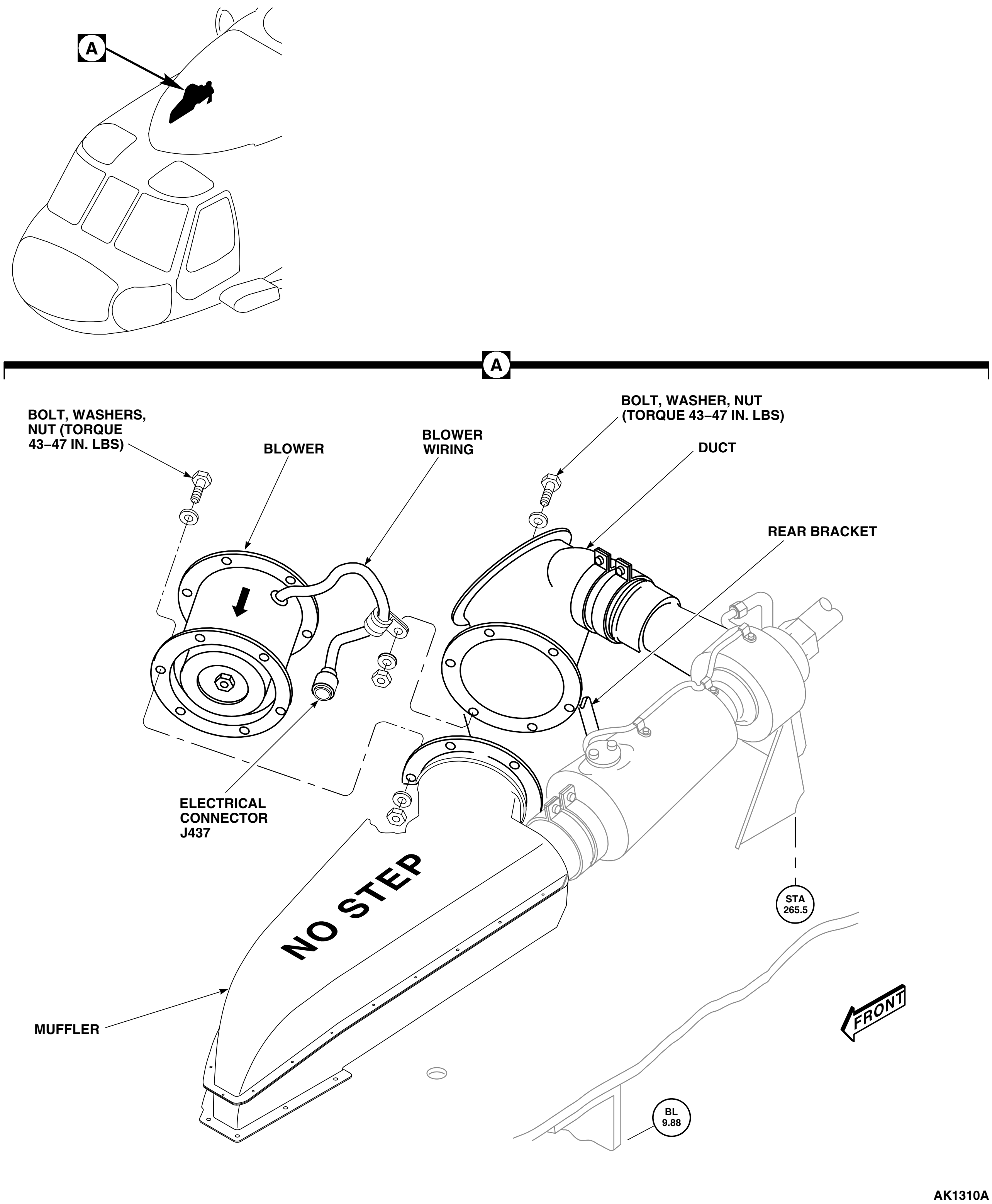


Figure 1. Replace Heating and Ventilation System Blower.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

HEATING AND VENTILATION SYSTEM AIR DUCT VALVE ASSEMBLIES

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

REMOVE

1. Turn off all electrical power.
2. Remove screw and washer and separate air duct valve assembly and outer duct assembly from center duct section (Figure 1, Detail A).
3. Remove screw and washer and separate air duct valve assembly from outer duct assembly.

INSPECT/DISASSEMBLE

1. Inspect air duct valve assembly for damage and proper operation.
2. If required, disassemble air duct valve assembly and replace any damaged components (Figure 1, Detail B and Detail C).

INSTALL

1. Turn off all electrical power.
2. **QA** Slide air duct valve assembly into outer duct assembly and install screw and washer (Figure 1, Detail A).
3. **QA** Slide air duct valve assembly and outer duct assembly into center duct section and install screw and washer.

INSTALL - Continued

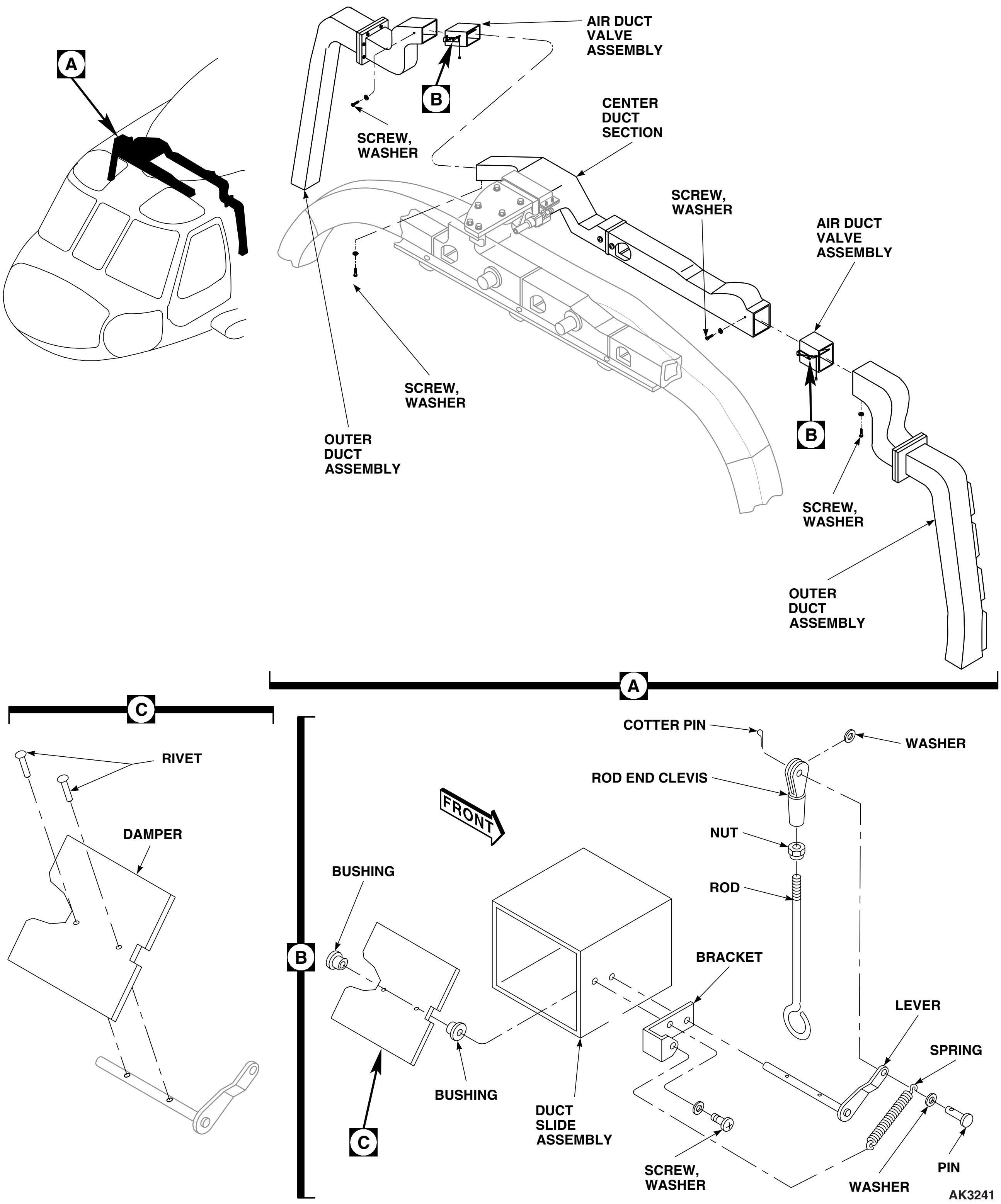


Figure 1. Replace Heating and Ventilation System Air Duct Valve Assemblies.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

HEATING AND VENTILATION SYSTEM REPAIR

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 10, WP 1803 00
Adhesive, Item 11, WP 1803 00
Adhesive, Item 13, WP 1803 00
Adhesive, Item 14, WP 1803 00
Adhesive, Item 29, WP 1803 00
Adhesive, Item 40, WP 1803 00
Adhesive Primer, Item 23, WP 1803 00
Cloth, Abrasive, Item 78, WP 1803 00
Cloth, Cleaning, Item 86, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

References

TM 1-1500-204-23
TM 55-1500-345-23
WP 0377 00
WP 0378 00
WP 0379 00
WP 0381 00
WP 1277 00
WP 1803 00

HEATING AND VENTILATION SYSTEM GENERAL REPAIRS

1. Heat and vent ducts are made from Kevlar/epoxy (reinforced plastic). Dampers, slide assemblies, brackets, and angles are aluminum alloy. Shafts for dampers are made from 4130 steel rod (Figure 1, Sheet 1, Detail B).
2. Evaluate and repair damage to Kevlar ducts (WP 0379 00).
3. Evaluate damage to aluminum parts (WP 0377 00).
4. Use WP 0378 00 for typical repairs to aluminum parts. Refer to TM 1-1500-204-23 for repair materials and fastener information.
5. Rebond seals. Refer to, REPLACE HEATING AND VENTILATION SYSTEM UPPER CABIN DUCT SEAL and REPLACE HEATING AND VENTILATION SYSTEM LOWER CABIN DUCT SEAL, in this work package.
6. Rebond ducts using adhesive primer, Item 23, WP 1803 00, and adhesive, Item 11, WP 1803 00. Apply adhesives per manufacturer's instructions.
7. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

REPLACE HEATING AND VENTILATION SYSTEM COCKPIT DUCT SEAL

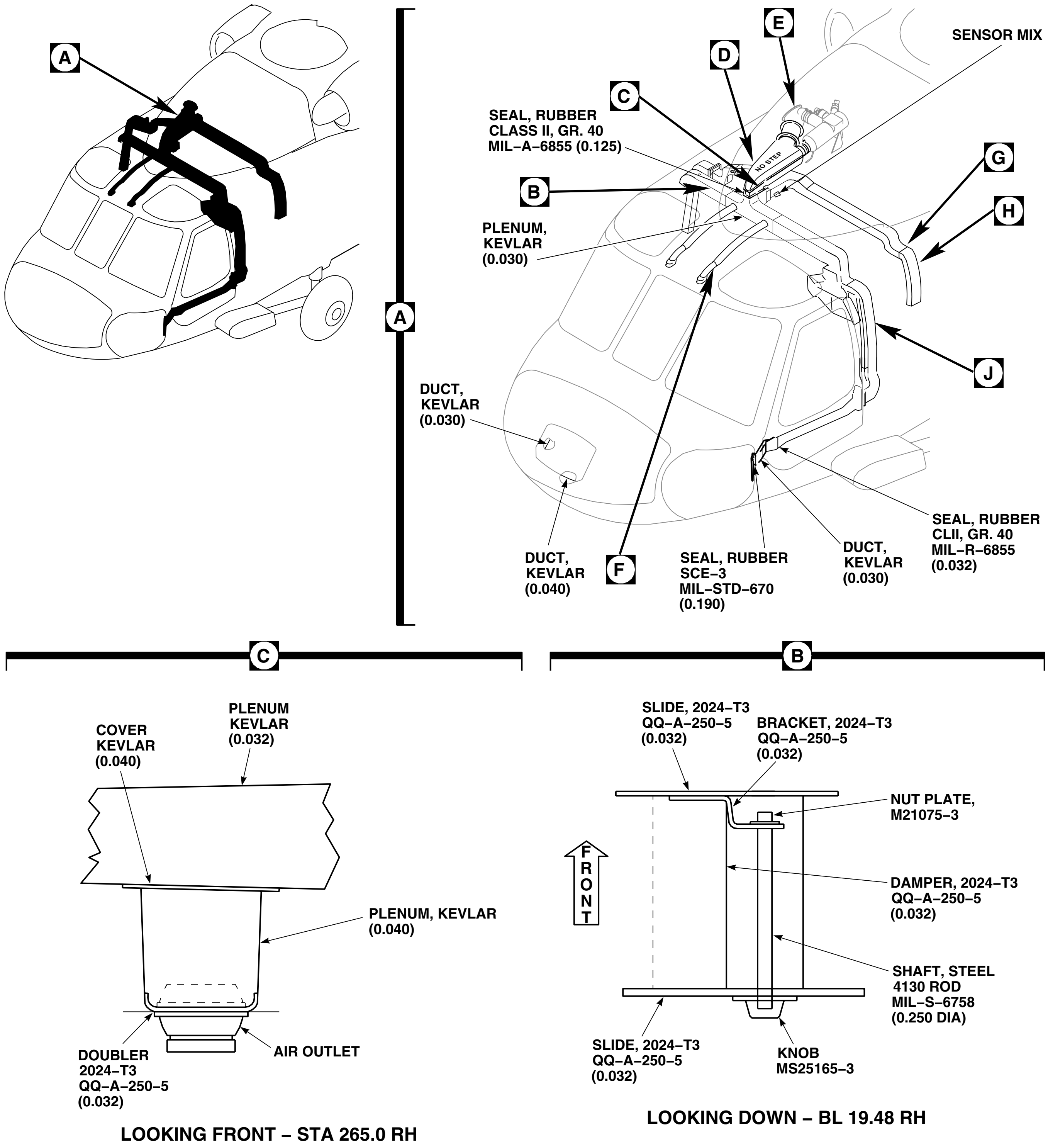
Remove

1. Turn off all electrical power.
2. Remove screws and washers holding duct to cockpit door (Figure 2, Sheet 2, Detail F).
3. Remove old seal from duct.

Install

1. Turn off all electrical power.

REPLACE HEATING AND VENTILATION SYSTEM COCKPIT DUCT SEAL - Continued



NOTES

1. POSITION SEAL JOINT ON CENTER OF FLANGE INBOARD SIDE.
2. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

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SA

Figure 1. Heating and Ventilation System Repair. (Sheet 1 of 3)

REPLACE HEATING AND VENTILATION SYSTEM COCKPIT DUCT SEAL - Continued

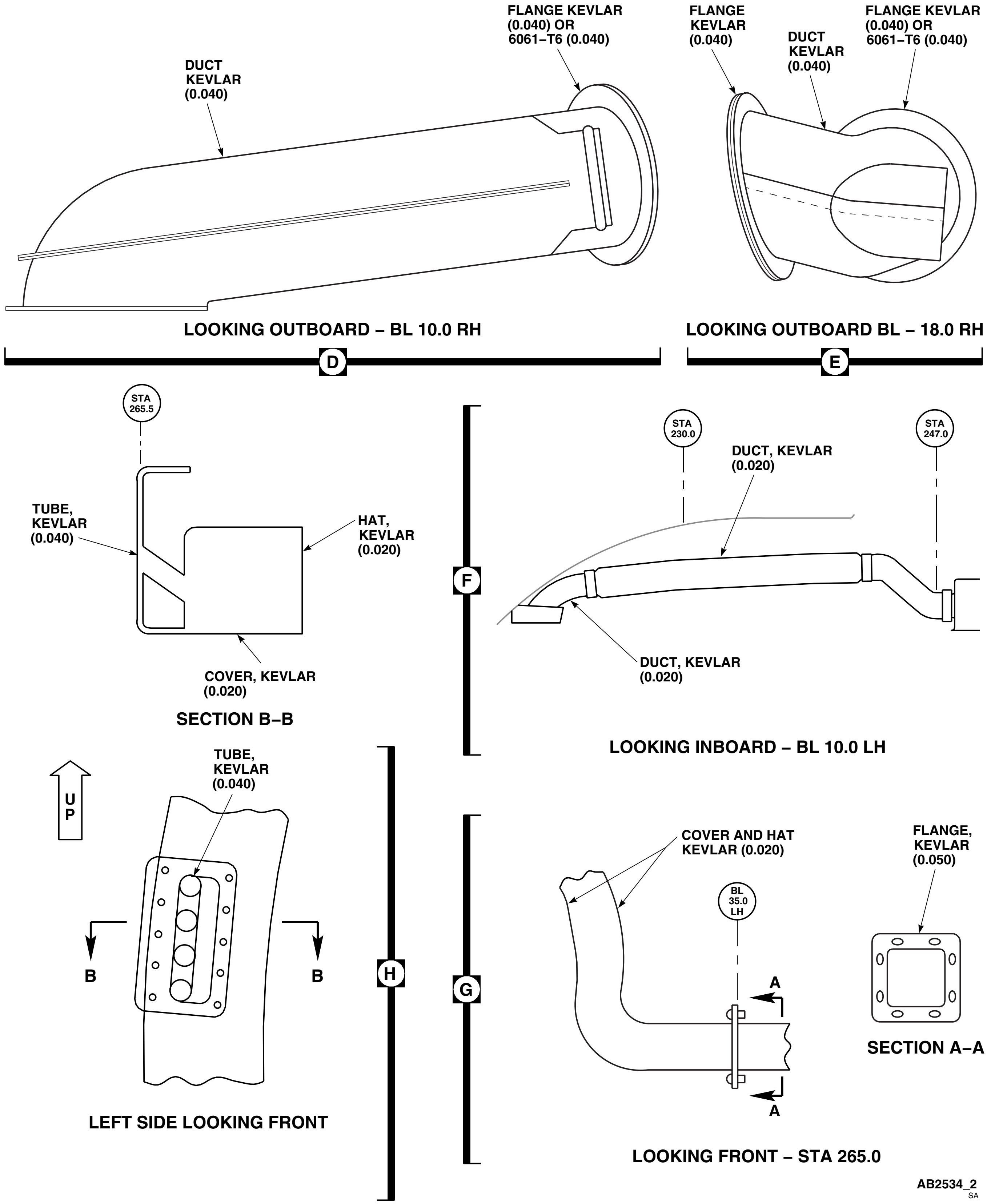


Figure 1. Heating and Ventilation System Repair. (Sheet 2 of 3)

REPLACE HEATING AND VENTILATION SYSTEM COCKPIT DUCT SEAL - Continued

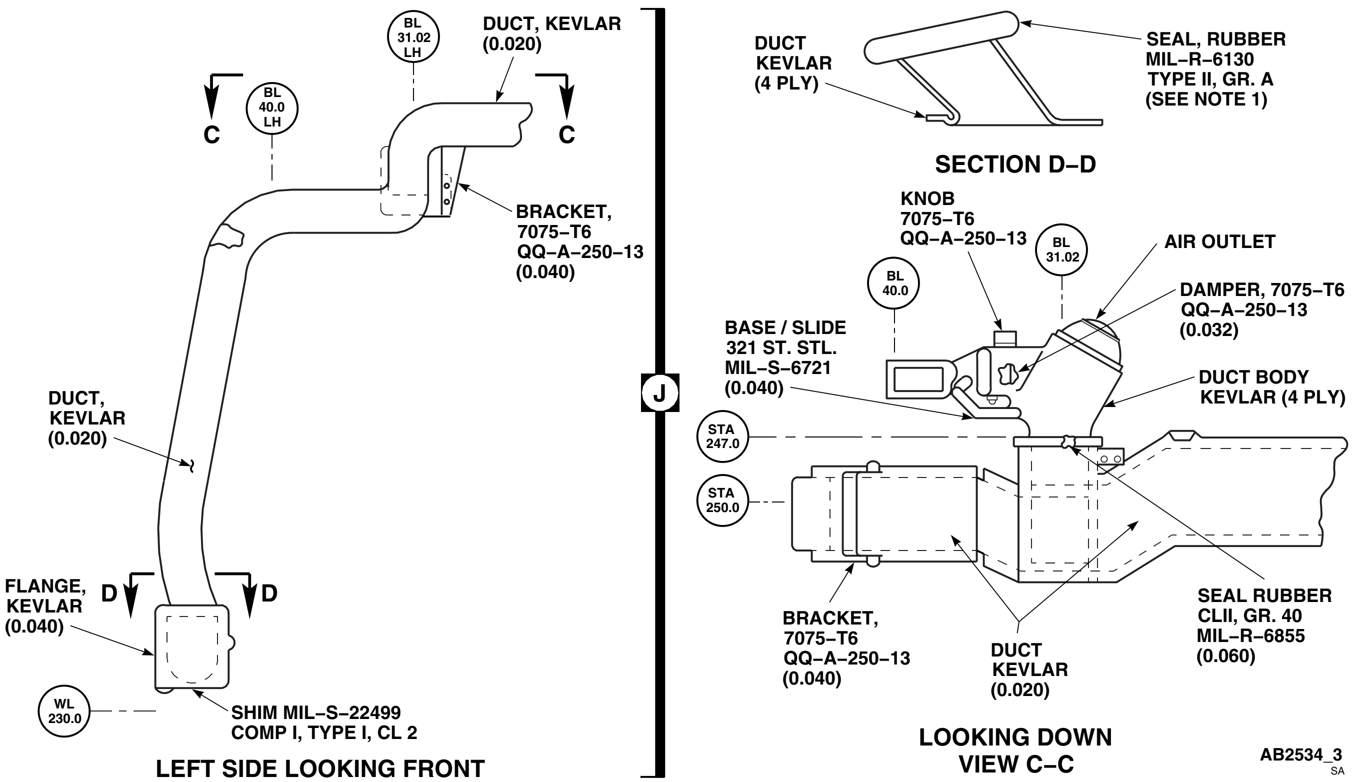


Figure 1. Heating and Ventilation System Repair. (Sheet 3 of 3)

WARNING

- Injury to personnel will result if area is not ventilated from volatile and toxic fumes when using solvents. Make sure area is vented properly to prevent a fire and a health hazard.
- Injury to personnel will result if proper ventilation and protective clothing, including eye shield, are not used when using solvents. Avoid breathing vapors and skin contact as much as possible. Wash contacted skin with soap and water. If solvent contacts eyes, flush them with clean water and get immediate medical help.

2. Clean bond area of old adhesive with cleaning cloth, Item 86, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
3. Roughen bonding surface of replacement seal with medium-grit abrasive cloth, Item 78, WP 1803 00.
4. Apply a smooth even coat of adhesive, Item 14, WP 1803 00, to mating surfaces of seal and duct. Allow adhesive to air-dry for about 10 minutes or until it becomes tacky.
5. Press seal firmly in place. Make sure all surfaces are in good contact. Allow seal to cure for 24 hours at 68°F(20°C) (Figure 2, Sheet 2, Detail F).
6. **QA** Install duct to cockpit door with screws, bolt, and washers.

REPLACE HEATING AND VENTILATION SYSTEM COCKPIT DUCT SEAL - Continued

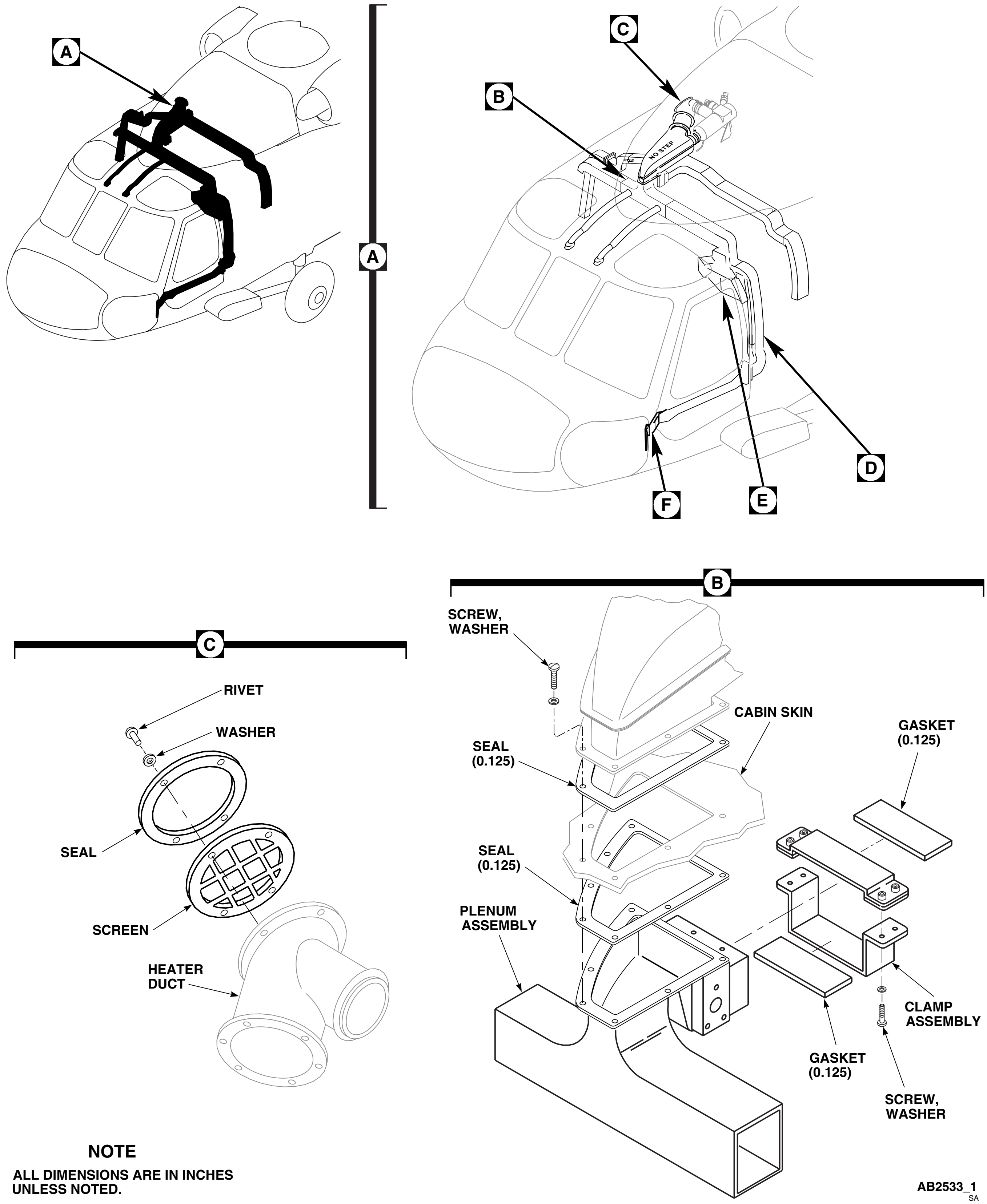


Figure 2. Replace Heating and Ventilation System Seals and Gaskets. (Sheet 1 of 2)

REPLACE HEATING AND VENTILATION SYSTEM COCKPIT DUCT SEAL - Continued

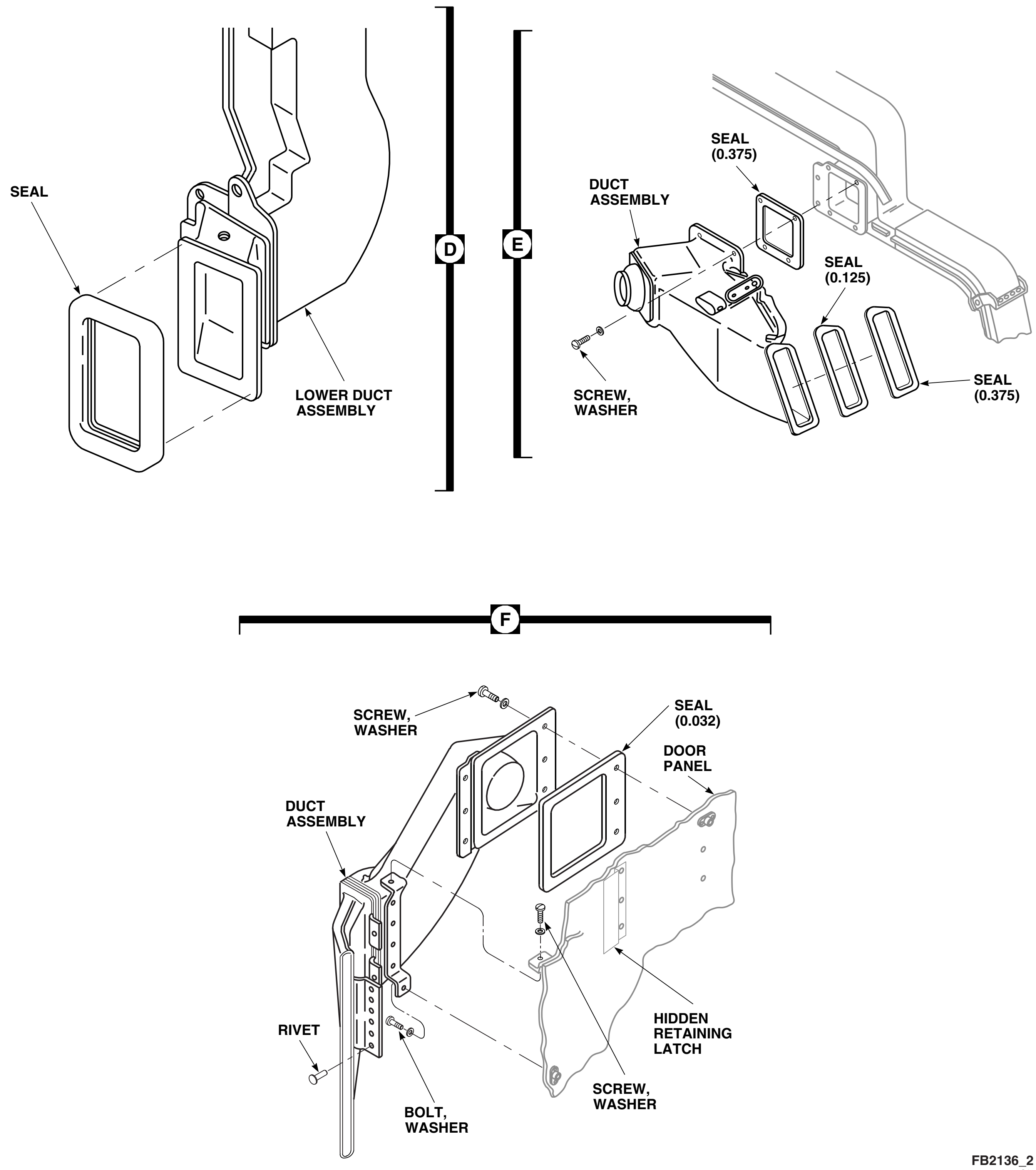


Figure 2. Replace Heating and Ventilation System Seals and Gaskets. (Sheet 2 of 2)

REPLACE HEATING AND VENTILATION SYSTEM UPPER CABIN DUCT SEAL

Remove

1. Turn off all electrical power.
2. Remove screws and washers holding duct (Figure 2, Sheet 2, Detail E).
3. Remove old seals from duct.

Install

1. Turn off all electrical power.

WARNING

- Injury to personnel will result if area is not ventilated from volatile and toxic fumes when using solvents. Make sure area is vented properly to prevent a fire and a health hazard.
- Injury to personnel will result if proper ventilation and protective clothing, including eye shield, are not used when using solvents. Avoid breathing vapors and skin contact as much as possible. Wash contacted skin with soap and water. If solvent contacts eyes, flush them with clean water and get immediate medical help.

2. Clean bond area of old adhesive with cleaning cloth, Item 86, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
3. Roughen bonding surface of replacement seal with medium-grit abrasive cloth, Item 78, WP 1803 00.
4. Apply a smooth even coat of adhesive, Item 14, WP 1803 00, to mating surfaces seal and duct. Allow adhesive to air-dry for about 10 minutes or until it becomes tacky.
5. Press seal firmly in place. Make sure all surfaces are in good contact. Allow seal to cure for 24 hours at 68°F(20°C) (Figure 2, Sheet 2, Detail E).
6. **QA** Install duct with screws and washers.

REPLACE HEATING AND VENTILATION SYSTEM LOWER CABIN DUCT SEAL

Remove

Remove old seal from duct (Figure 2, Sheet 2, Detail D).

Install

WARNING

- Injury to personnel will result if area is not ventilated from volatile and toxic fumes when using solvents. Make sure area is vented properly to prevent a fire and a health hazard.
- Injury to personnel will result if proper ventilation and protective clothing, including eye shield, are not used when using solvents. Avoid breathing vapors and skin contact as much as possible. Wash contacted skin with soap and water. If solvent contacts eyes, flush them with clean water and get immediate medical help.

1. Clean bond area of old adhesive with cleaning cloth, Item 86, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

REPLACE HEATING AND VENTILATION SYSTEM LOWER CABIN DUCT SEAL - Continued

2. Roughen bonding surface of replacement seal with abrasive cloth, Item 78, WP 1803 00.
3. Apply a smooth even coat of adhesive, Item 14, WP 1803 00, to mating surfaces of seal and duct. Allow adhesive to air-dry for about 10 minutes or until it becomes tacky.

NOTE

When bonding seal to duct locate split of seal at about midpoint of duct on inboard side.

4. Press seal firmly in place. Make sure all surfaces are in good contact. Allow seal to cure for 24 hours at 68°F(20°C) (Figure 2, Sheet 2, Detail D).

REPLACE HEATING AND VENTILATION SYSTEM MUFFLER/PLENUM SEAL AND GASKET

Remove

1. Turn off all electrical power.
2. Remove muffler from plenum (WP 1277 00).
3. Remove screws and washers from clamp assembly (Figure 2, Sheet 1, Detail B).
4. Remove plenum seal and gaskets from clamp assembly.
5. Remove seal from muffler cabin skin area.

Install

1. Turn off all electrical power.

WARNING

- Injury to personnel will result if area is not ventilated from volatile and toxic fumes when using solvents. Make sure area is vented properly to prevent a fire and a health hazard.
- Injury to personnel will result if proper ventilation and protective clothing, including eye shield, are not used when using solvents. Avoid breathing vapors and skin contact as much as possible. Wash contacted skin with soap and water. If solvent contacts eyes, flush them with clean water and get immediate medical help.

2. Clean bond area of old adhesive with cleaning cloth, Item 86, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
3. Roughen bonding surface of replacement seal and gaskets with abrasive cloth, Item 78, WP 1803 00. Wipe clean with cleaning cloth, Item 86, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
4. Apply a smooth even coat of adhesive, Item 13, WP 1803 00, to mating surfaces. Allow adhesive to air-dry for about 10 minutes or until it becomes tacky.
5. Press plenum seal and clamp gaskets firmly in place. Make sure all surfaces are in good contact. Allow adhesive to cure for 24 hours at 68°F(20°C) (Figure 2, Sheet 1, Detail B).
6. **QA** Install clamp with screws and washers.
7. Install muffler on plenum (WP 1277 00).

REPLACE HEATING AND VENTILATION SYSTEM HEATER DUCT SEAL AND SCREEN

Remove

1. Remove rivets and washers from screen and seal (TM 1-1500-204-23).
2. Remove screen and seal from heater duct (Figure 2, Sheet 1, Detail C).
3. Clean bonding surfaces of screen and heater duct using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.

Install

WARNING

- Injury to personnel will result if area is not ventilated from volatile and toxic fumes when using solvents. Make sure area is vented properly to prevent a fire and a health hazard.
- Injury to personnel will result if proper ventilation and protective clothing, including eye shield, are not used when using solvents. Avoid breathing vapors and skin contact as much as possible. Wash contacted skin with soap and water. If solvent contacts eyes, flush them with clean water and get immediate medical help.

1. Roughen bonding surfaces of replacement seal with abrasive cloth, Item 78, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
2. Apply a smooth even coat of adhesive, Item 11, WP 1803 00, to mating surface of heater duct.
3. Install screen on heater duct (Figure 2, Sheet 1, Detail C).
4. Apply a smooth even coat of adhesive, Item 10, WP 1803 00, adhesive, Item 29, WP 1803 00 or adhesive, Item 40, WP 1803 00, to mating surfaces of seal.
5. **QA** Install seal on heater duct and screen assembly and attach with washers and rivets (TM 1-1500-204-23). Allow adhesive to cure for 24 hours at room temperature.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM EVAPORATOR BLOWER EH60A

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 30 - 200 in. lbs.

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0258 00
WP 1703 00

REMOVE

1. Turn off all electrical power.
2. Remove left soundproofing panel from rear cabin bulkhead (WP 0258 00).
3. Disconnect electrical connector, P9, from evaporator blower (Figure 1).
4. Remove bolts and washers holding evaporator blower to front support bracket.
5. Remove bolts and washers holding evaporator blower to rear support bracket and evaporator inlet transition duct.
6. Remove blower from pallet.

INSTALL

1. Turn off all electrical power.
2. **QA** Make certain gasket is in place between rear support bracket and evaporator inlet transition duct.
3. **QA** Install evaporator blower on rear support bracket and evaporator inlet transition duct with bolts and washers (Figure 1). **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.**
4. **QA** Install evaporator blower on front support bracket with bolts and washers. **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.**
5. Inspect and treat electrical connector (WP 1703 00).
6. **QA** Connect electrical connector, P9, to evaporator blower.
7. Install left soundproofing panel on rear cabin bulkhead (WP 0258 00).

INSTALL - Continued

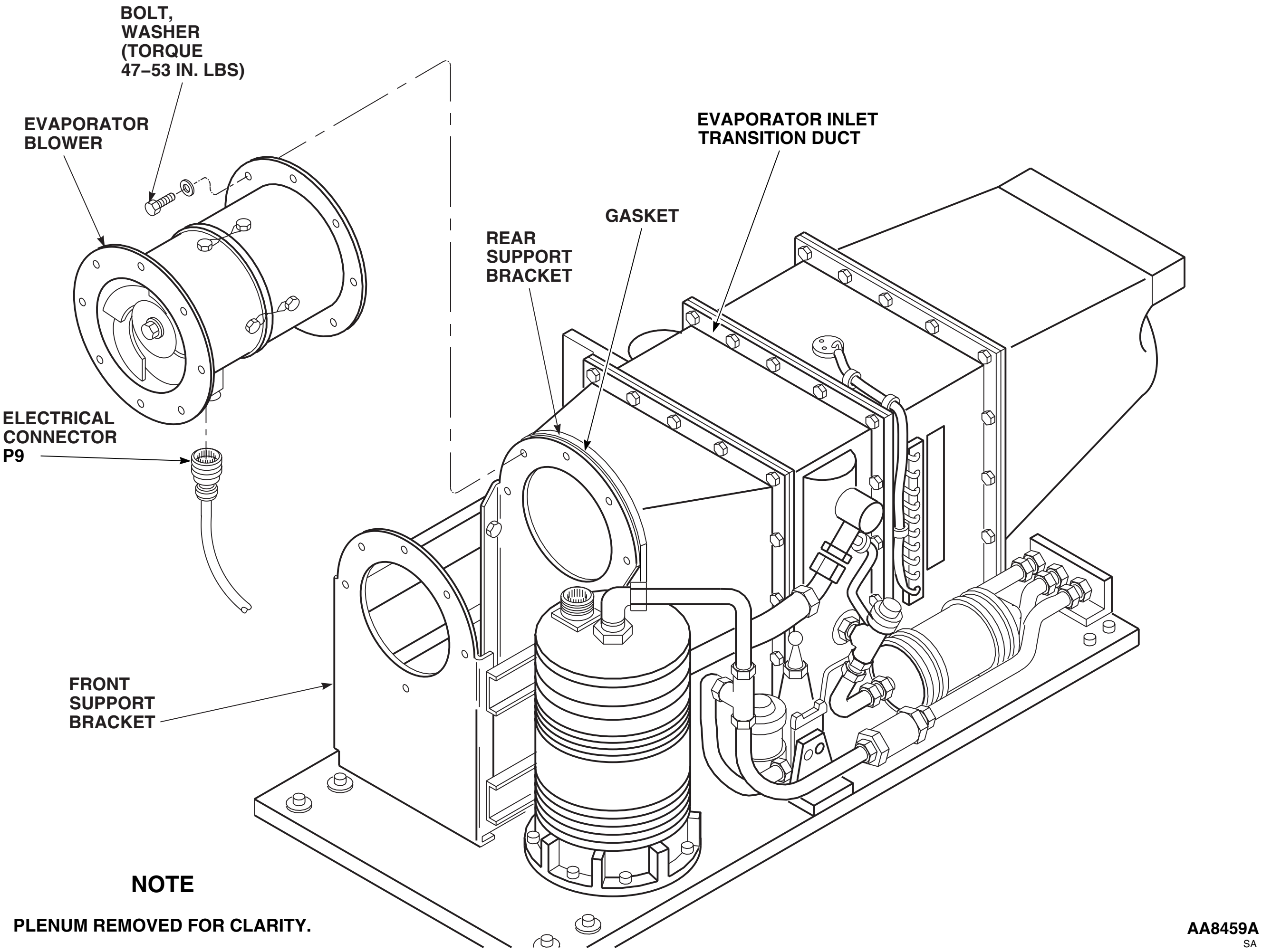


Figure 1. Replace Environmental Control System Evaporator Blower.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM EVAPORATOR ELECTRICAL HARNESS EH60A

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Electrical Repairer's Toolkit, SC 5180-99-B06
Torque Wrench, 0 - 30 in. lbs.

Material/Parts

Tags

Personnel Required

Aircraft Electrical Repairer MOS 15F (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0258 00
WP 1317 00
WP 1703 00

REMOVE

1. Turn off all electrical power.
2. Remove left soundproofing panel from rear cabin bulkhead (WP 0258 00).
3. Remove hot gas bypass valve (WP 1317 00).
4. Remove bolts, washers, and clamps holding harness to pallet and heater/demister (Figure 1).
5. Disconnect electrical connector, P9, from evaporator blower.
6. Disconnect electrical connector, P8, from compressor/motor.
7. Remove chaff guard and spiral wraps and separate electrical leads for the hot gas bypass valve, thermistor, expansion valve and the heater/demister temperature switches.
8. Disconnect and tag all electrical leads on heater/demister terminal board, TB-4.
9. Disconnect and tag electrical leads to heater elements.
10. Remove bolts, washers, clamps, and harness from pallet.

INSTALL

1. Turn off all electrical power.
2. **QA** Connect leads to heater elements. Remove tags.
3. **QA** Connect leads to heater/demister terminal board, TB-4. Remove tags.
4. Inspect and treat electrical connector (WP 1703 00).
5. **QA** Connect electrical connector, P9, to evaporator blower (Figure 1).
6. Inspect and treat electrical connector (WP 1703 00).
7. **QA** Connect electrical connector, P8, to compressor.

INSTALL - Continued

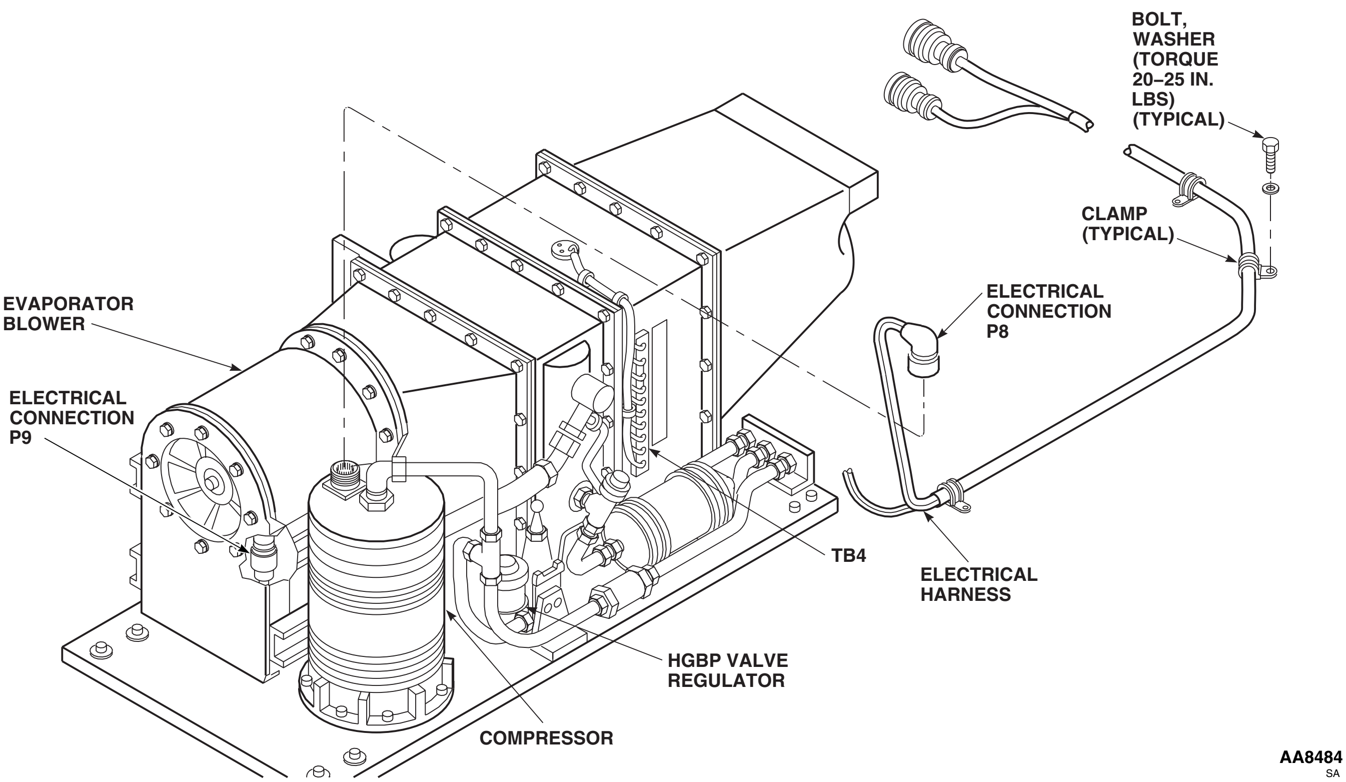


Figure 1. Replace Environmental Control System Evaporator Electrical Harness.

8. **QA** Add leads for the hot gas bypass valve, thermistor, expansion valve and the heater/demister temperature switches to harness. Install spiral wraps and chaff guards.
9. **QA** Install harness on pallet with bolts, washers, and clamps. **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**
10. Install hot gas bypass valve (WP 1317 00).
11. Install left soundproofing panel on rear cabin bulkhead (WP 0258 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM TEMPERATURE LIMITING SWITCH EH60A

INITIAL SETUP:

Tools and Special Tools
Electrical Repairer's Toolkit, SC 5180-99-B06
Torque Wrench, 0 - 30 in. lbs.

Material/Parts
Adhesive, Item 9, WP 1803 00
Tags

Personnel Required
Aircraft Electrical Repairer MOS 15F (1)

References
WP 0150 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Slit foam from around temperature limiting switch.
3. Remove screws, washers, and temperature limiting switch from heater/demister housing (Figure 1, Detail B).
4. Remove nuts, lockwashers, and switch wires from terminal 4 and terminal 7, of terminal board, TB-4. Tag wires.

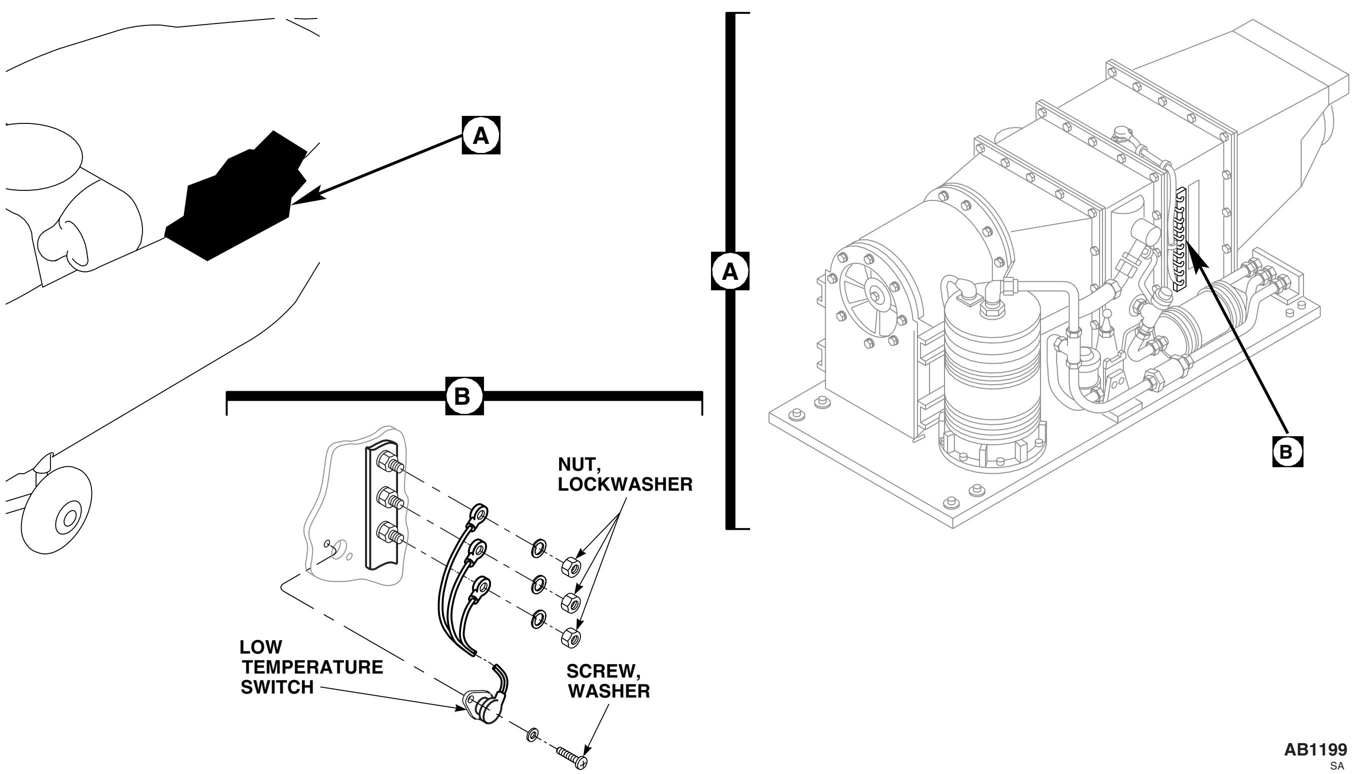


Figure 1. Replace Environmental Control System Temperature Limiting Switch.

INSTALL

1. Turn off all electrical power.
2. **QA** Connect switch wires to terminal 4 and terminal 7, of terminal board, TB-4 with nuts and lockwashers (Figure 1, Detail B). Remove tags.
3. **QA** Install switch to heater/demister housing with screws and washers. **TORQUE SCREWS TO 12 - 15 INCH-POUNDS.**
4. Repair foam insulation with adhesive, Item 9, WP 1803 00. Replace if required.
5. Do an operational check of ECS (WP 0150 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM LOW AND HIGH TEMPERATURE SWITCHES EH60A

INITIAL SETUP:

Tools and Special Tools
Electrical Repairer's Toolkit, SC 5180-99-B06
Torque Wrench, 0 - 30 in. lbs.

Material/Parts
Adhesive, Item 9, WP 1803 00
Tags

Personnel Required
Aircraft Electrical Repairer MOS 15F (1)

References
WP 0150 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Slit foam from around low temperature switch.
3. Remove screws, washers, and low temperature switch from heater/demister housing (Figure 1, Detail B).
4. Remove nuts, washers, and low temperature switch wires from terminal 8, terminal 9, and terminal 10, of terminal board, TB-4. Tag wires.
5. Remove screws, washers, clamps, nuts, and high temperature switch from heater/demister housing (Figure 1, Detail C).
6. Remove nuts, lockwashers, and high temperature switch wires from terminal 4, terminal 5, and terminal 6, of terminal board, TB-4. Tag wires.

INSTALL

1. Turn off all electrical power.
2. **QA** Connect low temperature switch wires to terminal 8, terminal 9, and terminal 10, of terminal board, TB-4 (Figure 1, Detail B). Remove tags.
3. **QA** Install low temperature switch to heater/demister housing with screws and washers. **TORQUE SCREWS TO 12 - 15 INCH-POUNDS.**
4. **QA** Connect high temperature switch wires to terminal 4, terminal 5, and terminal 6, of terminal board, TB-4 (Figure 1, Detail C). Remove tags.
5. **QA** Install high temperature switch to heater/demister housing with screws and washers. **TORQUE SCREWS TO 12 - 15 INCH-POUNDS.**
6. **QA** Clamp high temperature switch electrical harness to heater/demister flange brackets with screws, washers, and nuts.
7. Repair foam insulation with adhesive, Item 9, WP 1803 00. Replace if required.
8. Do an operational check of ECS (WP 0150 00).

INSTALL - Continued

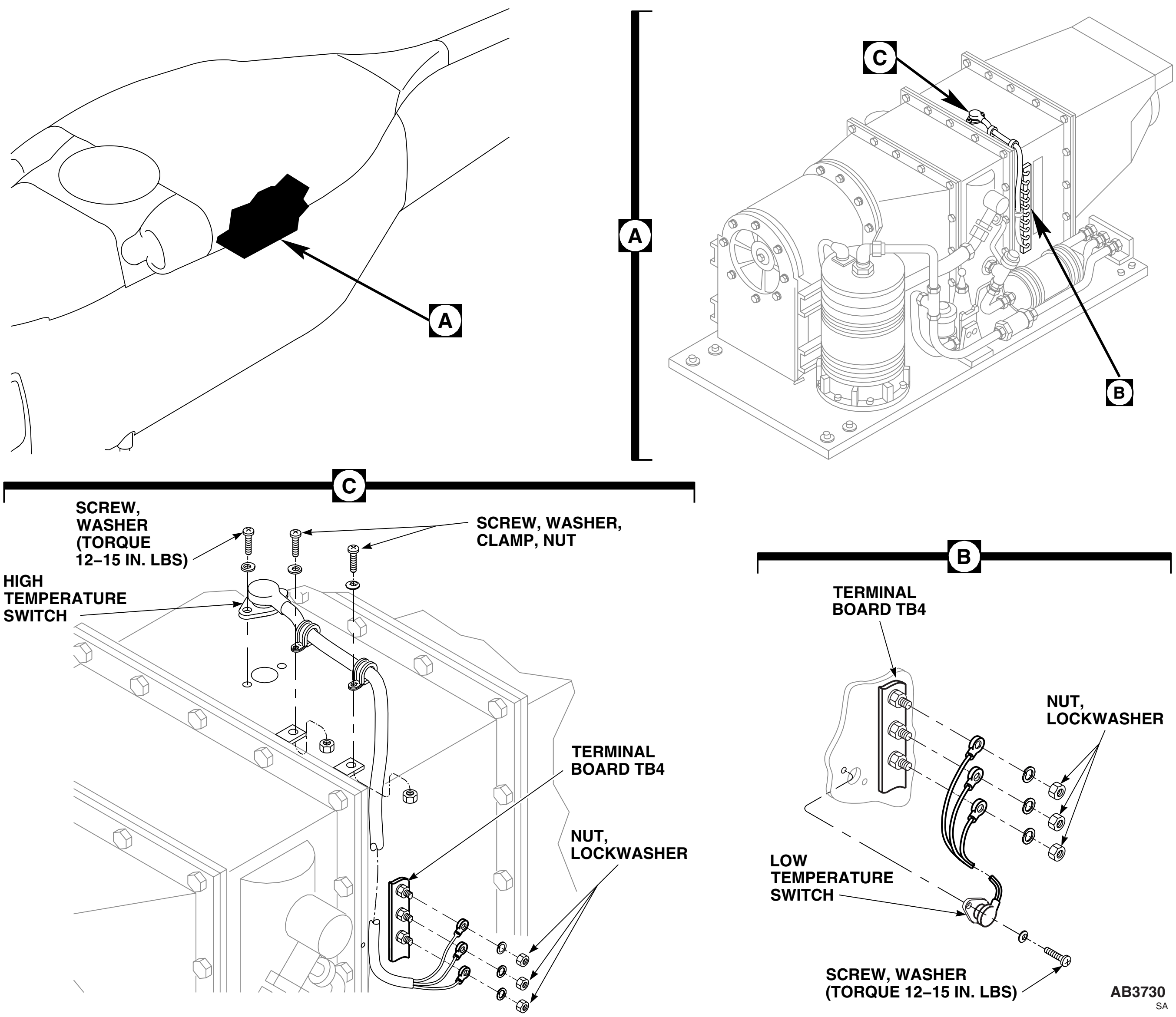


Figure 1. Replace Environmental Control System Low and High Temperature Switches.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM CONDENSER BLOWER EH60A

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Torque Wrench, 0 - 30 in. lbs.

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0258 00
WP 1289 00
WP 1292 00
WP 1703 00

REMOVE

1. Turn off all electrical power.
2. Remove right soundproofing panel from rear cabin bulkhead (WP 0258 00).
3. Remove right hand plenum (WP 1289 00).
4. Remove condenser exhaust duct (WP 1292 00).
5. Disconnect electrical connector, P10, from condenser blower (Figure 1, Detail A).
6. Remove clamp holding forward end of blower and duct sleeve to cradle bracket.
7. Remove bolts and washers holding condenser blower to condenser transition duct and support bracket. Remove blower from pallet.
8. Remove duct sleeve and install on new blower.

INSTALL

1. Turn off all electrical power.
2. **QA** Position blower on pallet (Figure 1, Detail A). Make certain that gasket is positioned between support bracket and transition duct. Install blower on transition duct and support bracket with bolts and washers. **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**
3. **QA** Install duct sleeve and forward end of blower on cradle bracket with clamp.
4. Inspect and treat electrical connector (WP 1703 00).
5. **QA** Connect electrical connector, P10, to condenser blower.
6. Install condenser exhaust duct (WP 1292 00).
7. Install right hand plenum (WP 1289 00).
8. Install right soundproofing panel from rear cabin bulkhead (WP 0258 00).

INSTALL - Continued

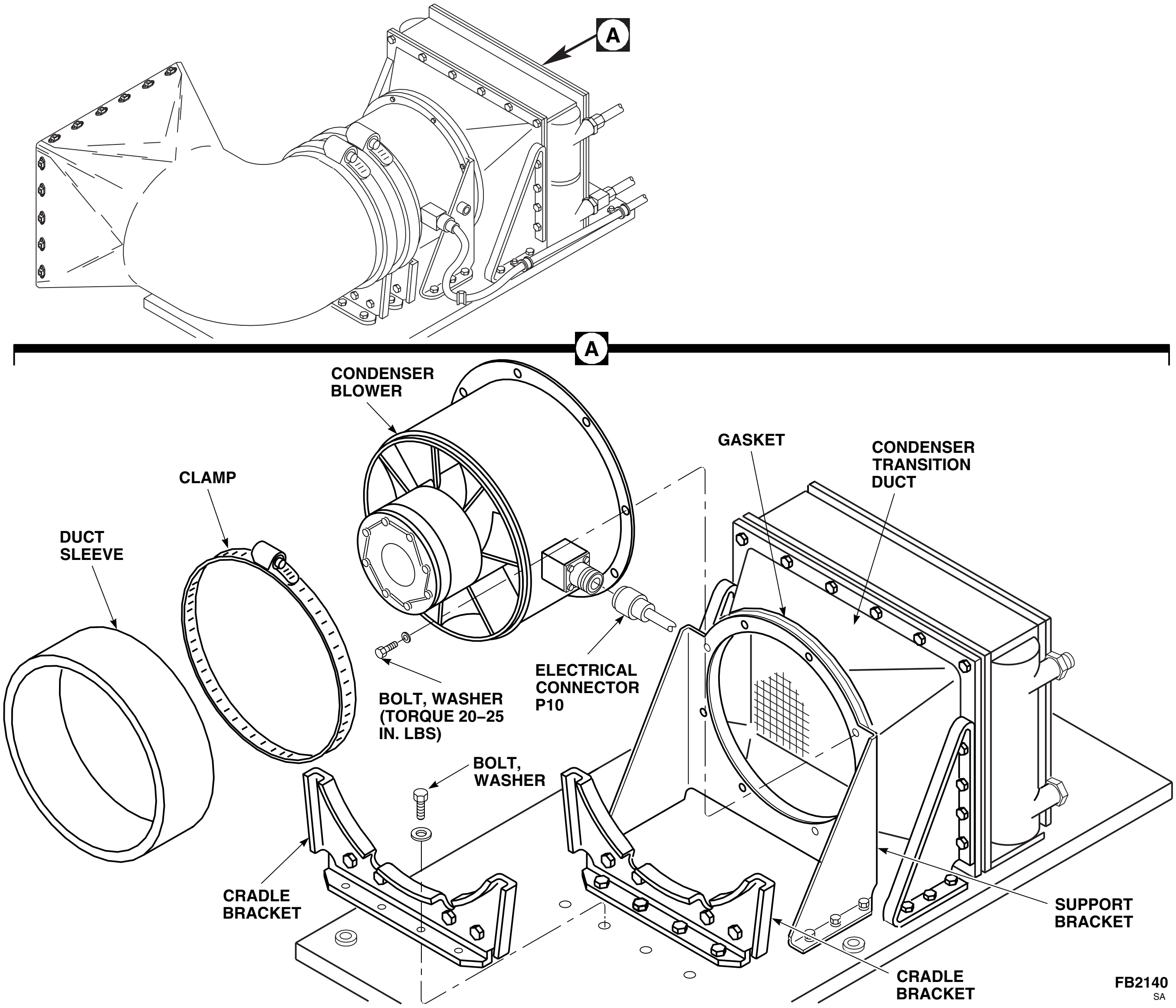


Figure 1. Replace Environmental Control System Condenser Blower.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM CONDENSER ELECTRICAL HARNESS EH60A

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 0 - 30 in. lbs.

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0258 00
WP 1292 00
WP 1703 00

REMOVE

1. Turn off all electrical power.
2. Remove right soundproofing panel from rear cabin bulkhead (WP 0258 00).
3. Remove condenser exhaust duct (WP 1292 00).
4. Disconnect electrical connector, P10, from condenser blower (Figure 1, Detail A).
5. Remove screws, washers, bolts, and clamps holding harness to pallet.
6. Disconnect electrical connector, P4, from electrical pallet.
7. Remove harness from pallet.

INSTALL

1. Inspect and treat electrical connector (WP 1703 00).
2. **QA** Connect electrical connector, P4, to electrical pallet (Figure 1, Detail A).
3. **QA** Install harness on pallet with screws, bolts, washers, and clamps. **TORQUE SCREWS AND BOLTS TO 20 - 25 INCH-POUNDS.**
4. Inspect and treat electrical connector (WP 1703 00).
5. **QA** Connect electrical connector, P10, to condenser blower.
6. Install condenser exhaust duct (WP 1292 00).
7. Install right soundproofing panel from rear cabin bulkhead (WP 0258 00).

INSTALL - Continued

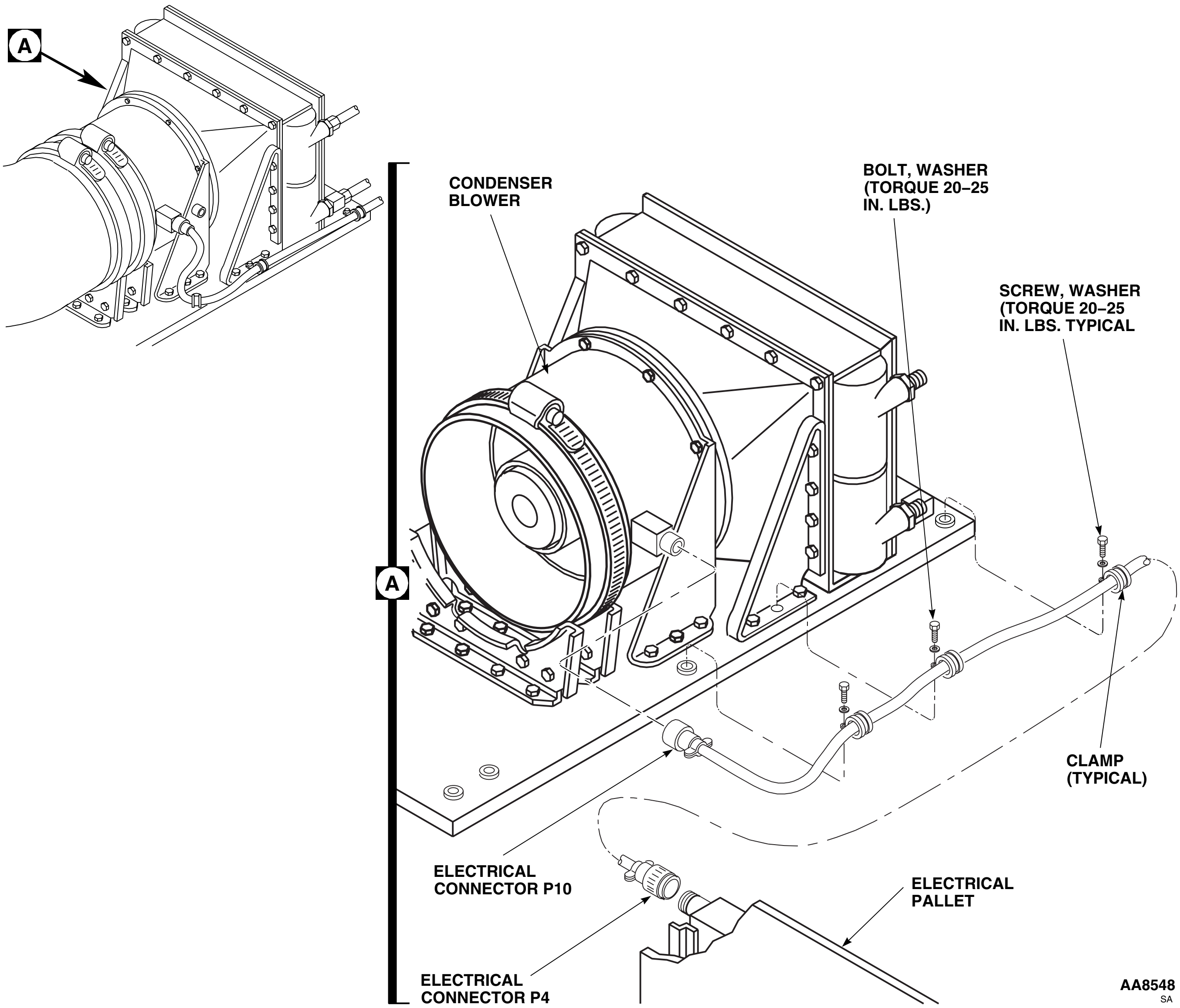


Figure 1. Replace Environmental Control System Condenser Electrical Harness.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM PLENUM EH-60A

This WP supersedes WP 1289 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Gloves
Goggles/Face Shield
Torque Wrench, 30 - 200 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 26, WP 1803 00
Cloth, Cleaning, Item 86, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
WP 0258 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Remove left soundproofing panel from rear cabin bulkhead (WP 0258 00).
3. Remove screw, washers, nut, and clamp holding duct temperature sensor to right side of plenum (Figure 1, Detail A).
4. Remove nut, washers, and temperature sensor from plenum.
5. Loosen clamp holding flex duct to ambient air valve and pull duct off valve.
6. Remove cotter pin and valve operating lever that protrudes through bulkhead from ambient air valve.
7. Remove screws, washers, nuts, and clamp from left side of plenum on airframe.
8. Remove screws and cover from plenum.
9. Remove bolts and washers holding plenum to evaporator blower.
10. Using a putty knife, carefully separate plenum from evaporator blower.
11. Remove plenum from helicopter and discard old seal.
12. Remove rivets and washers holding ambient air valve assembly to plenum (TM 1-1500-204-23).

INSTALL

1. Turn off all electrical power.
2. **QA** Install ambient air valve assembly on new plenum with rivet manufactured head near side (Figure 1, Detail A). Install washer under rivet bucked head.
3. To bond new seals, do this:

INSTALL - Continued

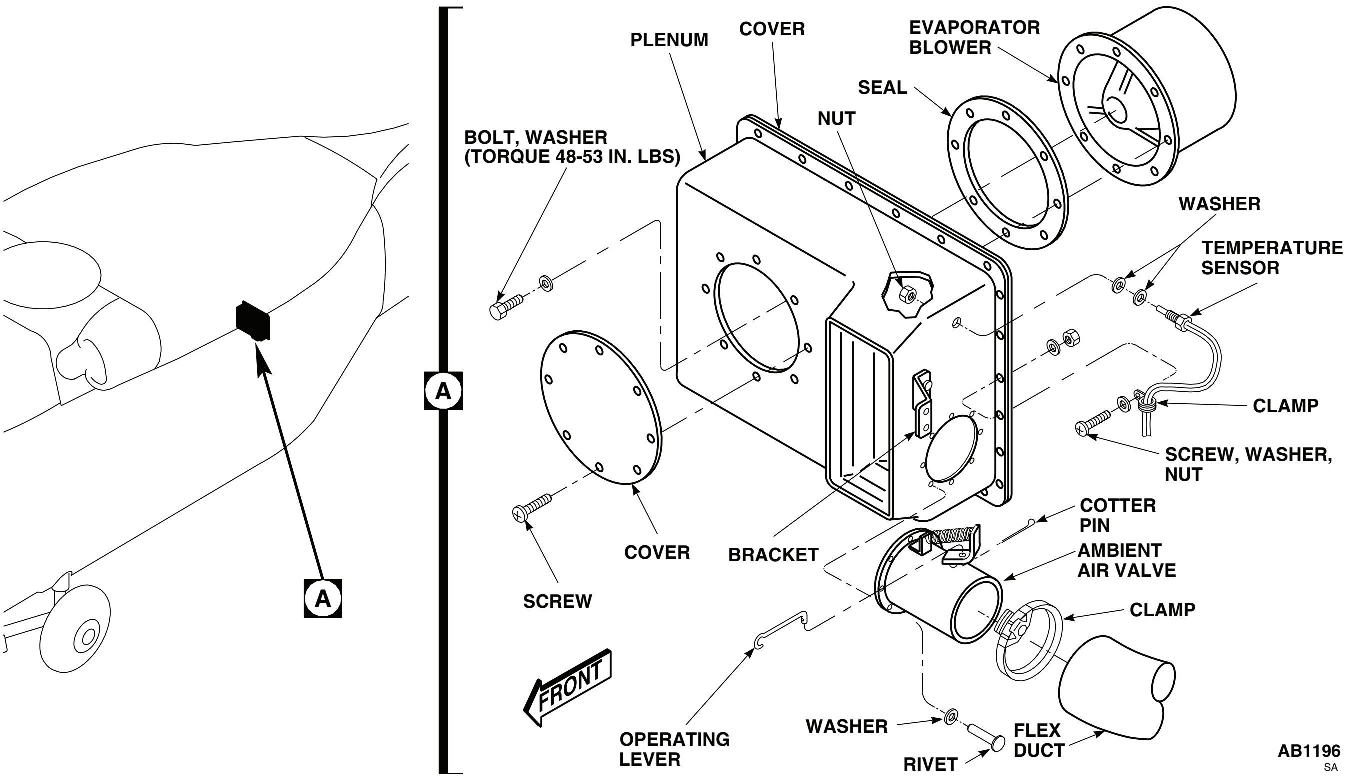


Figure 1. Replace Environmental Control System Plenum.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

a. Clean both sides of seals and sealing surfaces using cleaning cloth, Item 86, WP 1803 00 dampened with technical acetone, Item 5, WP 1803 00. Wipe dry with clean cloth.

b. Apply a light coat of adhesive, Item 26, WP 1803 00, to sealing surfaces and both sides of seals. Allow to air-dry for about 10 minutes or until adhesive is tacky. Install seal on evaporator blower.

4. **QA** Install plenum on evaporator blower with bolts and washers. **TORQUE BOLTS TO 48 - 53 INCH-POUNDS.**
5. **QA** Install plenum cover with screws.
6. **QA** Install clamp, on left side of plenum, on airframe with screws, washers, and nuts.
7. Install operating lever with cotter pin.
8. **QA** Install flex duct on ambient air inlet valve with clamp.
9. Install clamp holding temperature sensor to right side of plenum with screw, washers, and nut.
10. **QA** Install temperature sensor to plenum with nut and washers. **TIGHTEN NUT ON SENSOR 1/6 TO 1/3 TURN BEYOND POINT WHERE SHARP RISE IN TORQUE IS FELT.**

INSTALL - Continued

11. Install left soundproofing panel on rear cabin bulkhead (WP 0258 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM CABIN DUCTING EH60A

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0258 00

REMOVE

NOTE

Except as noted, removal procedure is the same for the right and left ducts.

1. Turn off all electrical power.
2. Remove overhead and rear cabin soundproofing panels as needed (WP 0258 00).
3. Remove screw and strap holding cabin duct to supply duct (Figure 1).

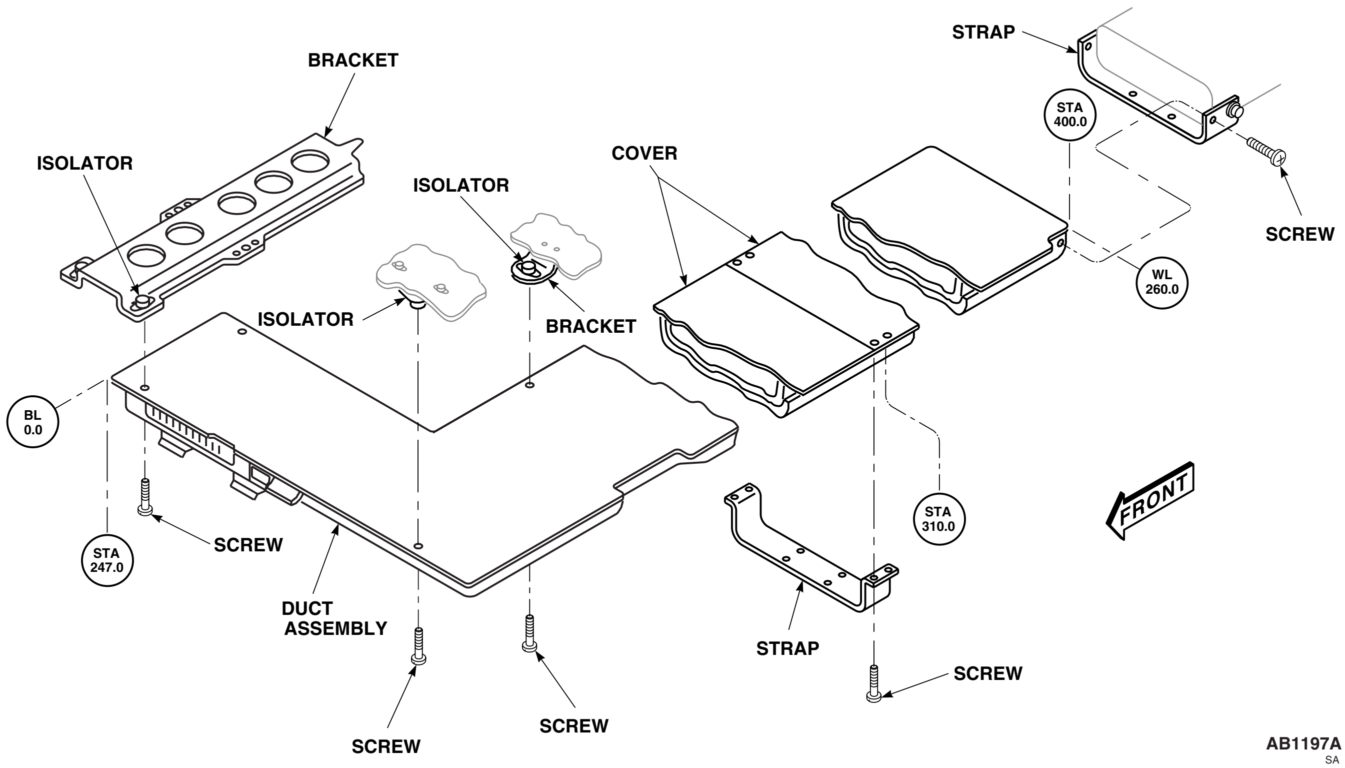


Figure 1. Replace Environmental Control System Cabin Ducting.

REMOVE - Continued

NOTE

Remove ducts from rear to front.

4. Support cabin duct assembly and remove screws holding duct to isolator mounts
5. Remove duct by carefully working cabin duct from supply duct connections.

INSTALL

NOTE

Except as noted, procedure is the same for installing the right and left ducts.

1. Turn off all electrical power.
2. **QA** Position duct in cabin overhead (Figure 1). Slip cabin duct over supply duct. Install strap with screw.
3. **QA** Install duct on isolator mounts with screws.
4. Install overhead and rear cabin soundproofing panels (WP 0258 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM SUPPLY DUCTING EH-60A

This WP supersedes WP 1291 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Gloves
Goggles/Face Shield

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 14, WP 1803 00
Adhesive, Item 40, WP 1803 00
Cloth, Cleaning, Item 86, WP 1803 00
Tape, Pressure Sensitive, Adhesive, Item 332, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0258 00
WP 1803 00

REMOVE

NOTE

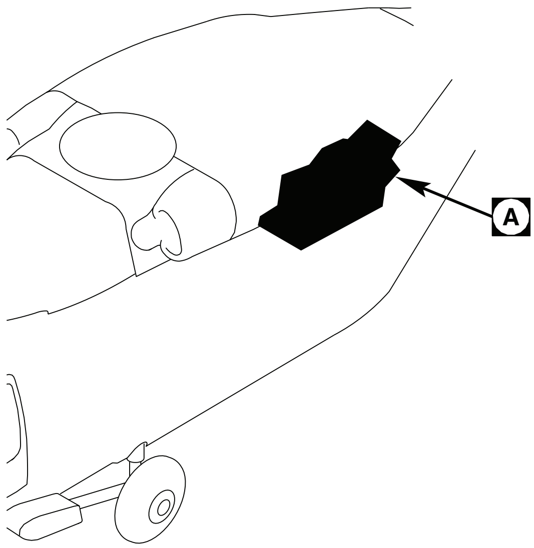
Except as noted, removal procedure is the same for right and left ducts.

1. Turn off all electrical power.
2. Open avionics compartment access door.
3. Remove right and left soundproofing panels from rear cabin bulkhead (WP 0258 00).
4. Remove strap holding cabin ducting to supply ducting (Figure 1, Detail A).
5. In transition section, remove two clamps holding right flex duct to airframe, lower evaporator outlet duct and right supply duct.
6. Remove flex duct from helicopter.
7. Remove screws holding strap to left supply duct to upper evaporator outlet duct.
8. Remove screws, washers, and nuts which hold support straps and hangers to airframe. Remove supply ducts from helicopter (Figure 1).

REPAIR/REPLACE INSULATION

1. To replace removed insulation, do this:

REPAIR/REPLACE INSULATION - Continued



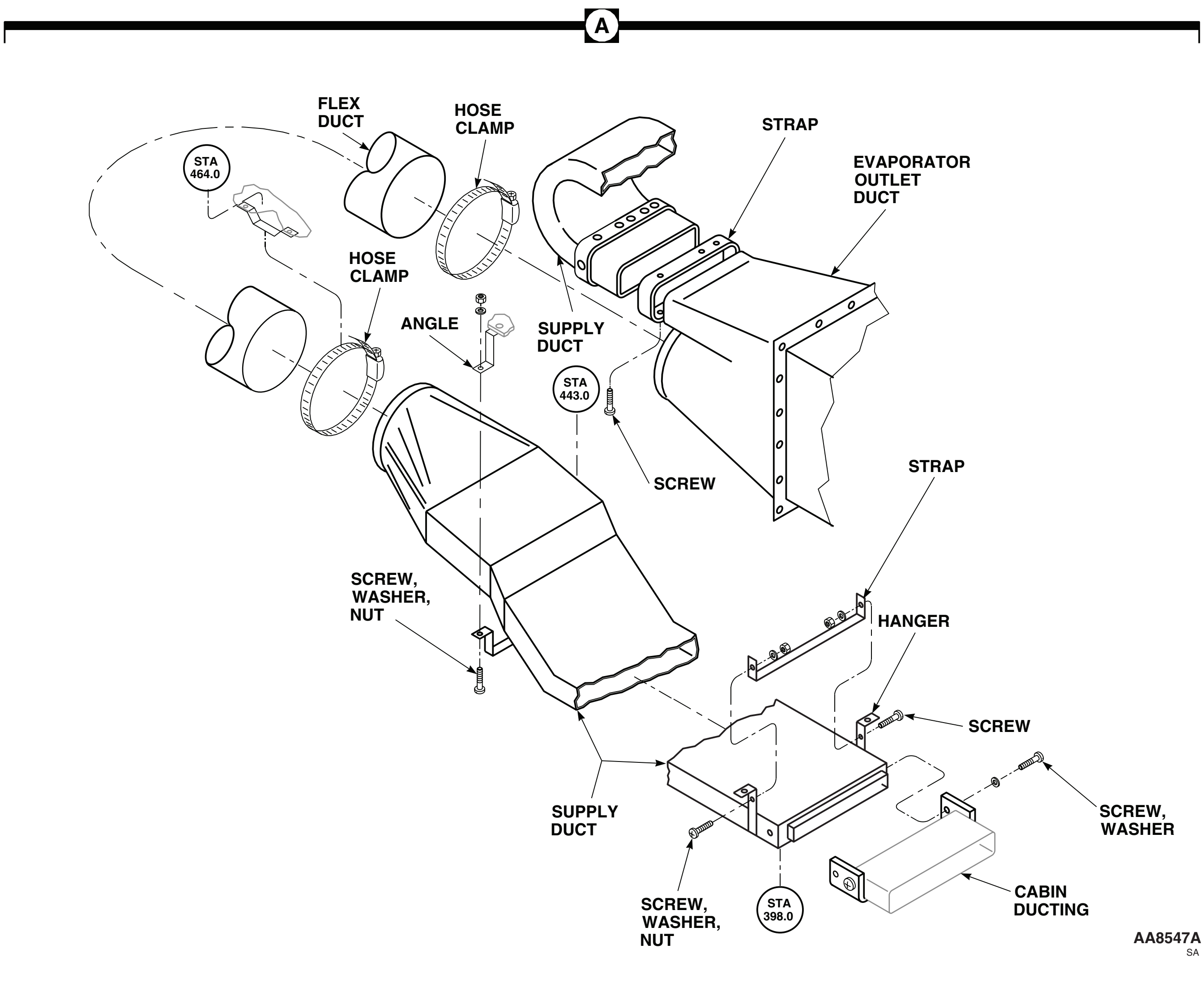


Figure 1. Replace Environmental Control System Supply Ducting.

REPAIR/REPLACE INSULATION - Continued

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

a. Clean surface of duct using cleaning cloth, Item 86, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00. Wipe dry with clean cloth.

b. Seal all edges of new insulation with adhesive, Item 40, WP 1803 00.

c. Apply a thin coat of adhesive, Item 14, WP 1803 00, to surfaces of duct and insulation.

d. Allow to air-dry for 10 minutes or until tacky.

e. Line up insulation and press into place.

f. Tape all joints with pressure sensitive adhesive tape, Item 332, WP 1803 00.

2. To repair damaged insulation, do this:

a. Wrap insulation and patch with reinforced plastic film.

b. Apply a thin coat of adhesive, Item 14, WP 1803 00, to surfaces of duct and insulation.

c. Allow to air-dry for 10 minutes or until tacky.

d. Press film against insulation.

e. Tape joints and splices with pressure sensitive adhesive tape, Item 332, WP 1803 00.

INSTALL

NOTE

Except as noted, procedure is same for installing left and right ducts.

1. Turn off all electrical power.
2. **QA** Install supply ducts on helicopter airframe with straps, hangers, screws, washers, and nuts (Figure 1, Detail A).
3. **QA** Install left supply duct on upper evaporator outlet duct with strap and screws.
4. **QA** Install flex duct on right supply duct, lower evaporator outlet duct, and airframe with clamps.
5. **QA** Connect supply ducting and cabin ducting. Install strap.
6. Install right and left soundproofing panels on rear cabin bulkhead (WP 0258 00).
7. Close and latch avionics compartment access door.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM CONDENSER EXHAUST DUCT EH60A

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 0 - 30 in. lbs.

Material/Parts
Sealing Compound, Item 273, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0258 00
WP 1289 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Remove right soundproofing panel from rear cabin bulkhead (WP 0258 00).
3. Remove right hand plenum (WP 1289 00).
4. Open avionics compartment access door.
5. Loosen clamps holding condenser intake flex duct and remove duct.
6. Disconnect drain tube from bottom of duct (Figure 1, Detail A).
7. Remove screws and washers holding outlet end of duct to outlet screen flange.
8. Loosen clamp holding exhaust duct to condenser duct sleeve.
9. Remove condenser pallet mounting bolts.
10. Slide condenser pallet to rear position.
11. Remove exhaust duct from helicopter.

INSTALL

1. Turn off all electrical power.
2. Apply a bead of sealing compound, Item 273, WP 1803 00, to outboard side of duct.
3. **QA** Position exhaust duct in helicopter and loosely install screws and washers in duct and outlet screen (Figure 1, Detail A).
4. **QA** Connect drain tube to bottom of duct.
5. **QA** Connect duct sleeve to condenser with clamp.
6. **QA** Tighten screws at outlet screen. **TORQUE SCREWS TO 20 - 25 INCH-POUNDS.**
7. **QA** Install condenser intake flex duct and tighten clamps.
8. Install right hand plenum (WP 1289 00).

INSTALL - Continued

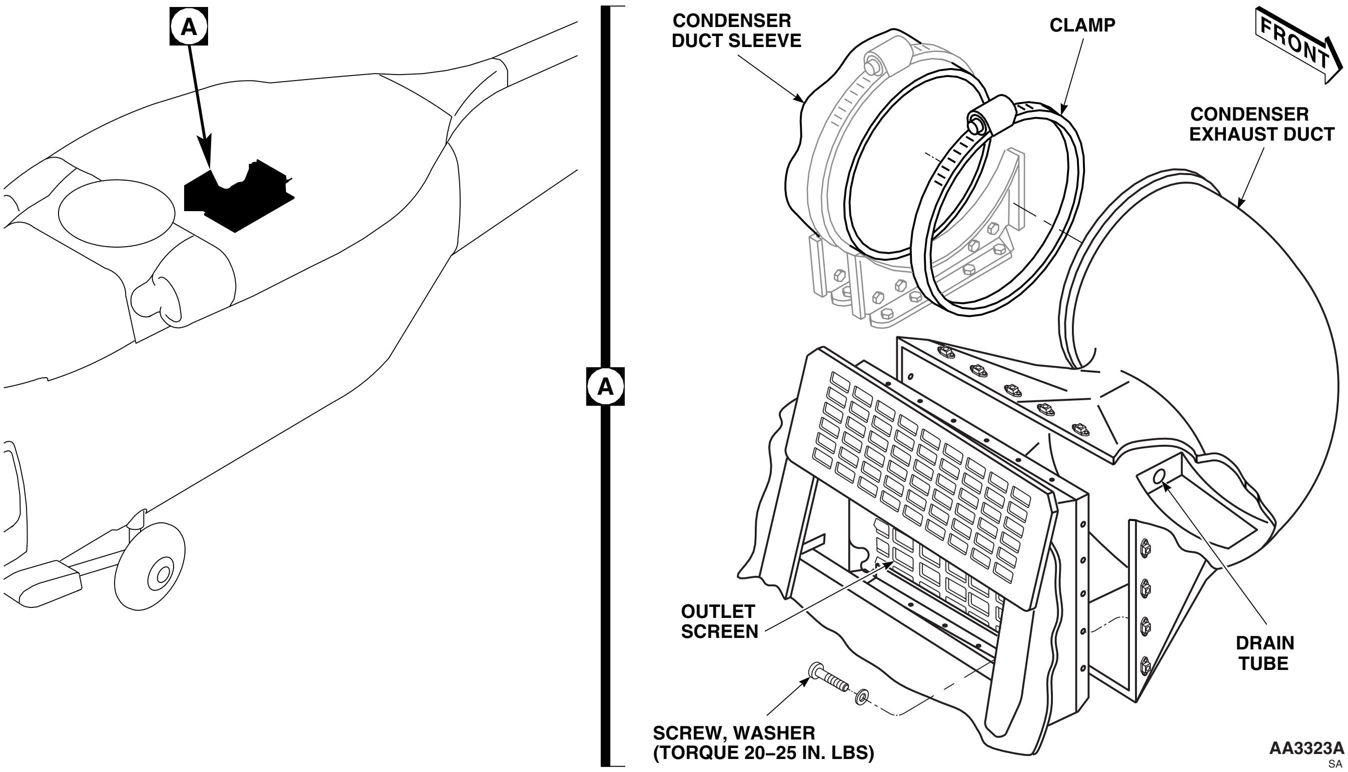


Figure 1. Replace Environmental Control System Condenser Exhaust Duct.

9. Install right soundproofing panel on rear cabin bulkhead (WP 0258 00).
10. Close and latch avionics compartment access door.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM TEMPERATURE SENSOR EH60A

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Material/Parts
Strap, Tiedown, Electrical, Item 302, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0150 00
WP 1284 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Remove screws and access cover from plenum (Figure 1, Detail A).
3. Remove screw, washer, nut, and clamp holding temperature sensor to right side of plenum.
4. Remove nut, washers, and temperature sensor from plenum.
5. Remove evaporator electrical harness (WP 1284 00).
6. Remove electrical tiedown straps from harness.
7. Remove temperature sensor wires from pin, N and pin P, of connector, P3.
8. Remove temperature sensor and its wires from electrical harness.

INSTALL

1. Turn off all electrical power.
2. **QA** Position sensor wires in harness wire bundle and install in pin, N and pin P, of connector, P3 (Figure 1, Detail A).
3. Install new electrical tiedown straps, Item 302, WP 1803 00.
4. Install evaporator electrical harness (WP 1284 00).
5. **QA** Install clamp holding temperature sensor to right side of plenum with screw, washers, and nut.
6. **QA** Install temperature sensor in plenum with washers and nut. **TIGHTEN NUT 1/6 TO 1/3 TURN BEYOND POINT WHERE SHARP RISE IN TORQUE IS FELT.**
7. **QA** Install cover on plenum with screws.
8. Do an operational check of ECS (WP 0150 00).

INSTALL - Continued

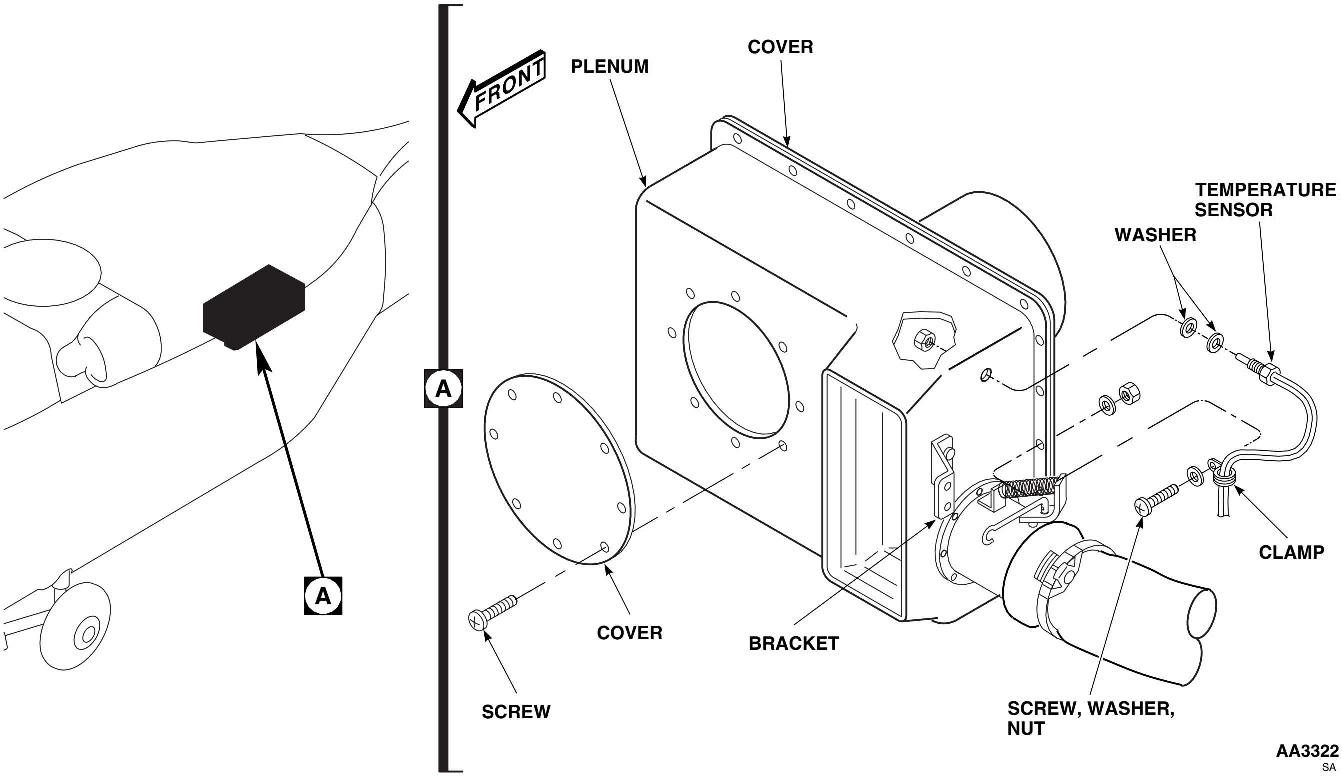


Figure 1. Replace Environmental Control System Temperature Sensor.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM BAFFLE EH60A

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

ADJUSTMENT

NOTE

The following ducting balance adjustment does not prohibit the use of the integral heater.

1. To adjust baffle assembly when air conditioning is to be used, do this:
   a. Loosen adjustment screw on right supply duct baffle assembly. Move screw so that center of screw is 1 inch from COLD end of adjustment slot (Figure 1, Detail A). Tighten screw.
   b. Loosen adjustment screw on left supply duct baffle assembly. Move screw until it touches COLD end of adjustment slot (Figure 1, Detail B). Tighten screw.

NOTE

- Make sure grip length of screws is not too long to prevent positive lock of baffles. Shim screws with washers if required.
- The following ducting balance adjustment does not prohibit the use of air conditioning.
- Adjustment procedures for right and left baffle assemblies is the same.

2. To adjust right and left baffle assemblies when heat is to be used, loosen adjustment screw on supply duct baffle assembly. Move screw so that center of screw is 2.50-inches from COLD end of adjustment slot (Figure 1, Detail C or Detail D). Tighten screw.

ADJUSTMENT - Continued

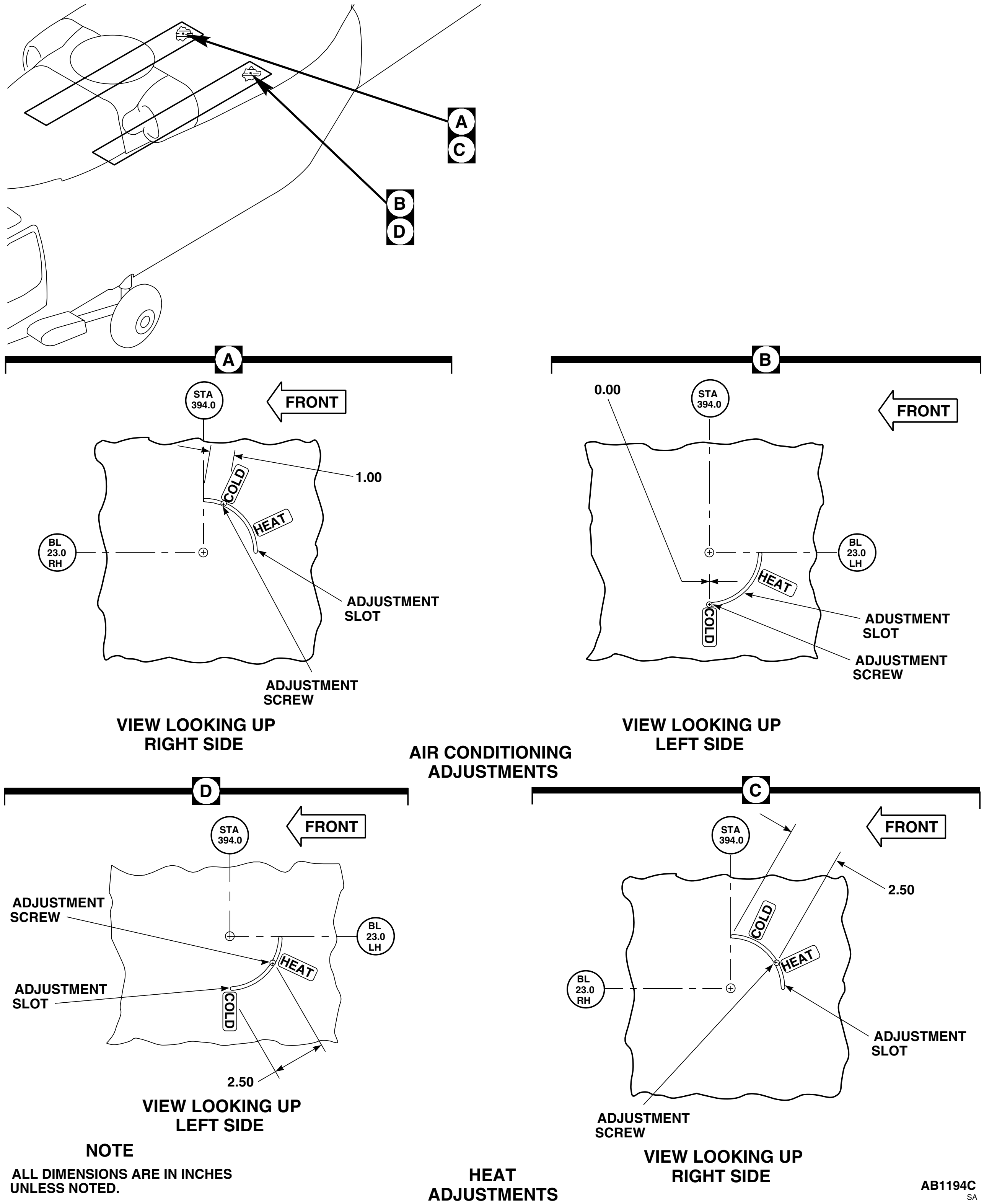


Figure 1. Adjust Environmental Control System Baffle.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM ELECTRICAL CONTROL UNIT (ECU) EH60A

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Electrical Repairer's Toolkit, SC 5180-99-B06

Personnel Required
Aircraft Electrical Repairer MOS 15F (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0150 00
WP 1703 00

REMOVE

1. Turn off all electrical power.
2. Disconnect all electrical connectors, P456 to J1, P457 to J2, P3 to J3, P4 to J4, P5 to J5, P6 to J6, and P7 to J7, from electrical control unit (Figure 1, Detail A).

CAUTION

Damage to ECS lines, fittings, or assemblies will occur if they are used as a hand hold. Do not use ECS lines, fittings, or assemblies as a hand hold.

3. Support electrical control unit and remove bolts and washers.
4. Remove electrical control unit from pallet.

INSTALL

1. Turn off all electrical power.
2. **QA** Position electrical control unit on electrical pallet assembly and secure with bolts and washers (Figure 1, Detail A).
3. Inspect and treat electrical connectors (WP 1703 00).
4. **QA** Connect electrical connectors, P456 to J1, P457 to J2, P3 to J3, P4 to J4, P5 to J5, P6 to J6, and P7 to J7, to electrical control unit.
5. Make sure area is clean and free of foreign material.
6. Do an operational check of ECS (WP 0150 00).

INSTALL - Continued

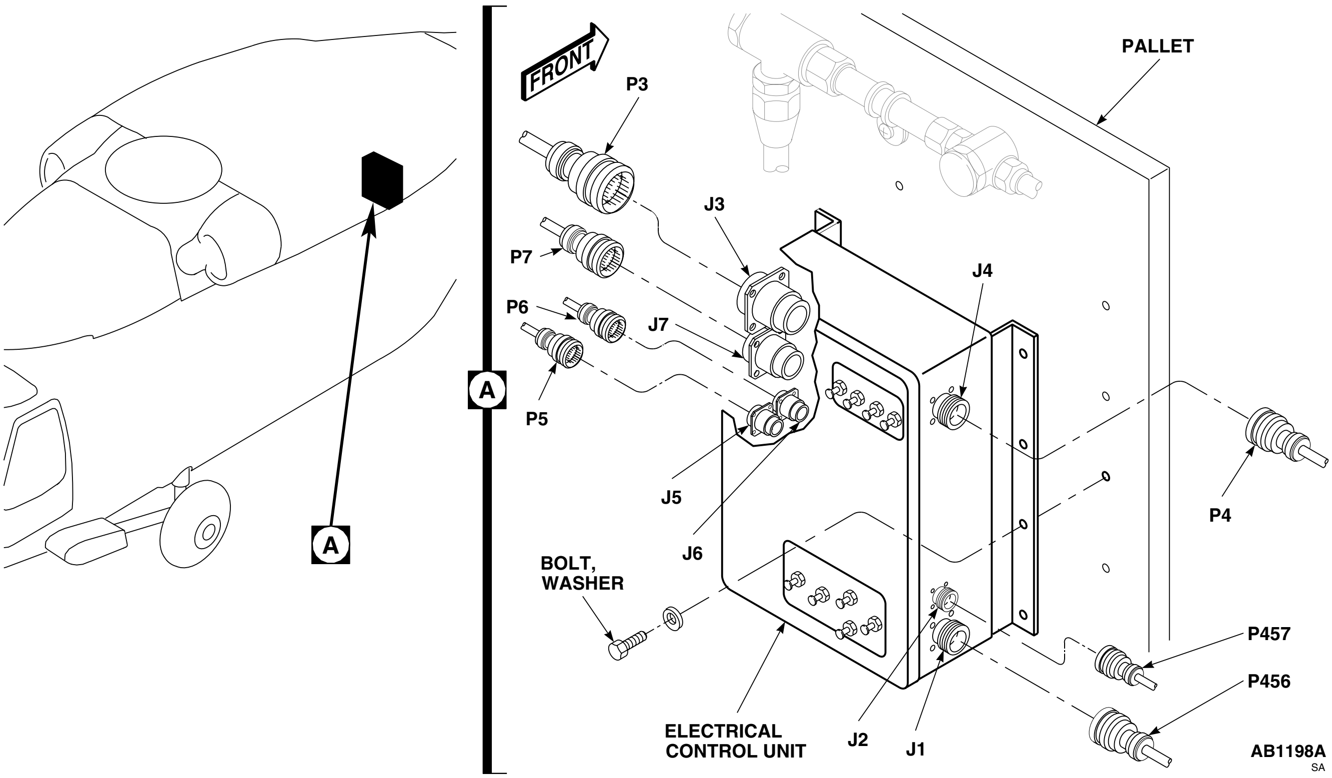


Figure 1. Replace Environmental Control System Electrical Control Unit (ECU).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM DUCTING REPAIR EH60A

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02

Material/Parts

Epoxy Primer Coating Kit, Item 120, WP 1803 00
Lacquer, Item 171, WP 1803 00
Sealing Compound, Item 273, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

References

TM 1-1500-204-23
WP 0377 00
WP 0378 00
WP 0379 00
WP 1803 00

REPAIR

1. The environmental control system ducting is composed of a diving cell foam and fiberglass duct section with 7075-T6 aluminum alloy cover which is welded and riveted together. Mounting provisions consist of bracket assemblies, made of 7075-T6 aluminum alloy, to which are installed rubber isolators which suspend the ducting from the overhead cabin structure. The ducting mounted along the upper cabin between STA 398.0 and STA 449.0 is suspended on aluminum alloy hangers. All ducting is secondary structure (Figure 1, Sheet 1, Detail A).
2. Evaluate damage to ducting (WP 0377 00).
3. Refer to WP 0378 00 for aluminum repairs.
4. Refer to WP 0379 00 for damage evaluation and repair of fiberglass parts.
5. Refer to TM 1-1500-204-23 for repair materials and fastener information.
6. When making riveted patch repairs, seal for water integrity corrosion protection with sealing compound, Item 273, WP 1803 00 (WP 0377 00).
7. Touch up repair areas with epoxy primer coating kit, Item 120, WP 1803 00, and paint with lacquer, Item 171, WP 1803 00.
8. Make sure area is clean and free of foreign material.
9. Close main rotor pylon sliding cover.

REPAIR - Continued

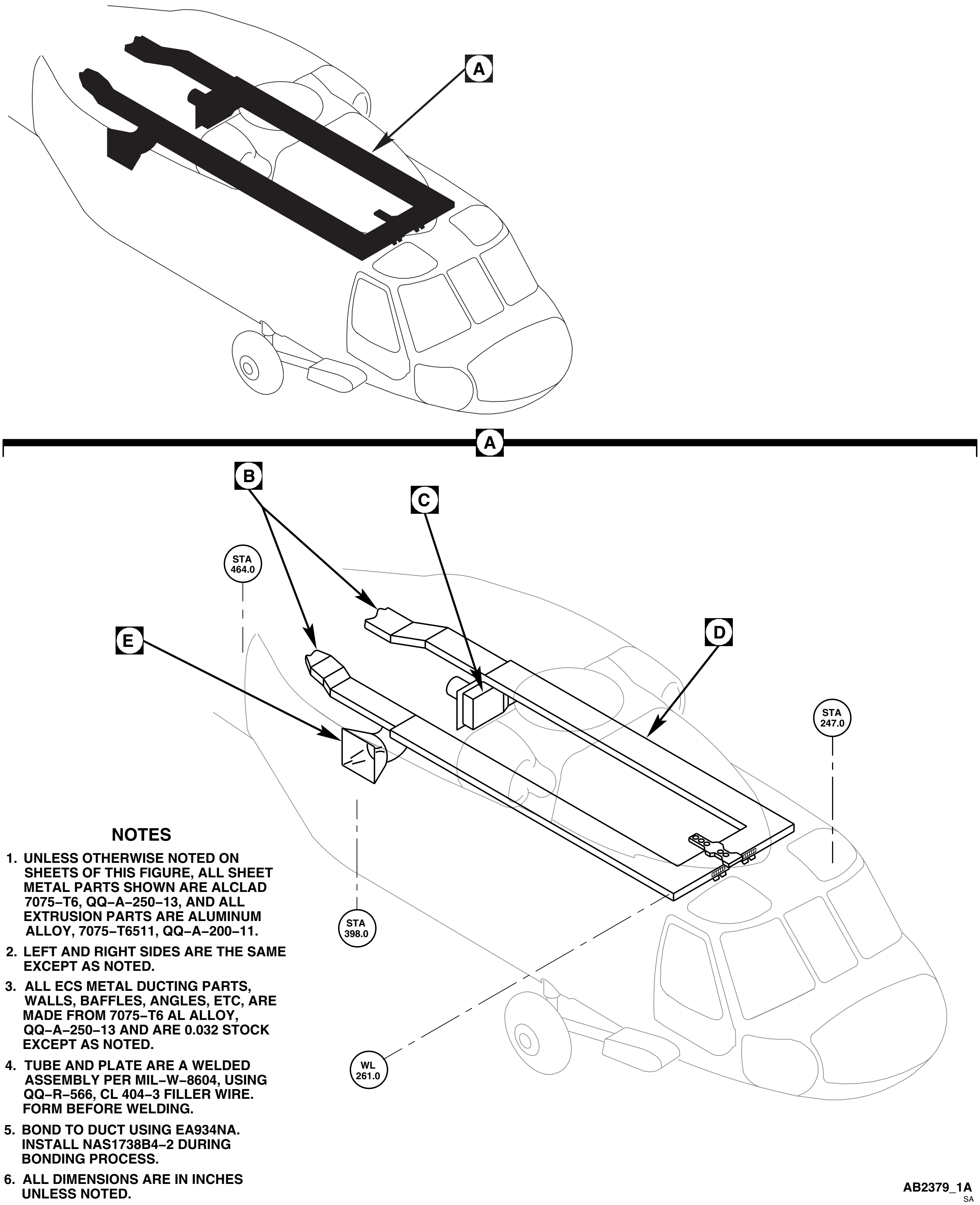
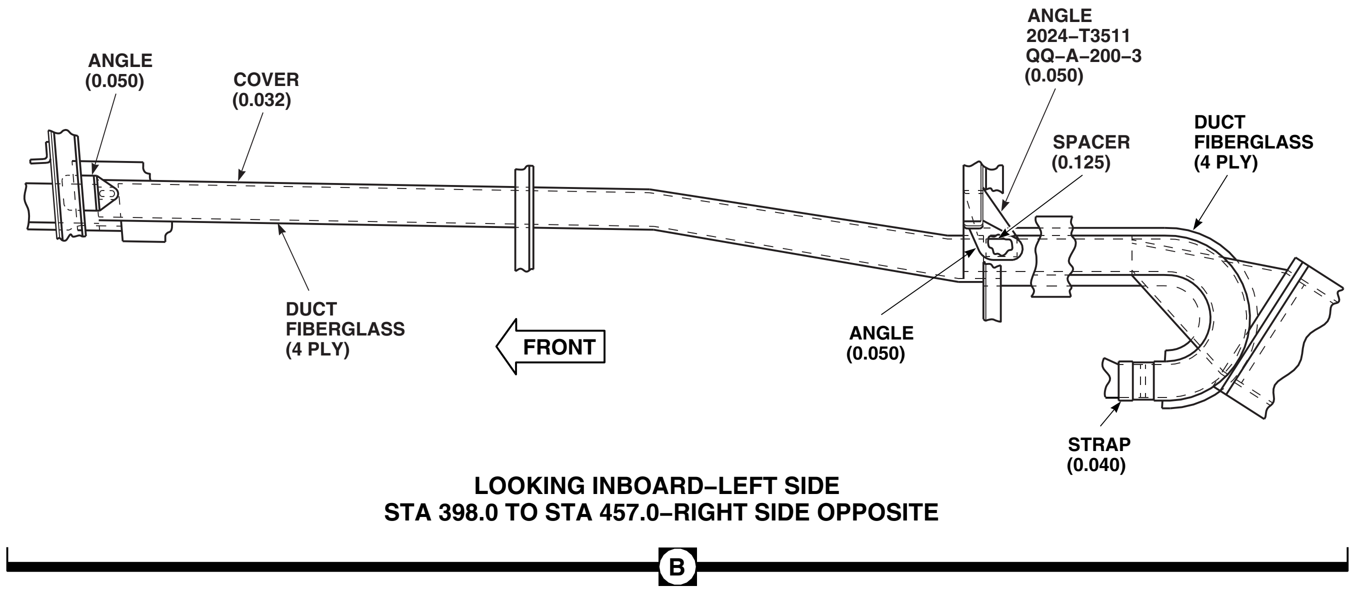


Figure 1. Environmental Control System Ducting Repair. (Sheet 1 of 3)

REPAIR - Continued



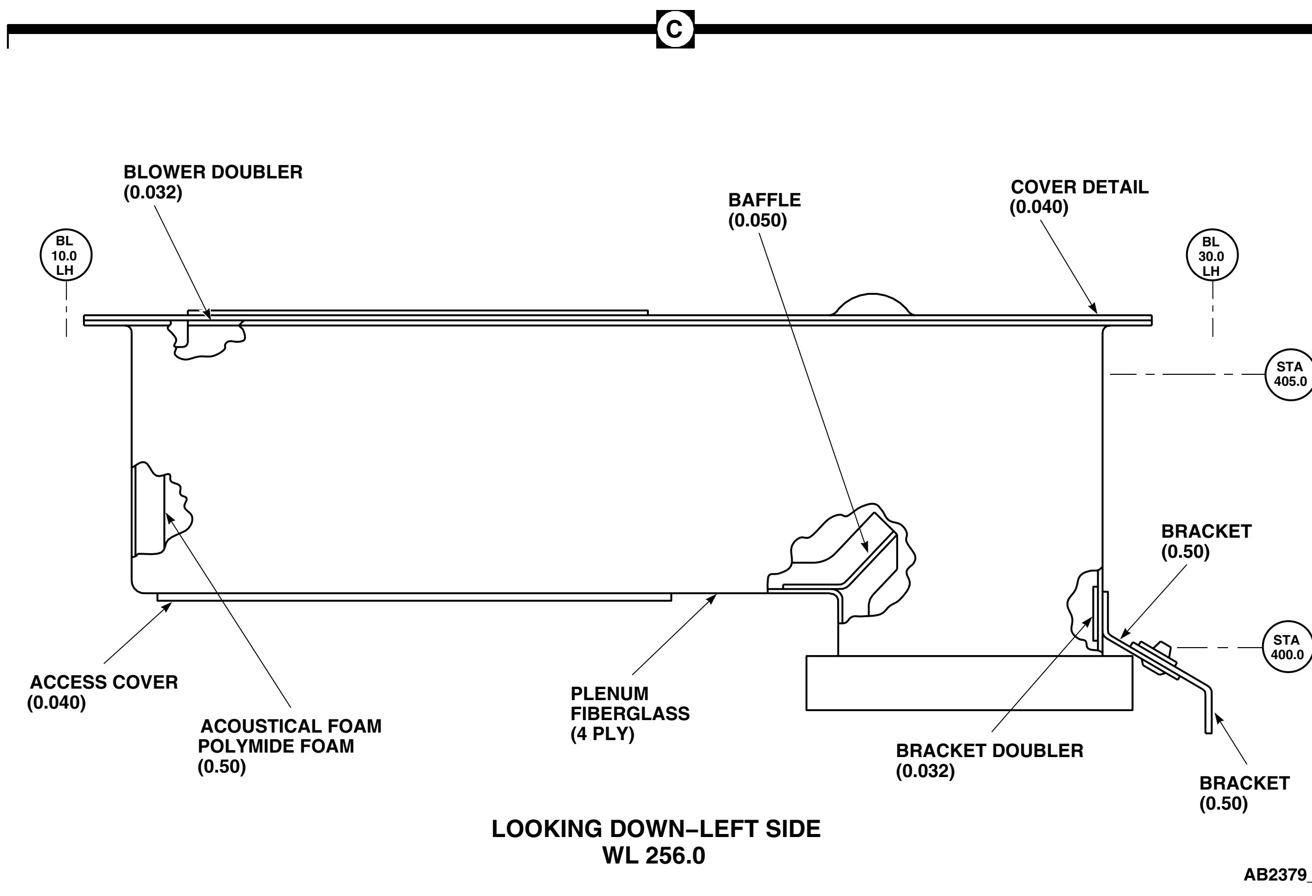


Figure 1. Environmental Control System Ducting Repair. (Sheet 2 of 3)

REPAIR - Continued

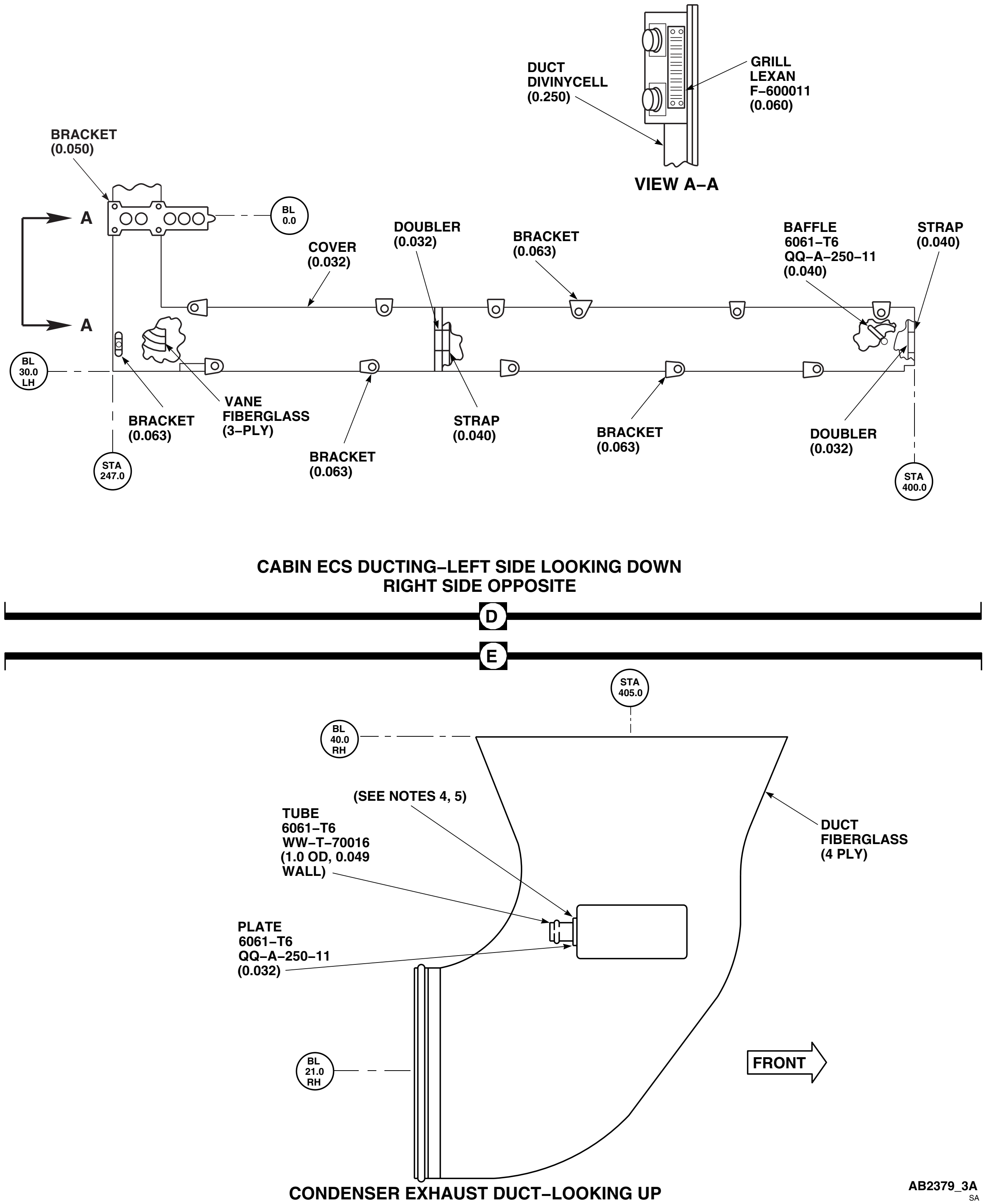


Figure 1. Environmental Control System Ducting Repair. (Sheet 3 of 3)

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM EVAPORATOR BLOWER HH-60A HH-60L

This WP supersedes WP 1297 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 30 - 200 in. lbs.

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 1304 00
WP 1703 00

REMOVE

1. Turn off all electrical power.
2. Remove evaporator pallet (WP 1304 00).
3. Disconnect electrical connector, P11, from evaporator blower (Figure 1, Detail A).
4. Remove bolts and washers holding evaporator blower to rear support bracket and inlet transition duct.
5. Remove evaporator blower from evaporator pallet.

INSTALL

1. Turn off all electrical power.
2. **QA** Make certain gasket is in place between rear support bracket and evaporator inlet transition duct.
3. **QA** Install evaporator blower on inlet transition duct with bolts and washers (Figure 1, Detail A). **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.**
4. Inspect and treat electrical connector (WP 1703 00).
5. **QA** Connect electrical connector, P11, to evaporator blower.
6. Install evaporator pallet (WP 1304 00).

INSTALL - Continued

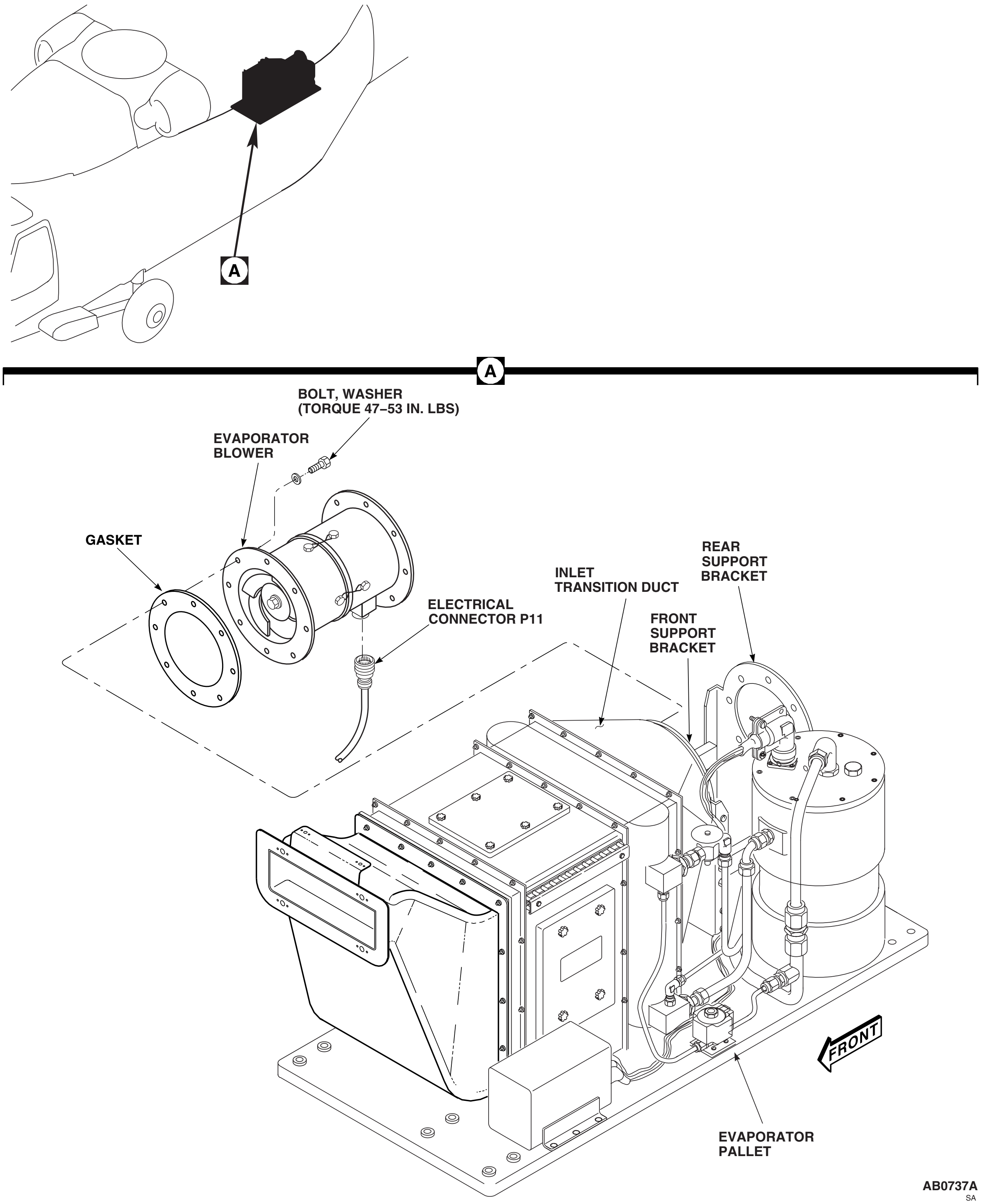


Figure 1. Replace Environmental Control System Evaporator Blower.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM CONDENSER BLOWER HH-60A HH-60L

This WP supersedes WP 1298 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 0 - 30 in. lbs.

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0258 00
WP 1299 00
WP 1703 00

REMOVE

1. Turn off all electrical power.
2. Slide right medical attendant seat forward and move seat back forward.
3. Remove right soundproofing panel from rear cabin bulkhead (WP 0258 00).
4. Remove condenser exhaust duct (WP 1299 00).
5. Disconnect electrical connector, P353, from condenser blower connector, J10 (Figure 1, Detail A).
6. Remove clamp holding forward end of blower and duct sleeve to cradle bracket.
7. Remove bolts and washers holding condenser blower to condenser transition duct and support bracket. Remove blower from pallet.
8. Remove duct sleeve and install on new blower.

INSTALL

1. Turn off all electrical power.
2. **QA** Position blower on pallet (Figure 1, Detail A). Make certain that gasket is positioned between support bracket and transition duct. Install blower on transition duct and support bracket with bolts and washers. **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**
3. **QA** Install duct sleeve and forward end of blower on cradle bracket with clamp.
4. Inspect and treat electrical connector (WP 1703 00).
5. **QA** Connect electrical connector, P353, to condenser blower connector, J10.
6. Install condenser exhaust duct (WP 1299 00).
7. Install right soundproofing panel on rear cabin bulkhead (WP 0258 00).
8. **QA** Slide right medical attendant seat aft and move seat back to upright position.

INSTALL - Continued

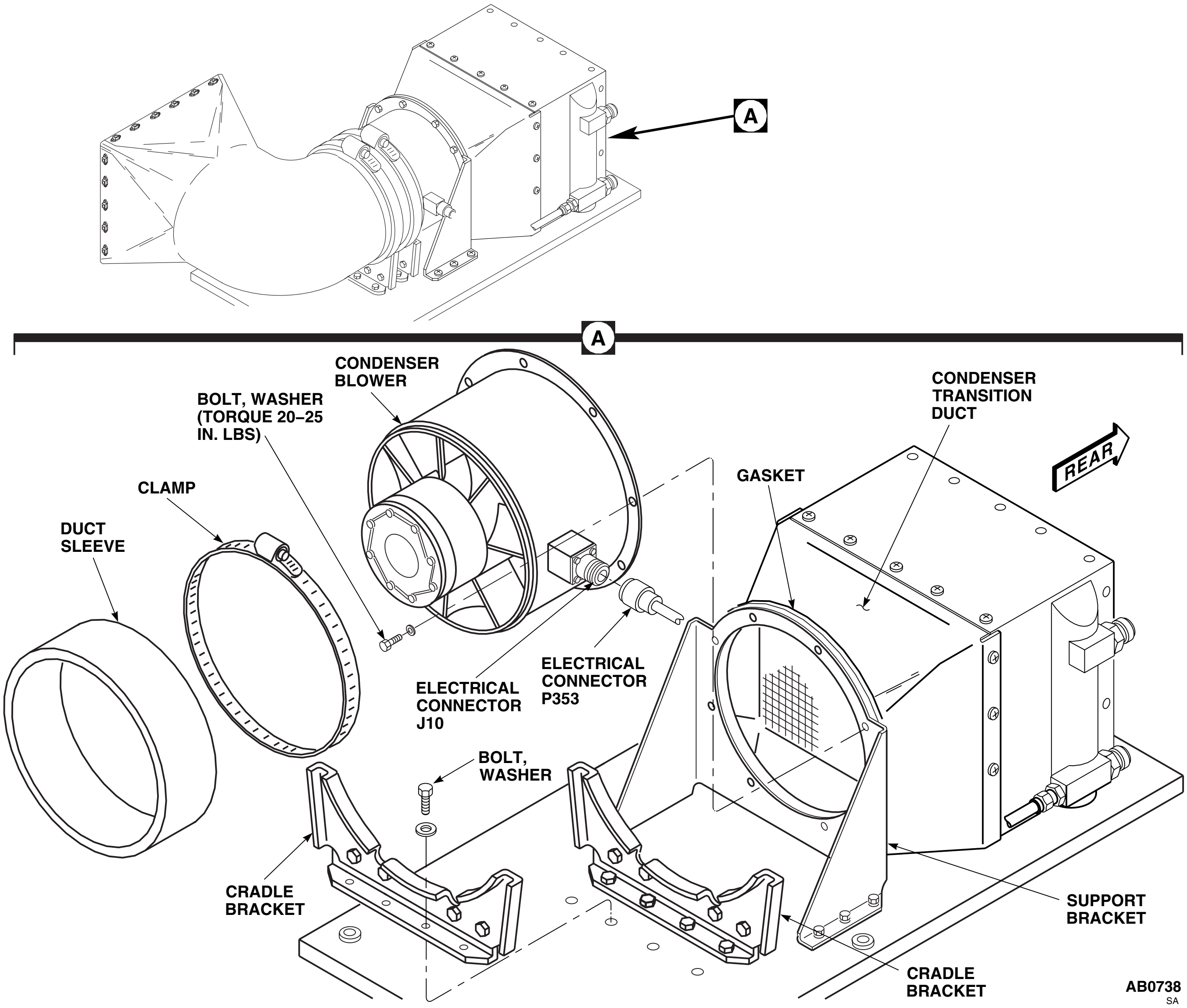


Figure 1. Replace Environmental Control System Condenser Blower.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM CONDENSER EXHAUST DUCT HH-60A HH-60L

This WP supersedes WP 1299 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.

Material/Parts

Sealing Compound, Item 273, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0258 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Open avionics compartment access door.
3. Remove right soundproofing panel from rear cabin bulkhead (WP 0258 00).
4. Loosen clamp holding evaporator inlet duct to bracket. Position evaporator inlet duct out of way.
5. Remove bolts and washers holding evaporator inlet duct bracket to airframe. Remove bracket.
6. Disconnect drain tube from bottom of duct (Figure 1, Detail A).
7. Remove bolts and washers holding outlet end of duct to outlet screen flange.
8. Loosen clamp holding exhaust duct to condenser duct sleeve.
9. Remove exhaust duct from helicopter by rotating the aft duct opening 90 degrees counterclockwise until the opening faces the cabin ceiling, and then pulling forward.

INSTALL

1. Turn off all electrical power.
2. Apply a bead of sealing compound, Item 273, WP 1803 00, to outboard side of condenser exhaust duct that mates with outlet screen.
3. **QA** Position exhaust duct in helicopter and loosely install bolts and washers in duct and outlet screen (Figure 1, Detail A).
4. **QA** Connect drain tube to bottom of duct.
5. **QA** Connect duct sleeve to condenser with clamp.
6. **QA** Tighten bolts at outlet screen. **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**
7. **QA** Install condenser intake flex duct and tighten clamps.
8. **QA** Position evaporator inlet duct bracket on airframe and secure with bolts and washers. **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.**

INSTALL - Continued

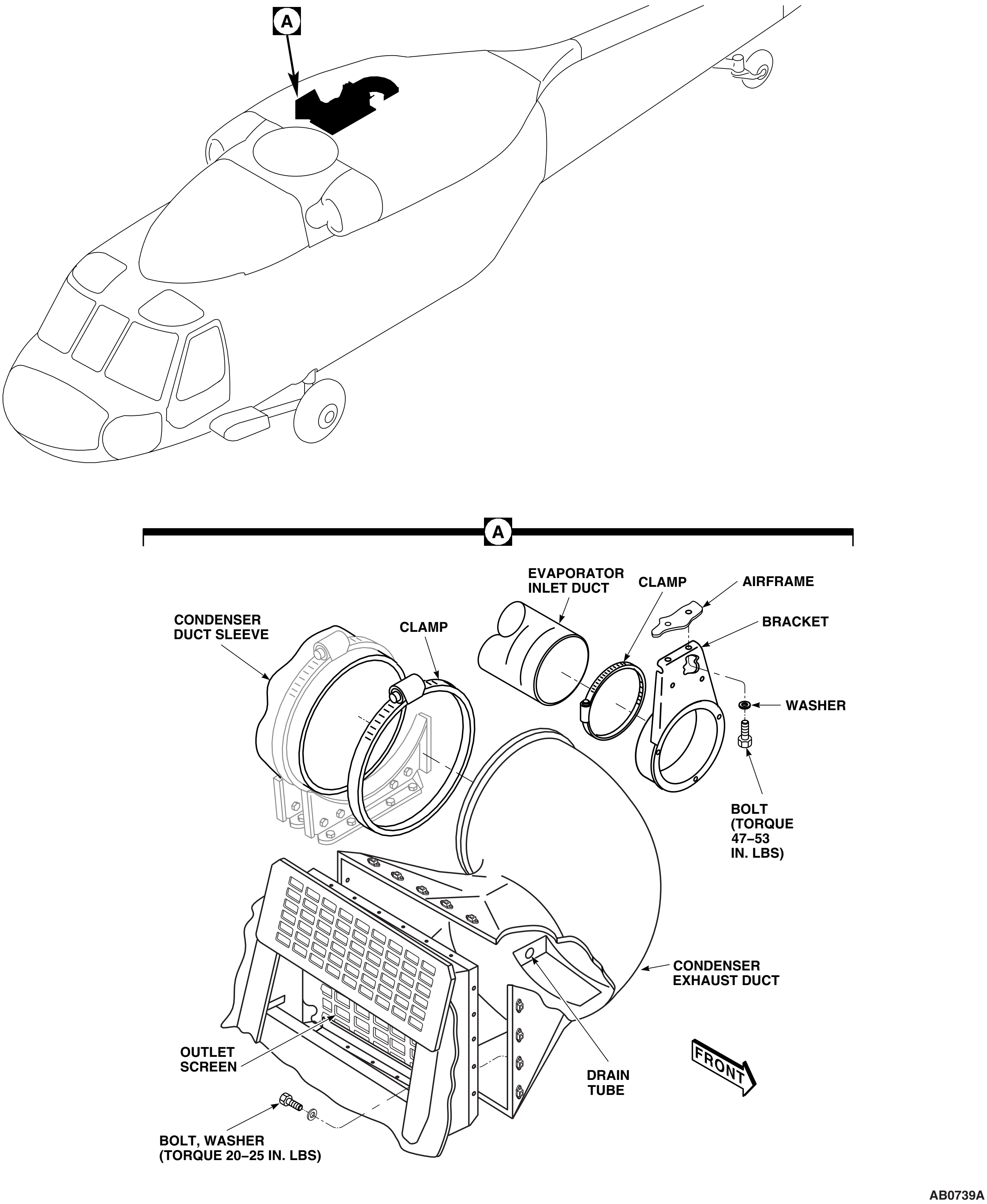


Figure 1. Replace Environmental Control System Condenser Exhaust Duct.

INSTALL - Continued

9. **QA** Position evaporator inlet duct on bracket and secure with clamp.
10. Install right soundproofing panel on rear cabin bulkhead (WP 0258 00).
11. Close and latch avionics compartment access door.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM ELECTRICAL CONTROL UNIT HH-60A HH-60L

This WP supersedes WP 1300 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Electrical Repairer's Toolkit, SC 5180-99-B06
Torque Wrench, 30 - 200 in. lbs.

Personnel Required
Aircraft Electrical Repairer MOS 15F (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0151 00
WP 1703 00

REMOVE

1. Turn off all electrical power.
2. Open avionics compartment access door.
3. Disconnect all electrical connectors, P351 to J1, P352 to J2, P3 to J3, P349 to J13, and P350 to J14 from electrical control unit (Figure 1, Detail A).

CAUTION

Damage to ECS lines, fittings, or assemblies will occur if they are used as a hand hold. Do not use ECS lines, fittings, or assemblies as a hand hold.

4. Support electrical control unit and remove bolts and washers.
5. Remove electrical control unit from pallet.

INSTALL

1. Turn off all electrical power.
2. **QA** Position electrical control unit on electrical pallet assembly and secure with bolts and washers (Figure 1, Detail A). **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.**
3. Inspect and treat electrical connectors (WP 1703 00).
4. **QA** Connect electrical connectors, P351 to J1, P352 to J2, P3 to J3, P349 to J13, and P350 to J14 on electrical control unit.
5. Make sure area is clean and free of foreign material.
6. Do an operational check of ECS (WP 0151 00).
7. Close and latch avionics compartment access door.

INSTALL - Continued

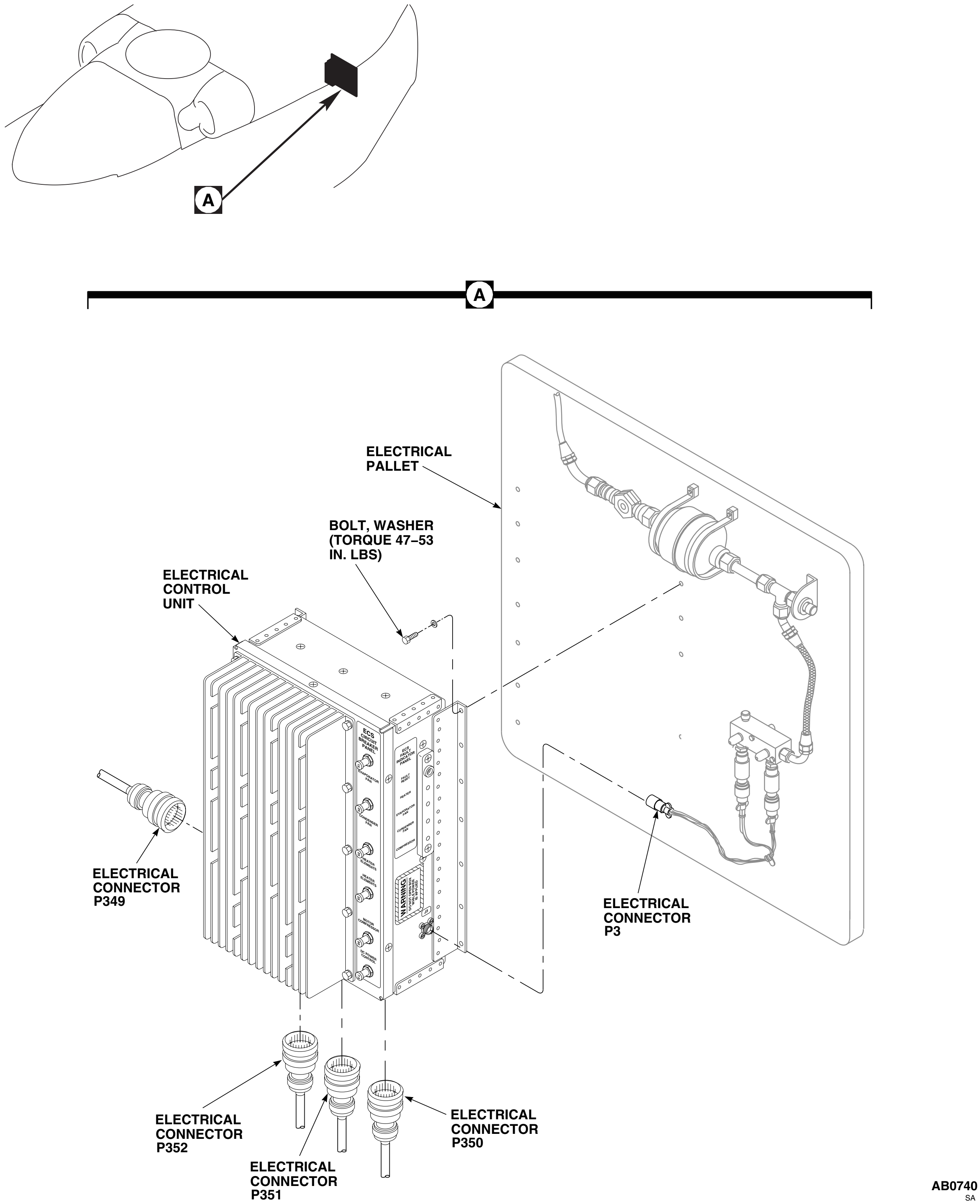


Figure 1. Replace Environmental Control System Electrical Control Unit.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM DUCTING REPAIR HH-60A HH-60L

This WP supersedes WP 1301 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Airframe Repairer's Toolkit, SC 5180-99-B02

Material/Parts
Epoxy Primer Coating Kit, Item 120, WP 1803 00
Lacquer, Item 171, WP 1803 00
Sealing Compound, Item 273, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)

References
WP 0377 00
WP 0379 00
WP 1803 00

REPAIR

1. The condenser exhaust duct is composed of fiberglass with a drain assembly bonded to the bottom.
2. Evaluate damage to ducting (WP 0377 00).
3. Refer to WP 0379 00 for damage evaluation and repair of fiberglass parts.
4. When making riveted patch repairs, seal for water integrity corrosion protection with sealing compound, Item 273, WP 1803 00 (WP 0377 00).
5. Touch up repair areas with epoxy primer coating kit, Item 120, WP 1803 00, and paint with lacquer, Item 171, WP 1803 00.
6. Make sure area is clean and free of foreign material.

REPAIR - Continued

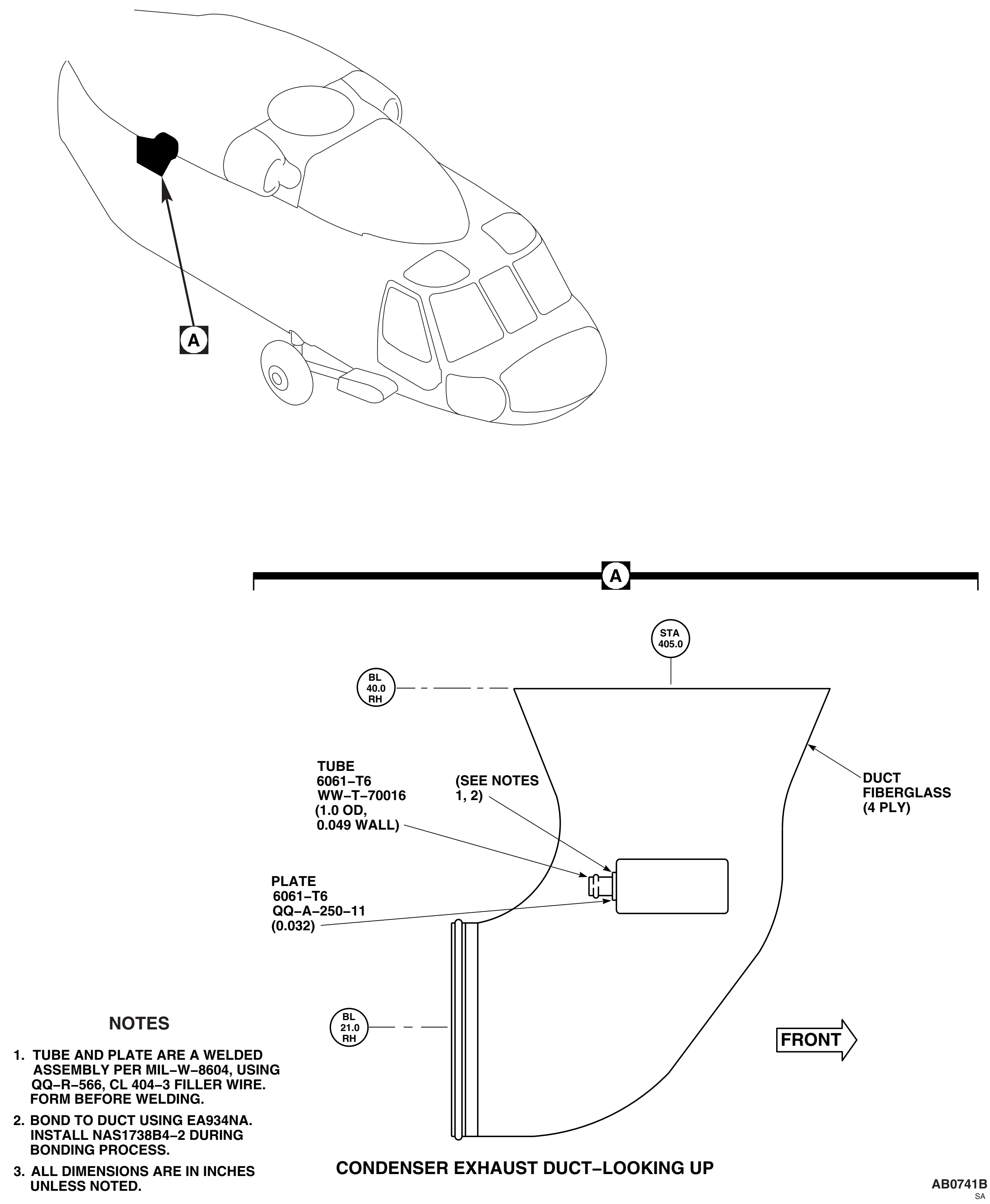


Figure 1. Environmental Control System Ducting Repair.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM FILTER/DRYER HH-60A HH-60L

This WP supersedes WP 1302 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Material/Parts

Antiseize Compound, Item 49, WP 1803 00
Lubricating Oil, Refrigerant, Item 191, WP 1803 00
Protective Caps and Plugs

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0151 00
WP 1626 00
WP 1803 00

REMOVE

1. Open avionics compartment access door.
2. Deservice environmental control system (ECS) (WP 1626 00).

CAUTION

- Damage to air-conditioning parts will occur if moisture is allowed to enter air-conditioning system. To prevent moisture from entering air-conditioning system, make sure to cap all lines immediately after disconnecting them.
- Damage to filter/dryer will occur if moisture is allowed to enter. Filter/dryer is extremely sensitive to moisture and humidity. Replace filter/dryer if inlet and/or outlet connections are left uncapped for more than 10 minutes.

3. Disconnect line from right side of filter/dryer. Cap and plug filter/dryer and tee fitting (Figure 1, Detail A).
4. Disconnect inlet hose from sight glass. Cap and plug lines and fittings
5. Remove clamp, spacer, screw, and washer holding sight glass.
6. Remove bolts, washers, clamps, and brackets from filter/dryer.
7. Disassemble filter/dryer from sight glass.

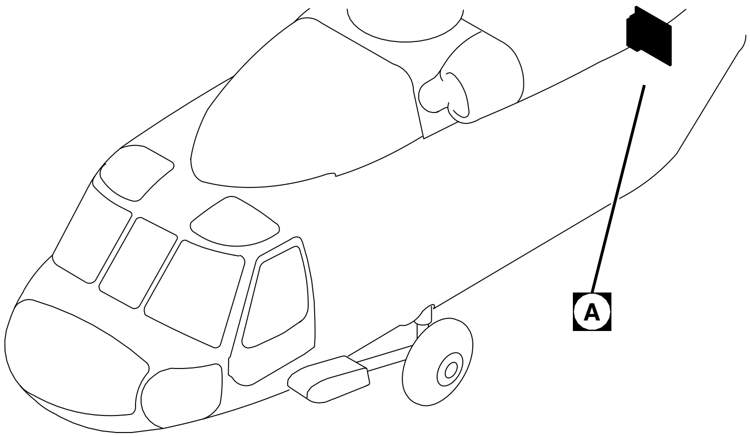
INSTALL

CAUTION

Damage to filter/dryer will occur if moisture is allowed to enter. Filter/dryer is extremely sensitive to moisture and humidity. Replace filter/dryer if inlet and/or outlet connections are left uncapped for more than 10 minutes.

1. Remove caps and plugs from lines and fittings immediately before assembly (Figure 1, Detail A).
2. **QA** Lubricate fittings on filter/dryer with antiseize compound, Item 49, WP 1803 00.

INSTALL - Continued



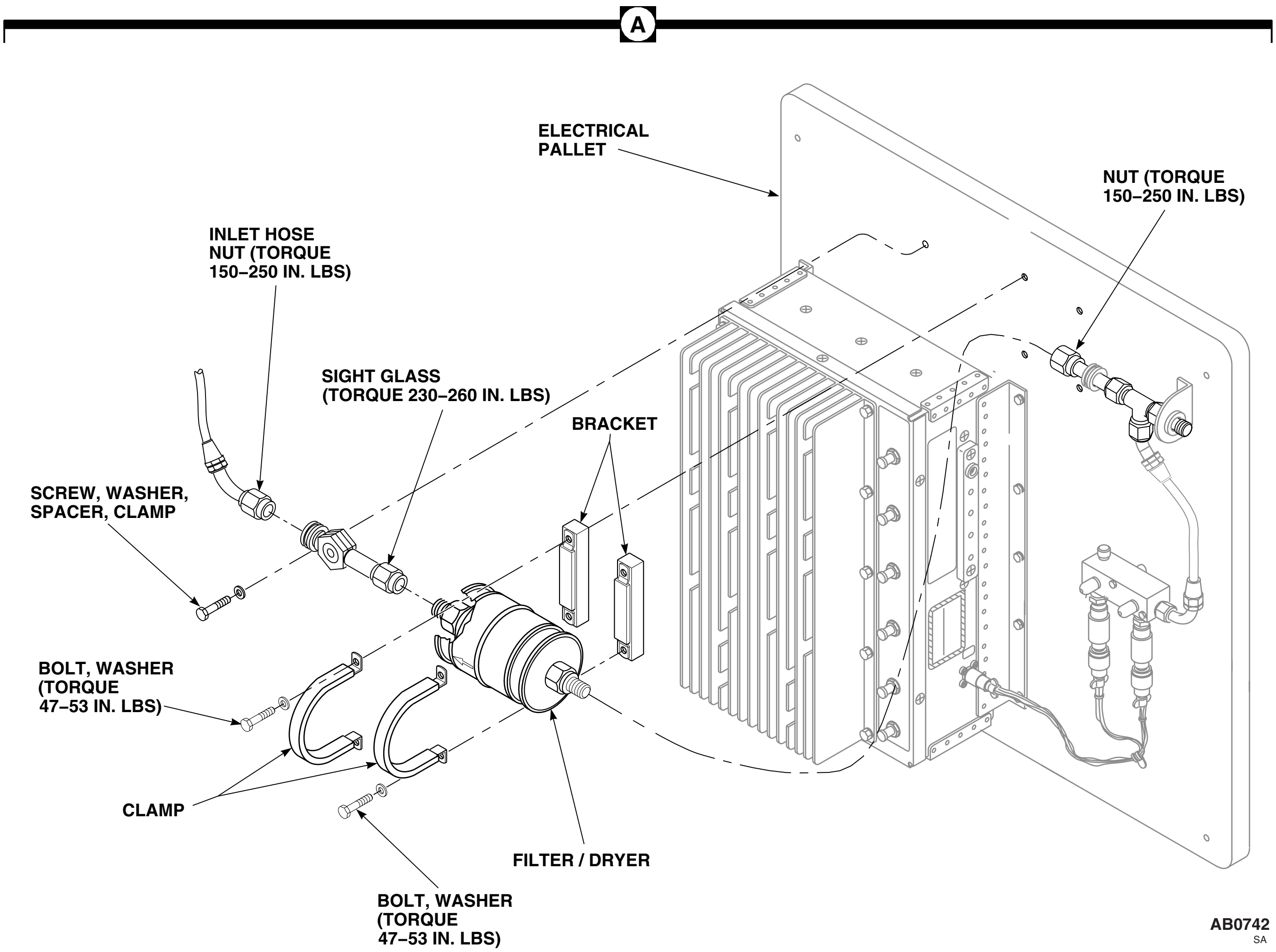


Figure 1. Replace Environmental Control System Filter/Dryer.

INSTALL - Continued

3. **QA** Assemble filter/dryer to sight glass. **TORQUE SIGHT GLASS TO 230 - 260 INCH-POUNDS.**
4. Position filter/dryer and install line nut to right of filter/dryer. Do not tighten nut at this time.
5. **QA** Install filter/dryer to pallet with bolts, washers, clamps, and brackets. **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.**
6. **QA** Lubricate threads with refrigerant lubricating oil, Item 191, WP 1803 00. Install inlet hose to sight glass. **TORQUE NUTS TO 150 - 250 INCH-POUNDS.**
7. Install clamp, spacer, screw, and washer on sight glass.
8. **QA** Tighten nut holding line to filter/dryer. **TORQUE NUT TO 150 - 250 INCH-POUNDS.**
9. Service ECS (WP 1626 00).
10. Do an operational check of ECS (WP 0151 00).
11. Close and latch avionics compartment access door.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM CONDENSER PALLET HH-60A HH-60L

This WP supersedes WP 1303 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanics Toolkit, SC 5180-99-B01
Refrigeration Unit Service Toolkit, SC 5180-90-N18
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.
Wrench, 1 inch, Open End Crowfoot Attachment, GGG-C-001507
Wrench, 7/8-inch, Open End Crowfoot Attachment, GGG-C-001507

Material/Parts

Cap, NAS817-8
Cap, NAS817-10
Lubricating Oil, Refrigerant, Item 191, WP 1803 00
Plug, NAS818-8
Plug, NAS818-10
Wire, Nonelectrical, Item 352, WP 1803 00

Personnel Required

Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)
Utility Equipment Repairer MOS 52C (1)

References

WP 0151 00
WP 0258 00
WP 1299 00
WP 1626 00
WP 1703 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Open avionics compartment access door.
3. Deservice ECS (WP 1626 00).
4. Fold or remove the crew chief seat to allow access to the condenser pallet assembly.

NOTE

If installed, remove troop seat installed on aft right bulkhead to allow access to the condenser pallet assembly.

5. Remove right soundproofing panel from rear cabin bulkhead (WP 0258 00).
6. Remove condenser exhaust duct (WP 1299 00).
7. Loosen clamps holding condenser intake flex duct and remove duct.
8. Remove lockwire and disconnect electrical connector, P353, from condenser blower (Figure 1, Detail A).

CAUTION

Damage to air-conditioning parts will occur if moisture is allowed to enter air-conditioning system. To prevent moisture from entering air-conditioning system, make sure to cap all lines immediately after disconnecting them.

9. Disconnect and remove hoses from condenser inlet and outlet fittings. Plug, NAS818-8 and NAS818-10, hoses and cap, NAS817-8 and NAS817-10, fittings.

REMOVE - Continued

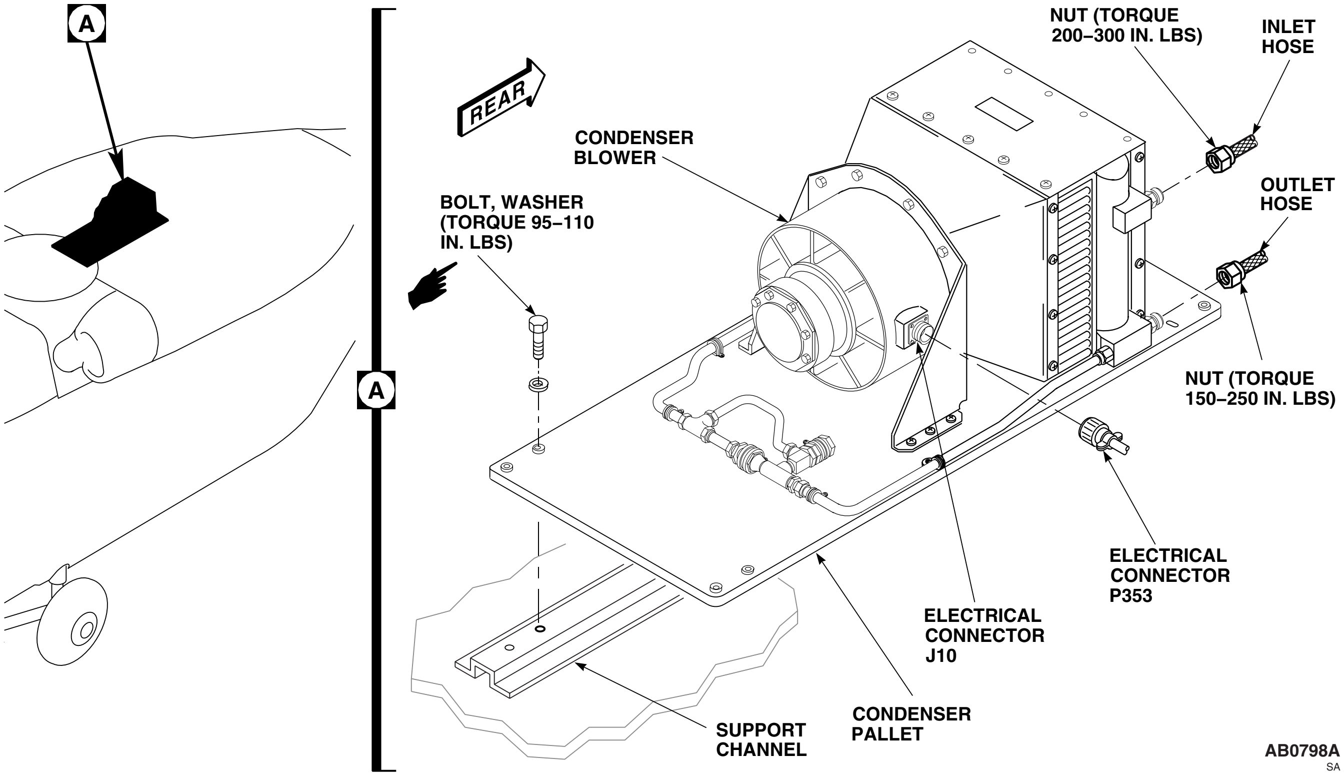


Figure 1. Replace Environmental Control System Condenser Pallet.

NOTE

Make sure aft bolt from elongated pallet hole is removed last.

10. Remove bolts and washers holding pallet to support channel.

CAUTION

To prevent damage to ECS, do not use lines, fittings, or other components as a hand-hold. Handle pallet by base only.

NOTE

If medical cabinet is installed, care must be taken during removal of the condenser pallet to prevent damage to both components.

11. Carefully remove condenser pallet from helicopter.

INSTALL

CAUTION

To prevent damage to ECS, do not use lines, fittings, or other components as a handhold. Handle pallet by base only.

NOTE

- If medical cabinet is installed, care must be taken during removal of the condenser pallet to prevent damage to both components.
- Insert evaporator inlet duct through aft bulkhead opening before installing the condenser pallet.
- Make sure aft bolt from elongated pallet hole is removed last.

1. Turn off all electrical power.
2. **QA** Install condenser pallet over fuel cells and attach to support channels with bolts and washers. **TORQUE BOLTS TO 95 - 110 INCH-POUNDS.**
3. Remove caps and plugs.
4. Lubricate threads of nuts on hoses with refrigerant lubricating oil, Item 191, WP 1803 00.
5. **QA** Install inlet hose on condenser inlet fitting (UPPER). **TORQUE INLET NUT TO 200 - 300 INCH-POUNDS.**
6. **QA** Install outlet hose on condenser outlet fitting (LOWER). **TORQUE OUTLET NUT TO 150 - 250 INCH-POUNDS.**
7. Inspect and treat electrical connector (WP 1703 00).
8. **QA** Connect electrical connector, P353, to condenser blower. Using nonelectrical wire, Item 352, WP 1803 00, lockwire, connector.
9. Install condenser exhaust duct (WP 1299 00).
10. Service ECS (WP 1626 00).
11. Do operational check on ECS (WP 0151 00).
12. Install right soundproofing panel on rear cabin bulkhead (WP 0258 00).
13. Close and latch avionics compartment access door.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM EVAPORATOR PALLET HH-60A HH-60L

This WP supersedes WP 1304 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanics Toolkit, SC 5180-99-B01
Refrigeration Unit Service Toolkit, SC 5180-90-N18
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Material/Parts

Cap, NAS817-4
Cap, NAS817-8
Cap, NAS817-10
Cloth, Cleaning, Item 86, WP 1803 00
Lubricating Oil, Refrigerant, Item 191, WP 1803 00
Plug, NAS818-4
Plug, NAS818-8
Plug, NAS818-10
Sealing Compound, Item 288, WP 1803 00

Personnel Required

Assistant - (2)
UH-60 Helicopter Repairer MOS 15T (1)
Utility Equipment Repairer MOS 52C (1)

References

TM 1-1520-253-13&P
WP 0151 00
WP 0258 00
WP 1626 00
WP 1627 00
WP 1703 00
WP 1803 00

REMOVE

WARNING

Injury to personnel and damage to equipment will result if evaporator pallet is not supported during removal. Evaporator pallet weighs approximately 165 pounds. Use two assistants to support evaporator pallet during removal.

1. Turn off all electrical power.
2. Open avionics compartment access door.
3. Deservice ECS (WP 1626 00).
4. Remove overhead soundproofing panel (WP 0258 00).
5. Remove OBOGS status panel (TM 1-1520-253-13&P).
6. Slide left medical attendant seat forward and move seat back forward.
7. Remove medical cabinet (TM 1-1520-253-13&P).
8. Disconnect and remove electrical connectors, P354, P355, and P357, from electrical receptacles J5, J6, and J12 on evaporator pallet (Figure 1, Detail B).
9. Remove clamp and inlet duct hose from the evaporator pallet (Figure 1, Detail A).

REMOVE - Continued

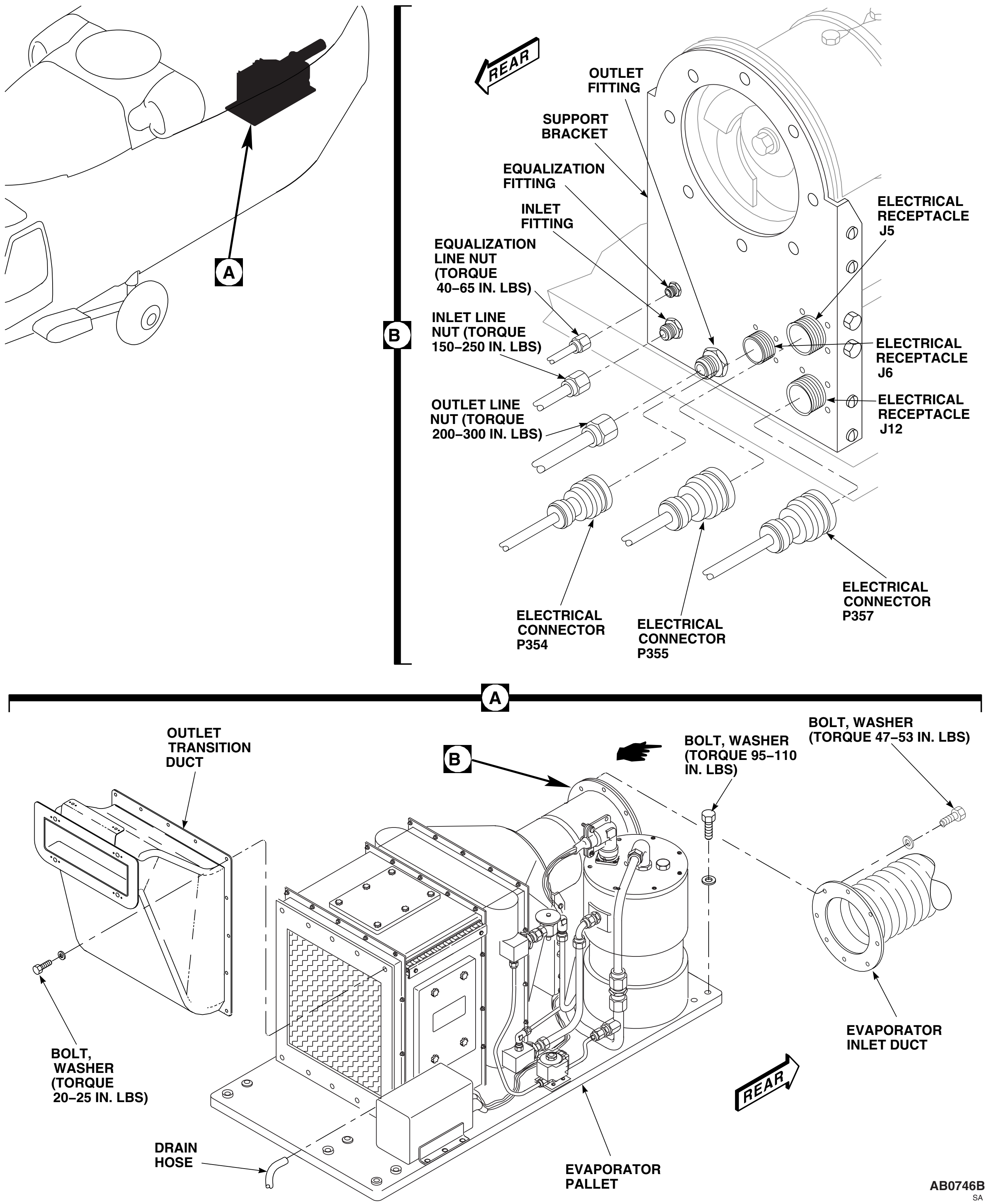


Figure 1. Replace Environmental Control System Evaporator Pallet.

REMOVE - Continued

CAUTION

Damage to air-conditioning parts will occur if moisture is allowed to enter air-conditioning system. To prevent moisture from entering air-conditioning system, make sure to cap all lines immediately after disconnecting them.

10. Disconnect outlet line, inlet line and equalization line (Figure 1, Detail B). Cap, NAS817-4, NAS817-8, and NAS817-10, fittings and plug, NAS818-4, NAS818-8, and NAS818-10, hoses.
11. Remove drain hose from front of evaporator pallet and position out of way.
12. Remove bolts and washers holding evaporator pallet to support channels.

CAUTION

Damage to evaporator pallet will result if components are used as handholds. To prevent damage to evaporator pallet, do not use lines, fittings, or other components as a handhold. Handle evaporator pallet by base only.

13. Remove evaporator pallet from helicopter.
14. Remove bolts and washers holding outlet transition duct to evaporator pallet. Remove outlet transition duct.

INSTALL

WARNING

Injury to personnel and damage to equipment will result if evaporator pallet is not supported during installation. Evaporator pallet weighs approximately 165 pounds. Use two assistants to support evaporator pallet during installation.

CAUTION

Damage to evaporator pallet will result if components are used as handholds. To prevent damage to evaporator pallet, do not use lines, fittings, or other components as a handhold. Handle evaporator pallet by base only.

1. Turn off all electrical power.
2. Service environmental control system compressor prior to installing evaporator pallet (WP 1627 00).
3. **QA** Position evaporator pallet over fuel cells and attach to support channels with bolts and washers. **TORQUE BOLTS TO 95 - 110 INCH-POUNDS.**
4. **QA** Install drain hose on front of evaporator pallet.
5. Clean flare surface of inlet, outlet and equalization fittings with a cleaning cloth, Item 86, WP 1803 00.
6. Apply sealing compound, Item 288, WP 1803 00, to flare surface only of the inlet, outlet and equalization male fittings.
7. Lubricate line nuts with refrigerant lubricating oil, Item 191, WP 1803 00 prior to connecting.
8. **QA** Connect outlet line to fitting on panel (Figure 1, Detail B). **TORQUE OUTLET LINE NUT TO 200 - 300 INCH-POUNDS.**
9. **QA** Connect inlet line to fitting on panel. **TORQUE INLET LINE NUT TO 150 - 250 INCH-POUNDS.**
10. **QA** Connect equalization line to fitting on panel. **TORQUE EQUALIZATION LINE NUT TO 40 - 65 INCH-POUNDS.**
11. Inspect and treat electrical connectors (WP 1703 00).

INSTALL - Continued

12. **QA** Connect electrical connector (P354, P355, and P357) to electrical receptacle (J6, J5, and J12) respectively, on evaporator pallet.
13. **QA** Install return duct to evaporator pallet and secure with clamp.
14. Install OBOGS status panel (TM 1-1520-253-13&P).
15. Install overhead soundproofing panel (WP 0258 00).
16. **QA** Slide left medical attendant seat aft and move seat back to upright position.
17. Service ECS (WP 1626 00).
18. Do operational check on ECS (WP 0151 00).
19. Close and latch avionics compartment access door.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM ELECTRICAL PALLET HH-60A HH-60L

This WP supersedes WP 1305 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanics Toolkit, SC 5180-99-B01
Refrigeration Unit Service Toolkit, SC 5180-90-N18
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Material/Parts
Lubricating Oil, Refrigerant, Item 191, WP 1803 00
Protective Caps and Plugs

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)
Utility Equipment Repairer MOS 52C (1)

References
WP 0151 00
WP 1626 00
WP 1703 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Open avionics compartment access door.
3. Deservice ECS (WP 1626 00).
4. Disconnect electrical connectors, P351, P352, P3, P349, and P350, from electrical control unit.
5. Remove any bolts, washers, spacers, and clamps holding wiring to electrical pallet.

CAUTION

Damage to air-conditioning parts will occur if moisture is allowed to enter air-conditioning system. To prevent moisture from entering air-conditioning system, make sure to cap all lines immediately after disconnecting them.

6. Disconnect inlet hose at sight glass. Plug and cap hoses and fittings.
7. Disconnect low pressure hose from top of servicing manifold. Cap and plug hose fittings (Figure 1, Detail A).
8. Disconnect high pressure hose from upper tee fitting on electrical pallet. Cap and plug hose fittings.
9. Remove bolt, washers and clamp securing low pressure line to electrical pallet.
10. Remove bolts and washers holding electrical pallet to airframe.

CAUTION

To prevent damage to ECS, do not use lines, fittings, and other components as a hand-hold. Handle electrical pallet by base only.

11. Remove electrical pallet from helicopter.

INSTALL

1. Turn off all electrical power.

INSTALL - Continued

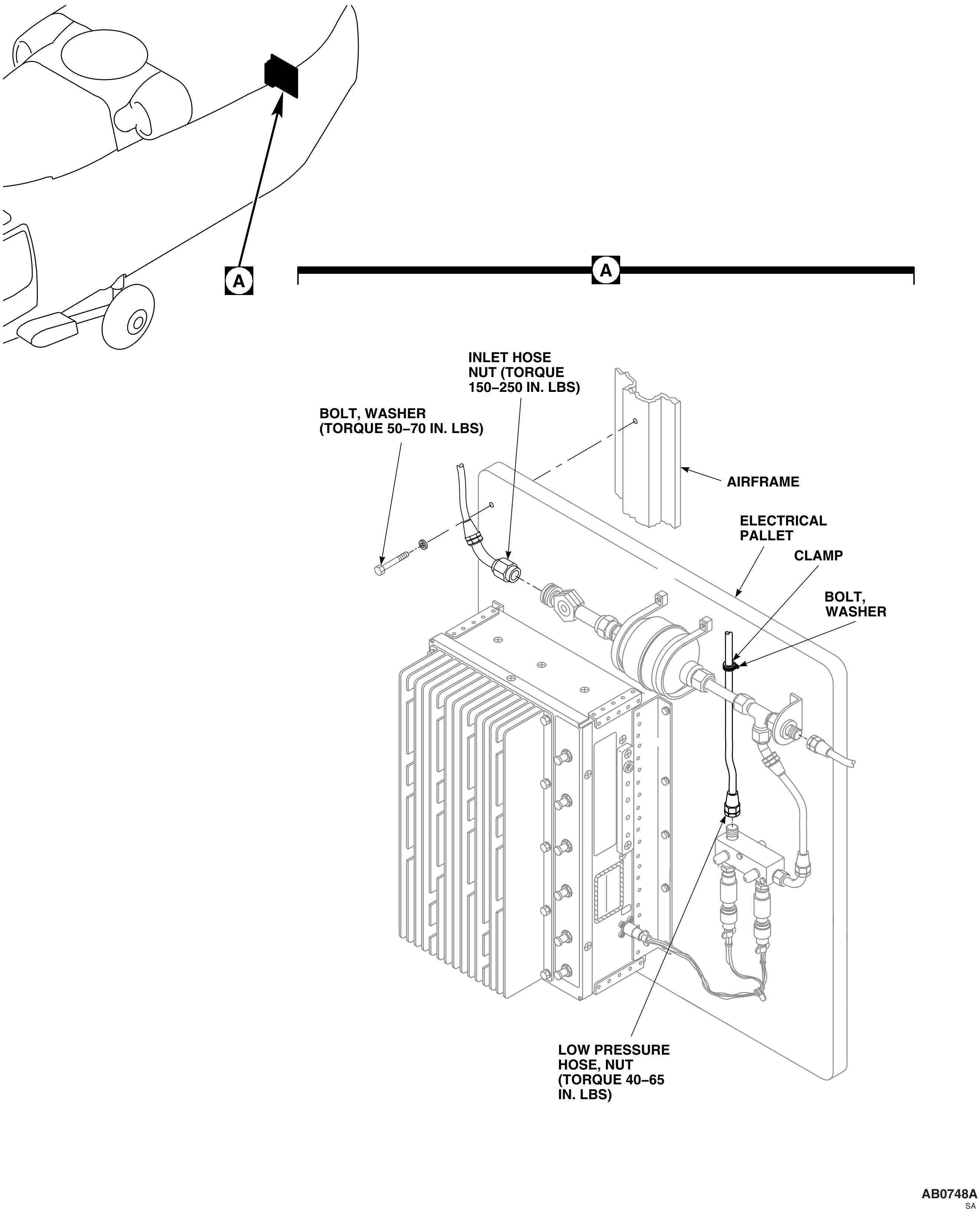


Figure 1. Replace Environmental Control System Electrical Pallet.

INSTALL - Continued

2. **QA** Install electrical pallet in helicopter with bolts and washers (Figure 1, Detail A). **TORQUE BOLTS TO 50 - 70 INCH-POUNDS.**
3. Lubricate threads on hoses with refrigerant lubricating oil, Item 191, WP 1803 00.
4. **QA** Remove plug and cap and connect inlet hose to sight glass. **TORQUE HOSE NUT TO 150 - 250 INCH-POUNDS.**
5. Remove cap and plugs and connect low pressure hose to low pressure fitting on top of servicing manifold. **TORQUE HOSE NUT TO 40 - 65 INCH-POUNDS.**
6. **QA** Remove cap and plugs and connect high pressure hose to upper tee fitting on electrical pallet. **TORQUE HOSE NUT TO 150 - 250 INCH-POUNDS.**
7. **QA** Install clamp on low pressure line and secure to electrical pallet with bolt and washers.
8. Inspect and treat electrical connectors (WP 1703 00).
9. **QA** Install electrical connectors, P351, P352, P3, P349, and P350, to electrical control unit.
10. **QA** Install any bolts, washers, spacers, and clamps holding wiring to electrical pallet.
11. Service ECS (WP 1626 00).
12. Do an operational check of ECS (WP 0151 00).
13. Close and latch avionics compartment access door.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM SERVICING MANIFOLD HH-60A HH-60L

This WP supersedes WP 1306 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.

Material/Parts
Lubricating Oil, Refrigerant, Item 191, WP 1803 00
Protective Caps and Plugs

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0151 00
WP 1308 00
WP 1626 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Open avionics compartment access door.
3. Deservice ECS (WP 1626 00).
4. Remove high or low pressure switches from servicing manifold (WP 1308 00).

CAUTION

Damage to air-conditioning parts will occur if moisture is allowed to enter air-conditioning system. To prevent moisture from entering air-conditioning system, make sure to cap all lines immediately after disconnecting them.

5. Remove line from servicing manifold (Figure 1, Detail A).
6. Remove low pressure hose from servicing manifold.
7. Remove bolts, washers, and spacers holding servicing manifold to electrical pallet.

INSTALL

1. Turn off all electrical power.
2. Remove any caps and plugs just prior to lubricating and installing hoses (Figure 1, Detail A).
3. Lubricate threads on elbow with refrigerant lubricating oil, Item 191, WP 1803 00.
4. **QA** Install servicing manifold on pallet with bolts, washers, and spacers. **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**
5. **QA** Install line on servicing manifold. **TORQUE NUT TO 40 - 65 INCH-POUNDS.**
6. **QA** Install low pressure hose on servicing manifold. **TORQUE NUT TO 40 - 65 INCH-POUNDS.**
7. Install pressure switches on servicing manifold (WP 1308 00).
8. Service ECS (WP 1626 00).

INSTALL - Continued

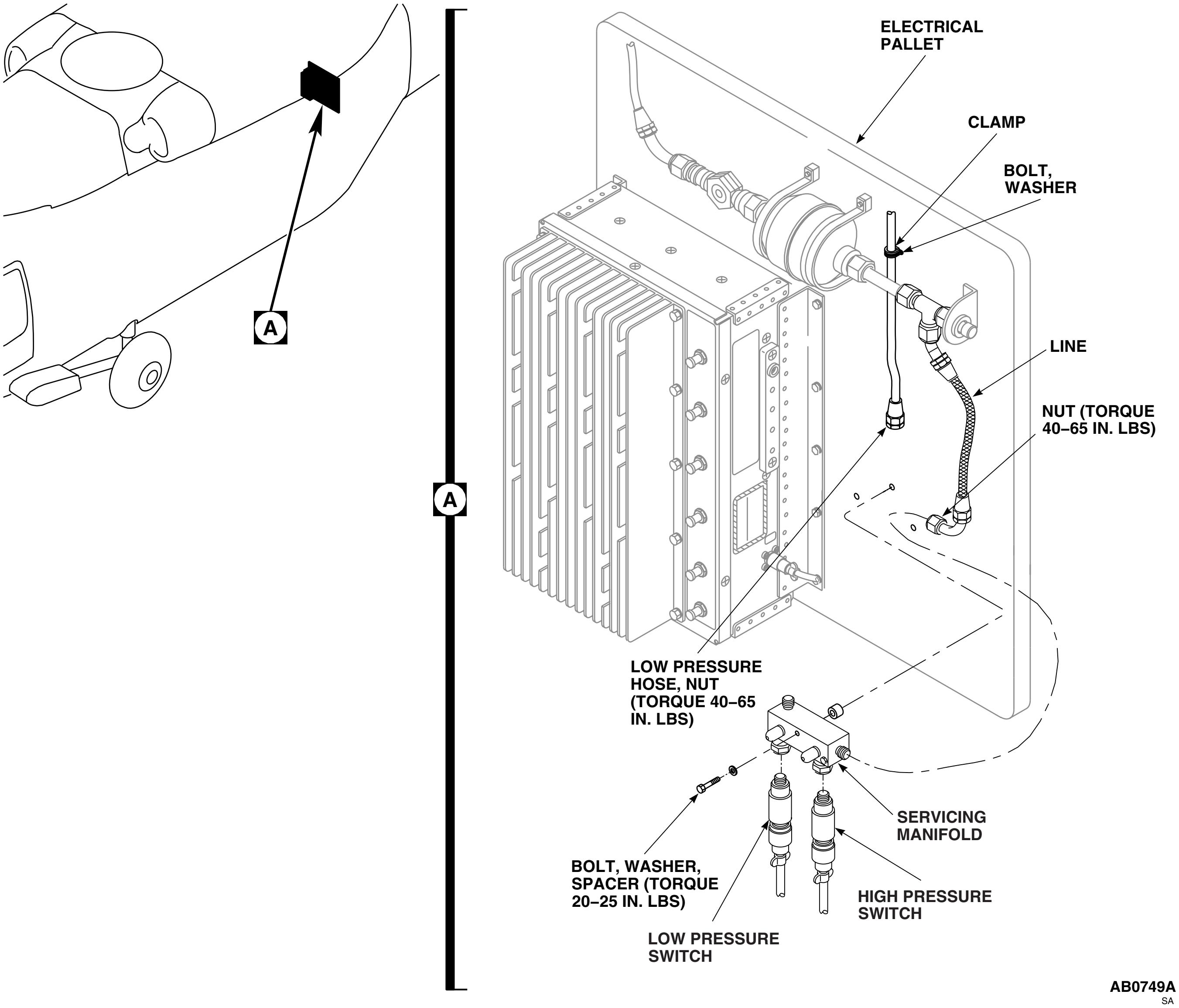


Figure 1. Replace Environmental Control System Servicing Manifold.

9. Do an operational check of ECS (WP 0151 00).
10. Close and latch avionics compartment access door.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM SIGHT GLASS HH-60A HH-60L

This WP supersedes WP 1307 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Refrigeration Unit Service Toolkit, SC 5180-90-N18
Torque Wrench, 150 - 1000 in. lbs.

Material/Parts

Lubricating Oil, Refrigerant, Item 191, WP 1803 00
Protective Caps and Plugs

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)
Utility Equipment Repairer MOS 52C (1)

References

WP 1626 00
WP 1803 00

REMOVE

1. Open avionics compartment access door.
2. Deservice ECS (WP 1626 00).
3. Remove clamp, spacer, screw, and washer holding sight glass.

CAUTION

Damage to air-conditioning parts will occur if moisture is allowed to enter air-conditioning system. To prevent moisture from entering air-conditioning system, make sure to cap all lines immediately after disconnecting them.

4. Disconnect inlet hose from sight glass. Remove sight glass from filter dryer. Plug hoses and cap fittings (Figure 1, Detail A).

INSTALL

1. Remove plugs and caps just before parts are assembled.
2. Lubricate threads on sight glass with refrigerant lubricating oil, Item 191, WP 1803 00.
3. **QA** Install sight glass on filter/dryer (Figure 1, Detail A). **TORQUE SIGHT GLASS 230 - 260 INCH-POUNDS.**
4. **QA** Connect inlet hose to sight glass. **TORQUE NUT 150 - 250 INCH-POUNDS.**
5. **QA** Install clamp, spacer, screw, and washer on sight glass.
6. Service ECS (WP 1626 00).
7. Close and latch avionics compartment access door.

INSTALL - Continued

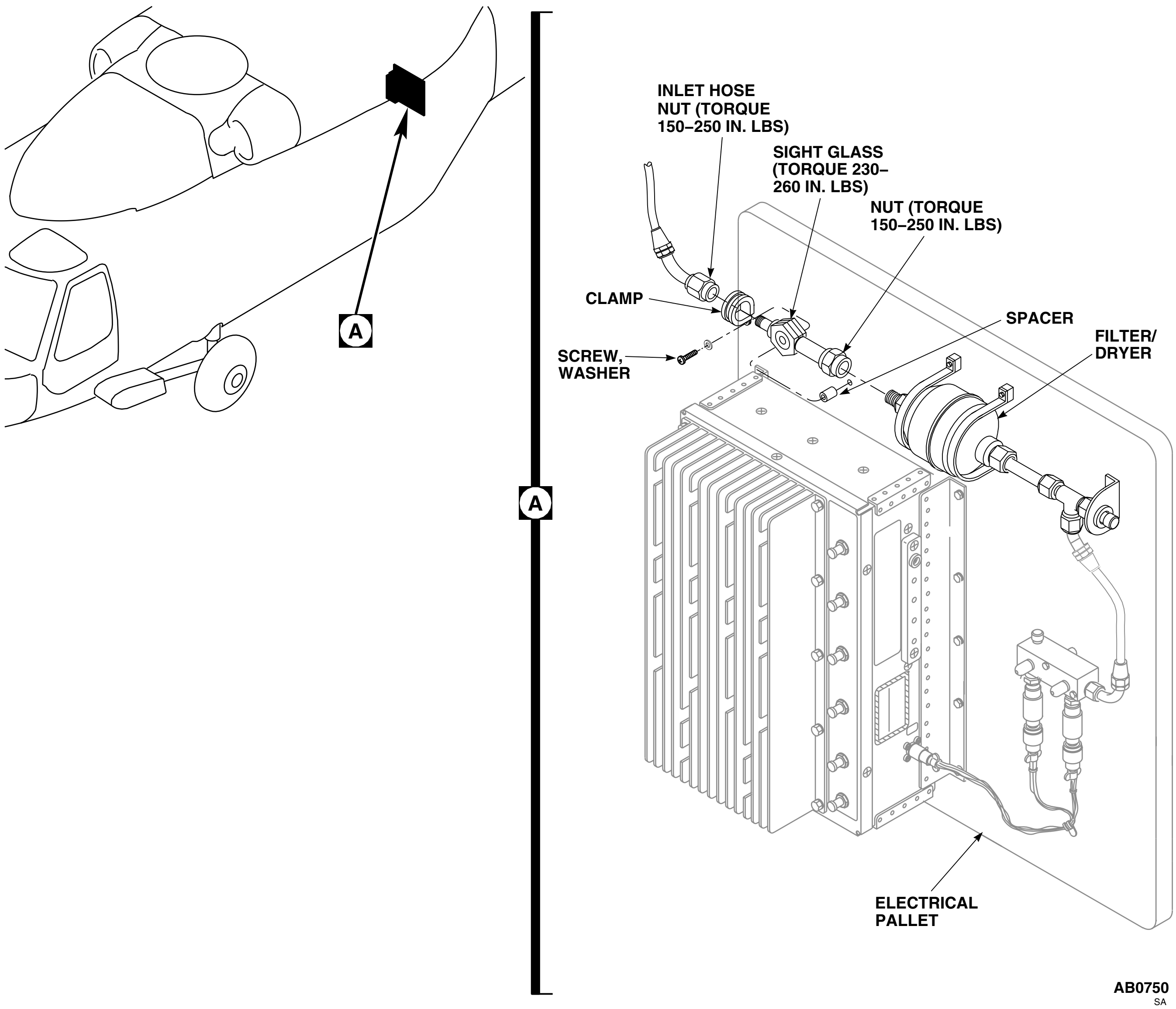


Figure 1. Replace Environmental Control System Sight Glass.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM HIGH AND LOW PRESSURE SWITCH HH-60A HH-60L

This WP supersedes WP 1308 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0151 00
WP 1626 00
WP 1703 00

REMOVE

1. Turn off all electrical power.
2. Open avionics compartment access door.
3. Deservice ECS (WP 1626 00).
4. Disconnect electrical connector, SW1, from high pressure switch (Figure 1, Detail A).
5. Disconnect electrical connector, SW16, from low pressure switch.

NOTE

One or both fittings may come out with switches. Fitting must be removed from switch for use with replacement switch.

6. Remove high and low pressure switches from fittings on servicing manifold.

INSTALL

1. Turn off all electrical power.
2. **QA** If either fitting was removed from servicing manifold, reinstall fittings, with new packings, in servicing manifold.
3. **QA** Install switches in fittings (Figure 1, Detail A).
4. Inspect and treat electrical connectors (WP 1703 00).
5. **QA** Install electrical connector, SW1, to high pressure switch.
6. **QA** Install electrical connector, SW16, to low pressure switch.
7. Service ECS (WP 1626 00).
8. Do an operational check of ECS (WP 0151 00).
9. Close and latch avionics compartment access door.

INSTALL - Continued

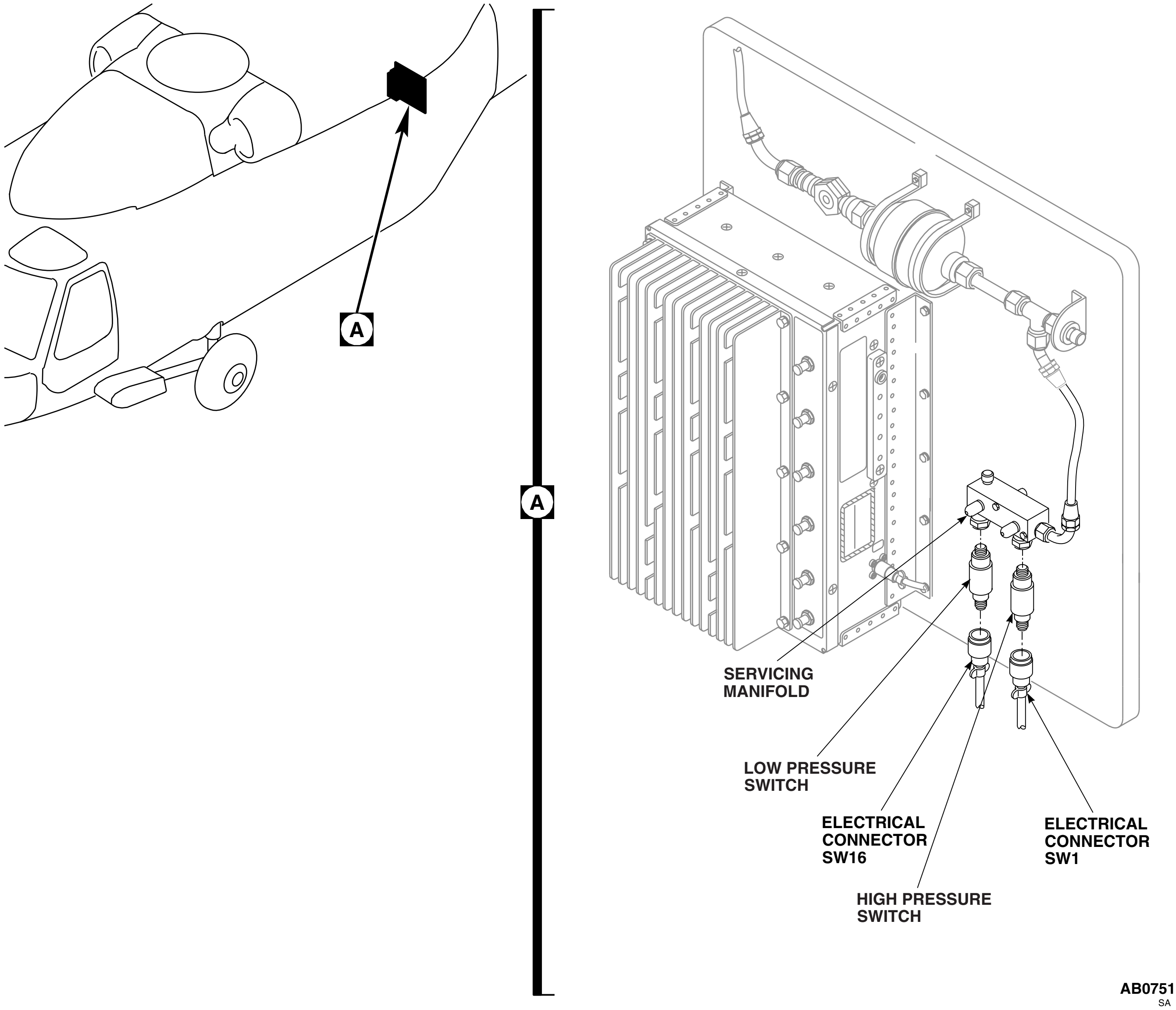


Figure 1. Replace Environmental Control System High and Low Pressure Switches.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM CONTROL PANEL HH-60A HH-60L

This WP supersedes WP 1309 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Refrigeration Unit Service Toolkit, SC 5180-90-N18

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)
Utility Equipment Repairer MOS 52C (1)

References
WP 0151 00
WP 1703 00

REMOVE

1. Turn of all electrical power.
2. Remove captive screws securing ECS control panel to mounting rails (Figure 1).
3. Disconnect electrical connector, P356, from ECS control panel.
4. Remove ECS control panel from helicopter.

INSTALL

1. Turn of all electrical power.
2. inspect and treat electrical connectors (WP 1703 00).
3. **QA** Connect electrical connector, P356 to ECS control panel (Figure 1).
4. Secure ECS control panel to mounting rails using captive screws.
5. Make sure area is clean and free of foreign material.
6. Do an operational check of ECS control panel (WP 0151 00).

INSTALL - Continued

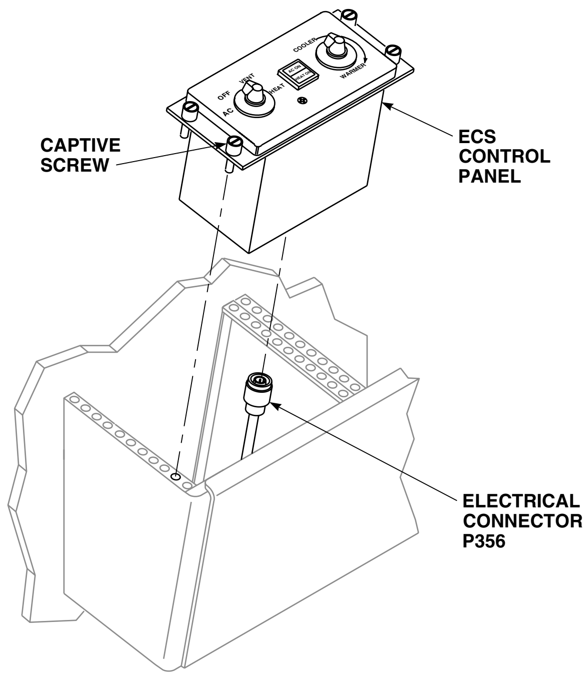


Figure 1. Replace Environmental Control System Control Panel.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

MIXTURE TEMPERATURE SENSOR PRESSURE BELLOWS REPAIR (AVIM)

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

REMOVE

1. Remove screws holding pressure bellows to mixture temperature sensor body (Figure 1).
2. Remove pressure bellows and actuating pin.

INSTALL

1. **QA** Insert actuating pin in mixture temperature sensor body (Figure 1).
2. **QA** Install pressure bellows over actuating pin on mixture temperature sensor body with screws.

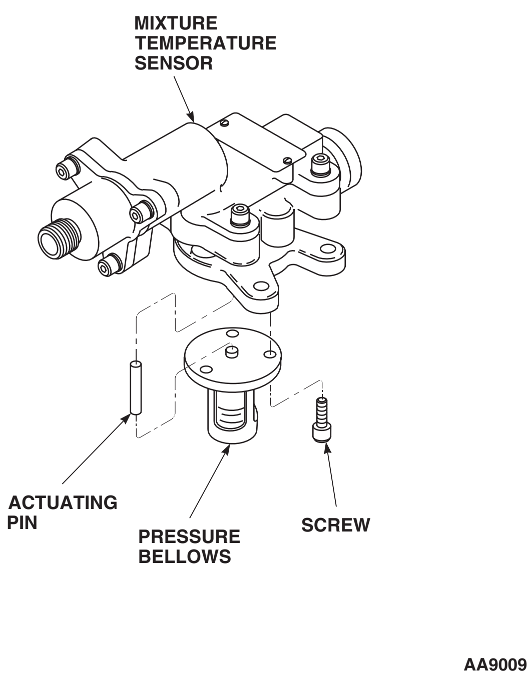


Figure 1. Repair Mixture Temperature Sensor Pressure Bellows (AVIM).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

BLOWER UNIT REPAIR (AVIM)

INITIAL SETUP:

Tools and Special Tools
Electrical Repairer's Toolkit, SC 5180-99-B06
Torque Wrench, 30 - 200 in. lbs.

Material/Parts
Wire, Nonelectrical, Item 354, WP 1803 00

Personnel Required
Aircraft Electrician MOS 15F (1)

References
TM 55-1500-323-24
WP 0149 00
WP 1803 00

DISASSEMBLE

1. Remove nut, washer, and rotor from motor shaft (Figure 1).
2. Remove screw, washer, retainer, and valve-plate.
3. Remove Woodruff key from motor shaft.
4. Disassemble electrical connector (TM 55-1500-323-24).
5. Remove screws and washers holding motor in stator.
6. Remove motor from stator.
7. Remove wiring harness through grommet.

INSPECT

1. Check rotor for bent, worn, or cracked blades.
2. Check wiring for signs of overheating. Check grommet for cracks.
3. Check stator for damage and cracks.
4. Check valve plate for broken springs and dented plates.
5. Check motor for signs of overheating. Motor shaft should turn freely.
6. Replace any unserviceable parts.

ASSEMBLE

1. Install wiring harness through grommet in stator (Figure 1).
2. **QA** Install motor in stator with screw and washer. Using nonelectrical wire, Item 354, WP 1803 00, lockwire screws.
3. Assemble electrical connector (TM 55-1500-323-24).
4. **QA** Install Woodruff key and rotor on motor shaft with nut and washer. **TORQUE NUT TO 60 - 100 INCH-POUNDS.**

ASSEMBLE - Continued

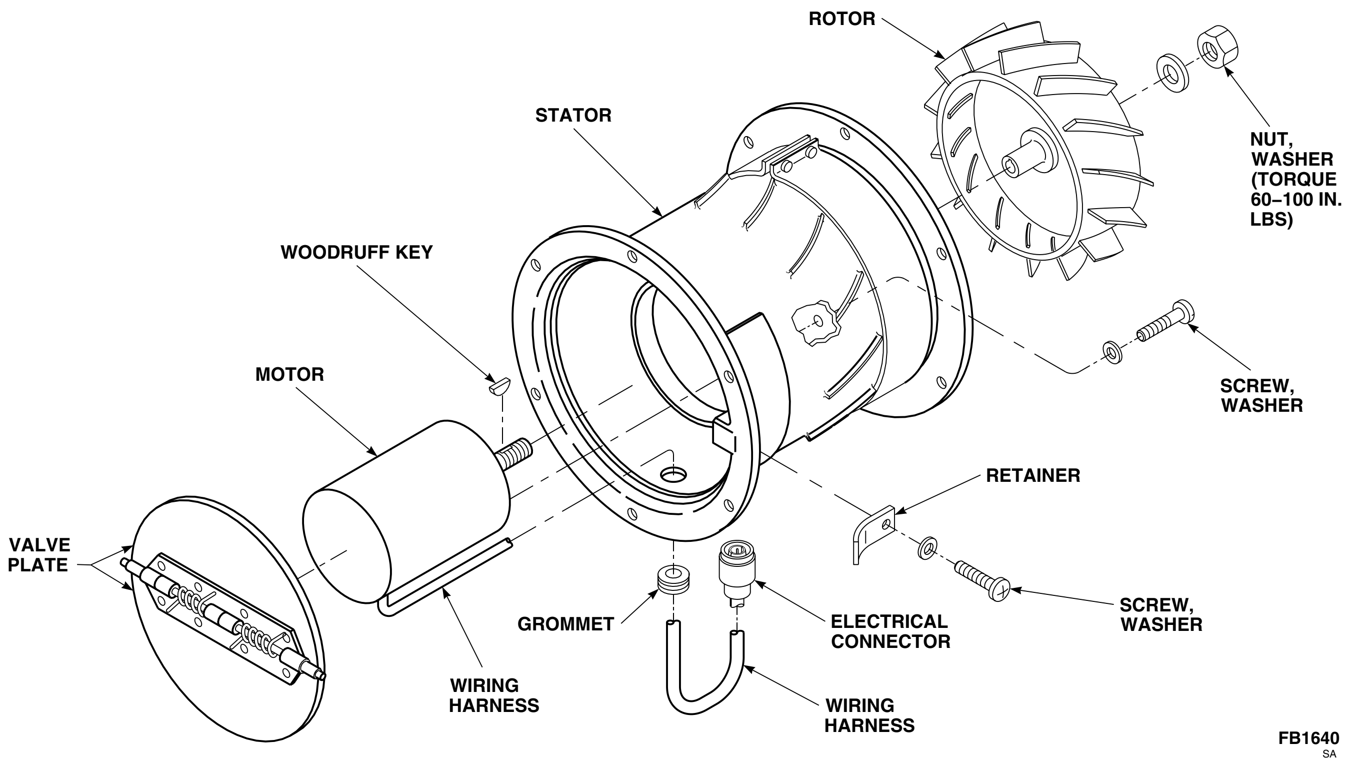


Figure 1. Repair Blower Unit (AVIM).

5. **QA** Install valve plate with retainer, screw, and washer.
6. Do an operational check of heating/ventilation system (WP 0149 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM CONDENSER PALLER (AVIM) EH60A

INITIAL SETUP:

Tools and Special Tools
Container, 5-Gallon
Halogen Leak Detector
Refrigerant Recovery and Recycling Station, 17500B
Refrigeration Unit Service Toolkit, SC 5180-90-N18
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Material/Parts
Cap, NAS817-8
Cap, NAS817-10
Lubricating Oil, Refrigerant, Item 190, WP 1803 00
Nitrogen, Item 199, WP 1803 00
Plug, NAS818-8
Plug, NAS818-10

Personnel Required
Utility Equipment Repairer MOS 52C (1)

References
WP 0150 00
WP 0258 00
WP 1289 00
WP 1292 00
WP 1622 00
WP 1703 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Open avionics compartment access door.

NOTE

Make sure the refrigerant recovery and recycling station, 17500B, is used during any deservicing of the air conditioning system.

3. Deservice ECS (WP 1622 00).
4. Remove right soundproofing panel from rear cabin bulkhead (WP 0258 00).
5. Remove right hand plenum (WP 1289 00).
6. Remove condenser exhaust duct (WP 1292 00).
7. Disconnect electrical connector, P4, from right side of electrical pallet (Figure 1, Detail A).

CAUTION

Damage to air-conditioning parts will occur if moisture is allowed to enter air-conditioning system. To prevent moisture from entering air-conditioning system, make sure to cap all lines immediately after disconnecting them.

8. Disconnect and remove hoses from condenser inlet and outlet fittings. Plug, NAS818-8 and NAS818-10, hoses and cap, NAS817-8 and NAS817-10, fittings.
9. Remove bolts and washers holding pallet to support channel.

REMOVE - Continued

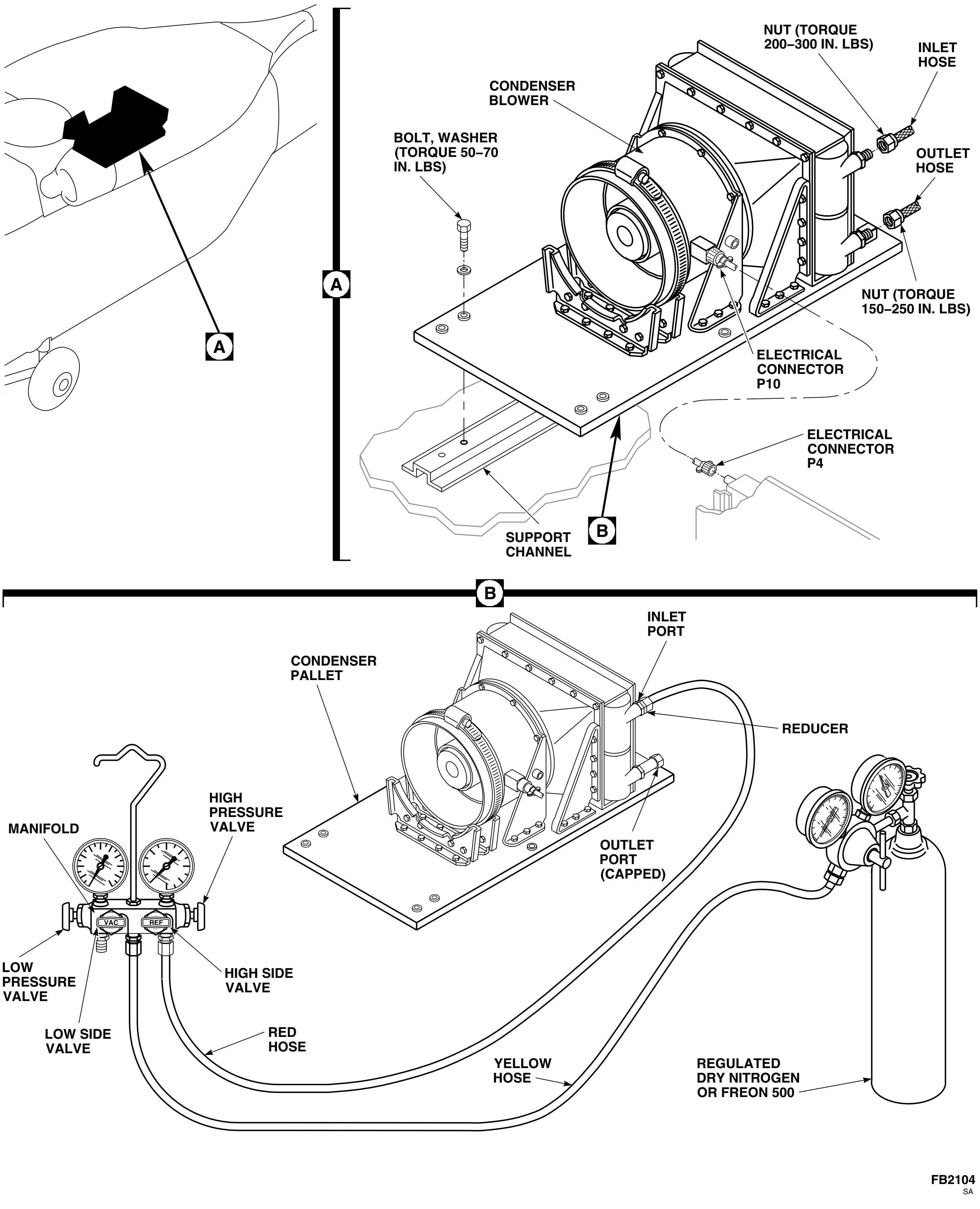


Figure 1. Replace Environmental Control System Condenser Pallet (AVIM).

REMOVE - Continued

CAUTION

Damage to ECS lines, fittings, or assemblies will occur if they are used as a hand hold. Do not use ECS lines, fittings, or assemblies as a hand hold. Handle pallet by base only.

10. Carefully remove condenser pallet from helicopter.

PRESSURE TEST

WARNING

- Injury to personnel will occur, if safety precautions are not observed when working on any part of the air conditioning system. Observe the following safety precautions when working on any part of the air conditioning system.
- Protect skin and eyes from possible contact with liquid refrigerant. Wear gloves and goggles. Keep your body as well protected as you can from possible contact with liquid refrigerant.
- If liquid refrigerant comes in contact with the eyes or skin, get immediate medical aid.
- The refrigerant in a sealed system is always under pressure. Do not disconnect any line or component of the pressurized cooling system without first carefully releasing the pressure.
- Freon-500 will displace air and can cause suffocation, even though it is nontoxic. It will produce highly toxic phosgene gas if exposed to open flame. Make sure work area is well ventilated, and that there are no open flames, or unshielded heaters in immediate area.
- The use of heat on any part of the pressurized system will cause a pressure buildup and could cause an explosion. Do not try to solder or weld any part of the system while the system is charged with refrigerant.

1. Connect high pressure (red) hose to high side of servicing manifold (Figure 1, Detail B). Make sure manifold hand wheels are closed.
2. Uncap condenser inlet and connect other end of high pressure (red) hose.
3. Install cap on condenser outlet port.
4. Connect yellow hose to Freon-500 bottle and vac side of pressure gage.

NOTE

Make sure Freon-500 bottle is kept upright so that only vapors will transfer.

5. Open valve on Freon-500 bottle and record the bottle pressure.
6. Close valve on Freon-500 bottle and disconnect yellow hose from vac side of gage.
7. Connect end of yellow hose to center supply port of servicing manifold.
8. Open Freon-500 bottle valve.
9. Purge yellow hose of air by loosening fitting at servicing manifold until a little Freon-500 vapor comes out. Immediately tighten fitting.
10. Open high side hand wheel until high side pressure gage equals Freon-500 bottle pressure.

PRESSURE TEST - Continued

11. Close high side hand wheel and Freon-500 bottle valve. Disconnect yellow hose from Freon-500 bottle.

WARNING

Injury to personnel and damage to equipment will result if system pressure exceeds 300 psig during leak test. Make sure that nitrogen pressure gages are properly calibrated and regulator adjusting screw is set for 0 psig. Close manifold valves prior to opening supply valve on nitrogen bottle.

12. Connect yellow hose to regulated dry nitrogen, Item 199, WP 1803 00, source.
13. Open valve on nitrogen bottle and slowly increase regulator pressure until it equals Freon-500 system pressure.
14. Open REF hand wheel.
15. Slowly increase nitrogen pressure until system pressure indicated on manifold high pressure gage reaches 300 psi.
16. Using a Halogen leak detector calibrated to detect leaks of 2 oz./year or less, check all fittings, connections, hoses and components for leaks. Repair as necessary.
17. Repeat step 16. until no leaks are found.
18. Close high side hand wheel on servicing manifold.
19. Close nitrogen bottle valve and slowly crack connection fitting.
20. Disconnect yellow hose from nitrogen regulator fitting and place loose end into a clean 5 gallon container, 1/4 full of water.
21. Slowly open high side hand wheel until a small stream of bubbles appears in water.
22. Continue to drain system until no bubbles appear in container. Pressure gage should read 0 psig.
23. Close high side hand wheel and disconnect yellow hose.
24. Disconnect high pressure (red) hose from condenser inlet.
25. Immediately cap inlet fitting.

INSTALL

CAUTION

Damage to ECS lines, fittings, or assemblies will occur if they are used as a hand hold. Do not use ECS lines, fittings, or assemblies as a hand hold.

1. Turn off all electrical power.
2. **QA** Install condenser pallet over fuel cells and attach to support channels with bolts and washers (Figure 1, Detail A). **TORQUE BOLTS TO 50 - 70 INCH-POUNDS.**
3. Remove caps and plugs.
4. Lubricate threads of nuts on hoses with refrigerant lubricating oil, Item 190, WP 1803 00.
5. **QA** Install inlet hose on condenser inlet fitting (UPPER). **TORQUE INLET NUT TO 200 - 300 INCH-POUNDS.**
6. **QA** Install outlet hose on condenser outlet fitting (LOWER). **TORQUE OUTLET NUT TO 150 - 250 INCH-POUNDS.**
7. Inspect and treat electrical connector (WP 1703 00).
8. **QA** Connect electrical connector, P4, to right side of electrical pallet.

INSTALL - Continued

9. Install condenser exhaust duct (WP 1292 00).
10. Install right hand plenum (WP 1289 00).
11. Service ECS (WP 1622 00).
12. Do operational check on ECS (WP 0150 00).
13. Install right soundproofing panel on rear cabin bulkhead (WP 0258 00).
14. Close and latch avionics compartment access door.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM CONDENSER LINES, BURST DISC, AND RELIEF VALVE (AVIM)

EH60A

INITIAL SETUP:

Tools and Special Tools

Refrigeration Unit Service Toolkit, SC 5180-90-N18
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.

Material/Parts

Cap, NAS817-6
Lubricating Oil, Refrigerant, Item 190, WP 1803 00
Plug, NAS818-6
Sealing Compound, Item 279, WP 1803 00

Personnel Required

Utility Equipment Repairer MOS 52C (1)

References

WP 1289 00
WP 1292 00
WP 1622 00
WP 1803 00

REMOVE

1. Open avionics compartment door.
2. Deservice ECS (WP 1622 00).
3. Remove right hand plenum (WP 1289 00).
4. Remove condenser exhaust duct (WP 1292 00).
5. Remove bolts, washers, clamps, and spacer holding over pressurization line to pallet (Figure 1, Detail B).
6. Loosen nut at tee on condenser outlet.
7. Slide outboard end of line from grommet in transition duct.
8. Remove line assembly from pallet.

CAUTION

Damage to air-conditioning parts will occur if moisture is allowed to enter air-conditioning system. To prevent moisture from entering air-conditioning system, make sure to cap all lines immediately after disconnecting them.

9. Immediately install plug, NAS818-6, on line and place cap, NAS817-6, on tee connection at condenser outlet.

INSTALL

1. Apply sealing compound, Item 279, WP 1803 00, to burst disc inlet and outlet fittings and relief valve inlet fitting.
2. Lubricate threads of nuts on hoses with refrigerant lubricating oil, Item 190, WP 1803 00.
3. Loosely assemble lines, tees, burst disc, and valve (Figure 1, Detail B).
4. Slide outboard end of line assembly into grommet in transition duct.
5. **QA** Install lines, burst disc, and relief valve to pallet with bolts, washers, clamps and spacer. **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**

INSTALL - Continued

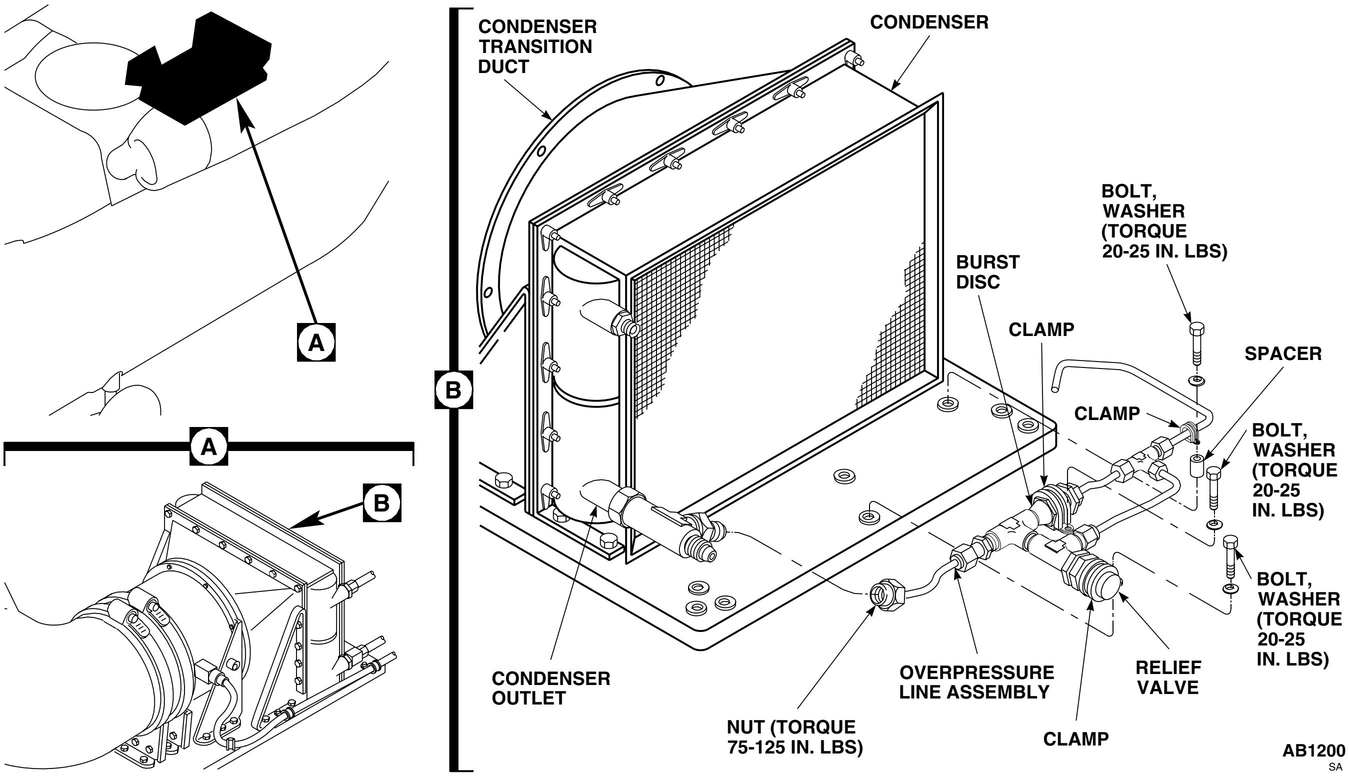


Figure 1. Replace Environmental Control System Condenser Lines, Burst Disc, and Relief Valve (AVIM).

6. Lubricate remaining nut with refrigerant lubricating oil, Item 190, WP 1803 00, and loosely connect to condenser outlet tee.
7. **QA TORQUE ALL TUBING NUTS TO 75 - 125 INCH-POUNDS.**
8. Install right hand plenum (WP 1289 00).
9. Install condenser exhaust duct (WP 1292 00).
10. Service ECS (WP 1622 00).
11. Close and latch avionics compartment door.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM CONDENSER AND TRANSITION DUCT (AVIM) EH60A

INITIAL SETUP:

Tools and Special Tools

Nonmetallic Scraper
Refrigeration Unit Service Toolkit, SC 5180-90-N18
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.

Material/Parts

Cap, NAS817-6
Lubricating Oil, Refrigerant, Item 190, WP 1803 00
Plug, NAS818-6
Sealing Compound, Item 273, WP 1803 00

Personnel Required

Utility Equipment Repairer MOS 52C (1)

References

WP 1312 00
WP 1622 00
WP 1803 00

REMOVE

1. Open avionics compartment door.
2. Deservice ECS (WP 1622 00).
3. Remove condenser pallet (WP 1312 00).

CAUTION

Damage to air-conditioning parts will occur if moisture is allowed to enter air-conditioning system. To prevent moisture from entering air-conditioning system, make sure to cap all lines immediately after disconnecting them.

4. Loosen nut at tee on condenser outlet (Figure 1, Detail B). Cap, NAS817-6, fitting and plug, NAS818-6, line.
5. Remove bolt, washer, spacer, and clamp holding over pressurization line at outboard side of pallet.
6. Loosen nut at tee and remove over pressurization line from grommet in transition duct. Cap, NAS817-6, fitting and plug, NAS818-6, line.
7. Remove bolts, washers, and brackets from pallet (Figure 1, Detail A).
8. Break seal between condenser and pallet, and remove condenser and transition duct from pallet.
9. Remove bolts, washers, and transition duct from condenser.
10. Remove sealing compound from transition duct mating flange and pallet with a nonmetallic scraper.

INSTALL

1. Seal interface of condenser and transition duct with sealing compound, Item 273, WP 1803 00.
2. **QA** Install transition duct on condenser with bolts and washers (Figure 1, Detail A). **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**
3. Seal interface of condenser and pallet with sealing compound, Item 273, WP 1803 00.

INSTALL - Continued

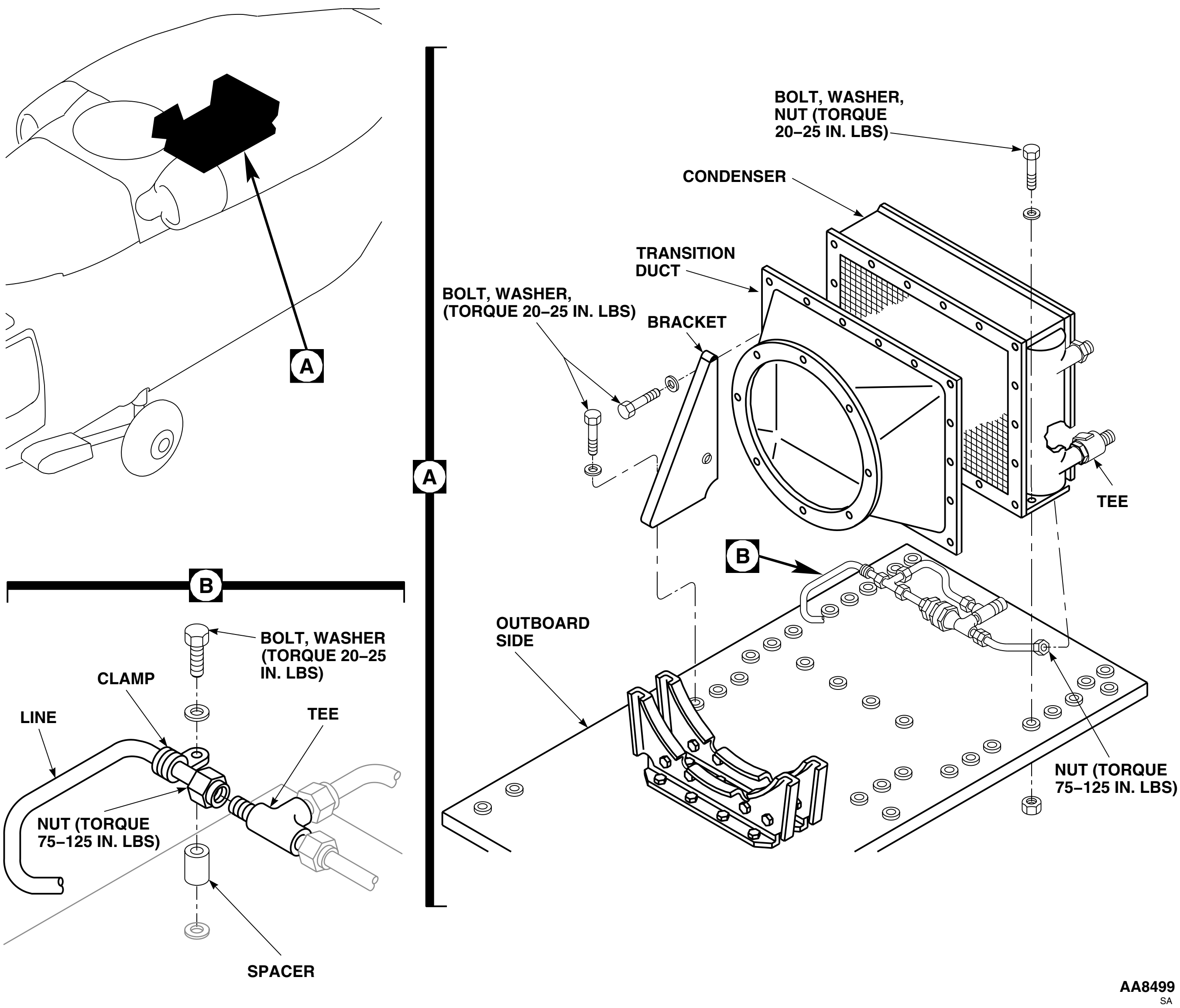


Figure 1. Replace Environmental Control System Condenser and Transition Duct (AVIM).

4. **QA** Install brackets on pallet with bolts and washers (Figure 1, Detail A). **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**

5. **QA** Install brackets on condenser with bolts and washers. **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**

6. Slide over pressurization line into grommet in transition duct.

7. **QA** Lubricate nut with refrigerant lubricating oil, Item 190, WP 1803 00 (Figure 1, Detail B). Install nut on tee. **TORQUE NUT TO 75 - 125 INCH-POUNDS.**

8. **QA** Install over pressurization line on pallet with bolt, washer, spacer, and clamp. **TORQUE BOLT TO 20 - 25 INCH-POUNDS.**

9. **QA** Lubricate remaining nut with refrigerant lubricating oil, Item 190, WP 1803 00, and connect to tee at condenser outlet. **TORQUE NUT TO 75 - 125 INCH-POUNDS.**

INSTALL - Continued

10. Pressure test condenser pallet (WP 1312 00).
11. Install condenser pallet (WP 1312 00).
12. Service ECS (WP 1622 00).
13. Close and latch avionics compartment door.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM EVAPORATOR PALLET (AVIM) EH60A

INITIAL SETUP:

Tools and Special Tools
Container, 5-Gallon
Halogen Leak Detector
Refrigeration Unit Service Toolkit, SC 5180-90-N18
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Material/Parts
Cap, NAS817-4
Cap, NAS817-8
Cap, NAS817-10
Lubricating Oil, Refrigerant, Item 190, WP 1803 00
Nitrogen, Item 199, WP 1803 00
Plug, NAS818-4
Plug, NAS818-8
Plug, NAS818-10

Personnel Required
Utility Equipment Repairer MOS 52C (1)

References
WP 0150 00
WP 0258 00
WP 1289 00
WP 1622 00
WP 1703 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Open avionics compartment access door.
3. Deservice ECS (WP 1622 00).
4. Remove left soundproofing panel from rear cabin bulkhead (WP 0258 00).
5. Remove left hand plenum (WP 1289 00).
6. Disconnect and remove electrical connectors, P3 and P7, from left side of electrical pallet (Figure 1, Sheet 1, Detail B).
7. Loosen clamps holding flex duct to lower evaporator outlet transition duct and right supply duct (Figure 1, Sheet 1, Detail A). Remove flex duct.
8. Remove screw and strap holding upper supply duct to evaporator outlet transition duct. Slide duct off evaporator outlet transition duct.
9. Remove bolts and washers, and remove outlet transition duct.

CAUTION

Damage to air-conditioning parts will occur if moisture is allowed to enter air-conditioning system. To prevent moisture from entering air-conditioning system, make sure to cap all lines immediately after disconnecting them.

10. Disconnect outlet line, inlet line and equalization line at left rear of evaporator pallet (Figure 1, Sheet 1, Detail A and Detail C). Cap, NAS817-4, NAS817-8, and NAS817-10, fittings and plug, NAS818-4, NAS818-8, and NAS818-10, hoses.
11. Remove bolts and washers holding evaporator pallet to support channels.

REMOVE - Continued

CAUTION

Damage to evaporator pallet will result if components are used as handholds. To prevent damage to evaporator pallet, do not use lines, fittings, or other components as a handhold. Handle evaporator pallet by base only.

12. Remove evaporator pallet from helicopter.

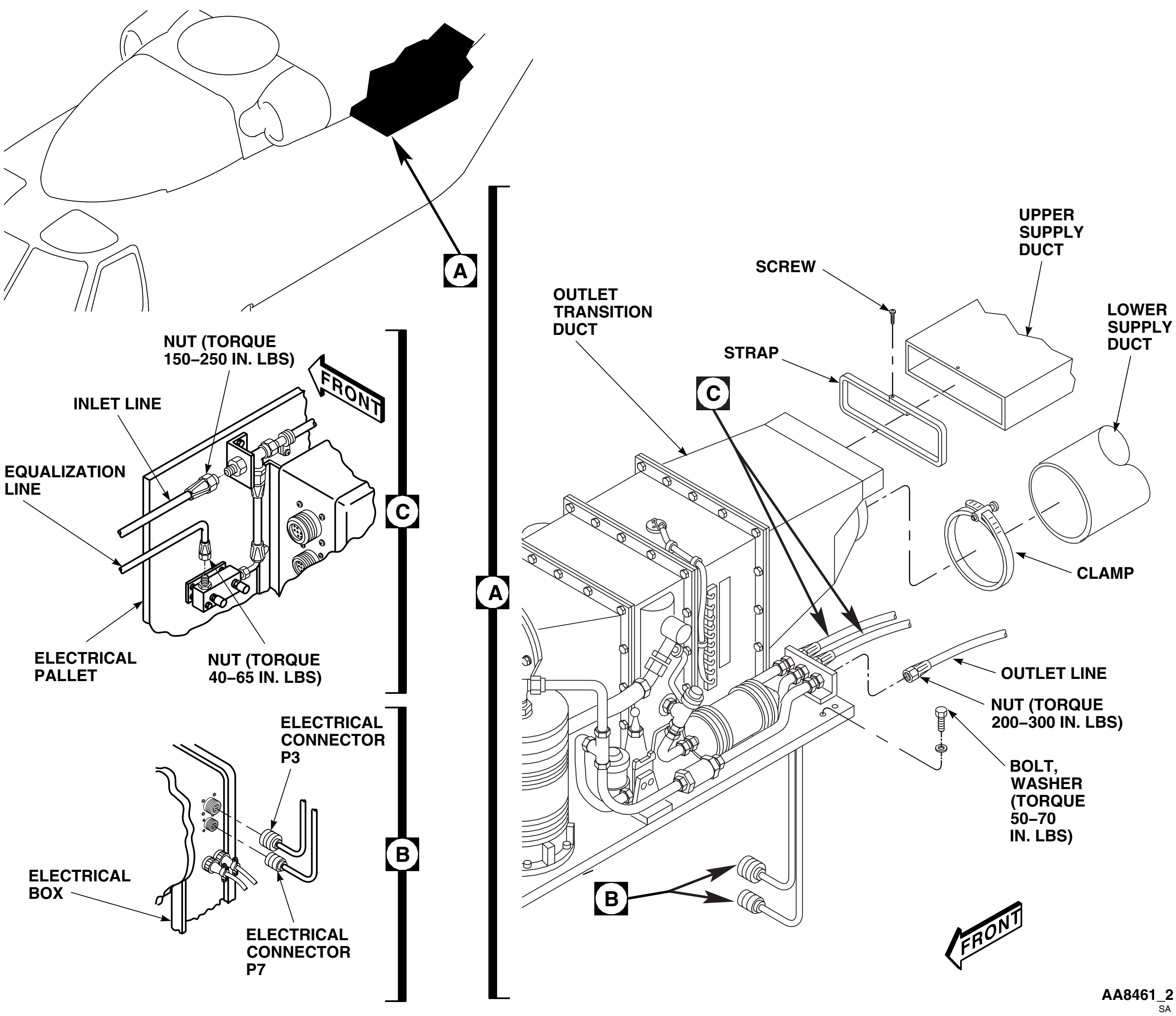


Figure 1. Replace Environmental Control System Evaporator Pallet (AVIM). (Sheet 1 of 2)

REMOVE - Continued

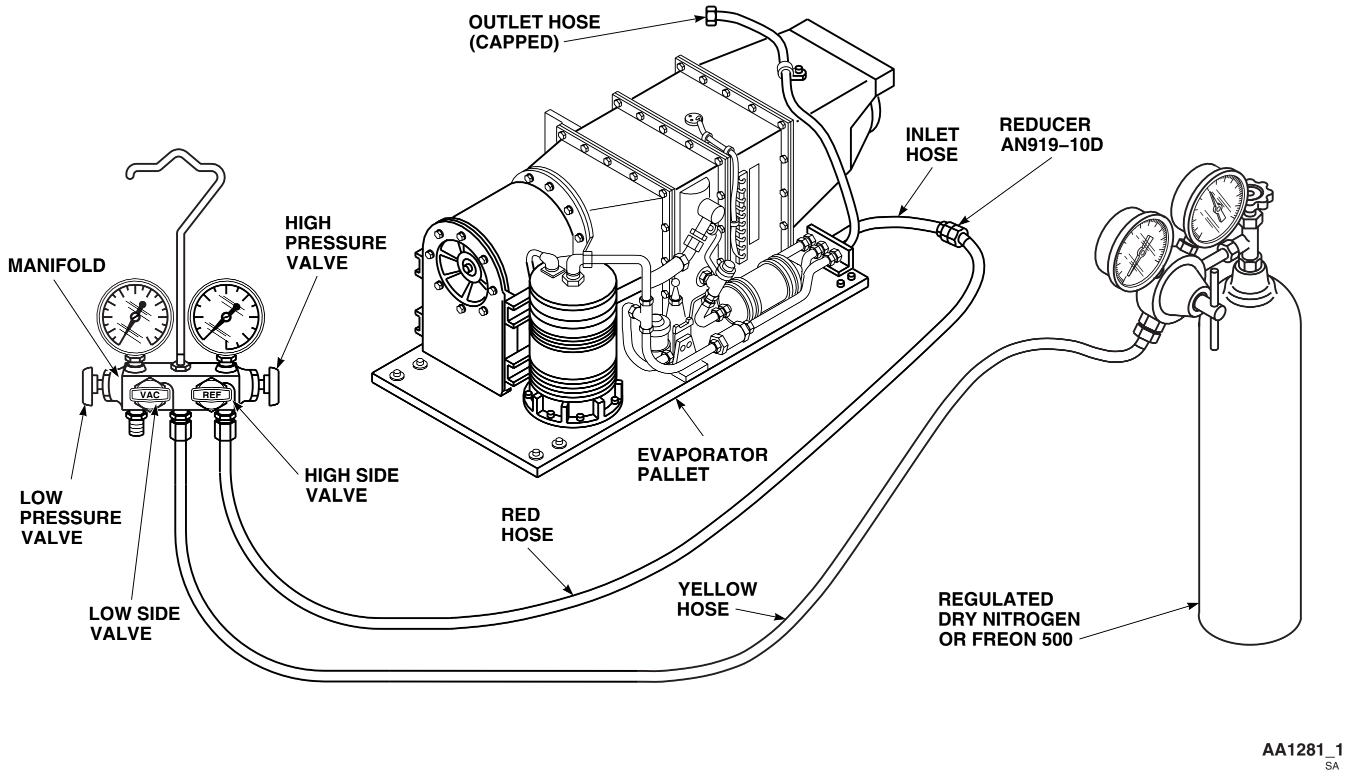


Figure 1. Replace Environmental Control System Evaporator Pallet (AVIM). (Sheet 2 of 2)

PRESSURE TEST

WARNING

- Injury to personnel will occur, if safety precautions are not observed when working on any part of the air conditioning system. Observe the following safety precautions when working on any part of the air conditioning system.
- Protect skin and eyes from possible contact with liquid refrigerant. Wear gloves and goggles. Keep your body as well protected as you can from possible contact with liquid refrigerant.
- If liquid refrigerant comes in contact with the eyes or skin, get immediate medical aid.
- The refrigerant in a sealed system is always under pressure. Do not disconnect any line or component of the pressurized cooling system without first carefully releasing the pressure.
- Freon-500 will displace air and can cause suffocation, even though it is non toxic. It will produce highly toxic phosgene gas if exposed to open flame. Make sure work area is well ventilated, and that there are no open flames, or unshielded heaters in immediate area.
- The use of heat on any part of the pressurized system will cause a pressure buildup and could cause an explosion. Do not try to solder or weld any part of the system while the system is charged with refrigerant.

1. Connect low pressure (yellow) hose to the VAC side of the servicing manifold and the high pressure (red) hose to the REF side of the servicing manifold. Make sure manifold hand wheels are closed (Figure 1, Sheet 2).
2. Uncap center fitting at rear of evaporator pallet and connect high pressure (red) hose. Install plug in hose outlet.
3. Connect yellow hose to Freon-500 bottle.

NOTE

Make sure Freon-500 bottle is positioned upright so only vapors will transfer.

4. Open valve on Freon-500 bottle and record the bottle pressure.
5. Close valve on Freon-500 bottle and disconnect yellow hose from vac side pressure gage.
6. Connect end of yellow hose to center supply port of servicing manifold.
7. Open Freon-500 bottle valve.
8. Purge yellow hose of air by loosening fitting at servicing manifold until a little Freon-500 vapor comes out. Immediately tighten fitting.
9. Open high side hand wheel until high side pressure gage equals Freon-500 pressure.
10. Close high side hand wheel and Freon-500 bottle valve. Disconnect yellow hose from Freon-500 bottle.

WARNING

Injury to personnel and damage to equipment will result if system pressure exceeds 300 psig during leak test. Make sure that nitrogen pressure gages are properly calibrated and regulator adjusting screw is set for 0 psig. Close manifold valves prior to opening supply valve on nitrogen bottle.

11. Connect yellow hose to regulated dry nitrogen, Item 199, WP 1803 00, source.

PRESSURE TEST - Continued

12. Open valve on nitrogen bottle and slowly increase regulator pressure until it equals Freon-500 system pressure.
13. Open REF hand wheel.
14. Slowly increase nitrogen pressure until system pressure indicated on manifold high pressure gage reaches 300 psi.
15. Using a halogen leak detector calibrated to detect leaks of 2 oz./year or less, check all fittings, connections, hoses, and components for leaks. Repair as necessary.
16. Repeat step 15. until no leaks are found.
17. Close high side hand wheel on servicing manifold.
18. Close nitrogen bottle valve and slowly crack connection fitting.
19. Disconnect yellow hose from nitrogen regulator fitting and place hose end into a clean 5 gallon container, 1/4 full of water.
20. Slowly open high side hand wheel until a small stream of bubbles appears in water.
21. Continue to drain system until no bubbles appear in container. Pressure gages should read 0 psig.
22. Close high side hand wheel and disconnect yellow hose.
23. Disconnect high pressure (red) hose from rear of evaporator pallet.
24. Immediately cap fitting.

INSTALL

CAUTION

Damage to evaporator pallet will result if components are used as handholds. To prevent damage to evaporator pallet, do not use lines, fittings, or other components as a handhold. Handle evaporator pallet by base only.

1. Turn off all electrical power.
2. **QA** Position evaporator pallet over fuel cells and attach to support channels with bolts and washers (Figure 1, Sheet 1, Detail A). **TORQUE BOLTS TO 50 - 70 INCH-POUNDS.**
3. Lubricate line nuts with refrigerant lubricating oil Item 190, WP 1803 00, prior to connecting.
4. **QA** Connect outlet line to outboard fitting on rear of evaporator pallet (Figure 1, Sheet 1, Detail A). **TORQUE OUTLET LINE NUT TO 200 - 300 INCH-POUNDS.**
5. **QA** Connect inlet line to middle fitting on rear of evaporator pallet (Figure 1, Sheet 1, Detail C). **TORQUE INLET LINE NUT TO 150 - 250 INCH-POUNDS.**
6. **QA** Connect equalization line to inboard fitting on rear of evaporator pallet (Figure 1, Sheet 1, Detail C). **TORQUE EQUALIZATION LINE NUT TO 40 - 65 INCH-POUNDS.**
7. Inspect and treat electrical connectors (WP 1703 00).
8. **QA** Connect electrical connectors P3 and P7 to left side of electrical pallet (Figure 1, Sheet 1, Detail B).
9. Install outlet transition duct on rear of evaporator pallet with bolts and washers.
10. **QA** Install upper supply duct on outlet transition duct with strap and screw (Figure 1, Sheet 1, Detail A).
11. **QA** Install flex duct on lower evaporator outlet transition duct and right supply duct with clamps.
12. Install left hand plenum (WP 1289 00).
13. Install left soundproofing panel on rear cabin bulkhead (WP 0258 00).

INSTALL - Continued

14. Service ECS (WP 1622 00).
15. Do operational check on ECS (WP 0150 00).
16. Close and latch avionics compartment access door.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM COMPRESSOR (AVIM) EH60A

INITIAL SETUP:

Tools and Special Tools

Container, 2-Quart
Refrigeration Unit Service Toolkit, SC 5180-90-N18
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Material/Parts

Alcohol, Denatured, Item 48, WP 1803 00
Cap, NAS817-10
Cap, NAS817-12
Lubricating Oil, Refrigerant, Item 190, WP 1803 00
Nitrogen, Item 199, WP 1803 00
Plug, NAS818-10
Plug, NAS818-12

Personnel Required

Utility Equipment Repairer MOS 52C (1)

References

WP 1312 00
WP 1315 00
WP 1317 00
WP 1319 00
WP 1320 00
WP 1622 00
WP 1703 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Remove evaporator pallet (WP 1315 00).
3. Disconnect electrical connector, P8, from compressor (Figure 1, Detail A).

CAUTION

Damage to air-conditioning parts will occur if moisture is allowed to enter air-conditioning system. To prevent moisture from entering air-conditioning system, make sure to cap all lines immediately after disconnecting them.

4. Remove nuts, and discharge line from top of compressor and tee. Plug, NAS818-10, line and cap, NAS817-10, fitting.
5. Loosen nut, and remove suction line from side of compressor. Plug, NAS818-12, line and cap, NAS817-12, fitting. Discard packing.
6. Remove bolts, washers, and nuts, noting location of clamp holding plenum temperature sensor wire. Remove compressor from pallet.

DISASSEMBLE

1. Turn off all electrical power.
2. Remove jamnut and elbow from top of compressor. Discard packing.
3. Using a funnel and an empty 2 quart container, remove sight gage from compressor, catching fluid as it exits compressor. Discard packing.

DISASSEMBLE - Continued

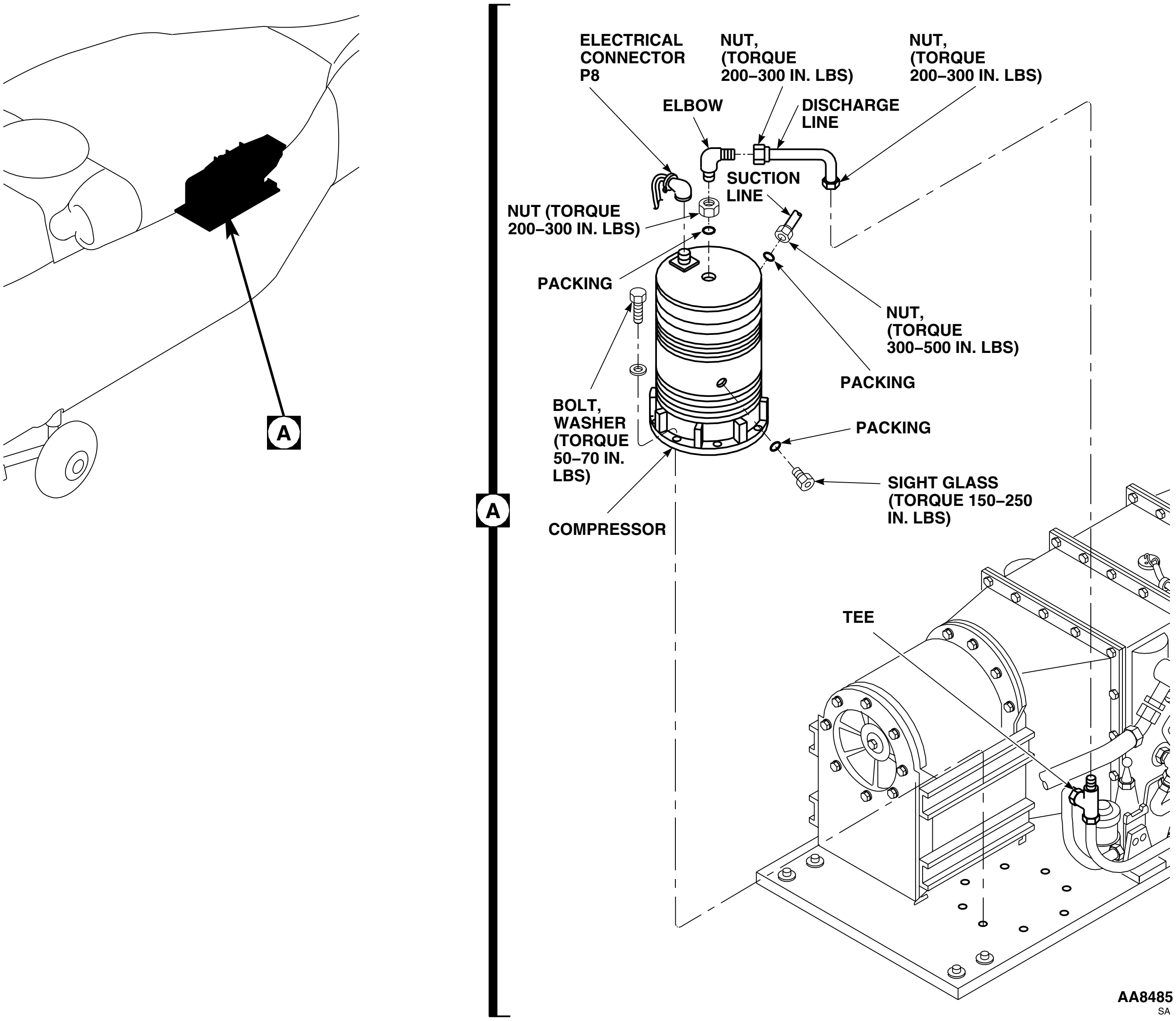


Figure 1. Replace Environmental Control System Compressor (AVIM).

CLEAN

CAUTION

Damage to the environmental control system will result if the system is not completely flushed during the replacement of the compressor. The failure of a compressor may release particulate matter into the refrigerant/compressor oil passages throughout the entire system. Make sure all parts, lines, and hoses which contain refrigerant/ compressor oil are flushed to remove any possible contaminates.

1. Remove condenser pallet (WP 1312 00).
2. Remove and disconnect hoses from electrical pallet (WP 1320 00).

CLEAN - Continued

3. Remove and discard filter/drier (WP 1319 00).
4. Remove hot gas bypass valve and associated lines from evaporator pallet (WP 1317 00).
5. Remove lines, burst disc, and relief valve from condenser pallet (WP 1312 00).
6. Flush condenser and evaporator heat exchanges with denatured alcohol, Item 48, WP 1803 00. Rotate heat exchangers several times on the forward and aft axis. Dump alcohol out, repeat until alcohol runs clear. Blow dry heat exchangers with dry nitrogen, Item 199, WP 1803 00.
7. Flush all plumbing lines and hoses with denatured alcohol, Item 48, WP 1803 00. Dump alcohol out, repeat until alcohol runs clear. Blow dry all plumbing with dry nitrogen, Item 199, WP 1803 00.
8. Flush expansion valve and check valve with denatured alcohol, Item 48, WP 1803 00. Dump alcohol out, repeat until alcohol runs clear. Blow dry valves with dry nitrogen, Item 199, WP 1803 00.
9. Install lines, burst disc, and relief valve on condenser pallet (WP 1312 00).
10. Install hot gas bypass valve and associated lines on evaporator pallet (WP 1317 00).
11. Install filter/dryer (WP 1319 00).
12. Install condenser pallet (WP 1312 00).
13. Install and connect hoses on electrical pallet (WP 1320 00).

ASSEMBLE

1. Turn off all electrical power.
2. **QA** Install sight glass on compressor with packing lubricated with refrigerant lubricating oil, Item 190, WP 1803 00. Do not torque sight gage.
3. Service compressor with compressor oil as follows:
   a. Check compressor oil level shown in sight glass. If completely full, overfilled condition is indicated: drain compressor oil until level appears in sight glass.
   b. Proper compressor oil level is no more than 1/2 full. If required, add refrigerant lubricating oil, Item 190, WP 1803 00, in 4-ounce increments with 2 minutes between for setting, through discharge line.
   c. **QA TORQUE SIGHT GLASS TO 150 - 250 INCH-POUNDS.**
4. Lubricate packing with refrigerant lubricating oil, Item 190, WP 1803 00.
5. **QA** Install elbow with packing on compressor. **TORQUE ELBOW TO 200 - 300 INCH-POUNDS.**

INSTALL

1. Turn off all electrical power.
2. **QA** Install compressor on pallet with bolts, washers, and nuts (Figure 1, Detail A). **TORQUE NUTS TO 50 - 70 INCH-POUNDS.**
3. Lubricate threads of suction line nut with refrigerant lubricating oil, Item 190, WP 1803 00.
4. **QA** Remove cap and install suction line on side of compressor with packing. **TORQUE NUT TO 300 - 500 INCH-POUNDS.**
5. Lubricate threads of discharge line nuts with refrigerant lubricating oil, Item 190, WP 1803 00.
6. **QA** Remove cap and install discharge line between compressor elbow and tee. **TORQUE NUTS TO 200 - 300 INCH-POUNDS.**

INSTALL - Continued

7. Inspect and treat electrical connector (WP 1703 00).
8. **QA** Install electrical connector, P8, on compressor.
9. Pressure test evaporator pallet (WP 1315 00).
10. Evacuate system for a minimum of two hours (WP 1622 00).
11. Service ECS system with refrigerant (WP 1622 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM HOT GAS BYPASS (HGBP) VALVE (AVIM) EH60A

INITIAL SETUP:

Tools and Special Tools

Refrigeration Unit Service Toolkit, SC 5180-90-N18
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Material/Parts

Cap, NAS817-4
Cap, NAS817-6
Cap, NAS817-10
Lubricating Oil, Refrigerant, Item 190, WP 1803 00
Plug, NAS818-4
Plug, NAS818-6
Plug, NAS818-10
Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

Utility Equipment Repairer MOS 52C (1)

References

WP 1315 00
WP 1319 00
WP 1803 00

REMOVE

1. Remove check valve as follows:
   a. Remove bolt, washer, spacer, and clamp, holding check valve to pallet (Figure 1, Sheet 1, Detail A).
   b. Loosen nuts at fitting bracket on rear of pallet and at compressor discharge line tee.

CAUTION

Damage to air-conditioning parts will occur if moisture is allowed to enter air-conditioning system. To prevent moisture from entering air-conditioning system, make sure to cap all lines immediately after disconnecting them.

   c. Remove lines, with check valve attached. Plug, NAS818-10, lines and cap, NAS817-10, fittings.
2. Remove filter/dryer (WP 1319 00).
3. Remove equalization line as follows:

CAUTION

Damage to air-conditioning parts will occur if moisture is allowed to enter air-conditioning system. To prevent moisture from entering air-conditioning system, make sure to cap all lines immediately after disconnecting them.

   a. Loosen nuts holding equalization lines on HGBP valve, evaporator outlet fitting, and fitting bracket at rear of pallet (Figure 1, Sheet 2, Detail C).
   b. Remove line assembly from pallet. Plug, NAS818-4, lines and cap, NAS817-4, fittings.
4. Remove HGBP valve as follows:

REMOVE - Continued

CAUTION

Damage to air-conditioning parts will occur if moisture is allowed to enter air-conditioning system. To prevent moisture from entering air-conditioning system, make sure to cap all lines immediately after disconnecting them.

a. Loosen nuts holding line on outlet of HGBP valve and evaporator inlet (Figure 1, Sheet 2, Detail D). Remove and plug, NAS818-6 and NAS818-10, line. Cap, NAS817-6, and NAS817-10, fittings.

b. Remove nut holding line on inlet of HGBP valve. Plug, NAS818-10, lines and cap, NAS817-10, fittings.

c. Loosen clamps holding electrical harness to pallet. Free harness enough to remove chafe guards and spiral wrap.

d. Disconnect and free the electrical leads of HGBP valve regulator from terminal 11 and terminal 12, of terminal board, TB4.

e. Remove bolts washers, HGBP valve, and bracket from pallet.

INSTALL

1. Install HGBP valve as follows:

   a. **QA** Install HGBP valve on bracket with bolts and washers (Figure 1, Sheet 2, Detail D). **TORQUE BOLTS TO 104 - 115 INCH-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire boltheads.

   b. **QA** Connect electrical leads of HGBP valve regulator to terminal 11 and terminal 12, of terminal board, TB4.

   c. **QA** Install spiral wrap and chafing strip around harness. **TORQUE CLAMP BOLTS TO 20 - 25 INCH-POUNDS.**

   d. Lubricate threads of compressor discharge line tee and HGBP valve inlet with refrigerant lubricating oil, Item 190, WP 1803 00.

   e. **QA** Install line between compressor discharge line tee and HGBP valve inlet. **TORQUE NUTS TO 200 - 300 INCH-POUNDS.**

   f. Lubricate threads of evaporator line and HGBP outlet with refrigerant lubricating oil, Item 190, WP 1803 00.

   g. **QA** Install line between evaporator inlet and HGBP valve outlet. **TORQUE NUTS TO 150 - 250 INCH-POUNDS.**

2. Install equalization line as follows:

   a. Lubricate threads of equalization line with refrigerant lubricating oil, Item 190, WP 1803 00.

   b. **QA** Install equalization line on HGBP valve, evaporator outlet fitting and fitting bracket at rear of pallet. **TORQUE ALL NUTS TO 40 - 65 INCH-POUNDS (Figure 1, Sheet 2, Detail C).**

3. Install filter/dryer (WP 1319 00).

4. Install check valve as follows:

   a. Lubricate threads of check valve line nuts with refrigerant lubricating oil, Item 190, WP 1803 00.

   b. **QA** Position check valve line assembly on pallet. Install on fitting bracket at rear of pallet and compressor discharge line tee (Figure 1, Sheet 1, Detail A). **TORQUE NUTS TO 200 - 300 INCH-POUNDS.**

   c. **QA** Install bolt, washer, spacers, and clamp on check valve. **TORQUE BOLT TO 20 - 25 INCH-POUNDS.**

   d. Pressure test evaporator pallet (WP 1315 00).

INSTALL - Continued

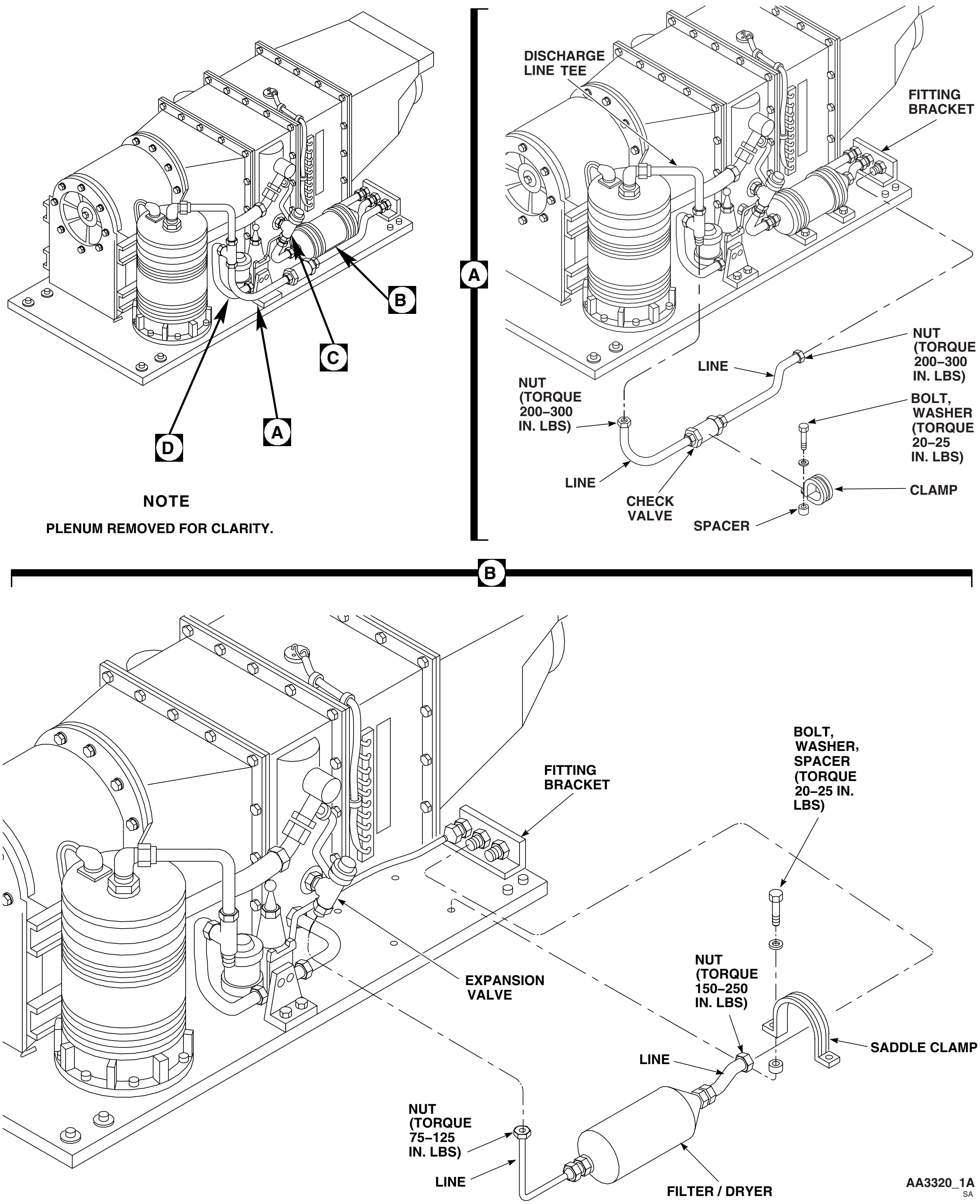
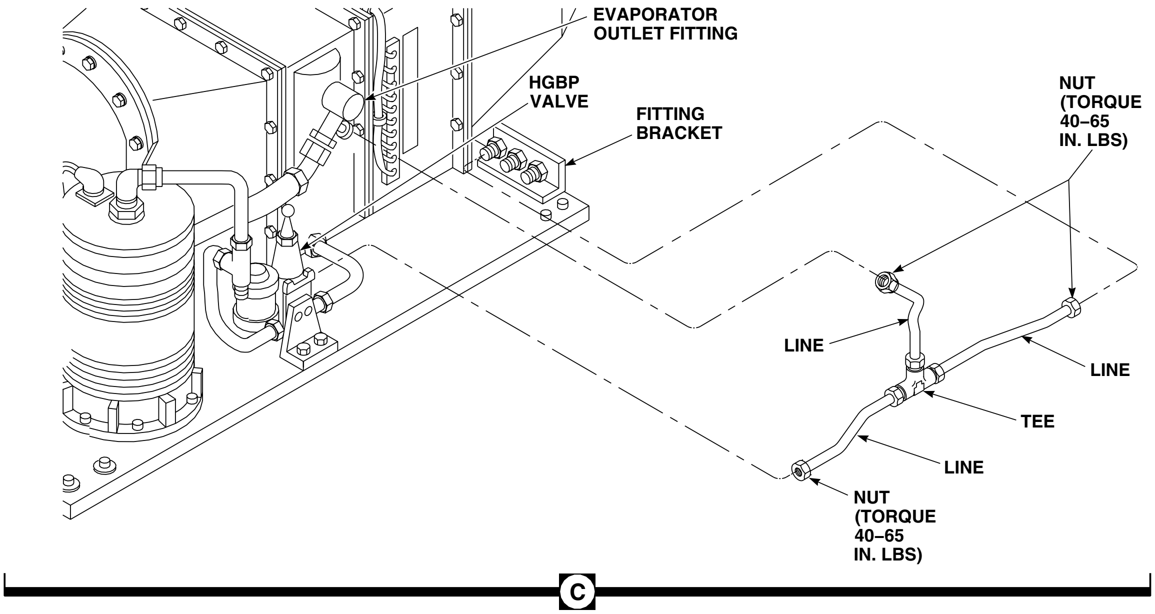


Figure 1. Replace Environmental Control System Hot Gas Bypass (HGBP) Valve (AVIM). (Sheet 1 of 2)

INSTALL - Continued



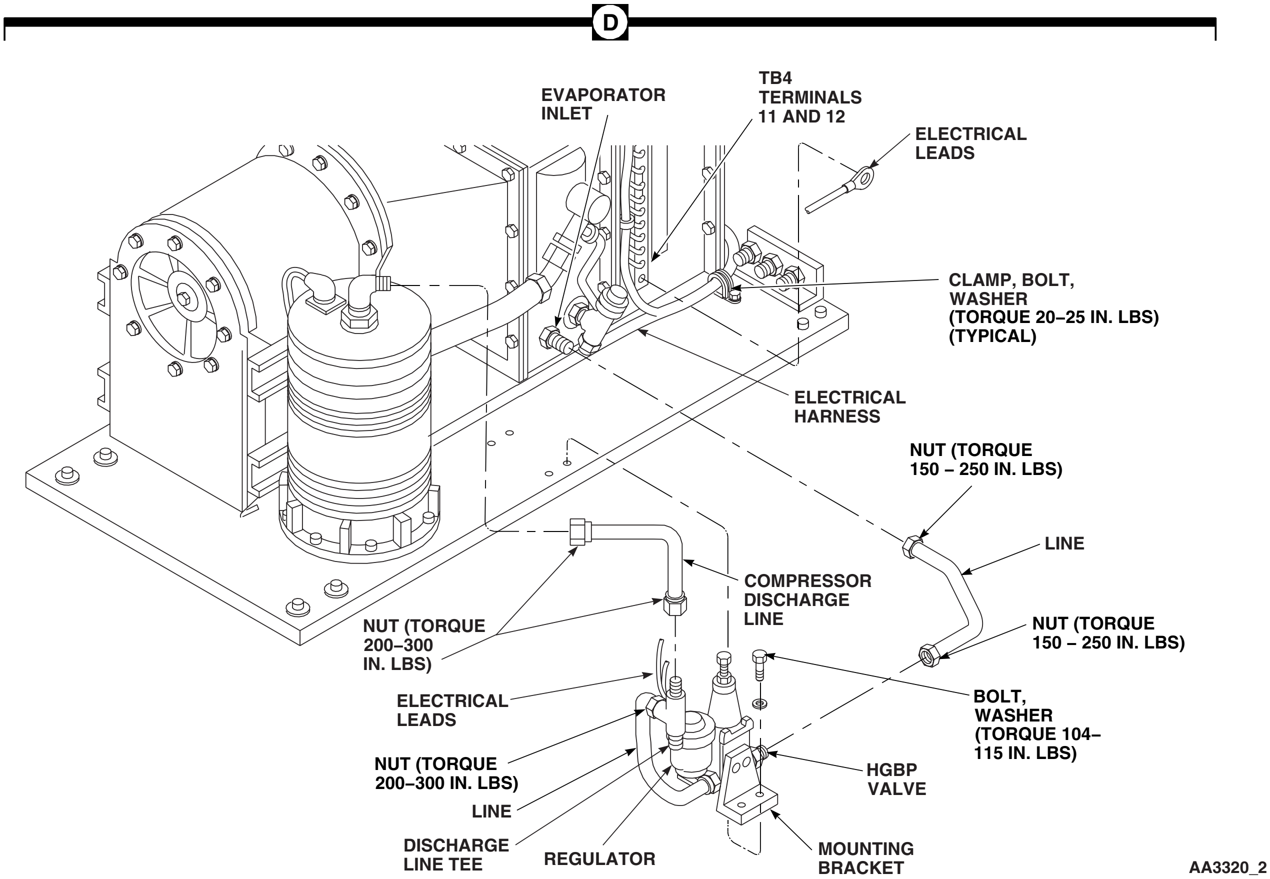


Figure 1. Replace Environmental Control System Hot Gas Bypass (HGBP) Valve (AVIM). (Sheet 2 of 2)

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM DUCTS, EVAPORATOR, HEATET/DEMISTER (AVIM) EH60A

INITIAL SETUP:

Tools and Special Tools

Electrical Repairer's Toolkit, SC 5180-99-B06
Nonmetallic Scraper
Refrigeration Unit Service Toolkit, SC 5180-90-N18
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Material/Parts

Adhesive, Item 40, WP 1803 00
Cap, NAS817-8
Cap, NAS817-12
Loctite, Item 178, WP 1803 00
Lubricating Oil, Refrigerant, Item 190, WP 1803 00
Plug, NAS818-4
Plug, NAS818-8
Plug, NAS818-12
Sealing Compound, Item 273, WP 1803 00
Tags

Personnel Required

Aircraft Electrical Repairer MOS 15F (1)
Utility Equipment Repairer MOS 52C (1)

References

WP 1283 00
WP 1315 00
WP 1317 00
WP 1703 00
WP 1803 00

REMOVE

NOTE

The inlet transition duct, evaporator heat exchanger, and the heater/demister are removed as an assembly from the pallet and disassembled thereafter.

1. Remove bolts and washers holding evaporator outlet transition duct to heater/demister (Figure 1, Sheet 1, Detail C).
2. Remove evaporator blower (WP 1283 00).

CAUTION

Damage to air-conditioning parts will occur if moisture is allowed to enter air-conditioning system. To prevent moisture from entering air-conditioning system, make sure to cap all lines immediately after disconnecting them.

3. Loosen nut holding compressor suction line to evaporator outlet (Figure 1, Sheet 2, Detail E). Plug, NAS818-12, line and cap, NAS817-12, fitting.
4. Remove hot gas bypass valve (WP 1317 00).
5. Remove thermistor from evaporator outlet, plug, NAS818-4, fitting.
6. Disconnect and tag wire from expansion valve.
7. Slit foam from around low temperature switch. Remove and tag switch (Figure 1, Sheet 1, Detail B).
8. Slit foam from around high temperature switch. Remove and tag switch.
9. Slit foam from around limiting switch. Remove and tag switch.

REMOVE - Continued

10. Disconnect and tag all electrical leads on heater/demister terminal board, TB4 (Figure 1, Sheet 1, Detail A).
11. Disconnect and tag heater element leads on heater/demister (Figure 1, Sheet 2, Detail G).
12. Remove electrical harness from clamps (Figure 1, Sheet 1, Detail D).
13. Disconnect demister drain tube.
14. Remove bolts, washers, and nuts holding assembly to pallet (Figure 1, Sheet 1, Detail B).
15. Remove bolts and washers holding heater/demister to evaporator.
16. Remove sealing compound from evaporator, heater/demister, and pallet with a nonmetallic scraper.

DISASSEMBLE

1. To disassemble heater/demister, do this:

NOTE

Each heat element is different. Do not attempt to interchange them.

   a. Remove nuts, washers, lockwashers, and fiber washers (inside housing) holding heater elements to housing. Remove heater elements. Discard hardware (Figure 1, Sheet 2, Detail E).

   b. Slit foam from around heater/demister and remove screws and nuts holding retaining brackets to heater/demister housing (Figure 1, Sheet 3, Detail H).

   c. Remove demister pad and screen.

2. Remove bolts and washers holding evaporator to inlet transition duct (Figure 1, Sheet 2, Detail E).
3. To disassemble evaporator, do this:

CAUTION

Damage to air-conditioning parts will occur if moisture is allowed to enter air-conditioning system. To prevent moisture from entering air-conditioning system, make sure to cap all lines immediately after disconnecting them.

   a. Remove union from outlet fitting. Plug, NAS818-4, fitting (Figure 1, Sheet 2, Detail E). Discard packing.

CAUTION

Damage to fitting will occur if fittings are not handled properly. To avoid cracking fitting, be careful when loosening nut holding expansion valve to evaporator inlet.

   b. Loosen nut holding expansion valve to evaporator. Plug, NAS818-8, fitting and cap, NAS817-8, expansion valve.

4. Remove bolts, washers, and evaporator inlet transition duct from bracket (Figure 1, Sheet 2, Detail F).
5. Remove bolts, washers, and baffles from evaporator inlet transition duct.

ASSEMBLE

1. **QA** Install baffles to evaporator inlet transition duct with bolts and washers (Figure 1, Sheet 2, Detail F). **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**
2. **QA** Install evaporator inlet transition duct to bracket with bolts and washers. **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**
3. To assemble evaporator, do this:

ASSEMBLE - Continued

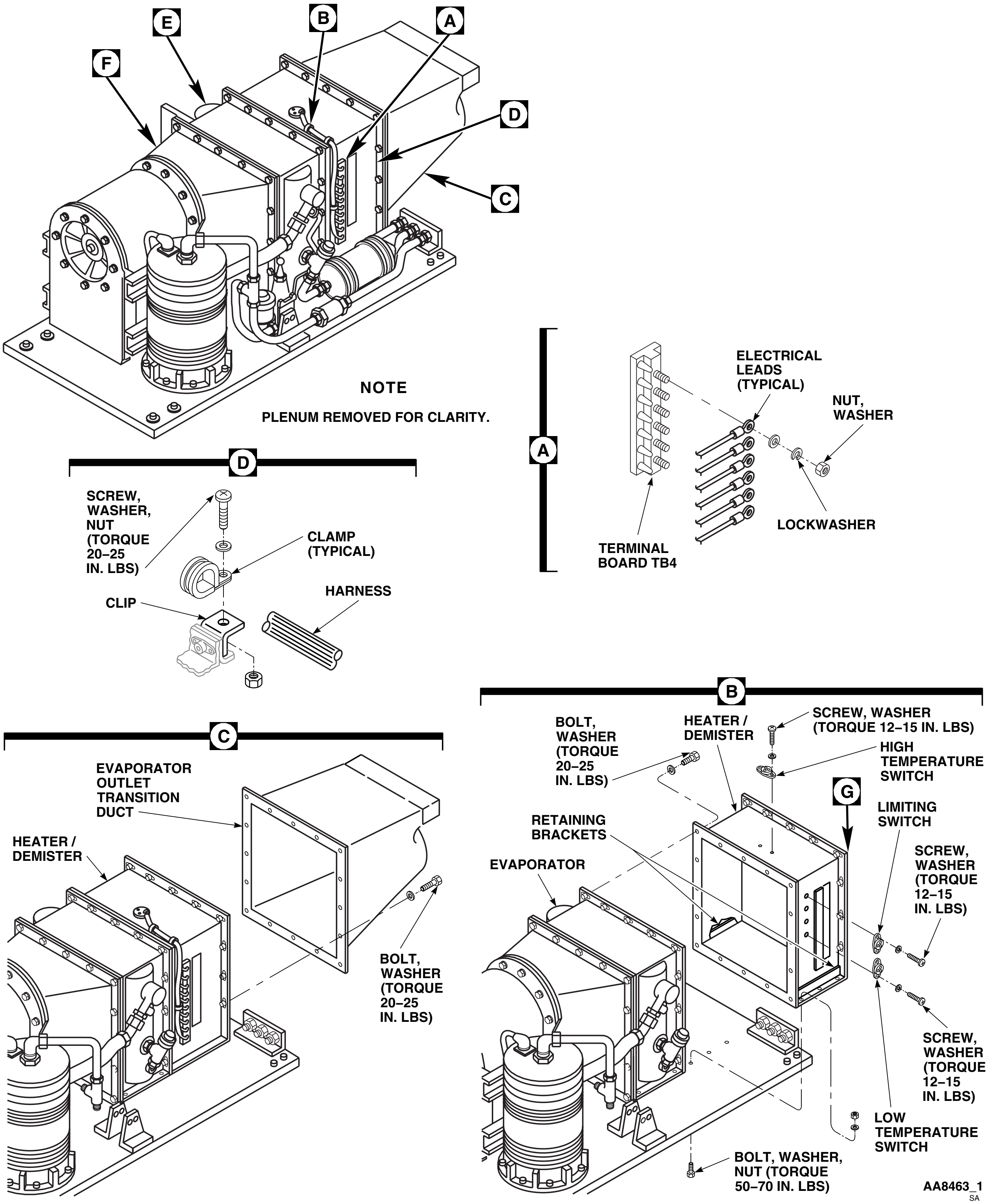


Figure 1. Replace Environmental Control System Ducts, Evaporator, Heater/Demister (AVIM). (Sheet 1 of 3)

ASSEMBLE - Continued

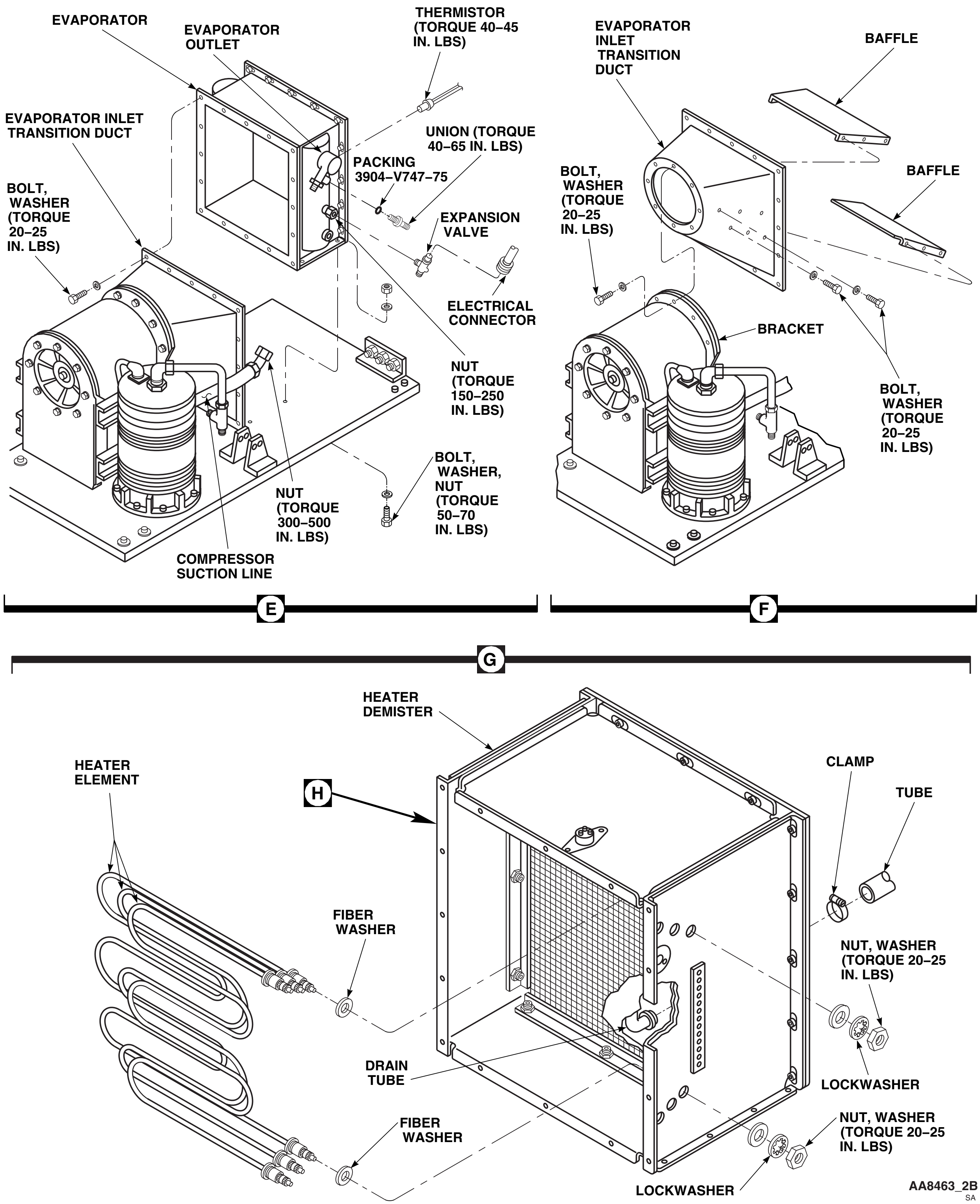


Figure 1. Replace Environmental Control System Ducts, Evaporator, Heater/Demister (AVIM). (Sheet 2 of 3)

ASSEMBLE - Continued

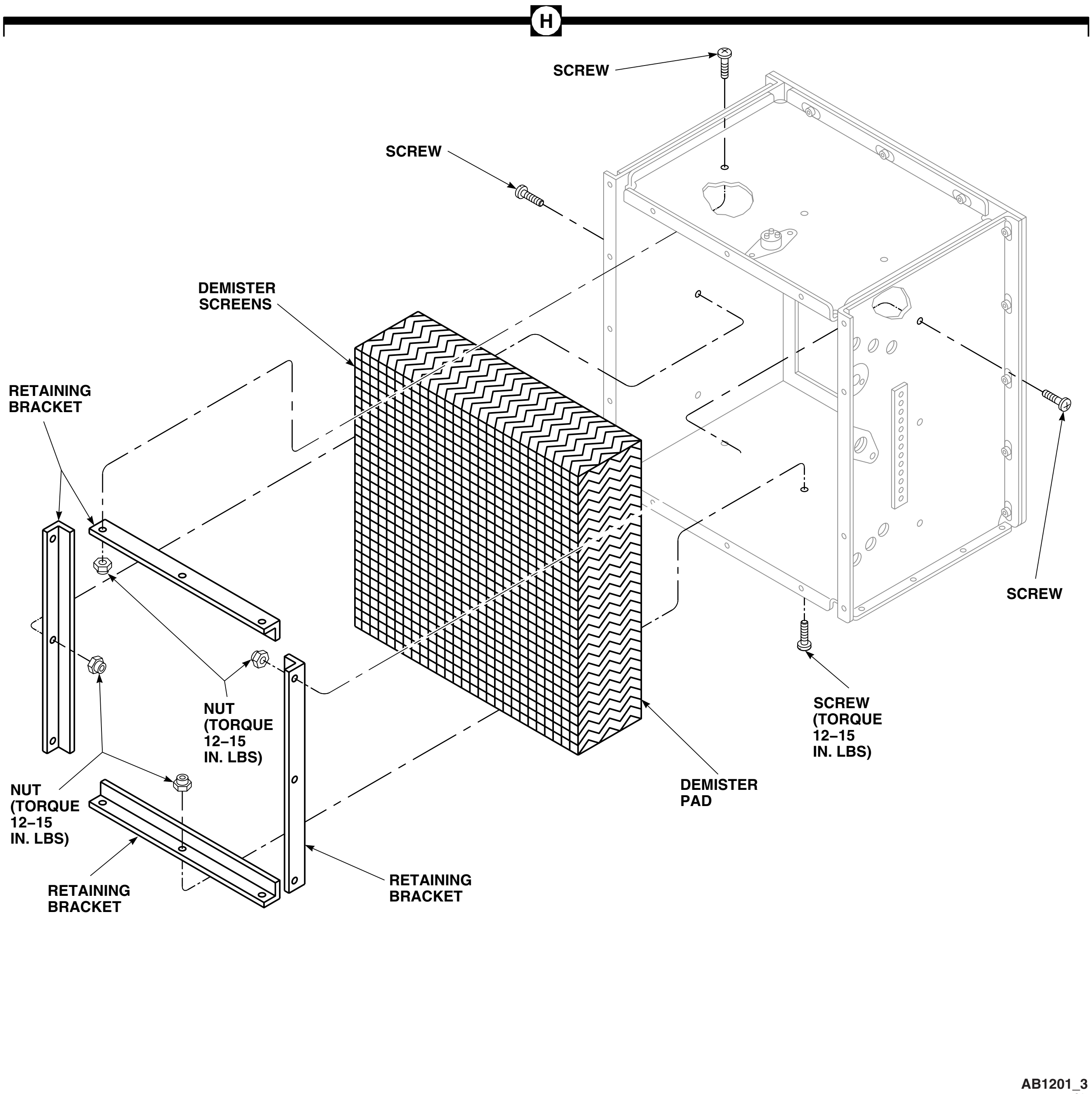


Figure 1. Replace Environmental Control System Ducts, Evaporator, Heater/Demister (AVIM). (Sheet 3 of 3)

ASSEMBLE - Continued

a. Remove cap and plug from expansion valve and fitting.

CAUTION

Damage to fitting will occur if fittings are not handled properly. To avoid cracking fitting, be careful when loosening nut holding expansion valve to evaporator inlet.

b. Lubricate nut on evaporator fittings with refrigerant lubricating oil, Item 190, WP 1803 00.

c. **QA** Connect expansion valve to evaporator fitting (Figure 1, Sheet 2, Detail E). **TORQUE NUT TO 150 - 250 INCH-POUNDS.**

d. Lubricate threads on union with refrigerant lubricating oil, Item 190, WP 1803 00.

e. **QA** Remove plug and connect union to evaporator outlet fitting using packing. **TORQUE UNION TO 40 - 65 INCH-POUNDS.**

4. **QA** Install evaporator on inlet transition duct with bolts and washers. **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**

5. To assemble heater/demister, do this:

a. **QA** Position demister pad and screen in housing. Install with retaining brackets, screws, and nuts (Figure 1, Sheet 3, Detail H). **TORQUE NUTS TO 12 - 15 INCH-POUNDS.**

NOTE

Each heater element is different. Do not attempt to interchange them.

b. **QA** Install heater elements in housing with new nuts, washers, lockwashers, and fiber washers (inside housing) (Figure 1, Sheet 2, Detail G). **TORQUE NUTS TO 20 - 25 INCH-POUNDS.**

6. **QA** Install heater/demister on evaporator with bolts and washers (Figure 1, Sheet 1, Detail B). **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**

INSTALL

1. **QA** Install and seal evaporator and heater/demister to pallet with sealing compound, Item 273, WP 1803 00, positioning gasket between transition duct and evaporator blower rear support bracket. Install evaporator mount bolts, washers and nuts (Figure 1, Sheet 1, Detail B). **TORQUE NUTS TO 50 - 70 INCH-POUNDS.**
2. **QA** Connect heater/demister drain tube (Figure 1, Sheet 2, Detail G).
3. Remove plug and cap.
4. Lubricate threads of compressor line with refrigerant lubricating oil, Item 190, WP 1803 00.
5. **QA** Connect compressor suction line to evaporator outlet (Figure 1, Sheet 2, Detail E). **TORQUE NUTS TO 300 - 500 INCH-POUNDS.**
6. **QA** Connect electrical harness to clips with clamps, screws, washers, and nuts (Figure 1, Sheet 1, Detail D). **TORQUE NUTS TO 20 - 25 INCH-POUNDS.**
7. **QA** Install limiting switch on housing with screws and washers (Figure 1, Sheet 1, Detail B). **TORQUE SCREWS TO 12 - 15 INCH-POUNDS.**
8. **QA** Install high temperature switch on housing with screws and washers. **TORQUE SCREWS TO 12 - 15 INCH-POUNDS.**
9. **QA** Install low temperature switch on housing with screws and washers. **TORQUE SCREWS TO 12 - 15 INCH-POUNDS.**

INSTALL - Continued

10. Repair foam insulation with adhesive, Item 40, WP 1803 00. Replace if required.
11. Remove plug and install thermistor into evaporator heat exchanger outlet fitting (Figure 1, Sheet 2, Detail E). Coat threads with Loctite, Item 178, WP 1803 00. Tighten as necessary.
12. **QA** Connect all electrical leads to terminal board, TB4, on heater/demister (Figure 1, Sheet 1, Detail A).
13. **QA** Connect heater element leads to heater/demister.
14. Inspect and treat electrical connector (WP 1703 00).
15. **QA** Install expansion valve electrical connector (Figure 1, Sheet 2, Detail E).
16. Install hot gas bypass valve (WP 1317 00).
17. Install evaporator blower (WP 1283 00).
18. **QA** Install evaporator outlet transition duct on heater/demister with bolts and washers (Figure 1, Sheet 1, Detail C). **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**
19. Pressure test evaporator pallet (WP 1311 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM FILTER/DRYER (AVIM) EH60A

INITIAL SETUP:

Tools and Special Tools

Refrigeration Unit Service Toolkit, SC 5180-90-N18
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Material/Parts

Cap, NAS817-8
Lubricating Oil, Refrigerant, Item 190, WP 1803 00
Plug, NAS818-8

Personnel Required

Utility Equipment Repairer MOS 52C (1)

References

WP 1622 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Deservice ECS (WP 1622 00).
3. Remove bolts, washers, and saddle clamps from filter/dryer.

CAUTION

Damage to filter/dryer will occur if moisture is allowed to enter. Filter/dryer is extremely sensitive to moisture and humidity. Replace filter/dryer if inlet and/or outlet connections are left uncapped for more than 10 minutes.

4. Loosen nuts that hold filter/dryer lines on expansion valve and fitting bracket on rear of pallet (Figure 1, Detail A).
5. Remove filter/dryer from pallet. Plug, NAS818-8, lines and cap, NAS817-8, fittings.

INSTALL

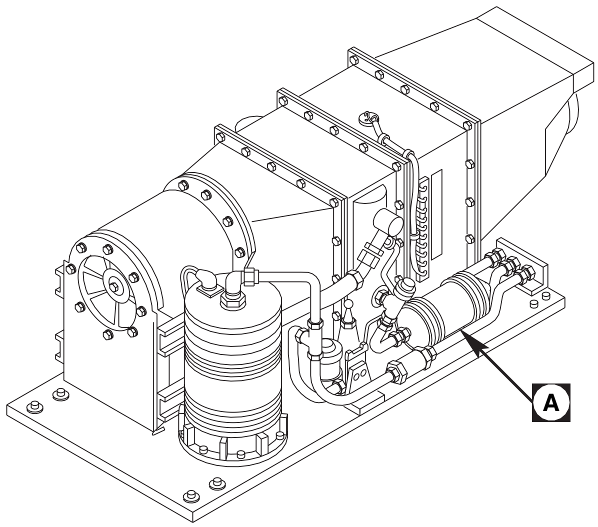
1. Turn off all electrical power.

CAUTION

Damage to filter/dryer will occur if moisture is allowed to enter. Filter/dryer is extremely sensitive to moisture and humidity. Replace filter/dryer if inlet and/or outlet connections are left uncapped for more than 10 minutes.

2. Remove caps and plugs.
3. Lubricate threads with refrigerant lubricating oil, Item 190, WP 1803 00.
4. **QA** Install filter/dryer on pallet with bolts, washers, spacers, and saddle clamps (Figure 1, Detail A). **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**
5. **QA** Install lines on expansion valve and fitting bracket at rear of pallet. **TORQUE NUT ON EXPANSION VALVE TO 75 - 125 INCH-POUNDS. TORQUE NUT ON FITTING BRACKET TO 150 - 250 INCH-POUNDS.**

INSTALL - Continued



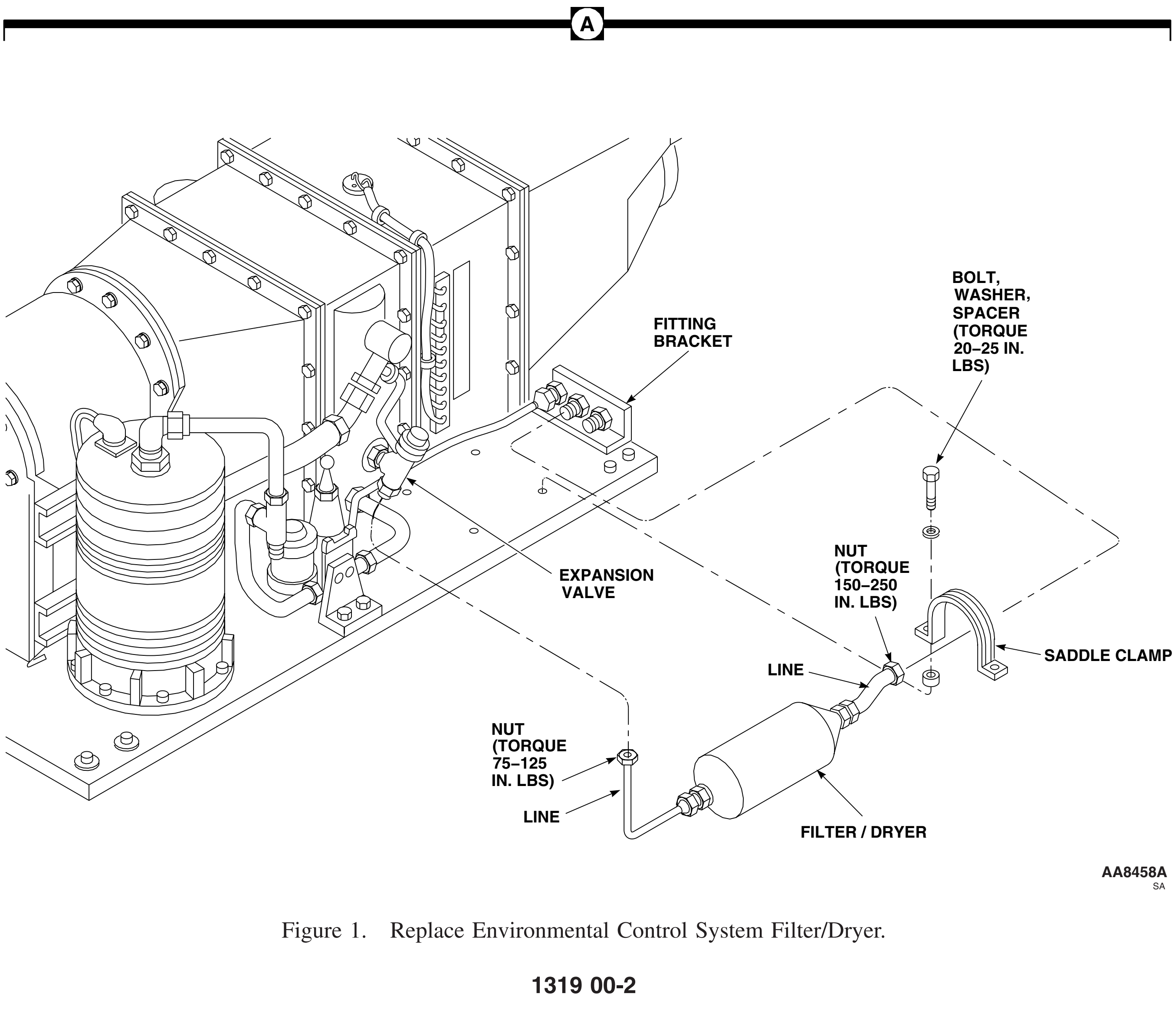


Figure 1. Replace Environmental Control System Filter/Dryer.

INSTALL - Continued

6. Service ECS (WP 1622 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM ELECTRICAL PALLET (AVIM) EH60A

INITIAL SETUP:

Tools and Special Tools

Refrigeration Unit Service Toolkit, SC 5180-90-N18
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Material/Parts

Cap, NAS817-4
Cap, NAS817-8
Lubricating Oil, Refrigerant, Item 190, WP 1803 00
Plug, NAS818-4
Plug, NAS818-8

Personnel Required

Utility Equipment Repairer MOS 52C (1)

References

WP 0150 00
WP 1622 00
WP 1703 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Deservice ECS (WP 1622 00).
3. Disconnect electrical connectors, P3, P4, P7, P456, and P457, from electrical box assembly (Figure 1).
4. Remove bolt, washer, spacer, and clamp holding inlet hose to pallet.

CAUTION

Damage to air-conditioning parts will occur if moisture is allowed to enter air-conditioning system. To prevent moisture from entering air-conditioning system, make sure to cap all lines immediately after disconnecting them.

5. Disconnect hoses at inlet and outlet side of sight glass assembly. Plug, NAS818-8, hoses and cap, NAS817-8, fittings.
6. Disconnect hose at servicing manifold. Plug, NAS818-4, hose and cap, NAS817-4, fitting.
7. Remove bolts and washers holding pallet to airframe.

CAUTION

Damage to ECS lines, fittings, or assemblies will occur if they are used as a hand hold. Do not use ECS lines, fittings, or assemblies as a hand hold. Handle pallet by base only.

8. Remove pallet from helicopter.

INSTALL

1. Turn off all electrical power.
2. **QA** Install electrical pallet in helicopter with bolts and washers (Figure 1). **TORQUE BOLTS TO 50 - 70 INCH-POUNDS.**

INSTALL - Continued

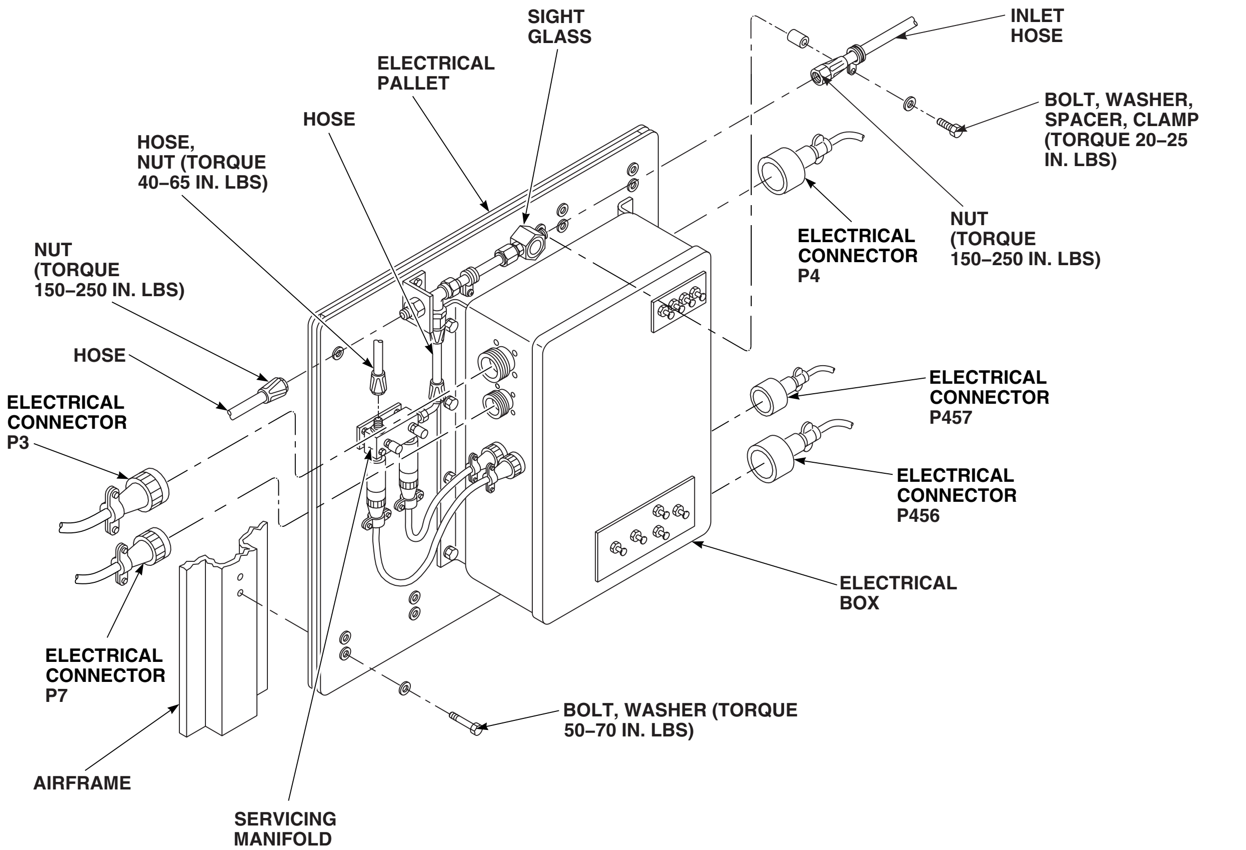


Figure 1. Replace Environmental Control System Electrical Pallet (AVIM).

INSTALL - Continued

3. Lubricate threads on hose with refrigerant lubricating oil, Item 190, WP 1803 00.
4. **QA** Remove plug and cap and connect hose to servicing manifold. **TORQUE NUT TO 40 - 65 INCH-POUNDS.**
5. Lubricate threads of hoses with refrigerant lubricating oil, Item 190, WP 1803 00.
6. Remove plugs and caps and connect hoses to sight glass assembly. **TORQUE NUT TO 150 - 250 INCH-POUNDS.**
7. **QA** Install inlet hose on pallet with bolts, washer, spacer, and clamp. **TORQUE BOLT TO 20 - 25 INCH-POUNDS.**
8. Inspect and treat electrical connectors (WP 1703 00).
9. **QA** Install electrical connectors, P3, P4, P7, P456, and P457, to electrical box assembly.
10. Service ECS (WP 1622 00).
11. Do an operational check of ECS (WP 0150 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM SERVICING MANIFOLD (AVIM) EH60A

INITIAL SETUP:

Tools and Special Tools

Refrigeration Unit Service Toolkit, SC 5180-90-N18
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.

Material/Parts

Cap, NAS817-4
Lubricating Oil, Refrigerant, Item 190, WP 1803 00
Plug, NAS818-4

Personnel Required

Utility Equipment Repairer MOS 52C (1)

References

WP 0150 00
WP 1323 00
WP 1622 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Deservice ECS (WP 1622 00).
3. Remove high and low pressure switches from servicing manifold (WP 1323 00).

CAUTION

Damage to air-conditioning parts will occur if moisture is allowed to enter air-conditioning system. To prevent moisture from entering air-conditioning system, make sure to cap all lines immediately after disconnecting them.

4. Remove hose assemblies from servicing manifold. Plug, NAS818-4, hoses and cap, NAS817-4, fittings (Figure 1, Detail B).
5. Remove bolts, washers, and spacers holding manifold to pallet.
6. Remove manifold from pallet.

INSTALL

1. Turn off all electrical power.
2. Remove any caps and plugs just prior to lubricating and installing hoses.
3. Lubricate threads of hose nuts with refrigerant lubricating oil, Item 190, WP 1803 00 prior to assembly.
4. **QA** Install manifold on pallet with bolts, washers, and spacers (Figure 1, Detail B). **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**
5. **QA** Install hose assemblies on manifold fittings. **TORQUE NUTS TO 40 - 65 INCH-POUNDS.**
6. Install high and low pressure switches in servicing manifold (WP 1323 00).
7. Service ECS (WP 1622 00).

INSTALL - Continued

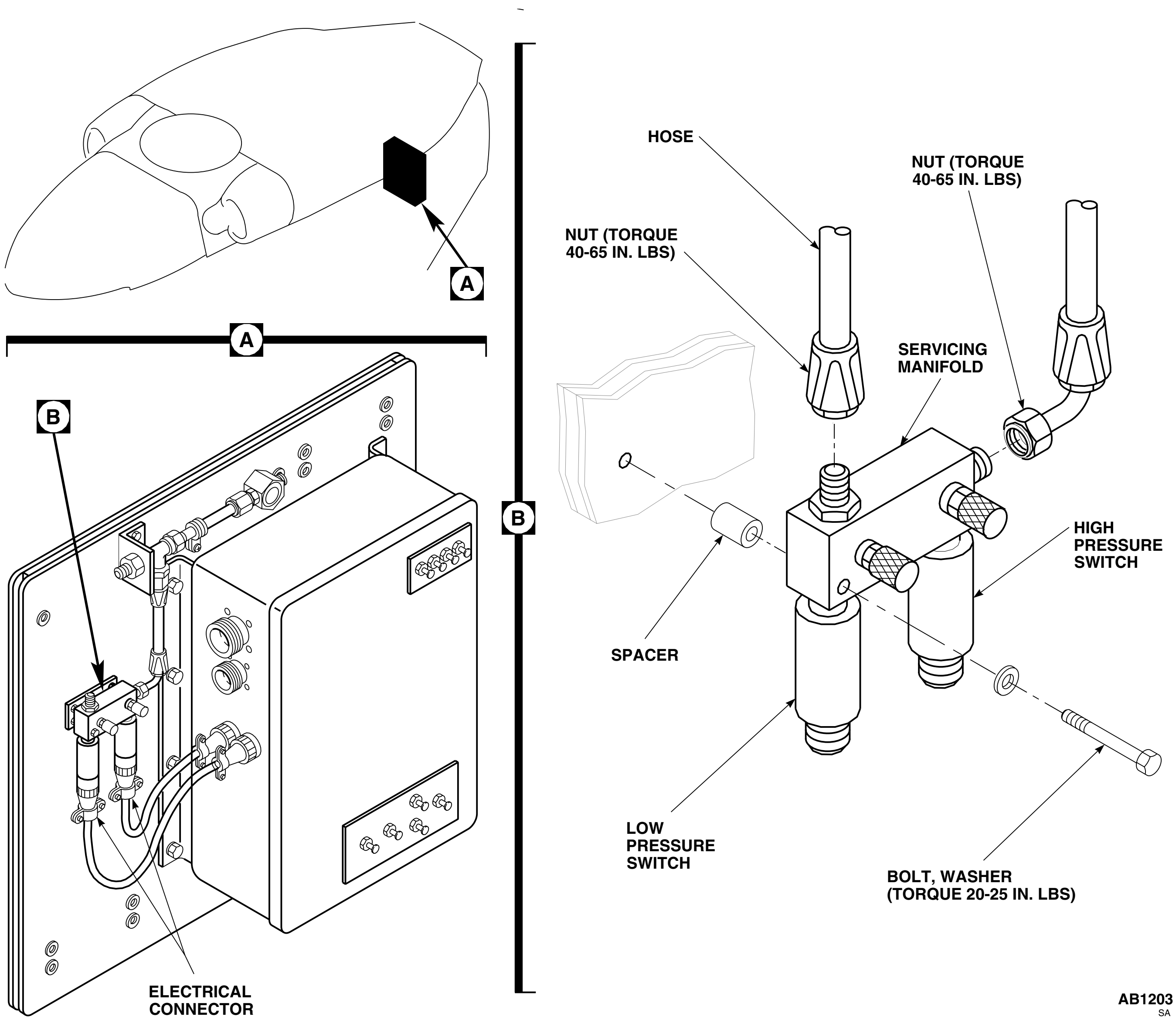


Figure 1. Replace Environmental Control System Servicing Manifold (AVIM).

8. Do an operational check of ECS (WP 0150 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM SIGHT GLASS (AVIM) EH60A

INITIAL SETUP:

Tools and Special Tools
Refrigeration Unit Service Toolkit, SC 5180-90-N18
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Material/Parts
Cap, NAS817-8
Lubricating Oil, Refrigerant, Item 190, WP 1803 00
Plug, NAS818-8

Personnel Required
Utility Equipment Repairer MOS 52C (1)

References
WP 1622 00
WP 1803 00

REMOVE

1. Deservice ECS (WP 1622 00).

CAUTION

Damage to air-conditioning parts will occur if moisture is allowed to enter air-conditioning system. To prevent moisture from entering air-conditioning system, make sure to cap all lines immediately after disconnecting them.

2. Disconnect hose from ends of sight glass. Plug, NAS818-8, hoses and cap, NAS817-8, fittings (Figure 1).
3. Remove bolt, washer, and spacer from clamp and slide hose back.
4. Remove sight glass.

INSTALL

1. Position sight glass between ends of hoses.
2. Remove plugs from hoses and caps from fittings.
3. Lubricate threads on hoses with refrigerant lubricating oil, Item 190, WP 1803 00.
4. Loosely connect hoses to sight glass (Figure 1).
5. **QA** Install bolt, washer, and spacer in clamp. **TORQUE BOLT TO 20 - 25 INCH-POUNDS.**
6. **QA TORQUE NUTS ON HOSES TO 150 - 250 INCH-POUNDS.**
7. Service ECS (WP 1622 00).

INSTALL - Continued

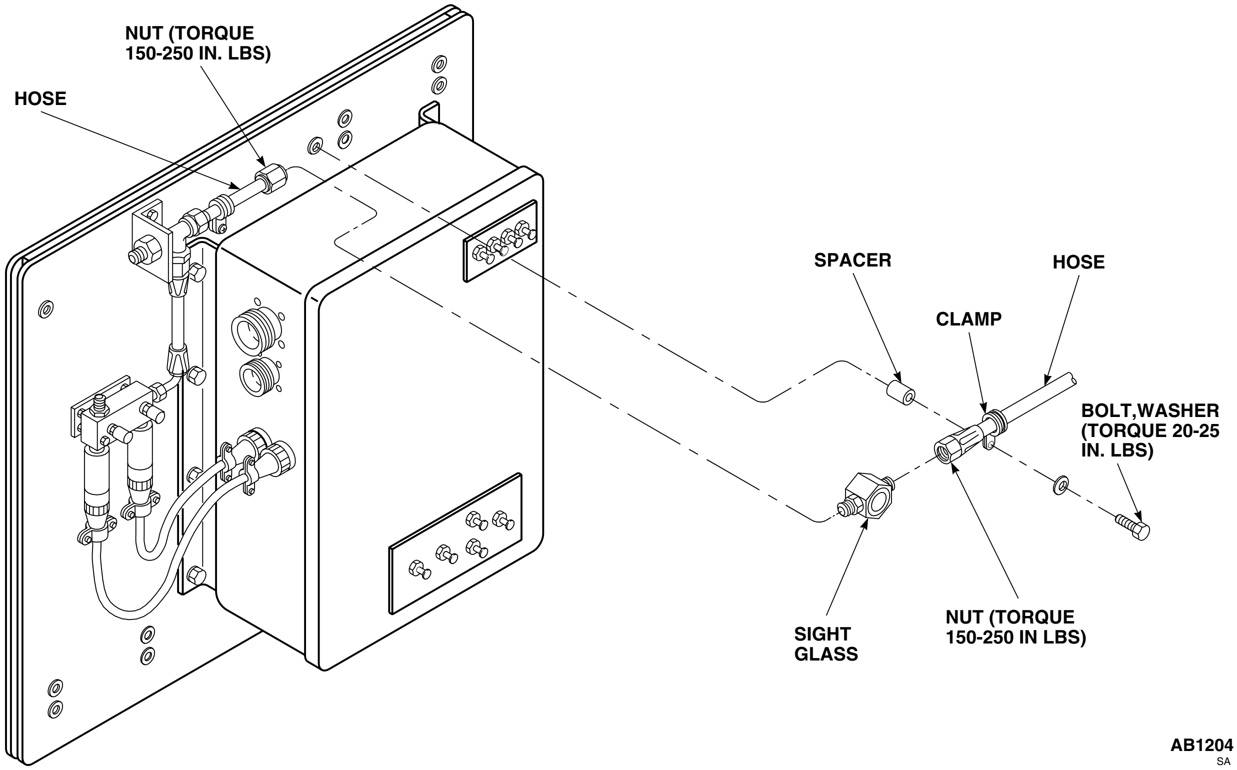


Figure 1. Replace Environmental Control System Sight Glass (AVIM).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM HIGH AND LOW PRESSURE SWITCHES (AVIM) EH60A

INITIAL SETUP:

Tools and Special Tools
Refrigeration Unit Service Toolkit, SC 5180-90-N18

Personnel Required
Utility Equipment Repairer MOS 52C (1)

References
WP 0150 00
WP 1622 00
WP 1703 00

REMOVE

1. Turn off all electrical power.
2. Deservice ECS (WP 1622 00).
3. Disconnect electrical connector, P11, from high pressure switch (Figure 1, Detail A).
4. Disconnect electrical connector, P12, from low pressure switch.

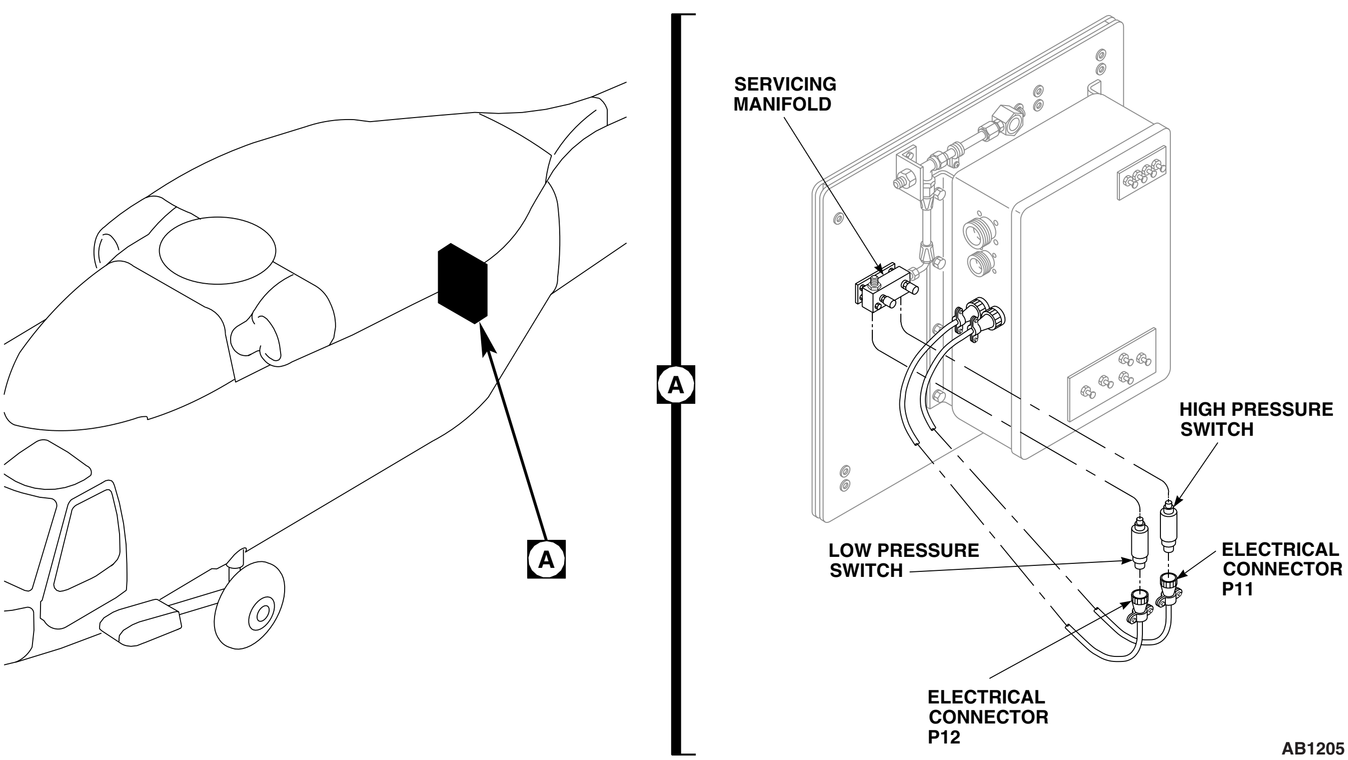


Figure 1. Replace Environmental Control System High and Low Pressure Switches (AVIM).

REMOVE - Continued

NOTE

One or both fittings may come out with switches. The fitting must be removed from switch for use with replacement switch.

5. Remove high and low pressure switches from fittings on servicing manifold.

INSTALL

1. Turn off all electrical power.
2. If either fitting was removed from servicing manifold, reinstall fittings, with new packings, in servicing manifold.
3. **QA** Install switches in fittings (Figure 1, Detail A).
4. Inspect and treat electrical connectors (WP 1703 00).
5. **QA** Connect electrical connector, P11, to high pressure switch
6. **QA** Connect electrical connector, P12, to low pressure switch.
7. Service ECS (WP 1622 00).
8. Do an operational check of ECS (WP 0150 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM CONDENSER PALLET PRESSURE TEST (AVIM) HH-60A HH-60L

This WP supersedes WP 1324 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Container, 5-Gallon
Halogen Leak Detector
Refrigeration Unit Service Toolkit, SC 5180-90-N18

Material/Parts
Lubricating Oil, Refrigerant, Item 191, WP 1803 00
Nitrogen, Item 199, WP 1803 00
Protective Caps and Plugs

Personnel Required
Utility Equipment Repairer MOS 52C (1)

References
WP 1803 00

CONDENSER PALLET PRESSURE TEST

WARNING

- Injury to personnel will occur, if safety precautions are not observed when working on any part of the air conditioning system. Observe the following safety precautions when working on any part of the air conditioning system.
- Protect skin and eyes from possible contact with liquid refrigerant. Wear gloves and goggles. Keep your body as well protected as you can from possible contact with liquid refrigerant.
- If liquid refrigerant comes in contact with eyes or skin, get immediate medical aid.
- Refrigerant in a sealed system is always under pressure. Do not disconnect any line or component of pressurized cooling system without first carefully releasing pressure.
- Refrigerant, R-407C will displace air and can cause suffocation, even though it is nontoxic. It will produce highly toxic phosgene gas if exposed to open flame. Make sure work area is well ventilated, and that there are no open flames, or unshielded heaters in immediate area.
- Use of heat on any part of pressurized system will cause a pressure buildup and could cause an explosion. Do not try to solder or weld any part of system while system is charged with refrigerant.

1. Connect high pressure (red) hose to high side of servicing manifold (Figure 1, Detail B). Make sure manifold hand wheels are closed.
2. Uncap condenser inlet port and connect other end of high pressure (red) hose.
3. Install cap on condenser outlet port.
4. Connect yellow hose to refrigerant or nitrogen bottle and vac side pressure gage.

CONDENSER PALLET PRESSURE TEST - Continued

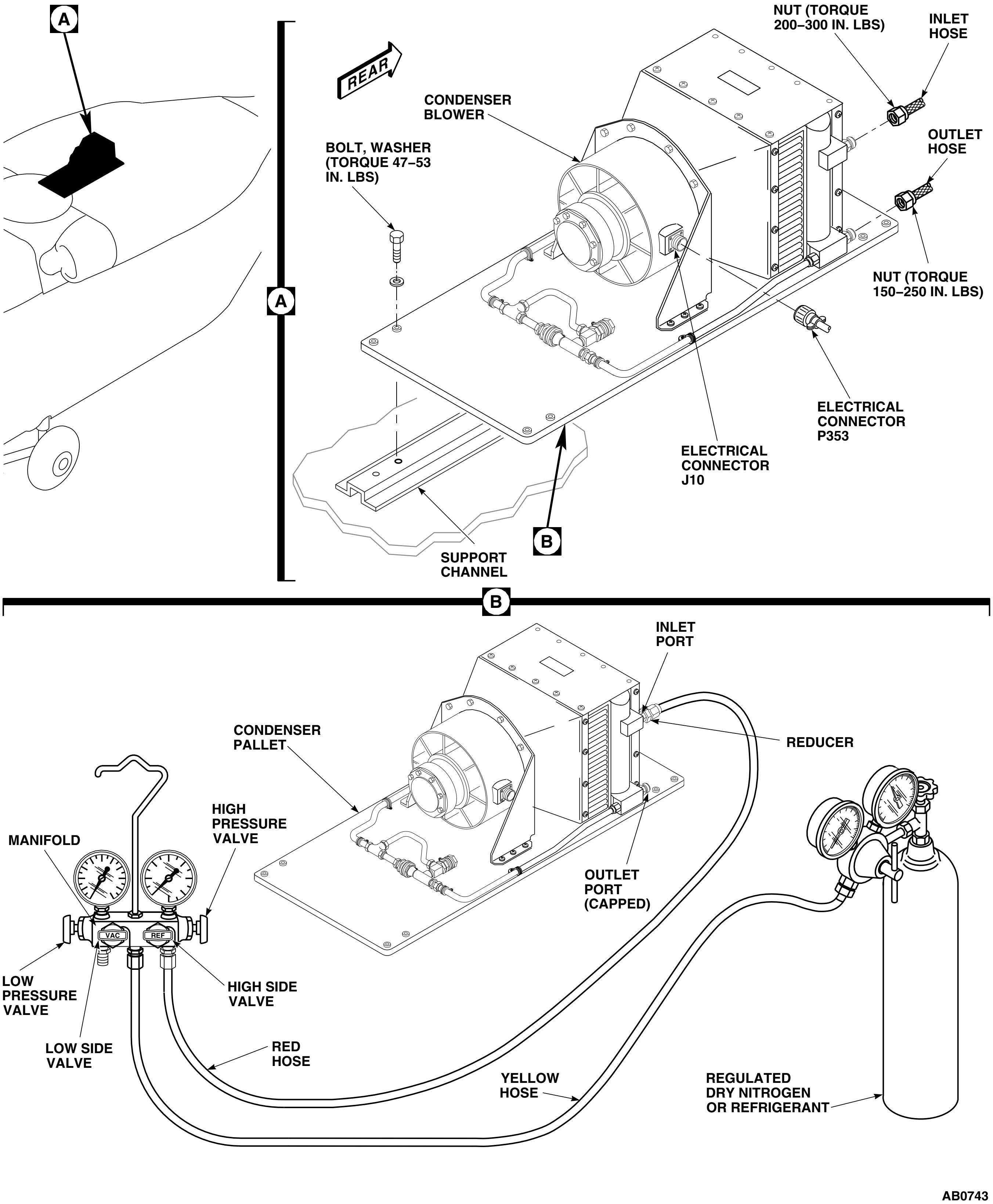


Figure 1. Environmental Control System Condenser Pallet Pressure Test.

CONDENSER PALLET PRESSURE TEST - Continued

NOTE

Make sure refrigerant or nitrogen bottle is kept upright so that only vapors will transfer.

5. Open valve on refrigerant or nitrogen bottle and record bottle pressure.
6. Close valve on refrigerant or nitrogen bottle and disconnect yellow hose from vac side gage.
7. Connect end of yellow hose to center supply port of servicing manifold.
8. Open refrigerant or nitrogen bottle valve.
9. Purge yellow hose of air by loosening fitting at servicing manifold. A small amount of refrigerant vapor may come out. Immediately tighten fitting.
10. Open high side hand wheel until high side pressure gage equals refrigerant or nitrogen bottle pressure.
11. Close high side hand wheel and bottle valve. Disconnect yellow hose from refrigerant or nitrogen bottle.

WARNING

Injury to personnel and damage to equipment will result if system pressure exceeds 300 psig during leak test. Make sure that nitrogen pressure gages are properly calibrated and regulator adjusting screw is set for 0 psig. Close manifold valves prior to opening supply valve on nitrogen bottle.

12. Unscrew adjustment on refrigerant lubricating oil, Item 191, WP 1803 00, or nitrogen, Item 199, WP 1803 00, source, until it is loose. Connect yellow hose to regulator.
13. Open valve on refrigerant or nitrogen bottle and slowly increase regulator pressure until it equals refrigerant system pressure.
14. Open REF hand wheel.
15. Slowly increase refrigerant or nitrogen bottle pressure until system pressure indicated on manifold high pressure gage reaches 300 psi.
16. Using a Halogen leak detector calibrated to detect leaks of 2 oz./year or less, check all fittings, connections, hoses and components for leaks. Repair as necessary.
17. Repeat step 16. until no leaks are found.
18. Close high side hand wheel on servicing manifold.
19. Close refrigerant or nitrogen bottle valve and slowly crack connection fitting.
20. Disconnect yellow hose from refrigerant or nitrogen regulator fitting and place loose end into a clean 5 gallon container, 1/4 full of water.
21. Slowly open high side hand wheel until a small stream of bubbles appears in water.
22. Continue to drain system until no bubbles appear in container. Pressure gage should read 0 psig.
23. Close high side hand wheel and disconnect yellow hose.
24. Disconnect high pressure (red) hose from condenser inlet port.
25. Immediately cap condenser inlet port.

CONDENSER PALLET PRESSURE TEST - Continued

EVAPORATOR PALLET PRESSURE TEST

WARNING

- Injury to personnel will occur, if safety precautions are not observed when working on any part of the air conditioning system. Observe the following safety precautions when working on any part of the air conditioning system.
- Protect skin and eyes from possible contact with liquid refrigerant. Wear gloves and goggles. Keep your body as well protected as you can from possible contact with liquid refrigerant.
- If liquid refrigerant comes in contact with the eyes or skin, get immediate medical aid.
- The refrigerant in a sealed system is always under pressure. Do not disconnect any line or component of the pressurized cooling system without first carefully releasing the pressure.
- Refrigerant, R407C will displace air and can cause suffocation, even though it is non toxic. It will produce highly toxic phosgene gas if exposed to open flame. Make sure work area is well ventilated, and that there are no open flames, or unshielded heaters in immediate area.
- The use of heat on any part of the pressurized system will cause a pressure buildup and could cause an explosion. Do not try to solder or weld any part of the system while the system is charged with refrigerant.

1. Connect low pressure (yellow) hose to vac side of servicing manifold and high pressure (red) hose to REF side of servicing manifold. Make sure manifold hand wheels are closed (Figure 2, Detail B).
2. Uncap center fitting at rear of evaporator pallet and connect high pressure (red) hose.
3. Install cap on outlet fitting and equalization fitting.
4. Connect yellow hose to refrigerant or nitrogen bottle.

NOTE

Make sure refrigerant or nitrogen bottle is positioned upright so only vapors will transfer.

5. Open valve on refrigerant or nitrogen bottle and record bottle pressure.
6. Close valve on refrigerant or nitrogen bottle and disconnect yellow hose from vac side pressure gage.
7. Connect end of yellow hose to center supply port of servicing manifold.
8. Open refrigerant or nitrogen bottle valve.
9. Purge yellow hose of air by loosening fitting at servicing manifold. A small amount of refrigerant vapor may come out. Immediately tighten fitting.
10. Open high side hand wheel until high side pressure gage equals refrigerant or nitrogen bottle pressure.
11. Close high side hand wheel and refrigerant or nitrogen bottle valve. Disconnect yellow hose from refrigerant or nitrogen bottle.

EVAPORATOR PALLET PRESSURE TEST - Continued

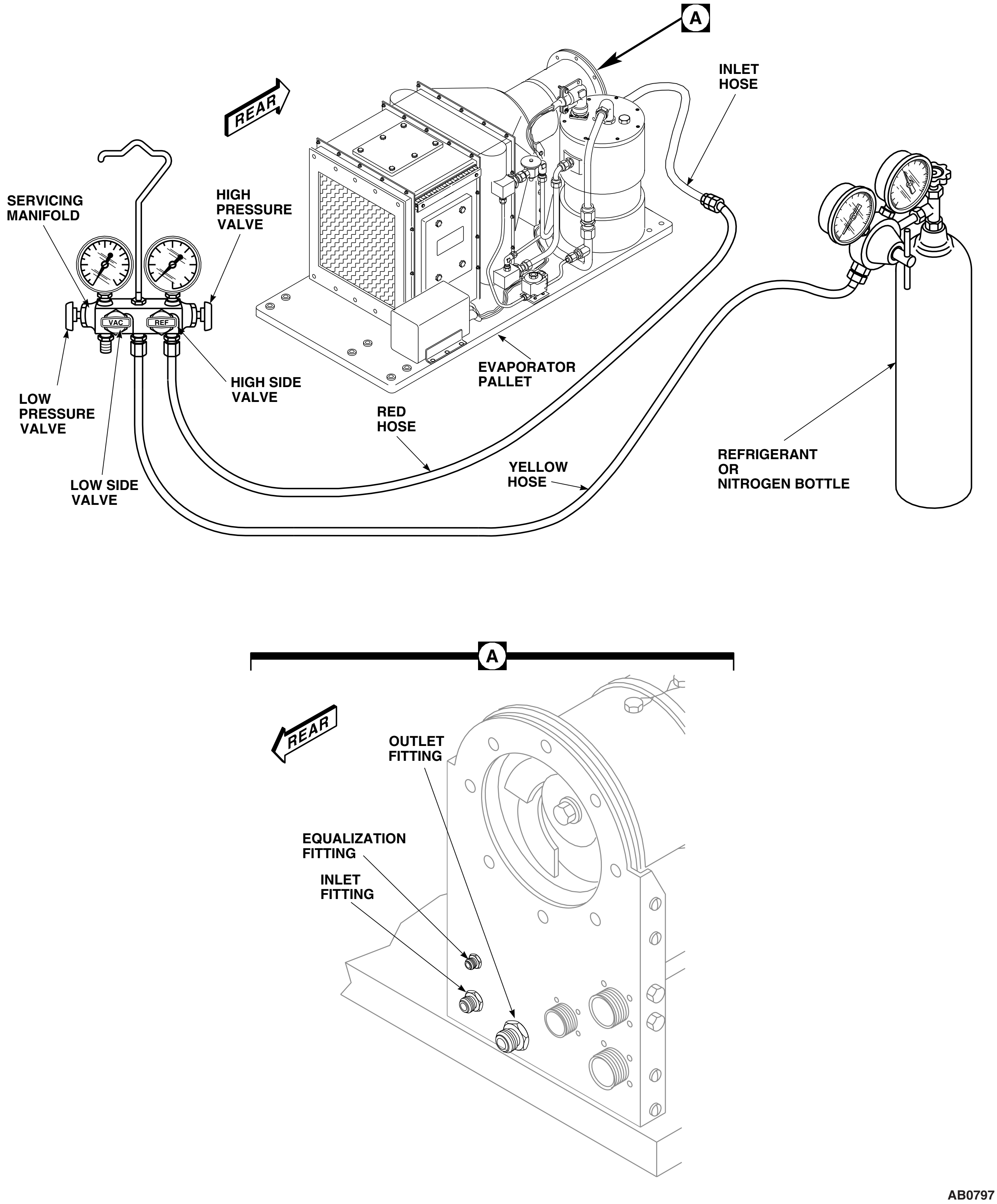


Figure 2. Environmental Control System Evaporator Pallet Pressure Test.

EVAPORATOR PALLET PRESSURE TEST - Continued

WARNING

Injury to personnel and damage to equipment will result if system pressure exceeds 300 psig during leak test. Make sure that nitrogen pressure gages are properly calibrated and regulator adjusting screw is set for 0 psig. Close manifold valves prior to opening supply valve on nitrogen bottle.

12. Unscrew adjustment on regulated refrigerant lubricating oil, Item 191, WP 1803 00, or dry nitrogen, Item 199, WP 1803 00, source, until it is loose. Connect yellow hose to regulator.
13. Open valve on refrigerant or nitrogen bottle and slowly increase regulator pressure until it equals refrigerant system pressure.
14. Open REF hand wheel.
15. Slowly increase refrigerant or nitrogen pressure until system pressure indicated on manifold high pressure gage reaches 300 psi.
16. Using a Halogen leak detector calibrated to detect leaks of 2 oz./year or less, check all fittings, connections, hoses, and components for leaks. Repair as necessary.
17. Repeat step 16. until no leaks are found.
18. Close high side hand wheel on servicing manifold.
19. Close refrigerant or nitrogen bottle valve and slowly crack connection fitting.
20. Disconnect yellow hose from refrigerant or nitrogen regulator fitting and place hose end into a clean 5 gallon container, 1/4 full of water.
21. Slowly open high side hand wheel until a small stream of bubbles appears in water.
22. Continue to drain system until no bubbles appear in container. Pressure gages should read 0 psig.
23. Close high side hand wheel and disconnect yellow hose.
24. Disconnect high pressure (red) hose from evaporator inlet fitting.
25. Immediately cap inlet fitting.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM CONDENSER LINES, BURST DISC, AND RELIEF VALVE CONDENSER PALLET (AVIM) HH-60A HH-60L

This WP supersedes WP 1325 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Container, 5-Gallon
Nonmetallic Scraper
Refrigeration Unit Service Toolkit, SC 5180-90-N18
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.

Material/Parts

Cap, NAS817-6
Lubricating Oil, Refrigerant, Item 191, WP 1803 00
Plug, NAS818-6
Sealing Compound, Item 279, WP 1803 00

Personnel Required

Utility Equipment Repairer MOS 52C (1)

References

WP 1299 00
WP 1626 00
WP 1803 00

REMOVE

1. Open avionics compartment access door.
2. Deservice ECS (WP 1626 00).
3. Remove soundproofing panels from rear cabin bulkhead.
4. Remove condenser exhaust duct (WP 1299 00).
5. Remove screws, washers, clamps and spacer holding over pressurization line to pallet (Figure 1, Detail A).
6. Loosen nut at tee on condenser outlet.
7. Slide outboard end of line from grommet in transition duct.
8. Remove line assembly from pallet.

CAUTION

Damage to air-conditioning parts will occur if moisture is allowed to enter air-conditioning system. To prevent moisture from entering air-conditioning system, make sure to cap all lines immediately after disconnecting them.

9. Immediately install plug, NAS818-6, on line and place cap, NAS817-6, on tee connection at condenser outlet.

INSTALL

1. Apply sealing compound, Item 279, WP 1803 00, to burst disc inlet and outlet fittings and relief valve inlet fitting.
2. Lubricate other nuts with refrigerant lubricating oil, Item 191, WP 1803 00.
3. Loosely assemble lines, tees, burst disc, and valve (Figure 1, Detail A).
4. Slide outboard end of line assembly into grommet in transition duct.

INSTALL - Continued

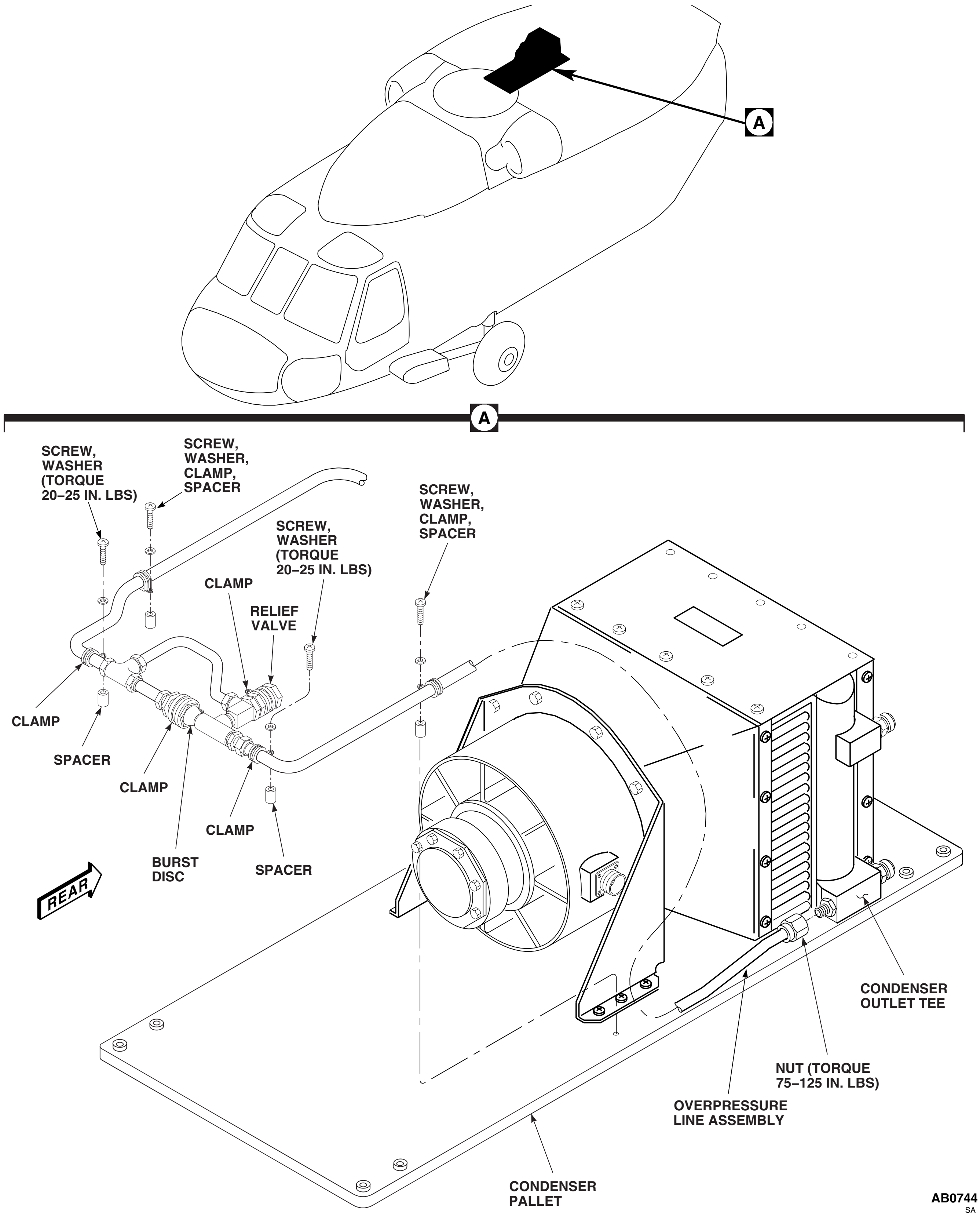


Figure 1. Replace Environmental Control System Lines, Burst Disc, and Relief Valve.

INSTALL - Continued

5. **QA** Install lines, burst disc, and relief valve to pallet with screws, washers, clamps and spacer. **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**
6. Lubricate remaining nut with refrigerant lubricating oil, Item 191, WP 1803 00, and loosely connect to condenser outlet tee.
7. **QA TORQUE ALL TUBING NUTS TO 75 - 125 INCH-POUNDS.**
8. Install condenser exhaust duct (WP 1299 00).
9. Install soundproofing panels on rear cabin bulkhead.
10. Service ECS (WP 1626 00).
11. Close and latch avionics compartment access door.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENVIRONMENTAL CONTROL SYSTEMS

ENVIRONMENTAL CONTROL SYSTEM CONDENSER AND TRANSITION DUCT (AVIM) HH-60A HH-60L

This WP supersedes WP 1326 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Container, 5-Gallon
Nonmetallic Scraper
Refrigeration Unit Service Toolkit, SC 5180-90-N18
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.

Material/Parts
Cap, NAS817-6
Lubricating Oil, Refrigerant, Item 191, WP 1803 00
Plug, NAS818-6
Sealing Compound, Item 273, WP 1803 00

Personnel Required
Utility Equipment Repairer MOS 52C (1)

References
WP 1303 00
WP 1324 00
WP 1626 00
WP 1803 00

REMOVE

1. Open avionics compartment access door.
2. Deservice ECS (WP 1626 00).
3. Remove condenser pallet (WP 1303 00).

CAUTION

Damage to air-conditioning parts will occur if moisture is allowed to enter air-conditioning system. To prevent moisture from entering air-conditioning system, make sure to cap all lines immediately after disconnecting them.

4. Loosen nut at tee on condenser outlet (Figure 1). Cap, NAS817-6, fitting and plug, NAS818-6, line.
5. Remove bolt, washer, spacer, and clamp holding over pressurization line at outboard side of pallet.
6. Loosen nut at tee and remove over pressurization line from grommet in transition duct. Cap, NAS817-6, fitting and plug, NAS818-6, line.
7. Remove bolts, washers, and brackets from pallet.
8. Break seal between condenser and pallet, and remove condenser and transition duct from pallet.
9. Remove screws, washers, intake transition duct and exhaust transition duct.
10. Remove sealing compound from transition duct mating flange and pallet with a nonmetallic scraper.

INSTALL

1. Seal interface of condenser and transition duct with sealing compound, Item 273, WP 1803 00.
2. **QA** Install exhaust transition duct to condenser with screws and washers. Tighten screws. Install intake transition duct to condenser with screws and washers. Tighten screws.

INSTALL - Continued

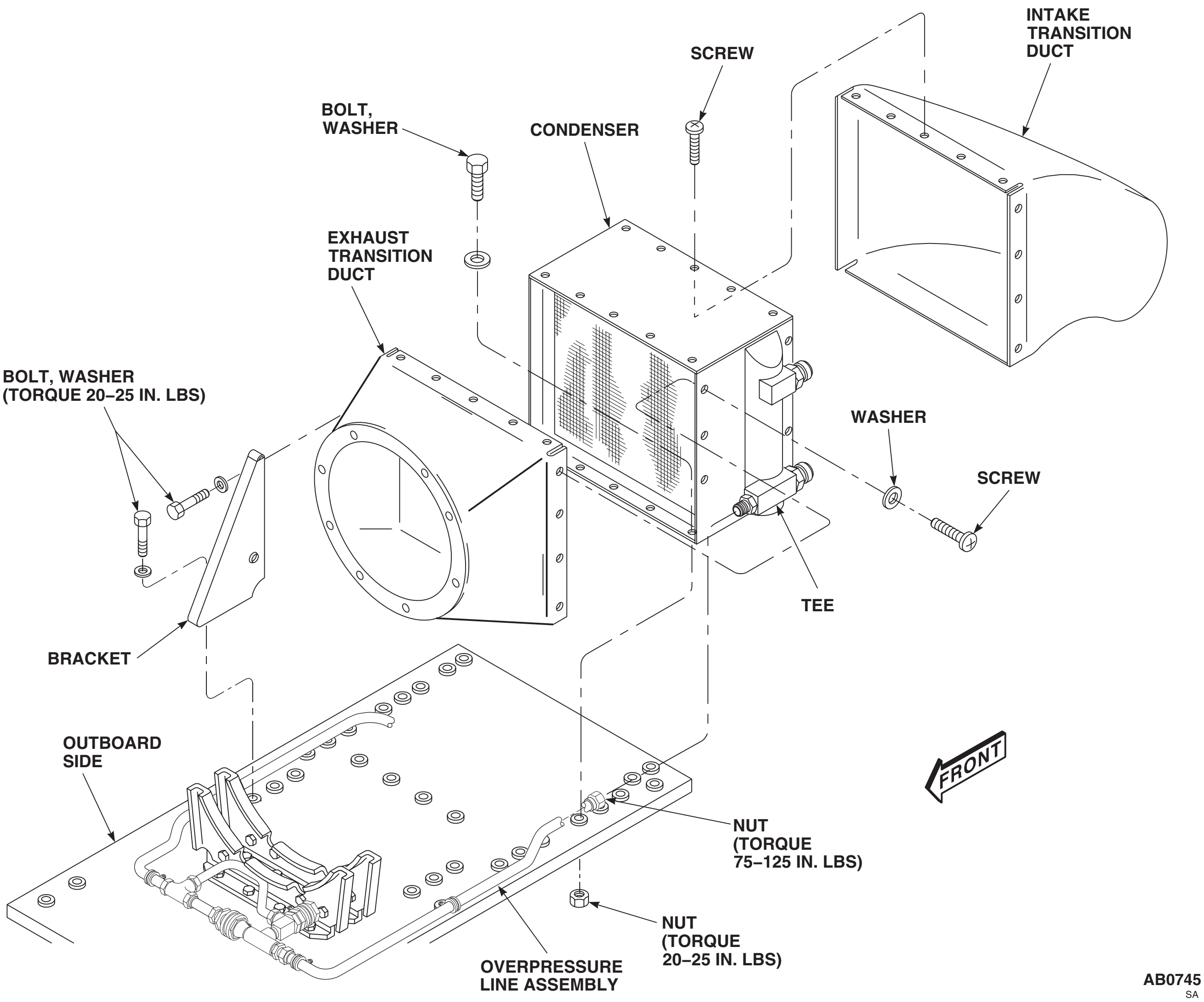


Figure 1. Replace Environmental Control System Condenser and Transition Duct.

3. Seal interface of condenser and pallet with sealing compound, Item 273, WP 1803 00.
4. **QA** Install brackets on pallet with bolts and washers. **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**
5. **QA** Install brackets on condenser with bolts and washers. **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**
6. Slide over pressurization line into grommet in transition duct.
7. **QA** Install nut on tee. Lubricate nut with refrigerant lubricating oil, Item 191, WP 1803 00 (Figure 1). **TORQUE NUT TO 75 - 125 INCH-POUNDS.**
8. **QA** Install over pressurization line on pallet with bolt, washer, spacer, and clamp. **TORQUE BOLT TO 20 - 25 INCH-POUNDS.**

INSTALL - Continued

9. **QA** Lubricate remaining nut with refrigerant lubricating oil, Item 191, WP 1803 00, and connect to tee at condenser outlet. **TORQUE NUT TO 75 - 125 INCH-POUNDS.**
10. Pressure test condenser pallet (WP 1324 00).
11. Install condenser pallet (WP 1303 00).
12. Service ECS (WP 1626 00).
13. Close and latch avionics compartment access door.

END OF WORK PACKAGE

UNIT LEVEL

HOISTS AND WINCHES

EXTERNAL RESCUE HOIST INSPECTIONS HH-60A HH-60L HOIST BL-29900-30

This WP supersedes WP 1327 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Caliper Set, Outside Micrometer, GGG-C-105
Container, 55-Gallon

Material/Parts
Cloth, Cleaning Item 86, WP 1803 00

Personnel Required
Assistants - (2)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 1803 00

IN-FLIGHT LOAD TEST

NOTE

Hoist operator should put on the safety harness and attach and secure before opening the sliding door.

1. The hoist operator's safety harness should be adjusted and secured.
2. The sliding door should be open.
3. Activate hoist power switch ON.
4. With the aircraft in hover at an altitude of 300 ft., lower the hoist cable using the control pendant.

WARNING

Injury to personnel and damage to equipment will occur if static electricity is not dissipated before ground personnel touch the hook or cable.

5. Perform a load test as follows:
   a. Operate the rescue hoist unit for one duty cycle. For each cycle, check the cable speed and the time required to travel the full cable stroke. Verify proper limit switch operation.

IN-FLIGHT LOAD TEST - Continued

WARNING

Injury to personnel and damage to equipment will occur if hoist operator is not wearing gloves to protect operators hand from broken cable strands.

NOTE

- The hoist operator should control the hoist load in forward flight to minimize cable contact with the side of the helicopter. When practical, extension or retraction of the load should be restricted to a hover or as slow a speed as possible.
- Swinging loads may cause erratic radar altimeter readings.

b. During cycle 3 (variable speed operation), check for proper variable speed operation and smoothness of control. There shall be no excessive control non-linearity.

DIMENSIONAL CHECK

Tension Roller Replacement Criteria

1. The elastomeric tension roller should be visually inspected for any obvious defects such as cracks, tears, abrasion or degradation of material. A monthly measurement of the roller outside diameter is recommended. Using outside micrometer, measure rollers. If the roller wears below 0.801-inches, at any cable contact point, the roller should be replaced. Refer to Figure 1 for visual examination criteria.
2. After tension roller replacement, the cable tension system must be tested to ensure the cable does not wrap loosely on the drum.

Crushable Bumper Replacement Criteria

The rescue hoist assembly uses a crushable bumper assembly mounted above the hook assembly to dissipate kinetic energy within the hoist moving parts. This is accomplished by a controlled crushing of the sacrificial element within the bumper assembly when the bumper assembly contacts the bellmouth of the cable guide and the full-in switch. Therefore, it is recommended that the bumper assembly be visually and dimensionally inspected before and after each flight or ground practice or maintenance cycling as described in Figure 2.

BEFORE EACH MISSION USE

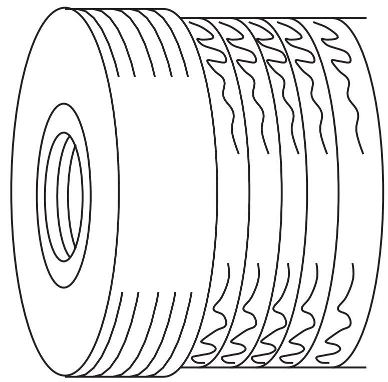
Preflight Check

NOTE

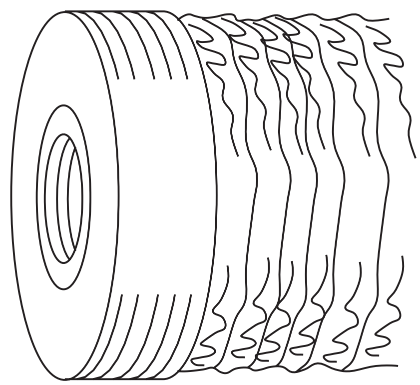
- This inspection is only required if rescue hoist is going to be used during a mission.
- Preflight check requires ac power applied.

1. Place HOIST POWER switch to ON position.

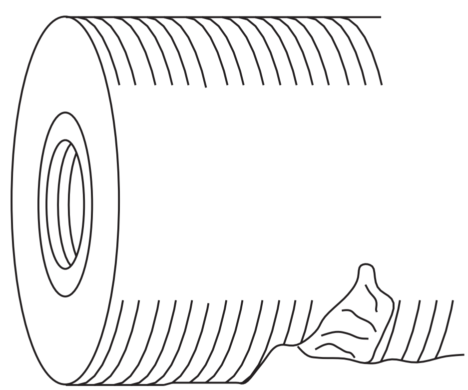
BEFORE EACH MISSION USE - Continued



(A) NORMAL SLIGHT WEAR. CONTINUE USE.



(B) NORMAL EXCESSIVE WEAR. PERFORM DIMENSIONAL CHECK.
REPLACE IF DIAMETER MEASURES BELOW 0.801.



(C) ABNORMAL WEAR. REPLACE ROLLER.

NOTE

ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AB0638A
SA

Figure 1. Tension Roller Replacement Criteria.

BEFORE EACH MISSION USE - Continued

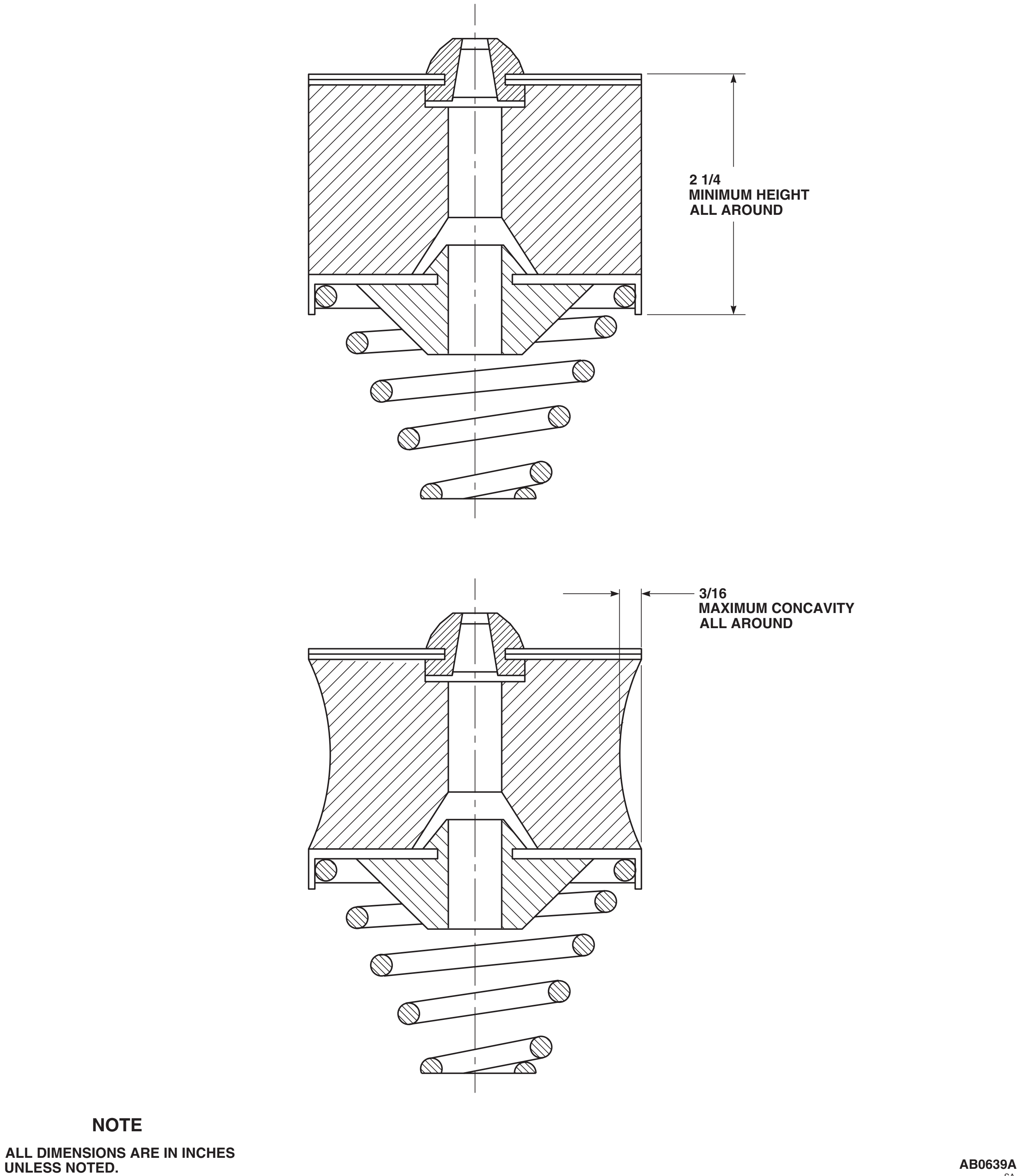


Figure 2. Crushable Bumper Replacement Criteria.

BEFORE EACH MISSION USE - Continued

WARNING

- Injury to personnel and damage to equipment will occur if work gloves are not used whenever handling rescue hoist cable with broken cable strands. Care should be taken not to damage cable by kinking when handling on ground.
- Electronic Controller contains internal circuitry that monitors status of ″up slow″ limit switches. The ″up slow″ limit switch, when actuated, slows retracting cable speed to 50 ft per minute fixed. This limit switch is actuated when retracting cable is 10 ± 2 ft from fully reeled in position. If Electronic Controller detects a ″full in″ switch actuation with no ″up slow″ actuation the electronic controller inhibits hoist operation. This feature is needed because of the high speed capabilities of hoist and will prevent possibility of breaking cable. It is important that ground testing full in limit switch be done after ″up slow″ switch is actuated. In the event that ″up stop″ switch is actuated manually with no ″up slow″ actuation the system can be reset by turning master hoist switch ″OFF″, and then ″ON″. This will reset system.

2. Using rescue hoist pendant, rotate thumbwheel downward to maximum position and hold. Let cable payout on a clean ground without interference and proceed with the following:

NOTE

Be sure cable is wrapping smoothly and evenly without bird caging or gaps.

a. Allow cable to reel out at full speed.

b. Midway of total payout, operate pilot control in lower direction. Cable speed shall reduce to approximately 200 ft/min. Verify override function by rotating pendant thumbwheel in both directions with no change.

c. Set pilot control switch to off.

d. Hoist shall decelerate then stop.

e. Operate pendant in lower direction.

f. Cable shall reduce speed to approximately 33 ft/min at 10 ± 2 ft.

g. Cable stops with 4 ± 1/2 wraps on drum.

h. Verify cable payout display reads 290 ± 5 and full-out lamp indicators illuminate.

i. Perform cable inspection. Refer to, INSPECTION AND MAINTENANCE, in this work package.

j. Repeat above procedure in hoisting direction.

k. Repeat pilot control functions as described in step b. to verify override function in hoisting mode.

l. Be sure that cable reduces speed and stops at full in position.

m. Check cable payout displays, both should read 000 ± 2/-0 and full-in lamp indicators illuminate.

3. Visually check for oil leaks.

4. Visually check wiring and electrical connectors for damage.

5. Check hook assembly for freeness of swivel and latch. If hook was exposed to salt water or if swivel does not operate freely, disassemble and clean hook assembly.

6. Check crushable bumper.

FLIGHTLINE CABLE HANDLING TECHNIQUES

1. The following ground procedures are vital in achieving reliable rescue hoist operation:

WARNING

Do not wind load cable from hoist drum onto a take up reel. This can induce twist into the load cable and cause miswrapping when winding back onto the drum.

a. Reeling In: It is important that cable is reeled in under a reasonably heavy, even pull so that it does not wrap loosely on the drum. A drag load must be applied using a gloved hand or clean heavy cloth on the cable to achieve tight, even layers on the drum. Reeling in should be accomplished at a slow speed so that when loops form they can be straightened out and not form kinks. A kink in the cable is caused by a loop in the cable being pulled up tight, resulting in a sharp, permanent bend in the cable. Kinks can lead to fouling failures as they cause a retarding force on the cable when the kink gets hung up going through the cable guide, especially when reeling out with no load. If it is required to reel in the cable at full speed for acceptance test purposes, the cable should be reeled out in a straight line for it's full length so that no kinks or permanent cable damage occur reeling in. If the purpose of the test is to measure hoist speed, timing of a specified shorter length is advisable.

WARNING

It is important that there is no retarding load on the cable while reeling out. If the action of the tension roller is retarded, the cable will immediately loosen on the drum and foul.

b. Reeling Out: The rescue hoist unit incorporates a tension roller system, the purpose of which is to make sure that there is always tension on the cable. During reeling out, the peripheral speed of the tension roller is slightly higher than that of the drum so the cable is always under tension between the roller and the drum. The load is in the order of seventeen pounds. When reeling out, the cable may be allowed to coil on the ground. It may also be coiled in a drum, however, caution must be exercised while reeling in to avoid kinks.

INSPECTION AND MAINTENANCE

1. In order to be effective, cable inspection must be performed on a regularly scheduled basis. Cable used for personnel lifting should be inspected carefully and frequently and rigid standards of acceptance and rejection requirements must be adhered to. Cable inspection should include careful visual checks for the following defects:

a. Evidence of Dragging Over Obstacles or Worn Cable Guides: Abrasion of cable is caused by contact with other cable sections or any material that may contact the cable such as aircraft structural members, auxiliary equipment, or almost any abrasive surface. Abrasion removes metal from the external wires in the outer strands and thus weakens the cable. Nicking, scarring, and scrubbing are also referred to as abrasion. Severe abrasion resulting from dragging a cable across an aircraft structural member can result in catastrophic cable failure and possibly loss of life. Severe abrasion as evidenced by a worn or damaged rough surface cable texture resulting in multiple broken wires in an outer strand in any section requires immediate cable replacement.

b. Improperly Adjusted or Defective Cable Guide: Cables evidencing damage such as broken wires at the turnaround or at cable crossovers are an indication that the hoist needs to have the levelwind readjusted or repaired. Smiles (gaps between adjacent turns of cable) are the result of the cable not being tightly and evenly wrapped and/or the levelwind leading the cable too much. If the levelwind is adjusted to lag the cable too much, excessive abrasion of the cable as it wraps on the drum occurs, particularly at the crossovers. If loud snapping is heard, shimming of the levelwind shafts relative to the drum may be improper. All of the above inspections are subjective and require an experienced inspector to make the inspection and readjustment if required. If the condition is corrected prior to excessive damage occurring, the cable can be continued in service if no more than one wire break is discovered and snipped and the necessary adjustments are properly carried out.

c. Severe Overload: A cable that has been severely overloaded may or may not show immediate results of the

INSPECTION AND MAINTENANCE - Continued

overload. Broken strands are the most obvious defects. However, broken core wires may also result from overload. ″Necking Down″ of a section of cable is an indication of broken core wires. A cable that has been knowingly overloaded should be replaced immediately even if no obvious physical damage is detected. Any obvious reduction in diameter that is noted is cause for immediate cable replacement.

d. Kinking: Kinks are man-made defects in cables caused by a loop in a cable being pulled up tight, resulting in a sharp, permanent bend in the cable. Kinks are identified as ″open″ or ″closed″. An ″open″ kink is caused by pulling a cable loop tight, creating a sharp permanent bend which tends to open the lay of the cable. A ″closed″ kink is caused in the same manner, but creates a sharp permanent bend which tends to close the lay of the cable. Continued use of a kinked cable can result in the kink being caught in the cable guide assembly upon paying out and will result in cable fouling. A kinked cable should not be run back onto a hoist for any reason.

e. Internal Wear Caused By Grit or Dirt Penetration: Internal wear caused by grit, dirt, sand, etc., embedded between cable strands is extremely difficult to detect. Preventive maintenance is the best procedure. The cable should be carefully wiped to remove gritty material as it is reeled in during inspection. Prolonged use of a cable loaded with gritty material will result in broken wires.

f. Improperly Attached or Defective Fittings: Carefully inspect the cable in the area of the hook ball end attachment. Do not bend the cable in the process of the inspection. The ball end of the cable must not evidence any broken wires at the swaged ball fitting. Any necking down, broken wires, loose wires in the immediate vicinity of the ball end is a cause for cable replacement. This condition may have resulted from the cable not being snugged up against the cable guide bell mouth properly or a full-in limit switch that is activating too fast. The continual vibration of a helicopter and a loosely ″homed″ hook can result in a fatigue failure of the cable immediately above the ball end fitting. This inspection should be performed prior to every flight. If a loose hook is noted, the necessary adjustment to the full-in limit switch shall be performed after the cable inspection.

g. Broken Wires: The number of broken wires on the outside of a wire rope is an index of it’s general condition and whether or not it must be considered for replacement. Tests have shown that a large number of broken wires can exist without significant reduction on the cable’s braking strength. Wires that are broken may cause some other problems that may lead to more severe situations. It is for this reason that no broken wires are allowed for the hoist cable. If broken wires are detected, the cable shall be removed and replaced as soon as possible. If it is required to operate with broken wires, the operator should pay extra attention to the hoist operation with broken protruding wires from the cable.

h. Bird Caging: Bird caging results from torsional imbalance that occurs because of abuse such as sudden stops, the rope being worked through tight sheaves, or a hook bearing that is not rotating. Any evidence of bird caging requires immediate replacement of the load cable.

i. Breaking In A New Wire Rope: In the event a wire rope needs to be replaced, the new rope requires careful installation and close adherence to all the appropriate procedures previously noted. After a new cable has been installed, the hoist should be run through a cycle of operation with no load at slow speed. During this trial operation, a very close watch should be kept of all working parts to make sure the cable runs freely, the levelwind is adjusted properly, and the tensioning system is operating as it should. One may witness some tendency for the cable to twist, which may result in a smile(s) developing. This may not be a problem, since the cable is not broken in yet and is still inflexible. If no other problems are noted, the next step is to run several cycles under a light load at a reduced speed. This procedure allows the component parts of the new rope to make a gradual adjustment to the actual operating conditions of the hoist.

j. After each use in salt water environments: Cable will be reeled out and washed with fresh water. Cable may be reeled out into a 55 gallon container filled with fresh water, and dried by passing cable through a cleaning cloth, Item 86, WP 1803 00, when reeling in.

INSPECTION AND MAINTENANCE - Continued

WARNING

Injury to personnel and damage to equipment will occur if lubricant is applied to cable or hoist. Tension roller system works on basis of friction. If oil or grease gets on cable, clean it with a cleaning cloth and fresh water. Solvents may be detrimental to tension roller, so a cable cleaned in this manner must be dry before reeling in.

NOTE

Cloth should be held firmly to impart a steady, reasonably heavy tension load to cable.

k. During final rewind of hoist cable, a cleaning cloth, Item 86, WP 1803 00, should be held firmly around cable. Cloth will help remove foreign particles (grit, dirt, etc.) and, in event a broken wire was overlooked during visual inspection, wire end will snag on cloth, preventing break from being overlooked.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

HOISTS AND WINCHES

RESCUE HOIST INSTALLATION HOIST 42305R1

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Crimping Tool, RR-G-619
Electrical Repairer's Toolkit, SC 5180-99-B06
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Material/Parts

Hydraulic Fluid, Automatic Transmission Item 143, WP 1803 00
Hydraulic Fluid, Petroleum Base Item 145, WP 1803 00
Protective Caps and Plugs
Wire, Nonelectrical Item 351, WP 1803 00

Personnel Required

Aircraft Electrician MOS 15F (1)
Aircraft Structural Repairer MOS 15G (1)
Assistants - (2)
UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0152 00
WP 0233 00
WP 0258 00
WP 1703 00
WP 1803 00

NOTE

The use of a BL-8300 series Breeze internal rescue hoist is prohibited.

PREPARE TO INSTALL RESCUE HOIST

1. Turn off all electrical power.
2. UH60L UH60A Remove troop seats (WP 0233 00).
3. Remove soundproofing panels (WP 0258 00).

INSTALL RESCUE HOIST HARNESS

1. Remove No. 2 junction box cover.
2. **QA** Pass wire, No. M3432A06, through junction box grommet and install in No. 2 junction box with screw, washer, nut, and clamp (Figure 1, Sheet 1, Detail A).
3. **QA** Double clamp wire, No. M3432A06, at STA 285.0, RBL 27.0, using screw, clamp, spacer, washer and nut (Figure 1, Sheet 2, Detail C).
4. **QA** Install mounting bracket and cable strap for support to adjoining harness.

INSTALL RESCUE HOIST HARNESS - Continued

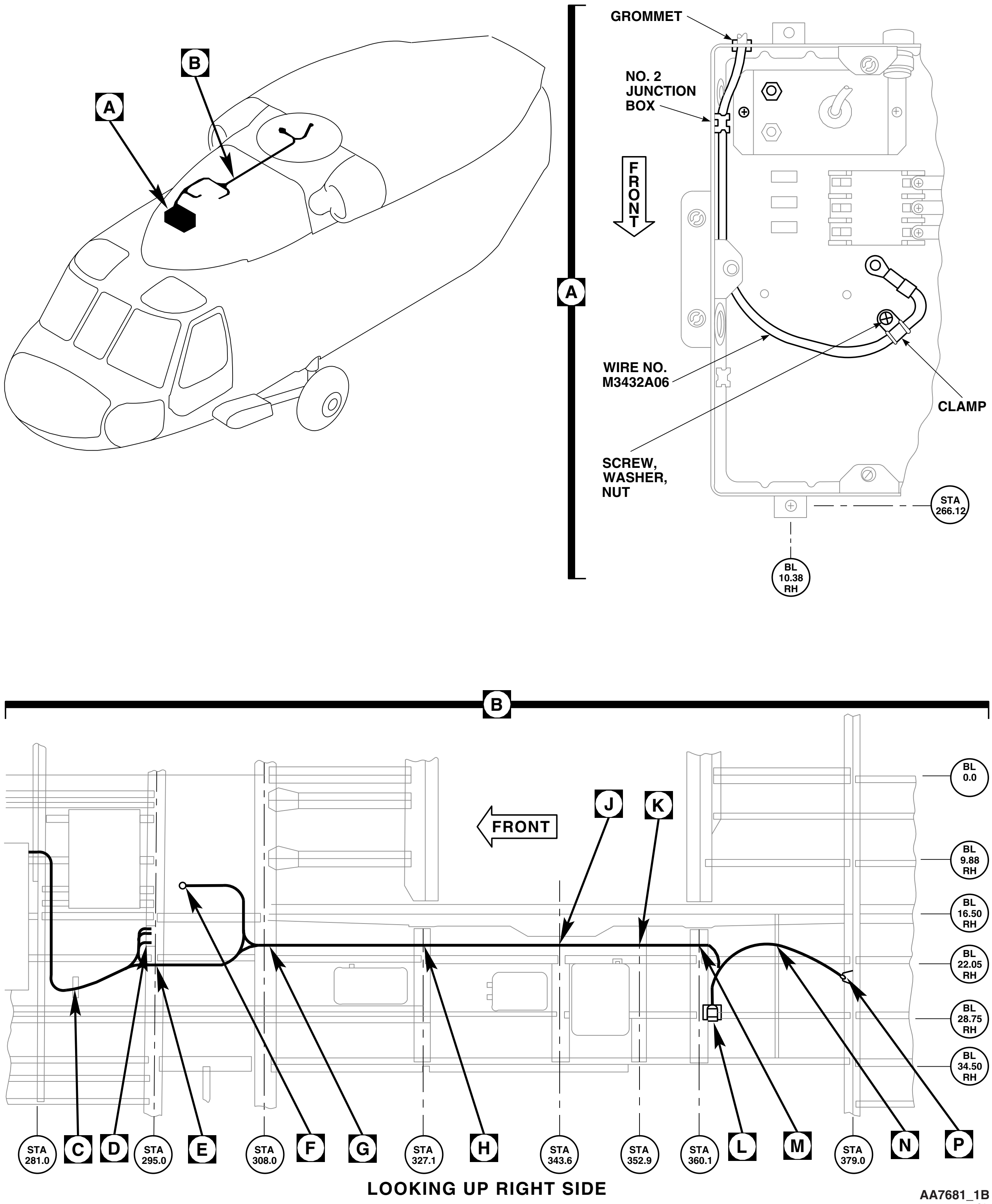


Figure 1. Remove/Install Rescue Hoist Harness. (Sheet 1 of 3)

INSTALL RESCUE HOIST HARNESS - Continued

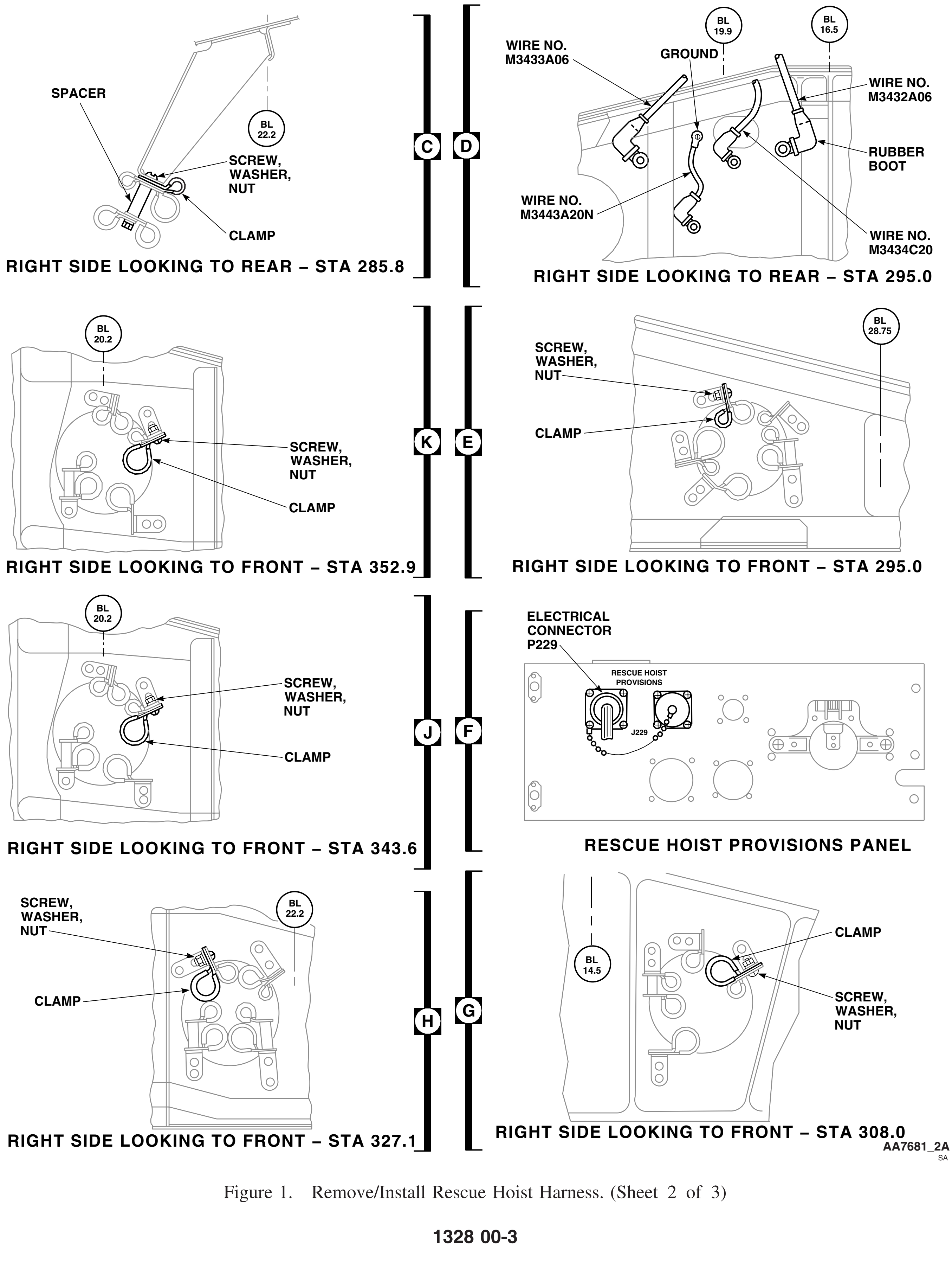


Figure 1. Remove/Install Rescue Hoist Harness. (Sheet 2 of 3)

INSTALL RESCUE HOIST HARNESS - Continued

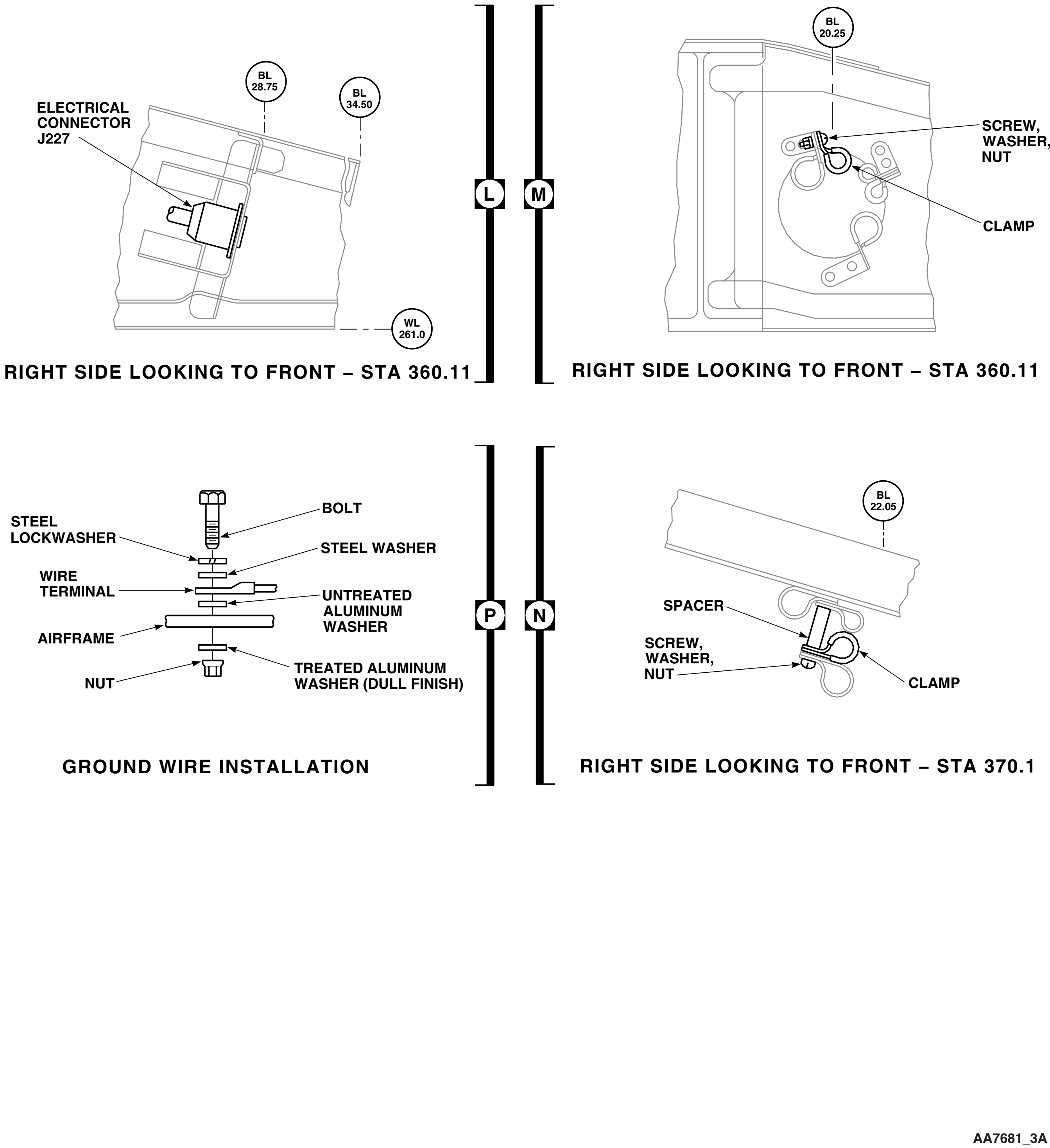


Figure 1. Remove/Install Rescue Hoist Harness. (Sheet 3 of 3)

INSTALL RESCUE HOIST HARNESS - Continued

NOTE

Wires, No. M3433A06 and No. M3443A20N, are loose until relay K14 installation is made.

5. **QA** Pass wires, No. M3433A06 and No. M3434C20, through frame hole at STA 295.0, RBL 22.0, and secure with screw, clamp, washer and nut (Figure 1, Sheet 2, Detail E).

NOTE

If necessary remove backshell from electrical connector, P229, to fit wiring and connector through intercoastal.

6. **QA** Pass electrical connector, P229, through intercostal at STA 306.0, RBL 16.5. Reinstall backshell if removed (Figure 1, Sheet 2, Detail G).
7. **QA** Remove protective cover from electrical connector, J229, on rescue hoist provisions panel, and install on adjacent dummy connector (Figure 1, Sheet 2, Detail F).
8. Inspect and treat electrical connector (WP 1703 00).
9. **QA** Connect electrical connector, P229, to electrical connector, J229, of rescue hoist provisions panel.
10. Install split grommet around wire bundle in intercostal at RBL 16.5 between STA 295.0 and STA 308.0.
11. **QA** Pass wiring harness through these frames, clamping harness in each frame hole using screw, clamp, washer and nut: STA 308.0, RBL 20.0; STA 327.0, RBL 22.0; STA 343.0, RBL 20.0; STA 352.0, RBL 20.0 and STA 360.0, RBL 20.0 (Figure 1, Sheet 2, Detail G, Detail H, Detail I, Detail J) and (Figure 1, Sheet 3, Detail L).
12. **QA** Install electrical connector, J227, in bracket with screws, washers, and nuts at STA 362.0, RBL 28.0, WL 262.0 (Figure 1, Sheet 3, Detail K).
13. **QA** Clamp harness at STA 370.0 and RBL 24.0, using screw, clamp, spacer, washer and nut (Figure 1, Sheet 3, Detail M).
14. Make sure airframe ground is clean for good electrical connection.
15. **QA** Connect wire, No. M3442A06N, to helicopter frame at STA 379.0, RBL 24.0, using bolt, lockwasher, steel washer, untreated aluminum washer, treated aluminum washer (dull finish), and nut (Figure 1, Sheet 3, Detail N).

NO. 2 JUNCTION BOX MODIFICATION

1. **QA** Install fuse holder in No. 2 junction box with screws (Figure 2, Detail A).
2. **QA** Place fuse in fuse holder and connect bus bar to fuse holder with washer and nut.
3. **QA** Connect bus bar to terminal A1 of relay K6 with washer, lockwasher, and nut.
4. **QA** Connect wire, No. M3432A06, to fuse holder, with washer and nut.
5. **QA** Install cover on No. 2 junction box.

INSTALL K14 BRACKET, RELAY, AND SUPPRESSOR

1. **QA** Position relay K14 on bracket and install with screws, washers, and nuts (Figure 3, Detail A).
2. **QA** Connect ground wire, No. M3432A06, to relay K14, terminal A1, and install nut and washer. Install nipple on terminal.

INSTALL K14 BRACKET, RELAY, AND SUPPRESSOR - Continued

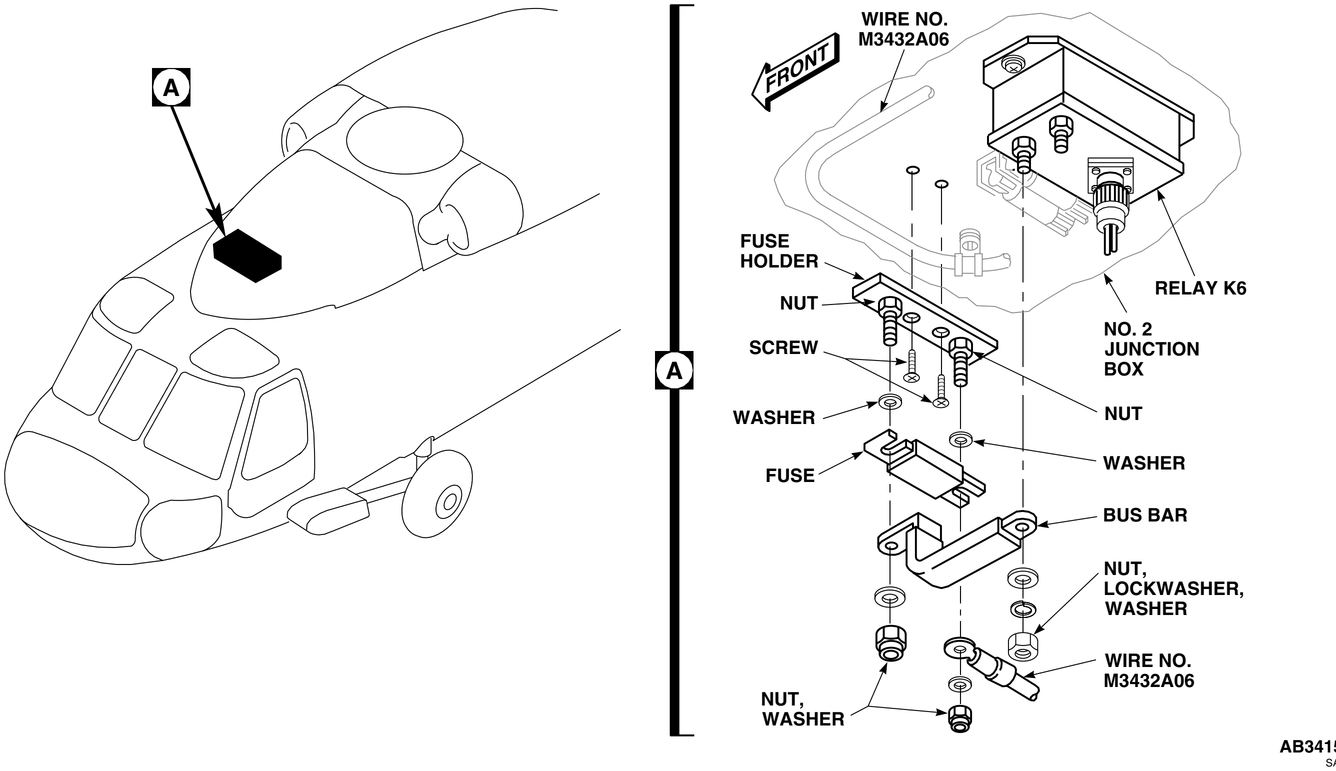


Figure 2. Rescue Hoist No. 2 Junction Modification.

3. **QA** Connect ground wire, No. M3433A06, to relay K14, terminal A2, and install nut and washer. Install nipple on terminal.
4. If installing new suppressor, do this:
   a. Cut suppressor POS and NEG leads 3 inches from end of cap.
   b. Strip 1/4-inch of insulation from leads.
   c. Using crimping tool, crimp terminals, to suppressor leads.
5. Place POS lead from suppressor on relay K14, terminal X1. Do not fasten.
6. Place NEG lead from suppressor on relay K14, terminal X2. Do not fasten.
7. **QA** Place ground wire, No. M3434C20, on relay K14, terminal X1, and install washer and nut. Install nipple on terminal.
8. **QA** Place ground wire, No. M3443A20N, on relay K14, terminal X2, and install washer and nut. Install nipple on terminal.
9. Make sure airframe ground is clean for good electrical connection.
10. **QA** Connect ground wire, No. M3443A20N, to helicopter frame using screw, lockwasher, steel washer, untreated aluminum washer, treated aluminum washer (dull finish), and nut.
11. **QA** Position bracket assembly on structure and install with screws and washers.

INSTALL K14 BRACKET, RELAY, AND SUPPRESSOR - Continued

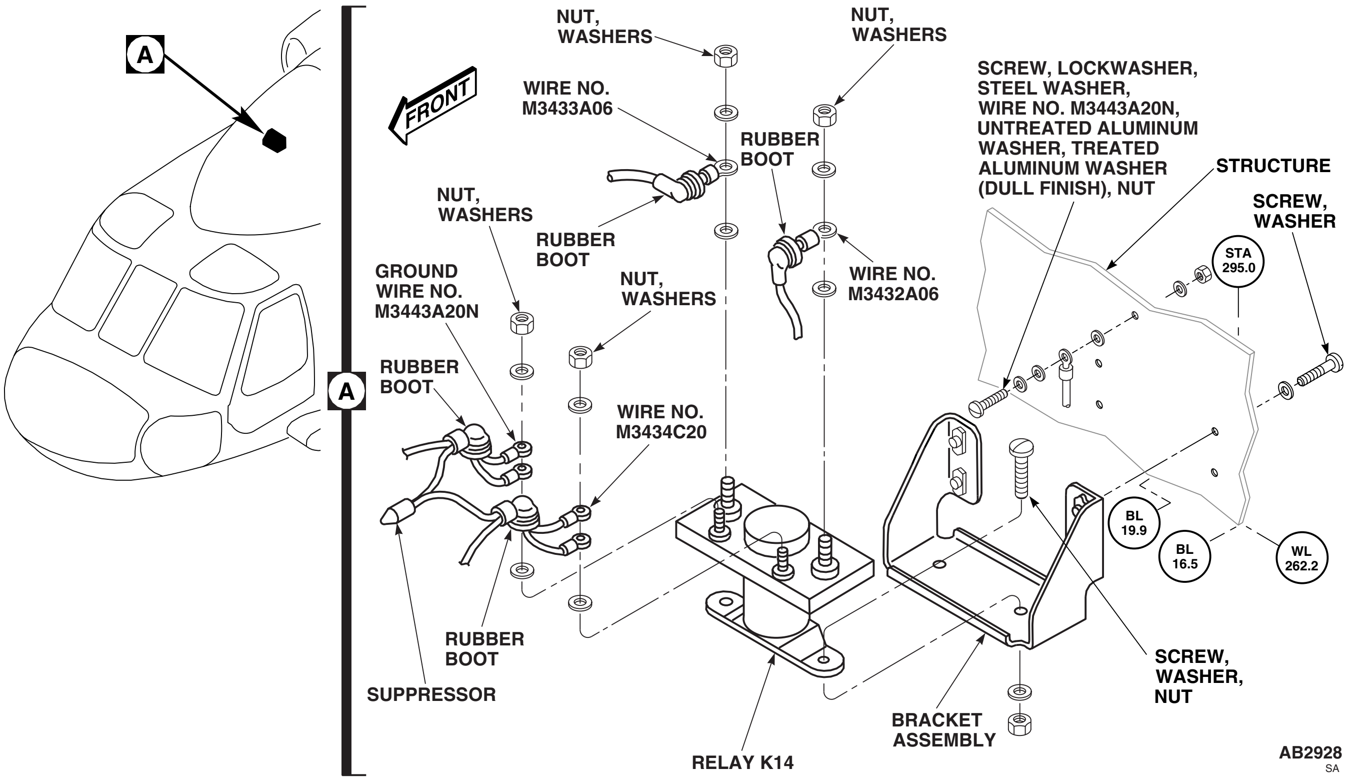


Figure 3. Remove/Install K14 Bracket, Relay, and Suppressor.

INSTALL RESCUE HOIST CONTROL PANEL, BRACKET, AND CABLE ASSEMBLY

1. **QA** Attach bracket assembly on cockpit lower console with screws and washers (Figure 4, Detail A).
2. **QA** Install rescue hoist control panel in bracket assembly with screws and washers.
3. Inspect and treat electrical connector (WP 1703 00).
4. **QA** Connect electrical connector, P228, to electrical connector, J228, on rescue hoist control panel.
5. Inspect and treat electrical connector (WP 1703 00).
6. **QA** Connect electrical connector, P226, to electrical connector, J226, right side of lower console.
7. Secure cable assembly to side of lower console with clamps, screws, and washers.
8. Using nonelectrical wire, Item 351, WP 1803 00, lockwire CABLE SHEAR switch guard to SAFE.

INSTALL RESCUE HOIST

1. Slide slip plate over floor fitting, while positioning quick-disconnect fitting over other floor fitting (Figure 5, Detail B).
2. **QA** Secure adapter plate to floor fitting with quick-disconnect fitting.
3. Loosely attach bracket assembly to channel assembly.
4. Loosely attach channel assembly and bracket to cabin ceiling with bolts and washers.
5. **QA** Tighten screws of bracket on helicopter frame member located at RBL 34.5.

INSTALL RESCUE HOIST - Continued

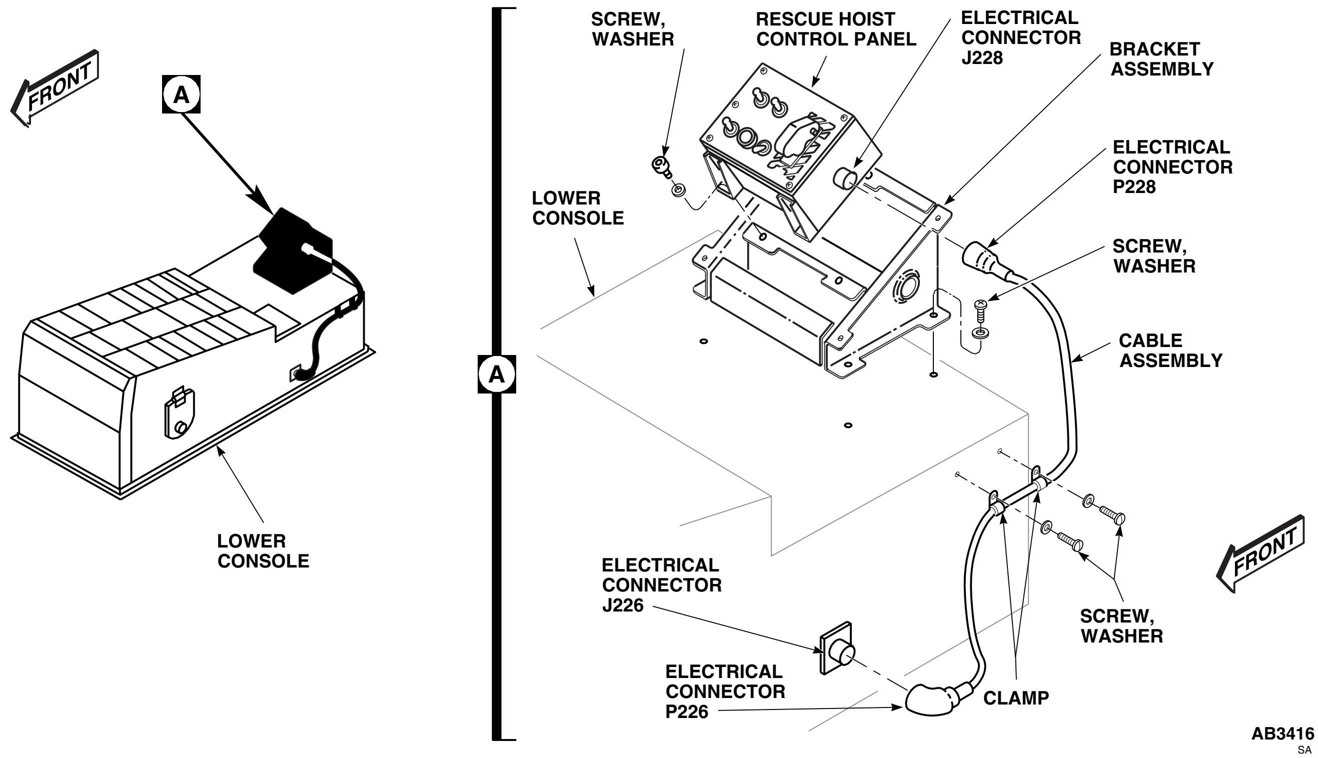


Figure 4. Remove/Install Rescue Hoist Control Panel, Bracket, and Cable Assembly.

6. Verify clearance of bracket to airframe with feeler gage at the heel and toe edges. **MAXIMUM OF 0.005-INCH IS ALLOWED. NO OTHER GAPS ARE ALLOWED.**

NOTE

Verify hole location of bracket to assure proper alignment to helicopter frame. Shim if required. Redrill bolt holes if needed.

7. **QA TORQUE BOLTS TO 47 - 53 INCH POUNDS.**
8. **QA** Install bracket to cabin ceiling with bolts and washers. **TORQUE BOLTS TO 47 - 53 INCH POUNDS (Figure 5, Detail A).**
9. **QA** Secure lower inboard edge of channel to structure and bracket with screws and washers.
10. **QA** Install stud in channel and bracket with washers and nut. **TORQUE NUT TO 165 - 275 INCH POUNDS.** Install cotter pin.
11. Release hoist reaction arm from stud on hoist post. Remove release pin connecting reaction arm assembly to post (Figure 6, Sheet 1, Detail A).
12. Place reaction arm in horizontal position. Install release pin.
13. Remove release pins from bottom of baseplate.

INSTALL RESCUE HOIST - Continued

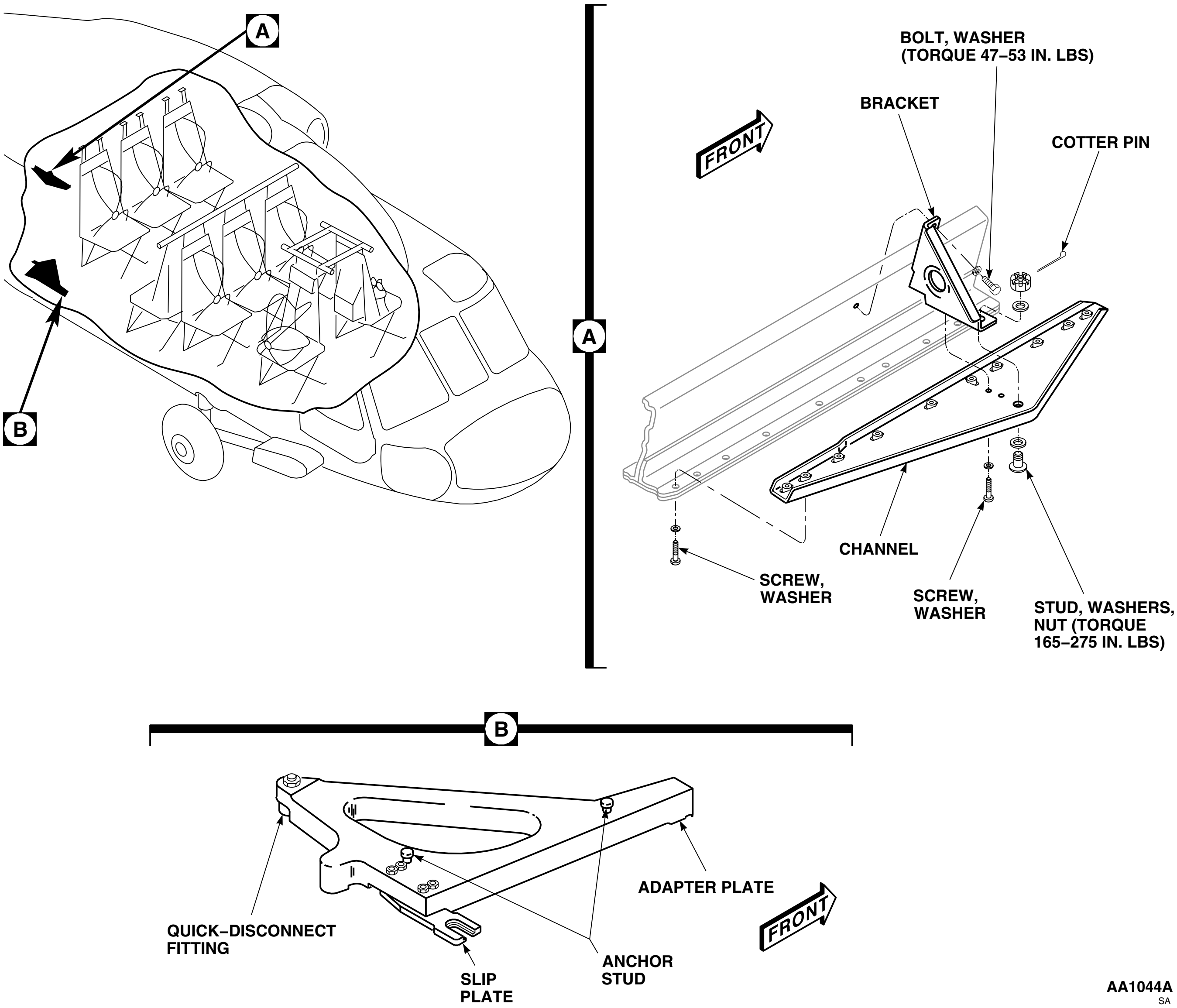


Figure 5. Remove/Install Rescue Hoist Mounts.

CAUTION

Damage to reaction arm may result if reaction arm is moved and hits mechanical stops before lining up UH-60 (Aircraft Position 3) holes. Move reaction arm in other direction until holes line up. Do not force arm.

14. Move reaction arm until UH-60 (Aircraft Position 3) holes in lower baseplate line up with holes in reaction arm. Install release pins.

INSTALL RESCUE HOIST - Continued

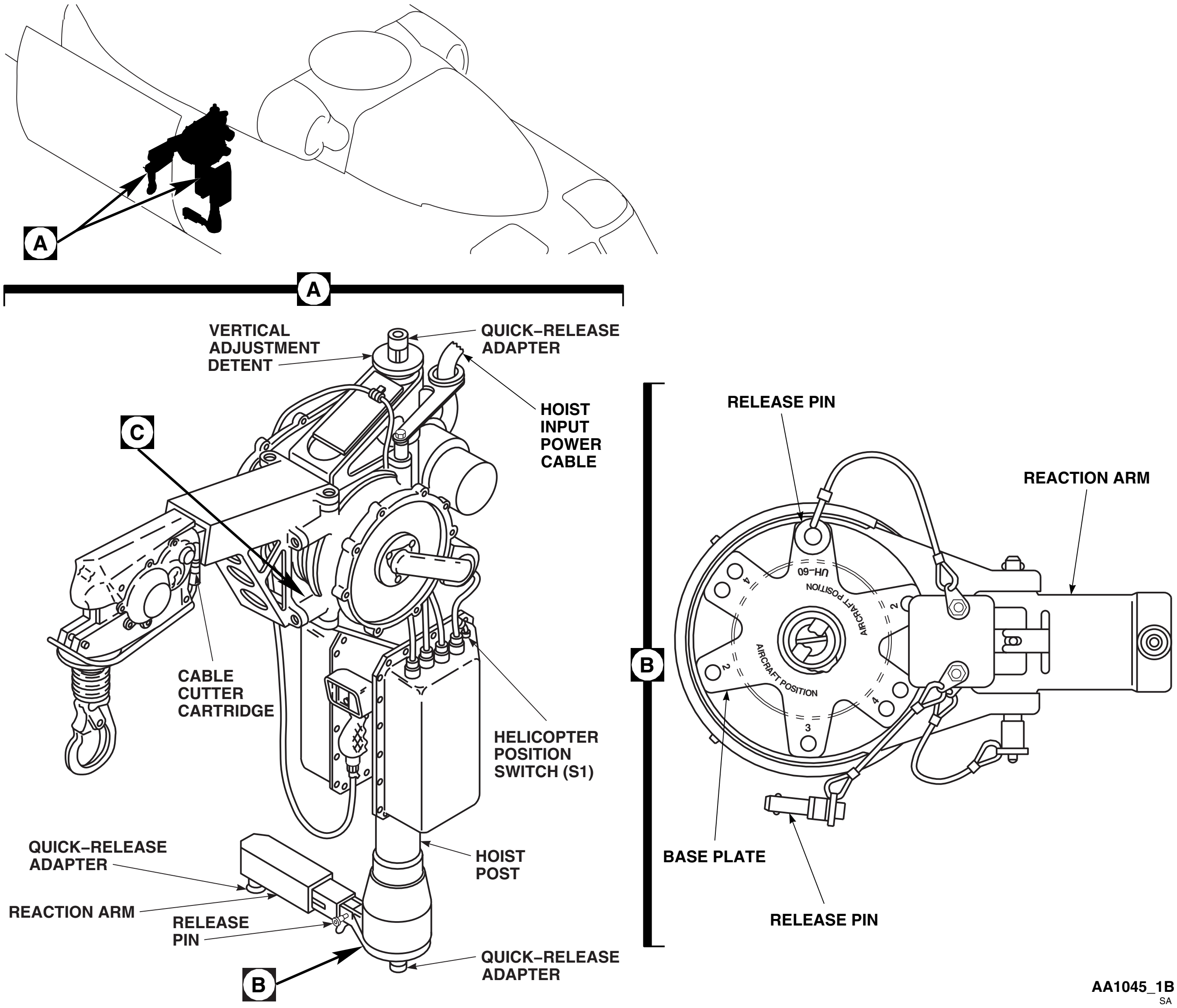


Figure 6. Remove/Install Rescue Hoist. (Sheet 1 of 2)

WARNING

Hoist is heavy and awkward to handle. Injury to personnel may occur while lifting and positioning hoist during installation. Two or more persons shall handle hoist during installation.

15. Lift hoist into helicopter. Release quick-release adapters at top and bottom of hoist.
16. Position hoist so quick-release adapters mate with adapter plate and are directly under channel anchor stud.

INSTALL RESCUE HOIST - Continued

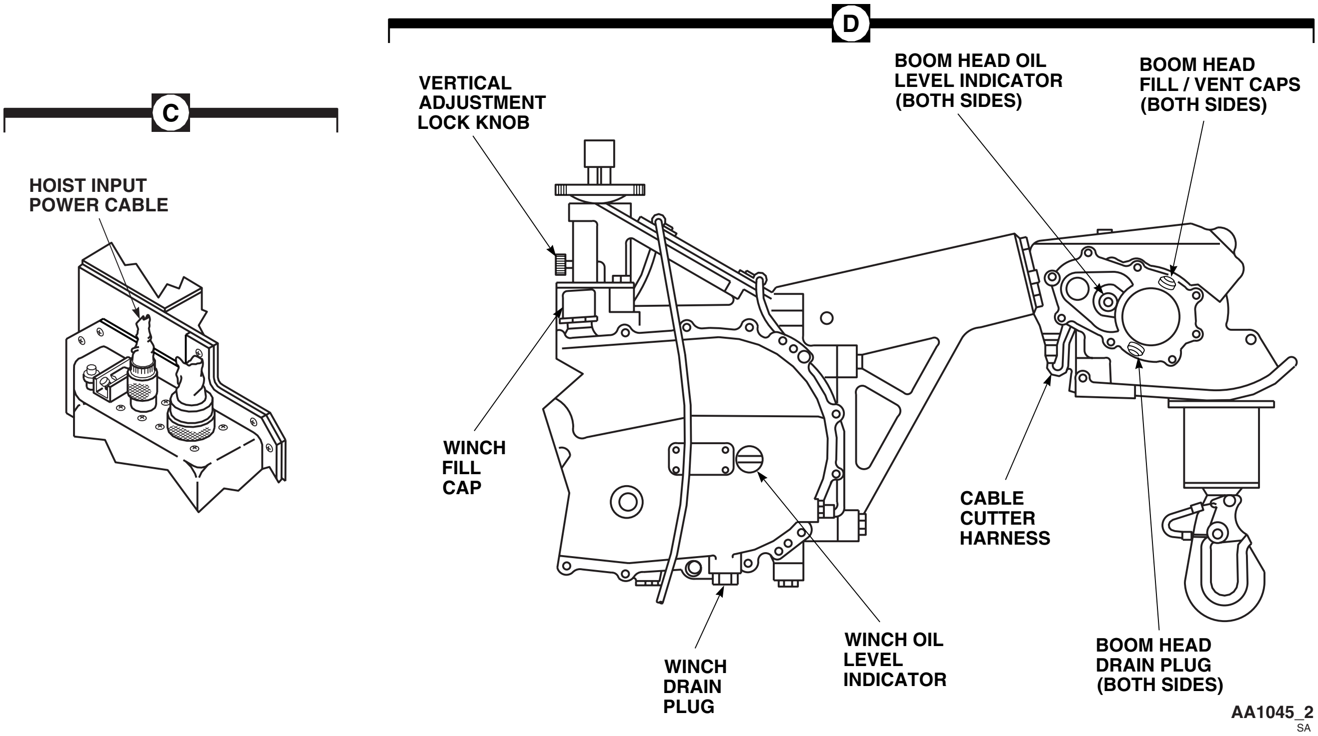


Figure 6. Remove/Install Rescue Hoist. (Sheet 2 of 2)

CAUTION

Make sure that reaction arm and hoist base quick-release adapters connect to adapter plate anchor studs. Do not overextend vertical adjustment as it may cause hoist binding and damage helicopter structure.

17. Pull vertical adjustment lock knob, turn vertical adjustment detent until quick-release adapter can be locked over channel anchor stud. **DO NOT LIFT QUICK RELEASE ADAPTER (PINTLE) TO ENGAGE THE STUD. ADJUST HEIGHT BY USE OF THE VERTICAL ADJUSTMENT DETENT ONLY (WHICH WILL PROVIDE NECESSARY VERTICAL PLAY IN THE PINTLE).**
18. Lock all quick-release adapters.

NOTE

Use petroleum base hydraulic fluid, Item 145, WP 1803 00, when using hoist in temperature below -40°F (-40°C).

19. Fill winch and boom head with automatic transmission hydraulic fluid, Item 143, WP 1803 00, until all three oil level indicators show full (Figure 6, Sheet 2, Detail D).
20. Remove protective cap from cable cutter cartridge and remove aluminum shorting strips installed between cartridge electrical connector pins. Save protective cap and shorting strips.
21. Inspect and treat electrical connector (WP 1703 00).
22. **QA** Connect cable cutter harness to cable cutter cartridge electrical connector.

INSTALL RESCUE HOIST - Continued

23. **QA** Place hoist control box helicopter position switch S1 to 2-3.
24. Disconnect cable cutter harness from cable cutter cartridge electrical connector. Install aluminum shorting strips between pins of cartridge electrical connector, and install protective cap on connector.

INSTALL RESCUE HOIST UMBILICAL CABLE

1. Disengage quick-disconnect on anti-torque arm and lift arm up.
2. Swing hoist to extend position.
3. Inspect and treat electrical connector (WP 1703 00).
4. **QA** Connect umbilical cable electrical connector, P227, to electrical connector, J227 (Figure 7, Detail C).

NOTE

Do not tighten clamp connections until final umbilical cable adjustment has been made.

5. **QA** Loosely install umbilical cable with clamp, screw, washer and nut at STA 362.85 and RBL 34.5 (Figure 7, Detail D).
6. **QA** Loosely install umbilical cable with clamp, screw, washer, spacer, and nut at STA 369.0 and RBL 34.5 (Figure 7, Detail B).
7. **QA** Loosely install umbilical cable with clamp, screw, washer, spacers, and nut at STA 379.0 and RBL 20.0 (Figure 7, Detail A).
8. Inspect and treat electrical connector (WP 1703 00).
9. **QA** Connect electrical connector on free end of umbilical cable to rescue hoist control box (Figure 7, Detail F).
10. **QA** Twist umbilical cable 1 to 1 1/2 turns counterclockwise (looking outboard) between ring on hoist and clamp. Allow 24 inches of cable between ring on hoist and clamp. Secure umbilical cable with clamp, screw, washer, and nut at STA 379.0, RBL 25.0, and WL 262.0. Tighten clamp (Figure 7, Detail E and Detail F).
11. Tighten all clamps securing umbilical cable to helicopter airframe.

COMPLETE INSTALLATION

1. Make sure area is clean and free of foreign objects.
2. Install soundproofing panels (WP 0258 00).
3. **UH60L UH60A** Install troop seats (WP 0233 00).
4. Update helicopter weight and balance records.
5. Do an operational check of rescue hoist system (WP 0152 00).

PREPARE TO REMOVE RESCUE HOIST

1. Turn off all electrical power.
2. Remove soundproofing panels (WP 0258 00).

PREPARE TO REMOVE RESCUE HOIST - Continued

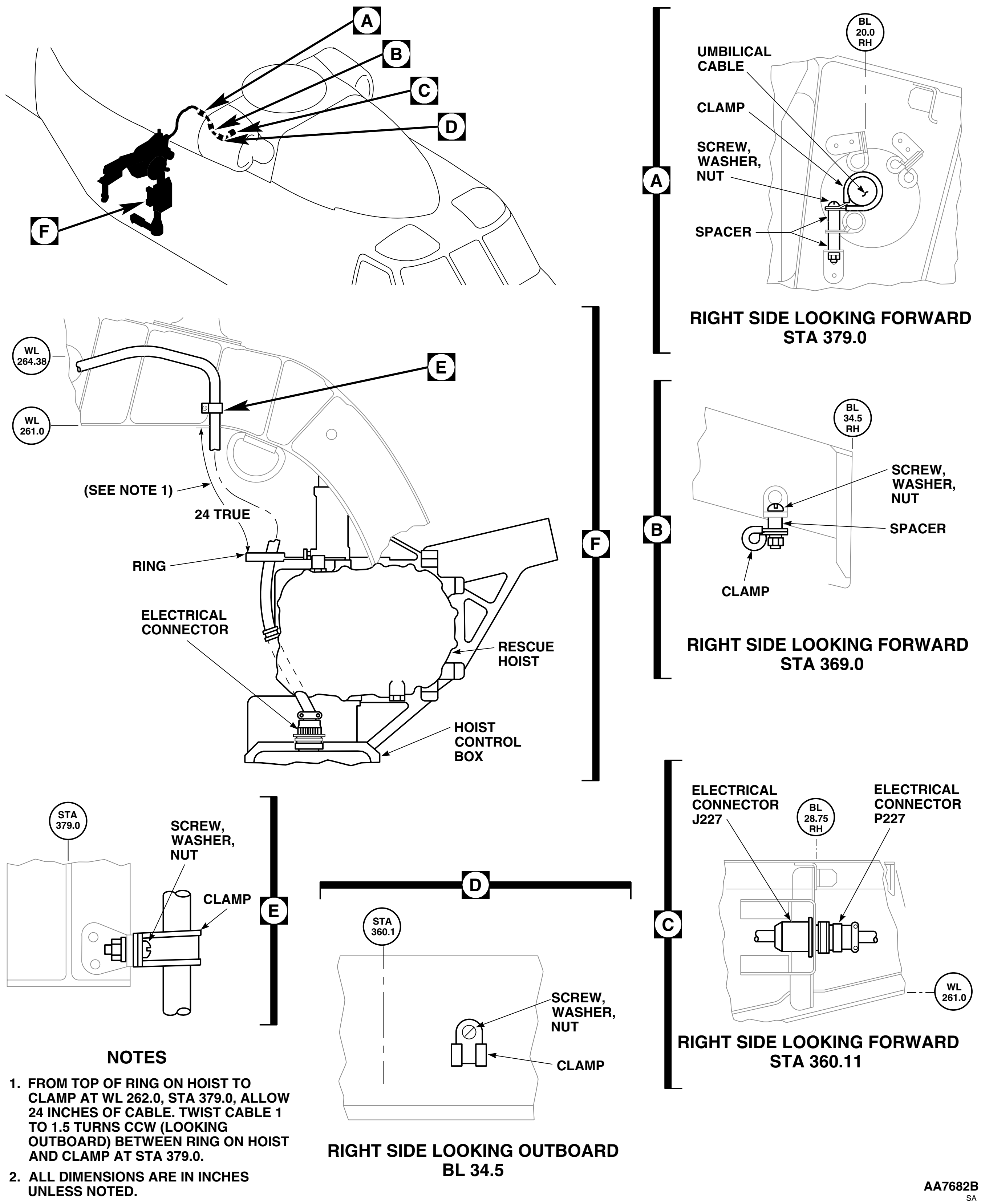


Figure 7. Remove/Install Rescue Hoist Umbilical Cable.

REMOVE RESCUE HOIST UMBILICAL CABLE

1. Disconnect umbilical cable from electrical connector, J227 (Figure 7, Detail C).
2. Remove clamps at STA 379.0, STA 369.0, STA 362.85, RBL 20.0, and RBL 34.5.
3. Disconnect umbilical cable electrical connector from hoist control box and remove umbilical cable.

REMOVE RESCUE HOIST

1. Disconnect cable cutter harness from cable cutter cartridge electrical connector. Install aluminum shorting strips between pins of cartridge electrical connector. Install a protective cap on connector (Figure 6, Sheet 1, Detail A).
2. Drain fluid from all three winch and boom head drains (Figure 6, Sheet 2, Detail D).

NOTE

Two or more persons shall handle hoist during removal.

3. Disconnect quick-release adapters attaching hoist base and reaction arm to adapter plate.
4. Disconnect quick-release adapter attaching hoist to upper channel anchor stud.
5. Pull vertical adjustment lock knob, turn vertical adjustment detent so that upper quick-release adapter will clear channel anchor stud, and remove hoist from helicopter.
6. Remove release pin attaching reaction arm to post.
7. Remove reaction arm, turn upside down, and reattach to post. Swing reaction arm up and lock quick-release adapter onto stud on post.
8. Remove adapter plate from floor of helicopter.
9. Remove adapter stud from channel.
10. Remove channel and bracket from cabin ceiling.

REMOVE RESCUE HOIST CONTROL PANEL, BRACKET, AND CABLE ASSEMBLY

1. Disconnect electrical connector, P226, from right side of lower console (Figure 4, Detail A).
2. Remove screws, washers, and clamps securing cable assembly to lower console.
3. Disconnect electrical connector, P228, from rescue hoist control panel and remove cable assembly.
4. Remove screws, washers and rescue hoist control panel from bracket assembly.
5. Remove screws, washers, and rescue hoist control panel bracket assembly.

REMOVE K14 BRACKET, RELAY, AND SUPPRESSOR

1. Remove screws and remove bracket assembly (Figure 3, Detail A).
2. Remove rubber boots from relay K14, terminals X1, X2, A1, and A2.
3. Remove wires from relay K14, terminals X1, X2, A1, and A2.
4. Remove suppressor from relay K14, terminals X1 and X2.
5. Remove screws and remove relay K14 from bracket.

REMOVE K14 BRACKET, RELAY, AND SUPPRESSOR - Continued

6. Remove nut and treated aluminum washer securing wire, No. M3443A20N, from ground at STA 295.0, RBL 24.0. Tag treated aluminum washer.
7. Remove wire, No. M3443A20N, from ground.

NO. 2 JUNCTION BOX DEMODIFICATION

1. Remove No. 2 junction box cover.
2. Remove nut, clamp and wire, No. M3432A06, from fuse holder (Figure 2, Detail A).
3. Remove nut and remove bus bar and fuse from fuse holder.
4. Remove nut from terminal A1 of relay K6 and remove bus bar.
5. Remove screws and remove fuse holder from No. 2 junction box.
6. Install cover on No. 2 junction box.

REMOVE RESCUE HOIST HARNESS

1. Remove nut and treated aluminum washer securing wire, No. M3442A06N, from ground at STA 379.0, and RBL 24.0. Tag treated aluminum washer (Figure 1, Sheet 3, Detail N).
2. Remove bolt, lockwasher, washers and wire terminal from ground (Figure 1, Sheet 3, Detail N).
3. Remove screw, washer, spacer, nut, and clamp attaching harness to structure at STA 370.0, RBL 24.0 (Figure 1, Sheet 3, Detail M).
4. Remove screw, washer, nut, and clamp attaching harness to structure at STA 362.0, RBL 24.0 (Figure 1, Sheet 3, Detail L).
5. Remove screws, nuts, and electrical connector, J227 (Figure 1, Sheet 3, Detail K).
6. Remove screws and clamps at STA 360.0, RBL 20.0; STA 352.0, RBL 20.0; STA 343.0, RBL 20.0; STA 327.0, RBL 22.0; STA 308.0, RBL 22.0, STA 295.0, BL 22.0 (Figure 1, Sheet 2, Details G, H, I, and J and Figure 1, Sheet 3, Detail L).
7. Remove electrical connector, P229, from rescue hoist provisions panel (Figure 1, Sheet 2, Detail F).
8. Remove electrical protective cover from dummy receptacle and attach to electrical connector, J229, of rescue hoist provisions panel.
9. Install protective caps and plugs on all open electrical connectors on harness.
10. Remove electrical harness from helicopter.

COMPLETE REMOVAL

1. Make sure area is clean and free of foreign objects.
2. Install soundproofing panels (WP 0258 00).
3. **UH60L UH60A** Install troop seats (WP 0233 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

HOISTS AND WINCHES

RESCUE HOIST FUSE CL16 HOIST 42305R1

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Electrical Repairer's Toolkit, SC 5180-99-B06

Personnel Required

Aircraft Electrician MOS 15F (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0152 00
WP 0258 00

REMOVE

1. Turn off all electrical power.
2. Remove soundproofing panels (WP 0258 00).
3. Remove No. 2 junction box cover.
4. Loose nuts and remove fuse from fuse holder (Figure 1, Detail A).

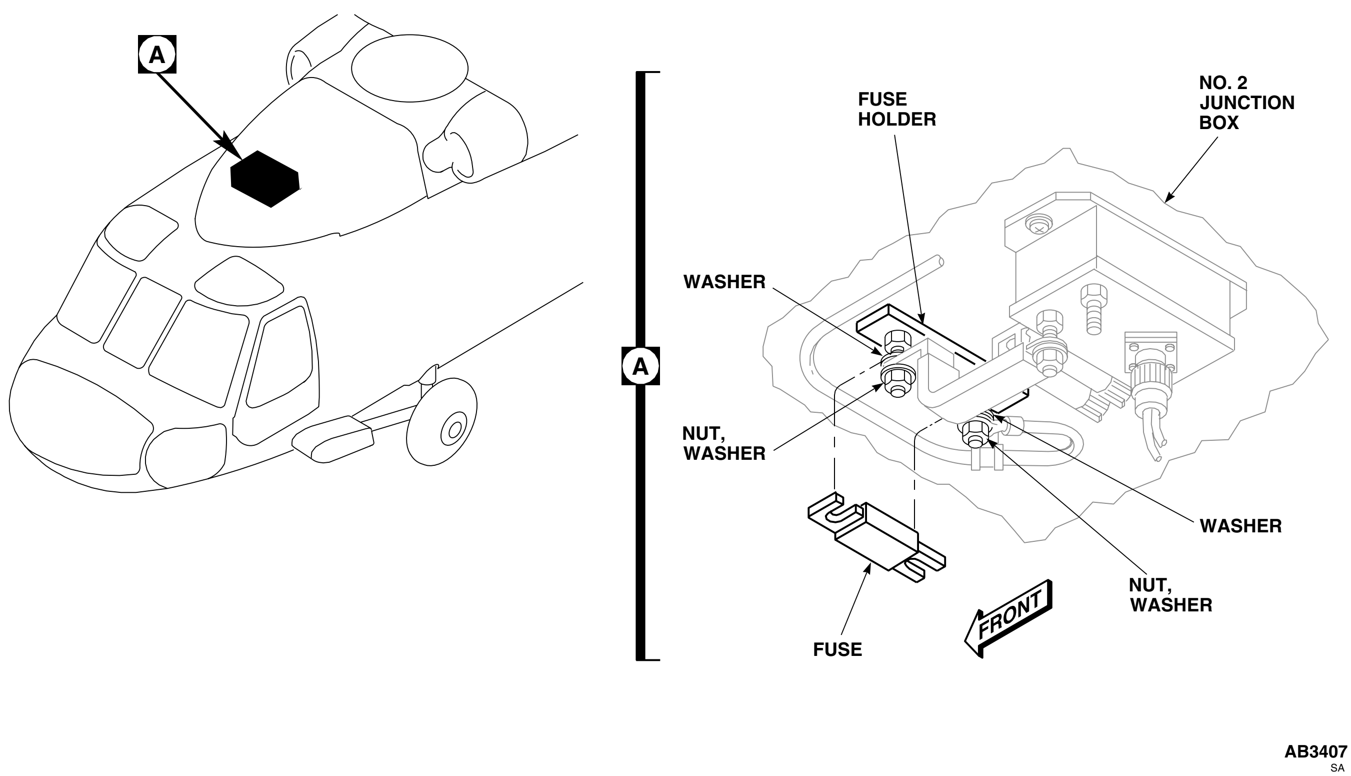


Figure 1. Replace Rescue Hoist Fuse CL16.

INSTALL

1. Turn off all electrical power.
2. **QA** Place fuse in fuse holder and tighten nuts. Make sure one end of fuse is placed between wire, No. M3432A06, and washer nearest fuse holder, and other end of fuse is placed between bus bar and washer nearest fuse holder (Figure 1, Detail A).
3. **QA** Install cover on No. 2 junction box.
4. Make sure area is clean and free of foreign material.
5. Install soundproofing panels (WP 0258 00).
6. Do an operational check of rescue hoist system (WP 0152 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

HOISTS AND WINCHES

RESCUE HOIST FUSE HOLDER HOIST 42305R1

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Electrical Repairer's Toolkit, SC 5180-99-B06

Personnel Required
Aircraft Electrician MOS 15F (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0152 00
WP 0258 00

REMOVE

1. Turn off all electrical power.
2. Remove soundproofing panels (WP 0258 00).
3. Remove No. 2 junction box cover.
4. Remove nut and remove wire, No. M3432A06, from fuse holder (Figure 1, Detail A).
5. Remove nut and remove bus bar and fuse from fuse holder.
6. Remove nut from terminal A1 of relay K6 and remove bus bar.
7. Remove screws and remove fuse holder from No. 2 junction box.

INSTALL

1. Turn off all electrical power.
2. **QA** Install fuse holder in No. 2 junction box with screws (Figure 1, Detail A).
3. **QA** Place fuse in fuse holder and connect bus bar to fuse holder with washer and nut.
4. **QA** Connect bus bar to terminal A1 of relay K6 with washer, lockwasher, and nut.
5. **QA** Connect wire, No. M3432A06, to fuse holder with washer and nut.
6. **QA** Install cover on No. 2 junction box.
7. Make sure area is clean and free of foreign material.
8. Install soundproofing panels (WP 0258 00).
9. Do an operational check of rescue hoist system (WP 0152 00).

INSTALL - Continued

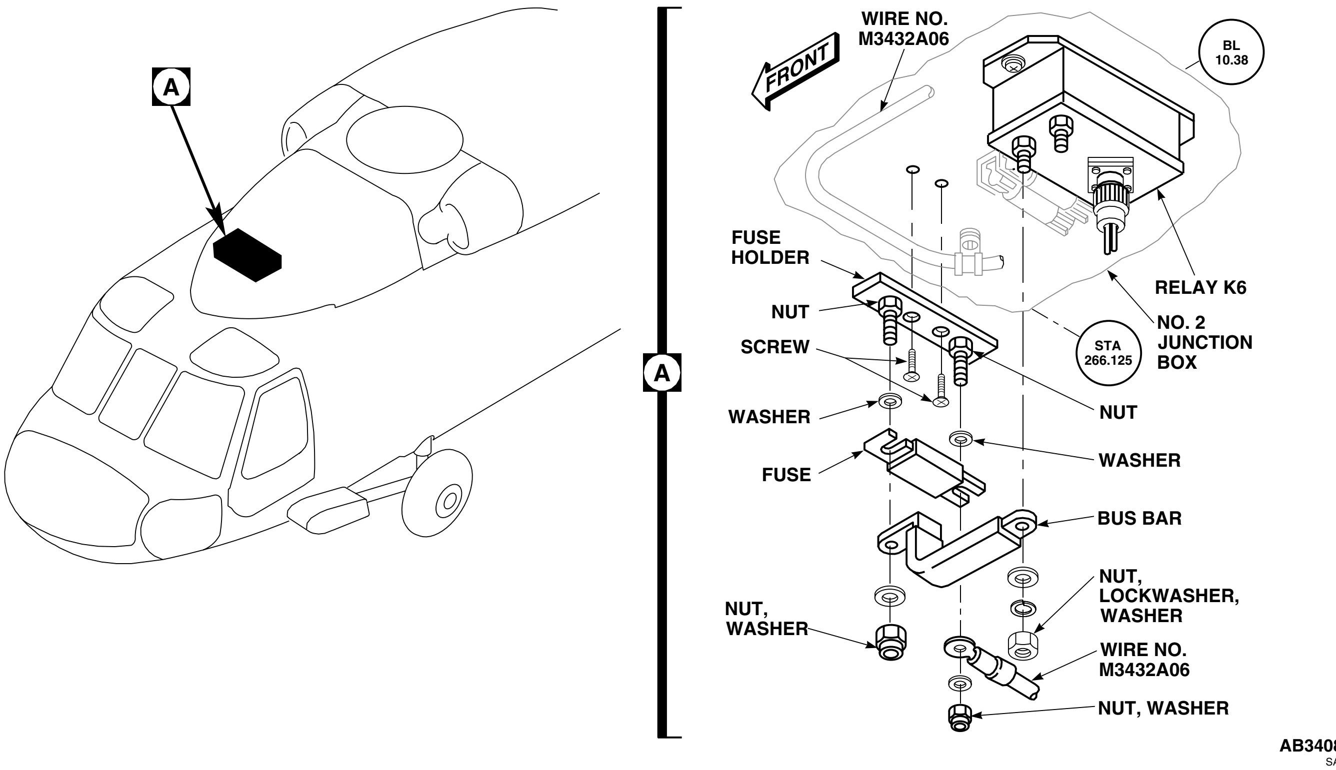


Figure 1. Replace Rescue Hoist Fuse Holder.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

HOISTS AND WINCHES

RESCUE HOIST BUS BAR HOIST 42305R1

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Electrical Repairer's Toolkit, SC 5180-99-B06

References
WP 0152 00
WP 0258 00

Personnel Required
Aircraft Electrician MOS 15F (1)
UH-60 Helicopter Repairer MOS 15T (1)

REMOVE

1. Turn off all electrical power.
2. Remove soundproofing panels (WP 0258 00).
3. Remove No. 2 junction box cover.
4. Remove washer and nut and remove bus bar from fuse holder (Figure 1, Detail A).

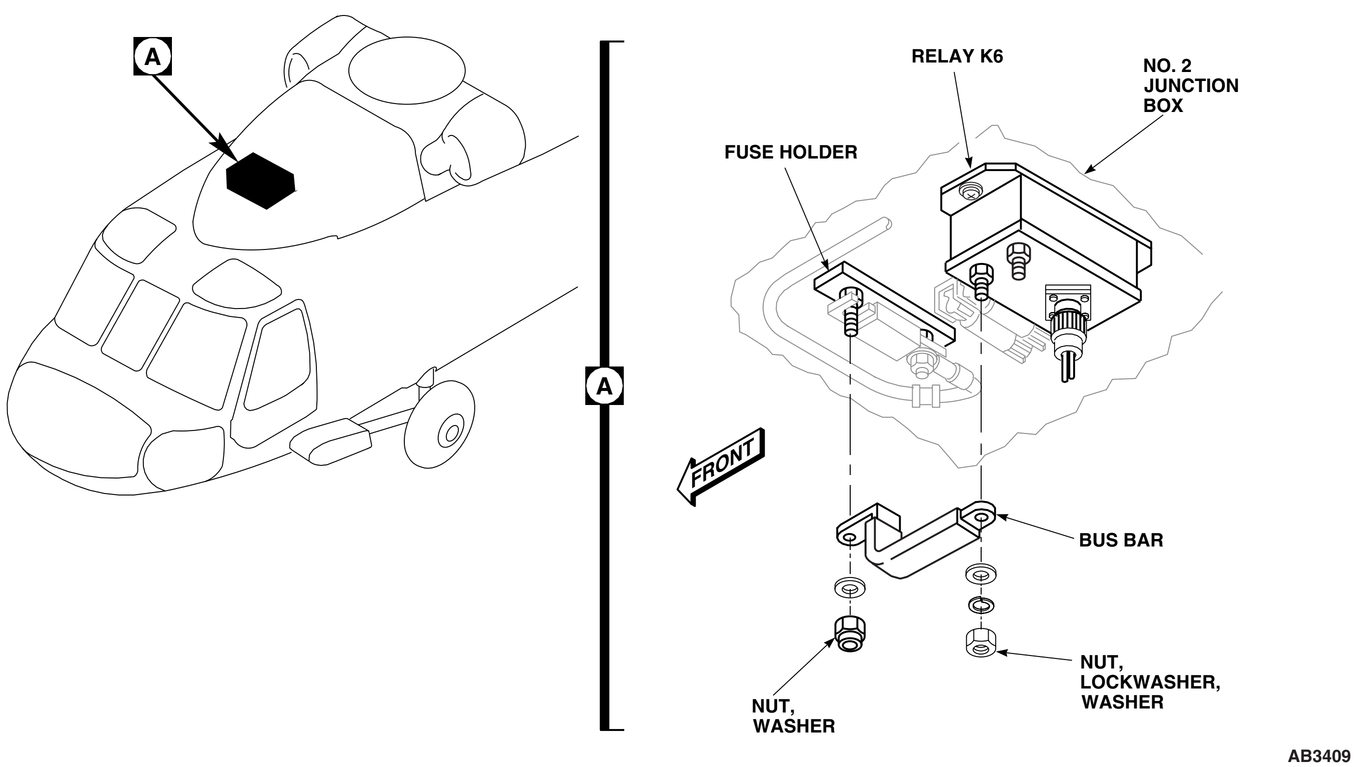


Figure 1. Replace Rescue Hoist Bus Bar.

REMOVE - Continued

5. Remove washer, lockwasher, and nut from terminal A1 of relay K6 and remove bus bar.

INSTALL

1. Turn off all electrical power.
2. **QA** Connect bus bar to terminal A1 of relay K6 with washer, lockwasher, and nut (Figure 1, Detail A).
3. **QA** Connect other end of bus bar to fuse holder with washer and nut.
4. **QA** Install cover on No. 2 junction box.
5. Make sure area is clean and free of foreign material.
6. Install soundproofing panels (WP 0258 00).
7. Do an operational check of rescue hoist system (WP 0152 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

HOISTS AND WINCHES

RESCUE HOIST RELAY K14 BRACKET ASSEMBLY HOIST 42305R1

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0152 00
WP 0258 00

REMOVE

1. Turn off all electrical power.
2. Remove soundproofing panels (WP 0258 00).
3. While supporting relay, remove screws, washers and nuts securing relay K14 to bracket assembly (Figure 1, Detail A). Remove K14 relay from bracket assembly.
4. Remove screws and washers securing bracket assembly to structure. Remove bracket assembly from structure.

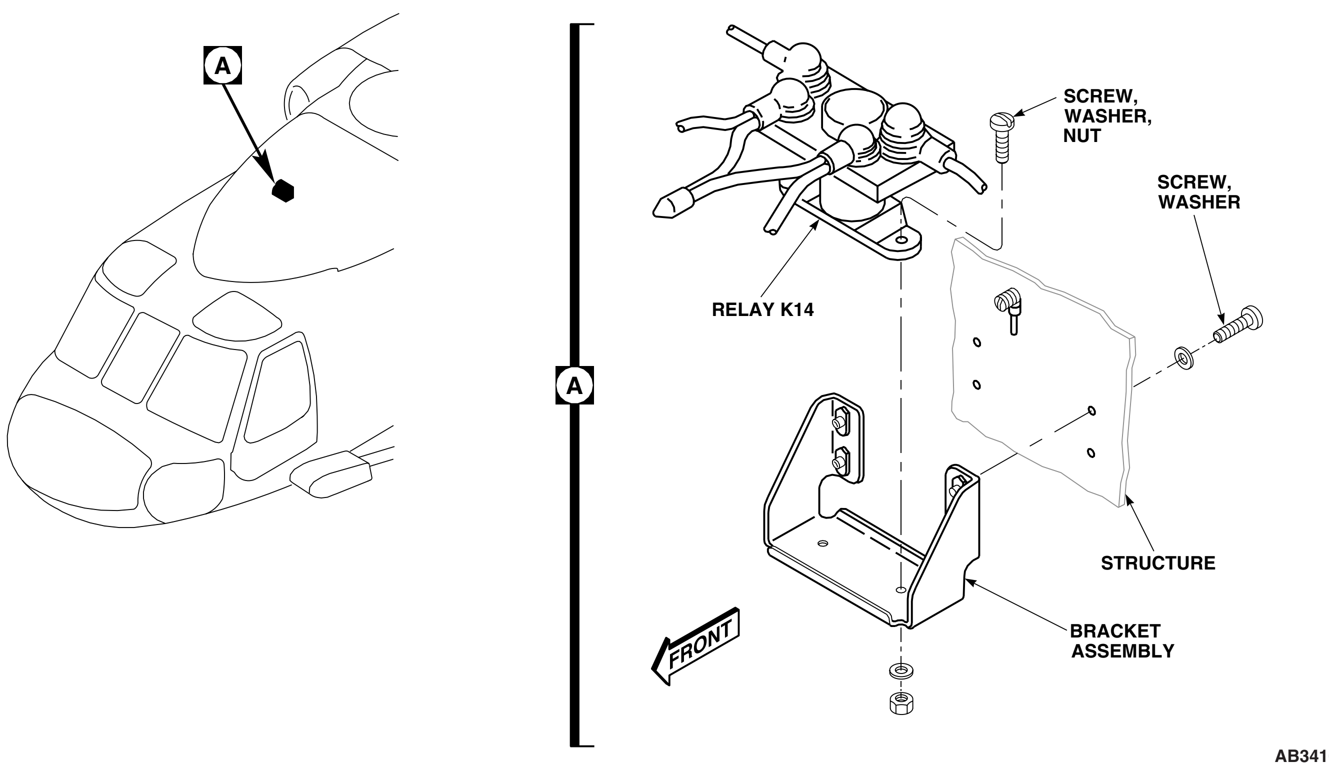


Figure 1. Replace Rescue Hoist Relay K14 Bracket Assembly.

INSTALL

1. Turn off all electrical power.
2. **QA** Secure bracket assembly to structure using screws and washers (Figure 1, Detail A).
3. **QA** Secure relay K14 to bracket assembly using screws, washers, and nuts.
4. Make sure area is clean and free of foreign material.
5. Install soundproofing panels (WP 0258 00).
6. Do an operational check of rescue hoist system (WP 0152 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

HOISTS AND WINCHES

RESCUE HOIST RELAY K14 HOIST 42305R1

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Electrical Repairer's Toolkit, SC 5180-99-B06

Personnel Required
Aircraft Electrician MOS 15F (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0152 00
WP 0258 00

REMOVE

1. Turn off all electrical power.
2. Remove soundproofing panels (WP 0258 00).
3. Remove rubber boot from relay K14, terminals X1, X2, A1, and A2 (Figure 1, Detail A).
4. Remove wires from relay K14, terminals X1, X2, A1, and A2.

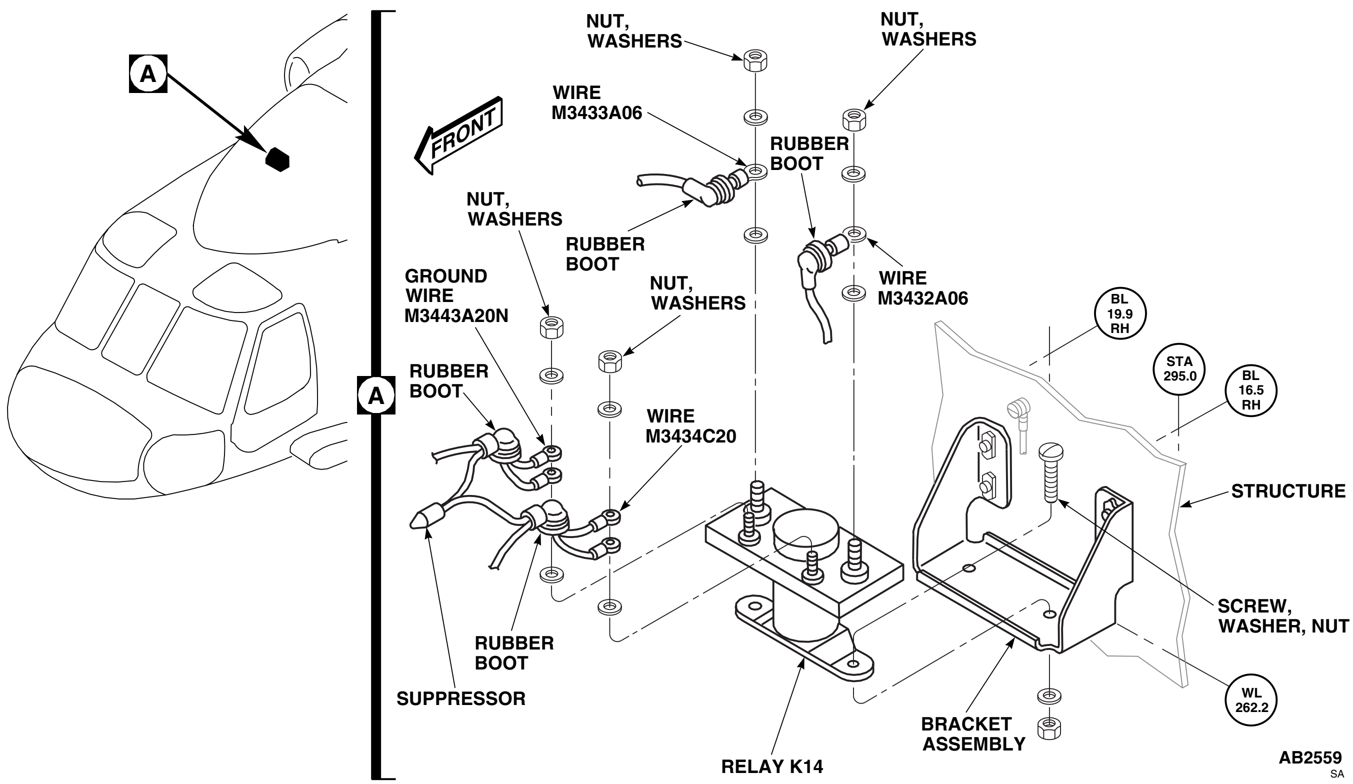


Figure 1. Replace Rescue Hoist Relay K14.

REMOVE - Continued

5. Remove suppressor from relay K14, terminals X1 and X2.
6. Remove screws, washers and nuts securing relay K14 to bracket assembly. Remove relay K14 from bracket assembly.

INSTALL

1. Turn off all electrical power.
2. **QA** Secure relay K14 to bracket assembly using screws, washers, and nuts (Figure 1, Detail A).
3. **QA** Connect wire, No. M3432A06, to relay K14, terminal A1, and install nut and washer. Install rubber boot on terminal.
4. **QA** Connect wire, No. M3433A06, to relay K14, terminal A2, and install nut and washer. Install nipple on terminal.
5. Place POS lead from suppressor on relay K14, terminal X1. Do not fasten.
6. Place NEG lead from suppressor on relay K14, terminal X2. Do not fasten.
7. **QA** Place wire, No. M3434C20, on relay K14, terminal X1, and install washer and nut. Install rubber boot on terminal.
8. **QA** Place wire, No. M3443A20N, on relay K14, terminal X2, and install washer and nut. Install rubber boot on terminal.
9. Do an operational check of rescue hoist system (WP 0152 00).
10. Make sure area is clean and free of foreign material.
11. Install soundproofing panels (WP 0258 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

HOISTS AND WINCHES

RESCUE HOIST RELAY K14 SUPPRESSOR HOIST 42305R1

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Crimping Tool, RR-G-619
Electrical Repairer's Toolkit, SC 5180-99-B06

Personnel Required
Aircraft Electrician MOS 15F (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0152 00
WP 0258 00

REMOVE

1. Turn off all electrical power.
2. Remove soundproofing panels (WP 0258 00).
3. Remove rubber boot from relay K14, terminals X1 and X2 (Figure 1, Detail A).

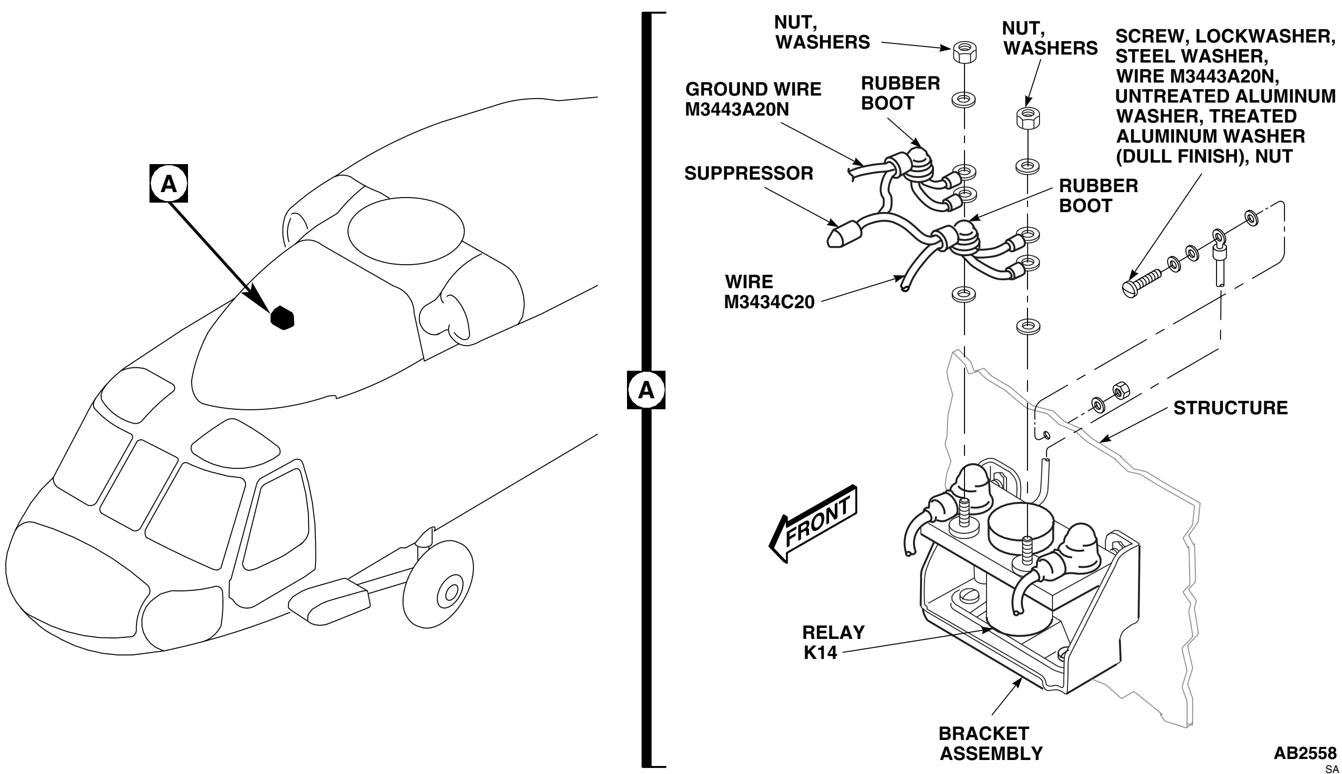


Figure 1. Replace Rescue Hoist Relay K14 Suppressor.

REMOVE - Continued

4. Remove wires from relay K14, terminals X1 and X2.
5. Remove suppressor from relay K14, terminals X1 and C2.
6. Remove screw, lockwasher, steel washer, untreated aluminum washer, treated aluminum washer (dull finish), and nut securing wire, No. M3443A20N, from ground at STA 295.0, RBL 24.0. Tag treated aluminum washer.
7. Remove wire, No. M3443A20N, from ground.

INSTALL

1. Turn off all electrical power.
2. **QA** Cut suppressor POS and NEG leads at 3 inches from end of cap.
3. Strip 1/4-inch of insulation from leads.
4. Using crimping tool, crimp terminals to suppressor leads.
5. Place POS lead from suppressor on relay K14, terminal X1. Do not fasten.
6. Place NEG lead from suppressor on relay K14, terminal X2. Do not fasten.
7. **QA** Place ground wire, No. M3434C20, on relay K14, terminal X1, and install washer and nut. Install rubber boot on terminal (Figure 1, Detail A).
8. **QA** Place ground wire, No. M3443A20N, on relay K14, terminal X2, and install washer and nut. Install rubber boot on terminal.
9. Make sure airframe ground is clean for good electrical connection.
10. **QA** Connect ground wire, No. M3443A20N, to helicopter frame using screw, lockwasher, steel washer, untreated aluminum washer, tagged treated aluminum washer (dull finish), and nut.
11. Do an operational check of rescue hoist system (WP 0152 00).
12. Make sure area is clean and free of foreign material.
13. Install soundproofing panels (WP 0258 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

HOISTS AND WINCHES

RESCUE HOIST UMBILICAL CABLE HOIST 42305R1

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-CL-B01

Material/Parts
Protective Caps and Plugs

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0152 00
WP 0258 00
WP 1703 00

REMOVE

1. Turn off all electrical power.
2. Remove soundproofing panels (WP 0258 00).

NOTE

Install protective caps and plugs as electrical connections are removed.

3. Disconnect cable electrical connector, P227, from electrical connector, J227 (Figure 1, Detail C).
4. Remove clamps at STA 379.0, STA 369.0, STA 362.85, RBL 34.5, and RBL 20.0 (Figure 1, Detail A, Detail B, Detail D and Detail E).
5. Disconnect cable electrical connector from hoist control box (Figure 1, Detail F).
6. Remove cable.

INSTALL

1. Turn off all electrical power.
2. **QA** From top of ring on hoist to WL 261.0, allow 24 inches of cable. Twist cable 1 to 1 1/2-turns counterclockwise (looking outboard) between ring on hoist and clamp on STA 379.0. Clamp cable at STA 379.0, RBL 25.0, and WL 262.0 (Figure 1, Detail F).
3. **QA** Route cable to electrical connector, J227.
4. Inspect and treat electrical connector (WP 1703 00).

NOTE

Remove protective caps and plugs prior to installation.

5. **QA** Connect cable electrical connector, P227, to electrical connector, J227 (Figure 1, Detail C).
6. **QA** Secure cable with clamps at STA 379.0, STA 369.0, STA 362.85, RBL 34.5, and RBL 20.0 (Figure 1, Detail A, Detail B, Detail D and Detail E).
7. **QA** Make sure other cable connector is connected to hoist control box (Figure 1, Detail F).

INSTALL - Continued

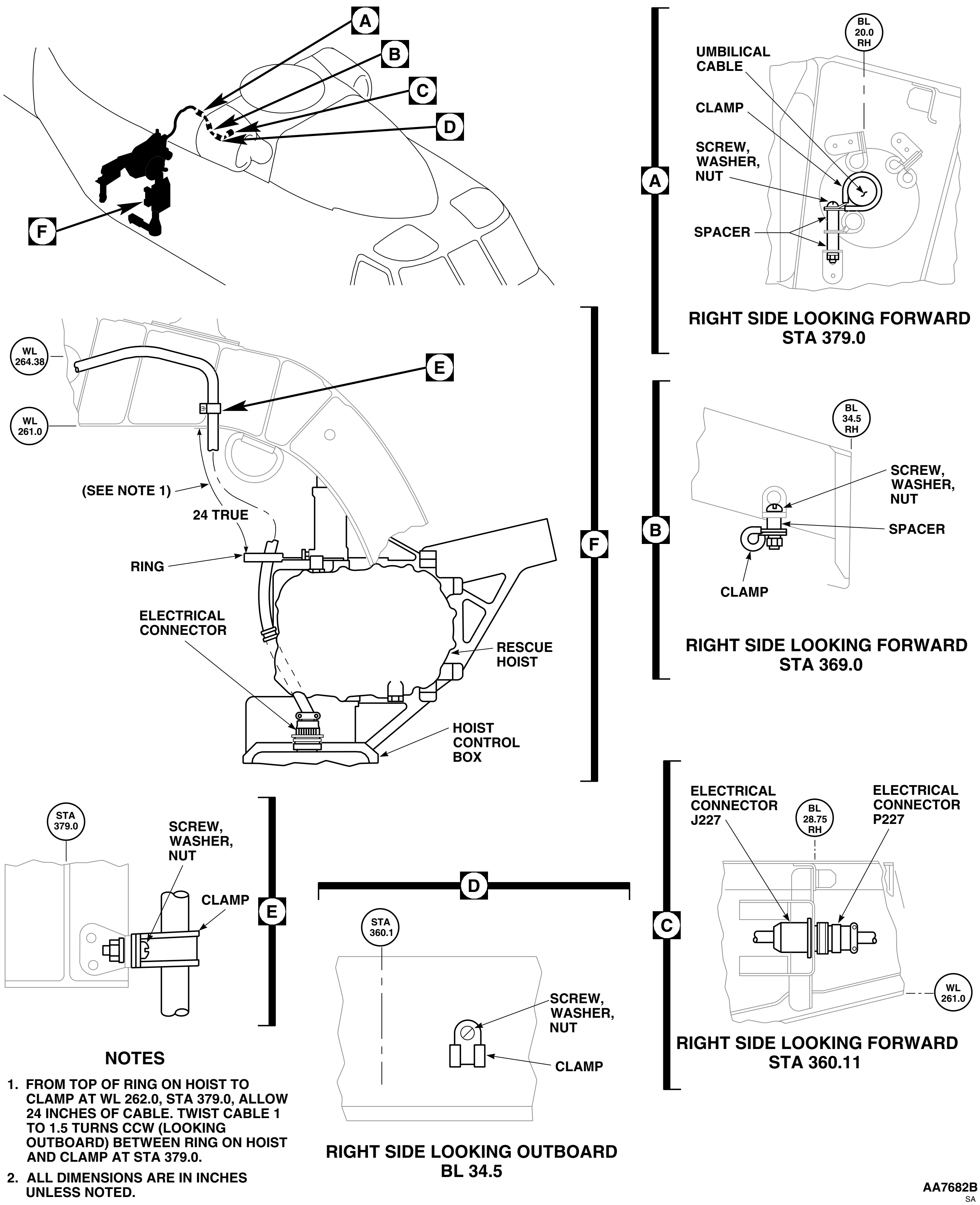


Figure 1. Replace Rescue Hoist Umbilical Cable.

INSTALL - Continued

8. Do an operational check of rescue hoist system (WP 0152 00).
9. Make sure area is clean and free of foreign material.
10. Install soundproofing panels (WP 0258 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

HOISTS AND WINCHES

RESCUE HOIST CONTROL PANEL BRACKET ASSEMBLY HOIST 42305R1

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Material/Parts
Protective Caps and Plugs

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0152 00
WP 1703 00

REMOVE

1. Turn off all electrical power.

NOTE

Install protective caps and plugs as electrical connections are removed.

2. Disconnect electrical connector, P228, from rescue hoist control panel (Figure 1, Detail A).
3. Remove screws and washers securing rescue hoist control panel to bracket assembly. Remove rescue hoist control panel from bracket assembly.
4. Remove screws and washers securing bracket assembly to lower console. Remove bracket assembly from lower console.

INSTALL

1. Turn off all electrical power.
2. **QA** Secure bracket assembly to lower console using screws and washers (Figure 1, Detail A).
3. **QA** Secure rescue hoist control panel to bracket assembly using screws and washers.
4. Inspect and treat electrical connector (WP 1703 00).

NOTE

Remove protective caps and plugs prior to installation.

5. **QA** Connect electrical connector, P228, to rescue hoist control panel.
6. Do an operational check of rescue hoist system (WP 0152 00).

INSTALL - Continued

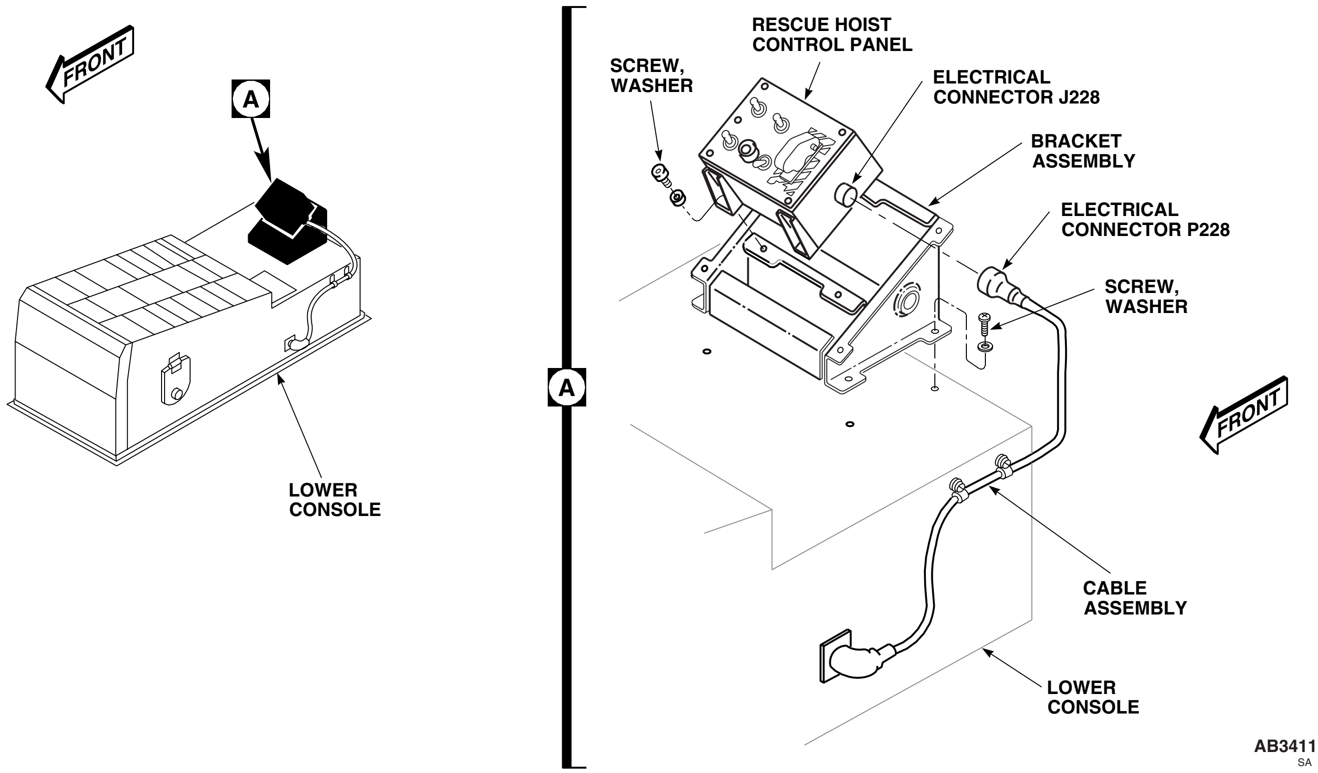


Figure 1. Replace Rescue Hoist Control Panel Bracket Assembly.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

HOISTS AND WINCHES

RESCUE HOIST CONTROL PANEL HOIST 42305R1

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Material/Parts
Protective Caps and Plugs

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0152 00
WP 1703 00

REMOVE

1. Turn off all electrical power.

NOTE

Install protective caps and plugs as electrical connections are removed.

2. Disconnect electrical connector, P228, electrical connector, J228, from rescue hoist control panel (Figure 1, Detail A).
3. Remove screws and washers securing rescue hoist control panel to bracket assembly. Remove rescue hoist control panel from bracket assembly.

INSTALL

1. Turn off all electrical power.
2. **QA** Place rescue hoist control panel in bracket on lower console and attach with screws and washers (Figure 1, Detail A).

NOTE

Remove protective caps and plugs prior to inspection.

3. Inspect and treat electrical connector (WP 1703 00).
4. **QA** Connect electrical connector, P228, to electrical connector, J228 on rescue hoist control panel.
5. Do an operational check of rescue hoist system (WP 0152 00).

INSTALL - Continued

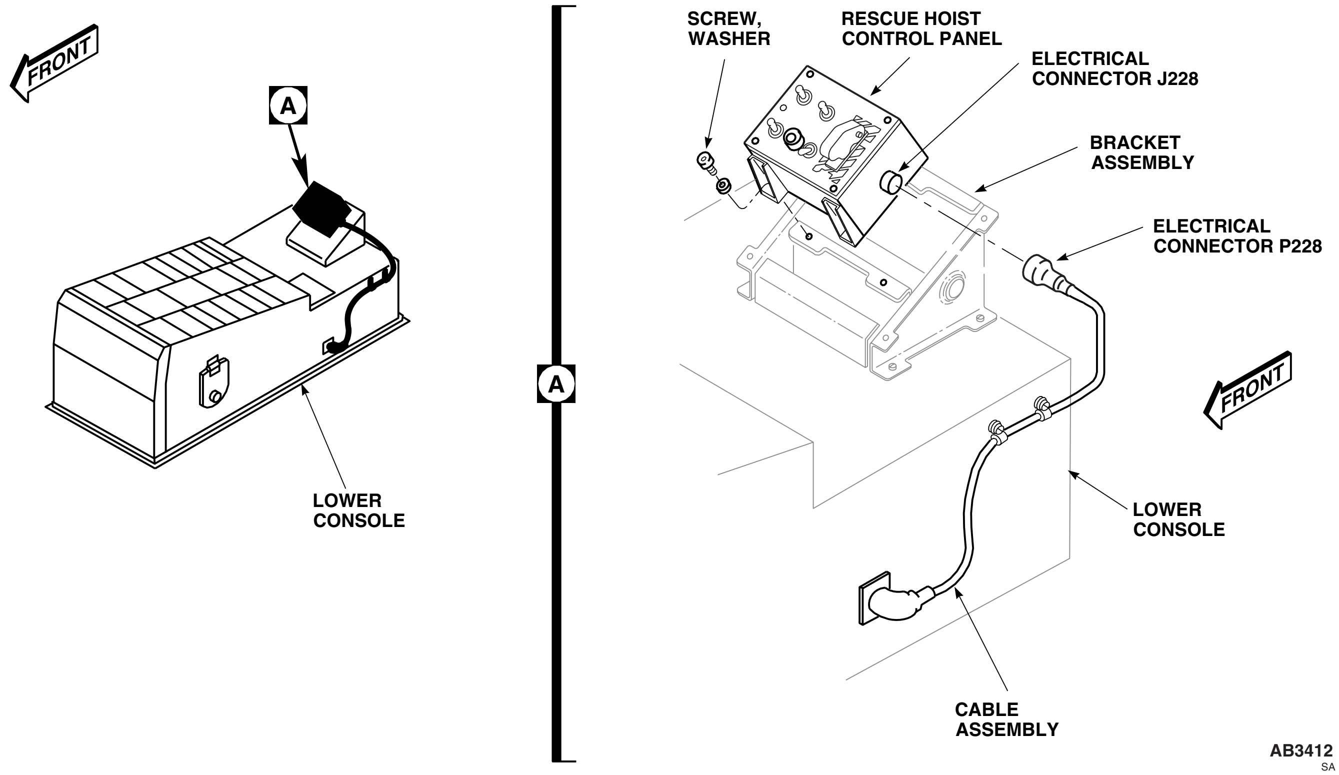


Figure 1. Replace Rescue Hoist Control Panel.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

HOISTS AND WINCHES

RESCUE HOIST CONTROL PANEL SQUIB TEST LAMP HOIST 42305R1

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0152 00

REMOVE

1. Turn off all electrical power.
2. Unscrew and remove SQUIB IND lamp indicator assembly from rescue hoist control panel (Figure 1, Detail A).
3. Remove lamp from SQUIB IND lamp indicator assembly.

INSTALL

1. Turn off all electrical power.

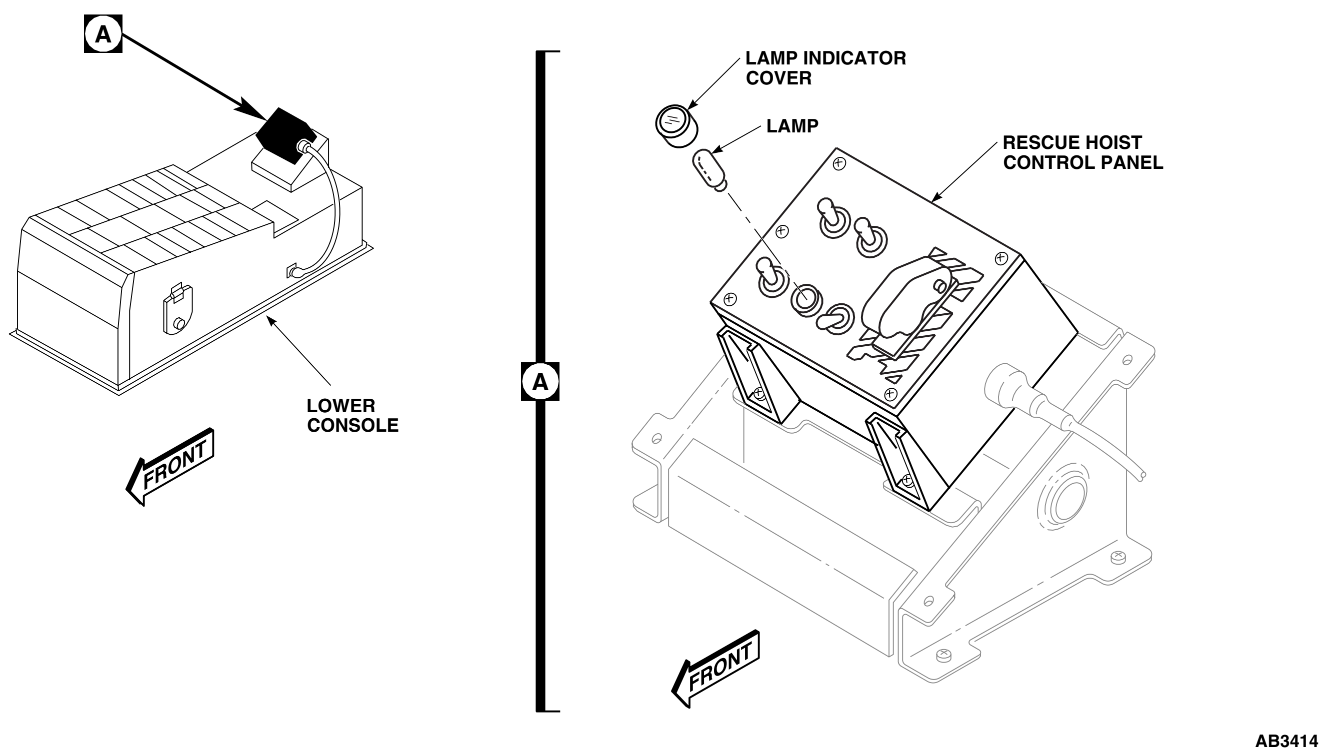


Figure 1. Replace Rescue Hoist Control Panel Squib Test Lamp.

INSTALL - Continued

2. Replace lamp in SQUIB IND lamp indicator assembly (Figure 1, Detail A).
3. **QA** Install SQUIB IND lamp indicator assembly in rescue hoist control panel.
4. Do an operational check of rescue hoist system (WP 0152 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

HOISTS AND WINCHES

RESCUE HOIST CONTROL PANEL CABLE ASSEMBLY HOIST 42305R1

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0152 00
WP 1703 00

REMOVE

1. Turn off all electrical power.
2. Disconnect electrical connector, P228, from electrical connector, J228, on rescue hoist control panel (Figure 1, Detail A).
3. Remove screws and clamps securing cable assembly to lower console.
4. Disconnect electrical connector, P226, from electrical connector, J226, on left side of lower console, and remove cable assembly.

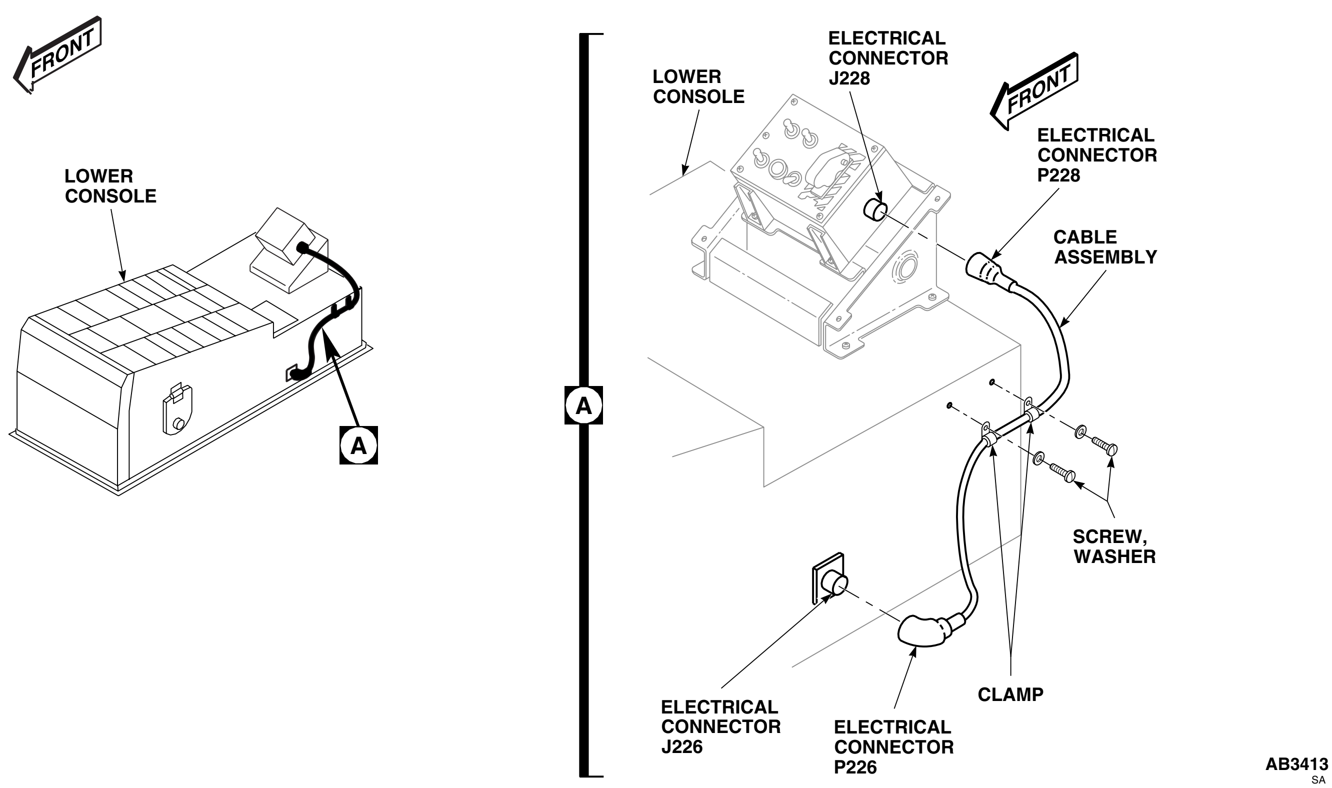


Figure 1. Replace Rescue Hoist Control Panel Cable Assembly.

INSTALL

1. Turn off all electrical power.
2. Inspect and treat electrical connector (WP 1703 00).
3. **QA** Connect cable assembly electrical connector, P228, to electrical connector, J228, on rescue hoist control panel (Figure 1, Detail A).
4. Inspect and treat electrical connector (WP 1703 00).
5. **QA** Connect cable assembly electrical connector, P226, to electrical connector, J226, on left side of lower console.
6. **QA** Install cable assembly to lower console with clamps, screws and washers.
7. Do an operational check of rescue hoist system (WP 0152 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

HOISTS AND WINCHES

RESCUE HOIST CONTROL PANEL INFORMATION PLATE HOIST 42305R1

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Material/Parts
Wire, Nonelectrical Item 351, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0152 00
WP 1685 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Remove screws securing switch guard to information plate (Figure 1, Detail A). Remove switch guard from information plate.
3. Unscrew lens from indicator light.

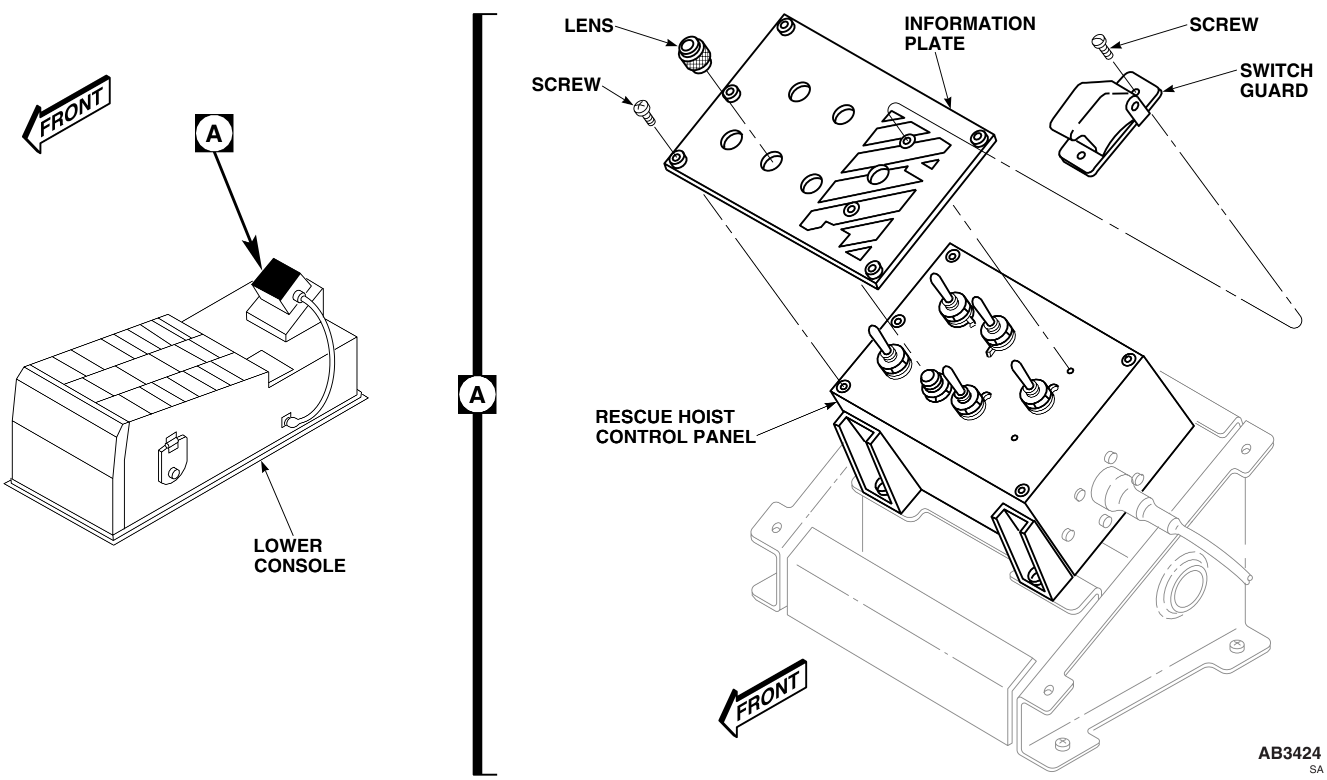


Figure 1. Replace Rescue Hoist Control Panel Information Plate.

REMOVE - Continued

4. Remove screws securing information plate to rescue hoist control panel. Remove information plate from rescue hoist control panel,

INSTALL

1. Turn off all electrical power.
2. **QA** Secure information plate to rescue hoist control panel using screws (Figure 1, Detail A).
3. Screw lens into light indicator.
4. Secure switch guard to information plate using screws.
5. **QA** Using nonelectrical wire, Item 351, WP 1803 00, lockwire switch guard into position.
6. Connect external electrical power to helicopter (WP 1685 00).
7. **QA** Make certain LIGHTS LWR CSL circuit breaker is in.
8. Turn CONSOLE LTS LOWER BRT-OFF control to BRT.
9. Press PNL LTS pushbutton on pilot's cyclic stick. Lights go on if off, and off if on. If result is not as specified, do an operational check of rescue hoist system (WP 0152 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

HOISTS AND WINCHES

RESCUE HOIST PENDANT HOIST 42305R1

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0152 00
WP 1685 00

REMOVE

1. Turn off all electrical power (WP 1685 00).
2. Disconnect rescue hoist pendant from cable assembly electrical connector (Figure 1, Detail B).

INSTALL

1. Turn off all electrical power (WP 1685 00).
2. **QA** Connect rescue hoist pendant to cable assembly electrical connector (Figure 1, Detail B).
3. Do an operational check of rescue hoist system (WP 0152 00).

INSTALL - Continued

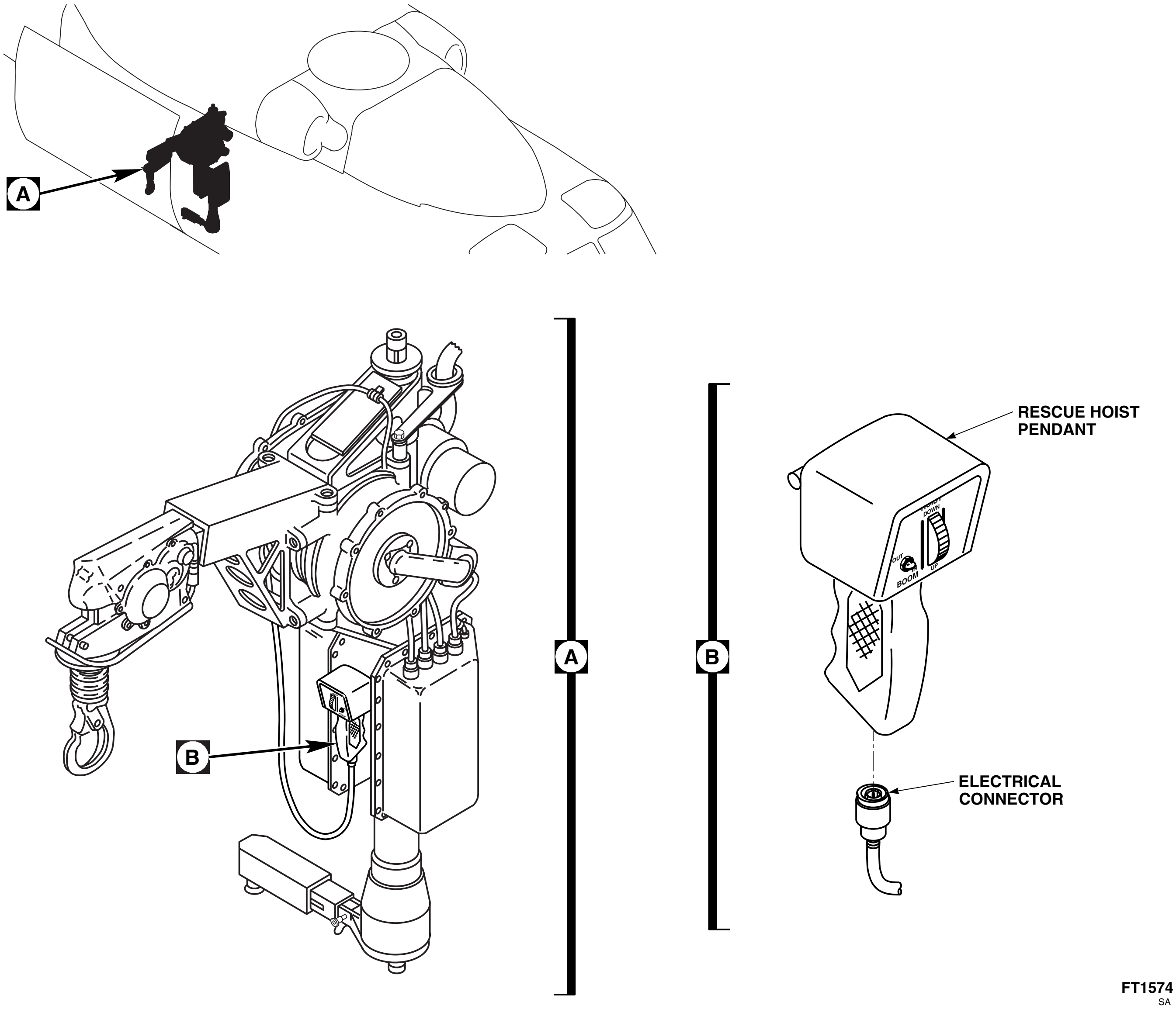


Figure 1. Replace Rescue Hoist Pendant.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

HOISTS AND WINCHES

RESCUE HOIST HH-60A HH-60L HOIST BL-29900-30

This WP supersedes WP 1342 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Fluorescent Penetrant Inspection Kit, XMA101
Leather Gloves
Nonmetallic Scraper
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.

Material/Parts

Abrasive Cloth, Assortment Item 1, WP 1803 00
Corrosion Resistant Coating Item 111, WP 1803 00
Epoxy Primer Coating Kit Item 120, WP 1803 00
Remover, Paint Item 252, WP 1803 00
Sealing Compound Item 273, WP 1803 00
Sealing Compound Item 257, WP 1803 00
Wire, Nonelectrical Item 354, WP 1803 00

Personnel Required

Assistants - (2)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-344-23
TM 1-1520-265-23
TM 55-1500-345-23
WP 0153 00
WP 0377 00
WP 0381 00
WP 1347 00
WP 1351 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Remove support arm fairings and connector cover (WP 1347 00).
3. Remove explosive cartridge (WP 1351 00).
4. Disconnect electrical connectors, P373 from J373 and P374 from J374, located on the helicopter roof (Figure 1, Sheet 2, Detail C).
5. Remove screws, washers, clamps, and electrical harness from base support (Figure 1, Sheet 2, Detail C).

WARNING

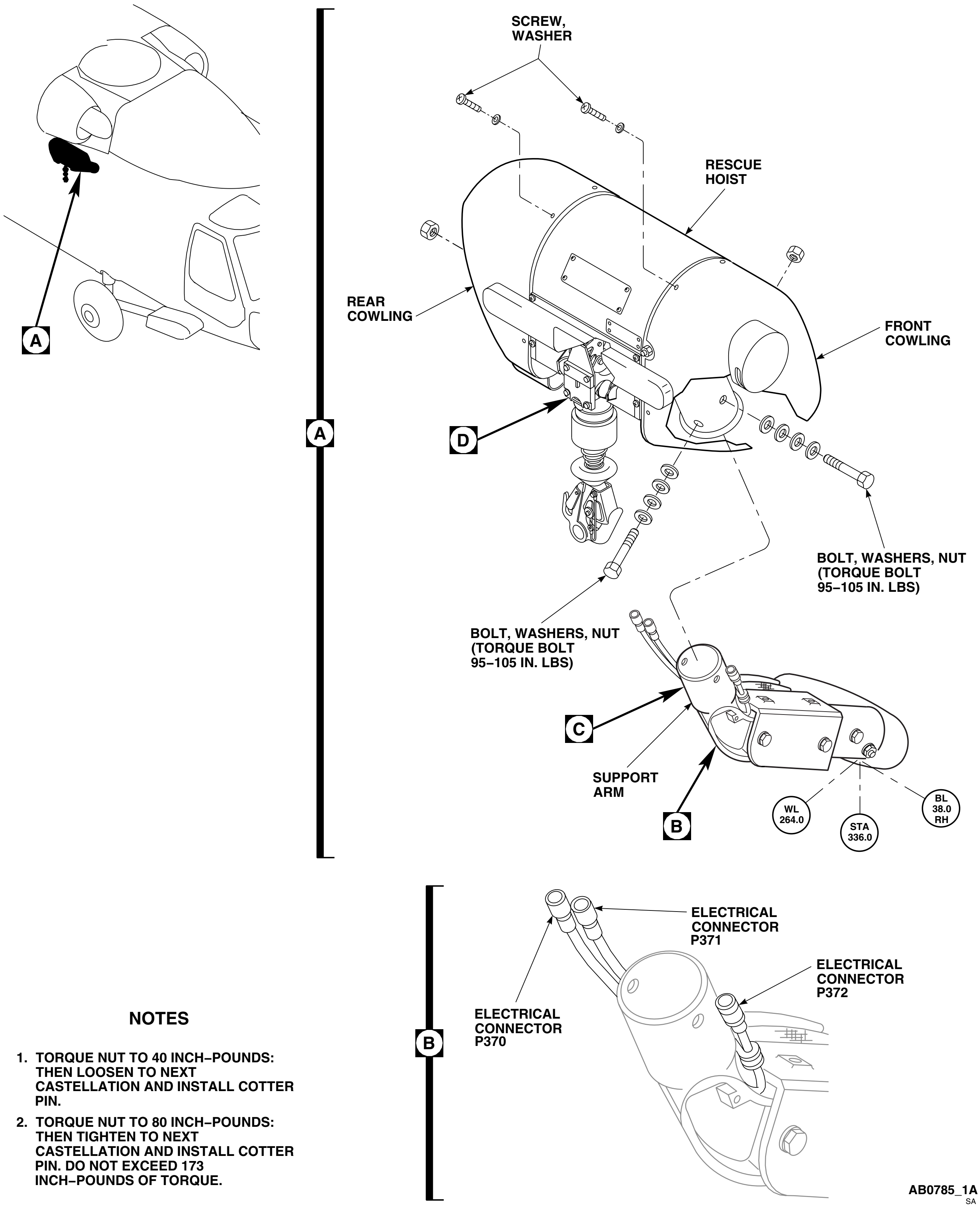
Injury to personnel and damage to equipment will result if rescue hoist is not supported during removal. Rescue hoist weighs 105 pounds. Use two assistants to support rescue hoist during removal.

6. Remove bolts, washers, and nuts securing support arm to base support. Remove hoist and support arm from helicopter.

DISASSEMBLE

1. Remove cuff fairing from rescue hoist (WP 1347 00).
2. Remove bolts, washers, and nuts securing support arm to hoist. Remove support arm.

DISASSEMBLE - Continued



NOTES

1. TORQUE NUT TO 40 INCH−POUNDS: THEN LOOSEN TO NEXT CASTELLATION AND INSTALL COTTER PIN.
2. TORQUE NUT TO 80 INCH−POUNDS: THEN TIGHTEN TO NEXT CASTELLATION AND INSTALL COTTER PIN. DO NOT EXCEED 173 INCH−POUNDS OF TORQUE.

Figure 1. Replace Rescue Hoist. (Sheet 1 of 3)

DISASSEMBLE - Continued

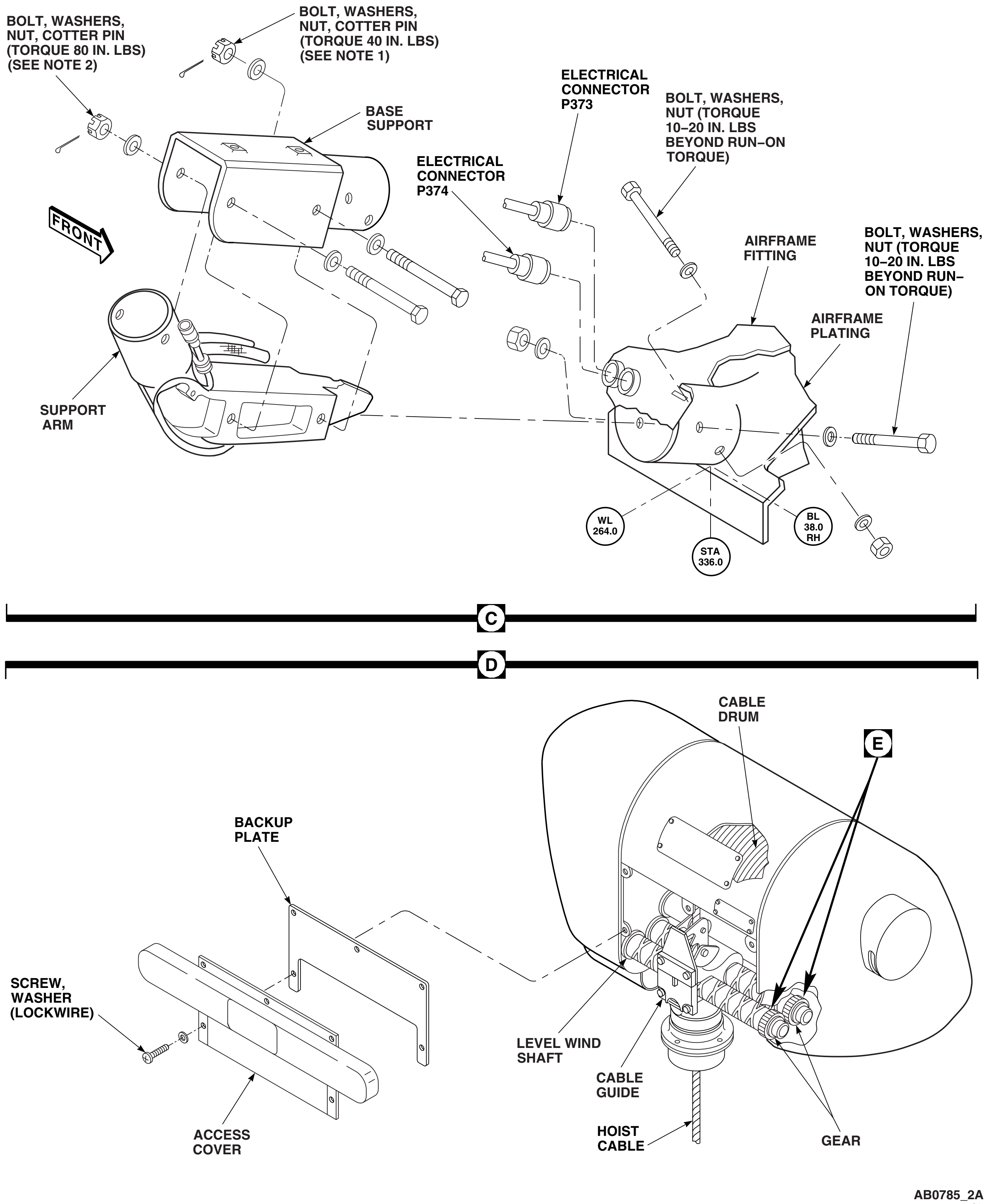


Figure 1. Replace Rescue Hoist. (Sheet 2 of 3)

DISASSEMBLE - Continued

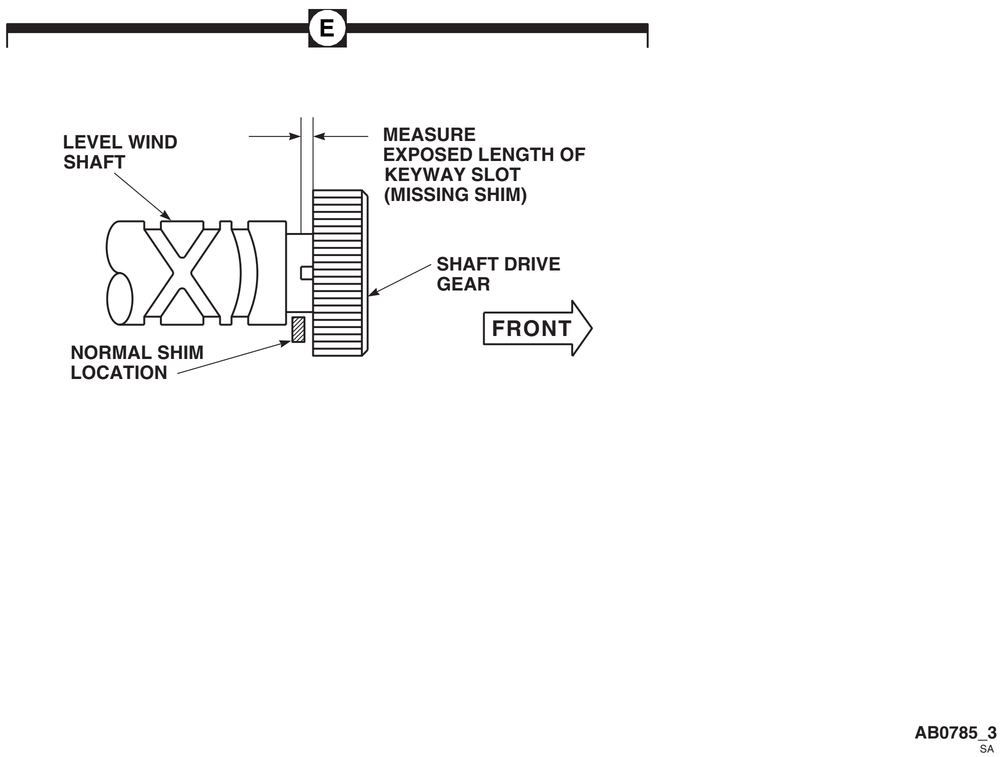


Figure 1. Replace Rescue Hoist. (Sheet 3 of 3)

3. Remove bolts, washers, and nuts securing base support to airframe fitting. Remove base support (Figure 1, Sheet 2, Detail C).
4. Remove sealing compound from base support and airframe fitting mating surface using nonmetallic scraper.

INSPECT

1. Turn off all electrical power.
2. Visually inspect hoist to determine if one or more shims are present between the forward ends of the level wind shafts and the shaft drive gears.

INSPECT - Continued

NOTE

Viewed from below, the level wind shafts are two cross-cut steel shafts in parallel with each other with a cable guide between them.

3. If no shims are present, proceed as follows:

   a. Remove screws and washers that secure the access cover and backup plate to the hoist and remove cover (Figure 1, Sheet 2, Detail D).

   b. Measure keyway slots in the forward ends of level wind shafts.

NOTE

It may be necessary to use a mirror and flashlight if the hoist is stopped with keyways facing inboard.

   c. If measurement of exposed length of keyway is 1/8-inch or less, install backup plate and access cover. Using nonelectrical wire, Item 354, WP 1803 00, lockwire screws.

   d. If measurement of exposed length of keyway is greater than 1/8-inch, replace hoist.

4. Inspect base support (Figure 2, Sheet 1) for cracks as follows:

   a. Using paint remover, Item 252, WP 1803 00, remove paint, sealing compound, and primer from inside and outside surfaces of cylindrical section that mates with channel and fitting section wall, Area A and Area B, respectively.

NOTE

Pay particular attention to areas around bolt holes during fluorescent penetrant inspection.

   b. Fluorescent penetrant inspect, Type 1, Method B, Method C, or Method D, Sensitivity Level 3 or Level 4, on cylindrical areas for cracks (TM 1-1520-265-23). **NO CRACKS ALLOWED**.

   c. Visually inspect all other areas for cracks. **NO CRACKS ALLOWED**.

5. Inspect base support (Figure 2, Sheet 1) for nicks, dents, scratches, pits, or gouges. Repair nicks, dents, scratches, pits, or gouges using abrasive cloth assortment, Item 1, WP 1803 00, as follows:

   a. Repair AREA A as follows:

      (1) **BLEND OUT NICKS, DENTS, SCRATCHES, PITS, OR GOUGES UP TO 0.010-INCH DEEP. BLENDED AREA AFTER REPAIR SHALL NOT EXCEED 0.750-INCH DIAMETER.**

      (2) **DAMAGE SHALL NOT BE CLOSER THAN ONE HOLE DIAMETER FROM ANY EXISTING FASTENER HOLES.**

      (3) Replace base support if damage or repairs are more than limits.

   b. Repair AREA B as follows:

      (1) **BLEND OUT NICKS, DENTS, SCRATCHES, PITS, OR GOUGES UP TO 0.002-INCH DEEP AND 0.750-INCH DIAMETER.**

      (2) Replace base support if damage or repairs are more than limits.

6. Inspect base support (Figure 2, Sheet 1) for corrosion. Repair using abrasive cloth assortment, Item 1, WP 1803 00, as follows:

   a. Repair AREA A as follows:

INSPECT - Continued

(1) **BLEND OUT CORROSION UP TO 0.0050-INCH DEEP OVER 100% OF SURFACE AREA.**

(2) Replace base support if damage or repairs are more than limits.

b. Repair AREA B as follows:

(1) **BLEND OUT CORROSION UP TO 0.0100-INCH DEEP OVER 100% OF INNER DIAMETER, AND UP TO 0.002-INCH DEEP INTO THE WALL THICKNESS TO MINIMUM OUTSIDE DIAMETER OF 2.966-INCHES.**

(2) Replace base support if damage or repairs are more than limits.

7. Inspect base support for incorporation of 0.375-inch slot at bottom of base support. If slot is not present do the following:

a. Add slot to base support as shown (Figure 2, Sheet 2, Detail A).

b. Treat repaired area with corrosion resistant coating, Item 111, WP 1803 00 (TM 1-1500-344-23).

c. Prime stripped and repaired area with epoxy primer coating kit, Item 120, WP 1803 00 (TM 1-1500-344-23).

8. Inspect support arm (Figure 2, Sheet 1) for cracks as follows:

a. Using paint remover, Item 252, WP 1803 00, remove paint, sealing compound, and primer from inside and outside surfaces of socket section that mates with channel and fitting section wall, Area A and Area B, respectively.

NOTE

Pay particular attention to areas around bolt holes during fluorescent penetrant inspection.

b. Fluorescent penetrant inspect, Type 1, Method B, Method C, or Method D, Sensitivity Level 3 or Level 4, on cylindrical areas for cracks (TM 1-1520-265-23). **NO CRACKS ALLOWED**.

c. Visually inspect all other areas for cracks. **NO CRACKS ALLOWED**.

9. Inspect support arm (Figure 2, Sheet 1) for nicks, dents, scratches, pits, or gouges. Repair nicks, dents, scratches, pits, or gouges using abrasive cloth assortment, Item 1, WP 1803 00, as follows:

a. Repair AREA A as follows:

(1) **BLEND OUT NICKS, DENTS, SCRATCHES, PITS, OR GOUGES UP TO 0.010-INCH DEEP. BLENDED AREA AFTER REPAIR SHALL NOT EXCEED 0.750-INCH DIAMETER.**

(2) **DAMAGE SHALL NOT BE CLOSER THAN ONE HOLE DIAMETER FROM ANY EXISTING FASTENER HOLES.**

(3) Replace support arm if damage or repairs are more than limits.

b. Repair AREA B as follows:

(1) **BLEND OUT NICKS, DENTS, SCRATCHES, PITS, OR GOUGES UP TO 0.020-INCH DEEP OVER 100% OF SURFACE AREA. BLENDED AREA AFTER REPAIR SHALL NOT EXCEED 0.750-INCH DIAMETER.**

(2) Replace support arm if damage or repairs are more than limits.

10. Inspect support arm (Figure 2, Sheet 1) for corrosion. Repair using abrasive cloth assortment, Item 1, WP 1803 00, as follows:

a. Repair AREA A as follows:

(1) **BLEND OUT CORROSION UP TO 0.0050-INCH DEEP OVER 100% OF SURFACE AREA.**

INSPECT - Continued

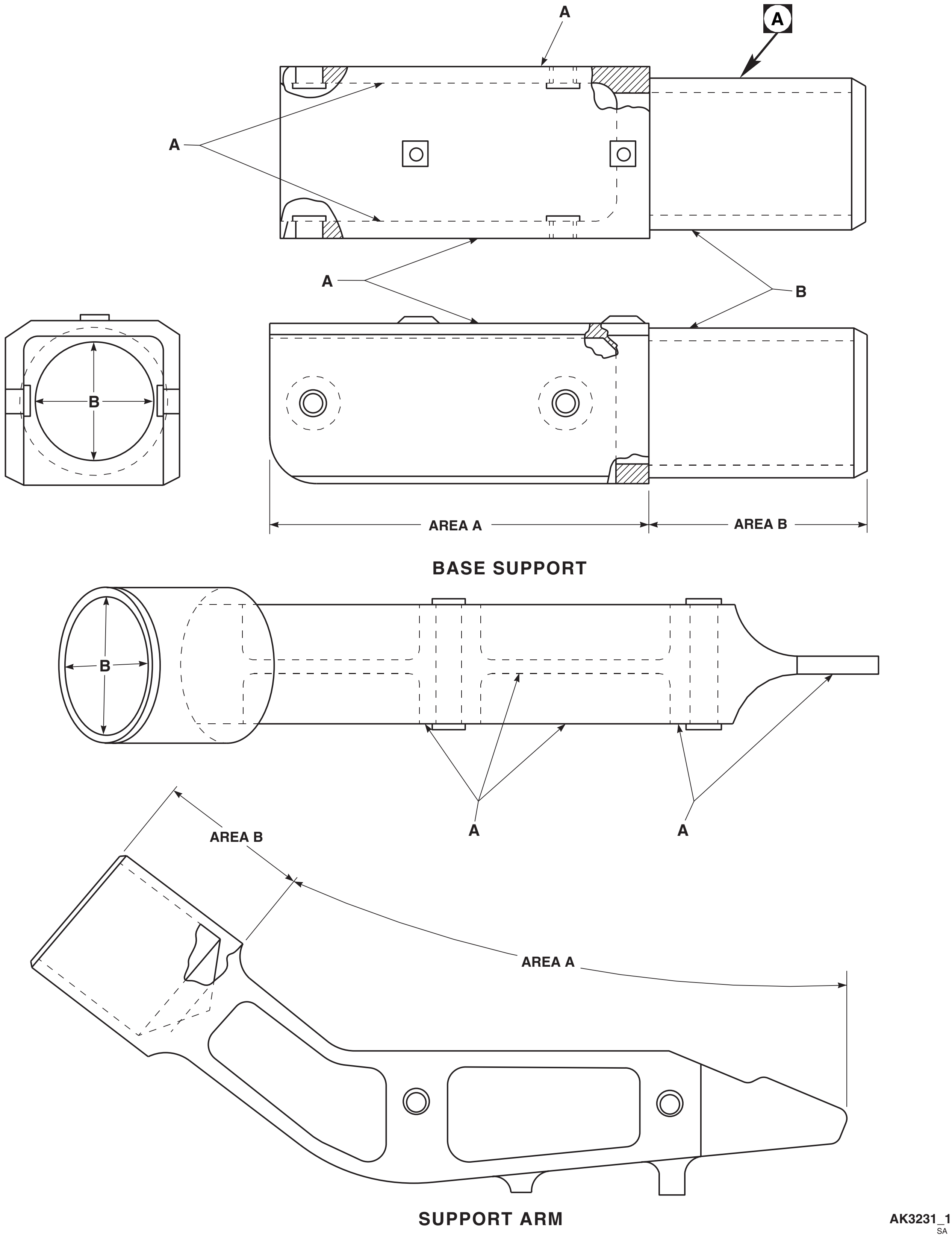


Figure 2. Inspect Rescue Hoist. (Sheet 1 of 2)

INSPECT - Continued

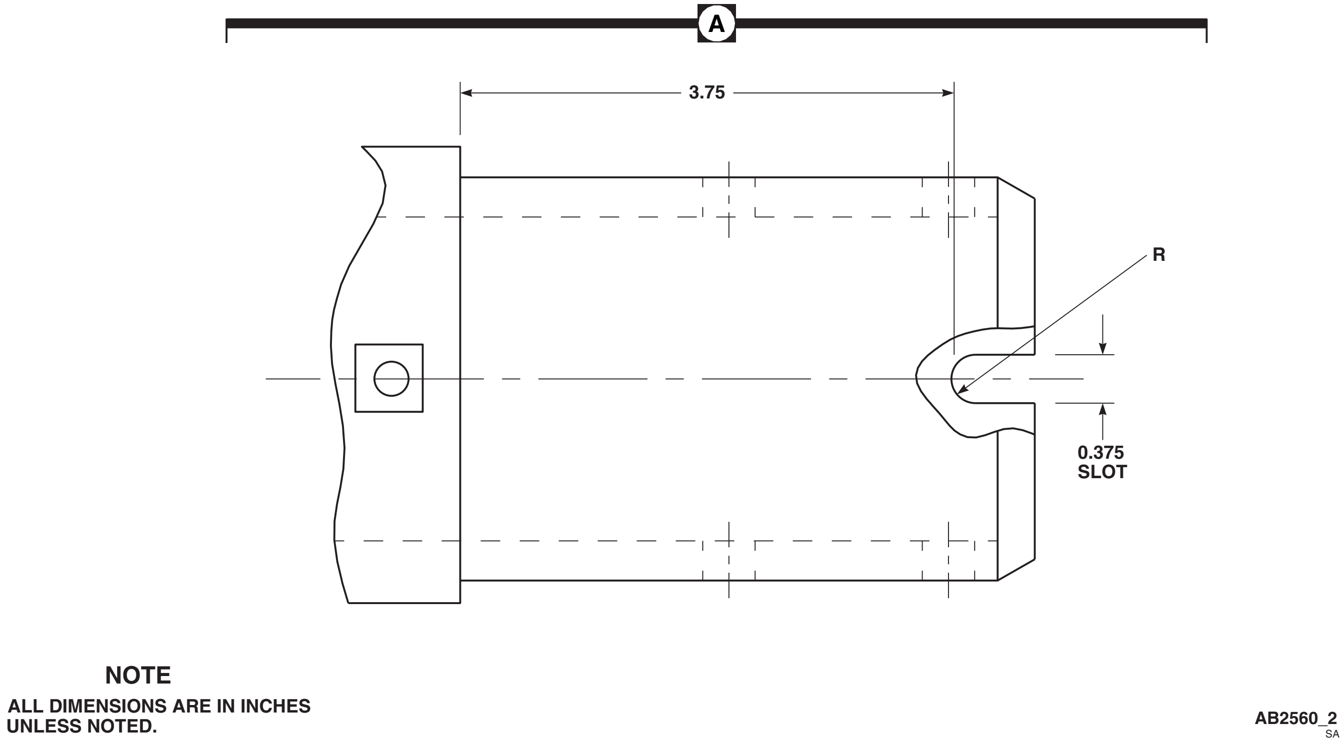


Figure 2. Inspect Rescue Hoist. (Sheet 2 of 2)

(2) Replace support arm if damage or repairs are more than limits.

b. Repair AREA B as follows:

(1) **BLEND OUT CORROSION UP TO 0.010-INCH DEEP OVER 100% OF INNER AND OUTER DIAMETER.**

(2) Replace support arm if damage or repairs are more than limits.

11. Inspect airframe fitting for cracks, corrosion, or deformation (WP 0377 00).

12. Inspect airframe fitting for plugged drain hole at bottom of fitting at 6 o'clock position as follows:

a. If drain hole is plugged, clean out drain hole and install drain tube (Figure 3).

b. If drain hole is drilled at 90 degree angle in fitting (Figure 4, Detail A). Plug hole with sealing compound, Item 257, WP 1803 00, and drill new hole as per step 13. b.

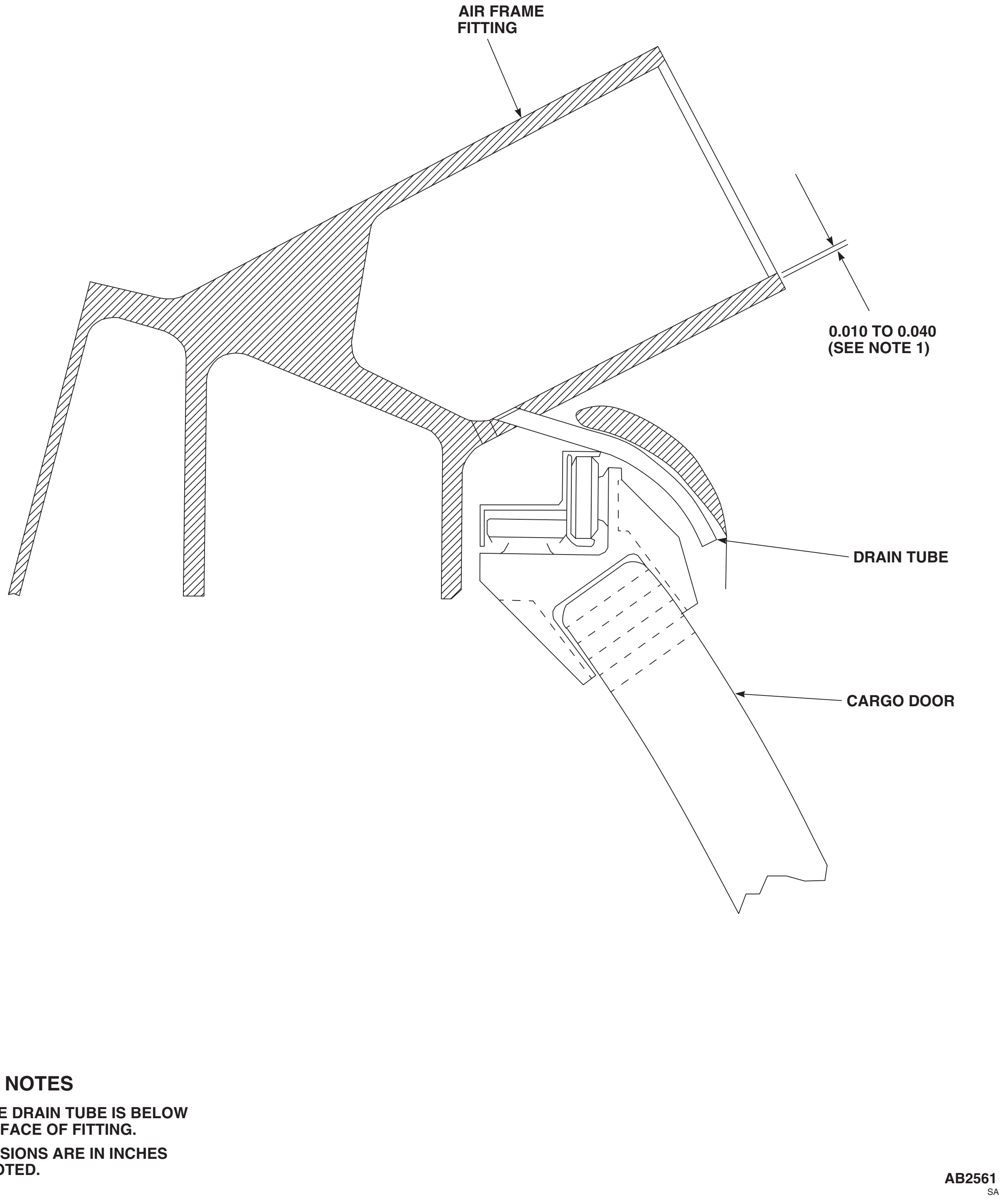
13. Inspect airframe fitting for missing drain hole at bottom of fitting at 6 o'clock position. If drain hole is missing or plugged per step 12. a. add hole (Figure 4, Detail A) as follows:

a. If 0.188-inch diameter drain hole exists at 45 degree angle at bottom of fitting at 6 o'clock position. Use hole as is, with 0.133-inch diameter drain tube.

b. If drain hole exists at 90 degree angle at bottom of fitting at 6 o'clock position. Plug hole with sealing compound, Item 257, WP 1803 00, as per step 12. a.

c. If no hole exists at bottom of fitting at 6 o'clock position. Drill new hole to 0.250-inch diameter at 45 degree angle.

INSPECT - Continued



NOTES

1. MAKE SURE DRAIN TUBE IS BELOW INNER SURFACE OF FITTING.
2. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

Figure 3. Inspect Rescue Hoist Drain Tube.

INSPECT - Continued

d. Manufacture drain tube from 0.133 ID or 0.186 ID plastic tubing, 3 inch long.

e. Install drain tubing into fitting with sealing compound, Item 257, WP 1803 00 (Figure 3). Make sure drain tube is below inner surface of fitting.

f. Seal drain tube to fitting with sealing compound, Item 257, WP 1803 00.

g. Make sure drain tube is clear of sealing compound.

14. Treat stripped and repaired areas with corrosion resistant coating, Item 111, WP 1803 00.

15. Prime stripped and repaired areas with epoxy primer coating kit, Item 120, WP 1803 00.

NOTE

Do not paint outside diameter of Area B, holes, bushings and inserts of support base assembly or outside diameter of Area B, holes, bushings and inserts of support arm assembly.

16. Paint stripped and repaired areas (WP 0381 00 and TM 55-1500-345-23).

ASSEMBLE

1. Turn off all electrical power.

NOTE

Avoid blocking drain hole of airframe fitting (cup area) with epoxy primer.

2. Apply epoxy primer coating kit, Item 120, WP 1803 00, to base support and airframe fitting mating surface. Allow primer to dry completely.

3. **QA** Install base support in airframe fitting with bolts, washers, and nuts. **TORQUE NUTS TO 10 - 20 INCH-POUNDS ABOVE RUN-ON TORQUE.**

4. Apply sealing compound, Item 273, WP 1803 00, around base support/airframe fitting mating edges and fastener locations.

5. Apply epoxy primer coating kit, Item 120, WP 1803 00, onto support arm and hoist mating surface. Allow primer to dry completely.

6. Install support arm into hoist support with bolts, washers, and nuts. **TORQUE BOLTS TO 95 - 105 INCH-POUNDS (Figure 1, Sheet 1, Detail A)**.

7. Install cuff fairing on rescue hoist (WP 1347 00).

INSTALL

1. Turn off all electrical power.

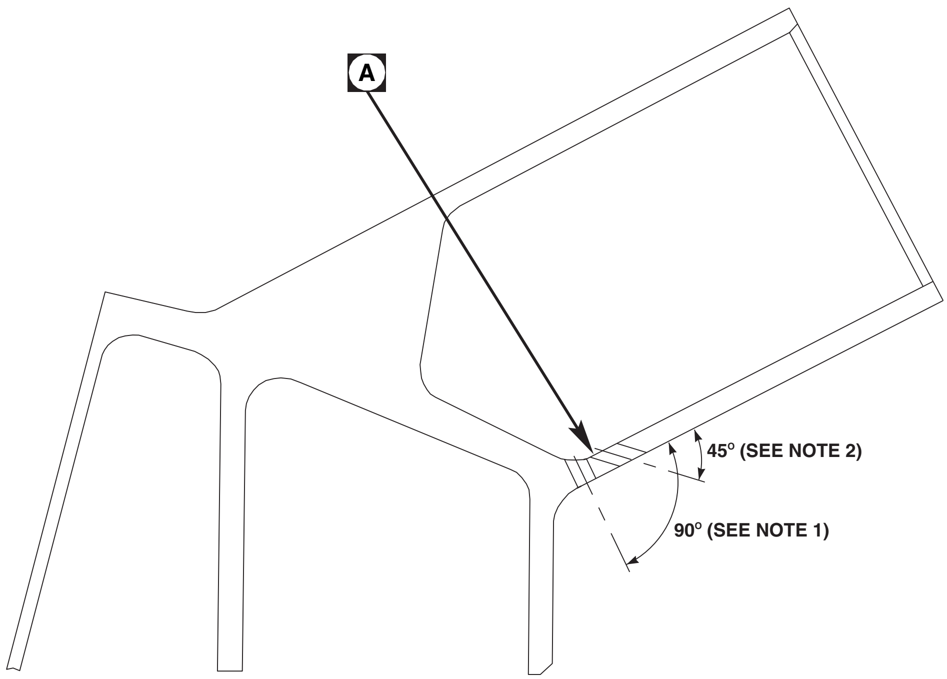
2. Apply epoxy primer coating kit, Item 120, WP 1803 00, onto support arm and base support mating surface. Allow primer to dry completely.

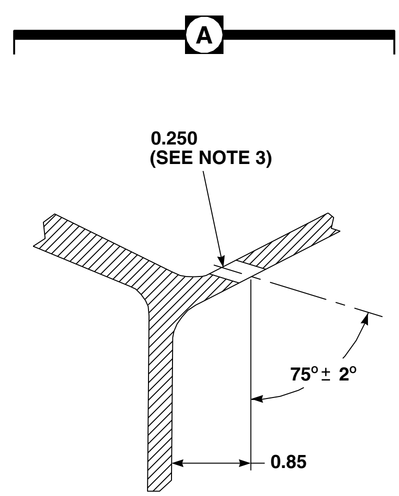
WARNING

Injury to personnel and damage to equipment will result if hoist is not supported during installation. Rescue hoist weighs 105 pounds. Use two assistants to support rescue hoist during installation.

3. **QA** Install hoist and support arm assembly to base support with bolts, washers, and nut (Figure 1, Sheet 2, Detail C).

INSTALL - Continued





NOTES

1. PLUG 90° HOLE WITH SEALING COMPOUND.
2. CLEAN EXISTING HOLE (0.188 INCH DIA). ADD 0.250 INCH DIA HOLE IF NO HOLE EXISTS.
3. AVOID GOUGING INNER WALL OF FITTING DURING REWORK.
4. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

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Figure 4. Inspect Rescue Hoist Drain Hole.

INSTALL - Continued

4. **QA TORQUE UPPER NUT TO 80 INCH-POUNDS. THEN TIGHTEN NUT TO NEXT CASTELLATION.** Install cotter pin. **DO NOT EXCEED 173 INCH-POUNDS TORQUE.**
5. **QA TORQUE LOWER NUT TO 40 INCH-POUNDS.** If required, loosen nut to next castellation. Install cotter pin.
6. Apply sealing compound, Item 273, WP 1803 00, around boltheads and nuts.
7. **QA** Install electrical harness on support arm with screw, washer, clamp, channel, spacer and block.
8. **QA** Install electrical harness to base support with screws, washers, and clamps.
9. **QA** Connect electrical connectors, P373 to J373 and P374 to J374, located on the helicopter roof.
10. **QA** Connect electrical connectors, P370 to J1, P371 to J5, and P372 to J8. Using nonelectrical wire, Item 354, WP 1803 00, lockwire connectors.
11. Make sure area is clean and free of foreign material.
12. Do an operational check of rescue hoist system (WP 0153 00).
13. Install explosive cartridge (WP 1351 00).
14. Install support fairings and connector cover (WP 1347 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

HOISTS AND WINCHES

RESCUE HOIST HH-60A, HH-60L RESCUE HOIST MOTOR EC-23050-1

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.

Personnel Required
Assistants - (2)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0153 00
WP 1347 00

REMOVE

1. Turn off all electrical power.
2. Remove forward fairing cover (WP 1347 00).
3. Disconnect two electrical connector harness at motor assembly connector.

NOTE

Support motor when removing top screws.

4. Remove 4 cap screws using 5/16-inch (7.938 mm) hex wrench with washers.
5. Separate motor splines from flange coupling by pulling motor straight out.
6. Cover exposed end flange and coupling to protect.

INSTALL

NOTE

Before installing motor, clean and inspect motor mounting plate and lubricate motor flange and splines.

1. Turn off all electrical power.
2. Position motor assembly shaft splines into hoist motor flange gear assembly.

CAUTION

Apply alternating torque to each cap screw.

3. Install four cap screws with washers. **TORQUE SCREWS 82 - 90-INCH POUNDS**.
4. Connect two electrical connector harness to motor assembly connectors.
5. Install forward fairing cover (WP 1347 00).
6. Apply power to aircraft.

INSTALL - Continued

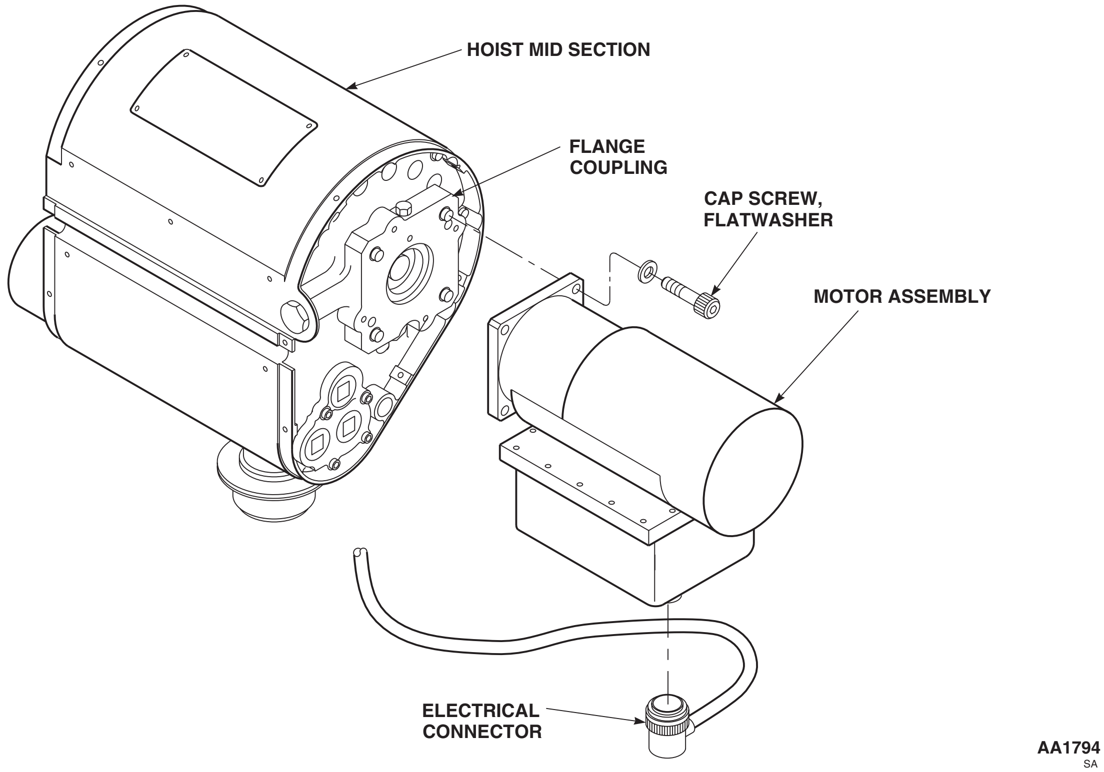

Figure 1. Replace Rescue Hoist Motor.

7. Do an operational check of rescue hoist system (WP 0153 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

HOISTS AND WINCHES

LOAD CABLE HH-60A HH-60L HOIST BL-29900-30

This WP supersedes WP 1343 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic’s Toolkit, SC 5180-99-B01
Allen Wrench, 1/16 X 6 in.
Container, 55-Gallon
Leather Gloves
Screwdriver Attachment Socket Wrench, TMA2E
Torque Wrench, 0 - 30 in. lbs.

Material/Parts

Wire, Nonelectrical Item 354, WP 1803 00

Personnel Required

Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

WP 1344 00
WP 1345 00
WP 1351 00
WP 1803 00

REMOVE

NOTE

If cable has been cut with the cable cutter, cable cutter and anvil must be replaced.

1. If applicable, remove cable cutter and anvil (WP 1345 00).
2. Turn off all electrical power.
3. Remove explosive cartridge (WP 1351 00).
4. Remove hook assembly (WP 1344 00).
5. Remove screws and washers securing access cover and backup plate to rescue hoist. Remove access cover (Figure 1, Detail B).
6. Position rescue hoist unit so that cable windings on drum are visible.

NOTE

Visually verify cable changing limit switch is aligned with access hole in front of aft cowling.

7. Turn on all electrical power.

REMOVE - Continued

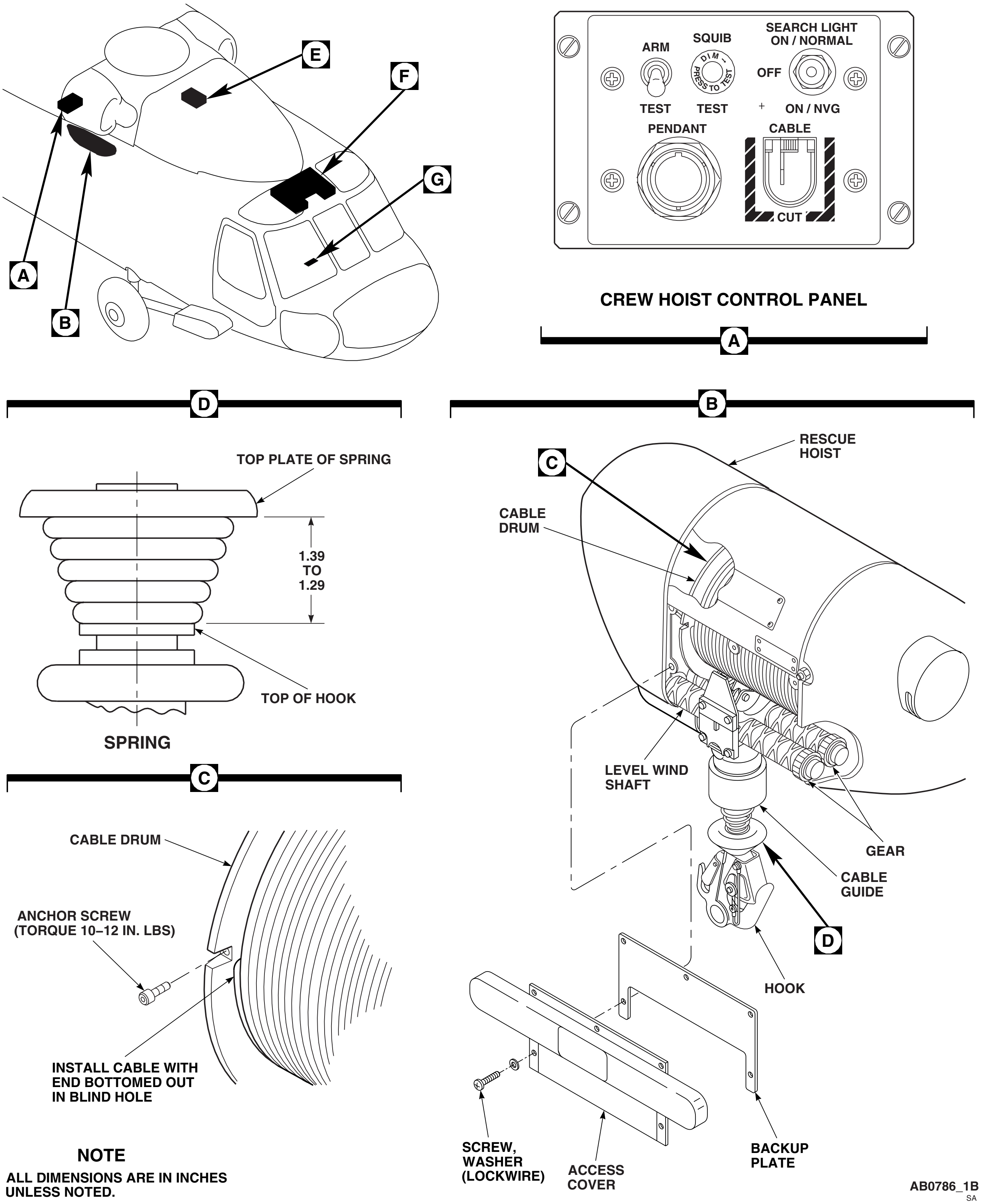

Figure 1. Replace Load Cable. (Sheet 1 of 2)

REMOVE - Continued

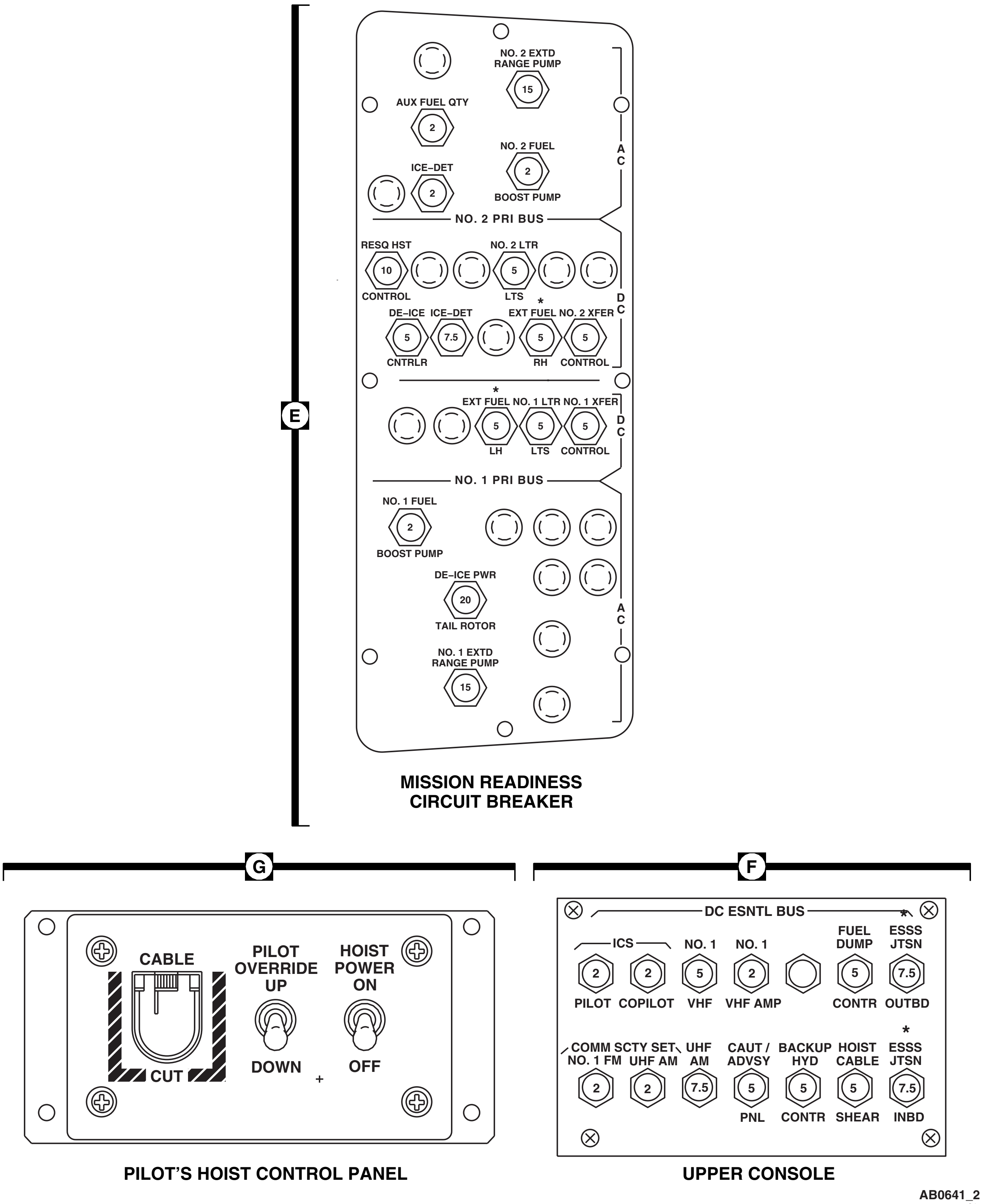

Figure 1. Replace Load Cable. (Sheet 2 of 2)

REMOVE - Continued

WARNING

- Injury to personnel and damage to equipment will occur if hands or clothing are caught in rotating machines.
- Injury to personnel will occur if gloves are not worn when reeling out cable. Leather gloves must be worn when reeling out cable.

CAUTION

Damage to cable or hoist can occur if cable is allowed to kink, twist, or ball up. Extreme care shall be taken not to allow cable to kink, twist, or ball up.

NOTE

- Tension shall always be applied to cable during hoist operation.
- Care should be taken to keep cable clean for proper hoist operations.

8. In cabin, operate rescue hoist pendant and reel out hoist cable while assistant, using leather gloves, coil cable into a clean 55-gallon container directly below rescue hoist until full-out limit switch is reached.

WARNING

Injury to personnel and damage to equipment will occur if drum is allowed to rotate beyond cable anchor set screw. Do not allow cable to begin ″backwinding″ on drum.

9. Insert a 1/16-inch Allen head wrench (Allen head wrench used for cable anchor screw) through access hole in front of aft cowling to actuate CABLE CHANGING LIMIT switch override switch. Actuate OVERRIDE switch and use the pendant control to slowly rotate the drum until no cable wraps remain on the drum and the cable anchor set screw is accessible.
10. Turn off all electrical power.
11. Loosen cable anchor screw, using 1/16-inch Allen head wrench and screwdriver attachment socket wrench. Retain screw for reuse.

CAUTION

Damage to hoist will occur if electrical hoist is operated once cable has been removed. This can cause mis-synchronization of drum and levelwind system.

12. Remove cable from drum by pulling it straight down at cable guide.

INSTALL

1. Turn off all electrical power.
2. **QA** Thread anchor end of cable through roller housing assembly cable guide, tension roller, and pressure roller.

INSTALL - Continued

CAUTION

Damage to cable integrity will occur if more than two reinstallations are allowed following the initial installation.

NOTE

Make sure that anchor end of cable is fully inserted into this hole. Proper insertion may be verified by viewing cable end through anchor screw hole using a good source of light. During this inspection, note that cable strands/wires only should be visible through the anchor screw hole; not the weld head on end of cable. If weld head is visible, reposition cable in hole until strands/wires only are visible. The cable anchor screw must bear against the cable strands/wires only, not against the weld head. Failure to conform with this requirement could compromise the integrity of load cable retention.

3. **QA** Insert anchor end of cable into hole in drum end flange.
4. **QA** Secure cable to drum with cable anchor screw, using 1/16-inch Allen head wrench and screwdriver attachment socket wrench. **TORQUE SCREW TO 10 - 12 INCH-POUNDS.**
5. Turn on all electrical power.

NOTE

The first few wraps of cable to be wound on the drum are critical to proper winding of subsequent wraps.

6. To ensure proper initial wrapping, do this:
   a. Insert a 1/16-inch Allen head wrench (Allen head wrench used for cable anchor screw) through access hole in front of aft cowling to actuate CABLE CHANGING LIMIT switch override switch. Actuate OVERRIDE switch.
   b. Apply tension to the cable by pulling downward with gloved hands while operating the pendant in the raise direction at very slow speeds. The first wrap of cable must lie in the first groove of the drum. Subsequent wraps must lie in their respective grooves (2nd wrap in 2nd groove, etc.).
   c. Observe for proper wrapping during this operation.
7. If proper wrapping is not being performed, do this:
   a. Turn off all electrical power.
   b. Loosen cable anchor screw using 1/16-inch Allen head wrench and screwdriver attachment socket wrench. Remove from drum flange. Retain screw for reuse.
   c. **QA** Secure cable to drum with cable anchor screw, using 1/16-inch Allen head wrench and screwdriver attachment socket wrench. **TORQUE SCREW TO 10 - 12 INCH-POUND.**
   d. Turn on all electrical power.
   e. Apply tension to the cable by pulling downward with gloved hands while operating the pendant in the raise direction at very slow speeds. The first wrap of cable must lie in the first groove of the drum. Subsequent wraps must lie in their respective grooves (2nd wrap in 2nd groove, etc.).
8. While maintaining tension on cable (gloved hands) wind cable on drum until 5 to 6 full wraps have been completed.

INSTALL - Continued

WARNING

Injury to personnel and damage to equipment will occur if tension is not maintained using gloved hands during this final operation to prevent loosening of cable wraps on drum. If cable is allowed to wrap loosely on drum, cable fouling problems may result during subsequent operation of hoist.

9. Using hand held pendant control, wind remaining cable on drum and observe for proper wrapping.
10. If installing a new cable, check for the following:
    a. Make sure installation step 1. through step 9. are complied with when installing cable.
    b. Run hoist through one cycle with no load on cable at slow speed. Make sure cable runs freely, levelwind is adjusted properly and tensioning system is operating properly.
    c. Check cable for twisting.
    d. Cycle hoist several times with light load on cable, applying load with gloved hands to cable, this allows cable to make adjustments to the actual operating conditions of the hoist.

NOTE

If cable cutter assembly was replaced, complete the installation of cable cutter assembly cotter pin (WP 1345 00).

11. Turn off all electrical power.
12. Install hook assembly (WP 1344 00).
13. If applicable, remove cable cutter and anvil (WP 1345 00).
14. Install explosive cartridge (WP 1351 00).
15. Make sure area is clean and free of foreign material.
16. **QA** Secure backup plate and access cover to rescue hoist using screws and washers. Using nonelectrical wire, Item 354, WP 1803 00, lockwire screws.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

HOISTS AND WINCHES

HOOK ASSEMBLY HH-60A HH-60L HOIST BL-29900-30

This WP supersedes WP 1344 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 1351 00

REMOVE

1. Turn off all electrical power.
2. Remove explosive cartridge (WP 1351 00).
3. Turn on all electrical power.
4. Operate rescue hoist unit in extend direction until cable is approximately 4 to 5 feet from full-in position.
5. Turn off all electrical power.
6. Separate bumper assembly from bushing guide nut and slide bumper assembly up cable toward drum (Figure 1, Sheet 1, Detail B).
7. Remove cotter pins and setscrews from bushing guide nut.

WARNING

Injury to personnel and damage to equipment will occur if hook is not supported when loosening bushing nut. Hook with attached parts will fall free when threads of bushing nut disengage from housing.

8. Unscrew hook housing from bushing guide nut and remove hook and hook housing (Figure 1, Sheet 2, Detail C). Remove bumper from hook housing.
9. Using punch, remove spring pin securing nut to hook and remove nut.
10. Remove thrust washer, ball bearing, hook housing, and washer from hook.

CAUTION

Damage to hook assembly will occur if hook was used in a salt water environment or exposed to corrosive substances. The hook assembly must be washed with fresh water and be free of any foreign objects.

NOTE

Two sections of clamp are matched set. Keep sections together after removal.

11. Slide bushing nut up cable to expose terminal clamp halves. Remove retaining ring and terminal clamp halves. Slide bushing guide nut off cable.

REMOVE - Continued

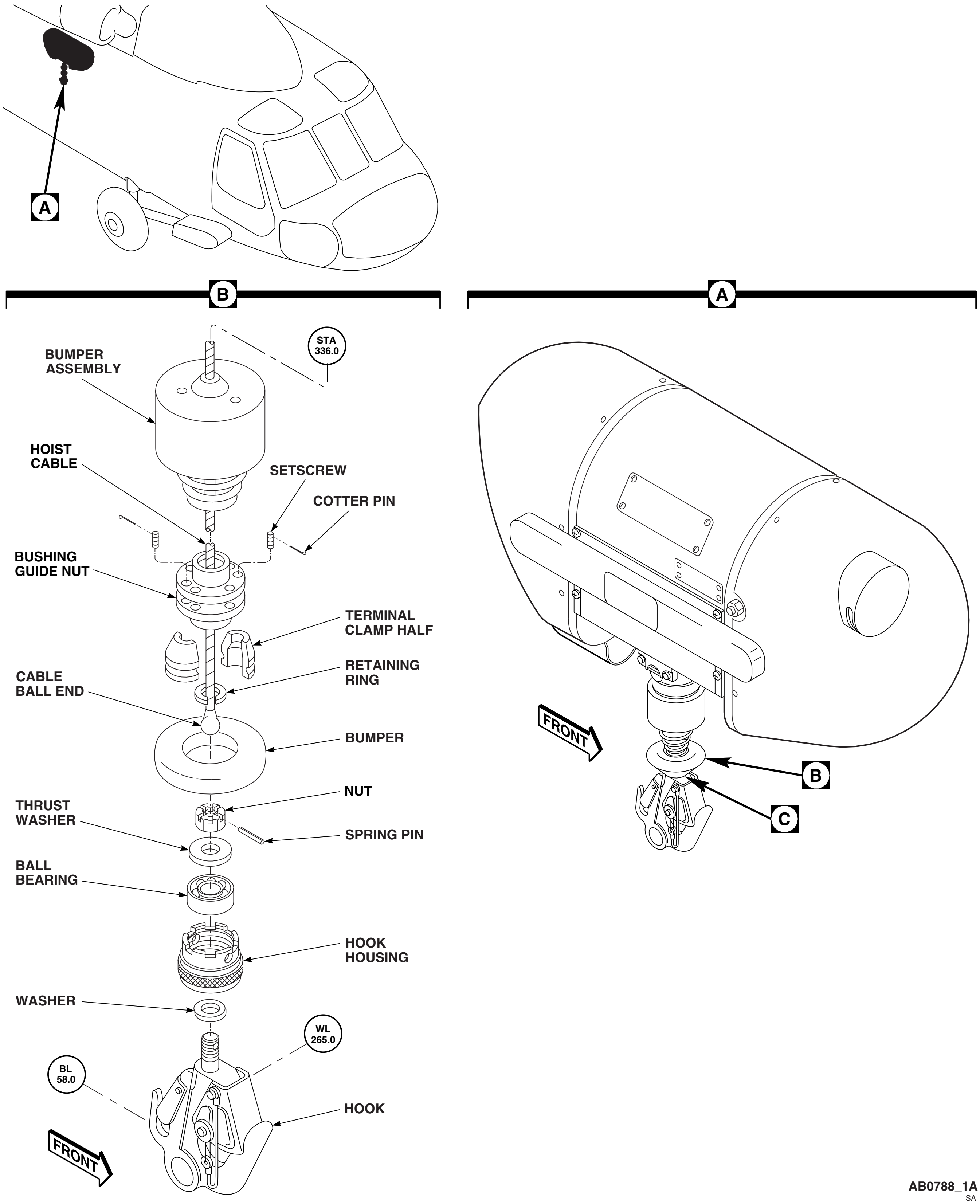

Figure 1. Replace Hook Assembly. (Sheet 1 of 2)

REMOVE - Continued

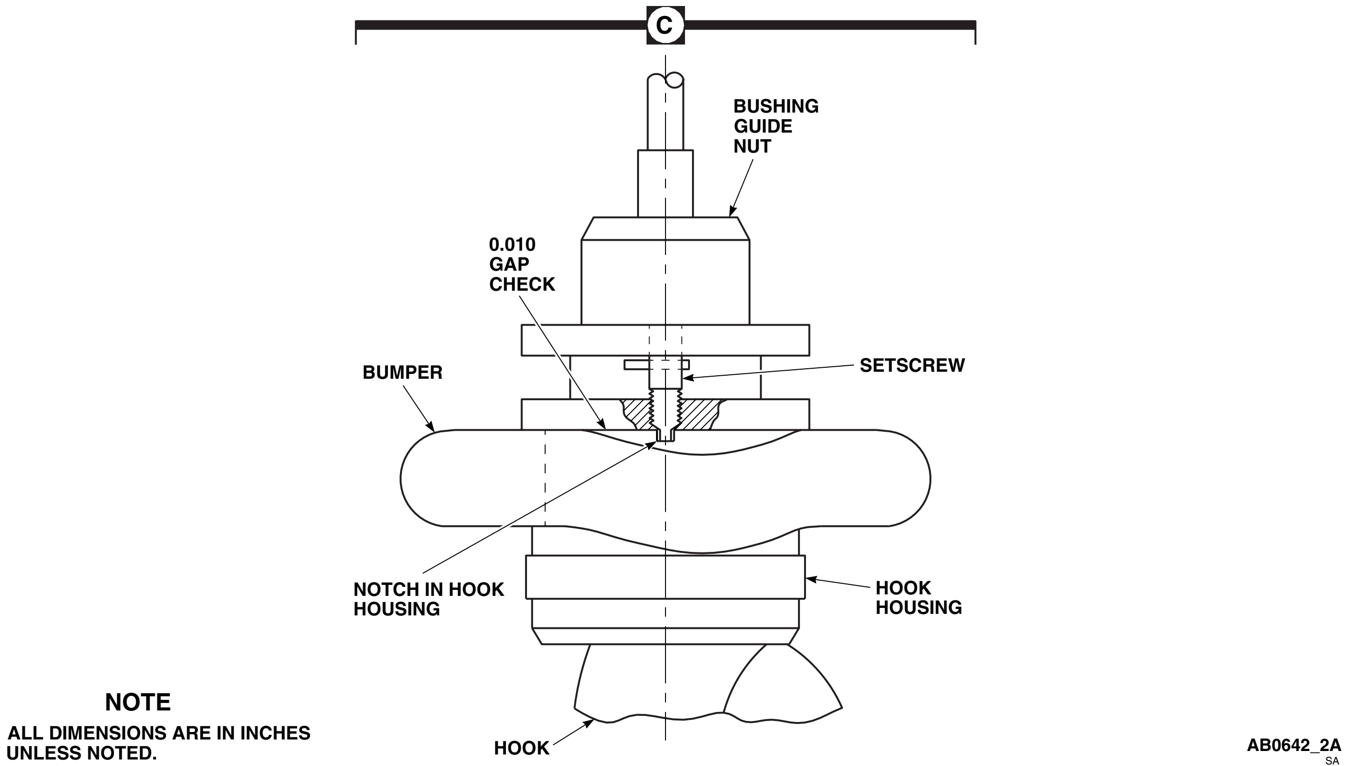

Figure 1. Replace Hook Assembly. (Sheet 2 of 2)

INSPECT

1. Check all components for wear or damage. **NO WEAR OR DAMAGE ALLOWED.**
2. Replace any worn or damaged components.
3. Inspect ball bearing for looseness, roughness, or stiffness caused by wear, deterioration of lubricants, or corrosion.

INSTALL

1. Turn off all electrical power.
2. Push bumper assembly over cable ball end on to hoist cable (Figure 1, Sheet 1, Detail B).

NOTE

Terminal clamp halves are matched set and must be replaced as set.

3. **QA** Slide bushing guide nut over cable ball end. Position terminal clamp halves over cable and secure with retaining ring. Slide bushing guide nut down over terminal clamp halves.
4. Place washer, hook housing, ball bearing, and thrust washer on hook and secure with nut.
5. Tighten nut on hook until nut bottoms. Back nut off one-half turn and align it with hole in hook.

INSTALL - Continued

NOTE

Unpainted spring pin may be used until all stock is gone, but it must be replaced after every exposure to salt water.

6. **QA** Check hook assembly for free rotation. If binding is felt, loosen nut until binding stops, and align nut with hole in hook again. Install spring pin in hole in hook.
7. Install bumper on hook housing.
8. **QA** Screw hook tightly onto bushing guide nut.

CAUTION

Be certain hook housing is screwed in all the way and after contact is made, back off to nearest castellation only. Setscrew must bottom into castellation of hook housing to lock.

9. **QA** Align setscrew holes with notches in hook housing. Install setscrew in bushing guide nut and tighten until it contacts and locks into hook housing. Back off setscrew only enough to install cotter pins through slot in setscrews.

WARNING

If gap between faces of hook housing and bushing guide nut is greater than 0.010-inch or hook housing can be backed off bushing guide nut, hook may separate from cable during operation.

10. Pull back bumper and using feeler gage, check gap between faces of hook housing and bushing guide nut is not more than 0.010-inch (Figure 1, Sheet 2, Detail C). If gap is greater than 0.010-inch or hook housing can be backed off bushing guide nut, check setscrews for proper installation.
11. **QA** Push spring on bushing guide nut.
12. Turn on all electrical power.
13. Reel in hoist cable.
14. Turn off all electrical power.
15. Install explosive cartridge (WP 1351 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

HOISTS AND WINCHES

CABLE CUTTER AND ANVIL HH-60A HH-60L HOIST BL-29900-30

This WP supersedes WP 1345 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-CL-A01

Material/Parts
Compressed Air
Wire, Nonelectrical Item 354, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 1351 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Remove explosive cartridge (WP 1351 00).
3. Remove anvil from cable guide assembly (Figure 1, Detail A).
4. Remove cable cutter by applying low air pressure (50 to 80 psi) to squib orifice.

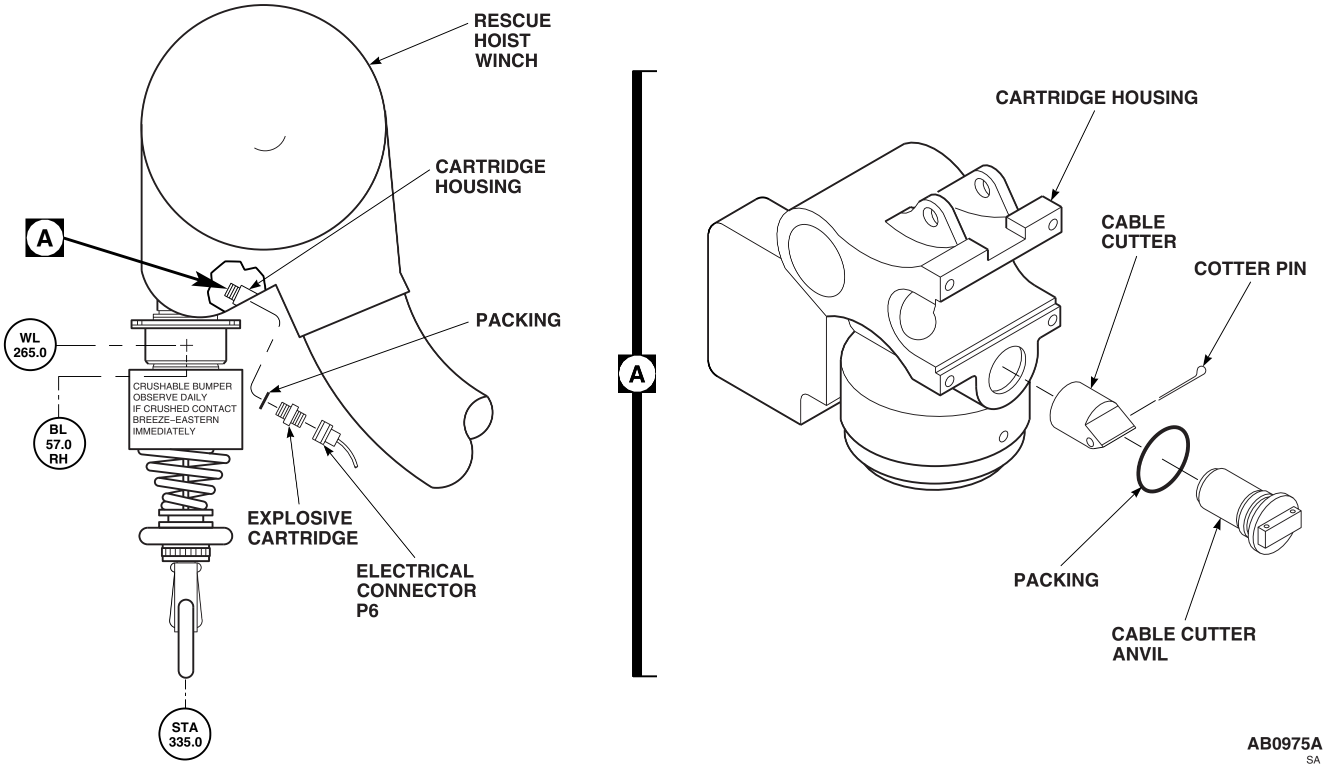

Figure 1. Replace Cable Cutter and Anvil.

REMOVE - Continued

5. After removal of cable cutter, remove severed cotter pin from cable cutter orifice and cable guide orifice.
6. Remove packing from cable guide cavity. Discard packing.

INSTALL

1. Turn off all electrical power (WP 1685 00).

WARNING

Injury to personnel and damage to equipment will occur if electrical power is not disconnected from helicopter when handling squib (cartridge) activated device.

NOTE

Prior to installation of cable cutter, make sure cable guide passageways and cable cutter are clean and free of excess discharge residue.

2. **QA** Install packing into cable guide cavity.
3. **QA** Install cable cutter into cable guide passageway ensuring alignment of cotter pin set holes (Figure 1, Detail A).

NOTE

The load cable must be reinstalled and the cable guide moved away from the hoist housing before complete installation of the cotter pin into the cable cutter set hole.

4. **QA** Install cotter pin in set hole.
5. **QA** Install anvil. Using nonelectrical wire, Item 354, WP 1803 00, lockwire anvil.
6. Install explosive cartridge (WP 1351 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

HOISTS AND WINCHES

RESCUE HOIST PENDANT HH-60A HH-60L HOIST BL-29900-30

This WP supersedes WP 1346 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0153 00

REMOVE

1. Turn off all electrical power.
2. Disconnect rescue hoist pendant cable electrical connector from crew hoist control panel (Figure 1, Detail A).
3. Open clip and remove rescue hoist pendant.

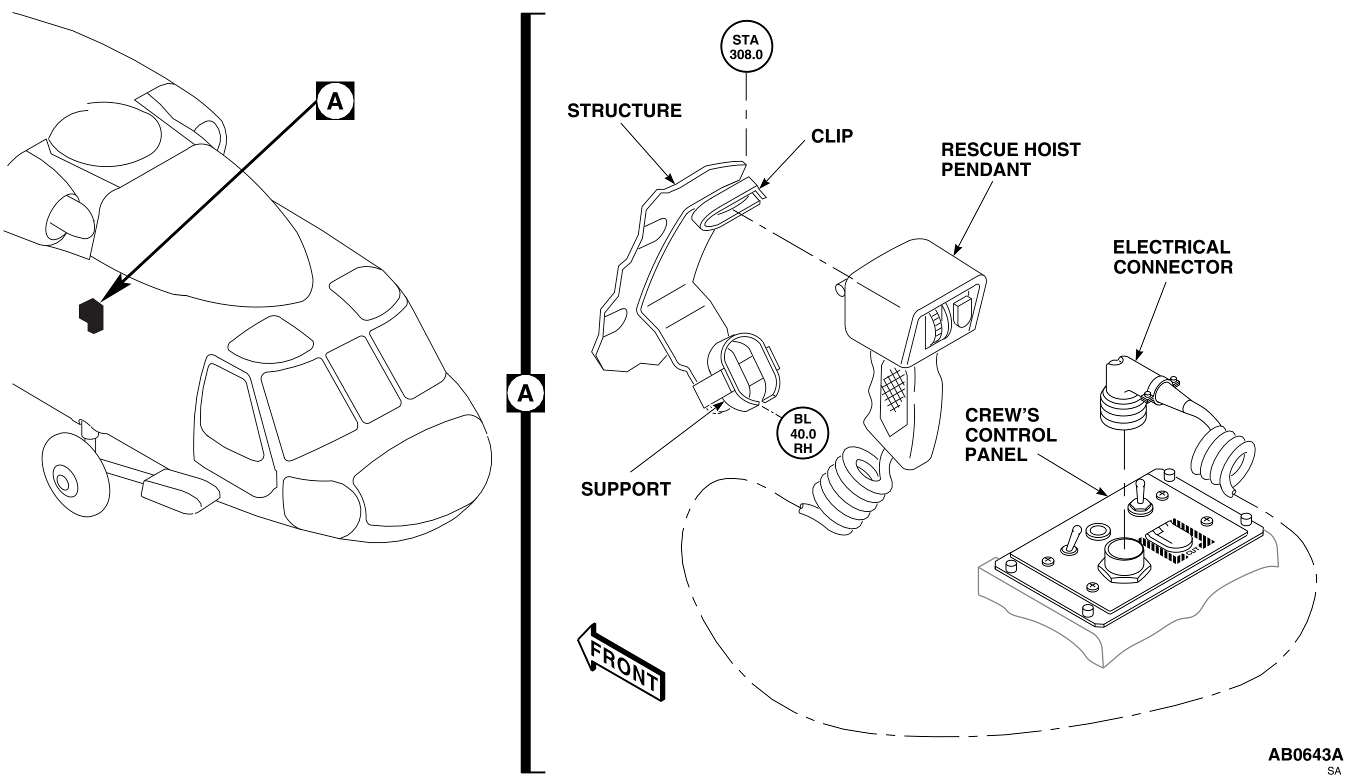

Figure 1. Replace Rescue Hoist Pendant.

INSTALL

1. Turn off all electrical power.
2. **QA** Position rescue hoist pendant in support and close clip (Figure 1, Detail A).
3. **QA** Connect rescue hoist pendant cable electrical connector to crew hoist control panel.
4. Do an operational check of rescue hoist system (WP 0153 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

HOISTS AND WINCHES

SUPPORT FAIRINGS AND COWLINGS HH-60A HH-60L HOIST BL-29900-30

This WP supersedes WP 1347 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Material/Parts
Wire, Nonelectrical Item 354, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Remove screws and washers securing access cover and backup plate to rescue hoist. Remove access cover (Figure 1, Detail A).
3. Remove screws, bolts and washers securing connector fairing, support arm fairings and cuff fairing to rescue hoist. Remove fairings.
4. Remove screws and washers securing front cowling to rescue hoist. Remove front cowling.
5. Remove screws and washers securing rear cowling to rescue hoist. Remove rear cowling.

INSTALL

1. Turn off all electrical power.
2. Make sure area is clean and free of foreign material before installing fairings and cowlings.
3. **QA** Secure rear cowling to rescue hoist using screws and washers. Using nonelectrical wire, Item 354, WP 1803 00, lockwire screws (Figure 1, Detail A).
4. **QA** Secure front cowling to rescue hoist using screws and washers. Using nonelectrical wire, Item 354, WP 1803 00, lockwire screws.
5. **QA** Secure cuff fairing to rescue hoist using screws and washers.
6. **QA** Secure support arm fairings to cuff fairing and connector fairing using screws and washers.
7. **QA** Secure backup plate to access cover and rescue hoist using screws and washers. Using nonelectrical wire, Item 354, WP 1803 00, lockwire screws.

INSTALL - Continued

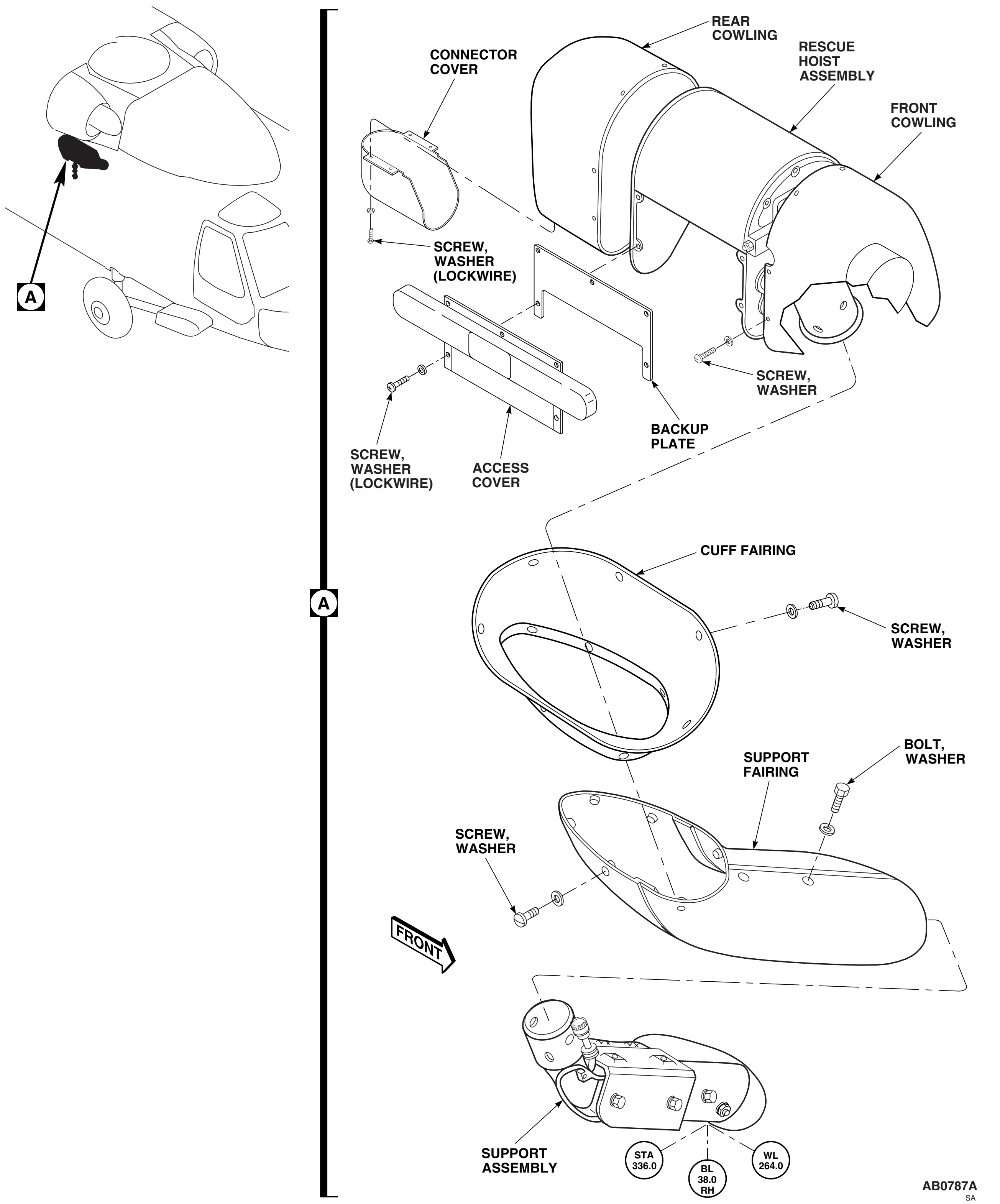

Figure 1. Replace Support Fairings and Cowlings.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

HOISTS AND WINCHES

PILOT HOIST CONTROL PANEL HH-60A HH-60L HOIST BL-29900-30

This WP supersedes WP 1348 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0153 00

REPLACE PILOT HOIST CONTROL PANEL

Remove

1. Turn off all electrical power.
2. Remove screws and washers securing control panel to center console (Figure 1, Detail A).
3. Carefully lift control panel to gain access to electrical connector.
4. Disconnect electrical connector, P218, from control panel.
5. Remove control panel from center console.

Install

1. Turn off all electrical power.
2. **QA** Connect electrical connector, P218, to control panel (Figure 1, Detail A).
3. **QA** Secure control panel to center console using screws and washers.
4. Make sure area is clean and free of foreign material.
5. Do an operational check of rescue hoist system (WP 0153 00).

REPLACE PILOT HOIST CONTROL PANEL INFORMATION PLATE

Remove

1. Turn off all electrical power.
2. Remove screws and washers securing information plate from control panel (Figure 2, Detail A). Remove information plate.

REPLACE PILOT HOIST CONTROL PANEL INFORMATION PLATE - Continued

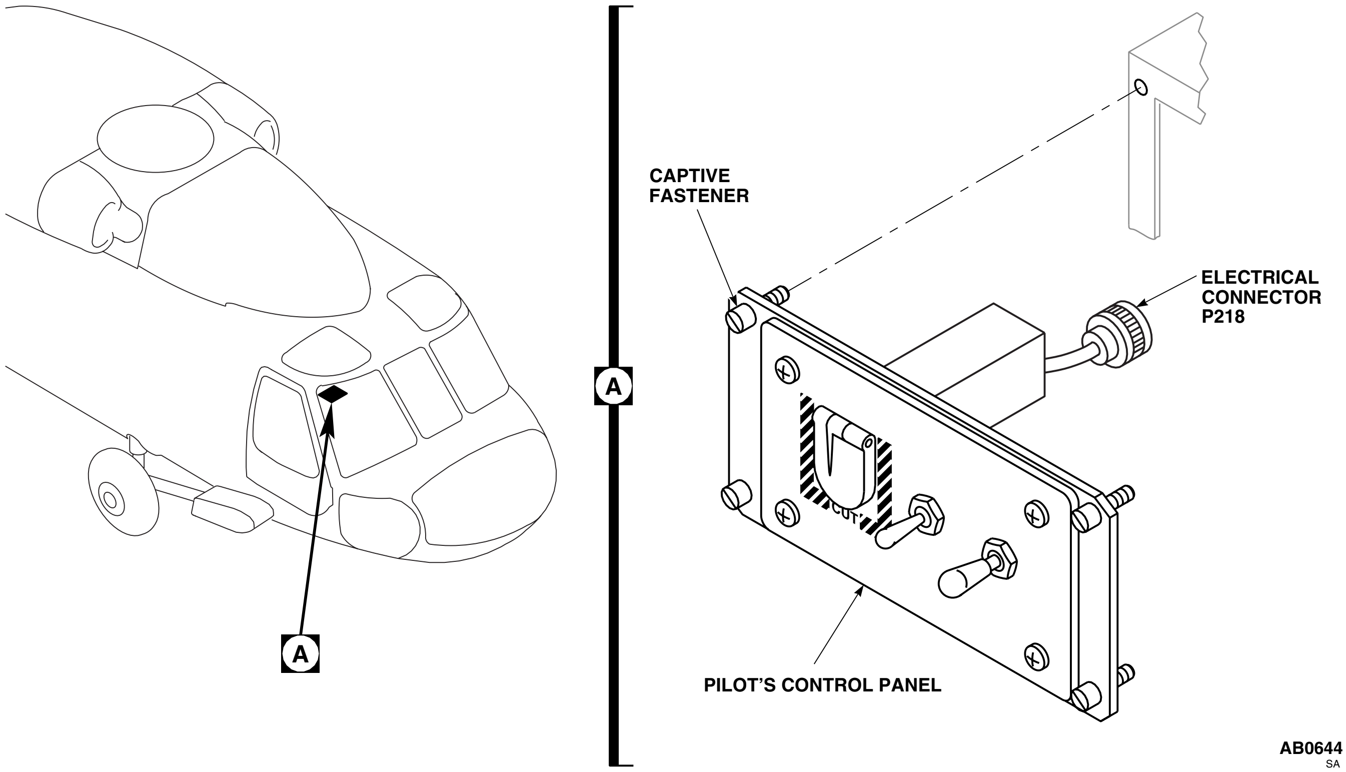

Figure 1. Replace Pilot Hoist Control Panel.

Install

1. Turn off all electrical power.
2. **QA** Secure information plate to control panel using screws and washers (Figure 2, Detail A).
3. Make sure area is clean and free of foreign material.
4. Do an operational check of utility systems rescue hoist (WP 0153 00).

REPLACE PILOT HOIST CONTROL PANEL INFORMATION PLATE - Continued

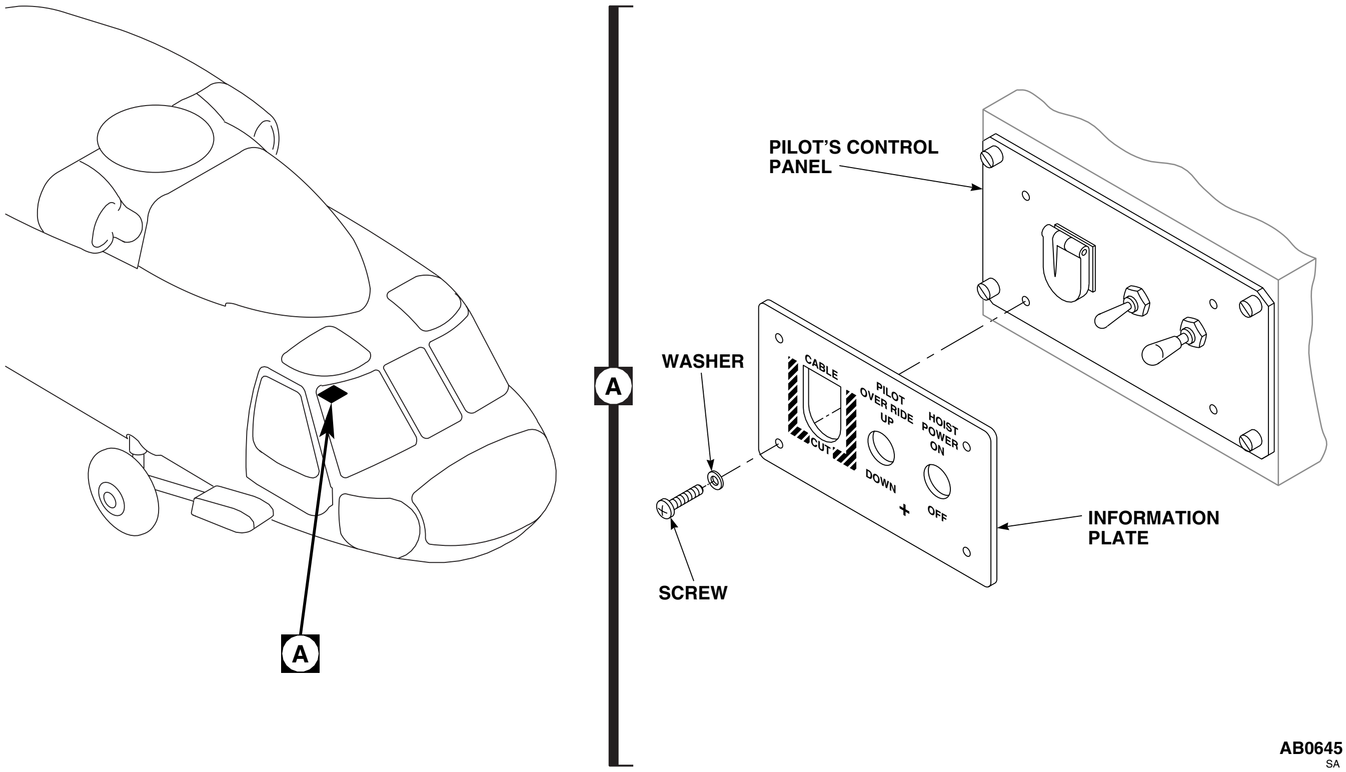

Figure 2. Replace Pilot Hoist Control Panel Information Plate.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

HOISTS AND WINCHES

CREW HOIST CONTROL PANEL HH-60A HH-60L HOIST BL-29900-30

This WP supersedes WP 1349 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0153 00
WP 0258 00

REPLACE CREW HOIST CONTROL PANEL

Remove

1. Turn off all electrical power.
2. Remove soundproofing panel (WP 0258 00).
3. Disconnect rescue hoist pendant cable electrical connector from crew hoist control panel (Figure 1, Detail A).
4. Disconnect electrical connectors, P23 and P226, from control panel.
5. Loosen captive screws and remove control panel from support.

Install

1. Turn off all electrical power.
2. **QA** Secure control panel to support using captive screws (Figure 1, Detail A).
3. **QA** Connect electrical connectors, P23 and P226, to control panel.
4. Install soundproofing panel (WP 0258 00).
5. **QA** Connect rescue hoist pendant cable electrical connector to control panel.
6. Do an operational check of rescue hoist system (WP 0153 00).

REPLACE CREW HOIST CONTROL PANEL INFORMATION PLATE

Remove

1. Turn off all electrical power.
2. Disconnect rescue hoist pendant cable electrical connector from crew hoist control panel (Figure 2, Detail A).

REPLACE CREW HOIST CONTROL PANEL INFORMATION PLATE - Continued

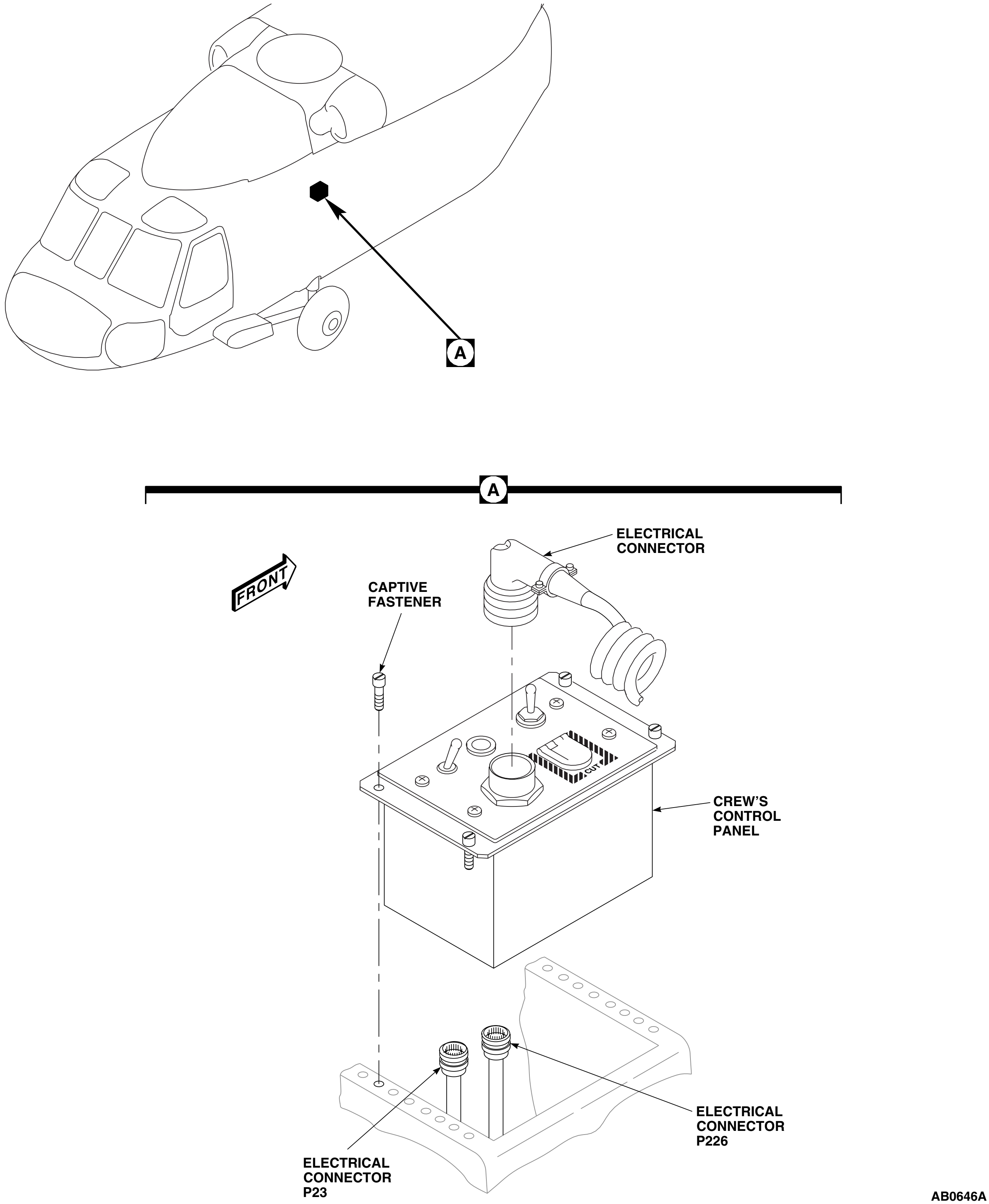

Figure 1. Replace Crew Hoist Control Panel.

REPLACE CREW HOIST CONTROL PANEL INFORMATION PLATE - Continued

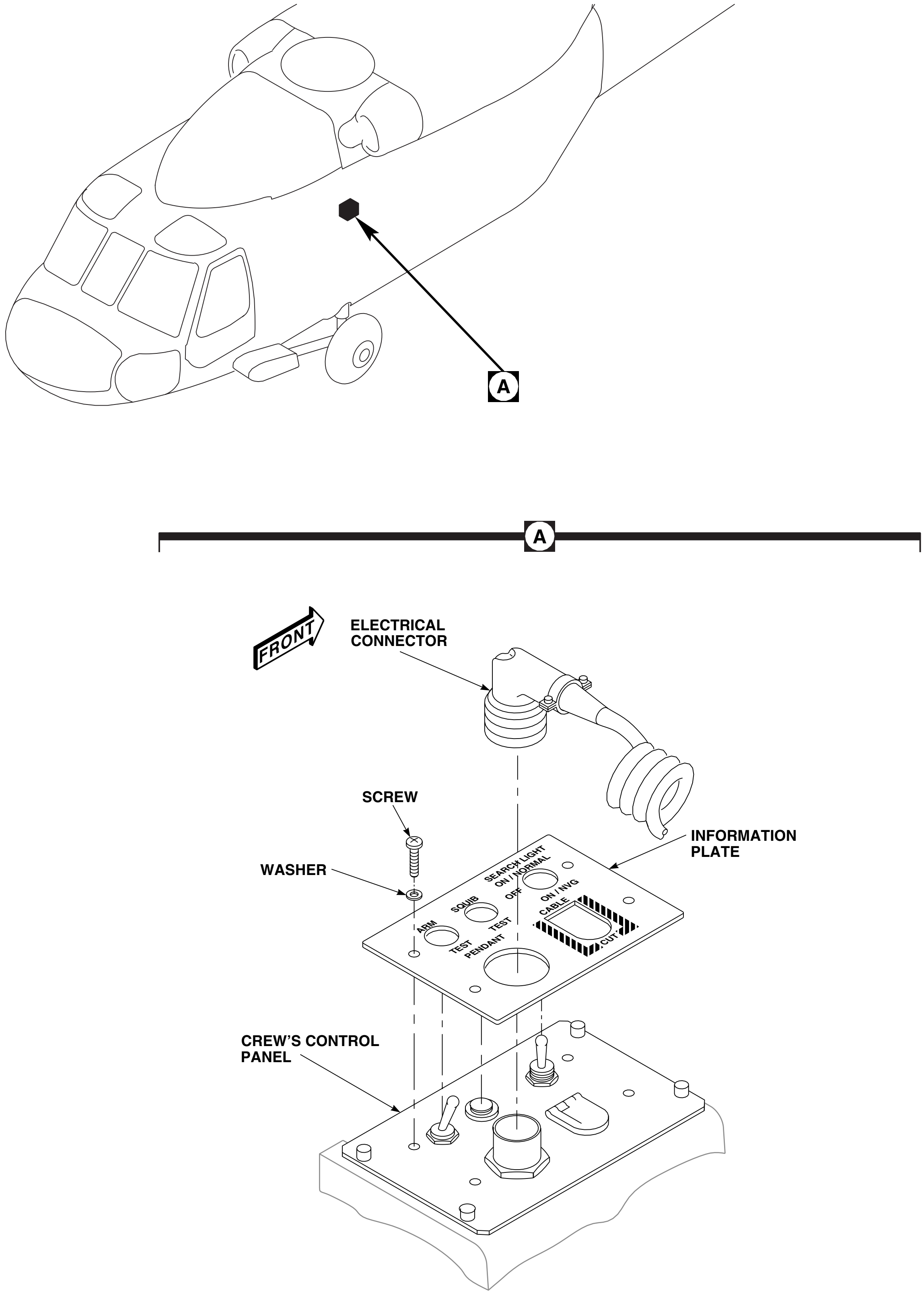

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Figure 2. Replace Crew Hoist Control Panel Information Plate.

REPLACE CREW HOIST CONTROL PANEL INFORMATION PLATE - Continued

3. Remove screws and washers securing information plate to control panel. Remove information plate from control panel.

Install

1. Turn off all electrical power.
2. **QA** Position information plate on control panel and secure with screws and washers (Figure 2, Detail A).
3. **QA** Connect rescue hoist pendant cable electrical connector to control panel.
4. Make sure area is clean and free of foreign material.
5. Do an operational check of utility systems rescue hoist (WP 0153 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

HOISTS AND WINCHES

LIMIT SWITCH SETTING/CHECK HH-60A HH-60L HOIST BL-29900-30

This WP supersedes WP 1350 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 1343 00

SETTING

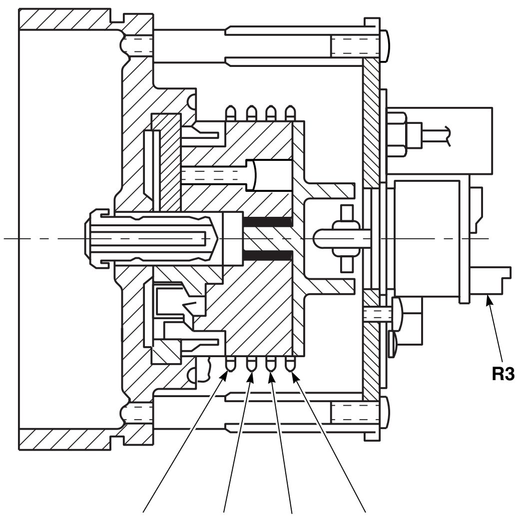
1. Before doing any performance testing, the limit switches must be set properly for operation. Use the following procedure:

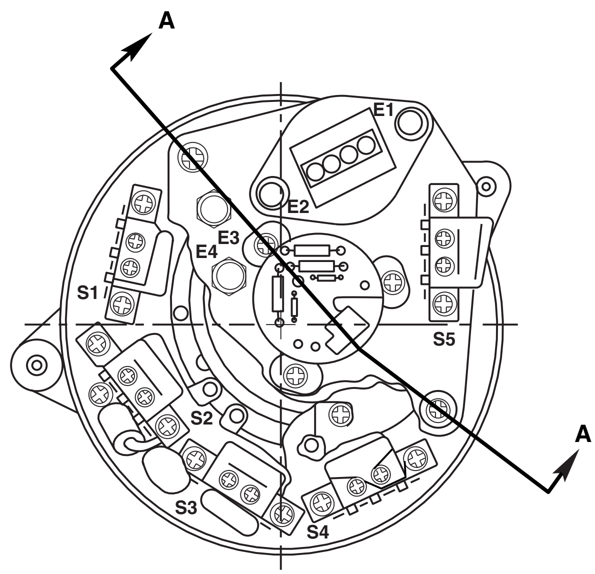
NOTE

Electrical power must be applied to the rescue hoist unit when performing this procedure and the test procedures that follow.

2. If the cable is installed, operate the rescue hoist unit in the lower mode and unwind cable from the drum.
3. Mark the cable at the following points:
   a. 4.5 feet from the weld end of the cable.
   b. 6 feet from the weld end of the cable.
   c. 16 feet from the weld end of the cable.
   d. 10 feet from the ball end of the cable.
4. Install the load cable (WP 1343 00).
5. Operate rescue hoist unit in the raise mode. Wind cable onto the drum until 10 feet from the ball end of the cable is reached. Remove the cam locking plate on the cyclocentric assembly (Figure 1) and adjust cam D so that switch S4 actuates within 12 inches of this point.
6. Apply a minimum load to the rescue hoist unit. Operate the rescue hoist unit in the lower mode and pay out cable until the 16 foot mark on cable is reached at the output point. Set CAM C so that switch S3 actuates within 12 inches of this point.
7. Operate the rescue hoist unit in the lower mode and payout cable until the 6 foot mark on the cable is at the hoist cable output point. Set cam B so that switch S2 actuates within 6 inches of this point.
8. Actuate the cable changing switch on the electrical control panel.
9. Using the pendant, operate the rescue hoist unit carefully in the lower mode using the control toggle switch, until the 4.5 foot mark on the cable is at the hoist cable output point. Set cam A to actuate switch S1 within 6 inches of this point.
10. With the cable changing switch actuated, raise using the pendant control slowly until 6 wraps of cable are on the drum.

SETTING - Continued





TABLE−FUNCTION (FOR REFERENCE)

FUNCTION	CAM	SWITCH	SWITCH ACTUATION
EMERGENCY FULL OUT	A	S1	3 ± 1/2 WRAPS LEFT ON DRUM
FULL OUT	B	S2	4 ± 1/2 WRAPS LEFT ON DRUM
DECELERATION (LOWERING)	C	S3	10 FEET ± 2 FEET FULLY REELED OUT
DECELERATION (HOISTING)	D	S4	10 FEET ± 2 FEET FULLY REELED IN

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Figure 1. Limit Switch Setting/Check.

CHECK

1. Check limit switch operation as follows (from the full-in position):
2. Operate the rescue hoist unit over its full cable travel in both the raise and lower modes. Verify that the limit switches operate as follows:
   a. Intermediate lower limit switch S3 shall actuate when the extending cable is 10 ± 2 feet from the fully reeled-out position.
   b. Full-out limit switch S2 shall actuate when the cable has been extended to the point where 3.5 to 4.5 wraps of cable remain on the drum.
3. Actuate the cable changing switch and using the pendant slowly rotate the cable in lower mode until the mark that was previously applied to the cable appears. Continue approximately 1/2 wrap further. Release the cable changing switch. 28 vdc power to the system shall be disconnected as indicated by no payout displays.

CAUTION

Damage to cable will occur if cable is allowed to begin backwinding on drum, Using pendant control, it is imperative that slow speed be used at this point to avoid back-winding cable on drum.

4. Intermediate hoist limit switch S4 shall actuate when the retracting cable is 10 ± 2 feet from the fully reeled-in position.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

HOISTS AND WINCHES

EXPLOSIVE CARTRIDGE HH-60A HH-60L HOIST BL-29900-30

This WP supersedes WP 1351 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Electrical Repairer's Toolkit, SC 5180-99-B06
Torque Wrench, 30 - 200 in. lbs.

Material/Parts

Cleaning Compound, Solvent Item 71, WP 1803 00
Marker, Identification Item 194, WP 1803 00
Wire, Nonelectrical Item 357, WP 1803 00

Personnel Required

Aircraft Electrician MOS 15F (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0153 00
WP 0857 00
WP 1803 00

REMOVE

1. Turn off all electrical and hydraulic power.
2. Unfasten and remove battery cover from battery support.
3. Disconnect battery connector, P140, from battery (WP 0857 00).

WARNING

- Injury to personnel will result if explosive cartridge accidentally fires. Explosive cartridges may be fired by radio or radar frequencies and static electricity. Short circuit explosive cartridge connector pins to prevent accidental firing.
- Injury to personnel and damage to equipment will result if cartridges are inadvertently activated. All cartridges and cartridge actuated devices shall be handled as live ammunition. Explosive cartridges which have been fired may retain an explosive residue capable of presenting a hazardous condition.

4. Disconnect electrical connector, P6, from explosive cartridge (Figure 1).
5. Install shorting jumper on cartridge receptacle.
6. Remove explosive cartridge from rescue hoist cartridge housing.
7. Clean inside of cartridge housing with cleaning compound solvent, Item 71, WP 1803 00.

REMOVE - Continued

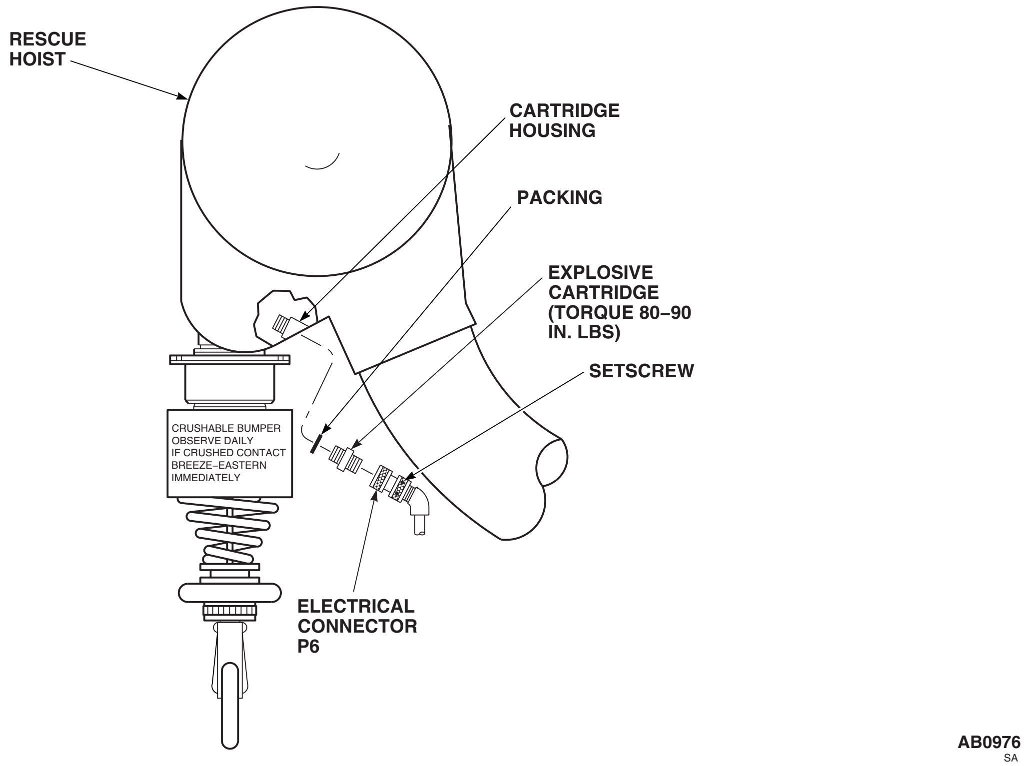


Figure 1. Replace Explosive Cartridge.

INSTALL

CAUTION

To prevent damage to connector (lower body/clamp area) and hoist fairing during operation make sure that electrical connector body is aligned parallel to support arm while connector clamp is aligned parallel to hoist assembly.

NOTE

Use identification marker, Item 194, WP 1803 00, for marking explosive cartridge. Mark each side (four places) with service life date (month and year). Do not scribe, scratch, or vibroetch these dates as damage will occur to the corrosion resistant surface.

1. Turn off all electrical and hydraulic power.
2. On replacement explosive cartridge, do this:
   a. Remove and discard packing.
   b. Remove and discard collar.
   c. **QA** Install packing.
3. **QA** Install cartridge in rescue hoist winch cartridge housing. **TORQUE CARTRIDGE TO 80 - 90 INCH-POUNDS (Figure 1).**
4. **QA** Using nonelectrical wire, Item 357, WP 1803 00, lockwire cartridge.

INSTALL - Continued

WARNING

Injury to personnel and damage to equipment will result if cartridge is fired by stray voltage. When installing cartridge, do not connect electrical connector until stray voltage check has been performed.

5. Perform stray voltage check (WP 0153 00).
6. Remove shorting jumper from cartridge receptacle.
7. **QA** Connect electrical connector, P6, to cartridge.
8. Connect battery connector, P140, to battery (WP 0857 00).
9. Fasten battery cover to battery support.
10. Do an operational check of rescue hoist system (WP 0153 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

HOISTS AND WINCHES

RESCUE HOIST CONTROL PANEL INFORMATION PLATE LAMPS (AVIM) HOIST 42305R1

INITIAL SETUP:

Tools and Special Tools
Electrical Repairer's Toolkit, SC 5180-99-B06

Personnel Required
Aircraft Electrical Repairer MOS 15F (1)

References
WP 0152 00
WP 1340 00

REMOVE

1. Turn off all electrical power.
2. Remove information plate from rescue hoist control panel (WP 1340 00).
3. Remove and discard moisture seal from back of information panel (Figure 1).
4. Remove screws securing lamp to information plate. Remove lamp from information plate.

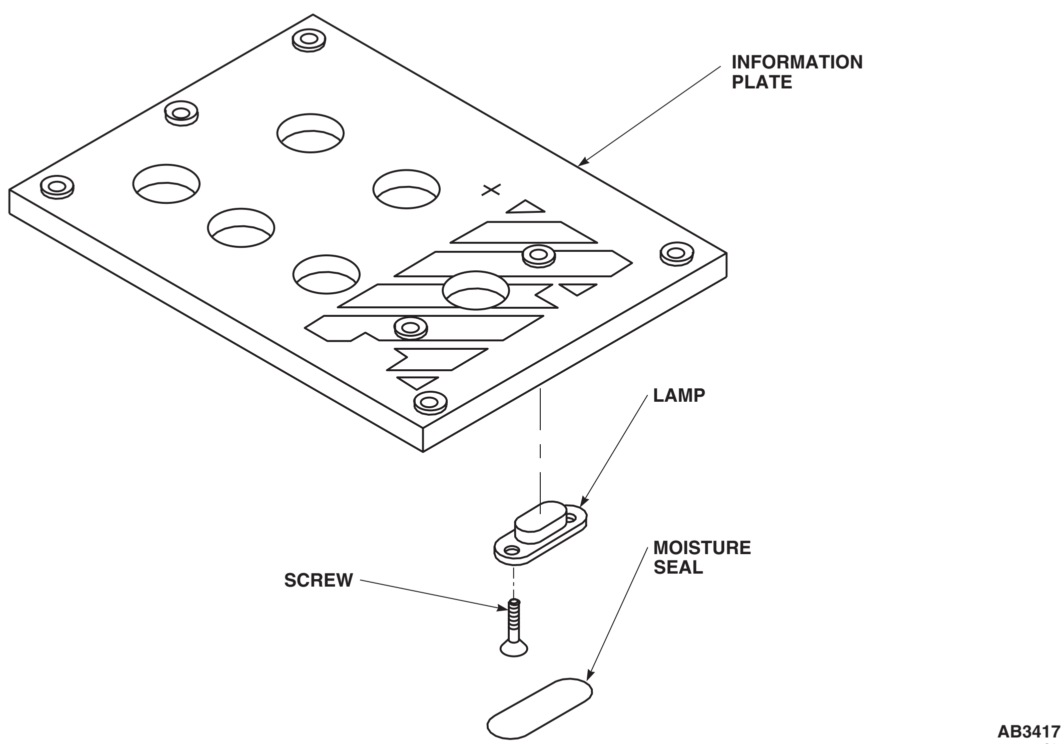


Figure 1. Replace Rescue Hoist Control Panel Information Plate Lamps (AVIM).

INSTALL

1. Turn off all electrical power.
2. **QA** Secure lamp to information plate using screws (Figure 1).
3. Install new moisture seal over lamp.
4. **QA** Install information plate on rescue hoist control panel (WP 1340 00)
5. Do an operational check of rescue hoist control panel (WP 0152 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

HOISTS AND WINCHES

RESCUE HOIST CONTROL PANEL TOGGLE SWITCHES (AVIM) HOIST 42305R1

INITIAL SETUP:

Tools and Special Tools
Electrical Repairer's Toolkit, SC 5180-99-B06

Material/Parts
Tags

Personnel Required
Aircraft Electrician MOS 15F (1)

References
WP 0152 00
WP 1340 00

NOTE

The rescue hoist control panel assembly has five toggle switches. Replacement of each switch is the same.

REMOVE

1. Turn off all electrical power.
2. Remove information plate from rescue hoist control panel assembly (WP 1340 00).
3. Remove screws and separate bottom cover assembly from rescue hoist control panel to access toggle switch (Figure 1).
4. Remove nut, lockwasher, and lockring. Remove toggle switch from rescue hoist control panel.
5. Tag wires. Remove screws, lockwashers and wiring from rear of switch.

INSTALL

1. Turn off all electrical power.
2. **QA** Connect wiring to switch terminals with screws and lockwashers (Figure 1). Remove tags.
3. **QA** Place toggle switch through opening in rescue hoist control panel. Install toggle switch to control panel using nut, lockwasher, and lockring.
4. Install information plate on rescue hoist control panel (WP 1340 00).
5. **QA** Install bottom cover assembly on rescue hoist control panel with screws.
6. Do an operational check of rescue hoist control panel (WP 0152 00).

INSTALL - Continued

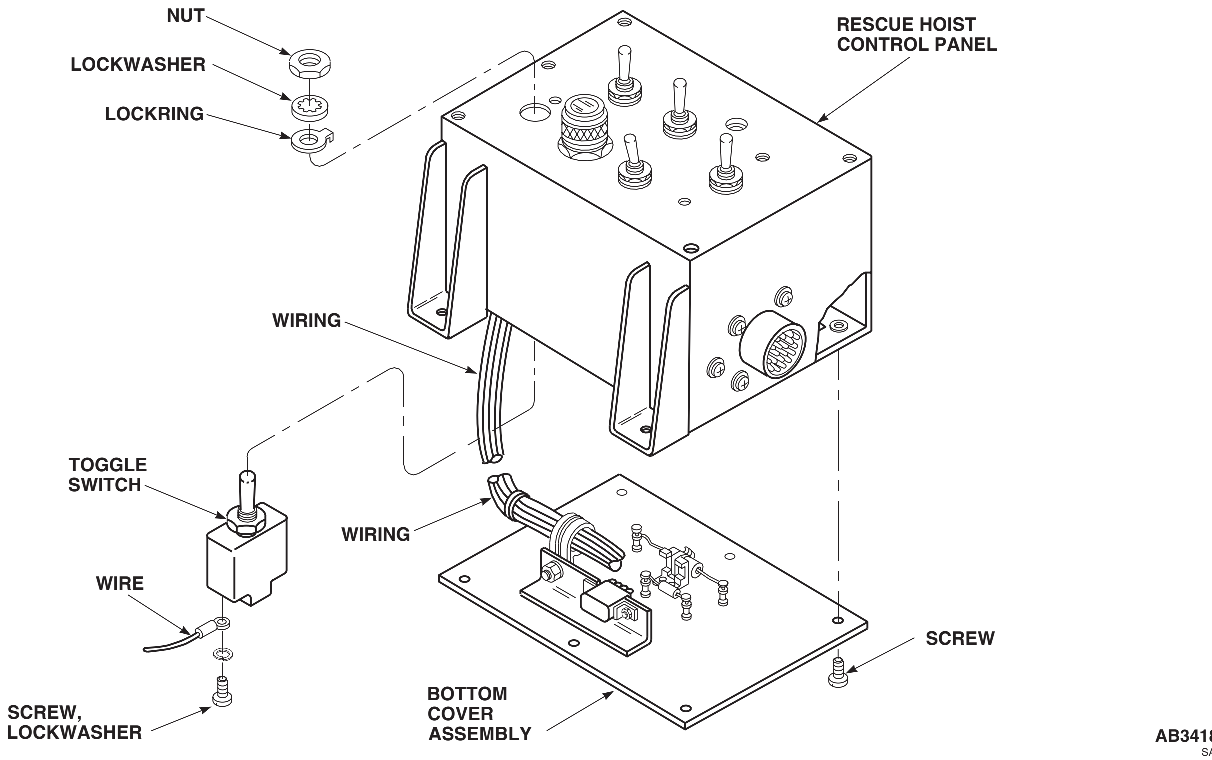


Figure 1. Replace Rescue Hoist Control Panel Toggle Switches (AVIM).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

HOISTS AND WINCHES

RESCUE HOIST CONTROL PANEL INDICATOR LIGHT (AVIM) HOIST 42305R1

INITIAL SETUP:

Tools and Special Tools
Electrical Repairer's Toolkit, SC 5180-99-B06

Material/Parts
Solder, Tin Alloy Item 299, WP 1803 00
Tags

Personnel Required
Aircraft Electrician MOS 15F (1)

References
WP 0152 00
WP 1340 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Remove lens and lamp from indicator light (WP 1340 00).
3. Remove information plate (WP 1340 00).
4. Remove screws and separate bottom cover assembly from rescue hoist control panel to access indicator light (Figure 1).
5. Tag wires. Unsolder wiring from indicator light.
6. Remove nut and washer. Remove indicator light from rescue hoist control panel.

INSTALL

1. Turn off all electrical power.
2. **QA** Place indicator light through opening in rescue hoist control panel (Figure 1). Secure indicator light to rescue hoist control panel using nut and lockwasher.
3. **QA** Solder wiring to indicator light using tin alloy solder, Item 299, WP 1803 00. Remove tags.
4. **QA** Install bottom cover assembly on rescue hoist control panel with screws.
5. Install information plate (WP 1340 00).
6. Install lamp and lens in indicator light (WP 1340 00).
7. Do an operational check of rescue hoist control panel (WP 0152 00).

INSTALL - Continued

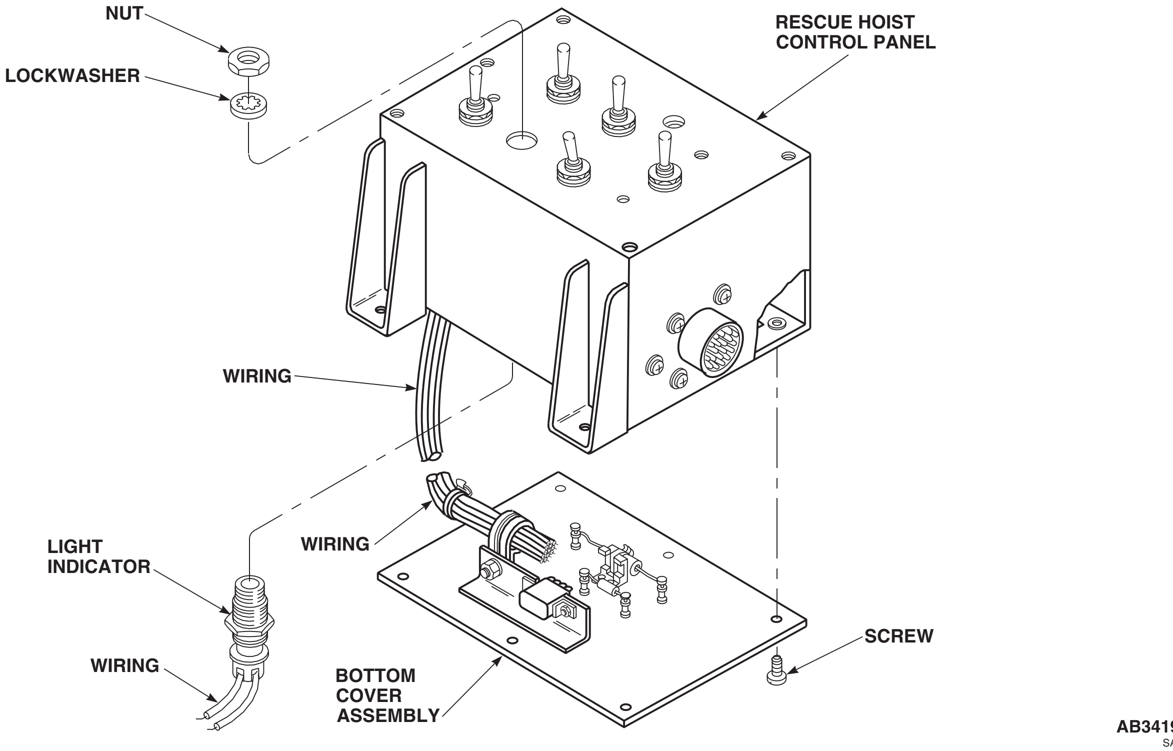


Figure 1. Replace Rescue Hoist Control Panel Indicator Light (AVIM).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

HOISTS AND WINCHES

RESCUE HOIST CONTROL PANEL DIODE (AVIM) HOIST 42305R1

INITIAL SETUP:

Tools and Special Tools
Electrical Repairer's Toolkit, SC 5180-99-B06

Material/Parts
Solder, Tin Alloy Item 299, WP 1803 00

Personnel Required
Aircraft Electrician MOS 15F (1)

References
WP 0152 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Remove screws and separate bottom cover assembly from rescue hoist control panel to access diode (Figure 1).
3. Unsolder leads from terminals and remove diode.

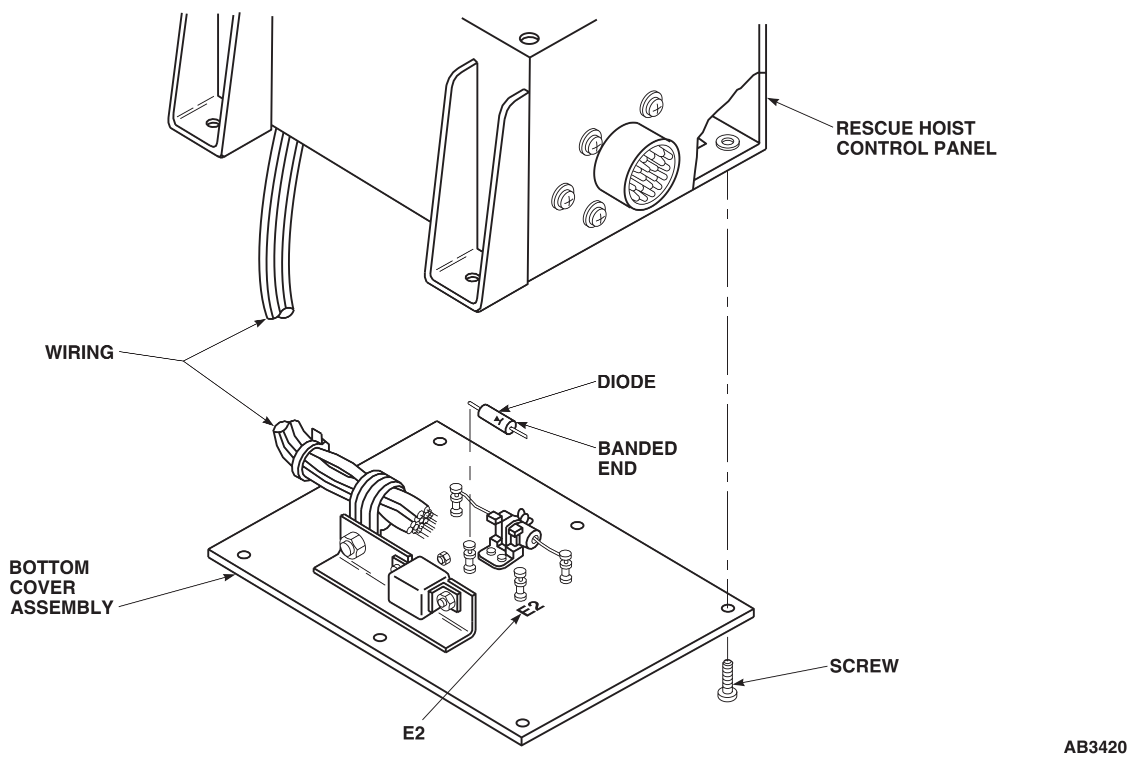


Figure 1. Replace Rescue Hoist Control Panel Diode (AVIM).

INSTALL

1. Turn off all electrical power.
2. **QA** Position diode with banded end lined up with terminal E2 (Figure 1).
3. **QA** Solder diode leads to terminals using tin alloy solder, Item 299, WP 1803 00.
4. Make sure area is clean and free of foreign material.
5. **QA** Install bottom cover assembly on rescue hoist control panel with screws.
6. Do an operational check of rescue hoist control panel (WP 0152 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

HOISTS AND WINCHES

RESCUE HOIST CONTROL PANEL RESISTOR (AVIM) HOIST 42305R1

INITIAL SETUP:

Tools and Special Tools
Electrical Repairer's Toolkit, SC 5180-99-B06

Material/Parts
Solder, Tin Alloy Item 299, WP 1803 00
Strap, Tiedown, Electrical Item 310, WP 1803 00
Tags

Personnel Required
Aircraft Electrician MOS 15F (1)

References
WP 0152 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Remove screws and separate bottom cover assembly from rescue hoist control panel to access resistor (Figure 1).
3. Remove tiedown strap on resistor bracket.
4. Tag resistor leads. Unsolder resistor leads from terminals and remove resistor.

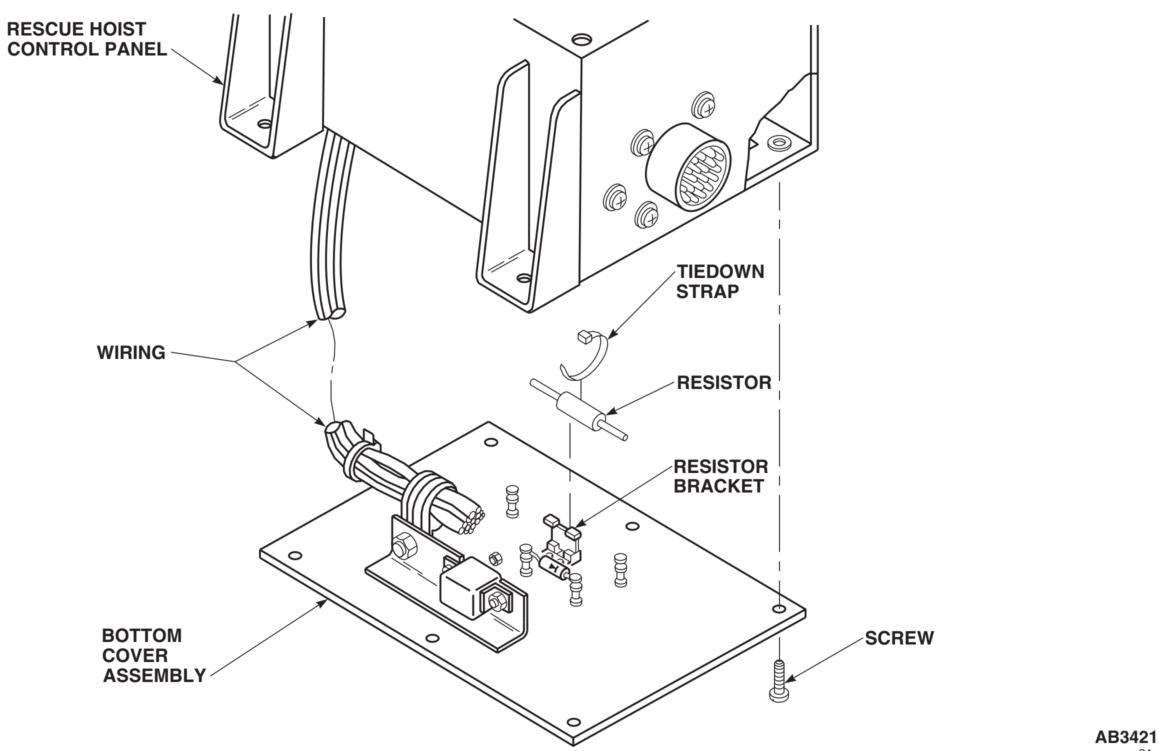


Figure 1. Replace Rescue Hoist Control Panel Resistor (AVIM).

INSTALL

1. Turn off all electrical power.
2. **QA** Secure resistor to bracket with electrical tiedown strap, Item 310, WP 1803 00 (Figure 1). Make sure terminals are lined up with correct leads.
3. **QA** Solder resistor leads to terminals using tin alloy solder, Item 299, WP 1803 00. Remove tags.
4. Make sure area is clean and free of foreign material.
5. **QA** Install bottom cover assembly on rescue hoist control panel assembly with screws.
6. Do an operational check of rescue hoist control panel (WP 0152 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

HOISTS AND WINCHES

RESCUE HOIST CONTROL PANEL ELECTRICAL CONNECTOR (AVIM) HOIST 42305R1

INITIAL SETUP:

Tools and Special Tools

Electrical Repairer's Toolkit, SC 5180-99-B06
Insert/Extract Tool, GGG-E-936

Material/Parts

Tags

Personnel Required

Aircraft Electrician MOS 15F (1)

References

WP 0152 00

REMOVE

1. Turn off all electrical power.
2. Remove screws and separate bottom cover assembly from rescue hoist control panel to access electrical connector (Figure 1).
3. Remove screws and nuts securing electrical connector to rescue hoist control panel.
4. Tag wires. Using insert/extract tool, remove wires from electrical connector.

INSTALL

1. Turn off all electrical power.
2. **QA** Using insert/extract tool insert wires into electrical connector. Remove tags.
3. **QA** Install electrical connector in mounting hole on side of rescue hoist control panel (Figure 1). Secure electrical connector to rescue hoist control panel using screws and nuts.
4. Make sure area is clean and free of foreign material.
5. **QA** Install bottom cover assembly on rescue hoist control panel with screws.
6. Do an operational check of rescue hoist control panel (WP 0152 00).

INSTALL - Continued

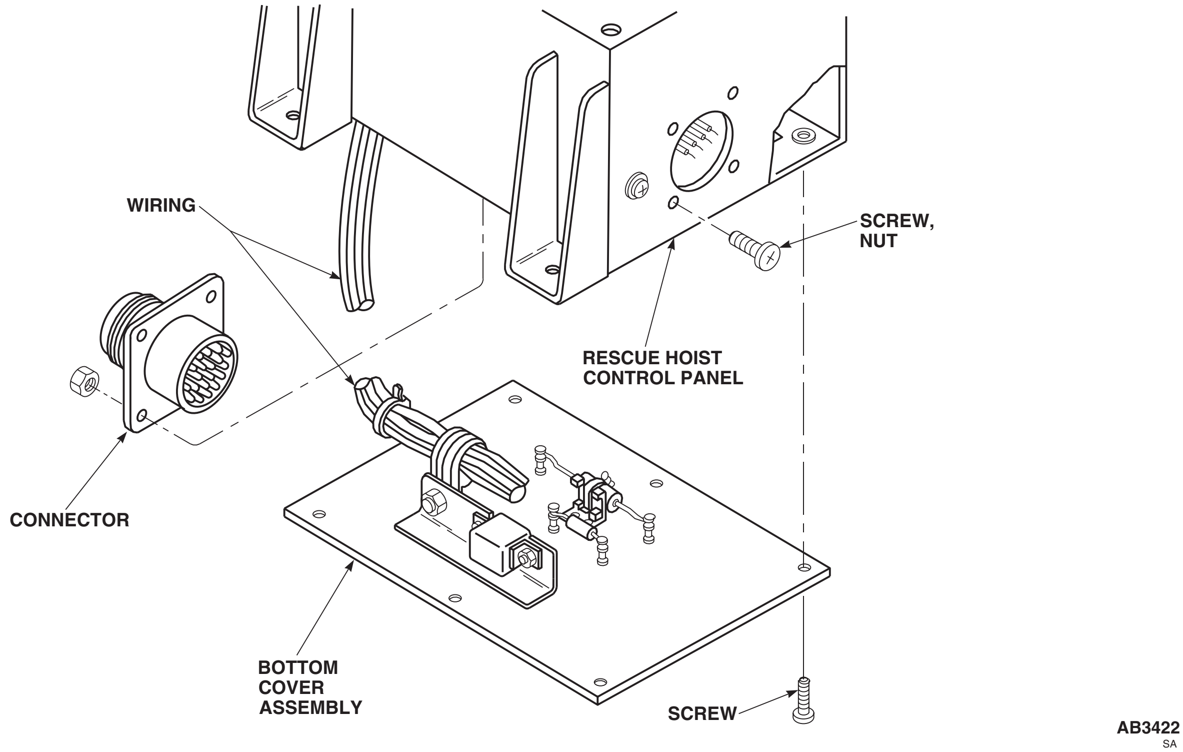


Figure 1. Replace Rescue Hoist Control Panel Electrical Connector (AVIM).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

HOISTS AND WINCHES

RESCUE HOIST CONTROL PANEL RELAY (AVIM) HOIST 42305R1

INITIAL SETUP:

Tools and Special Tools
Electric Gun Heater, LFT500
Electrical Repairer's Toolkit, SC 5180-99-B06

Material/Parts
Insulation Sleeving, Electrical Item 164, WP 1803 00
Solder, Tin Alloy Item 299, WP 1803 00
Tags

Personnel Required
Aircraft Electrician MOS 15F (1)

References
WP 0152 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Remove screws and separate bottom cover assembly from rescue hoist control panel to access electrical connector (Figure 1).
3. Remove screws, nuts, and washers securing relay and clamp to relay bracket.

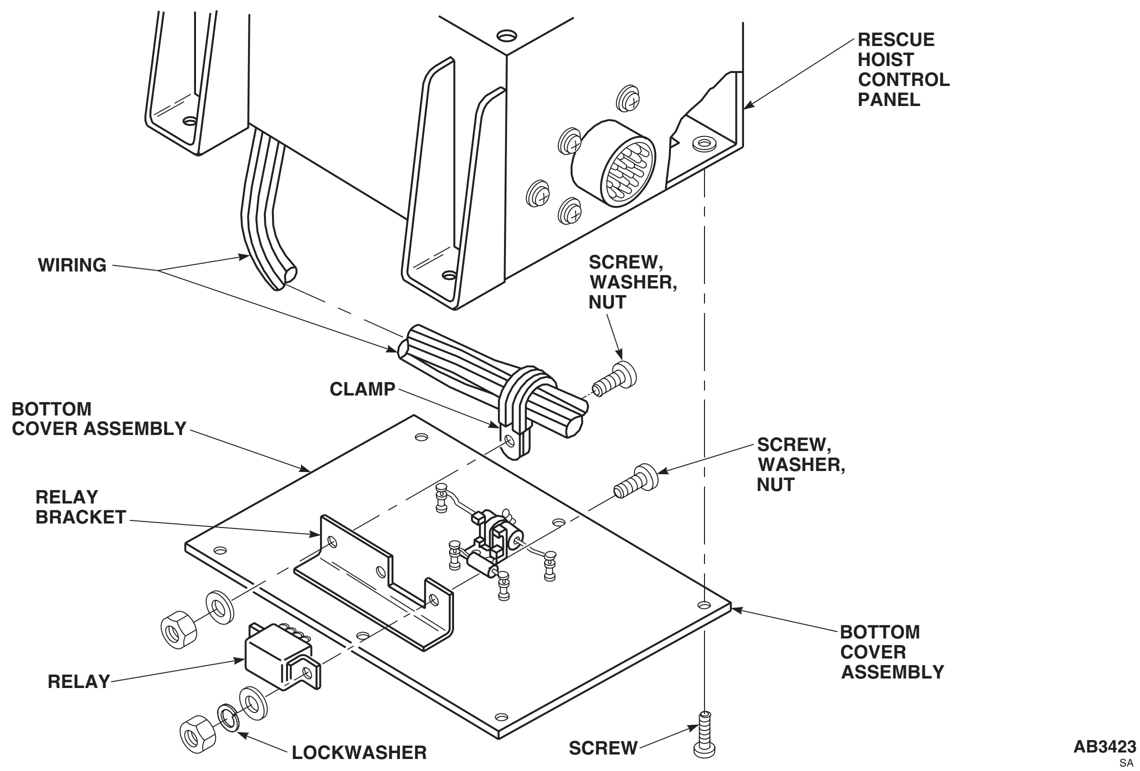


Figure 1. Replace Rescue Hoist Control Panel Relay (AVIM).

REMOVE - Continued

4. Tag wires. Unsolder wires from relay.
5. Remove relay from bottom cover assembly.

INSTALL

1. Turn off all electrical power.
2. **QA** Install relay onto relay bracket with screw, nut, and washers (Figure 1).
3. Place electrical insulation sleeving, Item 164, WP 1803 00, over relay wiring.
4. **QA** Solder wires to relay terminals using tin alloy solder, Item 299, WP 1803 00. Remove tags.
5. Slide electrical insulation sleeving down wires and over relay terminals.
6. Install electrical insulation sleeving over unused relay terminals.
7. **QA** Using electric gun heater, heat-shrink electrical insulation sleeving on relay.
8. Spread clamp and slide over wire harness.
9. **QA** Secure clamp and relay to relay bracket with screw, washer, and nut.
10. Make sure area is clean and free of foreign material.
11. **QA** Secure bottom cover assembly to rescue hoist control panel using screws.
12. Do an operational check of rescue hoist control panel (WP 0152 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AUXILIARY POWER PLANT SYSTEM

AUXILIARY POWER UNIT (APU)

This WP supersedes WP 1359 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Adapter Assembly, 70700-20423-041
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Hoist, to Support Weight of APU (150 lbs)
Hoisting Sling Assembly, 70700-20323-042
Maintenance Crane, (Alternate for Hoist) 70700-20300-042
Nonmetallic Mallet, 25714
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Material/Parts
Adhesive, Item 27, WP 1803 00
Antiseize Compound, Item 49, WP 1803 00
Grease, Molybdenium Disulfide, Item 138, WP 1803 00
Protective Caps and Plugs
Sealing Compound, Item 273, WP 1803 00
Tags
Wire, Nonelectrical, Item 354, WP 1803 00
Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required
Assistants - (2)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0155 00
WP 0156 00
WP 0303 00
WP 0692 00
WP 0698 00
WP 0742 00
WP 0842 00
WP 0865 00
WP 1360 00
WP 1365 00
WP 1366 00
WP 1367 00
WP 1590 00
WP 1604 00
WP 1605 00
WP 1703 00
WP 1745 00
WP 1803 00

REMOVE

NOTE

- When replacing APU, 116305-100 through 116305-300 series, with APU, 3800480-1 and 3800480-2, fuel line seal, 169037-1, firewall plate assembly, 169036-1, ignition seal assembly, 160362-100, and drain tube assembly, 70303-03016-044, must be put in a storage bag and kept with APU, 116305-100 through 116305-300 series.
- When replacing APU, 3800480-1 and 3800480-2, with APU, 116305-100 through 116305-300 series, fuel line seal, 3615184-1, firewall plate assembly, 3615185-1, and ignition seal assembly, 3615692-1, must be put in a storage bag and kept with APU, 3800480-1 and 3800480-2.

1. Turn off all electrical power.
2. If removing APU for storage or AVIM level maintenance, preserve APU in accordance with WP 1745 00.

REMOVE - Continued

3. If maintenance crane is to be used, assemble and set up crane (WP 1590 00).
4. Remove APU compartment access doors (WP 0303 00).
5. Remove clamp holding generator wiring harness to airframe.
6. Disconnect electrical connector, P414, from fire detector on rear panel (Figure 1, Sheet 1, Detail A).
7. Loosen stud fasteners holding fuel control upper cover to fuel control lower cover. Remove cover.

NOTE

Removal of fuel inlet hose bonding jumper is not required unless fuel inlet hose is being removed from helicopter.

8. Remove bonding jumper from fuel inlet hose.

CAUTION

Damage to fuel filter will result if union in fuel control base turns when removing fuel inlet hose. Hold union with wrench while removing fuel inlet hose to avoid damage to fuel filter.

9. Holding fuel inlet port union base with a wrench, disconnect fuel inlet hose.
10. Disconnect fuel control box drain hose, gear box vent hose, compartment drain hose, combustor drain hose, and fire extinguishing hose. Plug hoses and cap ports.
11. Depressurize APU accumulator (WP 0742 00).
12. Remove nuts, washers, hydraulic start motor, and gasket from APU.
13. Tie hydraulic start motor to beam. Make sure hoses are not kinked.
14. Disconnect electrical connectors, P252 and P279, from APU generator.
15. Disconnect electrical connector, P289, from side of APU.
16. Remove screws and washers holding exhaust duct to rear panel.
17. Remove bolts and saddle washers holding exhaust duct to airframe fairing.
18. Slide duct back past exhaust pipe.
19. Remove clamp holding check valve to pneumatic tube at front panel half firewall (Figure 1, Sheet 2, Detail B).
20. Loosen stud fasteners holding shroud assembly panels to exhaust firewall (Figure 1, Sheet 2, Detail D). Make sure APU (with shroud assembly) is separated from engine exhaust firewall.
21. Install hoisting sling on APU (Figure 1, Sheet 3, Detail E). Attach sling to maintenance crane or to a hoist with enough capacity to lift APU (150 pounds). Take up slack.
22. Remove bolts holding front mounting lugs to front supports (Figure 1, Sheet 3, Detail F). Operate hoist/crane and support full weight of APU.
23. Slide APU to rear, disengaging rear mounting lug from rear support mount. Raise APU up enough to disconnect generator wiring.
24. If terminal cover is installed, remove it, or slide terminal boots back on wiring harness leads (Figure 1, Sheet 2, Detail C). Tag and disconnect wiring harness leads from terminals on generator.

REMOVE - Continued

WARNING

Injury to personnel or damage to equipment will result if crane is used with excessive force to free APU from helicopter. Do not allow crane hook to retract fully against pulley when lifting APU from helicopter.

25. With assistant operating hoist/crane, remove APU from helicopter. Remove bolts and washers holding front mounting lugs to APU gear box sump. Install on maintenance trailer adapter assembly (Figure 1, Sheet 3, Detail G). If adapter assembly/maintenance trailer is not available, support APU to prevent damage to components on bottom of APU.
26. Remove screws holding cover halves around bleed air outlet flange on lower front panel half (Figure 1, Sheet 4, Detail H). Remove cover halves.
27. On helicopters with APU, 116305-100 through 116305-300 series, remove bolts and washers and remove firewall sleeves, grommet, and plate from front panel halves.
28. On helicopters with APU, 3800480-1 and 3800480-2, remove bolts and washers and remove ignition seals fuel line seal, and seal plate from front panel halves.
29. Remove band clamp holding front panel halves together.
30. Remove rear mounting lug from compressor (Figure 1, Sheet 3, Detail F).
31. Remove stud fasteners holding front panel halves to shroud (Figure 1, Sheet 4, Detail H). Remove front panel halves.
32. Remove shroud assembly (with rear panel) surrounding APU hot section.
33. On helicopters with APU, 116305-100 through 116305-300 series, remove combustor drain.
34. On helicopters with APU, 3800480-1 and 3800480-2, remove fuel drain tube and install protective plugs (Figure 1, Sheet 5, Detail J).

NOTE

If replacing APU, 116305-100, with APU, 116305-200, keep combustor drain line.

35. On helicopters with APU, 116305-100 through 116305-300 series, remove packing from APU combustor fuel drain check valve. Cap port. (Figure 1, Sheet 4, Detail H). Discard packing.
36. Loosen clamp and remove exhaust pipe from APU.
37. Remove APU generator (WP 0842 00).
38. Remove nuts, washers, QAD mount, and gasket from APU.
39. On helicopters with APU, 116305-100 through 116305-300 series, remove bolts and washers attaching tube to start bypass valve. Cover port.
40. On APU, 116305-100 through 116305-300 series, put the following parts in a storage bag and keep with APU:
    a. Fuel line seal, 169037-1.
    b. Firewall plate assembly, 169036-1.
    c. Ignition seal assembly, 160362-100.
    d. Drain tube assembly, 70303-03016-044.
41. On APU, 3800480-1 and 3800480-2, put the following parts in a storage bag and keep with APU.
    a. Fuel line seal, 3615184-1.

REMOVE - Continued

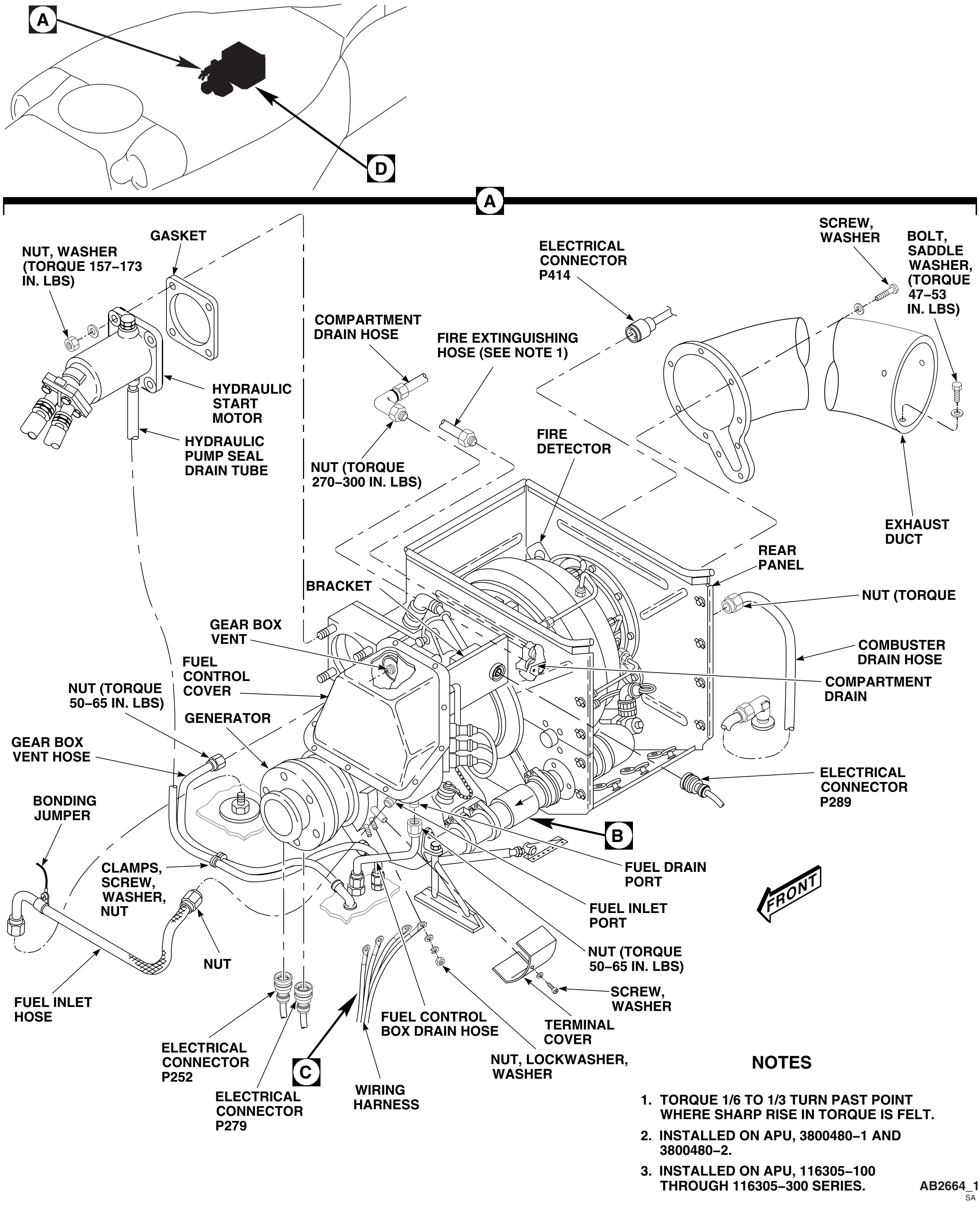


Figure 1. Replace Auxiliary Power Unit (APU). (Sheet 1 of 5)

REMOVE - Continued

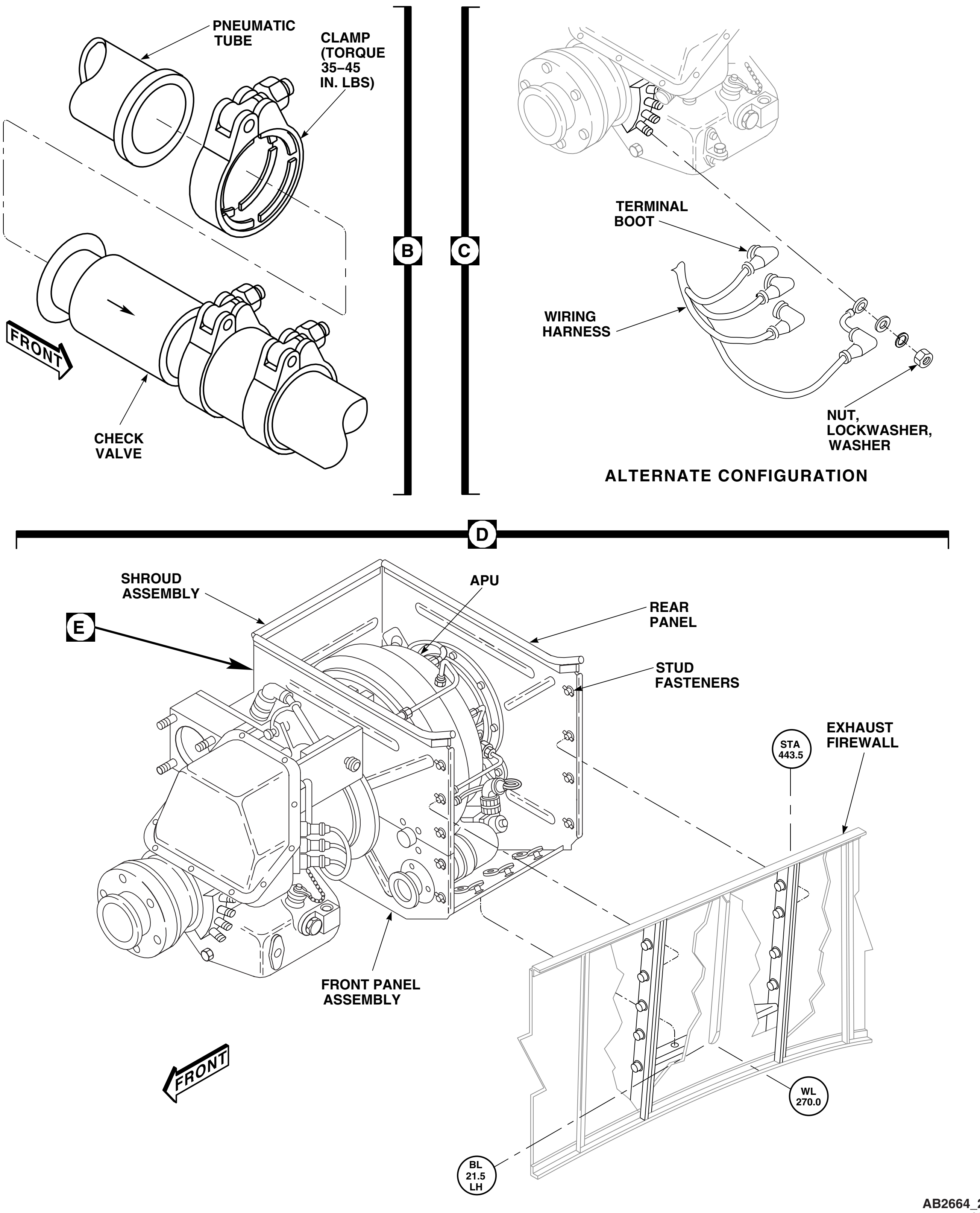


Figure 1. Replace Auxiliary Power Unit (APU). (Sheet 2 of 5)

REMOVE - Continued

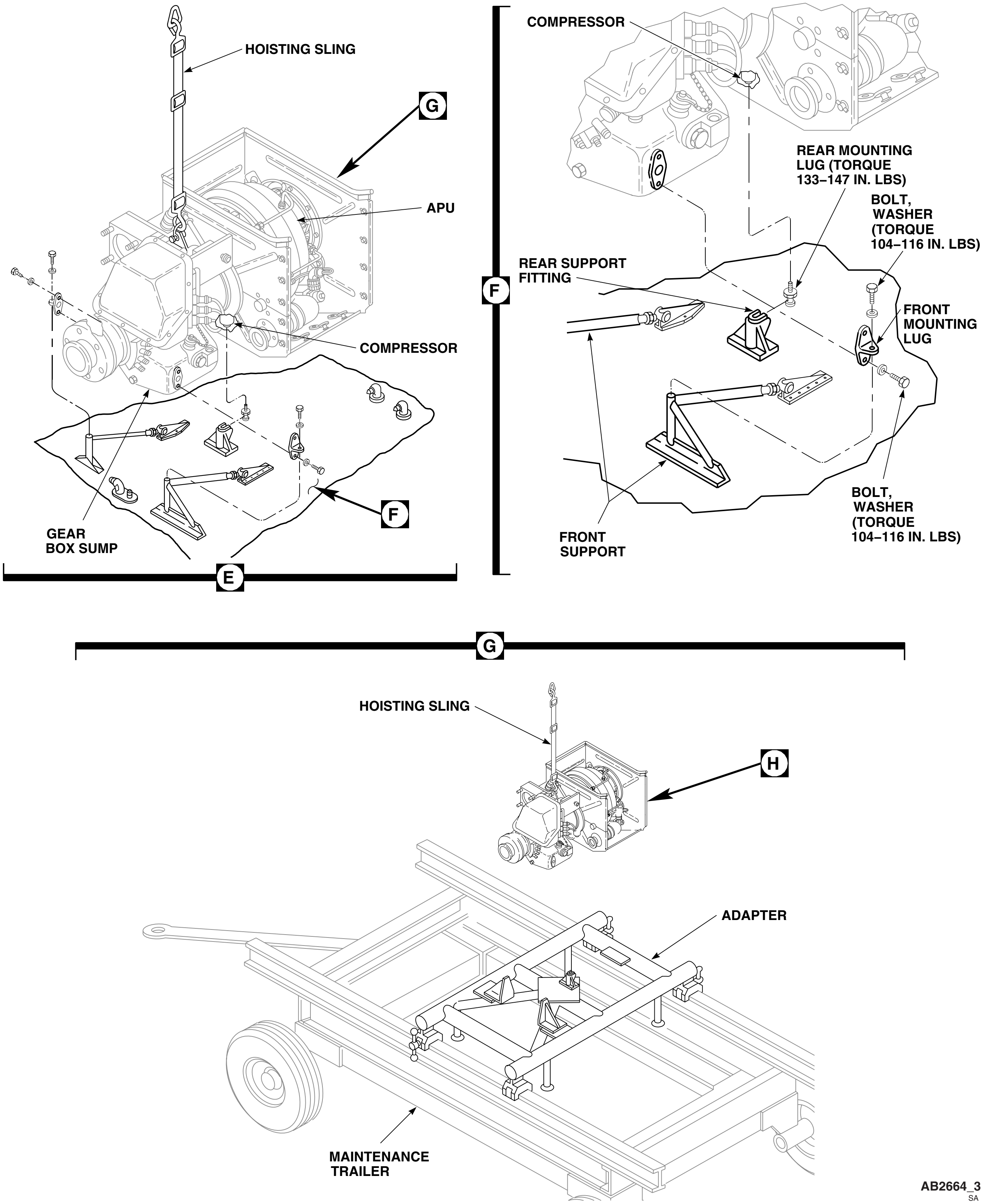


Figure 1. Replace Auxiliary Power Unit (APU). (Sheet 3 of 5)

REMOVE - Continued

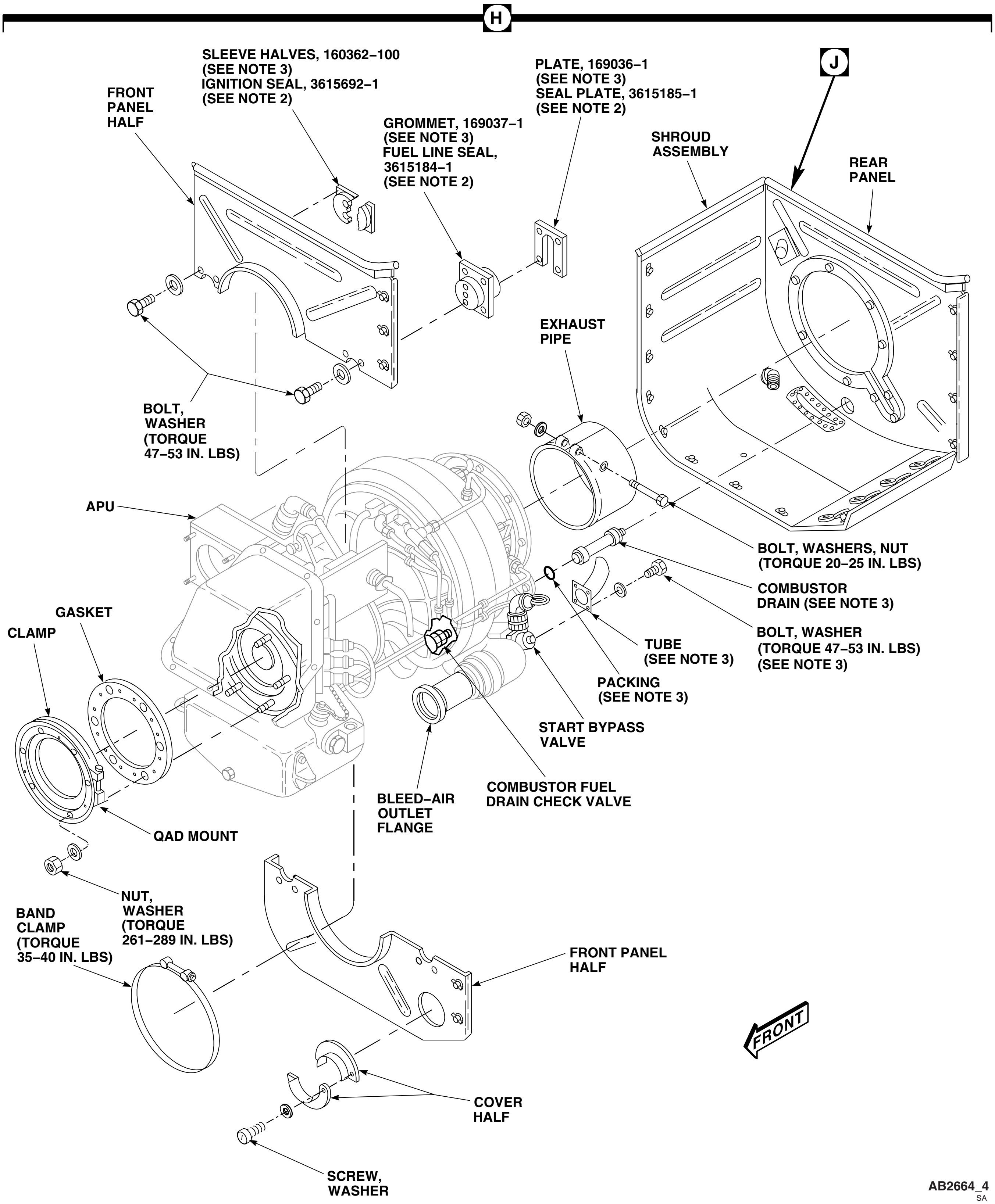


Figure 1. Replace Auxiliary Power Unit (APU). (Sheet 4 of 5)

REMOVE - Continued

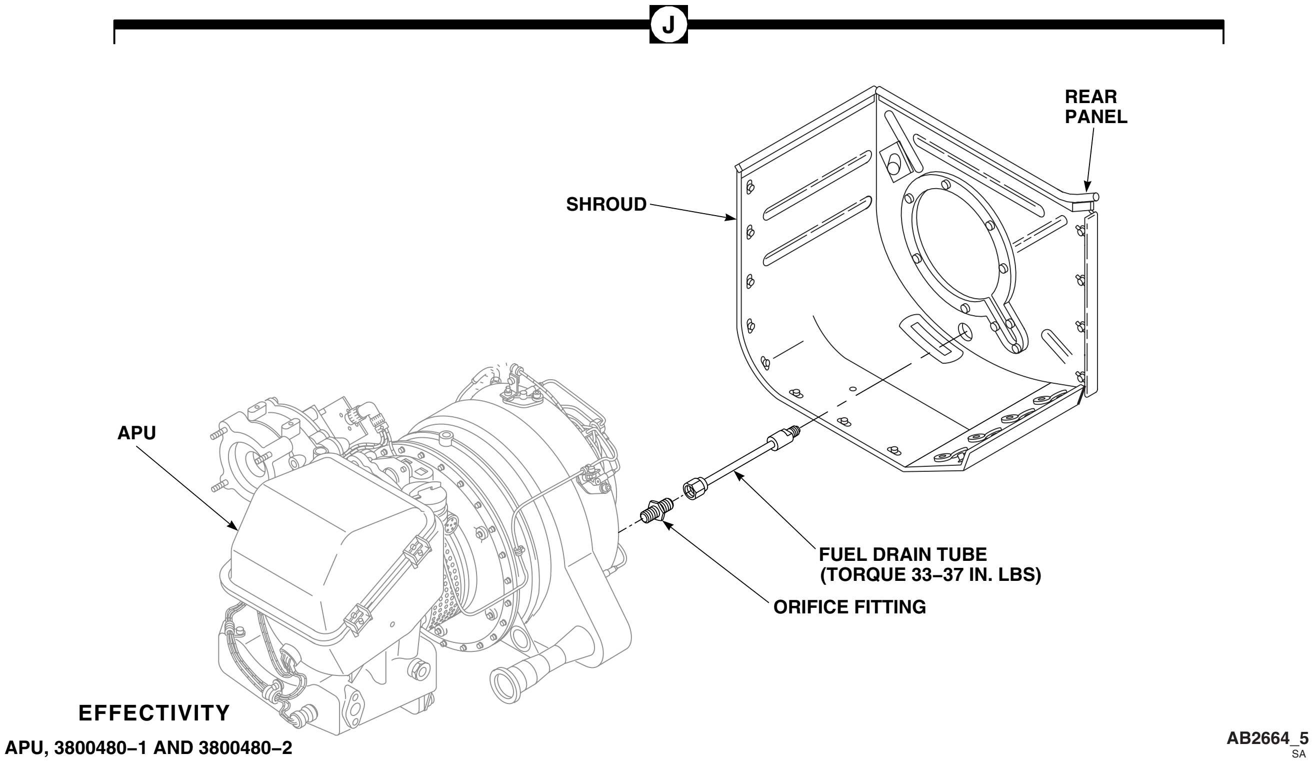


Figure 1. Replace Auxiliary Power Unit (APU). (Sheet 5 of 5)

b. Firewall plate assembly, 3615185-1.

c. Ignition seal assembly, 3615692-1.

INSPECT

1. Inspect APU check valve (WP 0698 00).
2. If APU is being replaced because of internal failure, check pneumatic system for possible metal contamination (WP 0692 00).
3. Inspect generator electrical terminal boots for cracks. Replace cracked boots.
4. Inspect APU generator spline adapter (WP 1366 00).
5. Inspect hydraulic start motor coupling shaft (WP 1365 00).

INSTALL

NOTE

- When installing APU, 3800480-1 and 3800480-2, make sure ESU, 211880606-2-1, is installed (WP 0865 00).
- When installing APU, 116305-100 through 116305-300 series, make sure ESU, 160200-200, 160200-201, 160200-600, or 163290-100, is installed (WP 0865 00).

1. Turn off all electrical power.

INSTALL - Continued

2. Depreserve replacement APU in accordance with WP 1745 00.
3. Place gasket and QAD mount on APU (Figure 1, Sheet 4, Detail H). Install with QAD mount bolt installed on left side of APU. Install washers and nuts. Torque nuts in a crisscross pattern in increments as follows:
   a. **TORQUE NUTS TO 50 INCH-POUNDS.**
   b. **TORQUE NUTS TO 150 INCH-POUNDS.**
   c. **TORQUE NUTS TO 250 INCH-POUNDS.**
   d. **QA TORQUE NUTS TO A FINAL TORQUE OF 261 - 289 INCH-POUNDS.**
4. Install APU generator (WP 0842 00).
5. **QA** On helicopters with APU, 116305-100 through 116305-300 series, remove port caps and bolt tube to start bypass valve. **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire bolts together.
6. Apply a coat of antiseize compound, Item 49, WP 1803 00, around inside clamping surface of exhaust pipe.
7. Place exhaust pipe on flange of APU.
8. Make sure exhaust pipe does not overlap APU exhaust case doubler. **TORQUE CLAMP BOLT TO 20 - 25 INCH-POUNDS.**
9. **QA** Tap clamping surface lightly with nonmetallic mallet. **CHECK CLAMP BOLT TORQUE FOR 20 - 25 INCH-POUNDS.**
10. Continue tapping clamping surface until two successive trials show no loss in torque.
11. **QA** On helicopters with APU, 116305-100 through 116305-300 series, remove port cover, and install packing on APU combustor fuel drain check valve.
12. On helicopters with APU, 116305-100 through 116305-300 series, insert combustor drain through hole in rear panel.
13. **QA** On helicopters with APU, 3800480-1 and 3800480-2, insert fuel drain tube through hole in rear panel (Figure 1, Sheet 5, Detail J). Thread fuel drain tube onto orifice fitting, angle facing down. **TORQUE FUEL DRAIN TUBE TO 33 - 37 INCH-POUNDS.**
14. **QA** Install shroud assembly (with rear panel) around APU (Figure 1, Sheet 4, Detail H).

CAUTION

Damage will occur if ignition lead, thermocouple lead, and wiring for valves and switches are installed incorrectly. To prevent chafing and pinching when panel halves are installed take extreme care when installing panel halves.

15. Place front panel halves on APU and install band clamp around front panel halves. **HANDTIGHTEN.**
16. On helicopters with APU, 116305-100 through 116305-300 series, do this:
   a. **QA** Install ignitor leads and electrical wiring inside sleeve halves.
   b. **QA** Install sleeve to front panel halves with bolts and washers. **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.**
   c. **QA** Using nonelectrical wire, Item 357, WP 1803 00, lockwire bolts together. Lockwire bolts only on right side sleeves (looking forward).
17. On helicopters with APU, 3800480-1 and 3800480-2, do this:
   a. **QA** Install ignitor leads and electrical wiring inside ignition seal.

INSTALL - Continued

b. **QA** Install ignition seal to front panel halves with bolts and washers. **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.**

18. On helicopters with APU, 116305-100 through 116305-300 series, do this:

a. **QA** Install grommet and plate over tube bundle.

b. **QA** Install grommet and plate to front panel halves with bolts and washers. **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.**

19. On helicopters with APU, 3800480-1 and 3800480-2, do this:

a. **QA** Install fuel line seal and seal plate over tube.

b. **QA** Install fuel line seal to front panel halves with bolts and washers. **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.**

20. **QA** Place cover halves around bleed air outlet flange, and install on lower front panel with washers and screws.
21. **QA TORQUE BAND CLAMP HOLDING FRONT PANEL HALVES TO 35 - 40 INCH-POUNDS.**
22. Seal firewall and shroud with adhesive, Item 27, WP 1803 00.
23. **QA** Thread rear mounting lug into APU compressor (Figure 1, Sheet 3, Detail F). **TORQUE LUG TO 133 - 147 INCH-POUNDS.** Using nonelectrical wire, Item 354, WP 1803 00, lockwire lug to shroud assembly.
24. Install hoisting sling on APU (Figure 1, Sheet 3, Detail E). Attach sling to hoist crane. Take up slack.

WARNING

Injury to personnel or damage to equipment will result if crane is used with excessive force to free APU from helicopter. Do not allow crane hook to retract fully against pulley when lifting APU from helicopter.

25. With assistants operating hoist/crane, support full weight of APU. Remove bolts and washers and disengage APU from adapter (Figure 1, Sheet 3, Detail G). Remove front mounting lugs from adapter.
26. **QA** Install front mounting lugs to APU gear box sump with bolts and washers (Figure 1, Sheet 3, Detail F). **TORQUE BOLTS TO 104 - 116 INCH-POUNDS.**
27. **QA** Using nonelectrical wire, Item 357, WP 1803 00, lockwire front mounting lug bolts together.
28. Raise APU and lower into compartment enough to connect wiring to generator.
29. **QA** Connect wiring harness leads to terminals on generator and remove tags (Figure 1, Sheet 2, Detail C).
30. **QA** Install terminal boots over terminals, or install terminal cover. Using nonelectrical wire, Item 354, WP 1803 00, lockwire screws.
31. Inspect and treat electrical connector (WP 1703 00).
32. **QA** Connect electrical connector, P289, to side of APU (Figure 1, Sheet 1, Detail A).
33. Inspect and treat electrical connector (WP 1703 00).
34. **QA** Connect electrical connectors, P252 and P279, to APU generator.
35. **QA** Install gasket on hydraulic start motor mounting pad.
36. **QA** Lubricate hydraulic start motor coupling splined shaft with molybdenium disulfide grease, Item 138, WP 1803 00, and install on mounting pad with washers and nuts. **TORQUE NUTS TO 157 - 173 INCH-POUNDS.**

INSTALL - Continued

37. Engage rear mounting lug in rear support fitting, and slide APU forward along engine exhaust firewall (Figure 1, Sheet 3, Detail F).
38. Attach APU shroud assembly panels to exhaust firewall, and lock stud fasteners (Figure 1, Sheet 2, Detail D).
39. **QA** Install front mounting lugs to front supports with bolts and washers (Figure 1, Sheet 3, Detail F). **TORQUE BOLTS TO 104 - 116 INCH-POUNDS.**
40. Remove hoisting sling from APU and hoist.
41. Install exhaust duct (WP 1360 00).
42. Seal all seams and joined areas of enclosure formed by shroud assembly and exhaust firewall with adhesive, Item 27, WP 1803 00.
43. Inspect and treat electrical connector (WP 1703 00).
44. **QA** Connect electrical connector, P414, to fire detector on rear panel (Figure 1, Sheet 1, Detail A).
45. **QA** Remove caps and plugs and connect APU fire extinguishing hose. **TORQUE NUT 1/6 - 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
46. **QA** Remove caps and plugs and connect APU compartment drain hose. **TORQUE NUT TO 270 - 300 INCH-POUNDS.**
47. **QA** Remove caps and plugs and connect APU combustor drain hose. **TORQUE NUT TO 135 - 150 INCH-POUNDS.**
48. **QA** Remove caps and plugs and connect APU gear box vent hose. **TORQUE NUT TO 50 - 65 INCH-POUNDS.**
49. **QA** Remove caps and plugs and connect APU fuel control box drain. **TORQUE NUT TO 50 - 65 INCH-POUNDS.**
50. **QA** Install APU check valve to pneumatic tube at front panel half firewall with coupling (Figure 1, Sheet 2, Detail B). **TORQUE CLAMPS TO 35 - 45 INCH-POUNDS.**
51. If fuel control cover was not removed, loosen stud fasteners holding fuel control upper cover to fuel control base. Remove cover (Figure 1, Sheet 1, Detail A).

CAUTION

Damage to fuel filter will result if union in fuel control base turns when installing fuel inlet hose. Hold union with wrench while installing fuel inlet hose to avoid damage to fuel filter.

52. Hold fuel inlet port union in fuel control base with a wrench. Remove plug and loosely connect fuel inlet hose to union (Figure 1, Sheet 1, Detail A).
53. **QA** If bonding jumper was removed from fuel inlet hose, install clamp on hose and connect bonding jumper to clamp with screw, washers, and nut.
54. **QA** Connect bonding jumper to airframe with screw, lockwasher, washers, and nut.
55. Seal area around screw and airframe with sealing compound, Item 273, WP 1803 00.
56. Prime APU fuel inlet hose (WP 1367 00).
57. **QA** Install fuel control cover on fuel control base and lock stud fasteners.
58. **QA** Install generator wiring harness to airframe with clamp.
59. **QA** Check wire harness, fuel lines, and drain lines for contact. If contact occurs, reposition wire harness, fuel lines, and drain lines. Install extra clamps as required.

INSTALL - Continued

60. Service APU (WP 1604 00).
61. Service APU accumulator (WP 1605 00).
62. Make sure area is clean and free of foreign material.
63. Install APU compartment access doors (WP 0303 00).
64. Do an operational check of APU (WP 0155 00 or WP 0156 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AUXILIARY POWER PLANT SYSTEM

APU EXHAUST DUCT

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 30 - 200 in. lbs.

Material/Parts
Adhesive, Item 40, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0303 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Open APU compartment access doors (WP 0303 00).
3. Remove screws and washers from APU exhaust duct mounting flange (Figure 1).
4. Remove bolts and saddle washers holding APU exhaust duct to airframe fairing.
5. Pull APU exhaust duct through fairing until mounting flange reaches airframe fairing. Rotate APU exhaust duct until flange finger points forward. Remove APU exhaust duct.

INSTALL

1. Turn off all electrical power.
2. **QA** With finger on mounting flange pointing forward, install APU exhaust duct through airframe fairing (Figure 1).
3. Apply a bead of adhesive, Item 40, WP 1803 00, to mounting flange, and install APU exhaust duct to rear shroud with screws and washers.
4. **QA** Install APU exhaust duct to airframe fairing with bolts and saddle washers. **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.**
5. Make sure area is clean and free of foreign material.
6. Close APU compartment access doors (WP 0303 00).

INSTALL - Continued

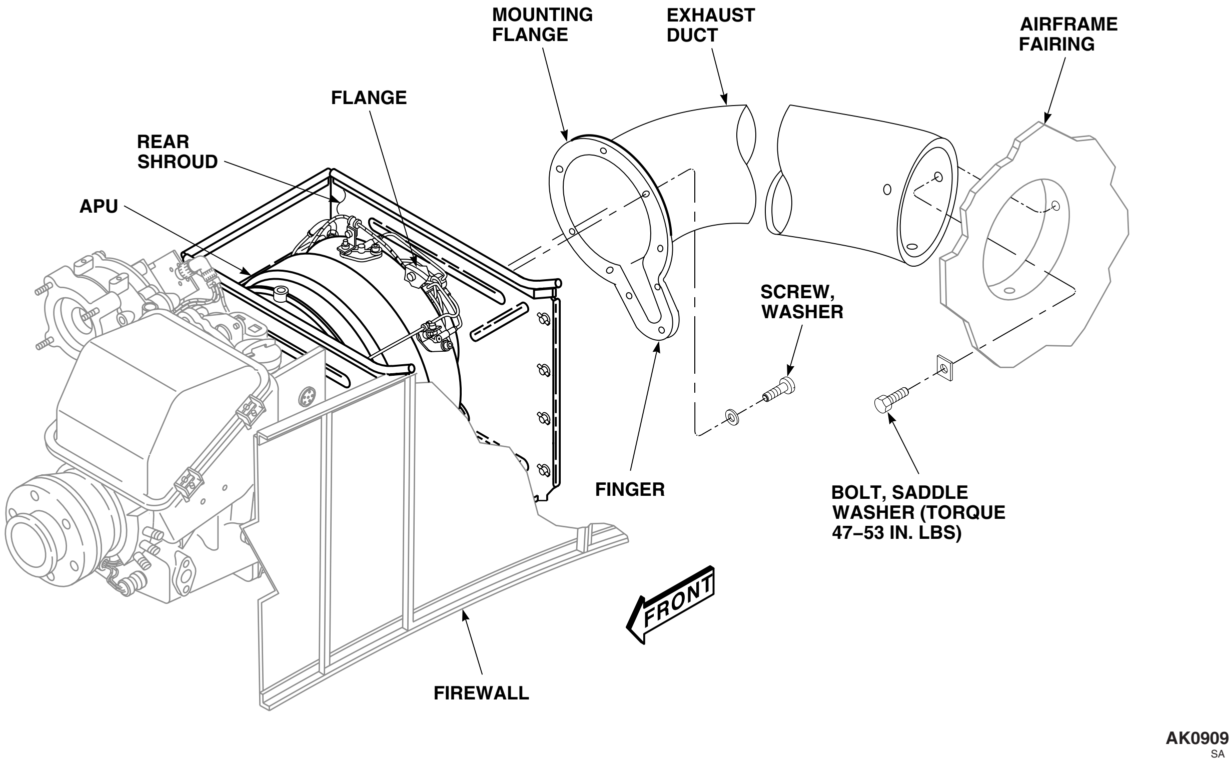


Figure 1. Replace APU Exhaust Duct.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AUXILIARY POWER PLANT SYSTEM

APU EXHAUST PIPE

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Nonmetallic Mallet
Torque Wrench, 0 - 30 in. lbs.

Material/Parts
Antiseize Compound, Item 49, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0303 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Open APU compartment access door (WP 0303 00).
3. Loosen exhaust pipe clamp bolt (Figure 1).
4. Slide exhaust pipe out through exhaust duct.

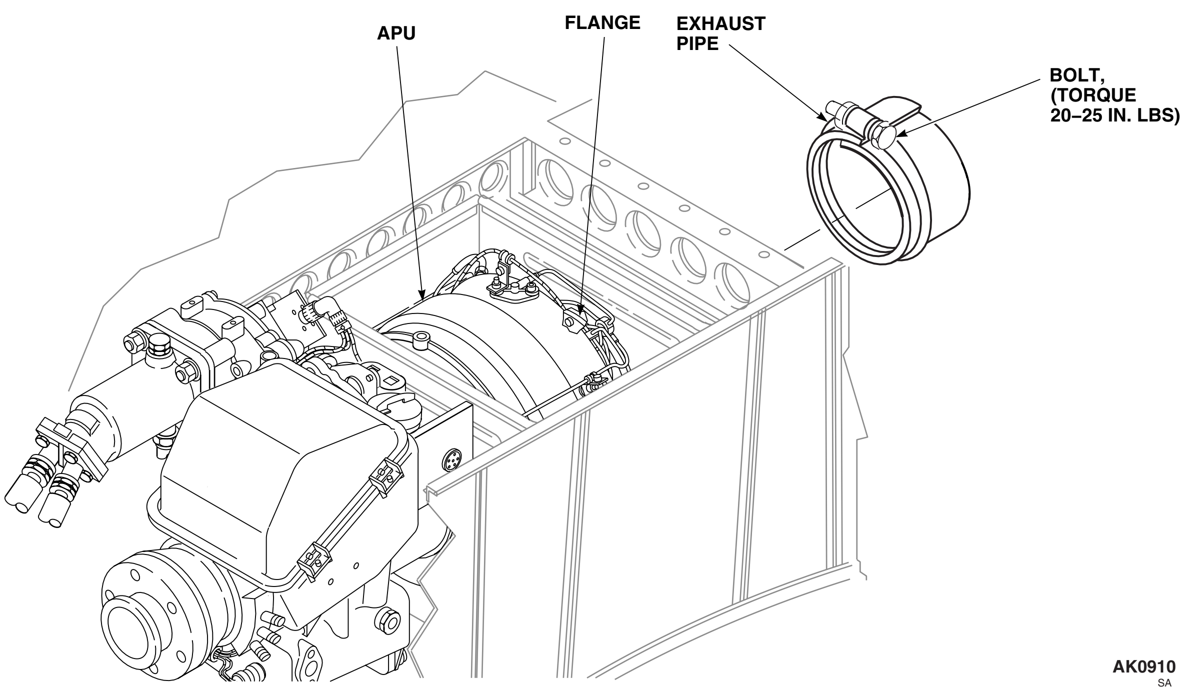


Figure 1. Replace APU Exhaust Pipe.

INSTALL

1. Turn off all electrical power.
2. Coat internal beaded surface of exhaust pipe with antiseize compound, Item 49, WP 1803 00.
3. Pass exhaust pipe up through exhaust duct. Install pipe on APU flange. **TORQUE CLAMP BOLT TO 20 - 25 INCH-POUNDS (Figure 1).**
4. **QA** Tap around clamp with nonmetallic mallet. **CHECK CLAMP BOLT TORQUE FOR 20 - 25 INCH-POUNDS.**
5. Continue tapping and checking clamp bolt torque until two successive trials show no loss of torque.
6. Make sure area is clean and free of foreign material.
7. Close APU compartment access doors (WP 0303 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AUXILIARY POWER PLANT SYSTEM

APU SHROUD ASSEMBLY

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Material/Parts

Adhesive, Item 27, WP 1803 00
Wire, Nonelectrical Item 354, WP 1803 00
Wire, Nonelectrical Item 357, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

WP 1359 00
WP 1803 00

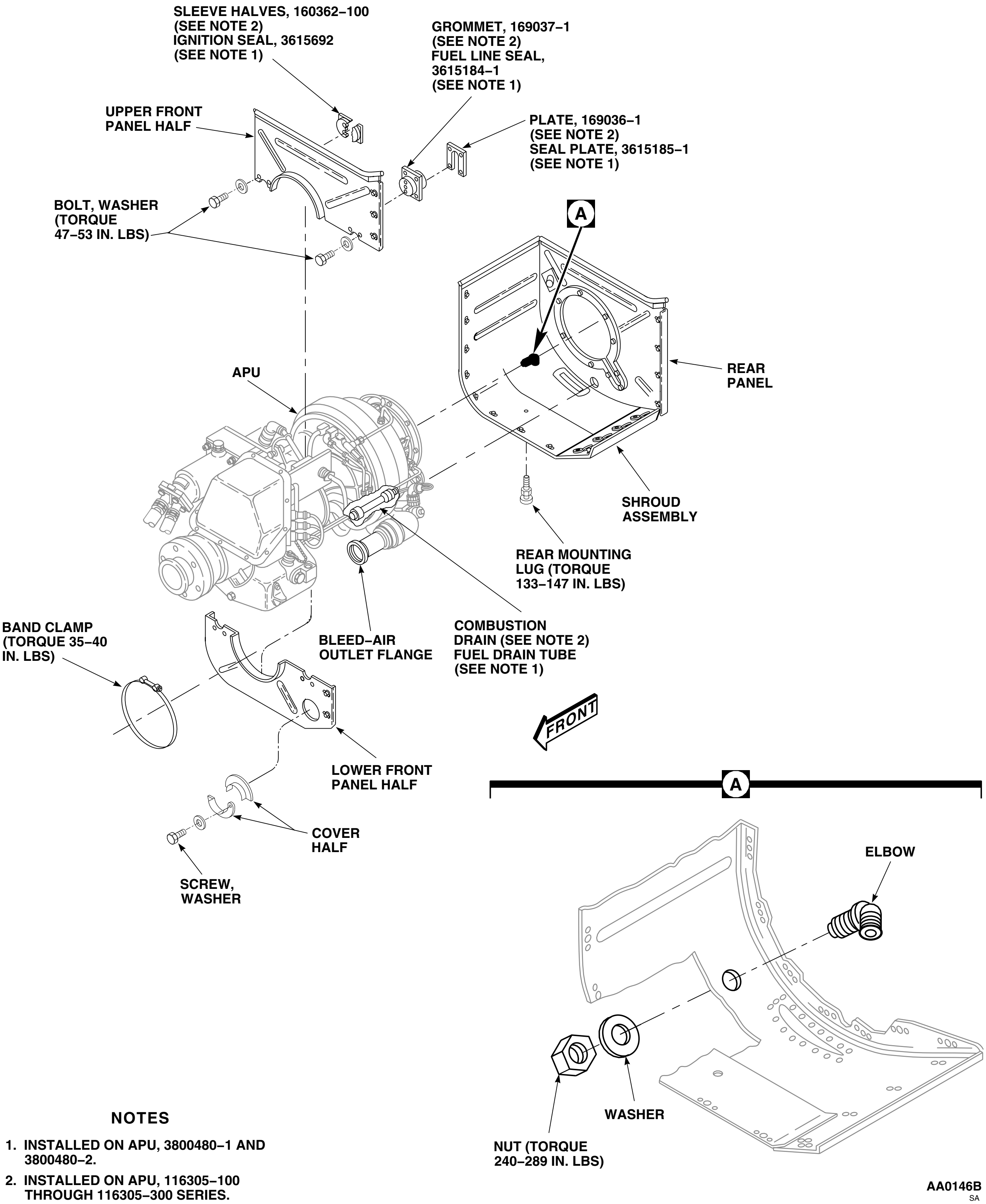
REMOVE

1. Turn off all electrical power.
2. Remove APU from helicopter (WP 1359 00).
3. On APU, 116305-100 through 116305-300 series, remove bolts, washers, firewall sleeves, grommet, and plate from front panel halves (Figure 1).
4. On APU, 3800480-1 and 3800480-2, remove bolts, washers, ignition seal, fuel line seal, and seal plate from front panel halves (Figure 1).
5. Remove band clamp holding upper and lower front panel halves together.
6. Remove screws holding cover halves around bleed-air outlet flange. Remove cover halves from lower front panel half.
7. Remove screws holding front panel halves to shroud assembly. Remove front panel halves.
8. On APU, 116305-100 through 116305-300 series, remove lockwire from combustor drain in rear panel.
9. Remove rear mounting lug from compressor.
10. On APU, 116305-100 through 116305-300 series, remove shroud assembly surrounding APU, carefully sliding off combustor drain.
11. On APU, 3800480-1 and 3800480-2, remove shroud assembly surrounding APU hot section, carefully sliding off fuel drain tube.
12. Remove elbow, washer, and nut from shroud assembly (Figure 1, Detail A).

INSTALL

1. Turn off all electrical power.
2. **QA** Install elbow in shroud assembly, facing forward, with washer, and nut (Figure 1, Detail A). **TORQUE NUT TO 240 - 289 INCH-POUNDS.**
3. **QA** Install shroud assembly around APU (Figure 1).
4. **QA** Install rear mounting lug in APU compressor. **TORQUE LUG TO 133 - 147 INCH-POUNDS.** Using nonelectrical wire, Item 354, WP 1803 00, lockwire lug to shroud assembly.

INSTALL - Continued



NOTES

1. INSTALLED ON APU, 3800480−1 AND 3800480−2.
2. INSTALLED ON APU, 116305−100 THROUGH 116305−300 SERIES.

Figure 1. Replace APU Shroud Assembly.

INSTALL - Continued

5. **QA** On APU, 116305-100 through 116305-300 series, using nonelectrical wire, Item 354, WP 1803 00, lockwire combustor drain to cover on rear panel.
6. Seal shroud assembly with adhesive, Item 27, WP 1803 00.

CAUTION

Damage will occur if ignition lead, thermocouple lead, and wiring for valves and switches are installed incorrectly. To prevent chafing and pinching when panel halves are installed take extreme care when installing panel halves.

7. **QA** Install upper and lower front panels on APU shroud with washers and screws.
8. **QA** Install band clamp around front panel halves. **TORQUE BAND CLAMP TO 35 - 40 INCH-POUNDS.**
9. **QA** On APU, 116305-100 through 116305-300 series, install ignitor leads and electrical wiring inside ignition seal. Install ignition seal to front panel halves with bolts and washers. **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.** Using nonelectrical wire Item 357, WP 1803 00, lockwire bolts together. Lockwire bolts only on right side of front panel (looking forward).
10. **QA** On APU, 3800480-1 and 3800480-2, install ignitor leads and electrical wiring inside ignition seal. Install ignition seal to front panel halves with bolts and washers. **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire bolts together. Lockwire bolts only on right side of front panel (looking forward).
11. **QA** On APU, 116305-100 through 116305-300 series, install grommet and plate over tube bundle. Install grommet and plate to front panel halves with bolts and washers. **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.**
12. **QA** On APU, 3800480-1 and 3800480-2, install fuel line seal and seal plate over fuel tube. Install fuel line seal and seal plate to front panel halves with bolts and washers. **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.**
13. **QA** Place cover halves around bleed-air outlet flange, and install on lower front panel with washers and screws.
14. Install APU in helicopter (WP 1359 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AUXILIARY POWER PLANT SYSTEM

APU FRONT MOUNTING LUGS

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Hoisting Sling, 70700-20323-042
Torque Wrench, 30 - 200 in. lbs.

Material/Parts
Wire, Nonelectrical Item 357, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0303 00
WP 1359 00
WP 1803 00

NOTE

Replacement of the front mounting lugs is the same for both right and left sides.

REMOVE

1. Turn off all electrical power.
2. Open APU compartment access doors (WP 0303 00).
3. Place wood block under APU sump or use hoisting sling to support APU (WP 1359 00).
4. Remove bolt and washer holding forward mounting lug to front support mount (Figure 1, Detail A).
5. Remove bolts and washers holding mounting lug to APU gear box sump, and remove mounting lug.

INSTALL

1. Turn off all electrical power.
2. **QA** Install front mounting lug (TOP side up) to APU gear box sump with washers and bolts (Figure 1, Detail A). **TORQUE BOLTS TO 104 - 116 INCH-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire bolts together.
3. Remove wood block or hoisting sling supporting APU (WP 1359 00).
4. **QA** Install front mounting lugs to front supports with washers and bolts. **TORQUE BOLTS TO 104 - 116 INCH-POUNDS.**
5. Make sure area is clean and free of foreign material.
6. Close and latch APU compartment access doors (WP 0303 00).

INSTALL - Continued

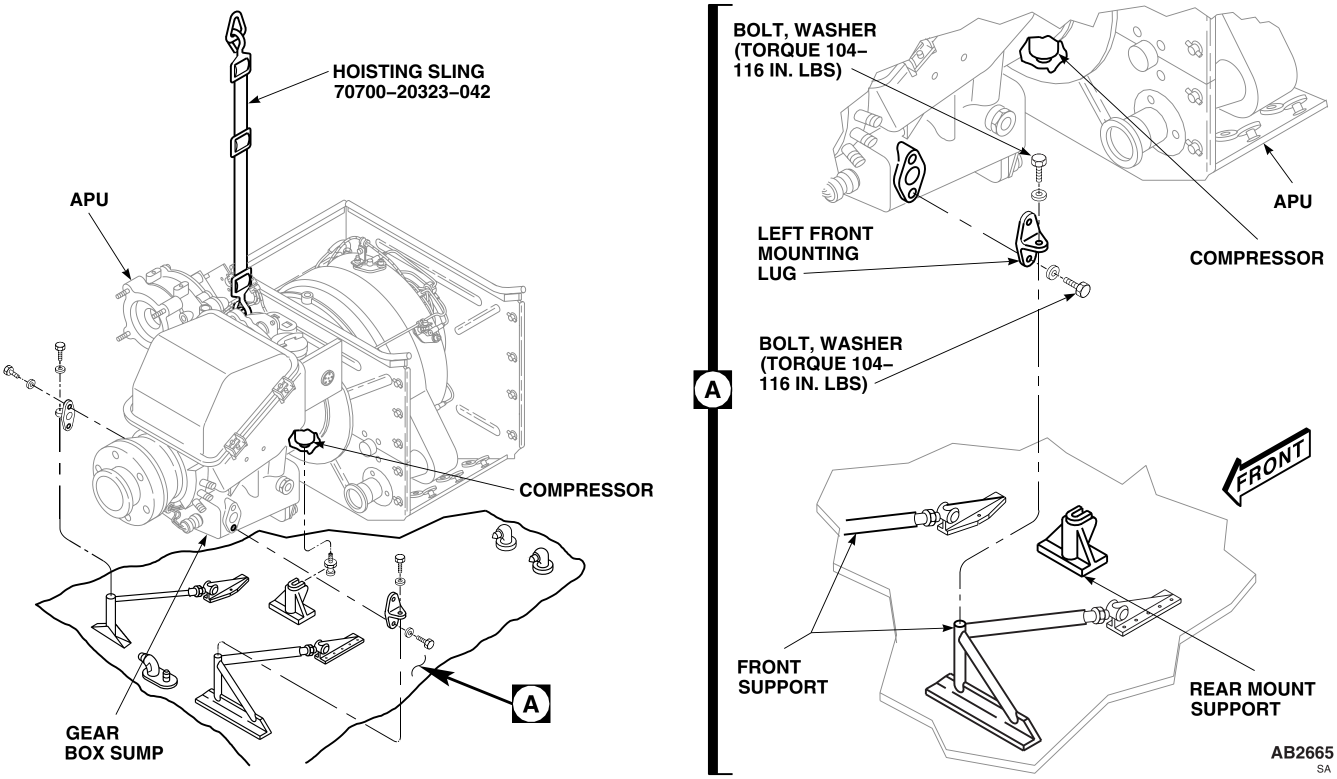


Figure 1. Replace APU Front Mounting Lugs.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AUXILIARY POWER PLANT SYSTEM

APU REAR MOUNTING LUG

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 30 - 200 in. lbs.

Material/Parts
Wire, Nonelectrical Item 354, WP 1803 00

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0303 00
WP 1359 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Open APU compartment access doors (WP 0303 00).
3. Remove APU (WP 1359 00).
4. Loosen and remove rear mounting lug from APU compressor (Figure 1, Detail A).

INSTALL

1. Turn off all electrical power.
2. **QA** Install rear mounting lug, in APU compressor (Figure 1, Detail A). **TORQUE LUG TO 133 - 147 INCH-POUNDS.** Using nonelectrical wire, Item 354, WP 1803 00, lockwire lug to shroud assembly.
3. Install APU (WP 1359 00).
4. Make sure area is clean and free of foreign material.
5. Close APU compartment access doors (WP 0303 00).

INSTALL - Continued

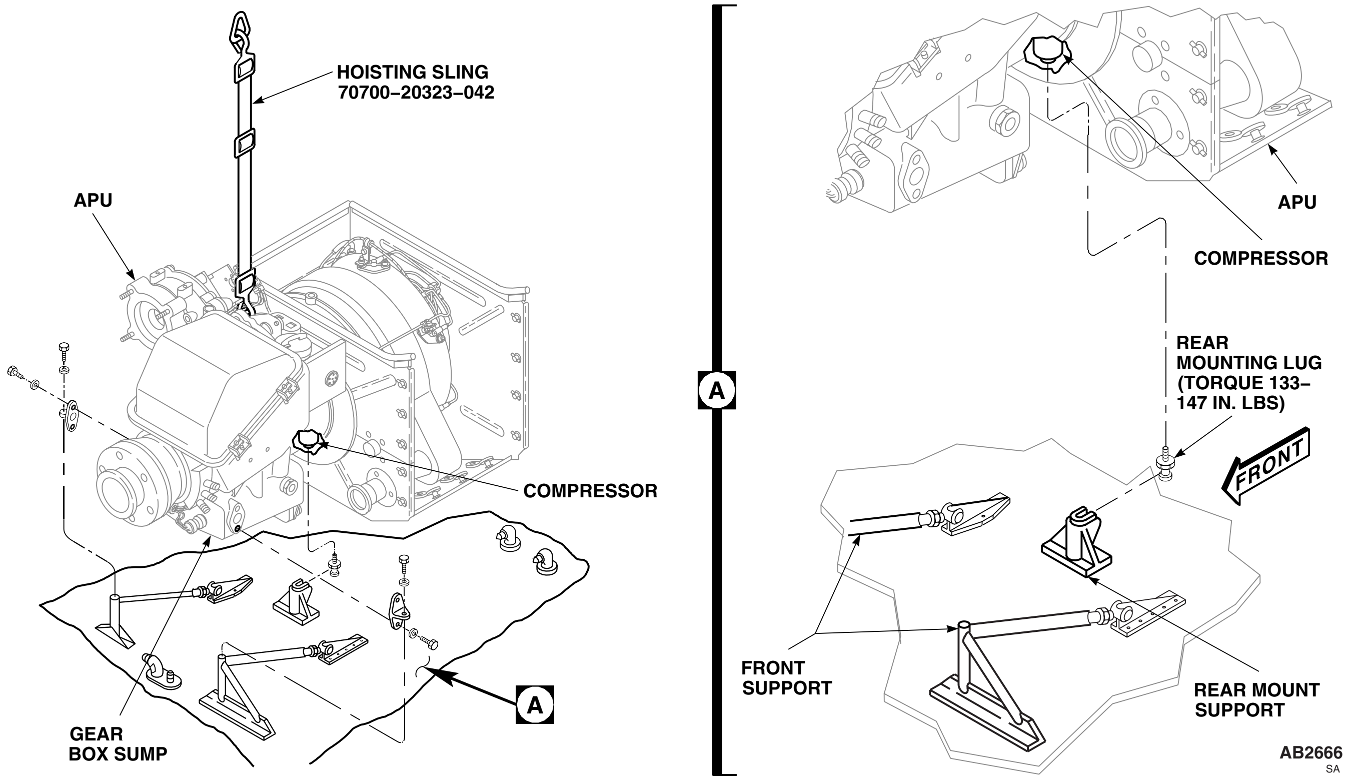


Figure 1. Replace APU Rear Mounting Lug.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AUXILIARY POWER PLANT SYSTEM

APU START MOTOR

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Micrometer, 0 to 1 Inch
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.
Torque Wrench, 300 - 2500 in. lbs.

Material/Parts

Grease, Molybdenium Disulfide Item 138, WP 1803 00
Hydraulic Fluid, Fire Resistant Item 144, WP 1803 00
Hydraulic Fluid, Petroleum Base Item 145, WP 1803 00
Petrolatum, Item 216, WP 1803 00
Protective Caps and Plugs
Tags
Wire, Nonelectrical Item 353, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0155 00
WP 0156 00
WP 0303 00
WP 0691 00
WP 1605 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Depressurize APU accumulator (WP 1605 00).

WARNING

Injury to personnel and damage to equipment will result if APU start valve is operated with personnel working in area of APU. Tag APU start valve warning personnel not to operate. Do not operate APU start valve when start motor is being replaced.

3. Open APU compartment access doors (WP 0303 00).
4. Remove hoses from PRESS and RETURN ports on start motor. Cap ports on motor and plug hoses (Figure 1, Detail A).
5. Disconnect drain tube from elbow on start motor. Cap elbow and plug drain tube.
6. Remove nuts, washers, start motor, and gasket from APU.
7. Remove jamnut, elbow, retainer, and packing from start motor. If plug interferes with fuel control unit cover, replace with low profile plug. Discard packing.

INSPECT

1. Inspect start motor coupling shaft for spline wear.
   a. Place 0.096-inch diameter pin between two splines.
   b. Place a second 0.096-inch diameter pin between two splines directly opposite first pin. Secure pins.
   c. Measure outside diameter of shaft splines, using micrometer, over pins. **OUTSIDE DIAMETER MUST NOT BE LESS THAN 0.735-INCH.** Replace start motor if coupling shaft is worn beyond limits.

INSPECT - Continued

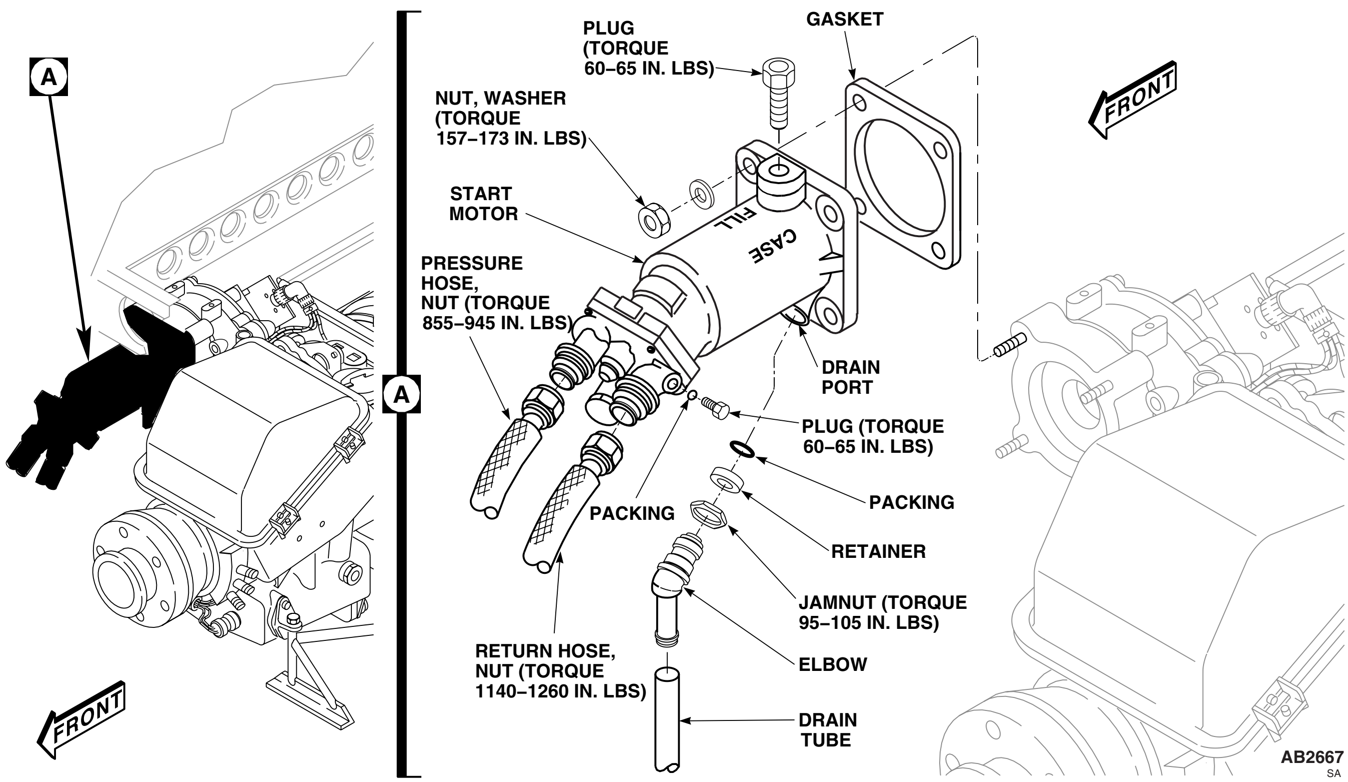


Figure 1. Replace APU Start Motor.

d. Repeat check with pins located in four different positions.

2. See WP 0691 00 for shaft seal leakage limits.

INSTALL

1. Turn off all electrical power.
2. Lubricate packing, retainer, and elbow threads with petrolatum, Item 216, WP 1803 00.
3. **QA** Install jamnut, retainer, and packing, onto elbow. Turn jamnut and retainer down until packing is pushed firmly against the lower threaded section of elbow (Figure 1, Detail A) .
4. **QA** Install elbow in start motor until a sharp rise in torque is felt while keeping jamnut from turning. **TIGHTEN ELBOW AND JAMNUT 1 1/2 TURNS PAST POINT WHERE SHARP RISE IN TORQUE IS FELT. TIGHTEN ELBOW NO MORE THAN 1 ADDITIONAL TURN TO POSITION ELBOW FORWARD.**
5. **QA TORQUE JAMNUT TO 95 - 105 INCH-POUNDS WHILE KEEPING ELBOW FROM TURNING.**

NOTE

For arctic operation, -29°F(-34°C), use petroleum base hydraulic fluid, Item 145, WP 1803 00.

6. Remove plug from CASE FILL port on start motor. Fill motor completely with fire resistant hydraulic fluid, Item 144, WP 1803 00. Install plug in CASE FILL port. **TORQUE PLUG TO 60 - 65 INCH-POUNDS.** Using nonelectrical wire Item 353, WP 1803 00, lockwire plug to starter motor body.
7. **QA** Install hydraulic start motor gasket, on hydraulic start motor mounting pad.

INSTALL - Continued

8. Lubricate motor coupling shaft with molybdenium disulfide grease, Item 138, WP 1803 00.
9. **QA** Install start motor on APU with washers and nuts. **TORQUE NUTS TO 157 - 173 INCH-POUNDS.**
10. Remove cap and plug and connect drain tube to elbow on start motor.
11. Remove caps from PRESS and RETURN ports on start motor.
12. **QA** Remove plug and connect pressure hose to PRESS port. **TORQUE PRESSURE HOSE NUT TO 855 - 945 INCH-POUNDS.**
13. **QA** Remove plug and connect return hose to RETURN port. **TORQUE RETURN HOSE NUT TO 1140 - 1260 INCH-POUNDS.**
14. Remove tag from APU start valve.
15. Service APU accumulator (WP 1605 00).
16. Make sure area is clean and free of foreign material.
17. Close and latch APU compartment door (WP 0303 00).
18. Do an operational check of APU (WP 0155 00 and WP 0156 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AUXILIARY POWER PLANT SYSTEM

APU SPLINE ADAPTER

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Powertrain Repairer's Toolkit, SC 5180-99-B13
Torque Wrench, 30 - 200 in. lbs.
Vespel Spline Adapter Installation Tool, 1106841-1
Vespel Spline Adapter Withdrawal Tool, 1106769-4

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0842 00

REMOVE

1. Turn off all electrical power.
2. Loosen bolt on QAD mount clamp and remove generator (WP 0842 00) from mount (Figure 1).
3. Move and support generator only far enough to get to spline adapter.

CAUTION

- Spline adapter is press fit and is difficult to remove and install. Use of hammer or excessive force to remove or install spline adapter could cause ring gear separation and damage APU. Do not use hammer or excessive force to remove or install APU spline adapter.
- Damage will occur if wrong spline adapter removal tool is used. Make sure removal tool is matched with appropriate APU.
- Do not reinstall a polymide spline adapter. Replace spline adapter, even if partially inserted and removed.

4. Remove spline adapter using vespel spline adapter withdrawal tool, 1106769-4.

INSPECT

1. Inspect spline in output drive gear for damage. **NONE ALLOWED.** If splines in output drive gear are damaged, replace APU.
2. Inspect output drive gear for any broken pieces of spline adapter. Remove any broken pieces of spline adapter using machinist scribe.

INSTALL

1. Turn off all electrical power.

INSTALL - Continued

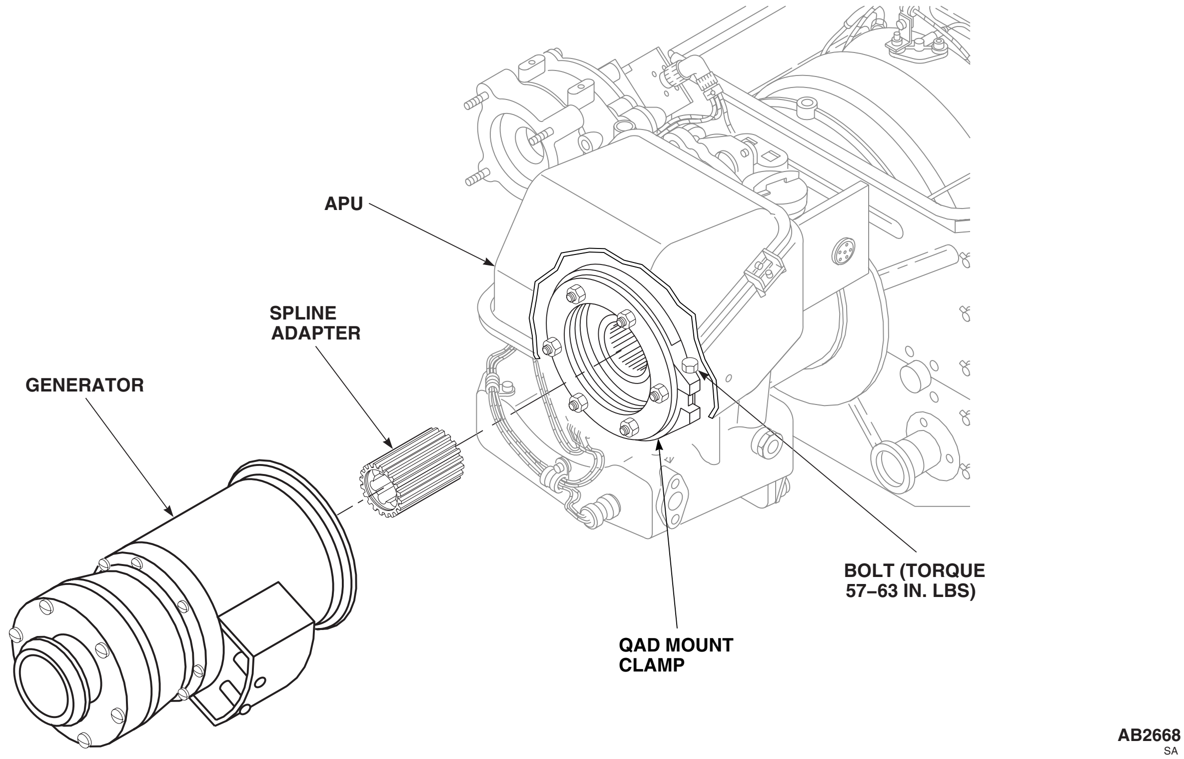


Figure 1. Replace APU Spline Adapter.

CAUTION

- Spline adapter is press fit and is difficult to remove and install. Use of hammer or excessive force to remove or install spline adapter could cause ring gear separation and damage APU. Do not use hammer or excessive force to remove or install APU spline adapter.
- Do not reinstall a polymide spline adapter. Replace spline adapter, even if partially inserted and removed.

2. **QA** Press tapered end of spline adapter onto output drive gear splines with Vespel Spline Adapter Installation Tool, 1106841-1.
3. Recess face of spline adapter no more than 1/8-inch from APU output drive spline front face.
4. **QA** Position generator on mount and secure with QAD mount clamp. **TORQUE CLAMP BOLT 57 - 63 INCH-POUNDS (Figure 1).**
5. Make sure area is clean and free of foreign material.
6. Install APU generator (WP 0842 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AUXILIARY POWER PLANT SYSTEM

APU FUEL LINE PRIME

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 30 - 200 in. lbs.

Material/Parts
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0303 00
WP 1685 00
WP 1803 00

APU FUEL LINE PRIME

1. Turn off all electrical power.
2. Open APU compartment access doors (WP 0303 00).
3. Loosen stud fasteners holding fuel control upper cover to fuel control lower cover. Remove fuel control upper cover.

CAUTION

Damage to fuel filter will result if union in fuel control lower cover turns while loosening fuel inlet hose. Hold union with a wrench while loosening fuel inlet hose to avoid damage to fuel filter.

4. Holding union inside fuel control lower cover with a wrench, loosen nut on fuel inlet hose to allow fuel to leak when priming fuel inlet hose (Figure 1, Detail A).
5. Place BATT switch on upper console ON, or connect external electrical power to helicopter (WP 1685 00).

WARNING

To prevent injury to personnel or damage to equipment, make sure APU T-handle on upper console is not pulled out.

6. **QA** Hold machinery wiping towel, Item 344, WP 1803 00, around fuel hose nut. Have assistant place FUEL PUMP switch on upper console in cockpit to APU BOOST. When fuel leaks from fuel hose nut, tighten nut and place FUEL PUMP switch OFF. **TORQUE NUT TO 110 - 130 INCH-POUNDS.**
7. Place BATT switch on upper console OFF, or disconnect external electrical power from helicopter (WP 1685 00).
8. Install fuel control upper cover.
9. Make sure area is clean and free of foreign material.
10. Close APU compartment access doors (WP 0303 00).

APU FUEL LINE PRIME - Continued

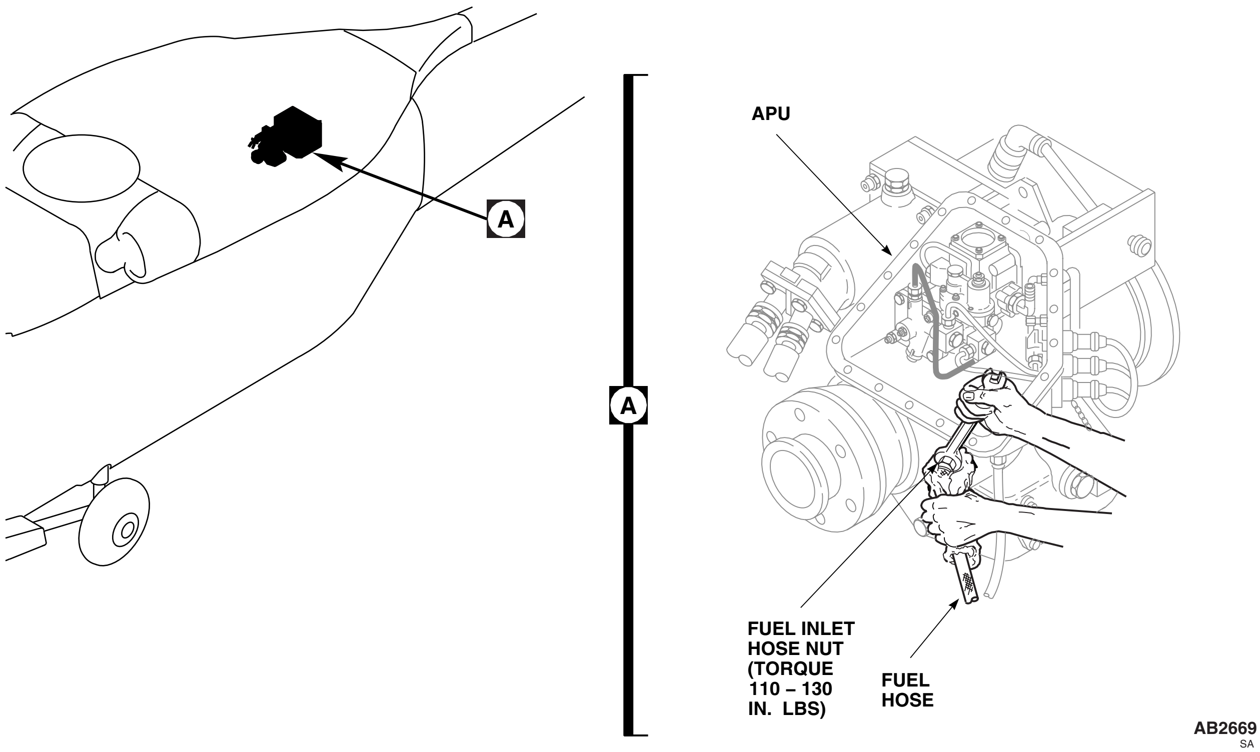


Figure 1. APU Fuel Line Prime.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AUXILIARY POWER PLANT SYSTEM

APU FUEL SHUTOFF VALVE

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 30 - 200 in. lbs.

Material/Parts

Antiseize Compound, Item 49, WP 1803 00
Protective Caps and Plugs
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0129 00
WP 0233 00
WP 0258 00
WP 0303 00
WP 1367 00
WP 1703 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. **UH60A UH60L** Gain access to fuel cell area as follows:
   a. Remove troop seats from rear cabin (WP 0233 00).
   b. Remove soundproofing panel to access area above No. 1 fuel cell (WP 0258 00).
   c. Unhook cargo netting to get into compartment.
3. **EH60A** Gain access to fuel cell area as follows:
   a. Remove soundproofing panel to access area above No. 1 fuel cell.
   b. If installed, remove access panel on right panel of fuel enclosure panel.
4. Unlock stud fasteners holding right panel enclosing fuel system components on top of fuel cells. Remove panel from compartment.
5. Disconnect electrical connector, P306, from fuel shutoff valve (Figure 1).
6. Disconnect fuel hose from fuel shutoff valve. Cap shutoff valve and plug hose.
7. Open APU compartment access doors (WP 0303 00).
8. Place machinery wiping towels, Item 344, WP 1803 00, on transition deck around fuel hose. Disconnect fuel hose from shutoff valve union protruding through deck. Plug fuel hose.
9. With assistant inside helicopter holding fuel shutoff valve, remove nut, insulated washer, and valve from transition deck.

INSTALL

1. Turn off all electrical power.

INSTALL - Continued

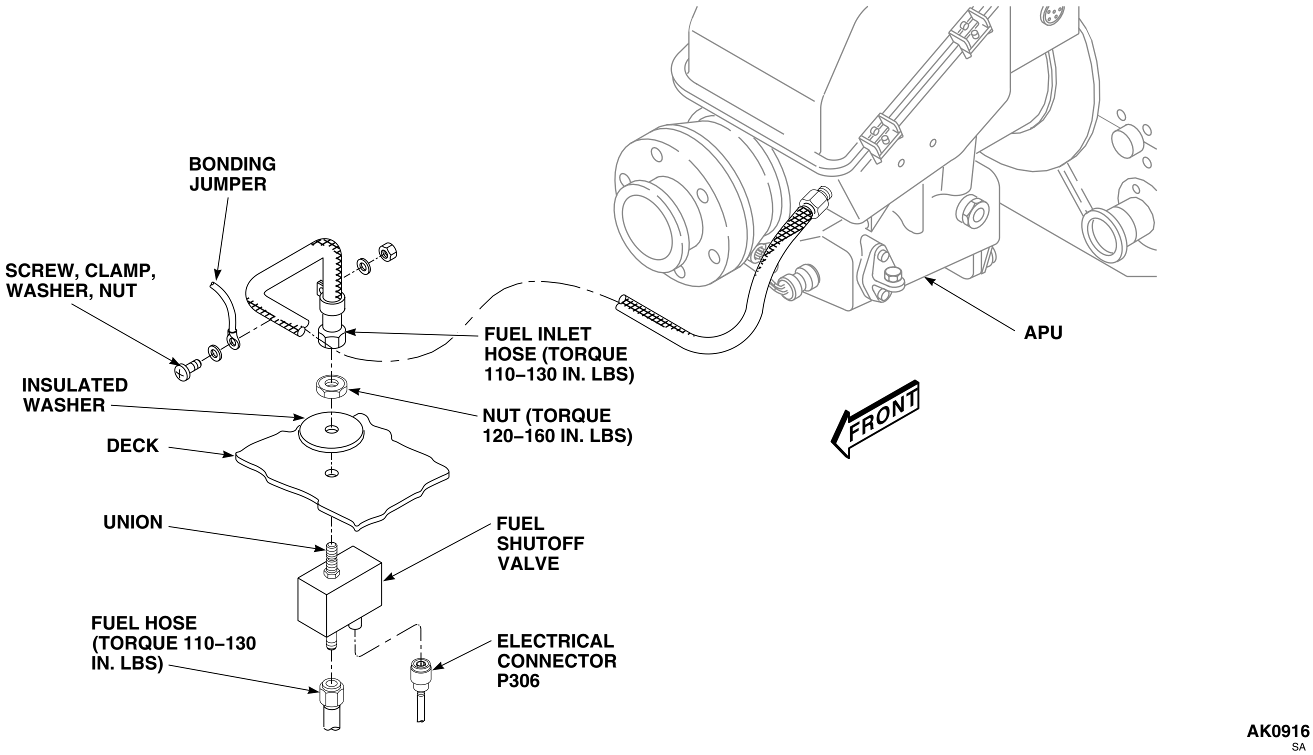


Figure 1. Replace APU Fuel Shutoff Valve.

2. **QA** Install fuel shutoff valve to transition deck with insulated washer and nut, using assistant to hold valve in place. **TORQUE NUT TO 120 - 160 INCH-POUNDS (Figure 1).**
3. Coat threads on fuel shutoff valve unions with antiseize compound, Item 49, WP 1803 00.
4. **QA** Remove plug from fuel hose and connect to fuel shutoff valve union. **TORQUE NUT TO 110 - 130 INCH-POUNDS.**
5. Inspect and treat electrical connector (WP 1703 00).
6. **QA** Connect electrical connector, P306, to shutoff valve.
7. Prime fuel inlet hose and check for leaks (WP 1367 00).
8. Do an operational check of fuel prime boost system (WP 0129 00).
9. Remove machinery wiping towels from APU compartment.
10. Make sure APU compartment area is clean and free of foreign materials.
11. Close and latch APU compartment access doors (WP 0303 00).
12. Make sure fuel cell area is clean and free of foreign material.
13. **UH60A UH60L** Do this:
    a. Hook cargo netting in stowage compartment access opening.
    b. Install soundproofing panel (WP 0258 00).
    c. Install troop seats (WP 0233 00).

INSTALL - Continued

14. **EH60A** Do this:

    a. Install access panel on right fuel enclosure panel.

    b. Install right panel on rear cabin bulkhead.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AUXILIARY POWER PLANT SYSTEM

APU FUEL LINE

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 30 - 200 in. lbs.

Material/Parts
Protective Caps and Plugs
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 0303 00
WP 1367 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Open APU compartment access doors (WP 0303 00).
3. Remove bonding jumper from fuel inlet hose (Figure 1, Detail A).
4. Loosen stud fasteners, holding fuel control upper cover to fuel control lower cover. Remove fuel control upper cover.

CAUTION

Damage to fuel filter will result if union in fuel control lower cover turns when removing fuel inlet hose. Hold union with a wrench while removing fuel inlet hose to avoid damage to fuel filter.

5. Place machinery wiping towels, Item 344, WP 1803 00, on transition deck around fuel hose.
6. Disconnect fuel inlet hose from union in fuel control lower cover. Cap union in lower cover and plug hose.
7. Disconnect fuel inlet hose from shutoff valve union protruding through deck. Cap fuel shutoff valve union and plug hose.

INSTALL

1. Turn off all electrical power.
2. **QA** Remove cap and plug and connect fuel inlet hose to fuel shutoff valve union. **TORQUE NUT TO 110 - 130 INCH-POUNDS (Figure 1, Detail A).**
3. Remove cap and plug and loosely install fuel inlet hose to union in fuel control lower cover.
4. Prime APU fuel inlet hose (WP 1367 00).
5. Check for leaks.
6. **QA** Install fuel control upper cover, and tighten stud fasteners.

INSTALL - Continued

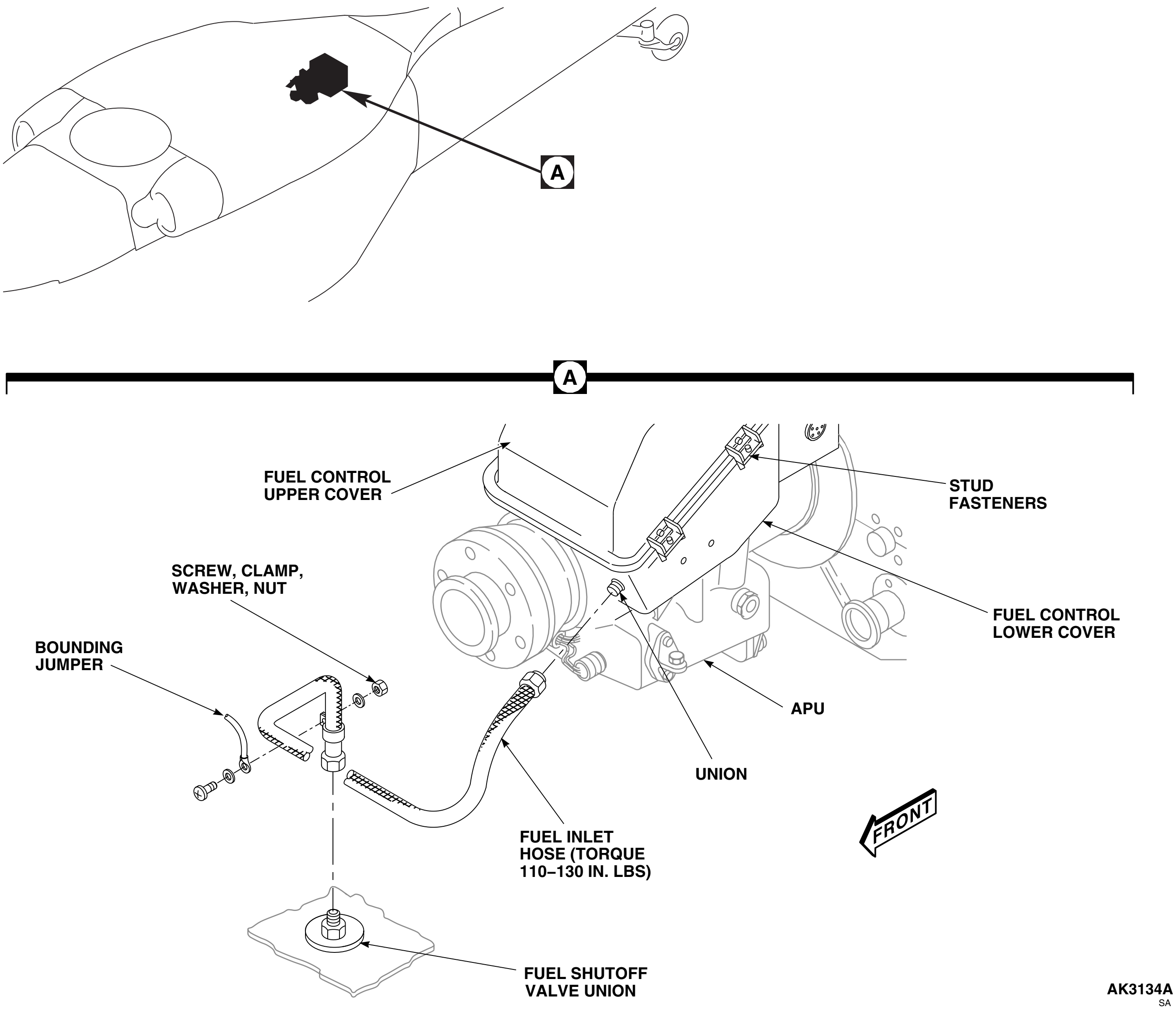


Figure 1. Replace APU Fuel Line.

7. **QA** Install bonding jumper on fuel inlet hose with clamp, screw, washers, and nut.
8. Make sure area is clean and free of foreign material.
9. Close APU compartment access doors (WP 0303 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AUXILIARY POWER PLANT SYSTEM

APU DRAIN LINES

This WP supersedes WP 1370 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Gloves
Goggles/Face Shield
Torque Wrench, 30 - 200 in. lbs.

Material/Parts
Acetone, Technical, Item 5, WP 1803 00

Personnel Required
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1500-204-23
WP 0233 00
WP 0258 00
WP 0303 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Open APU compartment access doors (WP 0303 00).
3. **UH-60A UH-60L** Gain access to fuel cell area as follows:
   a. Remove troop seats from rear cabin (WP 0233 00).
   b. Remove soundproofing panels to access area above No. 1 fuel cell (WP 0258 00).
   c. Unhook cargo netting.
4. **EH-60A** Open transition section avionics door.
5. Remove clamps, screws, washers, and nuts holding drain tube to rigid drain tube in cabin overhead. Leave clamps installed on rigid drain tube (Figure 1, Detail A).
6. In transition section remove clamps, screws, washers, and nuts holding drain tube to rigid drain tube on left side of bulkhead. Leave clamps installed on rigid drain tube.
7. Disconnect drain tube from elbow in cabin ceiling and from front of fuel cell bulkhead and remove from helicopter.
8. Disconnect drain tube from rear of bulkhead elbow and from grommet. Remove drain tube from helicopter.
9. Remove APU start motor drain tube from elbow in APU compartment.
10. Remove jamnut, washer, sealing washer and elbow from cabin ceiling.
11. Remove jamnut, washer, and elbow from bulkhead.
12. Check drain line identification tapes for condition. Replace missing or unreadable tapes, as required (TM 1-1500-204-23).
13. Check grommet for cracks and wear. Replace if damaged.

REMOVE - Continued

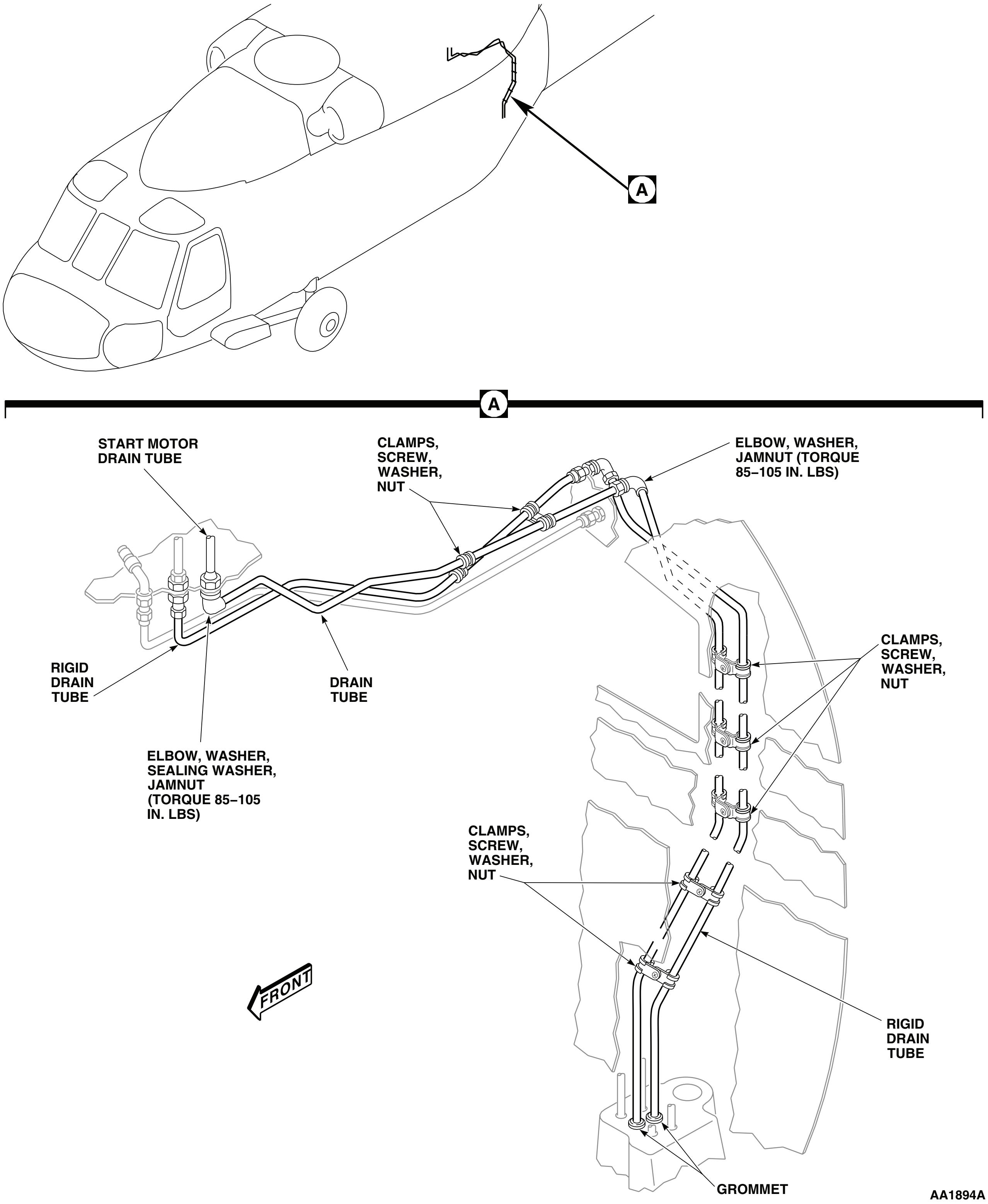


Figure 1. Replace APU Drain Lines.

INSTALL

1. Turn off all electrical power.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

2. Clean fittings and area around fittings with technical acetone, Item 5, WP 1803 00.
3. **QA** Install elbow, washer, sealing washer, and jamnut in cabin ceiling. Position elbow facing aft. **TORQUE JAMNUT TO 85 - 105 INCH-POUNDS (Figure 1, Detail A).**
4. **QA** Install start motor drain tube on elbow in APU compartment.
5. **QA** Install elbow, washer, and jamnut in bulkhead. Position elbow facing down and outboard. Do not torque jamnut at this time.
6. **QA** In transition section, install drain tube to elbow and through grommet in floor. Attach to existing clamps on rigid drain tube with clamps, screws, washers, and nuts down left side of bulkhead.
7. **QA TORQUE JAMNUT ON REAR BULKHEAD ELBOW TO 85 - 105 INCH-POUNDS.**
8. In cabin, install drain tube through ceiling beam and connect to elbows. Attach to existing clamps on rigid drain tube with clamps, screws, washers, and nuts.
9. **UH-60A UH-60L** Do this:
   a. Hook cargo netting in stowage compartment access opening.
   b. Install soundproofing panels (WP 0258 00).
   c. Install troop seats (WP 0233 00).
10. **EH-60A** Close transition section avionics door.
11. Close APU compartment access doors (WP 0303 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AUXILIARY POWER PLANT SYSTEM

APU FRONT INBOARD SUPPORT MOUNT

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Blind Rivet Gun Kit, 35351
Hoisting Sling, 70700-20323-042
Torque Wrench, 30 - 200 in. lbs.

Material/Parts
Sealing Compound, Item 280, WP 1803 00

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 1-1500-204-23
TM 55-1500-345-23
WP 0233 00
WP 0258 00
WP 0303 00
WP 0377 00
WP 0381 00
WP 1312 00
WP 1359 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Open APU compartment access doors (WP 0303 00).
3. Remove bolt and washer holding forward mounting lug to front inboard support mount (Figure 1, Detail A).
4. Place wood block under APU sump, or use hoisting sling to support APU (WP 1359 00).
5. Remove bolt, washer, and nut holding front inboard mount tube to bracket.
6. Remove cotter pin, nut, washer, and clevis bolt holding clevis end of mount tube to tee assembly.
7. Remove front inboard mount tube assembly.
8. **UH60A UH60L** Remove rear troop seats (WP 0233 00).
9. Remove soundproofing panel from left stowage compartment (WP 0258 00).
10. **UH60A UH60L** Remove cargo netting.
11. **EH60A** Remove condenser pallet (WP 1312 00).
12. Position assistant inside rear compartment to help remove and install fasteners.
13. Remove pin and collar assemblies holding bracket to airframe (TM 1-1500-204-23).
14. Remove bracket and radius block.
15. Remove rivets, locking swage pin and collar assemblies, and pin and collar assemblies holding tee assembly to airframe (TM 1-1500-204-23).
16. Remove tee assembly and radius blocks.
17. Touch up damaged protective finishes (WP 0381 00 and TM 55-1500-345-23).

REMOVE - Continued

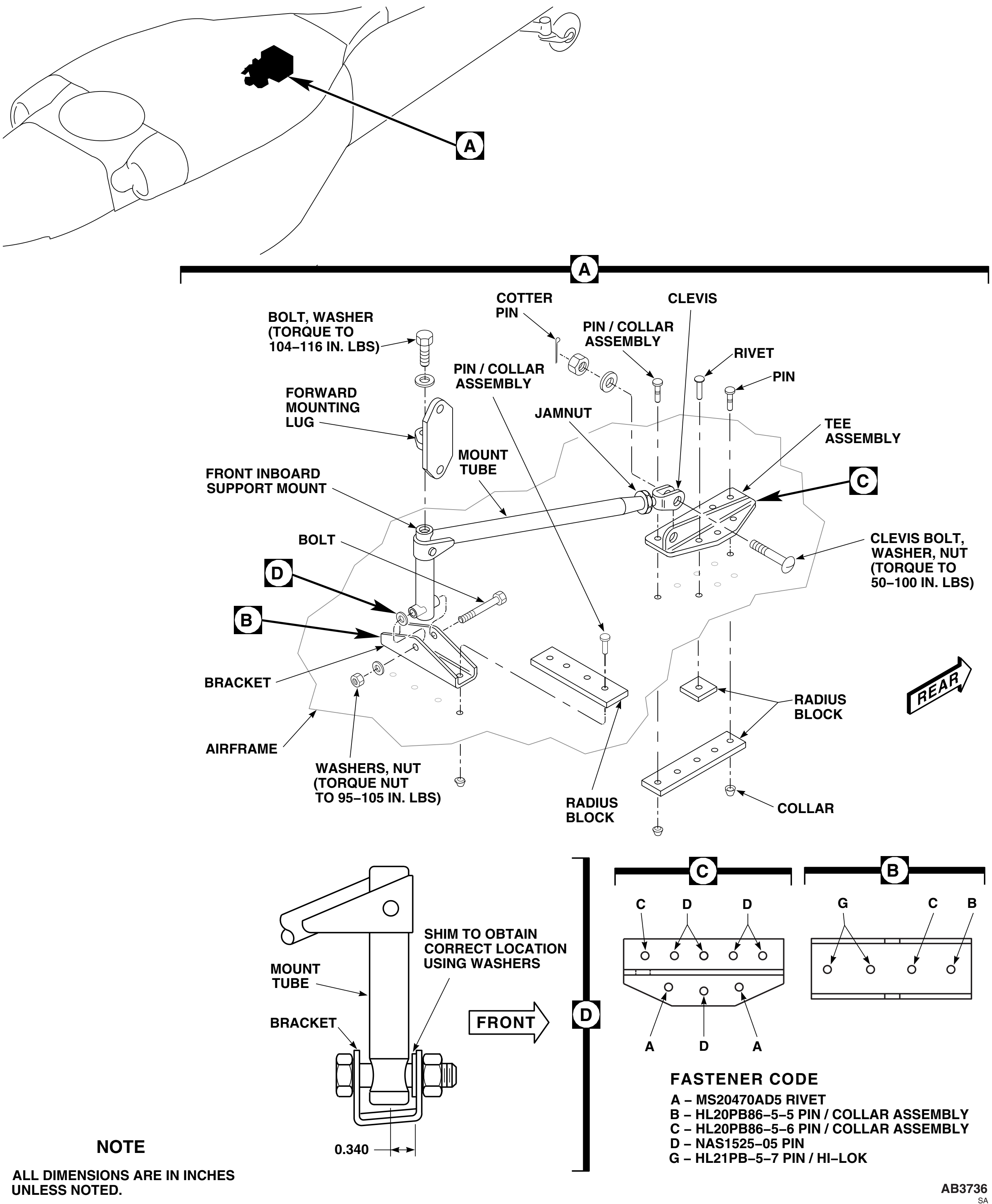


Figure 1. Replace APU Front Inboard Support Mount.

INSTALL

1. Turn off all electrical power.
2. Install all tee and bracket fasteners with sealing compound, Item 280, WP 1803 00 (WP 0377 00).
3. Position replacement bracket and radius block on airframe (Figure 1, Detail A).
4. **QA** Install required pin and collar assemblies holding bracket and radius block (TM 1-1500-204-23).
5. Position tee assembly and radius blocks on airframe (TM 1-1500-204-23).
6. **QA** Install required rivets, locking swage pin and collar assemblies, and pin and collar assemblies holding tee assembly and radius blocks to airframe.
7. **QA** Install front inboard mount tube in bracket with bolt. Shim front mount tube using washers to get 0.340-inch between centerline of tube and front of bracket (Figure 1, Detail D).
8. **QA** Install front inboard mount tube in bracket with bolt, washer(s), and nut. **TORQUE NUT 95 - 105 INCH-POUNDS.**
9. **QA** Loosen jamnut and adjust clevis end of mount tube to line up with hole in tee assembly. Attach clevis end to tee assembly with clevis bolt, washer, nut. **TORQUE NUT 50 - 100 INCH-POUNDS.** Install cotter pin.
10. **QA TIGHTEN JAMNUT 1/6 - 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
11. Line up front inboard support mount with front mounting lug on APU. Remove wood block or hoisting sling supporting APU (WP 1359 00).
12. **QA** Install bolt and washer holding forward mounting lug to front inboard support mount. **TORQUE BOLT TO 104 - 116 INCH-POUNDS.**
13. Clean and seal for corrosion protection and waterproofing (WP 0377 00).
14. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).
15. Make sure area is clean and free of foreign material.
16. **EH60A** Install condenser pallet (WP 1312 00).
17. **UH60A UH60L** Install cargo netting.
18. Install soundproofing panel to left stowage compartment (WP 0258 00).
19. **UH60A UH60L** Install rear troop seats (WP 0233 00).
20. Close and lock APU compartment access doors (WP 0303 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AUXILIARY POWER PLANT SYSTEM

APU FRONT OUTBOARD SUPPORT MOUNT

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Blind Rivet Gun Kit, 35351
Hoisting Sling, 70700-20323-042
Torque Wrench, 30 - 200 in. lbs.

Material/Parts

Sealing Compound, Item 280, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
TM 55-1500-345-23
WP 0233 00
WP 0258 00
WP 0303 00
WP 0377 00
WP 0381 00
WP 1312 00
WP 1359 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Open APU compartment access doors (WP 0303 00).
3. Remove bolt and washer holding forward mounting lug to front outboard support mount (Figure 1).
4. Place wood block under APU sump, or use hoisting sling to support APU (WP 1359 00).
5. Remove nut, washer, and clevis bolt holding clevis end to tee assembly.
6. **UH60A UH60L** Remove rear troop seats (WP 0233 00).
7. Remove soundproofing panel from left stowage compartment (WP 0258 00).
8. **UH60A UH60L** Remove cargo netting.
9. **EH60A** Remove condenser pallet (WP 1312 00).
10. Position assistant inside rear compartment to help remove and install fasteners.
11. Remove rivets, pin and collar assemblies holding welded assembly and radius blocks to airframe (TM 1-1500-204-23).
12. Remove front outboard support mount and radius blocks.
13. Remove locking swage pin and collar assemblies holding tee assembly and radius blocks to airframe (TM 1-1500-204-23).
14. Remove tee assembly and radius blocks.
15. Touch up damaged protective finishes (WP 0381 00 and TM 55-1500-345-23).

INSTALL

1. Turn off all electrical power.

INSTALL - Continued

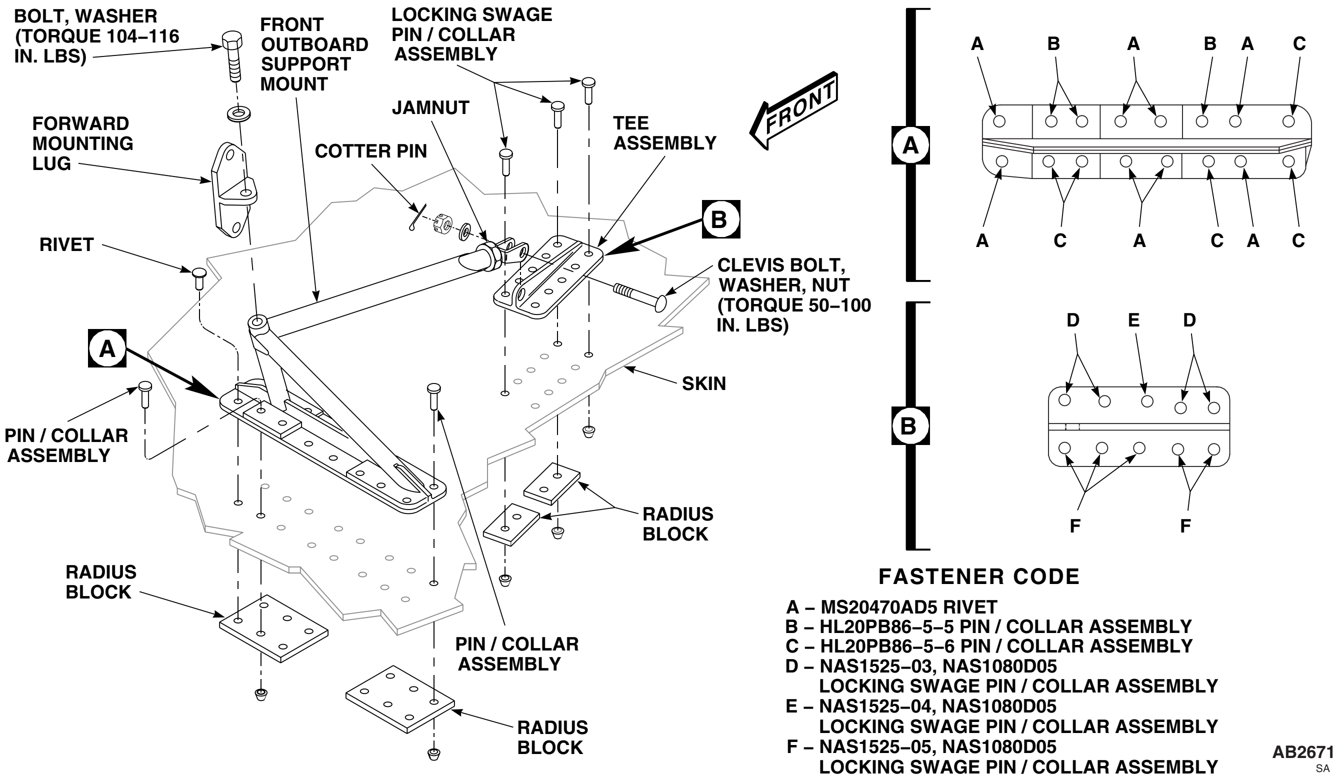


Figure 1. Replace APU Front Outboard Support Mount.

2. **QA** Install all tee assembly and front outboard support mount fasteners wet with sealing compound, Item 280, WP 1803 00 (WP 0377 00).
3. Position tee assembly and radius blocks on airframe (Figure 1).
4. **QA** Install required locking swage pin and collar assemblies holding tee and radius blocks to airframe (TM 1-1500-204-23).
5. Position front outboard support mount and radius blocks on airframe.
6. **QA** Install required pin and collars assemblies and rivets holding front outboard support mount and radius blocks to airframe (TM 1-1500-204-23).
7. **QA** Loosen jamnut and adjust clevis end of support mount to line up with hole in tee assembly. Attach clevis end to tee assembly with clevis bolt, washer, and nut. **TORQUE NUT TO 50 - 100 INCH-POUNDS.**Install cotter pin.
8. **QA TIGHTEN JAMNUT 1/6 - 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
9. Line up front outboard support mount with forward mounting lug on APU. Remove wood block or hoisting sling supporting APU (WP 1359 00).
10. **QA** Install bolt and washer holding forward mounting lug to front outboard support mount. **TORQUE BOLT TO 104 - 116 INCH-POUNDS.**
11. For corrosion protection and waterproofing, seal support mount parts to structure (WP 0377 00).
12. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).
13. Make sure area is clean and free of foreign material.

INSTALL - Continued

14. **EH60A** Install condenser pallet (WP 1312 00).
15. **UH60A UH60L** Install cargo netting.
16. Install soundproofing panel to left stowage compartment (WP 0258 00).
17. **UH60A UH60L** Install rear troop seats (WP 0233 00).
18. Close APU compartment access doors (WP 0303 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AUXILIARY POWER PLANT SYSTEM

APU FRONT INBOARD AND OUTBOARD SUPPORT MOUNTS REPAIR

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02

Material/Parts
Primer, Sealing Compound, Item 245, WP 1803 00

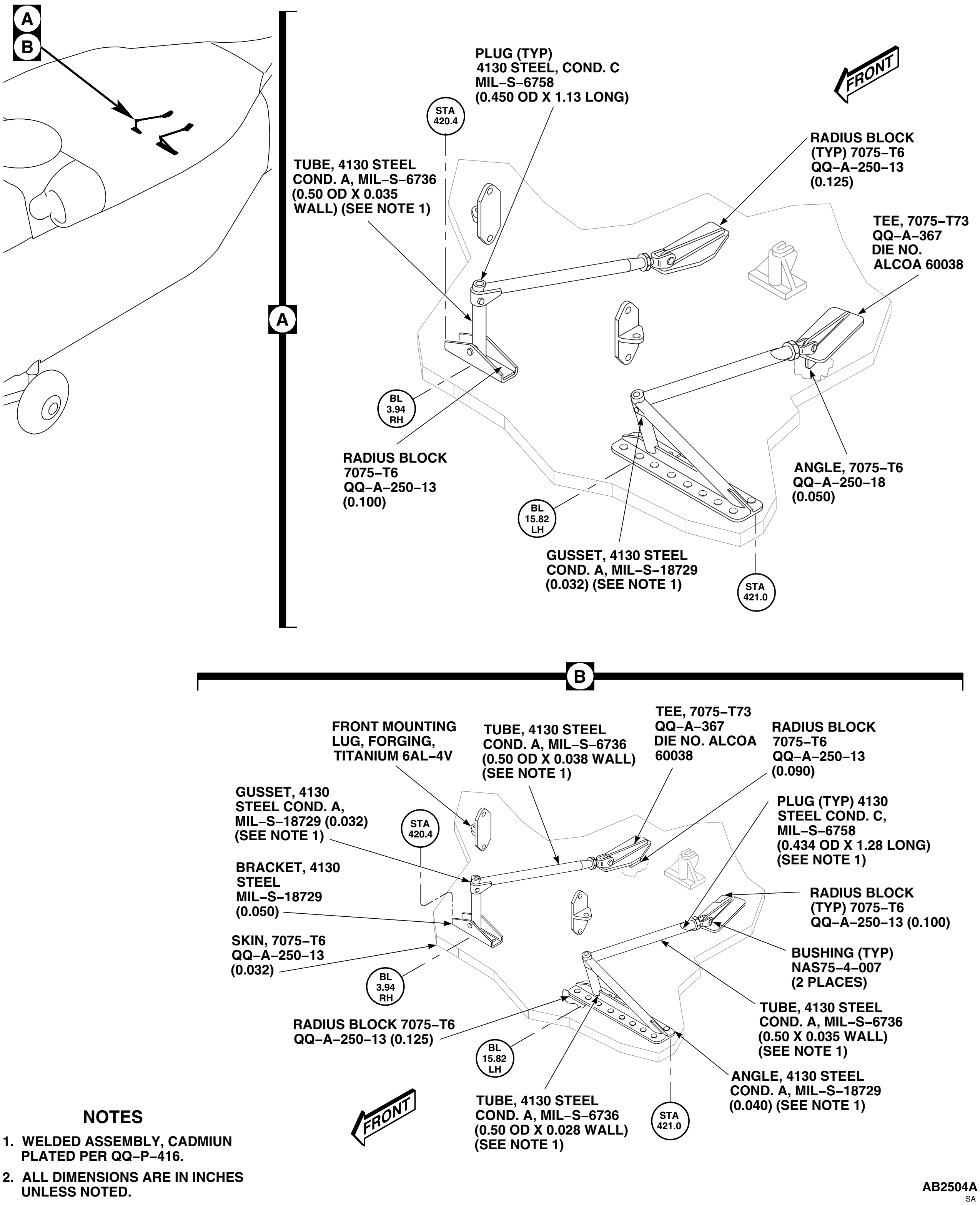
Personnel Required
Aircraft Structural Repairer MOS 15G (1)
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 55-1500-345-23
WP 0377 00
WP 0381 00
WP 1359 00
WP 1371 00
WP 1372 00
WP 1803 00

REPAIR

1. Front APU support mounts (inboard and outboard) include cadmium-plated 4130 steel tubular welded assemblies, 4130 steel attaching angles and bracket, and two aluminum alloy tee extrusion brackets (Figure 1, Detail A and Detail B).
2. Evaluate damage to aluminum parts (WP 0377 00). Damage greater than negligible limits to extrusion requires replacement of part. Replace worn bushing in tee brackets. Press-fit wet with sealing compound primer, Item 245, WP 1803 00.
3. Evaluate damage and make repairs to steel welded assemblies (WP 0377 00).
4. Refer to WP 1359 00 for hoisting, removing, and replacing APU.
5. Refer to WP 1371 00 and WP 1372 00 for removal of front support mounts.
6. Refer to WP 0377 00 for corrosion protection, watertightness, and sealing support parts to structure.
7. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

REPAIR - Continued



NOTES

1. WELDED ASSEMBLY, CADMIUN PLATED PER QQ−P−416.
2. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

Figure 1. Repair APU Front Inboard and Outboard Support Mounts.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AUXILIARY POWER PLANT SYSTEM

APU REAR SUPPORT MOUNT

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02

Material/Parts

Cloth, Abrasive, Item 83, WP 1803 00
Sealing Compound, Item 280, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
TM 55-1500-345-23
WP 0233 00
WP 0258 00
WP 0303 00
WP 0377 00
WP 0378 00
WP 0381 00
WP 1312 00
WP 1359 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Open APU compartment access doors (WP 0303 00).
3. Remove APU (WP 1359 00).
4. **UH60A UH60L** Remove rear troop seats (WP 0233 00).
5. Remove soundproofing panel from left stowage compartment (WP 0258 00).
6. **UH60A UH60L** Remove cargo netting.
7. **EH60A** Remove condenser pallet (WP 1312 00).
8. Position assistant inside rear compartment to help remove and install fasteners.
9. Remove pin and collar assemblies holding rear support mount to airframe (Figure 1, Detail A) (TM 1-1500-204-23).
10. Touch up damaged protective finishes (WP 0381 00 and TM 55-1500-345-23).

INSTALL

1. Turn off all electrical power.
2. **QA** Install all rear support mount pin and collar assemblies with sealing compound, Item 280, WP 1803 00 (WP 0377 00).
3. Position rear support mount on airframe.
4. **QA** Install required pin and collar assemblies holding rear support mount to airframe (Figure 1, Detail A) (TM 1-1500-204-23).

INSTALL - Continued

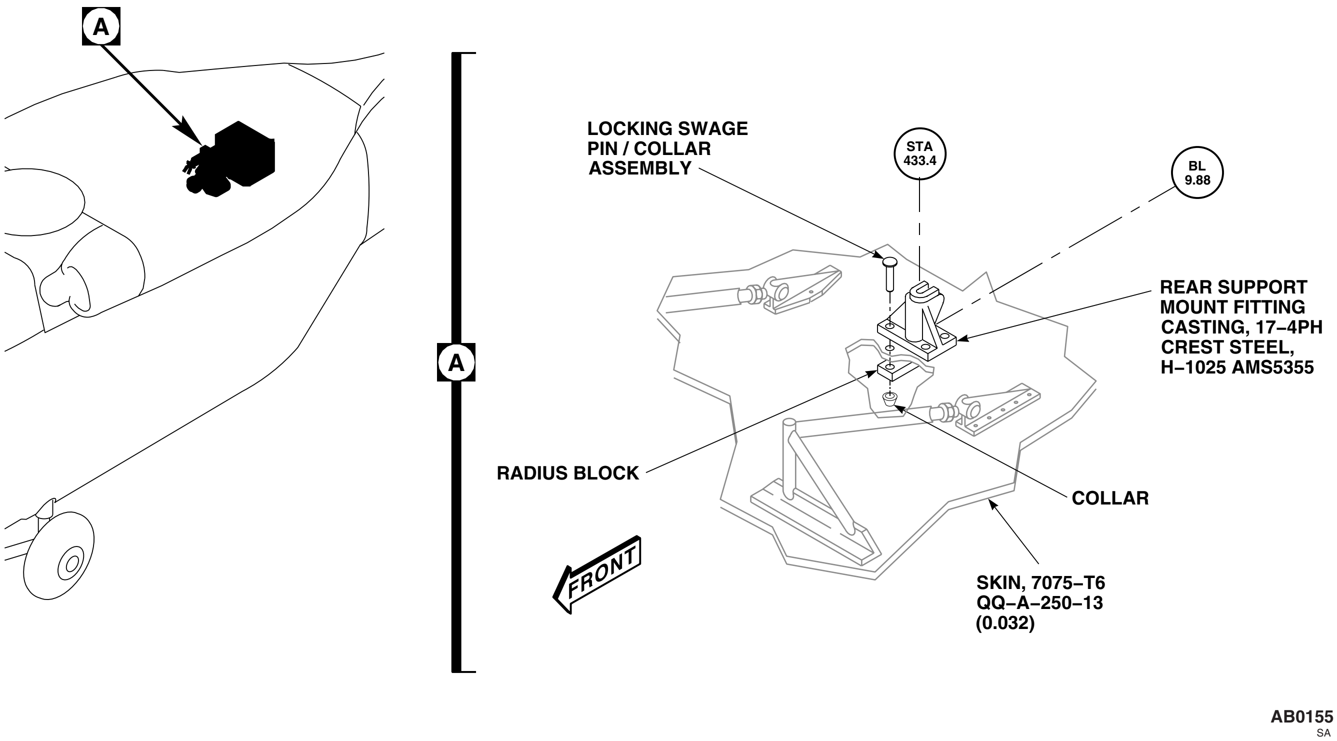


Figure 1. Replace APU Rear Support Mount.

5. For corrosion protection and waterproofing, seal support mount parts to structure (WP 0377 00).
6. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).
7. Install APU (WP 1359 00).
8. **EH60A** Install condenser pallet (WP 1312 00).
9. **UH60A UH60L** Install cargo netting.
10. Install soundproofing panel to left stowage compartment (WP 0258 00).
11. **UH60A UH60L** Install rear troop seats (WP 0233 00).
12. Make sure area is clean and free of foreign material.
13. Close APU compartment access doors (WP 0303 00).

REPAIR

1. The APU rear mount consists of a mounting lug attached to the APU and a support fitting installed on the transition section upper plating at STA 433.44. The mounting lug, which engages the support fitting, is made from 17-4PH corrosion-resistant steel, and threaded into the APU base. The support fitting, made from 17-4PH steel casting, is installed on the upper plating support structure with Hi-Lok fasteners, HL20PB5-3 and HL86PBTW-8. The support structure, made from formed aluminum alloy sheet stock, reinforces the area between stringers for installing the support fitting (Figure 2, Detail A and Detail B).
2. Evaluate damage to the support structure using (WP 0377 00).

REPAIR - Continued

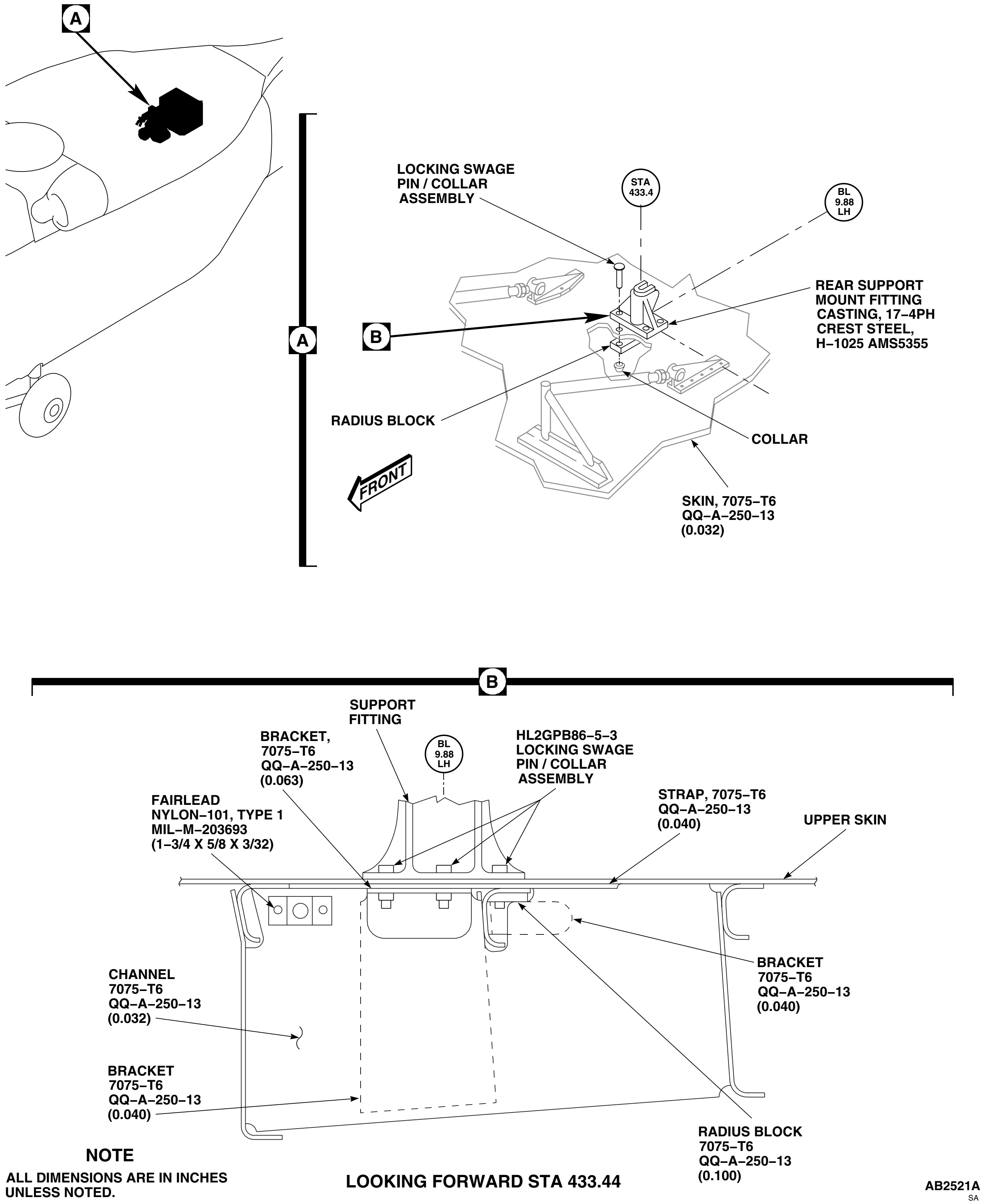


Figure 2. Repair APU Rear Support Mount.

REPAIR - Continued

3. Negligible damage to the mounting lug and support fitting is limited to nicks, scratches, and abrasions or wear, no deeper than 5% of material thickness at point of damage. Blend damage smoothly to surrounding area and polish with 240-grit abrasive cloth, Item 83, WP 1803 00. Damage greater than negligible limits must be reviewed by engineering authority for damage evaluation and either repair or replacement.
4. Refer to WP 0378 00 for typical support structure repair.
5. Refer to WP 1359 00 for hoisting, removing, or installing APU.
6. Refer to WP 0377 00 for corrosion protection, watertightness, and sealing support parts to structure.
7. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

AUXILIARY POWER PLANT SYSTEM

APU IGNITER PLUG

INITIAL SETUP:

Tools and Special Tools

Powerplant Repairer's Toolkit, SC 5180-99-B07
Torque Wrench, 30 - 200 in. lbs.

Material/Parts

Antiseize Compound, Item 51, WP 1803 00
Wire, Nonelectrical, Item 353, WP 1803 00
Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

Aircraft Powerplant Repairer MOS 15B (1)

References

WP 0303 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Open APU compartment access doors (WP 0303 00).

WARNING

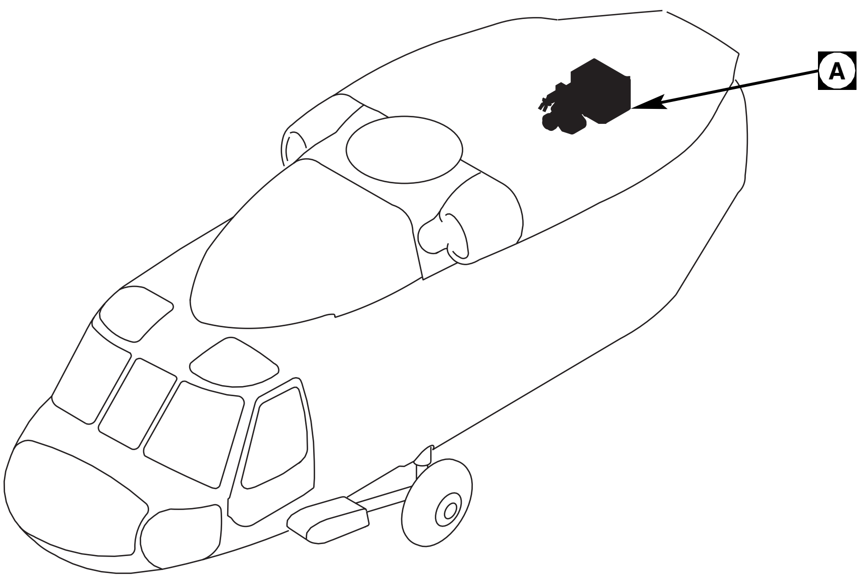
Injury or death and damage to equipment will result if ignition cable is disconnected within 30 minutes of running or attempting to start APU. Voltages used can cause arcing which may result in severe burns. Use extreme care when working with ignitions. Failure to observe all precautions may result in serious injury or death.

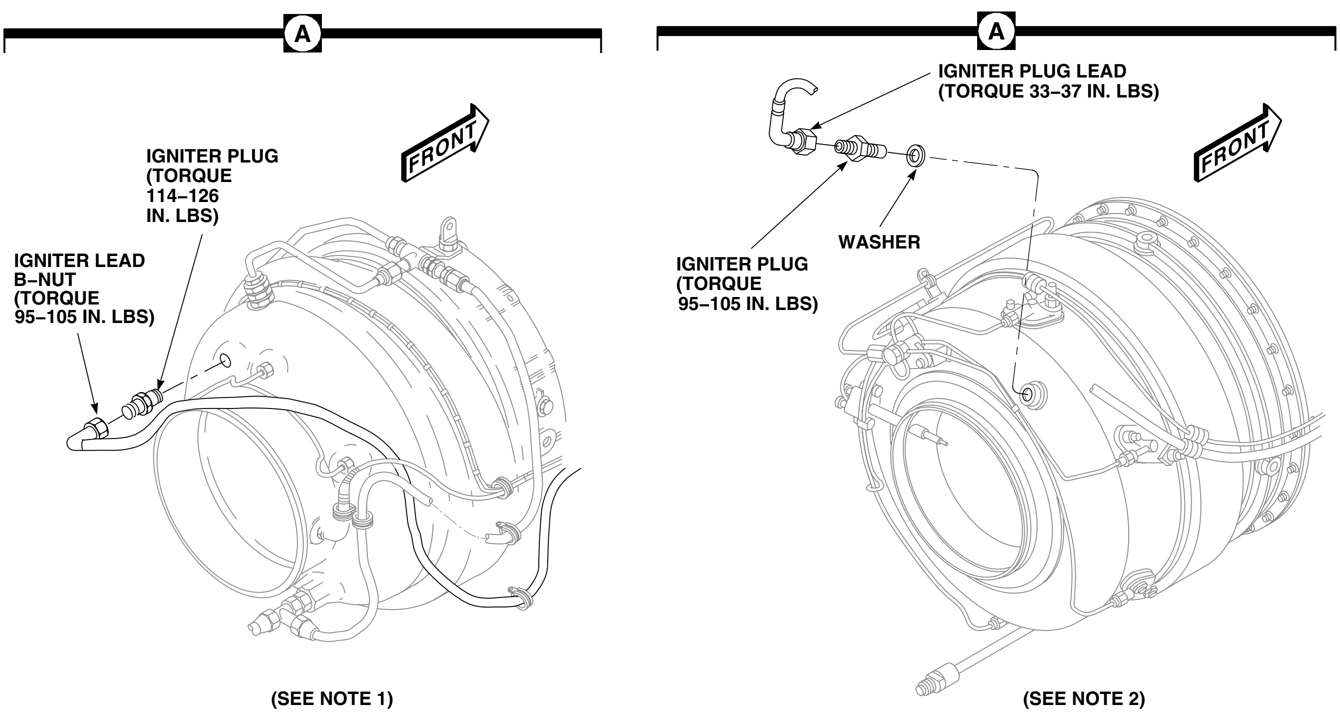
3. Disconnect igniter plug lead from igniter plug.
4. On APU, 116305-100 through 116305-300 series, loosen and remove igniter plug (Figure 1, Detail A).
5. On APU, 3800480-1 and 3800480-2, loosen and remove igniter plug and washer (Figure 1, Detail A).

INSTALL

1. Turn off all electrical power.
2. Open APU compartment access doors (WP 0303 00).
3. Apply antiseize compound, Item 51, WP 1803 00, to igniter plug threads.
4. **QA** On APU, 3800480-1 and 3800480-2, install washer on igniter plug (Figure 1, Detail A).
5. **QA** Install igniter plug in combustor case. **TORQUE IGNITER PLUG TO 95 - 105 INCH-POUNDS.**
6. **QA** Connect igniter plug lead to igniter plug. **TORQUE IGNITER PLUG LEAD TO 33 - 37 INCH-POUND.**
7. **QA** Using nonelectrical wire, Item 357, WP 1803 00, lockwire igniter plug lead.
8. **QA** On APU 116305-100 through 116305-300 series, install igniter plug in combustor case (Figure 1, Detail A). **TORQUE IGNITER PLUG TO 114 - 126 INCH-POUNDS.**
9. **QA** Connect igniter plug lead to igniter plug. **TORQUE IGNITER PLUG LEAD TO 95 - 105 INCH-POUNDS.**
10. **QA** Using nonelectrical wire, Item 353, WP 1803 00, lockwire igniter plug lead.
11. Make sure area is clean and free of foreign material.

INSTALL - Continued





NOTES

1. APU 116305–100 THROUGH 116305–300.
2. APU 3800480–1 THROUGH 3800480–2.

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Figure 1. Replace APU Igniter Plug.

INSTALL - Continued

12. Close APU compartment access doors (WP 0303 00).

END OF WORK PACKAGE

UNIT LEVEL

MISSION EQUIPMENT

AEROMEDICAL EVACUATION KIT INSPECTION

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

INSPECT

Inspect floor support for cracks and missing or loose rivets. If damage is found, repair subpart.

END OF WORK PACKAGE

UNIT LEVEL

MISSION EQUIPMENT

EXTERNAL STORES SUPPORT SYSTEM (ESSS) INSPECTIONS

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Spring Scale, AAA-S-13
Torque Wrench, 150 - 1000 in. lbs.

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23

INSPECT FUEL HOSE

1. Inspect fuel hose, fuel tee unions, elbows, and connections for signs of leaking or damage (TM 1-1500-204-23).
2. Replace damaged or leaking components as necessary.

INSPECT PNEUMATIC HOSE

1. Inspect pneumatic hose, check valves, and connections for signs of leaking or damage (TM 1-1500-204-23).
2. Replace damaged or leaking components as necessary.

INSPECT EXPANDABLE PIN AND BOLT

1. Inspect expandable pin for missing or damaged segments, for missing or damaged wedges (between segments and at top and bottom of pins) and for cracked or damaged washers. **NONE ALLOWED.** If any segments, wedges or washers are damaged or missing, replace expandable pin.
2. Inspect expandable pin closing torque whenever new pin is installed or when components that are connected by expandable pins are replaced.

WARNING

CSI

Verification of proper torque required to lock pin or bolt is a critical characteristic.

3. To check expandable pins and bolts, do this:
   a. Insert unexpanded pin or bolt in its bushing holes.
   b. Using torque wrench, turn hexhead until locking pin snaps into locking tab hole. Note torque needed to lock pin or bolt.

INSPECT EXPANDABLE PIN AND BOLT - Continued

NOTE

If earlier configuration pin without hex head is installed, torque must be measured with spring scale and tie wire attached to hex hole in handle end. Force required to close with scale pulling at right angle to long section of handle shall be 20 to 40 pounds.

c. **ON HSS TO FUSELAGE PINS, TORQUE MUST BE BETWEEN 150 - 300 INCH-POUNDS MEASURED BY TORQUING HEX HEAD MACHINED INTO HANDLE PIVOT.** Replace pins that are not within limits.

NOTE

If earlier configuration pin without hex head is installed torque can be measured with spring scale and tie wire attached to hex hole in handle end. Force required to close with scale pulling at right angle to long section of handle shall be between 36 to 72 pounds.

d. **ON SUPPORT STRUT PINS, TORQUE MUST BE BETWEEN 150 - 300 INCH-POUNDS MEASURED BY TORQUING HEX HEAD MACHINED INTO HANDLE PIVOT.** Replace pins that are not within limits.

e. **ON FITTING TO HSS FITTING (BL 80.0 AND BL 112.0) EXPANDABLE PINS, TORQUE MUST BE BETWEEN 150 - 300 INCH-POUNDS.** Replace pins that are not within limits.

f. **ON ELEVATION LINK EXPANDABLE BOLTS, TORQUE MUST BE BETWEEN 80 - 150 INCH-POUNDS.** Replace bolts that are not within limits.

END OF WORK PACKAGE

UNIT LEVEL

MISSION EQUIPMENT

MAIN ROTOR BLADE EROSION PROTECTION KIT INSPECTIONS

INITIAL SETUP:

Tools and Special Tools
Airframe Repairer's Toolkit, SC 5180-99-B02

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 1485 00
WP 1486 00

INSPECT POLYURETHANE TAPE

1. Inspect 2 inch tape for wear. **NO HOLES ALLOWED.**
2. Inspect 8 inch tape for wear. **WEAR SHALL NOT EXCEED 1.0 INCH ON UPPER OR LOWER SIDE OF BLADE.**
3. Inspect upper and lower trailing edge of tape for disbonding. **DISBONDING SHALL NOT EXCEED 0.25-INCH-WIDTH FOR ANY SPANWISE LENGTH OR 0.5-INCH-WIDTH FOR 6.0 INCH SPANWISE LENGTH OR 1.0 INCH-WIDTH FOR 2.0 INCH SPANWISE LENGTH.**
4. Inspect outboard end of tape segment for disbonding. **DISBONDING SHALL NOT EXCEED 1.0 INCH SPANWISE LENGTH.**
5. Inspect inboard end of tape segment for disbonding. **DISBONDING SHALL NOT EXCEED 0.5-INCH SPANWISE LENGTH.**
6. Inspect for internal disbonding of tape segment. **DISBONDING SHALL NOT EXCEED 6.0 SQUARE INCHES.** There is no limit to the number of disbonds providing material is not torn.

INSPECT TIP CAP POLYURETHANE TAPE

1. Inspect main rotor blade tip cap polyurethane coating. **WEAR THROUGH TO BLADE SURFACE NOT ALLOWED.**

NOTE

Do not reapply polyurethane coating on inspection zones (WP 1485 00, Figure 1).

2. If worn, repair polyurethane tape (WP 1486 00).

INSPECT TIP CAP POLYURETHANE BOOT

1. Inspect main rotor blade tip cap polyurethane boot. **WEAR THROUGH TO BLADE SURFACE NOT ALLOWED.**
2. Inspect upper and lower trailing edge of boot for disbonding. **DISBONDING SHALL NOT EXCEED 0.25-INCH-WIDTH FOR ANY SPANWISE LENGTH OR 0.5-INCH-WIDTH FOR 6.0 INCHES-SPANWISE LENGTH OR 1.0 INCH-WIDTH FOR 2.0 INCHES-SPANWISE LENGTH.**

INSPECT TIP CAP POLYURETHANE BOOT - Continued

3. Inspect inboard end of boot segment for disbonding. **DISBONDING SHALL NOT EXCEED 0.5-INCH SPAN-WISE LENGTH.**
4. Inspect for internal disbonding of boot segment. **DISBONDING SHALL NOT EXCEED 6.0 SQUARE INCHES.** There is no limit to the number of disbonds providing material is not torn.

END OF WORK PACKAGE

UNIT LEVEL

MISSION EQUIPMENT

TAIL ROTOR BLADE EROSION PROTECTION KIT INSPECTIONS

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required
UH-60 Helicopter Repairer MOS 15T (1)

References
WP 1490 00

INSPECT POLYURETHANE TAPE

1. Inspect 4 inch polyurethane tape for wear. **WEAR SHALL NOT EXCEED 0.5-INCH-WIDTH ON TRAILING EDGE. NO HOLES ALLOWED ON ANY OTHER SURFACE OF TAPE.**
2. Inspect outboard end of tape segment for disbonding. **DISBONDING SHALL NOT EXCEED 1.0 INCH.**
3. Inspect upper and lower trailing edge of tape for disbonding. **DISBONDING SHALL NOT EXCEED 0.25-INCH-WIDTH FOR ANY SPANWISE LENGTH OR 0.5-INCH-WIDTH FOR 6.0 INCHES SPANWISE LENGTH OR 1.0 INCH-WIDTH FOR 2 INCHES SPANWISE LENGTH.**
4. Inspect inboard end of tape segment for disbonding. **DISBONDING SHALL NOT EXCEED 0.5-INCH SPANWISE LENGTH.**
5. Inspect for internal disbonding of tape segment. **DISBONDING SHALL NOT EXCEED 6.0 SQUARE INCHES.** There is no limit to the number of disbonds providing material is not torn.
6. Inspect for blistering of polyurethane tape. **BLISTERING SHALL NOT EXCEED 0.2 SQUARE INCH AREA WITHIN 2 INCHES OF EACH OTHER.** If blistering exceeds limits, replace polyurethane coating (WP 1490 00).

INSPECT TIP CAP POLYURETHANE COATING

1. Inspect tail rotor blade tip cap polyurethane coating. **WEAR THROUGH TO BLADE SURFACE NOT ALLOWED.**
2. If worn, repair polyurethane coating (WP 1490 00).

END OF WORK PACKAGE

UNIT LEVEL

MISSION EQUIPMENT

VOLCANO MINE DISPENSING SYSTEM INSPECTIONS

INITIAL SETUP:

Tools and Special Tools
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02

Personnel Required
Aircraft Structural Repairer MOS 15G (1)
Assistants - (4)
UH-60 Helicopter Repairer MOS 15T (1)

References
TM 9-1095-208-13&P
WP 1540 00

TEST SYSTEM BIT

WARNING

Injury to personnel and damage to equipment will result if mine canisters are installed during Bit test. Remove canisters before performing test.

1. Remove mine canisters, if installed (WP 1540 00).

NOTE

If external power is used, make sure backup pump circuit breaker is pulled out. If APU is used, wait until backup pump goes off line then pull backup pump circuit breaker.

2. Turn on electrical power. Make sure upper console No. 1 Primary ESS JTSN INBD, No. 1 Primary ESS JTSN OUTBD, ESS BUS JTSN INBD and ESS BUS JTSN OUTBD circuit breakers are pushed in.
3. Remove DCU cover.
4. Ensure DCU Fire Circuit Switch is off.
5. Make sure safety pin and streamer assembly are installed.
6. Toggle DCU POWER switch to POWER and check for the following:
   a. For approximately 1 second, all digital indicators will flash 8s or 88s, except POWER and DCU FAILURE indicators. POWER indicator will display a steady ON. DCU FAILURE indicator will flash F. Check all displays for burned out segments. If any segments are burned out, replace DCU.
   b. All displays, except POWER display will then go blank.
   c. After approximately 15 seconds, CANISTER REMAINING indicators will display flashing 88s, POWER indicator will display a steady ON and all other indicators will be blank.
   d. Place TEST/OVERRIDE switch to TEST. Hold until 8888 goes out. After brief pause, QUANTITY indicator will flash 99 and ERROR indicator will display a flashing 9.
   e. If any of results do not occur, troubleshoot system (TM 9-1095-208-13&P).
7. Turn off DCU.
8. Turn off electrical power.

TEST SYSTEM BIT - Continued

9. Annotate DA Form 2408-13-1 (Aircraft Inspection and Maintenance Record) to show completion of test.

TEST SYSTEM ARM

WARNING

Injury to personnel and damage to equipment will result if mine canisters are installed during Arm test. Remove canisters before performing test.

1. Remove mine canisters, if installed (WP 1540 00).

NOTE

If external power is used, make sure backup pump circuit breaker is pulled out. If APU is used, wait until backup pump goes off line then pull backup pump circuit breaker.

2. Turn on electrical power. Make sure upper console No. 1 Primary ESS JTSN INBD, No. 1 Primary ESS JTSN OUTBD, ESS BUS JTSN INBD and ESS BUS JTSN OUTBD circuit breakers are pushed in.
3. Make sure ICP VOLCANO ARM switch is in safe position (DOWN) and P/F/ARMED is in off (OUT) position.
4. Depress both DRAG BEAM switches and hold.
5. Place ICP ARM switch to ARMED position (pull switch out and up).
6. Turn on DCU.
7. Verify ICP ARMED indicator is flashing ″ARMED″. Verify DCU ERROR indicator is flashing a 1. Disregard other DCU and ICP displays. If results are not indicated, troubleshoot system (TM 9-1095-208-13&P).
8. Place ICP ARM switch to safe (pull switch out and down).
9. Toggles DCU POWER switch to OFF then to power. Verify that no error code 1 is present on DCU. If error code appears, troubleshoot system (TM 9-1095-208-13&P).
10. Release DRAG BEAM switches.
11. Depress and hold Go Around switch on pilot cyclic stick.
12. Toggle DCU POWER switch to off then to POWER.
13. Verify ICP ARMED indicator is flashing ″ARMED″. Verify DCU ERROR indicator is flashing a 2. Disregard other DCU and ICP displays. If results are not indicated, troubleshoot system (TM 9-1095-208-13&P).
14. Release Go Around switch.
15. Repeat step 11. through step 13. for copilot stick.
16. Toggle DCU POWER switch to off.
17. Turn off DCU.
18. Turn off electrical power.
19. Annotate DA Form 2408-13-1 (Aircraft Inspection and Maintenance Record) to show completion of test.

TEST SYSTEM JETTISON

WARNING

Injury to personnel and damage to equipment will result if jettison safety pins are installed prior to performing checks. Make sure jettison safety pins are installed prior to performing checks.

1. Turn on electrical power. Make sure upper console No. 1 primary ESS JTSN INBD, No. 1 primary ESS JTSN OUTBD, ESS BUS JTSN INBD and ESS BUS JTSN OUTBD circuit breakers are pushed in.
2. Press ICP P/F/ARMED button and observe P and F indicators.
3. If a flashing or steady F is indicated, immediately pull above circuit breakers and troubleshoot system (TM 9-1095-208-13&P).
4. If P lamp lights, jettison system has passed continuity test.
5. Press ICP P/F/ARMED button to off position. P lamp goes out.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

MISSION EQUIPMENT

MICROCLIMATE COOLING SYSTEM (MCS)

INITIAL SETUP:

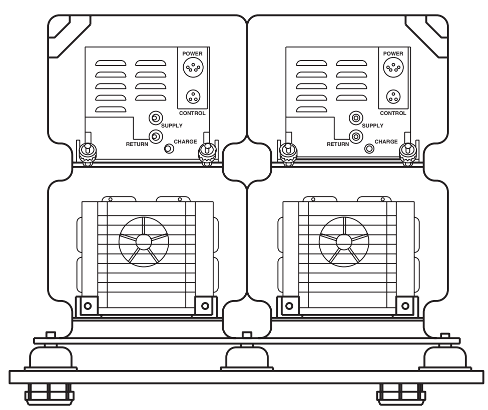
Tools and Special Tools
Aircraft Repairer's ToolKit, SC 5180-99-B02

Personnel Required
Tactical Transport Helicopter Repairer MOS 15T (1)

References
WP 0172 00

INSPECT

1. Check all connectors for corrosion.
2. Check all wiring harnesses, communication lines and cooling lines for excessive wear, fraying or breaks (Figure 1 or Figure 2).
3. Visually inspect connectors for loose connections and bent or broken pins.
4. Visually inspect wiring for broken insulation or crimped wiring.
5. Visually inspect wiring for discoloration indicating excessive heat.
6. Inspect MCU's for visible damage.
7. Inspect to verify all components and their mounts are securely mounted.
8. Visually inspect shock mounts for deterioration, free throw, bottoming and security.
9. Visually inspect controls and casings for breaks and malfunctions.
10. Make sure all electrical components power up (WP 0172 00).



GF0797
SA

Figure 1. Inspect Microclimate Cooling System (MCS).

INSPECT - Continued

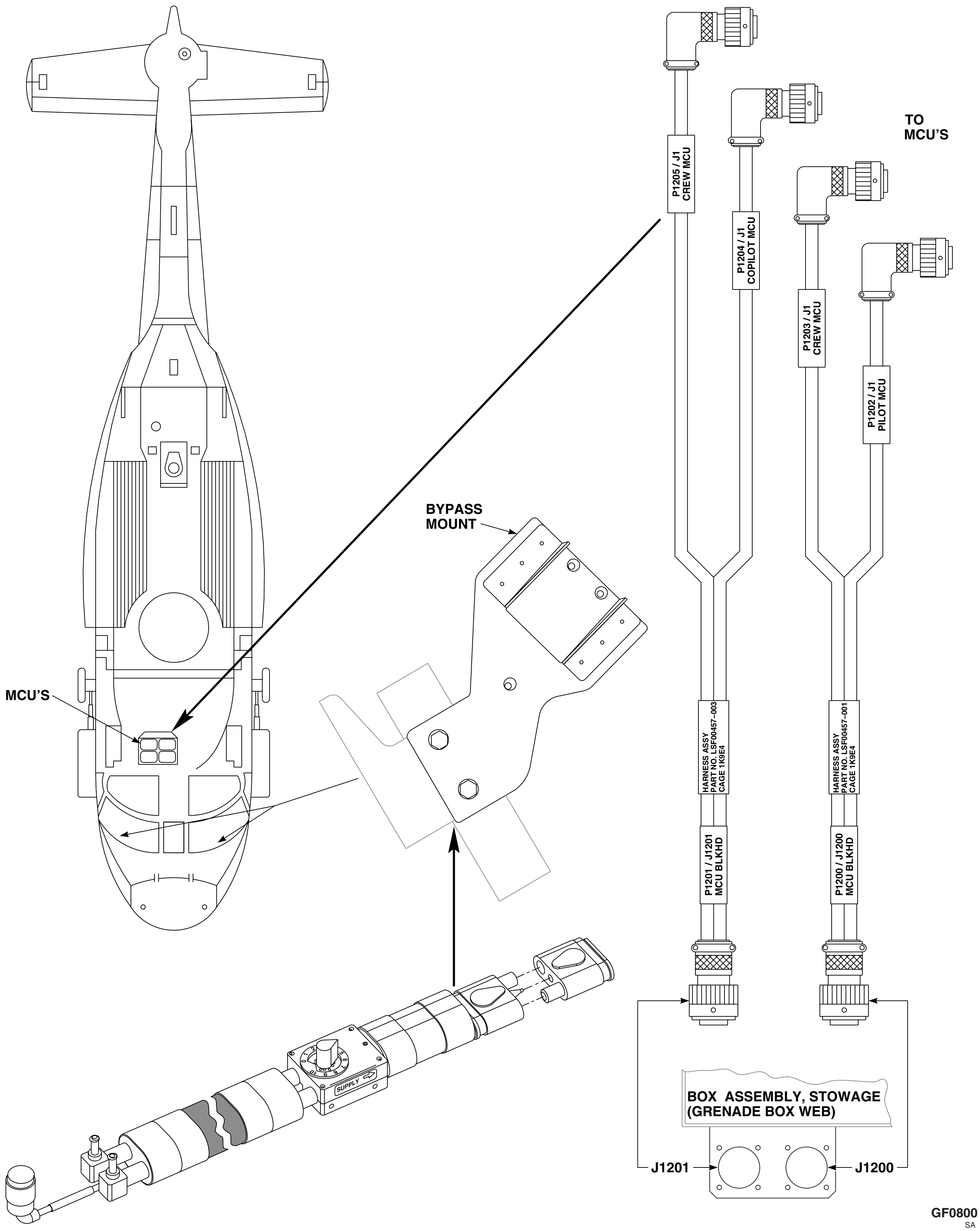


Figure 2. Inspect Microclimate Cooling System (MCS).

INSPECT - Continued

11. Check MCU for coolant leaks.

END OF WORK PACKAGE

By Order of the Secretary of the Army:

PETER J. SCHOOMAKER
General, United States Army
Chief of Staff

Official:

JOYCE E. MORROW
Administrative Assistant to the
Secretary of the Army
0609402

DISTRIBUTION:
To be distributed in accordance with Initial Distribution Number (IDN) 313749, requirements for TM 1-1520-237-23-10.

These are the instructions for sending an electronic 2028

The following format must be used if submitting an electronic 2028. The subject line must be exactly the same and all fields must be included; however only the following fields are mandatory: 1, 3, 4, 5, 6, 7, 8, 9, 10, 13, 15, 16, 17, and 27.

From: "Whomever" <whomever@wherever.army.mil>
To: 2028@redstone.army.mil

Subject: DA Form 2028

1. From: Joe Smith
2. Unit: home
3. Address: 4300 Park
4. City: Hometown
5. St: MO
6. Zip: 77777
7. Date Sent: 19--OCT–93
8. Pub no: 55--2840--229--23
9. Pub Title: TM
10. Publication Date: 04--JUL--85
11. Change Number: 7
12. Submitter Rank: MSG
13. Submitter FName: Joe
14. Submitter MName: T
15. Submitter LName: Smith
16. Submitter Phone: 123–123–1234
17. Problem: 1
18. Page: 2
19. Paragraph: 3
20. Line: 4
21. NSN: 5
22. Reference: 6
23. Figure: 7
24. Table: 8
25. Item: 9
26. Total: 123
27. Text:

This is the text for the problem below line 27.

RECOMMENDED CHANGES TO PUBLICATIONS AND BLANK FORMS For use of this form, see AR 25–30; the proponent agency is ODISC4.	Use Par reverse) for Repair Parts and Special Tool Lists (RPSTL) and Supply Catalogs/ Supply Manuals (SC/SM)	DATE 8/30/02
TO: (Forward to proponent of publication or form)(Include ZIP Code) Commander, U.S. Army Aviation and Missile Command ATTN: AMSAM–MMC–MA–NP Redstone Arsenal, 35898	FROM: (Activity and location)(Include ZIP Code) MSG, Jane Q. Doe 1234 Any Street Nowhere Town, AL 34565	

PART 1 – ALL PUBLICATIONS (EXCEPT RPSTL AND SC/SM) AND BLANK FORMS

PUBLICATION/FORM NUMBER	DATE	TITLE
TM 9–1005–433–24	16 Sep 2002	Organizational, Direct Support, And General Support Maintenance Manual for Machine Gun, .50 Caliber M3P and M3P Machine Gun Electrical Test Set Used On Avenger Air Defense Weapon System

ITEM NO.	PAGE NO.	PARA–GRAPH	LINE NO. *	FIGURE NO.	TABLE NO.	RECOMMENDED CHANGES AND REASON
1	W PG 3		2			Test or Corrective Action column should identify a different WP number.

EXAMPLE

* Reference to line numbers within the paragraph or subparagraph.

TYPED NAME, GRADE OR TITLE	TELEPHONE EXCHANGE/ AUTOVON, PLUS EXTENSION	SIGNATURE
MSG, Jane Q. Doe, SFC	788--1234	

TO: (Forward direct to addressee listed in publication)	FROM: (Activity and location) (Include ZIP Code)	DATE
Commander, U.S. Army Aviation and Missile Command ATTN: AMSAM–MMC–MA–NP Redstone Arsenal, 35898	MSG, Jane Q. Doe 1234 Any Street Nowhere Town, AL 34565	8/30/02

PART II – REPAIR PARTS AND SPECIAL TOOL LISTS AND SUPPLY CATALOGS/SUPPLY MANUALS

PUBLICATION NUMBER	DATE	TITLE

PAGE NO.	COLM NO.	LINE NO.	NATIONAL STOCK NUMBER	REFERENCE NO.	FIGURE NO.	ITEM NO.	TOTAL NO. OF MAJOR ITEMS SUPPORTED	RECOMMENDED ACTION

PART III – REMARKS (Any general remarks or recommendations, or suggestions for improvement of publications and blank forms. Additional blank sheets may be used if more space is needed.)

TYPED NAME, GRADE OR TITLE	TELEPHONE EXCHANGE/AUTOVON, PLUS EXTENSION	SIGNATURE
MSG, Jane Q. Doe, SFC	788–1234	

RECOMMENDED CHANGES TO PUBLICATIONS AND BLANK FORMS For use of this form, see AR 25-30; the proponent agency is ODISC4.	Use Par reverse) for Repair Parts and Special Tool Lists (RPSTL) and Supply Catalogs/ Supply Manuals (SC/SM)	DATE

TO: (Forward to proponent of publication or form)(Include ZIP Code) Commander, U.S. Army Aviation and Missile Command ATTN: AMSAM-MMC-MA-NP Redstone Arsenal, AL 35898 - 5000	FROM: (Activity and location)(Include ZIP Code)

PART 1 – ALL PUBLICATIONS (EXCEPT RPSTL AND SC/SM) AND BLANK FORMS

PUBLICATION/FORM NUMBER	DATE	TITLE

ITEM NO.	PAGE NO.	PARA– GRAPH	LINE NO. *	FIGURE NO.	TABLE NO.	RECOMMENDED CHANGES AND REASON

* Reference to line numbers within the paragraph or subparagraph.

TYPED NAME, GRADE OR TITLE	TELEPHONE EXCHANGE/ AUTOVON, PLUS EXTENSION	SIGNATURE

TO: (Forward direct to addressee listed in publication) Commander, U.S. Army Aviation and Missile Command ATTN: AMSAM-MMC-MA-NP Redstone Arsenal, AL 35898	FROM: (Activity and location) (Include ZIP Code)	DATE

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PART III – REMARKS (Any general remarks or recommendations, or suggestions for improvement of publications and blank forms. Additional blank sheets may be used if more space is needed.)

TYPED NAME, GRADE OR TITLE	TELEPHONE EXCHANGE/AUTOVON, PLUS EXTENSION	SIGNATURE

The Metric System and Equivalents

Linear Measure

1 centimeter = 10 millimeters = .39 inch
1 decimeter = 10 centimeters = 3.94 inches
1 meter = 10 decimeters = 39.37 inches
1 dekameter = 10 meters = 32.8 feet
1 hectometer = 10 dekameters = 328.08 feet
1 kilometer = 10 hectometers = 3,280.8 feet

Weights

1 centigram = 10 milligrams = .15 grain
1 decigram = 10 centigrams = 1.54 grains
1 gram = 10 decigram = .035 ounce
1 decagram = 10 grams = .35 ounce
1 hectogram = 10 decagrams = 3.52 ounces
1 kilogram = 10 hectograms = 2.2 pounds
1 quintal = 100 kilograms = 220.46 pounds
1 metric ton = 10 quintals = 1.1 short tons

Liquid Measure

1 centiliter = 10 milliters = .34 fl. ounce
1 deciliter = 10 centiliters = 3.38 fl. ounces
1 liter = 10 deciliters = 33.81 fl. ounces
1 dekaliter = 10 liters = 2.64 gallons
1 hectoliter = 10 dekaliters = 26.42 gallons
1 kiloliter = 10 hectoliters = 264.18 gallons

Square Measure

1 sq. centimeter = 100 sq. millimeters = .155 sq. inch
1 sq. decimeter = 100 sq. centimeters = 15.5 sq. inches
1 sq. meter (centare) = 100 sq. decimeters = 10.76 sq. feet
1 sq. dekameter (are) = 100 sq. meters = 1,076.4 sq. feet
1 sq. hectometer (hectare) = 100 sq. dekameters = 2.47 acres
1 sq. kilometer = 100 sq. hectometers = .386 sq. mile

Cubic Measure

1 cu. centimeter = 1000 cu. millimeters = .06 cu. inch
1 cu. decimeter = 1000 cu. centimeters = 61.02 cu. inches
1 cu. meter = 1000 cu. decimeters = 35.31 cu. feet

Approximate Conversion Factors

To change	***To***	***Multiply by***	***To change***	***To***	***Multiply by***
inches	centimeters	2.540	ounce-inches	Newton-meters	.007062
feet	meters	.305	centimeters	inches	.394
yards	meters	.914	meters	feet	3.280
miles	kilometers	1.609	meters	yards	1.094
square inches	square centimeters	6.451	kilometers	miles	.621
square feet	square meters	.093	square centimeters	square inches	.155
square yards	square meters	.836	square meters	square feet	10.764
square miles	square kilometers	2.590	square meters	square yards	1.196
acres	square hectometers	.405	square kilometers	square miles	.386
cubic feet	cubic meters	.028	square hectometers	acres	2.471
cubic yards	cubic meters	.765	cubic meters	cubic feet	35.315
fluid ounces	milliliters	29,573	cubic meters	cubic yards	1.308
pints	liters	.473	milliliters	fluid ounces	.034
quarts	liters	.946	liters	pints	2.113
gallons	liters	3.785	liters	quarts	1.057
ounces	grams	28.349	liters	gallons	.264
pounds	kilograms	.454	grams	ounces	.035
short tons	metric tons	.907	kilograms	pounds	2.205
pound-feet	Newton-meters	1.356	metric tons	short tons	1.102
pound-inches	Newton-meters	.11296			

Temperature (Exact)

°F	Fahrenheit temperature	5/9 (after subtracting 32)	Celsius temperature	°C

PIN: 083249-000